

The Thermodynamics of Phase and Reaction Equilibria

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by

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To the memory of my parents

Nesrin and Ahmet

Preface

The curriculum of most chemical engineering departments contains two thermodynamics courses. While the fundamental thermodynamics concepts (first and second laws) are introduced in the first one, the second course is devoted to mixtures and mixture phase equilibrium as well as reaction equilibrium.

In his article published in *Chemical Engineering Education* (Winter issue, pp. 284–287 and 291) in 2002, M.J. Misovich stated that: “I believe phase equilibrium thermodynamics is the most conceptually difficult undergraduate chemical engineering class. Even students who perform calculations satisfactorily seem confused over the meaning of what they have learned. Phase equilibrium is the single undergraduate chemical engineering class in which abstract concepts are presented to the near exclusion of practical applications.” I totally agree with Misovich that the lack of an intuitive feel for certain concepts, such as partial molar Gibbs energy, fugacity, and activity, makes the subject of phase and reaction equilibria extremely confusing for students. They are at a total loss in dealing with practical problems related to the chemical and petroleum industries. The lack of sound foundation in abstract concepts also hampers students when they take thermodynamics courses at graduate level.

My intention in writing this book is to equip students with a confident grasp of the fundamentals and numerical tools so that they can specialize in advanced topics at a later stage. Chapters 1–8 deal with the principles of phase and reaction equilibria. After briefly reviewing the first and second laws of thermodynamics in Chapter 1, general expressions to calculate changes in internal energy, enthalpy, and entropy are covered in Chapters 2 and 3. Chapter 4 talks about the equilibrium criteria and phase stability. The concept of fugacity is introduced for pure substances in Chapter 5. The subject of Chapter 6 is how to calculate mixture properties. Chapter 7 first presents equations for the calculation of component fugacities in a mixture, and then shows how to use the abstract term fugacity in phase equilibrium calculations. Excess mixture properties and activity coefficients are covered in Chapter 8. Once the first eight chapters are covered, it is up to the instructor to cover either phase equilibrium or reaction equilibrium first. Chapters 9–12 are devoted to phase equilibrium applications, and Chapters 13 and 14 deal with reaction equilibrium calculations. A one-semester course is sufficient to cover the material presented in this book.

Mathematics is an integral part of thermodynamics. Students need to understand the mathematics well enough to solve the problems related to phase and reaction equilibria. Engineering students’ weaknesses in mathematics is a common problem. Some textbooks avoid this problem by spoon-feeding the subject matter. Although this approach makes the life of students easier, when confronted with a

problem a little bit different from ones they have previously solved, they usually fail to solve it. Moreover, phase and reaction equilibria calculations (especially with cubic equations of state) require solutions of nonlinear equations. Some textbooks avoid these calculations by providing a software program in the form of a CD-ROM. Students may arrive at an answer, with little appreciation or interpretation of the result, by simply plugging in the given values. I am totally against this approach. In this book, students are encouraged to use MATHCAD[®] by writing their own programs. In this way, they can not only identify their mistakes but also get a feeling about the orders of magnitude of various quantities. In the example problems requiring MATHCAD[®], I have provided the results of the intermediate steps so that students can follow the solution easily.

Two of my colleagues, Güniz Gürüz and Halil Kalıpçılar, and one of my students, Özge Saatçı, kindly read the entire manuscript and made many helpful suggestions. My thanks are also extended to Ahmet Nedim Eraslan and Ali Çulfaz for their advice and comments. I appreciate the help provided by my students—Özgen Yalçın, Seval Gündüz, and Gökçe S. Avcıoğlu—for proof reading and checking the numerical calculations. Above all, I would like to thank my wife, Ayşe, for her support, patience, and understanding during the preparation of this book.

Suggestions and criticism from instructors and students using this book would be appreciated.

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NOTATION

Symbols

A	Helmholtz energy
A	dimensionless parameter, cubic equations of state
A_i	chemical symbol for species i
a	parameter, cubic equations of state
$\hat{a}_i$	activity of species i in a mixture
B	dimensionless parameter, cubic equations of state
B	second virial coefficient
$B^{(0)}, B^{(1)}$	functions, generalized second virial coefficient correlation
b	parameter, cubic equations of state
C	number of components, phase rule
C_P, C_V	heat capacity at constant pressure and at constant volume
C_P^*, C_V^*	ideal gas heat capacity
c_i	concentration of species i
E	total energy
$\mathcal{E}$	enhancement factor
E_j	chemical element
E_K	kinetic energy
E_P	potential energy
F	feed flow rate
$\mathcal{F}$	degrees of freedom, phase rule
f_i	fugacity of pure species i
$\hat{f}_i$	fugacity of species i in a mixture
G	Gibbs energy
ΔG_{mix}	Gibbs energy change on mixing
g	acceleration of gravity
H	enthalpy
ΔH_{mix}	enthalpy change on mixing
$\mathcal{H}_i$	Henry's law constant of solute i
i	van't Hoff factor
K_a	chemical equilibrium constant
K_b	boiling point elevation constant
K_f	freezing point depression constant
$K_{f/P}$	equilibrium ratio of pure component fugacity coefficients
K_i	K -factor, y_i/x_i
K_x	equilibrium ratio of liquid phase mole fractions
K_y	equilibrium ratio of vapor phase mole fractions

K_γ	equilibrium ratio of activity coefficients
K_ϕ	equilibrium ratio of fugacity coefficients
K_i^{ow}	octanol-water partition coefficient
k_{ij}	binary interaction parameter
L	liquid flow rate
M	molecular weight
m	mass
$\dot{m}$	mass flow rate
m_i	molality of solute i
N	total number of species, phase rule
n	number of moles
$\dot{n}$	molar flow rate
n_T	total number of moles
P	pressure
$\mathcal{P}$	number of phases, phase rule
P_{ex}	external pressure
P^{sub}	sublimation pressure
P^{vap}	vapor (saturation) pressure
p_i	partial pressure of species i
Q	heat
Q_a	apparent equilibrium constant (or reaction quotient)
R	gas constant
$\mathcal{R}$	number of independent reactions, phase rule
S	entropy
$\mathcal{S}$	number of stoichiometric constraints, phase rule
S_{gen}	entropy generation
T	temperature
T_b	boiling temperature
T_f	freezing temperature
T_m	melting temperature
t	time
t	total number of chemical species
U	internal energy
V	volume
V	vapor flow rate
ΔV_{mix}	volume change on mixing
W	work
W_s	shaft work
X_i	fractional conversion of reactant i
x_i	mole fraction of species i in vapor or liquid phase

y_i	mole fraction of species i in vapor phase
Z	compressibility factor
$Z^{(0)}, Z^{(1)}$	functions, generalized compressibility factor correlation
Z_{RA}	Rackett parameter
z_i	mole fraction of species i in feed stream

Greek symbols

α	function, cubic equations of state
$[\alpha]$	stoichiometric matrix
α_i	stoichiometric coefficient of species i
α_{ij}	relative volatility of i with respect to j
$\overline{\alpha}$	summation of stoichiometric coefficients
β	coefficient of thermal expansion
$[\beta]$	element-by-species matrix
γ_i	activity coefficient of species i
γ_i^*	asymmetric activity coefficient of species i
δ	solubility parameter
ϵ	molar extent of reaction
θ	surface area fraction
κ	isothermal compressibility
λ_i	molar Gibbs energy of pure species i at unit fugacity
ξ	intensive extent of reaction
Π	osmotic pressure
ρ	density
Φ	volume fraction
ϕ_i	fugacity coefficient of pure species i
$\widehat{\phi}_i$	fugacity coefficient of species i in a mixture
ω	acentric factor
ω_i	mass (weight) fraction of species i

Operators

δ	denotes a path dependent variation
$\Delta\varphi$	$\varphi_{\text{final}} - \varphi_{\text{initial}}$ (closed system); $\varphi_{\text{out}} - \varphi_{\text{in}}$ (open system)
d	total differential
∂	partial differential
$\int$	integral
$\ln$	natural logarithm

$\log$	common logarithm
$\prod$	cumulative product
$\sum$	cumulative summation

Overlines

$\tilde{\varphi}$	per mole
$\widehat{\varphi}$	per unit mass
$\overline{\varphi}$	partial molar property

Subscripts

atm	atmosphere
c	critical point property
eq	equilibrium
f	formation
i	species
in	incoming stream
k	number of species in a mixture
mix	mixing or mixture
o	initial
out	exiting stream
r	reduced property
ref	reference
rev	reversible process
rxn	reaction
soln	solution
surr	surroundings
sys	system
T	total
t	triple point property
α, β	generic phases

Superscripts

C	condensed phase (liquid or solid)
ex	excess property on mixing
fus	property change on fusion (or melting)
IG	ideal gas property

IGM	ideal gas mixture property
IM	ideal mixture property
L	liquid phase
o	standard state
S	solid phase
sat	saturated condition at phase equilibrium
soln	solution
sub	property change on sublimation
V	vapor phase
vap	property change on vaporization
α, β	generic phases
∞	at infinite dilution

Abbreviations

CST	critical solution temperature
LCST	lower critical solution temperature
PR	Peng-Robinson
RH	relative humidity
RK	Redlich-Kwong
rref	reduced row echelon form
SRK	Soave-Redlich-Kwong
UCST	upper critical solution temperature
VDW	van der Waals

Chapter 1

Review of the First and Second Laws of Thermodynamics

The purpose of this chapter is twofold: (i) to summarize the definitions and concepts used in thermodynamics and (ii) to explain what the first and second laws of thermodynamics are and to show how to simplify the equations expressing these laws for various systems.

1.1 Definitions

1.1.1 System

Any region that occupies a volume and has a boundary is called a *system*. The volume outside the boundary is called the *surroundings* of the system. The sum of the system and its surroundings is called the *universe*. Thermodynamics considers systems only at the macroscopic level. It is convenient to distinguish between three general types of systems:

- *Isolated system*: These are the set of systems that exchange neither mass nor energy with the surroundings. For example, the universe is an isolated system.
- *Closed system*: These are the set of systems that exchange energy (in the form of heat and work) but not mass with the surroundings.
- *Open system*: These are the set of systems that exchange both mass and energy with the surroundings.

The equations available to analyze closed and open systems are different from each other. Therefore, one should properly define the system before solving the problem.

1.1.2 Property and State

In order to describe and analyze a system, some of the quantities that are characteristic of it must be known. These quantities are called *properties* and comprise volume, mass, temperature, pressure, etc. Thermodynamic properties are considered to be either *intensive* or *extensive*. When the property is proportional to the mass of the system, the property is *extensive*, i.e. volume, kinetic energy, potential energy. On the other hand, when the property is independent of the mass of the system, the property is *intensive*, i.e. viscosity, refractive index, density, temperature, pressure, mole fraction.

Specific (or molar) properties are extensive properties divided by the total mass (or total moles) of the system, i.e.

$$\text{Specific property} = \frac{\text{Extensive property}}{\text{Total mass}}, \quad \text{Molar property} = \frac{\text{Extensive property}}{\text{Total moles}}. \quad (1.1-1)$$

If φ represents any extensive property, then Eqn (1.1-1) is expressed as

$$\hat{\varphi} = \frac{\varphi}{m} \quad \text{and} \quad \tilde{\varphi} = \frac{\varphi}{n}, \quad (1.1-2)$$

where m and n are the total mass and moles, respectively. Note that all specific (or molar) properties are intensive.

A complete list of the properties of a system describes its *state*. Consider a function

$$w = f(x, y) \quad (1.1-3)$$

in which there are three variables: w is dependent, and x and y are independent. In thermodynamics we would say that “*the state of the system, w , is fixed when the thermodynamic properties x and y are specified.*” Note that the mathematical term “point” is equivalent to the thermodynamic term “state.”

The number of independent intensive properties needed to fix the state of a system is called the *degrees of freedom*. The Gibbs phase rule specifies the number of degrees of freedom, $\mathcal{F}$, for a given system at equilibrium and is expressed in the form¹

$$\mathcal{P} + \mathcal{F} = \mathcal{C} + 2, \quad (1.1-4)$$

where $\mathcal{P}$ is the number of phases and $\mathcal{C}$ is the number of components.

Therefore, the state of a single-phase, single-component system can be specified by two independent intensive properties. Two properties are *independent* if one

¹The easiest way to remember Gibbs phase rule is to use the formula: *Police + Forces = Cops + 2*.

property can be varied while the other one is held constant. For example, temperature and density (or specific volume) are always independent properties, and together they can fix the state of a single-phase, single-component system.

A change of state is called a *process*, which can occur in a number of ways. Work and heat can occur only during processes and only across the boundary of the system. The path followed in going from one state to another is known as the *process path*.

1.1.3 Equilibrium

It is important to differentiate between the concepts of steady-state, uniform, and equilibrium:

- *Steady-state*: The term steady-state means that at a particular location in space the dependent variable φ does not change as a function of time, i.e.

$$\left(\frac{\partial \varphi}{\partial t}\right)_{x,y,z} = 0. \quad (1.1-5)$$

- *Uniform*: The term uniform means that at a particular instant in time the dependent variable φ is not a function of position, i.e.

$$\left(\frac{\partial \varphi}{\partial x}\right)_{y,z,t} = \left(\frac{\partial \varphi}{\partial y}\right)_{x,z,t} = \left(\frac{\partial \varphi}{\partial z}\right)_{x,y,t} = 0. \quad (1.1-6)$$

- *Equilibrium*: A system is in equilibrium if both the steady-state and uniform conditions are met simultaneously. This implies that the variables associated with the system, such as temperature, pressure, and density, are constant at all times and have the same magnitude at all positions within the system. A difference in any potential that causes a process to take place spontaneously is called a *driving force*. Driving force(s) turns out to be zero for a system in equilibrium. Thus, no work can be done by a system in equilibrium.

It should be kept in mind that thermodynamics considers systems only at equilibrium. By examining the system in its initial and final equilibrium states, it is possible to determine the heat and work interactions of the system with its surroundings during this process. Thermodynamics, however, does not consider the time it takes for the system in going from an initial equilibrium state to a final equilibrium state.

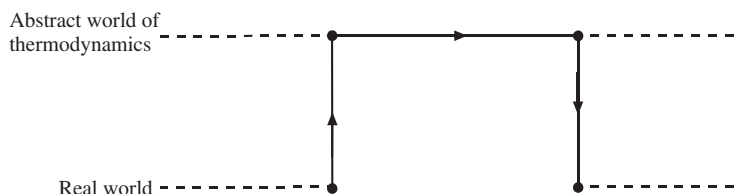


Figure 1.1: The path followed in the solution of thermodynamics problems (Perausnitz, 1986).

1.2 Concepts of the “Abstract World” of Thermodynamics

The steps followed in the solution of a thermodynamic problem are shown in Fig. 1.1. The first step is to transform the real problem into the abstract world of thermodynamics in which necessary equations are provided for solving it. Once the problem is solved in the abstract world, then it is transformed back into the real world.

In the abstract world of thermodynamics, the two concepts, i.e. “state/path function” and “reversible process,” play a vital role in the solution of problems.

1.2.1 State and Path Functions

If a system is caused to undergo a process, any property of the system whose value at the final state is the same no matter what path is used to carry out the process is called a *state function*.² On the other hand, if a property is a *path function*, its value at the final state will depend on what path is used and will be different for every path.

The quantities encountered in thermodynamics are all state functions except heat and work. The only exception to this statement is the work done by body forces, i.e. work done against a gravitational force. Body forces are *conservative forces*.³

The term “state function” in thermodynamics corresponds to the term “exact differential” in mathematics. The expression $M(x, y)dx + N(x, y)dy$ is called an *exact differential* if there exists some $\varphi = \varphi(x, y)$ for which this expression is the *total differential* $d\varphi$, i.e.

$$M(x, y)dx + N(x, y)dy = d\varphi(x, y). \quad (1.2-1)$$

²A property that is defined as the combination of other state functions is also a state function.

³A force, $\mathbf{F}$, is called *conservative* if the work done by the force is a state function. In mathematical terms, $\nabla \times \mathbf{F} = 0$.

A necessary and sufficient condition for the expression $M(x, y)dx + N(x, y)dy$ to be expressed as a total differential is that

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y. \quad (1.2-2)$$

1.2.2 Reversible Process

A process executed by a system is called *reversible* if the system and its surroundings can be restored to their initial states and leave *no net effects at all* on the physical world. Another term for reversible may be *completely erasable*. In the real world, all processes are irreversible. Processes are considered reversible only if they are executed very slowly, i.e. the driving force for the process is very small.

1.2.3 General Approach Used in the Solution of Thermodynamics Problems

Once the concepts of state/path function and reversible process are understood, the elaboration of Fig. 1.1 can be carried out as follows. Although all processes taking place in nature are irreversible, all thermodynamic properties are state functions. Thus, it is always possible to replace the path of a real process by a convenient hypothetical reversible path(s) as shown in Fig. 1.2. As long as the initial and final states are the same, the change in the state function φ is independent of the path followed during the process, i.e.

$$(\Delta\varphi)_{\text{actual irreversible path}} = (\Delta\varphi)_{\text{hypothetical reversible path}}. \quad (1.2-3)$$

1.3 Work

In thermodynamics, work can be broadly classified as expansion and non-expansion type of work. Expansion (or contraction) work is related to the change in the volume of system. Non-expansion work, on the other hand, includes shaft work, flow work, chemical work, electrical work, etc. By convention, work done on the system is considered positive.

In differential form,⁴ the *work associated with the volume change* is given by

$$\delta W = -P_{\text{ex}} dV_{\text{sys}}, \quad (1.3-1)$$

⁴Since W is a path function, it is expressed in differential form as δW rather than dW . Heat, in differential form, will also be expressed as δQ in the later stages.

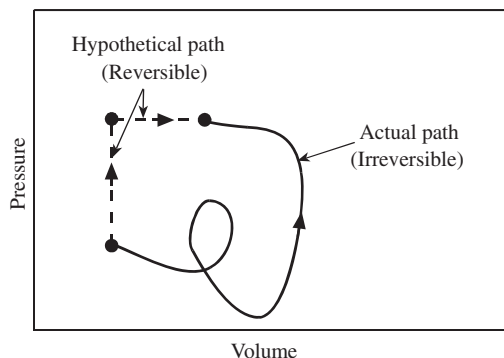


Figure 1.2: A hypothetical path replacing an actual path.

where P_{ex} represents the external pressure. Work is always done against an external force, i.e. there must be some resistance against which the force operates. Otherwise, no work is done. For example, the work done by a closed system in expanding into a vacuum is zero. For a reversible process, the difference in the driving force is very small, i.e. $P_{\text{sys}} \simeq P_{\text{ex}}$, and Eqn (1.3-1) becomes

$$\delta W_{\text{rev}} = -P_{\text{sys}} dV_{\text{sys}} \quad (1.3-2)$$

indicating that the work done by a system is a function of the properties of the system only when the work is done reversibly.

The *shaft work*, W_s , is the work done on the system by external means through a rotating shaft. Reversible shaft work in a steady-state flow process with negligible changes in kinetic and potential energies is given by

$$\delta W_{s,\text{rev}} = V_{\text{sys}} dP_{\text{sys}}. \quad (1.3-3)$$

The work associated with moving the fluid into and out of an open system is known as the *flow work*. The flow work done by a tangential stress (or shear stress) is usually considered negligible by choosing the boundaries of the system perpendicular to the flow direction. On the other hand, the flow work done by a normal stress, i.e. pressure force, is given by PV .

1.4 Paths Followed During a Process

In going from one state to another, a system may follow various paths. Depending on the path followed, the process may be:

- Isometric/isochoric (constant volume).
- Isobaric (constant pressure).
- Isothermal (constant temperature).
- Adiabatic.

An isothermal process is usually confused with an adiabatic process. An *isothermal process* is one in which the temperature of the system does not change *throughout* the process,⁵ i.e. $T_{\text{sys}} = \text{constant}$. An *adiabatic process*, on the other hand, is one in which there is no exchange of heat between the system and its surroundings, i.e. $Q = 0$. The adiabatic and isothermal processes are often opposite extremes and real processes fall in between.

It should be kept in mind that heat transfer is a “slow” process and can be considered negligible in processes taking place very “rapidly.” Since *slow* and *rapid* are relative terms, there is no clear-cut recipe to differentiate slow processes from rapid ones. One should use “engineering judgement” in the analysis of a given problem. For example, consider the following two cases in which a rigid tank filled with a high-pressure gas at ambient temperature is evacuated by:

- (i) Punching a tiny hole in the surface of the tank.
- (ii) Opening a large valve placed on the top of the tank.

Suppose that the tank is not insulated and it is required to find how the gas temperature within the tank changes with pressure. Since process (i) is rather slow, it allows heat transfer between the tank contents and ambient air to take place. Thus, the gas remaining in the tank may be considered to undergo an isothermal process. On the other hand, evacuation of the tank is very rapid in process (ii). Over the time scale of the evacuation process, heat transfer between the tank contents and ambient air is almost negligible. As a result, the gas remaining in the tank may be assumed to undergo an adiabatic process even though there is no insulation around the tank.

1.5 The First Law of Thermodynamics

The first law of thermodynamics is a statement of the conservation of energy, i.e. although energy can be transferred from one system to another in many forms, it can neither be created nor destroyed. Therefore, the total amount of energy available in the universe is constant. Consider an open system exchanging heat and work

⁵If the initial and final state temperatures are the same, this does not necessarily imply an isothermal process.

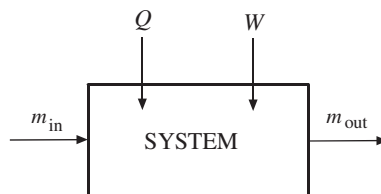


Figure 1.3: An open system exchanging mass and energy with its surroundings.

with its surroundings as shown in Fig. 1.3. The conservation statement for energy is expressed as

$$\text{Energy in} - \text{Energy out} = \text{Change in the energy of the system.} \quad (1.5-1)$$

In differential form, Eqn (1.5-1) takes the form

$$\begin{aligned} & [(\hat{H} + \hat{E}_K + \hat{E}_P)_{\text{in}} dm_{\text{in}} + \delta Q + \delta W] - (\hat{H} + \hat{E}_K + \hat{E}_P)_{\text{out}} dm_{\text{out}} \\ & = d[m(\hat{U} + \hat{E}_K + \hat{E}_P)]_{\text{sys}}, \end{aligned} \quad (1.5-2)$$

where H , E_K , E_P , and U represent enthalpy, kinetic energy, potential energy, and internal energy, respectively.

In terms of molar quantities, Eqn (1.5-2) is written as

$$\begin{aligned} & [(\tilde{H} + \tilde{E}_K + \tilde{E}_P)_{\text{in}} dn_{\text{in}} + \delta Q + \delta W] - (\tilde{H} + \tilde{E}_K + \tilde{E}_P)_{\text{out}} dn_{\text{out}} \\ & = d[n(\tilde{U} + \tilde{E}_K + \tilde{E}_P)]_{\text{sys}}. \end{aligned} \quad (1.5-3)$$

The internal energy is the sum of the kinetic energy (translational, rotational, and vibrational) and potential energies of attraction at the molecular level. Since thermodynamics is concerned with systems at macroscopic level, it is not concerned with the origin of internal energy but states that internal energy is *extensive*, i.e. its value is dependent on the extent or size of the system, and it is a *state function*.

The enthalpy is a made-up variable defined by

$$H = U + PV. \quad (1.5-4)$$

In an open flow system, enthalpy can be interpreted as the amount of energy transferred across a system boundary by a moving flow. In Eqn (1.5-4), the term PV represents the flow work.

The term W in Eqns (1.5-2) and (1.5-3) is the sum of the shaft work and the work associated with the volume change.

1.5.1 Simplification of the Energy Balance

1.5.1.1 Isolated System

Since there are no inlet and outlet streams, i.e. $dn_{\text{in}} = dn_{\text{out}} = 0$, and no exchange of heat and work, i.e. $\delta Q = \delta W = 0$, Eqn (1.5-3) reduces to

$$d[n(\tilde{U} + \tilde{E}_K + \tilde{E}_P)]_{\text{sys}} = dE_{\text{sys}} = 0 \Rightarrow E_{\text{sys}} = \text{constant}. \quad (1.5-5)$$

In an isolated system, energy is converted from one form to another but the total energy is always constant.

1.5.1.2 Closed System

Since there are no inlet or outlet streams, i.e. $dn_{\text{in}} = dn_{\text{out}} = 0$ and $n_{\text{sys}} = \text{constant}$, Eqn (1.5-3) reduces to

$$\delta Q + \delta W = n_{\text{sys}} d(\tilde{U} + \tilde{E}_K + \tilde{E}_P)_{\text{sys}}. \quad (1.5-6)$$

Integration of Eqn (1.5-6) gives⁶

$$\tilde{Q} + \tilde{W} = \Delta\tilde{U} + \Delta\tilde{E}_K + \Delta\tilde{E}_P. \quad (1.5-7)$$

1.5.1.3 Steady-State Flow System

In this case $dn_{\text{in}} = dn_{\text{out}} = dn$, $\delta W = \delta W_s$ (boundaries of the system are fixed in space) and $d[n(\tilde{U} + \tilde{E}_K + \tilde{E}_P)]_{\text{sys}} = 0$. Therefore, Eqn (1.5-3) reduces to

$$\frac{\delta Q}{dn} + \frac{\delta W_s}{dn} = (\tilde{H} + \tilde{E}_K + \tilde{E}_P)_{\text{out}} - (\tilde{H} + \tilde{E}_K + \tilde{E}_P)_{\text{in}} \quad (1.5-8)$$

or⁷

$$\tilde{Q} + \tilde{W}_s = \Delta\tilde{H} + \Delta\tilde{E}_K + \Delta\tilde{E}_P. \quad (1.5-9)$$

⁶For any *state function* φ :

$$\int_{\text{initial state}}^{\text{final state}} d\varphi = \varphi_{\text{final}} - \varphi_{\text{initial}} = \Delta\varphi.$$

For any *path function* φ :

$$\int_{\text{initial state}}^{\text{final state}} \delta\varphi = \varphi.$$

⁷In the case of a steady-state flow system, Δ implies “out – in.”

1.6 The Second Law of Thermodynamics

The second law of thermodynamics introduces a new property called *entropy*, S , which is an extensive property of a system. The entropy change of a closed system is equal to the heat added reversibly to it divided by the absolute temperature of the system, i.e.

$$dS = \frac{\delta Q_{\text{rev}}}{T}. \quad (1.6-1)$$

Note that although δQ_{rev} is a path function $\delta Q_{\text{rev}}/T$ is a state function. Therefore, $1/T$ can be interpreted as the integrating factor.

To determine the absolute value of entropy, it is necessary to define a reference state. Otherwise, only changes in entropy can be determined. The reference state of entropy is a perfect crystal at 0 K. Entropy of this reference state is arbitrarily fixed to 0 J/K. In this way, an absolute entropy can be attributed to every substance at specified conditions like temperature and pressure.

Although the second law of thermodynamics can be expressed in many ways, these statements are equivalent to each other. One of the statements of the second law of thermodynamics is that “*any spontaneous process in any isolated system always results in an increase in the entropy of that system.*” Since the universe is an isolated system, this statement is mathematically expressed in the form:

$$dS_{\text{universe}} = dS_{\text{sys}} + dS_{\text{surr}} \geq 0. \quad (1.6-2)$$

Note that the change in the entropy of the universe also gives the generation of entropy associated with the irreversibilities, i.e. $\Delta S_{\text{universe}} = S_{\text{gen}}$.

The second law of thermodynamics can also be expressed in terms of the *Clausius* and *Kelvin-Planck* statements:

- *Clausius statement:* As shown in Fig. 1.4, it is impossible to construct a device that, operating in a cycle, will produce no effect other than the transfer of heat from a cold body to a hot one.
- *Kelvin-Planck statement:* As shown in Fig. 1.5, it is impossible to construct a device that, operating in a cycle, will produce no effect other than converting heat completely into work.

The energy can be thought of as the ability of the system to perform work. The entropy, on the other hand, is a measure of how much this ability has been devaluated.⁸ An increase in entropy implies, (i) decrease in the ability of energy to do useful work, (ii) depreciation of energy, and (iii) increase in disorder.

⁸While inflation degrades money, conversion from one form to another degrades energy.

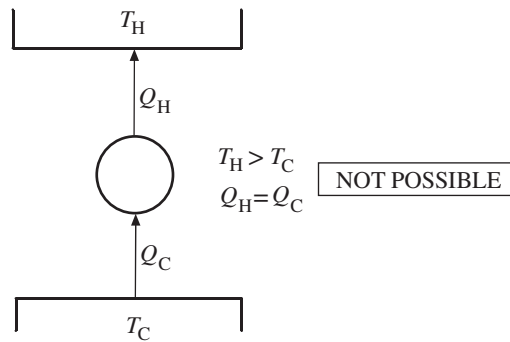


Figure 1.4: Schematic representation of the Clausius statement.

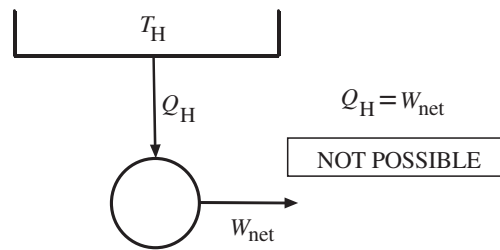


Figure 1.5: Schematic representation of the Kelvin-Planck statement.

While the generation of entropy is either equal to or greater than zero, the entropy of a system may decrease, increase, or remain the same. Therefore, a decrease in the entropy of a system does not necessarily imply the violation of the second law of thermodynamics. In general, the entropy of a system decreases as it changes to a more orderly state. Some examples are:

- During phase transformation to a more orderly state, i.e. Liquid $\rightarrow$ Solid or Gas $\rightarrow$ Liquid.
- As the temperature of a substance decreases.
- As the pressure of a gas increases.

It is important to note that there is no such thing as the conservation of entropy. Real processes always generate entropy and, as a result, entropy of the universe always increases. The change in the entropy of a system can be calculated from the following expression:

$$\begin{aligned}
 &\text{Entropy in} - \text{Entropy out} + \text{Entropy generation} \\
 &= \text{Change in the entropy of the system.}
 \end{aligned}
 \tag{1.6-3}$$

For an open system as shown in Fig. 1.3, the differential form of Eqn (1.6-3) becomes

$$\left(\widehat{S}_{\text{in}} dm_{\text{in}} + \frac{\delta Q}{T}\right) - \widehat{S}_{\text{out}} dm_{\text{out}} + \delta S_{\text{gen}} = d(m\widehat{S})_{\text{sys}}. \quad (1.6-4)$$

Rearrangement of Eqn (1.6-4) gives

$$\delta S_{\text{gen}} = d(m\widehat{S})_{\text{sys}} + \widehat{S}_{\text{out}} dm_{\text{out}} - \widehat{S}_{\text{in}} dm_{\text{in}} - \frac{\delta Q}{T} \geq 0, \quad (1.6-5)$$

which is also known as the *entropy balance*.

In terms of molar quantities, Eqn (1.6-5) becomes

$$\delta S_{\text{gen}} = d(n\widetilde{S})_{\text{sys}} + \widetilde{S}_{\text{out}} dn_{\text{out}} - \widetilde{S}_{\text{in}} dn_{\text{in}} - \frac{\delta Q}{T} \geq 0. \quad (1.6-6)$$

1.6.1 Simplification of the Entropy Balance

1.6.1.1 Isolated System

Since there are no inlet and outlet streams, i.e. $dn_{\text{in}} = dn_{\text{out}} = 0$, and no exchange of heat, i.e. $\delta Q = 0$, Eqn (1.6-6) reduces to

$$dS_{\text{sys}} \geq 0. \quad (1.6-7)$$

1.6.1.2 Closed System

Since there are no inlet or outlet streams, i.e. $dn_{\text{in}} = dn_{\text{out}} = 0$, and $n_{\text{sys}} = \text{constant}$, Eqn (1.6-6) reduces to

$$d\widetilde{S}_{\text{sys}} - \frac{\delta \widetilde{Q}}{T} \geq 0. \quad (1.6-8)$$

1.6.1.3 Steady-State Flow System

In this case $dn_{\text{in}} = dn_{\text{out}} = dn$ and $d(n\widetilde{S})_{\text{sys}} = 0$. Therefore, Eqn (1.6-6) reduces to

$$\widetilde{S}_{\text{out}} - \widetilde{S}_{\text{in}} - \frac{\widetilde{Q}}{T} \geq 0. \quad (1.6-9)$$

Reference

Prausnitz, J.M., 1986, *J. Non-Equil. Thermodyn.*, **11**, 49–66.

Chapter 2

Thermodynamic Properties of Real Substances

Thermodynamics deals with the work and heat interactions of the system with its surroundings as it undergoes a process. These quantities can be estimated once the changes in various state functions, namely, internal energy, enthalpy, and entropy, are known. The purpose of this chapter is to develop general expressions to calculate changes in internal energy, enthalpy, and entropy.

2.1 Work Functions

For a closed system undergoing a reversible isothermal process, integration of Eqn (1.6-8) gives

$$Q_{\text{rev}} = T \Delta S. \quad (2.1-1)$$

The use of Eqn (2.1-1) in Eqn (1.5-7) leads to

$$\Delta U = T \Delta S + W_{\text{rev}} \quad (2.1-2)$$

in which the changes in kinetic and potential energies are considered negligible. Rearrangement of Eqn (2.1-2) results in

$$-W_{\text{rev}} = -\Delta(U - TS) = -\Delta A = A_{\text{initial}} - A_{\text{final}}, \quad (2.1-3)$$

where the term A is called the *Helmholtz energy* and is defined by

$$A = U - TS. \quad (2.1-4)$$

Since U , T , and S are all state functions, Helmholtz energy is a state function. It is also an extensive property. As can be seen from Eqn (2.1-3), the decrease in the

Helmholtz energy indicates the maximum work that can be obtained from a closed system undergoing an isothermal process. The term A comes from the German word “Arbeit,” meaning work.

For a steady-state flow system undergoing a reversible isothermal process, combination of Eqns (1.5-9) and (1.6-9) yields

$$\Delta H = T \Delta S + (W_s)_{\text{rev}} \quad (2.1-5)$$

in which the changes in kinetic and potential energies are considered negligible. Rearrangement of Eqn (2.1-5) gives

$$-(W_s)_{\text{rev}} = -\Delta(H - TS) = -\Delta G = G_{\text{initial}} - G_{\text{final}}, \quad (2.1-6)$$

where the term G is called the *Gibbs¹ energy* and is defined by

$$G = H - TS. \quad (2.1-7)$$

Since H , T , and S are all state functions, Gibbs energy is a state function. It is also an extensive property. As can be seen from Eqn (2.1-6), the decrease in the Gibbs energy indicates the maximum work that can be obtained from a steady-state flow system under isothermal conditions. Helmholtz and Gibbs energies are sometimes referred to as *work functions*.

Examination of Eqns (2.1-4) and (2.1-7) indicates that $A < U$ and $G < H$ as a result of the subtraction of the term “ TS .” Hence, the term “ TS ” can be interpreted as a measure of unavailable energy.

2.2 Thermodynamic Properties of a Single-Phase System

2.2.1 Fundamental Equations

In differential form, the first law of thermodynamics for a closed system is given by

$$dU = \delta Q + \delta W. \quad (2.2-1)$$

If the process is reversible, substitution of Eqns (1.3-2) and (1.6-1) into Eqn (2.2-1) gives

$$dU = T dS - P dV. \quad (2.2-2)$$

¹In honor of Josiah Willard Gibbs, Professor of Mathematical Physics at Yale University. He was awarded the first PhD degree in engineering in the USA.

The definition of enthalpy is given by

$$H = U + PV. \quad (2.2-3)$$

The differential form of Eqn (2.2-3) becomes

$$dH = dU + P dV + V dP. \quad (2.2-4)$$

Substitution of Eqn (2.2-2) into Eqn (2.2-4) gives

$$dH = T dS + V dP. \quad (2.2-5)$$

The definition of Helmholtz energy is given by

$$A = U - TS. \quad (2.2-6)$$

The differential form of Eqn (2.2-6) becomes

$$dA = dU - T dS - S dT. \quad (2.2-7)$$

Substitution of Eqn (2.2-2) into Eqn (2.2-7) gives

$$dA = -P dV - S dT. \quad (2.2-8)$$

The definition of Gibbs energy is given by

$$G = H - TS. \quad (2.2-9)$$

The differential form of Eqn (2.2-9) becomes

$$dG = dH - T dS - S dT. \quad (2.2-10)$$

Substitution of Eqn (2.2-5) into Eqn (2.2-10) gives

$$dG = V dP - S dT. \quad (2.2-11)$$

It is also possible to express Eqns (2.2-2), (2.2-5), (2.2-8), and (2.2-11) in terms of mass or molar quantities, i.e.

$$d\widehat{U} = T d\widehat{S} - P d\widehat{V}, \quad d\widetilde{U} = T d\widetilde{S} - P d\widetilde{V}, \quad (2.2-12)$$

$$d\widehat{H} = T d\widehat{S} + \widehat{V} dP, \quad d\widetilde{H} = T d\widetilde{S} + \widetilde{V} dP, \quad (2.2-13)$$

$$d\widehat{A} = -P d\widehat{V} - \widehat{S} dT, \quad d\widetilde{A} = -P d\widetilde{V} - \widetilde{S} dT, \quad (2.2-14)$$

$$d\widehat{G} = \widehat{V} dP - \widehat{S} dT, \quad d\widetilde{G} = \widetilde{V} dP - \widetilde{S} dT. \quad (2.2-15)$$

These equations are called *fundamental equations*. Note that fundamental equations consist of only properties and their differential changes. These properties and their changes are *state functions* and are not dependent on the path or process involved. Therefore, once the initial and final states are specified, the expressions given in Eqns (2.2-12)–(2.2-15) hold for all processes (reversible or irreversible) and for a change of state in either a closed system or a steady-state flow system.

2.2.2 Maxwell Relations

For an exact differential, $d\phi$, expressed in the form

$$d\phi = M(x, y)dx + N(x, y)dy, \quad (2.2-16)$$

the following relationship holds:

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y. \quad (2.2-17)$$

Since internal energy, U , enthalpy, H , Helmholtz energy, A , and Gibbs energy, G , are all state functions, application of Eqn (2.2-17) to the expressions given in Eqns (2.2-12)–(2.2-15) leads to

$$\left(\frac{\partial T}{\partial \tilde{V}}\right)_{\tilde{S}} = -\left(\frac{\partial P}{\partial \tilde{S}}\right)_{\tilde{V}}, \quad (2.2-18)$$

$$\left(\frac{\partial T}{\partial P}\right)_{\tilde{S}} = \left(\frac{\partial \tilde{V}}{\partial \tilde{S}}\right)_P, \quad (2.2-19)$$

$$\left(\frac{\partial P}{\partial T}\right)_{\tilde{V}} = \left(\frac{\partial \tilde{S}}{\partial \tilde{V}}\right)_T, \quad (2.2-20)$$

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = -\left(\frac{\partial \tilde{S}}{\partial P}\right)_T. \quad (2.2-21)$$

These equations are known as the *Maxwell relations*.

2.2.3 Internal Energy Change

For a single-phase and single-component system, the Gibbs phase rule, Eqn (1.1-4), gives the number of degrees of freedom as two. Hence, the state of such a system is specified by two independent intensive properties. The dependence of internal energy on temperature and volume, i.e.

$$\tilde{U} = \tilde{U}(T, \tilde{V}) \quad (2.2-22)$$

leads to a convenient relationship to use in calculating the internal energy change.² The total differential of $\tilde{U}$ is given by

$$d\tilde{U} = \left(\frac{\partial \tilde{U}}{\partial T}\right)_{\tilde{V}} dT + \left(\frac{\partial \tilde{U}}{\partial \tilde{V}}\right)_T d\tilde{V} = \tilde{C}_V dT + \left(\frac{\partial \tilde{U}}{\partial \tilde{V}}\right)_T d\tilde{V}. \quad (2.2-23)$$

²The functional form of internal energy can also be expressed as $\tilde{U} = \tilde{U}(P, T)$ or $\tilde{U} = \tilde{U}(P, \tilde{V})$.

In order to express the term $(\partial \tilde{U}/\partial \tilde{V})_T$ in terms of measurable quantities, differentiation of Eqn (2.2-12) with respect to $\tilde{V}$ keeping T constant gives

$$\left(\frac{\partial \tilde{U}}{\partial \tilde{V}}\right)_T = T \left(\frac{\partial \tilde{S}}{\partial \tilde{V}}\right)_T - P. \quad (2.2-24)$$

The use of Eqn (2.2-20) in Eqn (2.2-24) yields

$$\left(\frac{\partial \tilde{U}}{\partial \tilde{V}}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_{\tilde{V}} - P. \quad (2.2-25)$$

Thus, substitution of Eqn (2.2-25) into Eqn (2.2-23) gives the change in internal energy as

$$d\tilde{U} = \tilde{C}_V dT + \left[T \left(\frac{\partial P}{\partial T}\right)_{\tilde{V}} - P \right] d\tilde{V}. \quad (2.2-26)$$

2.2.4 Enthalpy Change

The dependence of enthalpy on temperature and pressure, i.e.

$$\tilde{H} = \tilde{H}(T, P) \quad (2.2-27)$$

leads to a convenient relationship to use in calculating the enthalpy change.³ The total differential of $\tilde{H}$ is given by

$$d\tilde{H} = \left(\frac{\partial \tilde{H}}{\partial T}\right)_P dT + \left(\frac{\partial \tilde{H}}{\partial P}\right)_T dP = \tilde{C}_P dT + \left(\frac{\partial \tilde{H}}{\partial P}\right)_T dP. \quad (2.2-28)$$

In order to express the term $(\partial \tilde{H}/\partial P)_T$ in terms of measurable quantities, differentiation of Eqn (2.2-13) with respect to P keeping T constant gives

$$\left(\frac{\partial \tilde{H}}{\partial P}\right)_T = T \left(\frac{\partial \tilde{S}}{\partial P}\right)_T + \tilde{V}. \quad (2.2-29)$$

The use of Eqn (2.2-21) in Eqn (2.2-29) yields

$$\left(\frac{\partial \tilde{H}}{\partial P}\right)_T = \tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T}\right)_P. \quad (2.2-30)$$

Thus, substitution of Eqn (2.2-30) into Eqn (2.2-28) gives the change in enthalpy as

$$d\tilde{H} = \tilde{C}_P dT + \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T}\right)_P \right] dP. \quad (2.2-31)$$

³The functional form of enthalpy can also be expressed as $\tilde{H} = \tilde{H}(T, \tilde{V})$ or $\tilde{H} = \tilde{H}(P, \tilde{V})$.

2.2.5 Entropy Change

Before deriving the equations to calculate the change in entropy, it is necessary to express heat capacities at constant volume and pressure in terms of partial derivatives of entropy.

Heat capacity at constant volume, $\tilde{C}_V$, is defined by

$$\tilde{C}_V = \left(\frac{\partial \tilde{U}}{\partial T} \right)_{\tilde{V}}. \quad (2.2-32)$$

The use of Eqn (2.2-12) in Eqn (2.2-32) leads to

$$\tilde{C}_V = T \left(\frac{\partial \tilde{S}}{\partial T} \right)_{\tilde{V}}. \quad (2.2-33)$$

On the other hand, heat capacity at constant pressure, $\tilde{C}_P$, is defined by

$$\tilde{C}_P = \left(\frac{\partial \tilde{H}}{\partial T} \right)_P. \quad (2.2-34)$$

The use of Eqn (2.2-13) in Eqn (2.2-34) leads to

$$\tilde{C}_P = T \left(\frac{\partial \tilde{S}}{\partial T} \right)_P. \quad (2.2-35)$$

Entropy may be expressed as a function of temperature and volume, i.e.

$$\tilde{S} = \tilde{S}(T, \tilde{V}). \quad (2.2-36)$$

The total differential of $\tilde{S}$, $d\tilde{S}$, is given by

$$d\tilde{S} = \left(\frac{\partial \tilde{S}}{\partial T} \right)_{\tilde{V}} dT + \left(\frac{\partial \tilde{S}}{\partial \tilde{V}} \right)_T d\tilde{V}. \quad (2.2-37)$$

The use of Eqns (2.2-20) and (2.2-33) in Eqn (2.2-37) gives the change in entropy as

$$d\tilde{S} = \frac{\tilde{C}_V}{T} dT + \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} d\tilde{V}. \quad (2.2-38)$$

It is also possible to express entropy as a function of temperature and pressure, i.e.

$$\tilde{S} = \tilde{S}(T, P). \quad (2.2-39)$$

The total differential of $\tilde{S}$, $d\tilde{S}$, is given by

$$d\tilde{S} = \left(\frac{\partial \tilde{S}}{\partial T} \right)_P dT + \left(\frac{\partial \tilde{S}}{\partial P} \right)_T dP. \quad (2.2-40)$$

The use of Eqns (2.2-21) and (2.2-35) in Eqn (2.2-40) gives the change in entropy as

$$d\tilde{S} = \frac{\tilde{C}_P}{T} dT - \left(\frac{\partial \tilde{V}}{\partial T} \right)_P dP. \quad (2.2-41)$$

2.2.6 Relationship Between Partial Derivatives

Inverse rule: The total differential of the function $f(x, y)$ is expressed as

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy. \quad (2.2-42)$$

If both sides of Eqn (2.2-42) are differentiated with respect to f by keeping y constant, the result is

$$1 = \left(\frac{\partial f}{\partial x} \right)_y \left(\frac{\partial x}{\partial f} \right)_y \quad (2.2-43)$$

or

$$\left(\frac{\partial f}{\partial x} \right)_y = \frac{1}{\left(\frac{\partial x}{\partial f} \right)_y}, \quad (2.2-44)$$

which is known as the *inverse rule*.

Triple Product Rule: Differentiation of Eqn (2.2-42) with respect to y keeping f constant gives

$$0 = \left(\frac{\partial f}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_f + \left(\frac{\partial f}{\partial y} \right)_x. \quad (2.2-45)$$

The use of the inverse rule leads to

$$0 = \left(\frac{\partial f}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_f + \frac{1}{\left(\frac{\partial y}{\partial f} \right)_x} \quad (2.2-46)$$

or

$$\left(\frac{\partial f}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_f \left(\frac{\partial y}{\partial f} \right)_x = -1, \quad (2.2-47)$$

which is known as the *triple product rule*. It can be memorized with a cyclic relation as shown in Fig. 2.1, i.e. by marking the variables f , x , and y on a circle

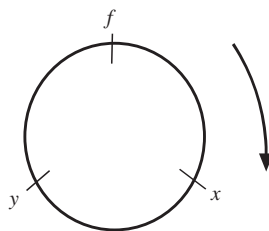


Figure 2.1: Cyclic relation for the triple product rule.

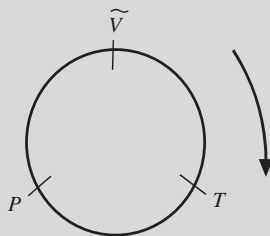
in the clockwise direction. Going in the clockwise direction, differentiate one of the variables (let us call this the first variable) with respect to the second one by keeping the third variable constant. Repeat this procedure for each of the variables going in the clockwise direction, and the product of three terms is equal to -1 .

Example 2.1 Evaluate $(\partial \tilde{V} / \partial T)_P$ if a gas is represented by the following equation of state:

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{T\tilde{V}^2}.$$

Solution

Since $\tilde{V}$ cannot be expressed explicitly in terms of T and P , direct differentiation is not possible. Marking $\tilde{V}$, T , and P on a circle in the clockwise direction and the application of the triple product rule give



$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_{\tilde{V}} \left(\frac{\partial P}{\partial \tilde{V}}\right)_T = -1 \quad (1)$$

or

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = -\frac{1}{\left(\frac{\partial T}{\partial P}\right)_{\tilde{V}} \left(\frac{\partial P}{\partial \tilde{V}}\right)_T}. \quad (2)$$

With the help of the inverse rule, Eqn (2) takes the form

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = -\frac{(\partial P/\partial T)_{\tilde{V}}}{(\partial P/\partial \tilde{V})_T}. \quad (3)$$

From the given equation of state

$$\left(\frac{\partial P}{\partial T}\right)_{\tilde{V}} = \frac{R}{\tilde{V} - b} + \frac{a}{T^2 \tilde{V}^2}, \quad (4)$$

$$\left(\frac{\partial P}{\partial \tilde{V}}\right)_T = -\frac{RT}{(\tilde{V} - b)^2} + \frac{2a}{T \tilde{V}^3}. \quad (5)$$

Substitution of Eqns (4) and (5) into Eqn (3) yields

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = \frac{\frac{R}{\tilde{V} - b} + \frac{a}{T^2 \tilde{V}^2}}{\frac{RT}{(\tilde{V} - b)^2} - \frac{2a}{T \tilde{V}^3}}. \quad (6)$$

2.2.7 Relationship Between Heat Capacities

Differentiation of Eqn (2.2-41) with respect to temperature keeping volume constant gives

$$\left(\frac{\partial \tilde{S}}{\partial T}\right)_{\tilde{V}} = \frac{\tilde{C}_P}{T} - \left(\frac{\partial \tilde{V}}{\partial T}\right)_P \left(\frac{\partial P}{\partial T}\right)_{\tilde{V}}. \quad (2.2-48)$$

The use of Eqn (2.2-33) in Eqn (2.2-48) leads to

$$\tilde{C}_P = \tilde{C}_V + T \left(\frac{\partial \tilde{V}}{\partial T}\right)_P \left(\frac{\partial P}{\partial T}\right)_{\tilde{V}}. \quad (2.2-49)$$

In general, it is difficult to express $\tilde{V}$ as an explicit function of T and P , i.e. $\tilde{V} = \tilde{V}(T, P)$.⁴ It is much easier to express P as an explicit function of $\tilde{V}$ and T , i.e. $P = P(\tilde{V}, T)$. Therefore, the quantity $(\partial \tilde{V}/\partial T)_P$ in Eqn (2.2-49) is difficult to evaluate and must be replaced by $(\partial P/\partial T)_{\tilde{V}}$ and $(\partial P/\partial \tilde{V})_T$. From Eqn (3) of Example 2.1

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = -\frac{(\partial P/\partial T)_{\tilde{V}}}{(\partial P/\partial \tilde{V})_T}. \quad (2.2-50)$$

⁴The reason for this statement will be apparent in the next chapter.

Substitution of Eqn (2.2-50) into Eqn (2.2-49) leads to the following relationship between heat capacities:

$$\tilde{C}_P = \tilde{C}_V - T \frac{(\partial P / \partial T)_{\tilde{V}}^2}{(\partial P / \partial \tilde{V})_T}. \quad (2.2-51)$$

2.2.8 Coefficient of Thermal Expansion and Isothermal Compressibility

Coefficient of thermal expansion, β , is the variation of volume with temperature at constant pressure per unit volume, i.e.

$$\beta = \frac{1}{\hat{V}} \left(\frac{\partial \hat{V}}{\partial T} \right)_P = \frac{1}{\tilde{V}} \left(\frac{\partial \tilde{V}}{\partial T} \right)_P. \quad (2.2-52)$$

Since $\rho = 1/\hat{V}$, it is also possible to express β in terms of density as

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P. \quad (2.2-53)$$

For most liquids, density increases with decreasing temperature. As a result, $\beta > 0$. However, for liquid water between 4°C and 0°C, β turns out to be negative. This is the reason why lakes start freezing from top to bottom.

Isothermal compressibility, κ , is the variation of volume with pressure at constant temperature per unit volume, i.e.

$$\kappa = -\frac{1}{\hat{V}} \left(\frac{\partial \hat{V}}{\partial P} \right)_T = -\frac{1}{\tilde{V}} \left(\frac{\partial \tilde{V}}{\partial P} \right)_T. \quad (2.2-54)$$

Since volume decreases with increasing pressure, $(\partial \hat{V} / \partial P)_T$ is a negative quantity. To make κ positive, a minus sign is used in Eqn (2.2-54). In terms of density, Eqn (2.2-54) becomes

$$\kappa = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T. \quad (2.2-55)$$

Both β and κ remain almost constant for liquids and solids.⁵ Thus, as will be shown in the next chapter, coefficient of thermal expansion and isothermal compressibility are especially useful in calculating changes in internal energy, enthalpy, and entropy for liquids and solids.

⁵This statement does not hold near the critical point.

Example 2.2 A 1 L bottle is completely filled with an equimolar mixture of benzene and cyclohexane at 298 K. What pressure will develop within the bottle if it is heated to 305 K as a result of the sunlight coming through the lab window?

The following data are provided for an equimolar mixture of benzene and cyclohexane:

$$\beta = 1.21 \times 10^{-3} \text{ K}^{-1} \quad \text{and} \quad \kappa = 108 \times 10^{-6} \text{ bar}^{-1}.$$

Solution

It is required to determine the variation of pressure with temperature at constant volume, i.e. $(\partial P/\partial T)_V$. From the triple product rule

$$\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T = -1.$$

Solving for $(\partial P/\partial T)_V$ gives

$$\left(\frac{\partial P}{\partial T}\right)_V = -\frac{1}{\left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T} = -\frac{(\partial V/\partial T)_P}{(\partial V/\partial P)_T} = \frac{\beta}{\kappa}. \quad (1)$$

Assuming β and κ to be constants, and the initial pressure within the bottle to be 1 bar, integration of Eqn (1) gives

$$P = 1 + \left(\frac{1.21 \times 10^{-3}}{108 \times 10^{-6}}\right) (305 - 298) = 79.4 \text{ bar}.$$

Comment: The bottle probably breaks before the pressure reaches 79.4 bar.

Example 2.3 At 573 K, the variation in the specific volume of steam as a function of pressure is given as follows:

P (MPa)	$\hat{V}$ (m ³ /kg)
0.8	0.3241
1.0	0.2579
1.2	0.2138

- Calculate the isothermal compressibility for steam at 1 MPa and 573 K.
- Compare this value with the one obtained by assuming that the steam obeys the ideal gas equation of state.

Solution

(a) From Eqn (2.2-54)

$$\begin{aligned}\kappa &= -\frac{1}{\widehat{V}} \left(\frac{\partial \widehat{V}}{\partial P} \right)_T \simeq -\frac{1}{\widehat{V}} \left(\frac{\Delta \widehat{V}}{\Delta P} \right) \\ &= -\frac{1}{0.2579} \left(\frac{0.2138 - 0.3241}{1.2 - 0.8} \right) = 1.07 \text{ MPa}^{-1}.\end{aligned}$$

(b) If steam is considered an ideal gas, then

$$\widetilde{V} = \frac{RT}{P} \Rightarrow \left(\frac{\partial \widetilde{V}}{\partial P} \right)_T = -\frac{RT}{P^2}. \quad (1)$$

Substitution of Eqn (1) into Eqn (2.2-54) gives

$$\kappa = -\frac{1}{\widetilde{V}} \left(\frac{\partial \widetilde{V}}{\partial P} \right)_T = -\left(\frac{P}{RT} \right) \left(-\frac{RT}{P^2} \right) = \frac{1}{P} = 1 \text{ MPa}^{-1}.$$

So far, the equations developed to calculate changes in internal energy, enthalpy, and entropy apply to all substances. However, they contain partial derivatives involving T , P , and $\widetilde{V}$. Thus, an equation of state must be known for a substance in question. The following example shows how to evaluate the partial derivatives to determine changes in internal energy or enthalpy once the equation of state is known. Determination of changes in internal energy, enthalpy, and entropy for various equations of state will be covered in detail in the next chapter.

Example 2.4 A piston-cylinder assembly contains 2 mol of a gas at 5 bar and 350 K. As a result of irreversible operations 3.5 kJ of work is done on the system, the final pressure and temperature being 25 bar and 400 K, respectively. Calculate the heat interaction of the gas with the surroundings.

Data: The gas obeys the following equation of state:

$$P(\widetilde{V} - b) = RT$$

with $b = 4 \times 10^{-5} \text{ m}^3/\text{mol}$.

The heat capacity at constant pressure is given as a function of temperature as

$$\widetilde{C}_P = 8.314 + 0.108T,$$

where $\widetilde{C}_P$ is in J/mol K and T is in K.

Solution**System:** Contents of the cylinder.

When changes in kinetic and potential energies are negligible, the first law of thermodynamics, i.e. Eqn (1.5-7), reduces to

$$\Delta U = Q + W. \quad (1)$$

Determination of Q requires the value of ΔU to be known. The variation in internal energy can be determined from Eqn (2.2-26), i.e.

$$d\tilde{U} = \tilde{C}_V dT + \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] d\tilde{V}. \quad (2)$$

From the given equation of state,

$$P = \frac{RT}{\tilde{V} - b} \Rightarrow \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{R}{\tilde{V} - b}. \quad (3)$$

Therefore, the term

$$T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P = \frac{RT}{\tilde{V} - b} - \frac{RT}{\tilde{V} - b} = 0 \quad (4)$$

and Eqn (2) simplifies to

$$\Delta \tilde{U} = \int_{T_1}^{T_2} \tilde{C}_V dT \quad (5)$$

in which subscripts 1 and 2 represent the initial and final states, respectively. To determine $\tilde{C}_V$, we have to use Eqn (2.2-51), i.e.

$$\tilde{C}_P = \tilde{C}_V - T \frac{(\partial P / \partial T)_{\tilde{V}}^2}{(\partial P / \partial \tilde{V})_T}. \quad (6)$$

Note that

$$\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{R}{\tilde{V} - b}, \quad \left(\frac{\partial P}{\partial \tilde{V}} \right)_T = -\frac{RT}{(\tilde{V} - b)^2}. \quad (7)$$

Substitution of the terms in Eqn (7) into Eqn (6) leads to

$$\tilde{C}_P = \tilde{C}_V + R. \quad (8)$$

Thus

$$\tilde{C}_V = 8.314 + 0.108T - 8.314 = 0.108T. \quad (9)$$

Substitution of Eqn (9) into Eqn (5) and integration give

$$\Delta \tilde{U} = \frac{0.108}{2} [(400)^2 - (350)^2] = 2025 \text{ J/mol}$$

$$\Delta U = (2)(2025) = 4050 \text{ J.}$$

Therefore, the heat interaction of the system with the surroundings is calculated from Eqn (1) as

$$Q = \Delta U - W = 4050 - 3500 = 550 \text{ J}$$

indicating that heat is added to the system.

Alternate solution: The change in enthalpy is given by Eqn (2.2-31), i.e.

$$d\tilde{H} = \tilde{C}_P dT + \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP. \quad (10)$$

From the given equation of state,

$$\tilde{V} = b + \frac{RT}{P} \Rightarrow \left(\frac{\partial \tilde{V}}{\partial T} \right)_P = \frac{R}{P}. \quad (11)$$

Substitution of Eqn (11) into Eqn (10) and integration yield

$$\Delta \tilde{H} = \int_{T_1}^{T_2} \tilde{C}_P dT + b(P_2 - P_1) \quad (12)$$

or

$$\Delta \tilde{H} = (8.314)(400 - 350) + \frac{0.108}{2} [(400)^2 - (350)^2]$$

$$+ 4 \times 10^{-5} [(25 - 5) \times 10^5] = 2520.7 \text{ J/mol.}$$

Since $\tilde{U} = \tilde{H} - P\tilde{V}$, the change in internal energy is given by

$$\Delta \tilde{U} = \Delta \tilde{H} - \Delta(P\tilde{V}). \quad (13)$$

From the given equation of state

$$P\tilde{V} = RT + bP. \quad (14)$$

The use of Eqn (14) in Eqn (13) gives

$$\Delta \tilde{U} = \Delta \tilde{H} - R \Delta T - b \Delta P = 2520.7 - (8.314)(400 - 350)$$

$$- 4 \times 10^{-5} [(25 - 5) \times 10^5] = 2025 \text{ J/mol.}$$

Once $\Delta \tilde{U}$ is known, Q can be calculated from Eqn (1).

Problems

Problems Related to Section 2.2.

2.1 For steady-state flow through a throttling valve with negligible changes in kinetic and potential energies, the first law states that the enthalpy remains constant, i.e. isenthalpic process. The variation of temperature with pressure in such a process is known as the *Joule-Thomson coefficient*, μ .

(a) Show that the Joule-Thomson coefficient is given by

$$\mu = \frac{1}{\tilde{C}_P} \left[T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P - \tilde{V} \right] \quad (1)$$

The throttling process may lead to an increase or decrease in temperature depending on the value of μ , i.e.

$$\mu \begin{cases} > 0, & \text{Temperature decreases on throttling,} \\ < 0, & \text{Temperature increases on throttling.} \end{cases} \quad (2)$$

The temperature at which the Joule-Thomson coefficient changes sign is called an *inversion temperature*. At higher temperatures $\mu < 0$, and at lower temperatures $\mu > 0$. As a result, cryogenic applications require gas temperature to be lower than the inversion temperature.

Most gases have an inversion temperature higher than room temperature. Hydrogen, however, has an inversion temperature of -80°C . In order to liquefy hydrogen, it is first necessary to decrease its temperature below -80°C using liquefied nitrogen and then decrease its pressure by a throttling process.

(b) Show that the inversion condition, $\mu = 0$, is given by the equation

$$T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} + \tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T = 0. \quad (3)$$

(c) Consider a gas obeying an equation of state of the form

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{\tilde{V}^2}, \quad (4)$$

where a and b are constants. Assuming $(\tilde{V} - b)/\tilde{V} \simeq 1$, show that the inversion temperature, T_{inv} , is approximated as

$$T_{\text{inv}} \simeq \frac{2a}{Rb}. \quad (5)$$

Although the inversion temperature turns out to be independent of pressure for this specific case, it is generally dependent on pressure.

- 2.2** Ammonia at 400 K and 5 MPa enters a throttling valve and leaves it at 100 kPa. Determine the exit temperature if the equation of state is given by

$$P(\tilde{V} - b) = RT,$$

where $b = 3.73 \times 10^{-5} \text{ m}^3/\text{mol}$. The molar heat capacity at constant pressure is given by

$$\tilde{C}_P = 25.895 + 0.033T,$$

where $\tilde{C}_P$ is in J/mol K and T is in K.

(Answer: 404.7 K)

- 2.3** A gas contained in a piston-cylinder assembly is represented by

$$P\left(\tilde{V} - a + \frac{b}{T}\right) = RT. \quad (1)$$

The gas is compressed irreversibly and isothermally from an initial pressure of P_1 to a final pressure of P_2 .

- (a) Use Eqn (2.2-31) and show that

$$\Delta\tilde{H} = \left(a - \frac{2b}{T}\right)(P_2 - P_1). \quad (2)$$

- (b) Using the identity

$$\Delta\tilde{U} = \Delta\tilde{H} - \Delta(P\tilde{V}) \quad (3)$$

show that

$$\Delta\tilde{U} = \frac{b(P_1 - P_2)}{T}. \quad (4)$$

- (c) Use Eqn (2.2-26) to obtain

$$\Delta\tilde{U} = -bR\left(\frac{1}{\tilde{V}_2 - a + \frac{b}{T}} - \frac{1}{\tilde{V}_1 - a + \frac{b}{T}}\right) \quad (5)$$

and show that the use of Eqn (1) in Eqn (5) leads to Eqn (4).

- (d) Show that the entropy change associated with this compression is given by

$$\Delta\tilde{S} = -R \ln\left(\frac{P_2}{P_1}\right) + \frac{b(P_1 - P_2)}{T^2}. \quad (6)$$

2.4 Consider a piston-cylinder assembly containing 3 mol of gas represented by

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{T\tilde{V}^2},$$

where

$$a = 35 \text{ J K m}^3/\text{mol}^2 \quad \text{and} \quad b = 4.8 \times 10^{-5} \text{ m}^3/\text{mol}.$$

The gas expands irreversibly from 0.01 m^3 to 0.02 m^3 at 350 K . Use Eqn (2.2-38) and calculate the change in entropy associated with this process.

(Answer: 17.6 J/K)

2.5

(a) Consider a function

$$y = 3x^2 - 1.$$

First evaluate dy/dx . Then express x as a function of y , i.e.

$$x = \sqrt{\frac{y+1}{3}}$$

and evaluate dx/dy . Conclude that

$$\frac{dy}{dx} = \frac{1}{dx/dy}.$$

(b) Consider a function

$$y = 3x^2 - \sin x + \ln x.$$

Show that

$$\frac{dx}{dy} = \frac{x}{6x^2 + 1 - x \cos x}.$$

(c) Consider a function

$$z = 6x^2y - \frac{1}{x} + \ln x.$$

Show that

$$\left(\frac{\partial x}{\partial y}\right)_z = -\frac{6x^4}{12x^3y + x + 1}.$$

2.6 Show that the ratio of the heat capacities is expressed in the form

$$\frac{\tilde{C}_V}{\tilde{C}_P} = 1 - \frac{T\beta^2}{\rho\hat{C}_P\kappa}.$$

Using the following values at 298 K and 1 bar, evaluate the term $T\beta^2/(\rho\hat{C}_P\kappa)$.

Substance	ρ (kg/m ³)	β (K ⁻¹)	κ (Pa ⁻¹)	$\hat{C}_P$ (J/kg K)
Water	1000	2.5×10^{-4}	4.53×10^{-10}	4186
Copper	8920	5×10^{-5}	9.1×10^{-8}	386

Although it is widely assumed that $\tilde{C}_V \simeq \tilde{C}_P$ for solids and incompressible liquids, this assumption is not generally correct for various liquids. For a more thorough discussion on the subject, see [Garvin \(2002\)](#).

- 2.7** Show that the Gibbs energy can be regarded as a generating function to estimate enthalpy, internal energy, Helmholtz energy, and heat capacity, i.e.

$$\tilde{H} = \tilde{G} - T \left(\frac{\partial \tilde{G}}{\partial T} \right)_P,$$

$$\tilde{U} = \tilde{G} - T \left(\frac{\partial \tilde{G}}{\partial T} \right)_P - P \left(\frac{\partial \tilde{G}}{\partial P} \right)_T,$$

$$\tilde{A} = \tilde{G} - P \left(\frac{\partial \tilde{G}}{\partial P} \right)_T,$$

$$\tilde{C}_P = -T \left(\frac{\partial^2 \tilde{G}}{\partial T^2} \right)_P.$$

- 2.8** Express the term $(\partial T/\partial P)_{\tilde{S}}$ in terms of measurable quantities as follows:

(a) Use the triple product rule to show that

$$\left(\frac{\partial T}{\partial P} \right)_{\tilde{S}} = - \frac{(\partial \tilde{S}/\partial P)_T}{(\partial \tilde{S}/\partial T)_P}. \quad (1)$$

(b) Use Eqn (2.2-41) and conclude that

$$\left(\frac{\partial \tilde{S}}{\partial P} \right)_T = - \left(\frac{\partial \tilde{V}}{\partial T} \right)_P, \quad (2)$$

$$\left(\frac{\partial \tilde{S}}{\partial T} \right)_P = \frac{\tilde{C}_P}{T}. \quad (3)$$

(c) Show that the substitution of Eqns (2) and (3) into Eqn (1) leads to

$$\left(\frac{\partial T}{\partial P} \right)_{\tilde{S}} = \frac{\beta \tilde{V} T}{\tilde{C}_P}. \quad (4)$$

(d) Also combine Eqn (4) with Eqn (1) of Problem 2.1 to obtain

$$\left(\frac{\partial T}{\partial P}\right)_{\tilde{S}} = \mu + \frac{\tilde{V}}{\tilde{C}_P}. \quad (5)$$

(e) A gas obeying

$$P(\tilde{V} - b) = RT \quad (6)$$

flows through an isentropic nozzle from high pressure to low pressure. Will the gas temperature decrease or increase?

(Answer: (e) Decreases)

2.9 Show that

(a)

$$\left(\frac{\partial \tilde{H}}{\partial \tilde{V}}\right)_T = \frac{\beta T - 1}{\kappa}.$$

(b)

$$\left(\frac{\partial \tilde{G}}{\partial T}\right)_{\tilde{V}} = \frac{\beta \tilde{V}}{\kappa} - \tilde{S}.$$

(c)

$$\left(\frac{\partial \tilde{H}}{\partial T}\right)_{\tilde{S}} = \frac{\tilde{C}_P}{\beta T}.$$

2.10 The volume of an aluminum sphere is reduced from 48.25 cm³ to 48.17 cm³ at 300 K by applying increased pressure. Use Eqns (2.2-38) and (1) of Example 2.2 to estimate the change in entropy. For aluminum

$$\beta = 6.9 \times 10^{-5} \text{ K}^{-1} \quad \text{and} \quad \kappa = 13.2 \times 10^{-12} \text{ Pa}^{-1}.$$

(Answer: -0.418 J/K)

2.11 Changes in potential energy are considered negligible in the development of the fundamental equations, i.e. Eqns (2.2-2), (2.2-5), (2.2-8), and (2.2-11). When the gravitational force is appreciable, the change in potential energy is represented by

$$dE_P = -mg \, dz, \quad (1)$$

where m is the mass, g is the acceleration of gravity, and z is the distance measured in the direction opposite to gravity. Under these circumstances, show that the fundamental equations take the following form:

$$dU = T dS - P dV + mg dz, \quad (2)$$

$$dH = T dS + V dP + mg dz, \quad (3)$$

$$dA = -P dV - S dT + mg dz, \quad (4)$$

$$dG = V dP - S dT + mg dz. \quad (5)$$

2.12 A closed system containing 1 mol of ethanol at 298 K and 1 bar receives 2700 J of heat from its surroundings.

- Calculate the final temperature and pressure if heating takes place at constant volume.
- Calculate the final temperature and molar volume if heating takes place at constant pressure.

Assume that $\tilde{C}_P$, β , and κ can be considered constant over the temperature ranges considered in this problem. Ethanol at 298 K and 1 bar exists in liquid form and has the following properties:

$$\begin{aligned} \tilde{V} &= 58.39 \text{ cm}^3/\text{mol}, & \tilde{C}_P &= 112.4 \text{ J/mol K}, \\ \beta &= 1.12 \times 10^{-3} \text{ K}^{-1}, & \kappa &= 1.11 \times 10^{-9} \text{ Pa}^{-1}. \end{aligned}$$

(Answer: (a) 327.1 K, 292 bar and (b) 322 K, 59.98 cm³/mol)

2.13 Specific volume of liquids can be expressed as a function of temperature and pressure in the form

$$\hat{V}(P, T) = \hat{V}(0, T) \left\{ 1 - C \ln \left[1 + \frac{P}{B(T)} \right] \right\}, \quad (1)$$

which is known as the *Tait equation*. Show that the isothermal compressibility is given by

$$\kappa = \frac{1}{[P + B(T)] \left\{ \frac{1}{C} - \ln \left[1 + \frac{P}{B(T)} \right] \right\}}. \quad (2)$$

2.14 Isentropic compressibility, τ , is the variation of volume with pressure at constant entropy per unit volume, i.e.

$$\tau = -\frac{1}{\tilde{V}} \left(\frac{\partial \tilde{V}}{\partial P} \right)_{\tilde{S}}. \quad (1)$$

(a) Use the triple product rule to show that

$$\left(\frac{\partial \tilde{V}}{\partial P} \right)_{\tilde{S}} = -\frac{(\partial \tilde{S} / \partial P)_{\tilde{V}}}{(\partial \tilde{S} / \partial \tilde{V})_P}. \quad (2)$$

(b) Use Eqn (2.2-38) and conclude that

$$\left(\frac{\partial \tilde{S}}{\partial P} \right)_{\tilde{V}} = \frac{\tilde{C}_V}{T} \left(\frac{\partial T}{\partial P} \right)_{\tilde{V}}. \quad (3)$$

(c) Use Eqn (2.2-41) and conclude that

$$\left(\frac{\partial \tilde{S}}{\partial \tilde{V}} \right)_P = \frac{\tilde{C}_P}{T} \left(\frac{\partial T}{\partial \tilde{V}} \right)_P. \quad (4)$$

(d) Substitute Eqns (3) and (4) into Eqn (2) to get

$$\left(\frac{\partial \tilde{V}}{\partial P} \right)_{\tilde{S}} = -\frac{\tilde{C}_V}{\tilde{C}_P} \left(\frac{\partial T}{\partial P} \right)_{\tilde{V}} \left(\frac{\partial \tilde{V}}{\partial T} \right)_P. \quad (5)$$

(e) Use the triple product rule to show that

$$\left(\frac{\partial T}{\partial P} \right)_{\tilde{V}} \left(\frac{\partial \tilde{V}}{\partial T} \right)_P = -\left(\frac{\partial \tilde{V}}{\partial P} \right)_T. \quad (6)$$

(f) Show that the use of Eqn (6) in Eqn (5) leads to

$$\tau = \frac{\tilde{C}_V}{\tilde{C}_P} \kappa. \quad (7)$$

(g) The speed of sound, u , is defined by

$$u^2 = \left(\frac{\partial P}{\partial \rho} \right)_{\tilde{S}}. \quad (8)$$

Express Eqn (8) in the form

$$u^2 = -\frac{\tilde{V}^2}{M} \left(\frac{\partial P}{\partial \tilde{V}} \right)_{\tilde{S}}, \quad (9)$$

where M is the molecular weight. Combine Eqn (9) with Eqn (1) to obtain

$$\tau = \frac{1}{u^2 \rho}. \quad (10)$$

Once the speed of sound and density are experimentally measured, the isentropic compressibility is calculated from Eqn (10).

Reference

Garvin, J., 2002, *Chem. Eng. Progr.*, **98** (7), 64–65.

Chapter 3

Calculation of Changes in Internal Energy, Enthalpy, and Entropy

In the previous chapter, general expressions for calculating changes in internal energy, enthalpy, and entropy are developed. These expressions contain partial derivatives involving temperature, pressure, and molar volume. The purpose of this chapter is to show how to evaluate these derivatives in a systematic manner.

3.1 Equations of State

Any mathematical relationship between temperature, pressure, and molar volume is called an *equation of state*, i.e.

$$f(T, P, \tilde{V}) = 0. \quad (3.1-1)$$

Equations of state can be expressed either in pressure-explicit form

$$P = P(T, \tilde{V}) \quad (3.1-2)$$

or in volume-explicit form

$$\tilde{V} = \tilde{V}(T, P). \quad (3.1-3)$$

Moreover, an equation of state can also be expressed in terms of the dimensionless *compressibility factor*, Z , as

$$Z = \frac{P\tilde{V}}{RT}. \quad (3.1-4)$$

3.1.1 Ideal Gas Equation of State

The ideal gas model is dependent on the following assumptions:

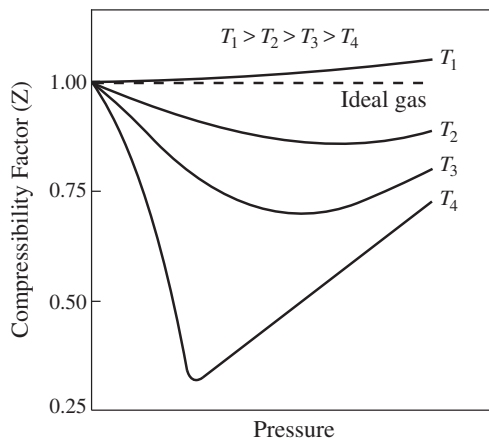


Figure 3.1: Variation of Z as a function of P and T .

- Molecules occupy no volume.
- Collisions of the molecules are totally elastic, i.e. energy of the molecules before a collision is equal to the energy of the molecules after a collision. In other words, there are no interactions between the molecules.

The equation of state for an ideal gas is given by

$$P\tilde{V} = RT. \quad (3.1-5)$$

Since $Z = 1$ for an ideal gas, the compressibility factor shows the deviation from ideal behavior. A representative plot of the compressibility factor as a function of pressure with temperature as a parameter is shown in Fig. 3.1.

3.1.2 The Virial Equation of State

The virial equation of state is useful for calculating thermodynamic properties in the gas phase. It can be derived from statistical mechanics and is given by an infinite series in molar volume, $\tilde{V}$, as

$$Z = \frac{P\tilde{V}}{RT} = 1 + \frac{B(T)}{\tilde{V}} + \frac{C(T)}{\tilde{V}^2} + \dots \quad (3.1-6)$$

The coefficient, $B(T)$, is called the *second virial coefficient*, $C(T)$ is called the *third virial coefficient*, and so on. All virial coefficients are dependent on temperature. In practice, terms above the third virial coefficient are rarely used in chemical thermodynamics.

An equivalent form of the virial expansion is an infinite series in pressure expressed as

$$Z = 1 + B'(T)P + C'(T)P^2 + \dots \quad (3.1-7)$$

The coefficients B' and C' can be expressed in terms of B and C as

$$B' = \frac{B}{RT} \quad \text{and} \quad C' = \frac{C - B^2}{(RT)^2}. \quad (3.1-8)$$

In practice, it is recommended to consider only the second virial coefficient for pressures up to 15 bar. Under these circumstances, Eqn (3.1-7) takes the form

$$Z = 1 + \frac{BP}{RT}. \quad (3.1-9)$$

Van Ness and Abbott (1982) proposed the following equation to estimate the second virial coefficient for nonpolar fluids:

$$B = \frac{RT_c}{P_c} \left[B^{(0)} + \omega B^{(1)} \right], \quad (3.1-10)$$

where

$$B^{(0)} = 0.083 - \frac{0.422}{T_r^{1.6}}, \quad (3.1-11)$$

$$B^{(1)} = 0.139 - \frac{0.172}{T_r^{4.2}}. \quad (3.1-12)$$

The terms T_r , *reduced temperature*, P_r , *reduced pressure*, and ω , *acentric factor*, are defined by

$$T_r = \frac{T}{T_c} \quad \text{and} \quad P_r = \frac{P}{P_c}, \quad (3.1-13)$$

$$\omega = -1.0 - \log P_r^{\text{vap}}|_{T_r=0.7}. \quad (3.1-14)$$

The definition of the acentric factor is based on the observation that $\log P_r^{\text{vap}}$ versus $1/T_r$ is approximately a straight line (Pitzer *et al.*, 1955).¹ For noble gases (Ar, Kr, Xe) $\log P_r^{\text{vap}} = -1$ at $1/T_r = 1/0.7 \simeq 1.43$, indicating that $\omega = 0$. With reference to Fig. 3.2, Pitzer (1977) explained the conceptual basis of the acentric factor as follows:

¹See Problem 5.21.

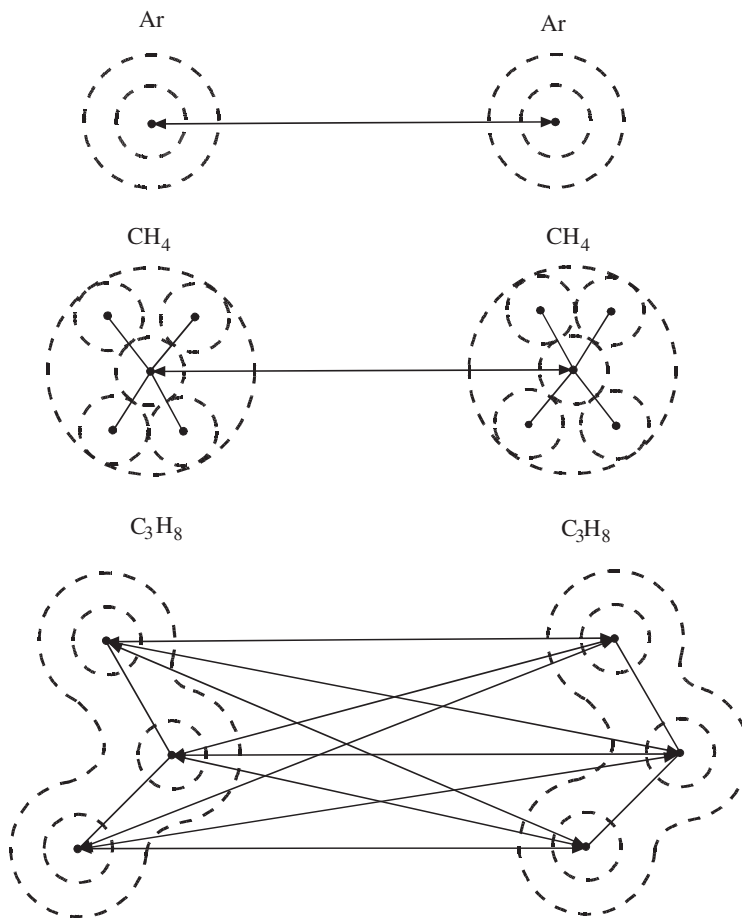


Figure 3.2: The conceptual basis of the acentric factor. (Reprinted with permission from [Pitzer \(1977\)](#). Copyright 1977 by the American Chemical Society.)

“Intermolecular forces operate between the centers of regions of substantial electron density. These centers are the molecular centers for Ar and (approximately) for CH₄, but are best approximated by the separate CH₃ and CH₂ groups in C₃H₈—hence the name acentric factor for the forces arising from points other than molecular centers.”

Example 3.1 Calculate the molar volume of ethylene at 350 K and 10 bar if it obeys the virial equation of state.

Solution

From Appendix A

$$T_c = 282.5 \text{ K}, \quad P_c = 50.6 \text{ bar}, \quad \omega = 0.089.$$

The reduced temperature and pressure values are

$$T_r = \frac{350}{282.5} = 1.239 \quad P_r = \frac{10}{50.6} = 0.198.$$

The terms $B^{(0)}$ and $B^{(1)}$ are calculated from Eqns (3.1-11) and (3.1-12) as

$$B^{(0)} = 0.083 - \frac{0.422}{(1.239)^{1.6}} = -0.217,$$

$$B^{(1)} = 0.139 - \frac{0.172}{(1.239)^{4.2}} = 0.069.$$

The use of Eqn (3.1-10) in Eqn (3.1-9) leads to

$$Z = 1 + \frac{P_r}{T_r} [B^{(0)} + \omega B^{(1)}]. \quad (1)$$

Substitution of numerical values into Eqn (1) results in

$$Z = 1 + \frac{0.198}{1.239} [-0.217 + (0.089)(0.069)] = 0.966.$$

Therefore, the molar volume is

$$\tilde{V} = \frac{ZRT}{P} = \frac{(0.966)(83.14)(350)}{10} = 2811 \text{ cm}^3/\text{mol}.$$

3.1.3 Cubic Equations of State

The cubic equations of state are expressed in pressure-explicit form, i.e. $P = P(\tilde{V}, T)$, some of which are given in Table 3.1. The cubic equations of state have been extensively used over the last three decades,² Soave-Redlich-Kwong (Soave, 1972) and Peng and Robinson (1976)³ being the most popular ones. The reason for this is twofold: (i) they are applicable over a wide range of pressures and temperatures and (ii) they are capable of describing substances in both liquid and vapor phases.

²For a comprehensive review on the state of the cubic equations of state, see Valderrama (2003).

³Peng-Robinson equation of state is one of the most widely used correlations in chemical engineering. To celebrate the twentieth anniversary of its publication, a symposium was organized in Edmonton, Canada, in 1996. Papers presented at the symposium were published in a special issue of *Industrial and Engineering Chemistry Research* (May 1998).

Table 3.1: Cubic equations of state.

Equation	P	a	b
van der Waals (1873)	$\frac{RT}{\tilde{V}-b}-\frac{a}{\tilde{V}^2}$	$\frac{27}{64}\left(\frac{R^2T_c^2}{P_c}\right)$	$\frac{1}{8}\left(\frac{RT_c}{P_c}\right)$
Redlich and Kwong (1949)	$\frac{RT}{\tilde{V}-b}-\frac{a}{\tilde{V}(\tilde{V}+b)\sqrt{T}}$	$0.42748\left(\frac{R^2T_c^{2.5}}{P_c}\right)$	$0.08664\left(\frac{RT_c}{P_c}\right)$
Soave-Redlich-Kwong	$\frac{RT}{\tilde{V}-b}-\frac{a}{\tilde{V}(\tilde{V}+b)}$	$0.42748\left(\frac{R^2T_c^2}{P_c}\right)\alpha$	$0.08664\left(\frac{RT_c}{P_c}\right)$
Peng-Robinson	$\frac{RT}{\tilde{V}-b}-\frac{a}{\tilde{V}(\tilde{V}+b)+b(\tilde{V}-b)}$	$0.45724\left(\frac{R^2T_c^2}{P_c}\right)\alpha$	$0.07780\left(\frac{RT_c}{P_c}\right)$

These equations are called “cubic” because they are of the third degree in molar volume, i.e.

$$\tilde{V}^3 + c_1 \tilde{V}^2 + c_2 \tilde{V} + c_3 = 0. \quad (3.1-15)$$

In each equation given in Table 3.1, the first term is identical, i.e. $RT/(\tilde{V} - b)$, and accounts for the repulsive forces between the molecules. The second term, on the other hand, is different in each equation and accounts for the attractive forces between the molecules. The parameters “ a ” and “ b ” are representative of attractive forces and volume of a molecule, respectively.⁴ Note that the parameter “ a ” in the van der Waals and Redlich-Kwong equations of state is a constant. In the case of Soave-Redlich-Kwong and Peng-Robinson equations of state, however, it is dependent on temperature through the term α defined by

$$\alpha = \begin{cases} \left[1 + (0.480 + 1.574\omega - 0.176\omega^2)(1 - \sqrt{T_r}) \right]^2 & \text{Soave-Redlich-Kwong,} \\ \left[1 + (0.37464 + 1.54226\omega - 0.26992\omega^2)(1 - \sqrt{T_r}) \right]^2 & \text{Peng-Robinson.} \end{cases} \quad (3.1-16)$$

A typical isotherm of a cubic equation of state is shown in Fig. 3.3a. The horizontal line joining the points L and V is drawn so that the areas N and M are equal, known as the *Maxwell equal area rule*. The points L and V represent the saturated liquid and saturated vapor, respectively, at the given temperature T . The intersection of the horizontal line on the pressure axis gives the vapor (or saturation) pressure⁵ of the substance at T . When this procedure is repeated for other isotherms as well, the resulting pressure versus specific volume diagram is shown in Fig. 3.3b. Vapor and liquid phases exist in equilibrium within the dome-shaped region. The critical isotherm is tangent and has a point of inflection at the critical point. The resulting mathematical expressions, i.e.

$$\left(\frac{\partial P}{\partial \tilde{V}} \right)_{T_c} = \left(\frac{\partial^2 P}{\partial \tilde{V}^2} \right)_{T_c} = 0 \quad (3.1-17)$$

are used to evaluate the parameters “ a ” and “ b ” appearing in the cubic equations of state given in Table 3.1.

At a given temperature and pressure, the solution of Eqn (3.1-15) gives three roots. When $T > T_c$, the only meaningful root is the one greater than b , the other two being complex conjugate numbers. On the other hand, when $T < T_c$, depending on the value of pressure we can conclude the following: (i) $P = P^{\text{vap}}$, the liquid and vapor phases are in equilibrium, the largest and the smallest roots correspond to the molar volumes of the saturated vapor and liquid phases, respectively; the

⁴The value of $\tilde{V}$ must be greater than b . Why?

⁵Vapor pressure can be thought of a measure of the escaping tendency of a pure substance.

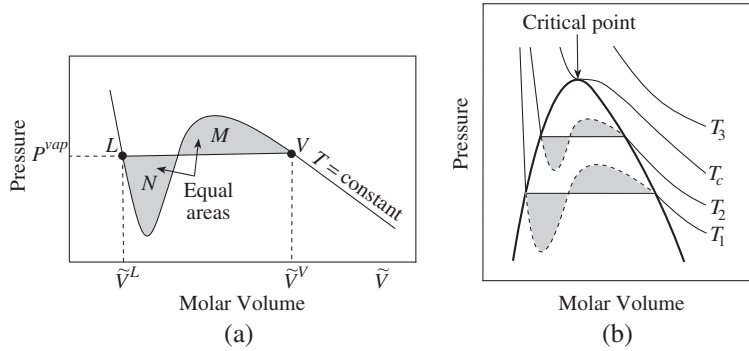


Figure 3.3: Pressure versus molar volume diagram for a pure substance.

intermediate root has no physical meaning, (ii) $P < P^{\text{vap}}$, the fluid is superheated vapor and the largest root gives its molar volume, and (iii) $P > P^{\text{vap}}$, the fluid is compressed liquid and the smallest root gives its molar volume.

It is also possible to express Eqn (3.1-15) in terms of the compressibility factor Z as

$$Z^3 + pZ^2 + qZ + r = 0, \quad (3.1-18)$$

where the coefficients p , q , and r are given in Table 3.2. Since the value of Z generally lies between 0 and 1, Eqn (3.1-18) is preferred over Eqn (3.1-15).

In engineering calculations, it is always more convenient to work with dimensionless quantities. For this reason, in Table 3.2, the parameters a and b are expressed in terms of the dimensionless parameters A and B defined by

$$A = \begin{cases} \frac{aP}{(RT)^2} & \text{van der Waals, Soave-Redlich-Kwong, Peng-Robinson} \\ \frac{aP}{R^2T^{2.5}} & \text{Redlich-Kwong} \end{cases} \quad (3.1-19)$$

and

$$B = \frac{bP}{RT}. \quad (3.1-20)$$

Example 3.2 Estimate the molar volume of methane at 340 K and 30 bar using the Peng-Robinson equation of state.

Table 3.2: Parameters in Eqn (3.1-18).^a

Equation of state	A	B	p	q	r
van der Waals	$\frac{27}{64} \left(\frac{P_r}{T_r} \right)$	$\frac{1}{8} \left(\frac{P_r}{T_r} \right)$	$-1 - B$	A	$-AB$
Redlich-Kwong	$0.42748 \left(\frac{P_r}{T_r^{2.5}} \right)$	$0.08664 \left(\frac{P_r}{T_r} \right)$	-1	$A - B - B^2$	$-AB$
Soave-Redlich-Kwong	$0.42748 \left(\frac{P_r}{T_r^2} \right) \alpha$	$0.08664 \left(\frac{P_r}{T_r} \right)$	-1	$A - B - B^2$	$-AB$
Peng-Robinson	$0.45724 \left(\frac{P_r}{T_r^2} \right) \alpha$	$0.07780 \left(\frac{P_r}{T_r} \right)$	$-1 + B$	$A - 2B - 3B^2$	$-AB + B^2 + B^3$

^aVDW—van der Waals, RK—Redlich-Kwong, SRK—Soave-Redlich-Kwong, and PR—Peng-Robinson.

Solution

An algorithm to obtain the molar volume is as follows:

1. Obtain T_c , P_c , and ω from Appendix A.
2. Calculate reduced temperature and reduced pressure.
3. Use Eqn (3.1-16) to determine the term α .
4. Determine the dimensionless parameters A and B (Table 3.2).
5. Evaluate the parameters p , q , and r appearing in Eqn (3.1-18) (Table 3.2).
6. Solve Eqn (3.1-18), i.e. $Z^3 + pZ^2 + qZ + r = 0$.
7. Use $\tilde{V} = ZRT/P$ and determine the molar volume.

From Appendix A

$$T_c = 190.6 \text{ K}, \quad P_c = 46.1 \text{ bar}, \quad \omega = 0.011.$$

The reduced temperature is

$$T_r = \frac{T}{T_c} = \frac{340}{190.6} = 1.784.$$

The reduced pressure is

$$P_r = \frac{P}{P_c} = \frac{30}{46.1} = 0.651.$$

The use Eqn (3.1-16) gives the term α as

$$\alpha = \left\{ 1 + [0.37464 + (1.54226)(0.011) - (0.26992)(0.011)^2] \left(1 - \sqrt{1.784} \right) \right\}^2 = 0.754.$$

The dimensionless parameters A and B are calculated as

$$A = 0.45724 \left(\frac{P_r}{T_r^2} \right) \alpha = \frac{(0.45724)(0.651)(0.754)}{(1.784)^2} = 0.071,$$

$$B = 0.07780 \left(\frac{P_r}{T_r} \right) = \frac{(0.07780)(0.651)}{1.784} = 0.028.$$

The parameters p , q , and r are calculated as

$$p = -1 + B = -1 + 0.028 = -0.972,$$

$$q = A - 2B - 3B^2 = 0.071 - (2)(0.028) - (3)(0.028)^2 = 0.011,$$

$$\begin{aligned} r &= -AB + B^2 + B^3 = -(0.071)(0.028) + (0.028)^2 + (0.028)^3 \\ &= -1.182 \times 10^{-3}. \end{aligned}$$

Equation (3.1-18) takes the form

$$Z^3 - 0.972Z^2 + 0.011Z - 1.182 \times 10^{-3} = 0 \Rightarrow Z = 0.962.$$

Therefore, the molar volume is given by

$$\tilde{V} = \frac{ZRT}{P} = \frac{(0.962)(83.14)(340)}{30} = 906.4 \text{ cm}^3/\text{mol}.$$

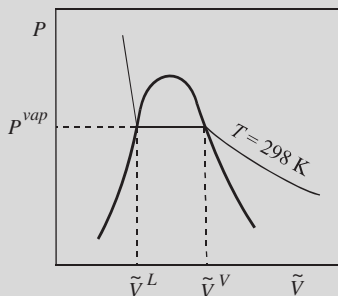
Example 3.3 A 1 m^3 rigid tank contains 25 kg of propylene at 298 K. Determine the pressure and the state of propylene in the tank using the Peng-Robinson equation of state. The vapor pressure of propylene at 298 K is 11.53 bar.

Solution

From Appendix A

$$T_c = 365.2 \text{ K}, \quad P_c = 46.0 \text{ bar}, \quad \omega = 0.144, \quad M = 42.08 \text{ g/mol}.$$

The state of a single-phase, single-component system can be specified by two independent intensive properties. The state of propylene lies on the 298 K isotherm as shown in the figure below. To identify the exact location, one more independent intensive property must be known besides temperature.



Since the tank volume and the amount of propylene are given, this additional intensive property is the molar volume, calculated as

$$\tilde{V} = \frac{1}{\frac{25}{42.08}} = 1.683 \text{ m}^3/\text{kmol}.$$

To determine the state of propylene, this value must be compared with the molar volumes of saturated liquid and vapor. To determine $\tilde{V}^L$ and $\tilde{V}^V$ at 298 K and 11.53 bar, it is first necessary to calculate Z^L and Z^V . The reduced temperature and pressure values are

$$T_r = \frac{T}{T_c} = \frac{298}{365.2} = 0.816 \quad \text{and} \quad P_r = \frac{P}{P_c} = \frac{11.53}{46} = 0.251.$$

The use Eqn (3.1-16) gives the term α as

$$\alpha = \left\{ 1 + [0.37464 + (1.54226)(0.144) - (0.26992)(0.144)^2] \left(1 - \sqrt{0.816} \right) \right\}^2 = 1.118.$$

The dimensionless parameters A and B are calculated as

$$A = 0.45724 \left(\frac{P_r}{T_r^2} \right) \alpha = \frac{(0.45724)(0.251)(1.118)}{(0.816)^2} = 0.193,$$

$$B = 0.07780 \left(\frac{P_r}{T_r} \right) = \frac{(0.07780)(0.251)}{0.816} = 0.024.$$

The cubic equation in terms of the compressibility factor, Eqn (3.1-18), takes the form

$$Z^3 - 0.976Z^2 + 0.143Z - 4.042 \times 10^{-3} = 0.$$

The solution gives

$$Z^L = 0.037 \quad \text{and} \quad Z^V = 0.804.$$

Therefore, molar volumes of the saturated liquid and vapor are

$$\tilde{V}^L = \frac{Z^L RT}{P} = \frac{(0.037)(8.314 \times 10^{-2})(298)}{11.53} = 0.080 \text{ m}^3/\text{kmol},$$

$$\tilde{V}^V = \frac{Z^V RT}{P} = \frac{(0.804)(8.314 \times 10^{-2})(298)}{11.53} = 1.728 \text{ m}^3/\text{kmol}.$$

Since $\tilde{V}^L < \tilde{V} < \tilde{V}^V$, propylene exists as a two-phase mixture and the pressure within the tank is 11.53 bar. The mole fraction of vapor in the tank, x , is given by

$$x = \frac{\tilde{V} - \tilde{V}^L}{\tilde{V}^V - \tilde{V}^L} = \frac{1.683 - 0.080}{1.728 - 0.080} = 0.97$$

3.2 Calculation of the Change in Internal Energy

In the previous chapter, the general equation to calculate the change in internal energy was given by Eqn (2.2-26), i.e.

$$d\tilde{U} = \tilde{C}_V dT + \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] d\tilde{V}. \quad (3.2-1)$$

This equation is valid for gases, liquids, and solids.

3.2.1 Change in Internal Energy for Liquids and Solids

For solids and liquids, it is more convenient to express Eqn (3.2-1) in terms of the coefficient of thermal expansion, β , and isothermal compressibility, κ . As shown in Example 2.2, the use of the triple product rule leads to

$$\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{\beta}{\kappa}. \quad (3.2-2)$$

Substitution of Eqn (3.2-2) into Eqn (3.2-1) and using $\tilde{C}_V \simeq \tilde{C}_P$ give⁶

$$d\tilde{U} = \tilde{C}_P dT + \left(\frac{\beta T}{\kappa} - P \right) d\tilde{V}. \quad (3.2-3)$$

The variation of internal energy with pressure at constant temperature is

$$\left(\frac{\partial \tilde{U}}{\partial P} \right)_T = \left(\frac{\beta T}{\kappa} - P \right) \left(\frac{\partial \tilde{V}}{\partial P} \right)_T. \quad (3.2-4)$$

The use of Eqn (2.2-54) in Eqn (3.2-4) results in

$$\left(\frac{\partial \tilde{U}}{\partial P} \right)_T = \tilde{V} (\kappa P - \beta T). \quad (3.2-5)$$

3.2.2 Change in Internal Energy for Gases

The term $(\partial P / \partial T)_{\tilde{V}}$ in Eqn (3.2-1) can be evaluated once the equation of state is known. It should be noted, however, that the heat capacity data are available only at ideal gas conditions, i.e. as either $P \rightarrow 0$ or $\tilde{V} \rightarrow \infty$. Therefore, it is necessary to develop an equation relating the actual value of $\tilde{C}_V$ to an ideal one, $\tilde{C}_V^*$. Since

⁶As shown in Problem 2.6, $\tilde{C}_V \simeq \tilde{C}_P$ if $T\beta^2/(\rho\hat{C}_P\kappa) \ll 1$.

$\tilde{U}$ is a state function, i.e. exact differential, the use of Eqn (1.2-2) in Eqn (3.2-1) leads to

$$\left(\frac{\partial \tilde{C}_V}{\partial \tilde{V}}\right)_T = \left\{ \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] \right\}_{\tilde{V}} \quad (3.2-6)$$

Carrying out the differentiation leads to

$$\left(\frac{\partial \tilde{C}_V}{\partial \tilde{V}}\right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}}. \quad (3.2-7)$$

Under isothermal conditions, integration of Eqn (3.2-7) from ideal gas state to real gas state gives

$$\int_{\text{ideal gas}}^{\text{real gas}} d\tilde{C}_V = T \int_{\text{ideal gas}}^{\text{real gas}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \quad (3.2-8)$$

or

$$\tilde{C}_V = \tilde{C}_V^* + T \int_{\infty}^{\tilde{V}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V}, \quad T = \text{constant}. \quad (3.2-9)$$

Substitution of Eqn (3.2-9) into Eqn (3.2-1) gives the change in internal energy as

$$d\tilde{U} = \left[\tilde{C}_V^* + T \int_{\infty}^{\tilde{V}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT + \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] d\tilde{V}. \quad (3.2-10)$$

Appendix B gives the molar heat capacities of gases in the ideal gas state as a function of temperature in the form

$$\tilde{C}_P^* = a + bT + cT^2 + dT^3 + eT^4. \quad (3.2-11)$$

The value of $\tilde{C}_V^*$ can be found from the relationship

$$\tilde{C}_V^* = \tilde{C}_P^* - R. \quad (3.2-12)$$

Let us consider a process in which the system goes from an initial state ($\tilde{V}_1, T_1$) to a final state ($\tilde{V}_2, T_2$). Since internal energy is a state function, calculation of $\Delta\tilde{U}$ is independent of the path chosen in going from state 1 to state 2. The most convenient paths to be followed for the integration of Eqn (3.2-10) in going from state 1 to state 2 are shown in Fig. 3.4. Thus, the change in internal energy is given by

Path A

$$\Delta\tilde{U} = \int_{\tilde{V}_1}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \left[\tilde{C}_V^* + T \int_{\infty}^{\tilde{V}_2} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT. \quad (3.2-13)$$

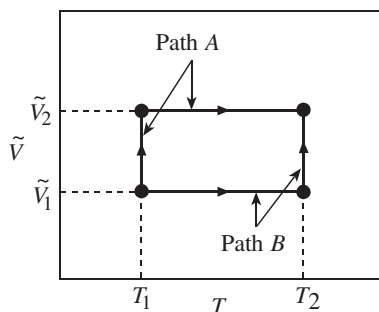


Figure 3.4: Paths of integration to calculate the change in internal energy.

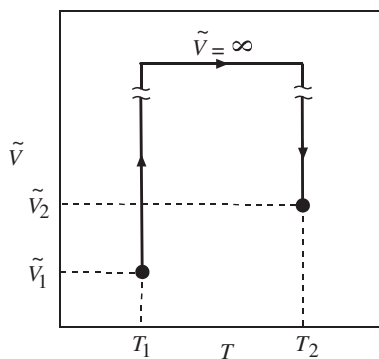


Figure 3.5: Alternative path of integration to calculate the change in internal energy.

Path B

$$\Delta \tilde{U} = \int_{T_1}^{T_2} \left[\tilde{C}_V^* + T \int_{\infty}^{\tilde{V}_1} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT + \int_{\tilde{V}_1}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] d\tilde{V}. \quad (3.2-14)$$

The use of Eqns (3.2-13) and (3.2-14) requires the value of $(\partial^2 P / \partial T^2)_{\tilde{V}}$ to be evaluated. Unless the equation of state is very simple, evaluation of this term may lead to a mathematically difficult expression to integrate. Such a difficulty can be avoided by choosing an alternative path for the integration of Eqn (3.2-1) as shown in Fig. 3.5, where $\tilde{V} = \infty$ indicates the ideal gas state. The evaluation of Eqn (3.2-1) along the path shown in Fig. 3.5 gives

$$\begin{aligned}\Delta \tilde{U} = & \int_{\tilde{V}_1}^{\infty} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \tilde{C}_V^* dT \\ & + \int_{\infty}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V}. \quad (3.2-15)\end{aligned}$$

Depending on the equation of state, equations for calculating the change in internal energy are given below.

van der Waals equation of state: The use of the van der Waals equation of state in Eqn (3.2-15) leads to

$$\Delta \tilde{U} = R \left(\frac{A_1 T_1}{Z_1} - \frac{A_2 T_2}{Z_2} \right) + \int_{T_1}^{T_2} \tilde{C}_V^* dT, \quad (3.2-16)$$

where the dimensionless parameter A is given in Table 3.2.

Redlich-Kwong equation of state: The use of the Redlich-Kwong equation of state in Eqn (3.2-15) leads to

$$\Delta \tilde{U} = \frac{3R}{2} \left[\frac{A_1 T_1}{B_1} \ln \left(1 + \frac{B_1}{Z_1} \right) - \frac{A_2 T_2}{B_2} \ln \left(1 + \frac{B_2}{Z_2} \right) \right] + \int_{T_1}^{T_2} \tilde{C}_V^* dT, \quad (3.2-17)$$

where the dimensionless parameters A and B are given in Table 3.2.

Peng-Robinson equation of state: The use of the Peng-Robinson equation of state in Eqn (3.2-15) leads to

$$\begin{aligned}\Delta \tilde{U} = & \left[\frac{T_2 (da/dT)_{T_2} - a_2}{\sqrt{8}b} \right] \ln \left[\frac{Z_2 + (1 + \sqrt{2}) B_2}{Z_2 + (1 - \sqrt{2}) B_2} \right] \\ & - \left[\frac{T_1 (da/dT)_{T_1} - a_1}{\sqrt{8}b} \right] \ln \left[\frac{Z_1 + (1 + \sqrt{2}) B_1}{Z_1 + (1 - \sqrt{2}) B_1} \right] + \int_{T_1}^{T_2} \tilde{C}_V^* dT, \quad (3.2-18)\end{aligned}$$

where

$$\frac{da}{dT} = -\frac{a\Gamma}{T} \quad (3.2-19)$$

in which

$$\Gamma = (0.37464 + 1.54226\omega - 0.26992\omega^2) \sqrt{\frac{T_r}{\alpha}}. \quad (3.2-20)$$

Substitution of Eqn (3.2-19) into Eqn (3.2-18) and rearrangement give

$$\begin{aligned} \Delta \tilde{U} = & \frac{RT_1 A_1 (1 + \Gamma_1)}{\sqrt{8} B_1} \ln \left[\frac{Z_1 + (1 + \sqrt{2}) B_1}{Z_1 + (1 - \sqrt{2}) B_1} \right] - \frac{RT_2 A_2 (1 + \Gamma_2)}{\sqrt{8} B_2} \\ & \times \ln \left[\frac{Z_2 + (1 + \sqrt{2}) B_2}{Z_2 + (1 - \sqrt{2}) B_2} \right] + \int_{T_1}^{T_2} \tilde{C}_V^* dT, \end{aligned} \quad (3.2-21)$$

where the dimensionless parameters A and B are given in Table 3.2.

Example 3.4 Show that Eqns (3.2-14) and (3.2-15) are equivalent to each other.

Solution

Equation (3.2-14) is given as

$$\begin{aligned} \Delta \tilde{U} = & \int_{T_1}^{T_2} \tilde{C}_V^* dT + \underbrace{\int_{T_1}^{T_2} \left[T \int_{\infty}^{\tilde{V}_1} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT}_X \\ & + \int_{\tilde{V}_1}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V}. \end{aligned} \quad (1)$$

Let us consider the term X in Eqn (1), i.e.

$$X = \int_{T_1}^{T_2} \left[\int_{\infty}^{\tilde{V}_1} T \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT. \quad (2)$$

Note that

$$T \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} = \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{\tilde{V}}. \quad (3)$$

Substitution of Eqn (3) into Eqn (2) gives

$$X = \int_{T_1}^{T_2} \left\{ \int_{\infty}^{\tilde{V}_1} \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{\tilde{V}} d\tilde{V} \right\} dT. \quad (4)$$

Changing the order of integration leads to

$$\begin{aligned} X &= \int_{\infty}^{\tilde{V}_1} \left\{ \int_{T_1}^{T_2} \frac{\partial}{\partial T} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right] dT \right\} d\tilde{V} \\ &= \int_{\infty}^{\tilde{V}_1} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V} - \int_{\infty}^{\tilde{V}_1} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V}. \quad (5) \end{aligned}$$

Substitution of Eqn (5) into Eqn (1) gives

$$\begin{aligned} \Delta \tilde{U} &= \int_{T_1}^{T_2} \tilde{C}_V^* dT + \int_{\infty}^{\tilde{V}_1} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V} \\ &\quad - \int_{\infty}^{\tilde{V}_1} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V} + \int_{\tilde{V}_1}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V} \end{aligned}$$

or

$$\begin{aligned} \Delta \tilde{U} &= \int_{T_1}^{T_2} \tilde{C}_V^* dT + \int_{\infty}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V} \\ &\quad - \int_{\infty}^{\tilde{V}_1} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V}, \end{aligned}$$

which is identical with Eqn (3.2-15).

Example 3.5 Two moles of acetylene is compressed isothermally but irreversibly in a piston-cylinder assembly from 0.03 m³ and 350 K to 0.002 m³. Calculate the work required if the heat removed from the system is 1.8 kJ. Acetylene is represented by the van der Waals equation of state.

Solution

From Appendix A

$$T_c = 308.3 \text{ K}, \quad P_c = 61.4 \text{ bar}.$$

System: Acetylene within the piston-cylinder assembly.

Application of the first law gives

$$W = \Delta U - Q. \quad (1)$$

Since the process is isothermal, i.e. $T_1 = T_2 = T$, Eqn (3.2-16) reduces to

$$\Delta \tilde{U} = RT \left(\frac{A_1}{Z_1} - \frac{A_2}{Z_2} \right). \quad (2)$$

The total change in internal energy is

$$\Delta U = n \Delta \tilde{U} = nRT \left(\frac{A_1}{Z_1} - \frac{A_2}{Z_2} \right). \quad (3)$$

Noting that

$$A_1 = \frac{27 P_{r1}}{64 T_r^2} \quad A_2 = \frac{27 P_{r2}}{64 T_r^2} \quad Z_1 = \frac{P_1 V_1}{nRT} \quad Z_2 = \frac{P_2 V_2}{nRT} \quad (4)$$

Eqn (3) becomes

$$\Delta U = n^2 \left(\frac{27 R^2 T_c^2}{64 P_c} \right) \left(\frac{1}{V_1} - \frac{1}{V_2} \right). \quad (5)$$

Substitution of the numerical values into Eqn (5) gives

$$\begin{aligned} \Delta U &= (2)^2 \left[\frac{27 (8.314 \times 10^{-5})^2 (308.3)^2}{64 \cdot 61.4} \right] \left(\frac{1}{0.03} - \frac{1}{0.002} \right) \\ &= -8.427 \times 10^{-3} \text{ bar m}^3 = -842.7 \text{ J.} \end{aligned}$$

The work done on the system can now be calculated from Eqn (1) as

$$W = -842.7 + 1800 = 957.3 \text{ J.}$$

Example 3.6 A well-insulated rigid tank is divided into two equal parts of $3 \times 10^{-4} \text{ m}^3$ each by a rigid partition. One side contains 10 mol of methane at 500 K, and the other side is evacuated. The partition is removed and methane fills the entire tank. Using the van der Waals equation of state, determine the final temperature of the gas.

Solution

From Appendix A

$$T_c = 190.6 \text{ K}, \quad P_c = 46.1 \text{ bar.}$$

From Appendix B

$$\begin{aligned}\tilde{C}_P^* &= 36.155 - 0.511 \times 10^{-1}T + 2.215 \times 10^{-4}T^2 \\ &\quad - 1.824 \times 10^{-7}T^3 + 4.899 \times 10^{-11}T^4.\end{aligned}$$

System: Contents of the tank.

Application of the first law gives

$$\Delta U = \underbrace{Q}_0 + \underbrace{W}_0 \Rightarrow \Delta \tilde{U} = 0. \quad (1)$$

If methane were an ideal gas, the temperature would remain constant at 500 K. In this case, however, methane is represented by the van der Waals equation of state and the change in internal energy is given by Eqn (3.2-16), i.e.

$$\Delta \tilde{U} = R \left(\frac{A_1 T_1}{Z_1} - \frac{A_2 T_2}{Z_2} \right) + \int_{T_1}^{T_2} \tilde{C}_V^* dT = 0, \quad (2)$$

where subscripts 1 and 2 represent the initial and final states, respectively. Noting that $n_1 = n_2 = n$ and making use of the substitution

$$A_1 = \frac{27 P_{r1}}{64 T_{r1}^2} \quad A_2 = \frac{27 P_{r2}}{64 T_{r2}^2} \quad Z_1 = \frac{P_1 V_1}{nRT_1} \quad Z_2 = \frac{P_2 V_2}{nRT_2} \quad (3)$$

Eqn (2) takes the form

$$n \left(\frac{27 R^2 T_c^2}{64 P_c} \right) \left(\frac{1}{V_1} - \frac{1}{V_2} \right) + \int_{T_1}^{T_2} \tilde{C}_V^* dT = 0. \quad (4)$$

The heat capacity at constant volume is

$$\begin{aligned}\tilde{C}_V^* &= \tilde{C}_P^* - R = 27.841 - 0.511 \times 10^{-1}T + 2.215 \times 10^{-4}T^2 \\ &\quad - 1.824 \times 10^{-7}T^3 + 4.899 \times 10^{-11}T^4.\end{aligned}$$

Substitution of the numerical values into Eqn (4) gives

$$\begin{aligned}(10) \quad &\left[\frac{27 (8.314 \times 10^{-5})^2 (190.6)^2}{64 \cdot 46.1} \right] \\ &\times \left(\frac{1}{3 \times 10^{-4}} - \frac{1}{6 \times 10^{-4}} \right) \left(\frac{1}{10^{-5} \text{ bar m}^3} \right) + \int_{500}^{T_2} (27.841 - 0.511 \\ &\times 10^{-1}T + 2.215 \times 10^{-4}T^2 - 1.824 \times 10^{-7}T^3 \\ &+ 4.899 \times 10^{-11}T^4) dT = 0.\end{aligned}$$

The solution of the above equation by MATHCAD[®] gives $T_2 = 390$ K.

Example 3.7 Suppose that the methane in Example 3.6 is replaced by gas A, which is represented by the following equation of state:

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{T \tilde{V}^2},$$

where

$$a = 35 \text{ J K m}^3/\text{mol}^2 \quad \text{and} \quad b = 4.8 \times 10^{-5} \text{ m}^3/\text{mol}.$$

If the ideal gas heat capacity of gas A is $\tilde{C}_V^* = 1.5R$, determine the final temperature.

Solution

Since both $\tilde{Q}$ and $\tilde{W}$ are zero, then

$$\Delta \tilde{U} = 0. \quad (1)$$

The change in internal energy is given by Eqn (3.2-15), i.e.

$$\begin{aligned} \Delta \tilde{U} = & \int_{\tilde{V}_1}^{\infty} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \tilde{C}_V^* dT \\ & + \int_{\infty}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_2} d\tilde{V}. \end{aligned} \quad (2)$$

The integrand in Eqn (2) is evaluated using the given equation of state as

$$\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{R}{\tilde{V} - b} + \frac{a}{T^2 \tilde{V}^2} \quad \Rightarrow \quad T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P = \frac{2a}{T \tilde{V}^2} \quad (3)$$

Substitution of Eqns (1) and (3) into Eqn (2) gives

$$0 = \frac{2a}{T_1} \int_{\tilde{V}_1}^{\infty} \frac{1}{\tilde{V}^2} d\tilde{V} + \int_{T_1}^{T_2} 1.5R dT + \frac{2a}{T_2} \int_{\infty}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V}. \quad (4)$$

Carrying out the integration leads to

$$\frac{2a}{T_1 \tilde{V}_1} + 1.5R(T_2 - T_1) - \frac{2a}{T_2 \tilde{V}_2} = 0. \quad (5)$$

Substitution of the numerical values into Eqn (5) results in

$$\frac{(2)(35)}{(500)(3 \times 10^{-4}/10)} + (1.5)(8.314)(T_2 - 500) - \frac{(2)(35)}{T_2(6 \times 10^{-4}/10)} = 0. \quad (6)$$

The solution of Eqn (6) gives $T_2 = 375.2 \text{ K}$.

Alternate solution: The solution is also possible by using either Eqn (3.2-13) or Eqn (3.2-14). Note that

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_{\tilde{V}} = -\frac{2a}{T^3 \tilde{V}^2}. \quad (3.2-7)$$

Substitution of Eqns (1), (3), and (3.2-7) into Eqn (3.2-13) leads to

$$0 = \frac{2a}{T_1} \int_{\tilde{V}_1}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V} + \int_{T_1}^{T_2} \left(1.5R - \frac{2a}{T^2} \int_{\infty}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V} \right) dT \quad (3.2-8)$$

or

$$0 = \frac{2a}{T_1} \int_{\tilde{V}_1}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V} + \int_{T_1}^{T_2} \left(1.5R + \frac{2a}{T^2 \tilde{V}_2} \right) dT \quad (3.2-9)$$

Once the integration is carried out, the result is equivalent to Eqn (5).

On the other hand, substitution of Eqns (1), (3), and (3.2-7) into Eqn (3.2-14) leads to

$$0 = \int_{T_1}^{T_2} \left(1.5R - \frac{2a}{T^2} \int_{\infty}^{\tilde{V}_1} \frac{1}{\tilde{V}^2} d\tilde{V} \right) dT + \frac{2a}{T_2} \int_{\tilde{V}_1}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V} \quad (3.2-10)$$

or

$$0 = \int_{T_1}^{T_2} \left(1.5R + \frac{2a}{T^2 \tilde{V}_1} \right) dT + \frac{2a}{T_2} \int_{\tilde{V}_1}^{\tilde{V}_2} \frac{1}{\tilde{V}^2} d\tilde{V} \quad (3.2-11)$$

Integration of Eqn (3.2-11) is also equivalent to Eqn (5).

Example 3.8 A piston-cylinder assembly contains carbon monoxide at 283 K and 1 bar. It is compressed to a pressure of 80 bar by a reversible and isothermal process. Calculate the work and heat interactions with the surroundings if carbon monoxide obeys the following equation of state:

$$\frac{P\tilde{V}}{RT} = 1 + \frac{B}{\tilde{V}} + \frac{C}{\tilde{V}^2}$$

with

$$B = -23.1 \text{ cm}^3/\text{mol} \quad \text{and} \quad C = 1560 \text{ cm}^6/\text{mol}^2.$$

Solution

System: CO within the piston-cylinder assembly.

The reversible work done in compressing the gas is given by

$$\tilde{W} = - \int_{\tilde{V}_1}^{\tilde{V}_2} P d\tilde{V}. \quad (1)$$

Once $\tilde{W}$ is determined, the heat interaction is calculated indirectly from the first law as

$$\tilde{Q} = \Delta\tilde{U} - \tilde{W}. \quad (2)$$

Substitution of the given equation of state into Eqn (1) gives

$$\begin{aligned} \tilde{W} &= -RT \int_{\tilde{V}_1}^{\tilde{V}_2} \left(\frac{1}{\tilde{V}} + \frac{B}{\tilde{V}^2} + \frac{C}{\tilde{V}^3} \right) d\tilde{V} \\ &= -RT \left[\ln \left(\frac{\tilde{V}_2}{\tilde{V}_1} \right) - B \left(\frac{1}{\tilde{V}_2} - \frac{1}{\tilde{V}_1} \right) - \frac{C}{2} \left(\frac{1}{\tilde{V}_2^2} - \frac{1}{\tilde{V}_1^2} \right) \right]. \end{aligned} \quad (3)$$

Using MATHCAD[®], the initial and final molar volumes are calculated from the equation of state as

$$\begin{aligned} \frac{(1) \tilde{V}_1}{(83.14)(283)} &= 1 - \frac{23.1}{\tilde{V}_1} + \frac{1560}{\tilde{V}_1^2} \Rightarrow \tilde{V}_1 = 23,506 \text{ cm}^3/\text{mol}, \\ \frac{(80) \tilde{V}_2}{(83.14)(283)} &= 1 - \frac{23.1}{\tilde{V}_2} + \frac{1560}{\tilde{V}_2^2} \Rightarrow \tilde{V}_2 = 275 \text{ cm}^3/\text{mol}. \end{aligned}$$

Substitution of the numerical values into Eqn (3) results in

$$\begin{aligned} \tilde{W} &= -(8.314)(283) \left\{ \ln \left(\frac{275}{23,506} \right) + 23.1 \left(\frac{1}{275} - \frac{1}{23,506} \right) \right. \\ &\quad \left. - \frac{1560}{2} \left[\frac{1}{(275)^2} - \frac{1}{(23,506)^2} \right] \right\} = 10,295 \text{ J/mol}. \end{aligned}$$

Since the process is isothermal Eqn (3.2-13) reduces to

$$\Delta\tilde{U} = \int_{\tilde{V}_1}^{\tilde{V}_2} \left[T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P \right]_{T_1} d\tilde{V}. \quad (4)$$

From the given equation of state

$$\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{R}{\tilde{V}} \left(1 + \frac{B}{\tilde{V}} + \frac{C}{\tilde{V}^2} \right) \Rightarrow T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - P = 0. \quad (5)$$

Therefore, $\Delta\tilde{U} = 0$ and the heat interaction of the system with its surroundings can be calculated from Eqn (2) as

$$\tilde{Q} = -10,295 \text{ J/mol}.$$

Example 3.9 Solve Example 3.5 if acetylene obeys the Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 308.3 \text{ K}, \quad P_c = 61.4 \text{ bar}, \quad \omega = 0.190.$$

The change in internal energy is calculated from Eqn (3.2-21). Since the process is isothermal, i.e. $T_1 = T_2 = T = 350 \text{ K}$ and $\Gamma_1 = \Gamma_2 = \Gamma$, it simplifies to

$$\Delta \tilde{U} = \frac{RT(1 + \Gamma)}{\sqrt{8}} \left\{ \frac{A_1}{B_1} \ln \left[\frac{Z_1 + (1 + \sqrt{2})B_1}{Z_1 + (1 - \sqrt{2})B_1} \right] - \frac{A_2}{B_2} \ln \left[\frac{Z_2 + (1 + \sqrt{2})B_2}{Z_2 + (1 - \sqrt{2})B_2} \right] \right\}. \quad (1)$$

The reduced temperature is

$$T_r = \frac{T}{T_c} = \frac{350}{308.3} = 1.135$$

and the parameter α is calculated from Eqn (3.1-16) as

$$\alpha = \left[1 + [0.37464 + (1.54226)(0.190) - (0.26992)(0.190)^2] (1 - \sqrt{1.135}) \right]^2 = 0.916.$$

Therefore,

$$\begin{aligned} a &= 0.45724 \left(\frac{R^2 T_c^2}{P_c} \right) \alpha = \frac{(0.45724)(8.314 \times 10^{-5})^2 (308.3)^2 (0.916)}{61.4} \\ &= 4.482 \times 10^{-6} \frac{\text{bar m}^6}{\text{mol}^2}, \\ b &= 0.07780 \frac{RT_c}{P_c} = \frac{(0.0778)(8.314 \times 10^{-5})(308.3)}{61.4} \\ &= 3.249 \times 10^{-5} \text{ m}^3/\text{mol}. \end{aligned}$$

The use of Eqn (3.2-20) gives

$$\begin{aligned} \Gamma_1 = \Gamma_2 &= [0.37464 + (1.54226)(0.190) - (0.26992)(0.190)^2] \\ &\quad \times \sqrt{\frac{1.135}{0.916}} = 0.732. \end{aligned}$$

State 1 ($T_1 = 350 \text{ K}$, $\tilde{V}_1 = 0.03/2 = 0.015 \text{ m}^3/\text{mol}$)

The pressure is calculated from the Peng-Robinson equation of state as

$$P_1 = \frac{(8.314 \times 10^{-5})(350)}{0.015 - 3.249 \times 10^{-5}} - \frac{4.482 \times 10^{-6}}{(0.015)(0.015 + 3.249 \times 10^{-5}) + (3.249 \times 10^{-5})(0.015 - 3.249 \times 10^{-5})} = 1.924 \text{ bar.}$$

The reduced pressure is

$$P_{r1} = \frac{1.924}{61.4} = 0.031.$$

The dimensionless parameters are calculated as

$$A_1 = 0.45724 \left(\frac{P_{r1}}{T_r^2} \right) \alpha = \frac{(0.45724)(0.031)(0.916)}{(1.135)^2} = 0.010,$$

$$B_1 = 0.07780 \left(\frac{P_{r1}}{T_r} \right) = \frac{(0.07780)(0.031)}{1.135} = 0.002.$$

The cubic equation for Z , Eqn (3.1-18), takes the form

$$Z_1^3 - 0.998Z_1^2 + 5.988 \times 10^{-3}Z_1 - 1.599 \times 10^{-5} = 0 \Rightarrow Z_1 = 0.992.$$

State 2 ($T_2 = 350 \text{ K}$, $\tilde{V}_2 = 0.002/2 = 0.001 \text{ m}^3/\text{mol}$)

The pressure is

$$P_2 = \frac{(8.314 \times 10^{-5})(350)}{0.001 - 3.249 \times 10^{-5}} - \frac{4.482 \times 10^{-6}}{(0.001)(1 \times 10^{-3} + 3.249 \times 10^{-5}) + (3.249 \times 10^{-5})(0.001 - 3.249 \times 10^{-5})} = 25.863 \text{ bar.}$$

Therefore, the reduced pressure is

$$P_{r2} = \frac{25.863}{61.4} = 0.421$$

The dimensionless parameters are calculated as

$$A_2 = 0.45724 \left(\frac{P_{r2}}{T_r^2} \right) \alpha = \frac{(0.45724)(0.421)(0.916)}{(1.135)^2} = 0.137,$$

$$B_2 = 0.07780 \left(\frac{P_{r2}}{T_r} \right) = \frac{(0.07780)(0.421)}{1.135} = 0.029.$$

The cubic equation for Z , Eqn (3.1-18), takes the form

$$Z_2^3 - 0.971Z_2^2 + 0.076Z_2 - 3.108 \times 10^{-3} = 0 \Rightarrow Z_2 = 0.889.$$

Substitution of the numerical values into Eqn (1) gives

$$\begin{aligned} \Delta \tilde{U} = & \frac{(8.314)(350)(1 + 0.732)}{\sqrt{8}} \\ & \times \left\{ \frac{0.010}{0.002} \ln \left[\frac{0.992 + (1 + \sqrt{2})(0.002)}{0.992 + (1 - \sqrt{2})(0.002)} \right] \right. \\ & \left. - \frac{0.137}{0.029} \ln \left[\frac{0.889 + (1 + \sqrt{2})(0.029)}{0.889 + (1 - \sqrt{2})(0.029)} \right] \right\} = -702 \text{ J/mol.} \end{aligned}$$

Therefore, the work interaction is

$$W = \Delta U - Q = (2)(-702) + 1800 = 396 \text{ J,}$$

which is quite different from the one calculated in Example 3.5.

Comment: The results are strongly dependent on the equation of state used in the calculations.

3.3 Calculation of the Change in Enthalpy

In the previous chapter, the general equation to calculate the change in enthalpy was given by Eqn (2.2-31), i.e.

$$d\tilde{H} = \tilde{C}_P dT + \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP. \quad (3.3-1)$$

This equation is valid for gases, liquids, and solids.

3.3.1 Change in Enthalpy for Liquids and Solids

For solids and liquids, it is more convenient to express Eqn (3.3-1) in terms of the coefficient of thermal expansion, β . The use of Eqn (2.2-52) in Eqn (3.3-1) leads to

$$d\tilde{H} = \tilde{C}_P dT + \tilde{V} (1 - \beta T) dP. \quad (3.3-2)$$

Example 3.10 One mole of methanol ($\beta = 12.0 \times 10^{-4} \text{ K}^{-1}$, $\tilde{V} = 40.47 \text{ cm}^3/\text{mol}$) at 293 K is compressed isothermally from 1 bar to 500 bar. Calculate the change in enthalpy.

Solution

Integration of Eqn (3.3-2) at constant temperature gives

$$\Delta \tilde{H} = \tilde{V} (1 - \beta T) \Delta P \quad (1)$$

in which variations in $\tilde{V}$ and β with pressure are considered negligible. Substitution of the values into Eqn (1) results in

$$\begin{aligned} \Delta \tilde{H} &= (40.47 \times 10^{-6}) [1 - (12.0 \times 10^{-4})(293)] \\ &\quad \times (500 - 1) \times 10^5 = 1309 \text{ J/mol.} \end{aligned}$$

3.3.2 Change in Enthalpy for Gases

Since the heat capacity data are available only at ideal gas conditions, i.e. as $P \rightarrow 0$ or $\tilde{V} \rightarrow \infty$, it is first necessary to develop an equation relating actual $\tilde{C}_P$ value to an ideal one, $\tilde{C}_P^*$. Since $\tilde{H}$ is a state function, i.e. exact differential, the use of Eqn (1.2-2) in Eqn (3.3-1) leads to

$$\left(\frac{\partial \tilde{C}_P}{\partial P} \right)_T = \left\{ \frac{\partial}{\partial T} \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] \right\}_P. \quad (3.3-3)$$

Carrying out the differentiation leads to

$$\left(\frac{\partial \tilde{C}_P}{\partial P} \right)_T = -T \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P. \quad (3.3-4)$$

Under isothermal conditions, integration of Eqn (3.3-4) from ideal gas state to real gas state gives

$$\int_{\text{ideal gas}}^{\text{real gas}} d\tilde{C}_P = -T \int_{\text{ideal gas}}^{\text{real gas}} \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \quad (3.3-5)$$

or

$$\tilde{C}_P = \tilde{C}_P^* - T \int_0^P \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP, \quad T = \text{constant.} \quad (3.3-6)$$

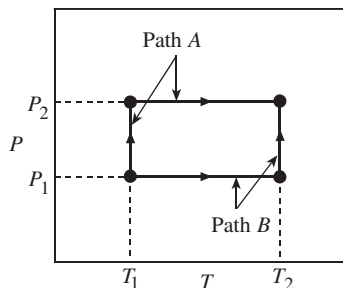


Figure 3.6: Paths of integration to calculate the change in enthalpy.

Substitution of Eqn (3.3-6) into Eqn (3.3-1) gives the change in enthalpy as

$$d\tilde{H} = \left[\tilde{C}_P^* - T \int_0^P \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT + \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP. \quad (3.3-7)$$

Let us consider a process in which the system goes from an initial state (P_1, T_1) to a final state (P_2, T_2) . Since enthalpy is a state function, calculation of $\Delta \tilde{H}$ is independent of the path chosen in going from state 1 to state 2. The most convenient paths to be followed for the integration of Eqn (3.3-7) in going from state 1 to state 2 are shown in Fig. 3.6 as paths A and B. Thus, the change in enthalpy is given by

Path A

$$\begin{aligned} \Delta \tilde{H} = & \int_{P_1}^{P_2} \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_1} dP \\ & + \int_{T_1}^{T_2} \left[\tilde{C}_P^* - T \int_0^{P_2} \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT. \end{aligned} \quad (3.3-8)$$

Path B

$$\begin{aligned} \Delta \tilde{H} = & \int_{T_1}^{T_2} \left[\tilde{C}_P^* - T \int_0^{P_1} \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT \\ & + \int_{P_1}^{P_2} \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_2} dP. \end{aligned} \quad (3.3-9)$$

Example 3.11 Ammonia is to be compressed from 350 K and 1 bar to 400 K and 50 bar. If the work supplied to the compressor is 1500 J/mol, calculate the

heat that must be supplied per mole of ammonia passing through the compressor. The equation of state is given by

$$P(\tilde{V} - b) = RT,$$

where $b = 3.73 \times 10^{-5} \text{ m}^3/\text{mol}$.

Solution

From Appendix B

$$\begin{aligned}\tilde{C}_P^* = & 34.236 - 0.221 \times 10^{-1}T + 1.213 \times 10^{-4}T^2 \\ & - 1.088 \times 10^{-7}T^3 + 3.203 \times 10^{-11}T^4.\end{aligned}$$

System: Compressor.

For a steady-state flow system, the first law gives

$$\tilde{Q} = \Delta\tilde{H} - \tilde{W}_s. \quad (1)$$

To calculate $\Delta\tilde{H}$, it is first necessary to evaluate the partial derivatives appearing in Eqns (3.3-8) and (3.3-9)

$$\left(\frac{\partial\tilde{V}}{\partial T}\right)_P = \frac{R}{P} \quad \left(\frac{\partial^2\tilde{V}}{\partial T^2}\right)_P = 0 \quad \tilde{V} - T\left(\frac{\partial\tilde{V}}{\partial T}\right)_P = b. \quad (2)$$

Substitution of Eqn (2) either into Eqn (3.3-8) or into Eqn (3.3-9) gives

$$\Delta\tilde{H} = b(P_2 - P_1) + \int_{T_1}^{T_2} \tilde{C}_P^* dT \quad (3)$$

or

$$\begin{aligned}\Delta\tilde{H} = & (3.73 \times 10^{-5}) \left[(50 - 1) \times 10^5 \right] \\ & + \int_{350}^{400} (34.236 - 0.221 \times 10^{-1}T + 1.213 \times 10^{-4}T^2 \\ & - 1.088 \times 10^{-7}T^3 + 3.203 \times 10^{-11}T^4) dT \\ = & 2078.2 \text{ J/mol}.\end{aligned}$$

The heat interaction is calculated from Eqn (1) as

$$\tilde{Q} = 2078.2 - 1500 = 578.2 \text{ J/mol}.$$

Example 3.12 An evacuated gas cylinder of 50 L volume is connected to a main line that supplies CO₂ at the constant conditions of 400 K and 2 MPa. The valve between the line and the cylinder is opened and the cylinder quickly fills with CO₂ until the pressure is 2 MPa, and then the valve is closed. Calculate the temperature immediately after the valve is closed. The equation of state for CO₂ is given by

$$P(\tilde{V} - b) = RT,$$

where $b = 4.41 \times 10^{-5} \text{ m}^3/\text{mol}$.

Solution

From Appendix B

$$\begin{aligned} \tilde{C}_P^* = & 29.268 - 0.224 \times 10^{-1}T + 2.653 \times 10^{-4}T^2 \\ & - 4.153 \times 10^{-7}T^3 + 20.057 \times 10^{-11}T^4. \end{aligned}$$

System: Contents of the cylinder.

The unsteady-state material and energy balances become

$$dn_{\text{in}} - \underbrace{dn_{\text{out}}}_0 = dn_{\text{sys}}, \quad (1)$$

$$\tilde{H}_{\text{in}} dn_{\text{in}} - \underbrace{\tilde{H}_{\text{out}} dn_{\text{out}}}_0 + \underbrace{\delta Q}_0 + \underbrace{\delta W}_0 = d(n\tilde{U})_{\text{sys}}. \quad (2)$$

Note that Q is considered zero since filling takes place quickly and heat transfer is a slow process. Combination of Eqns (1) and (2) leads to

$$\tilde{H}_{\text{in}} dn_{\text{sys}} = d(n\tilde{U})_{\text{sys}}. \quad (3)$$

Since $\tilde{H}_{\text{in}}$ is constant, integration of Eqn (3) gives

$$\tilde{H}_{\text{in}}(n_2 - n_1) = n_2\tilde{U}_2 - n_1\tilde{U}_1. \quad (4)$$

The cylinder is initially evacuated, i.e. $n_1 = 0$, and Eqn (4) simplifies to

$$\tilde{H}_{\text{in}} = \tilde{U}_2. \quad (5)$$

Expressing internal energy in terms of enthalpy, i.e. $\tilde{U}_2 = \tilde{H}_2 - P_2\tilde{V}_2$, reduces Eqn (5) to

$$P_2\tilde{V}_2 = \tilde{H}_2 - \tilde{H}_{\text{in}}. \quad (6)$$

The use of the equation of state results in

$$RT_2 + bP_2 = \tilde{H}_2(2 \text{ MPa}, T_2) - \tilde{H}_{\text{in}}(2 \text{ MPa}, 400 \text{ K}). \quad (7)$$

The right-hand side of Eqn (7) can be calculated from either Eqn (3.3-8) or Eqn (3.3-9). Both equations simplify to

$$\tilde{H}_2(2 \text{ MPa}, T_2) - \tilde{H}_{\text{in}}(2 \text{ MPa}, 400 \text{ K}) = \int_{400}^{T_2} \tilde{C}_P^* dT. \quad (8)$$

Substitution of Eqn (8) into Eqn (7) gives

$$RT_2 + bP_2 = \int_{400}^{T_2} \tilde{C}_P^* dT \quad (9)$$

or

$$\begin{aligned} 8.314T_2 + (4.41 \times 10^{-5})(2 \times 10^6) = \int_{400}^{T_2} (29.268 - 0.224 \times 10^{-1}T \\ + 2.653 \times 10^{-4}T^2 - 4.153 \times 10^{-7}T^3 \\ + 20.057 \times 10^{-11}T^4) dT. \end{aligned} \quad (10)$$

Solution of Eqn (10) by MATHCAD[®] gives $T_2 = 498 \text{ K}$.

Comment: It is implicitly assumed that there is no heat transfer between the cylinder surface and CO₂. Under what conditions is this assumption valid?

Since the cubic equations of state are not explicit in molar volume, evaluation of the terms $(\partial \tilde{V} / \partial T)_P$ and $(\partial^2 \tilde{V} / \partial T^2)_P$ in Eqns (3.3-8) and (3.3-9) causes difficulty. The term $(\partial \tilde{V} / \partial T)_P$ can be evaluated by the use of the triple product rule as

$$\left(\frac{\partial \tilde{V}}{\partial T} \right)_P = - \frac{(\partial P / \partial T)_{\tilde{V}}}{(\partial P / \partial \tilde{V})_T}. \quad (3.3-10)$$

For example, if a gas is represented by the Redlich-Kwong equation of state, i.e.

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{\tilde{V}(\tilde{V} + b)\sqrt{T}} \quad (3.3-11)$$

then

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = \frac{\frac{R}{\tilde{V} - b} + \frac{a}{2 \tilde{V} (\tilde{V} + b) T^{3/2}}}{\frac{RT}{(\tilde{V} - b)^2} - \frac{a (2 \tilde{V} + b)}{\tilde{V}^2 (\tilde{V} + b)^2 \sqrt{T}}} \quad (3.3-12)$$

Substitution of Eqn (3.3-12) together with the term $(\partial^2 \tilde{V} / \partial T^2)_P$ into Eqns (3.3-8) and (3.3-9) leads to a formidable equation. To circumvent mathematical complexity, an alternative path is chosen for the integration of Eqn (3.3-1) as shown in Fig. 3.7, where $P = 0$ indicates the ideal gas state. The evaluation of Eqn (3.3-1) along the path shown in Fig. 3.7 gives

$$\Delta \tilde{H} = \underbrace{\int_{P_1}^0 \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_1} dP}_X + \int_{T_1}^{T_2} \tilde{C}_P^* dT + \underbrace{\int_0^{P_2} \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_2} dP}_Y \quad (3.3-13)$$

It is apparent that the use of Eqn (3.3-13) is much simpler than that of Eqns (3.3-8) and (3.3-9). However, in the case of cubic equations of state, evaluation of the terms X and Y in Eqn (3.3-13) is not straightforward. While integrations in Eqn (3.3-13) are with respect to P , the term $(\partial \tilde{V} / \partial T)_P$ leads to an expression containing $\tilde{V}$ as can be seen from Eqn (3.3-12). Therefore, it is useful to have an integration over volume rather than pressure. For this purpose consider the integral

$$\int_0^P \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_T dP. \quad (3.3-14)$$

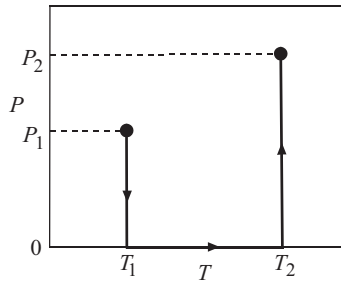


Figure 3.7: Alternative path of integration to calculate the change in enthalpy.

Since $P = P(T, \tilde{V})$, then

$$dP = \left(\frac{\partial P}{\partial \tilde{V}} \right)_T d\tilde{V} + \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} dT. \quad (3.3-15)$$

Note that the integrand in Eqn (3.3-14) is evaluated at constant temperature. Under the condition of constant temperature, Eqn (3.3-15) reduces to

$$dP = \left(\frac{\partial P}{\partial \tilde{V}} \right)_T d\tilde{V}. \quad (3.3-16)$$

Substitution of Eqn (3.3-16) into Eqn (3.3-14) gives

$$\int_0^P \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP = \int_{\infty}^{\tilde{V}} \left[\tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \left(\frac{\partial P}{\partial \tilde{V}} \right)_T \right] d\tilde{V}. \quad (3.3-17)$$

From the triple product rule

$$\left(\frac{\partial \tilde{V}}{\partial T} \right)_P \left(\frac{\partial P}{\partial \tilde{V}} \right)_T = - \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}}. \quad (3.3-18)$$

The use of Eqn (3.3-18) in Eqn (3.3-17) leads to

$$\int_0^P \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP = \int_{\infty}^{\tilde{V}} \left[\tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T + T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right] d\tilde{V}. \quad (3.3-19)$$

Thus, Eqn (3.3-13) takes the form

$$\begin{aligned} \Delta \tilde{H} &= \int_{\tilde{V}_1}^{\infty} \left[\tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T + T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \tilde{C}_P^* dT \\ &\quad + \int_{\infty}^{\tilde{V}_2} \left[\tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T + T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_2} d\tilde{V}. \end{aligned} \quad (3.3-20)$$

Depending on the equation of state, equations for calculating the change in enthalpy are given below.

van der Waals equation of state: The use of the van der Waals equation of state in Eqn (3.3-20) leads to

$$\Delta \tilde{H} = RT_2 \left(Z_2 - 1 - \frac{A_2}{Z_2} \right) - RT_1 \left(Z_1 - 1 - \frac{A_1}{Z_1} \right) + \int_{T_1}^{T_2} \tilde{C}_P^* dT, \quad (3.3-21)$$

where the dimensionless parameter A is given in Table 3.2.

Redlich-Kwong equation of state: The use of the Redlich-Kwong equation of state in Eqn (3.3-20) leads to

$$\begin{aligned} \Delta \tilde{H} = RT_2 \left[Z_2 - 1 - \frac{3 A_2}{2 B_2} \ln \left(1 + \frac{B_2}{Z_2} \right) \right] \\ - RT_1 \left[Z_1 - 1 - \frac{3 A_1}{2 B_1} \ln \left(1 + \frac{B_1}{Z_1} \right) \right] + \int_{T_1}^{T_2} \tilde{C}_P^* dT, \end{aligned} \quad (3.3-22)$$

where the dimensionless parameters A and B are given in Table 3.2.

Peng-Robinson equation of state: The use of the Peng-Robinson equation of state in Eqn (3.3-20) leads to

$$\begin{aligned} \Delta \tilde{H} = RT_2 \left\{ Z_2 - 1 + \frac{1}{RT_2} \left[\frac{T_2 (da/dT)_{T_2} - a_2}{\sqrt{8}b} \right] \ln \left[\frac{Z_2 + (1 + \sqrt{2}) B_2}{Z_2 + (1 - \sqrt{2}) B_2} \right] \right\} \\ - RT_1 \left\{ Z_1 - 1 + \frac{1}{RT_1} \left[\frac{T_1 (da/dT)_{T_1} - a_1}{\sqrt{8}b} \right] \ln \left[\frac{Z_1 + (1 + \sqrt{2}) B_1}{Z_1 + (1 - \sqrt{2}) B_1} \right] \right\} \\ + \int_{T_1}^{T_2} \tilde{C}_P^* dT, \end{aligned} \quad (3.3-23)$$

where the derivative da/dT is given by Eqn (3.2-19). Substitution of Eqn (3.2-19) into Eqn (3.3-23) and expressing the resulting equation in dimensionless form yield

$$\begin{aligned} \Delta \tilde{H} = RT_2 \left\{ Z_2 - 1 - \frac{A_2(1 + \Gamma_2)}{\sqrt{8}B_2} \ln \left[\frac{Z_2 + (1 + \sqrt{2}) B_2}{Z_2 + (1 - \sqrt{2}) B_2} \right] \right\} \\ - RT_1 \left\{ Z_1 - 1 - \frac{A_1(1 + \Gamma_1)}{\sqrt{8}B_1} \ln \left[\frac{Z_1 + (1 + \sqrt{2}) B_1}{Z_1 + (1 - \sqrt{2}) B_1} \right] \right\} + \int_{T_1}^{T_2} \tilde{C}_P^* dT. \end{aligned} \quad (3.3-24)$$

The dimensionless parameters A and B are given in Table 3.2, and Γ is defined by Eqn (3.2-20).

Example 3.13 Propane gas is to be compressed irreversibly from 5 bar and 323 K to 15 bar and 343 K by an adiabatic compressor in a petrochemical plant. Calculate the work input per mole of propane passing through the compressor if

- (a) Propane obeys the Redlich-Kwong equation of state.
- (b) Propane obeys the Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 369.9 \text{ K}, \quad P_c = 42.5 \text{ bar}, \quad \omega = 0.153.$$

From Appendix B

$$\begin{aligned} \tilde{C}_p^* = & 29.595 + 0.838 \times 10^{-1}T + 3.256 \times 10^{-4}T^2 \\ & - 3.958 \times 10^{-7}T^3 + 13.129 \times 10^{-11}T^4. \end{aligned}$$

System: Compressor.

This is a steady-state flow process and from the first law

$$\Delta \tilde{H} = \underbrace{\tilde{Q}}_0 + \tilde{W}_s. \quad (1)$$

Therefore, the work input to the compressor is equal to the change in enthalpy. Determination of enthalpy at the compressor inlet (state 1) and outlet (state 2) requires reduced temperature and pressure values to be known:

State	T_r	P_r
1	0.873	0.118
2	0.927	0.353

- (a) The enthalpy change is given by Eqn (3.2-17). Therefore, it is first necessary to determine A , B , and Z values at the compressor inlet and outlet.

State 1 [$T_1 = 323 \text{ K}$ ($P^{\text{vap}} = 17.4 \text{ bar}$), $P_1 = 5 \text{ bar}$]

The dimensionless parameters are calculated as

$$A_1 = 0.42748 \left(\frac{P_{r1}}{T_{r1}^{2.5}} \right) = \frac{(0.42748)(0.118)}{(0.873)^{2.5}} = 0.071,$$

$$B_1 = 0.08664 \left(\frac{P_{r1}}{T_{r1}} \right) = \frac{(0.08664)(0.118)}{0.873} = 0.012.$$

The cubic equation for Z , Eqn (3.1-18), takes the form

$$Z_1^3 - Z_1^2 + 0.059 Z_1 - 8.52 \times 10^{-4} = 0 \Rightarrow Z_1 = 0.938.$$

State 2 [$T_2 = 343$ K ($P^{\text{vap}} = 25.8$ bar), $P_2 = 15$ bar]

The dimensionless parameters are calculated as

$$A_2 = 0.42748 \left(\frac{P_{r2}}{T_{r2}^{2.5}} \right) = \frac{(0.42748)(0.353)}{(0.927)^{2.5}} = 0.182,$$

$$B_2 = 0.08664 \left(\frac{P_{r2}}{T_{r2}} \right) = \frac{(0.08664)(0.353)}{0.927} = 0.033.$$

The cubic equation for Z , Eqn (3.1-18), takes the form

$$Z_2^3 - Z_2^2 + 0.148 Z_2 - 6 \times 10^{-3} = 0 \Rightarrow Z_2 = 0.830.$$

The third term on the right-hand side of Eqn (3.3-22) is calculated as

$$\int_{323}^{343} \tilde{C}_P^* dT = 1612 \text{ J/mol}$$

Thus, substitution of numerical values into Eqn (3.3-22) gives

$$\begin{aligned} \Delta \tilde{H} = \tilde{W} &= (8.314)(343) \left[0.830 - 1 - \frac{3}{2} \left(\frac{0.182}{0.033} \right) \ln \left(1 + \frac{0.033}{0.830} \right) \right] \\ &\quad - (8.314)(323) \left[0.938 - 1 - \frac{3}{2} \left(\frac{0.071}{0.012} \right) \ln \left(1 + \frac{0.012}{0.938} \right) \right] \\ &\quad + 1612 = 676.9 \text{ J/mol.} \end{aligned}$$

(b) The enthalpy change is given by Eqn (3.3-24). Therefore, it is first necessary to determine A , B , Γ , and Z values at the compressor inlet and outlet.

State 1 [$T_1 = 323$ K ($P^{\text{vap}} = 17.4$ bar), $P_1 = 5$ bar]

The parameter α is calculated from Eqn (3.1-16) as

$$\begin{aligned} \alpha_1 &= \left[1 + [0.37464 + (1.54226)(0.153) \right. \\ &\quad \left. - (0.26992)(0.153)^2] (1 - \sqrt{0.873}) \right]^2 = 1.081. \end{aligned}$$

The use of Eqn (3.2-20) gives

$$\Gamma_1 = [0.37464 + (1.54226)(0.153) - (0.26992)(0.153)^2] \sqrt{\frac{0.873}{1.081}} = 0.543.$$

The dimensionless parameters are calculated as

$$A_1 = 0.45724 \left(\frac{P_{r1}}{T_{r1}^2} \right) \alpha_1 = \frac{(0.45724)(0.118)(1.081)}{(0.873)^2} = 0.077,$$

$$B_1 = 0.07780 \left(\frac{P_{r1}}{T_{r1}} \right) = \frac{(0.07780)(0.118)}{0.873} = 0.011.$$

The cubic equation for Z, Eqn (3.1-18), takes the form

$$Z_1^3 - 0.989Z_1^2 + 0.055Z_1 - 7.247 \times 10^{-4} = 0 \Rightarrow Z_1 = 0.931.$$

State 2 [$T_2 = 343 \text{ K}$ ($P^{\text{vap}} = 25.8 \text{ bar}$), $P_2 = 15 \text{ bar}$]

The parameter α is calculated from Eqn (3.1-16) as

$$\alpha_2 = \left[1 + [0.37464 + (1.54226)(0.153) - (0.26992)(0.153)^2] \left(1 - \sqrt{0.927} \right) \right]^2 = 1.046.$$

The use of Eqn (3.2-20) gives

$$\Gamma_2 = [0.37464 + (1.54226)(0.153) - (0.26992)(0.153)^2] \sqrt{\frac{0.927}{1.046}} = 0.569.$$

The dimensionless parameters are calculated as

$$A_2 = 0.45724 \left(\frac{P_{r2}}{T_{r2}^2} \right) \alpha_2 = \frac{(0.45724)(0.353)(1.046)}{(0.927)^2} = 0.196,$$

$$B_2 = 0.07780 \left(\frac{P_{r2}}{T_{r2}} \right) = \frac{(0.07780)(0.353)}{0.927} = 0.030.$$

The cubic equation for Z, Eqn (3.1-18), takes the form

$$Z_2^3 - 0.97Z_2^2 + 0.133Z_2 - 4.953 \times 10^{-3} = 0 \Rightarrow Z_2 = 0.814.$$

Finally, substitution of the numerical values into Eqn (3.3-24) gives the change in enthalpy as

$$\Delta\tilde{H} = \tilde{W} = -1570 + 524 + 1612 = 566 \text{ J/mol.}$$

3.4 Calculation of the Change in Entropy

In the previous chapter, the general equations to calculate the change in entropy are given by Eqns (2.2-38) and (2.2-41), i.e.

$$d\tilde{S} = \frac{\tilde{C}_V}{T} dT + \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} d\tilde{V}, \quad (3.4-1)$$

$$d\tilde{S} = \frac{\tilde{C}_P}{T} dT - \left(\frac{\partial \tilde{V}}{\partial T} \right)_P dP. \quad (3.4-2)$$

These equations are valid for gases, liquids, and solids.

3.4.1 Change in Entropy for Liquids and Solids

In terms of the coefficient of thermal expansion, β , and isothermal compressibility, κ , Eqns (3.4-1) and (3.4-2) take the form

$$d\tilde{S} = \frac{\tilde{C}_V}{T} dT + \frac{\beta}{\kappa} d\tilde{V}, \quad (3.4-3)$$

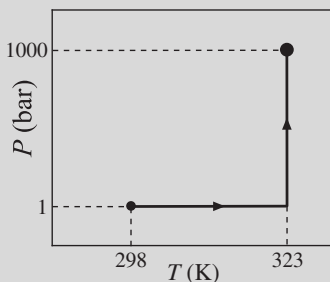
$$d\tilde{S} = \frac{\tilde{C}_P}{T} dT - \beta \tilde{V} dP. \quad (3.4-4)$$

Example 3.14 Determine the change in entropy when liquid water at 1 bar and 298 K is compressed to 1000 bar and 323 K. The following data for water are available:

T (K)	P (bar)	$\tilde{C}_P$ (J/mol K)	$\tilde{V}$ (cm ³ /mol)	β (K ⁻¹)
298	1	75.305	18.071	256×10^{-6}
298	1000		18.012	366×10^{-6}
323	1	75.314	18.234	458×10^{-6}
323	1000		18.174	568×10^{-6}

Solution

Since entropy is a state function, the path of integration is arbitrary. Let us consider the integration path as shown in the figure below:



Integration of Eqn (3.4-4) gives

$$\Delta \tilde{S} = \int_{T_1}^{T_2} \frac{\tilde{C}_P}{T} dT - \int_{P_1}^{P_2} \beta \tilde{V} dP. \quad (1)$$

Since $\tilde{C}_P$ is a weak function of temperature, it is possible to take the arithmetic average of $\tilde{C}_P$, $\langle \tilde{C}_P \rangle$, in going from 298 K to 323 K at a constant pressure of 1 bar. Similarly, it is possible to take the arithmetic averages of β and $\tilde{V}$ in going from 1 bar to 1000 bar at a constant temperature of 323 K. Thus, Eqn (1) reduces to

$$\Delta \tilde{S} = \langle \tilde{C}_P \rangle \ln \left(\frac{T_2}{T_1} \right) - \langle \beta \rangle \langle \tilde{V} \rangle (P_2 - P_1). \quad (2)$$

The average values are

$$\begin{aligned} \langle \tilde{C}_P \rangle &= \frac{75.305 + 75.314}{2} = 75.310 \text{ J/mol K}, \\ \langle \beta \rangle &= \left(\frac{458 + 568}{2} \right) \times 10^{-6} = 513 \times 10^{-6} \text{ K}^{-1}, \\ \langle \tilde{V} \rangle &= \frac{18.234 + 18.174}{2} = 18.204 \text{ cm}^3/\text{mol}. \end{aligned}$$

Substitution of the numerical values into Eqn (2) leads to

$$\begin{aligned} \Delta \tilde{S} &= (75.310) \ln \left(\frac{323}{298} \right) - (513 \times 10^{-6})(18.204 \times 10^{-6}) \\ &\quad \times [(1000 - 1) \times 10^5] = 5.13 \text{ J/mol K}. \end{aligned}$$

3.4.2 Change in Entropy for Gases

For gases, the change in entropy can be calculated from either Eqn (3.4-1) or Eqn (3.4-2) once the equation of state is known. To relate $\tilde{C}_V$ and $\tilde{C}_P$ values to $\tilde{C}_V^*$ and $\tilde{C}_P^*$, it is necessary to use Eqns (3.2-9) and (3.3-6), respectively. The result is

$$d\tilde{S} = \left[\frac{\tilde{C}_V^*}{T} + \int_{\infty}^{\tilde{V}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT + \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} d\tilde{V}, \quad (3.4-5)$$

$$d\tilde{S} = \left[\frac{\tilde{C}_P^*}{T} - \int_0^P \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT - \left(\frac{\partial \tilde{V}}{\partial T} \right)_P dP. \quad (3.4-6)$$

Integration of Eqn (3.4-5) over the paths shown in Fig. 3.4 leads to

Path A

$$\Delta\tilde{S} = \int_{\tilde{V}_1}^{\tilde{V}_2} \left[\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \left[\frac{\tilde{C}_V^*}{T} + \int_{\infty}^{\tilde{V}_2} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT. \quad (3.4-7)$$

Path B

$$\Delta\tilde{S} = \int_{T_1}^{T_2} \left[\frac{\tilde{C}_V^*}{T} + \int_{\infty}^{\tilde{V}_1} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} \right] dT + \int_{\tilde{V}_1}^{\tilde{V}_2} \left[\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_2} d\tilde{V}. \quad (3.4-8)$$

On the other hand, integration of Eqn (3.4-6) over the paths shown in Fig. 3.6 results in

Path A

$$\Delta\tilde{S} = - \int_{P_1}^{P_2} \left[\left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_1} dP + \int_{T_1}^{T_2} \left[\frac{\tilde{C}_P^*}{T} - \int_0^{P_2} \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT. \quad (3.4-9)$$

Path B

$$\Delta\tilde{S} = \int_{T_1}^{T_2} \left[\frac{\tilde{C}_P^*}{T} - \int_0^{P_1} \left(\frac{\partial^2 \tilde{V}}{\partial T^2} \right)_P dP \right] dT - \int_{P_1}^{P_2} \left[\left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_2} dP. \quad (3.4-10)$$

Example 3.15 A 0.7 m³ insulated, rigid tank contains ammonia at 350 K and 50 bar. A valve on the tank is opened, and the pressure inside quickly drops to 10 bar, at which point the valve is closed. Calculate the mass of ammonia vented to the atmosphere if the equation of state is given by

$$P(\tilde{V} - b) = RT,$$

where $b = 3.73 \times 10^{-2} \text{ m}^3/\text{kmol}$.

Solution

From Appendix B

$$\begin{aligned}\tilde{C}_P^* &= 34.236 - 0.221 \times 10^{-1}T + 1.213 \times 10^{-4}T^2 \\ &\quad - 1.088 \times 10^{-7}T^3 + 3.203 \times 10^{-11}T^4.\end{aligned}$$

The difference between the initial and final mass of ammonia gives the amount of ammonia vented to the atmosphere. For this purpose, it is first necessary to determine the final temperature in the tank.

System: Ammonia remaining in the tank when the pressure reaches 10 bar.

This is a closed system and if we assume that the system undergoes a reversible and adiabatic expansion then

$$\tilde{S}_1 = \tilde{S}_2 \Rightarrow \Delta\tilde{S} = 0.$$

The entropy change can be calculated from either Eqn (3.4-9) or Eqn (3.4-10). From the given equation of state, i.e.

$$\tilde{V} = \frac{RT}{P} + b$$

the following partial derivatives are obtained:

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = \frac{R}{P} \quad \text{and} \quad \left(\frac{\partial^2 \tilde{V}}{\partial T^2}\right)_P = 0.$$

Thus, Eqns (3.4-9) and (3.4-10) both reduce to

$$0 = \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - \int_{P_1}^{P_2} \frac{R}{P} dP.$$

Substitution of the numerical values results in

$$\begin{aligned}\int_{350}^{T_2} \left(\frac{34.236 - 0.221 \times 10^{-1}T + 1.213 \times 10^{-4}T^2 - 1.088 \times 10^{-7}T^3 + 3.203 \times 10^{-11}T^4}{T} \right) dT \\ - (8.314) \ln \left(\frac{10}{50} \right) = 0.\end{aligned}$$

The solution by MATHCAD[®] gives $T_2 = 240.6$ K. The initial and final molar volumes are calculated from the equation of state as

$$\begin{aligned}\tilde{V}_1 &= \frac{(8.314 \times 10^{-2})(350)}{50} + 3.73 \times 10^{-2} = 0.6193 \text{ m}^3/\text{kmol}, \\ \tilde{V}_2 &= \frac{(8.314 \times 10^{-2})(240.6)}{10} + 3.73 \times 10^{-2} = 2.0376 \text{ m}^3/\text{kmol}.\end{aligned}$$

The initial and final number of moles of ammonia in the tank are

$$\begin{aligned}n_1 &= \frac{V}{\tilde{V}_1} = \frac{0.7}{0.6193} = 1.13 \text{ kmol}, \\ n_2 &= \frac{V}{\tilde{V}_2} = \frac{0.7}{2.0376} = 0.34 \text{ kmol}.\end{aligned}$$

Therefore, the mass of ammonia vented to the atmosphere, m , is

$$\begin{aligned}m &= (n_1 - n_2)M \\ &= (1.13 - 0.34)(17) = 13.43 \text{ kg}\end{aligned}$$

The second-order derivatives in Eqns (3.4-7)–(3.4-10) may lead to mathematically complicated expressions to integrate. To eliminate such a problem, Eqn (3.4-1) can be integrated along the path shown in Fig. 3.5. The result is

$$\Delta\tilde{S} = \int_{\tilde{V}_1}^{\infty} \left[\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_1} d\tilde{V} + \int_{T_1}^{T_2} \frac{\tilde{C}_V^*}{T} dT + \int_{\infty}^{\tilde{V}_2} \left[\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right]_{T_2} d\tilde{V}. \quad (3.4-11)$$

On the other hand, integration of Eqn (3.4-2) along the path shown in Fig. 3.7 gives

$$\Delta\tilde{S} = - \int_{P_1}^0 \left[\left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_1} dP + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - \int_0^{P_2} \left[\left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right]_{T_2} dP. \quad (3.4-12)$$

Depending on the equation of state, equations for calculating the change in entropy are given below.

van der Waals equation of state: The use of the van der Waals equation of state in Eqn (3.4-11) leads to⁷

$$\Delta\tilde{S} = R \ln \left(\frac{Z_2 - B_2}{Z_1 - B_1} \right) + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right), \quad (3.4-13)$$

where the dimensionless parameter B is given in Table 3.2.

Redlich-Kwong equation of state: The use of the Redlich-Kwong equation of state in Eqn (3.4-11) leads to

$$\begin{aligned} \Delta\tilde{S} = R \ln \left(\frac{Z_2 - B_2}{Z_1 - B_1} \right) - \frac{R}{2} \left[\frac{A_2}{B_2} \ln \left(1 + \frac{B_2}{Z_2} \right) - \frac{A_1}{B_1} \ln \left(1 + \frac{B_1}{Z_1} \right) \right] \\ + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right), \end{aligned} \quad (3.4-14)$$

where the dimensionless parameters A and B are given in Table 3.2.

Peng-Robinson equation of state: The use of the Peng-Robinson equation of state in Eqn (3.4-11) leads to

$$\begin{aligned} \Delta\tilde{S} = R \ln \left(\frac{Z_2 - B_2}{Z_1 - B_1} \right) + \frac{(da/dT)_{T_2}}{\sqrt{8}b} \ln \left[\frac{Z_2 + (1 + \sqrt{2})B_2}{Z_2 + (1 - \sqrt{2})B_2} \right] \\ - \frac{(da/dT)_{T_1}}{\sqrt{8}b} \ln \left[\frac{Z_1 + (1 + \sqrt{2})B_1}{Z_1 + (1 - \sqrt{2})B_1} \right] + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right), \end{aligned} \quad (3.4-15)$$

where the derivative term da/dT is given by Eqn (3.2-19). Substitution of Eqn (3.2-19) into Eqn (3.4-15) and expressing the resulting equation in dimensionless form yield

$$\begin{aligned} \Delta\tilde{S} = R \ln \left(\frac{Z_2 - B_2}{Z_1 - B_1} \right) + \frac{RA_2\Gamma_2}{\sqrt{8}B_2} \ln \left[\frac{Z_2 + (1 + \sqrt{2})B_2}{Z_2 + (1 - \sqrt{2})B_2} \right] \\ + \frac{RA_1\Gamma_1}{\sqrt{8}B_1} \ln \left[\frac{Z_1 + (1 + \sqrt{2})B_1}{Z_1 + (1 - \sqrt{2})B_1} \right] + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right). \end{aligned} \quad (3.4-16)$$

The dimensionless parameters A and B are given in Table 3.2, and Γ is defined by Eqn (3.2-20).

Depending on the problem statement, one can use the following identity in Eqns (3.4-13), (3.4-14), and (3.4-16):

⁷See Problem 3.24.

$$\int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right) = \int_{T_1}^{T_2} \frac{\tilde{C}_V^*}{T} dT + R \ln \left(\frac{\tilde{V}_2}{\tilde{V}_1} \right). \quad (3.4-17)$$

Example 3.16 Three moles of isobutane at 40 bar and 450 K is contained in a piston-cylinder assembly. Calculate the work interaction with the surroundings using the Redlich-Kwong equation of state if

- (a) Isobutane expands reversibly and isothermally to 10 bar.
- (b) Isobutane expands reversibly and adiabatically to 10 bar.

Solution

From Appendix A

$$T_c = 407.7 \text{ K}, \quad P_c = 36.5 \text{ bar}, \quad \omega = 0.183.$$

From Appendix B

$$\begin{aligned} \tilde{C}_P^* &= 24.108 + 2.091 \times 10^{-1} T + 2.166 \times 10^{-4} T^2 \\ &\quad - 3.372 \times 10^{-7} T^3 + 11.619 \times 10^{-11} T^4. \end{aligned}$$

System: Isobutane within the piston-cylinder assembly.

- (a) The work done by isobutane during reversible expansion is calculated as

$$W = -n \int_{\tilde{V}_1}^{\tilde{V}_2} P d\tilde{V} = -n \int_{\tilde{V}_1}^{\tilde{V}_2} \left[\frac{RT}{\tilde{V} - b} - \frac{a}{\tilde{V}(\tilde{V} + b)\sqrt{T}} \right] d\tilde{V}. \quad (1)$$

Since the process is isothermal, i.e. temperature remains constant at 450 K throughout the process, integration of Eqn (1) is straightforward, with the result

$$W = n \left[RT \ln \left(\frac{\tilde{V}_1 - b}{\tilde{V}_2 - b} \right) + \frac{a}{b\sqrt{T}} \ln \left(1 + \frac{b}{\tilde{V}_1} \right) - \frac{a}{b\sqrt{T}} \ln \left(1 + \frac{b}{\tilde{V}_2} \right) \right]. \quad (2)$$

In terms of the dimensionless parameters, A and B , defined in Table 3.2, Eqn (2) takes the form

$$\begin{aligned} W &= nRT \left[\ln \left(\frac{Z_1 - B_1}{Z_2 - B_2} \right) + \ln \left(\frac{P_2}{P_1} \right) + \frac{A_1}{B_1} \ln \left(1 + \frac{B_1}{Z_1} \right) \right. \\ &\quad \left. - \frac{A_2}{B_2} \ln \left(1 + \frac{B_2}{Z_2} \right) \right]. \end{aligned} \quad (3)$$

Note that the work interaction can also be calculated indirectly from the first law as

$$W = n(\Delta\tilde{U} - \tilde{Q}) \quad (4)$$

or

$$W = n(\Delta\tilde{U} - T\Delta\tilde{S}). \quad (5)$$

Substitution of Eqns (3.2-17) and (3.4-14) into Eqn (5) also leads to Eqn (3).

State 1 ($T_1 = 450$ K, $P_1 = 40$ bar)

The reduced temperature and pressure are

$$T_{r1} = \frac{T_1}{T_c} = \frac{450}{407.7} = 1.104 \quad P_{r1} = \frac{P_1}{P_c} = \frac{40}{36.5} = 1.096.$$

The dimensionless parameters are calculated as

$$A_1 = 0.42748 \left(\frac{P_{r1}}{T_{r1}^{2.5}} \right) = \frac{(0.42748)(1.096)}{(1.104)^{2.5}} = 0.366,$$

$$B_1 = 0.08664 \left(\frac{P_{r1}}{T_{r1}} \right) = \frac{(0.08664)(1.096)}{1.104} = 0.086.$$

The cubic equation for Z, Eqn (3.1-18), takes the form

$$Z_1^3 - Z_1^2 + 0.273Z_1 - 0.031 = 0 \Rightarrow Z_1 = 0.656.$$

State 2 ($T_2 = 450$ K, $P_2 = 10$ bar)

The reduced temperature and pressure are

$$T_{r2} = \frac{T_2}{T_c} = \frac{450}{407.7} = 1.104 \quad P_{r2} = \frac{P_2}{P_c} = \frac{10}{36.5} = 0.274.$$

The dimensionless parameters are calculated as

$$A_2 = 0.42748 \left(\frac{P_{r2}}{T_{r2}^{2.5}} \right) = \frac{(0.42748)(0.274)}{(1.104)^{2.5}} = 0.091,$$

$$B_2 = 0.08664 \left(\frac{P_{r2}}{T_{r2}} \right) = \frac{(0.08664)(0.274)}{1.104} = 0.022.$$

The cubic equation for Z, Eqn (3.1-18), takes the form

$$Z_2^3 - Z_2^2 + 0.069Z_2 - 2 \times 10^{-3} = 0 \Rightarrow Z_2 = 0.928.$$

Substitution of the numerical values into Eqn (3) gives

$$W = (3)(8.314)(450) \left[\ln \left(\frac{0.656 - 0.086}{0.928 - 0.022} \right) + \ln \left(\frac{10}{40} \right) + \frac{0.366}{0.086} \right. \\ \left. \times \ln \left(1 + \frac{0.086}{0.656} \right) - \frac{0.091}{0.022} \ln \left(1 + \frac{0.022}{0.928} \right) \right] = -15,964 \text{ J}$$

(b) Since temperature does not remain constant in this case, integration of Eqn (1) can only be possible numerically. Hence, it is much more convenient to calculate work indirectly from the first law, i.e.

$$W = \Delta U - \underbrace{Q}_0. \quad (6)$$

To calculate the change in internal energy, it is first necessary to calculate the temperature at the final state. Since the process is reversible and adiabatic (or isentropic), the entropy remains constant, i.e.

$$\Delta \tilde{S} = 0 \quad (7)$$

or, from Eqn (3.4-14),

$$R \ln \left(\frac{Z_2 - B_2}{Z_1 - B_1} \right) - \frac{R}{2} \left[\frac{A_2}{B_2} \ln \left(1 + \frac{B_2}{Z_2} \right) - \frac{A_1}{B_1} \ln \left(1 + \frac{B_1}{Z_1} \right) \right] \\ + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right) = 0. \quad (8)$$

The final temperature T_2 should be determined so as to satisfy Eqn (8). The results of the trial-and-error solution are given below:

T_2 (K)	A_2	B_2	Z_2	$\Delta \tilde{S}$ (J/mol K)
425	0.106	0.023	0.913	9.282
400	0.123	0.024	0.894	1.312
390	0.131	0.025	0.885	-1.926
396	0.126	0.024	0.891	0.020

Thus, $T_2 \simeq 396$ K and the change in internal energy is calculated from Eqn (3.2-17) as

$$\Delta U = (3) \left\{ \frac{3(8.314)}{2} \left[\frac{(0.366)(450)}{0.086} \ln \left(1 + \frac{0.086}{0.656} \right) \right. \right. \\ \left. \left. - \frac{(0.126)(396)}{0.024} \ln \left(1 + \frac{0.024}{0.891} \right) \right] + \int_{450}^{396} \tilde{C}_V^* dT \right\}. \quad (9)$$

Noting that

$$\int_{450}^{396} \tilde{C}_V^* dT = \int_{450}^{396} (\tilde{C}_P^* - R) dT = -6544 \text{ J/mol}$$

the work interaction is calculated from Eqns (6) and (9) as

$$W = \Delta U = -12,873 \text{ J.}$$

3.5 The Principle of Corresponding States

According to the *principle (or law) of corresponding states*, the compressibility factor of fluids is dependent only on the reduced temperature and pressure (or volume), i.e.

$$Z = Z(T_r, P_r) \quad \text{or} \quad Z = Z(T_r, \tilde{V}_r). \quad (3.5-1)$$

In other words, fluids must have the same compressibility factor at the same reduced temperature and pressure (or volume). This generalization comes from the fact that the van der Waals equation of state can be expressed in the form given by Eqn (3.5-1) as will be shown in the following example.

Example 3.17 Show that the van der Waals equation of state can be expressed in the form given by Eqn (3.5-1).

Solution

The van der Waals equation of state is

$$P = \frac{RT}{\tilde{V} - b} - \frac{a}{\tilde{V}^2}. \quad (1)$$

The use of Eqn (3.1-17) gives

$$\left(\frac{\partial P}{\partial \tilde{V}} \right)_{T_c} = 0 \quad \Rightarrow \quad \frac{RT_c}{(\tilde{V}_c - b)^2} = \frac{2a}{\tilde{V}_c^3}, \quad (2)$$

$$\left(\frac{\partial^2 P}{\partial \tilde{V}^2} \right)_{T_c} = 0 \quad \Rightarrow \quad \frac{RT_c}{(\tilde{V}_c - b)^3} = \frac{3a}{\tilde{V}_c^4}. \quad (3)$$

Division of Eqn (2) by Eqn (3) yields

$$b = \frac{1}{3} \tilde{V}_c. \quad (4)$$

Substitution of Eqn (4) into Eqn (2) results in

$$a = \frac{9}{8} RT_c \tilde{V}_c. \quad (5)$$

The compressibility factor for the van der Waals equation of state is

$$Z = \frac{P\tilde{V}}{RT} = \frac{\tilde{V}}{\tilde{V} - b} - \frac{a}{\tilde{V}RT}. \quad (6)$$

Substitution of Eqns (4) and (5) into Eqn (6) leads to

$$Z = \frac{\tilde{V}_r}{\tilde{V}_r - \frac{1}{3}} - \frac{9}{8} \frac{1}{\tilde{V}_r T_r}. \quad (7)$$

Comment: At the critical point, $\tilde{V}_r = T_r = 1$. Thus, the critical compressibility factor, Z_c , is calculated from Eqn (7) as

$$Z_c = \frac{3}{8} = 0.375.$$

The critical compressibility factor, Z_c , is the value of Z at $P_r = T_r = 1$. If Eqn (3.5-1) were universally valid, all fluids would have to have the same Z_c value. In reality, however, Z_c values for real fluids range from 0.23 to 0.31. To make the principle of corresponding states applicable to all fluids, [Pitzer *et al.* \(1955\)](#) introduced a third parameter and modified Eqn (3.5-1) as

$$Z = Z(P_r, T_r, \omega), \quad (3.5-2)$$

where ω is the *acentric factor* defined by Eqn (3.1-14). It takes into account the geometry (deviation from spherical shape) and polarity of a molecule. For simple fluids, such as argon, krypton, and xenon, its value is zero. The functional form of Z is suggested in the form

$$Z(P_r, T_r, \omega) = Z^{(0)}(P_r, T_r) + \omega Z^{(1)}(P_r, T_r). \quad (3.5-3)$$

[Pitzer *et al.* \(1955\)](#) tabulated the values of $Z^{(0)}$ and $Z^{(1)}$ as functions of P_r and T_r .

These values were later improved by Lee and Kesler (1975) using the modified Benedict-Webb-Rubin equation of state as described in Appendix D.

Example 3.18 Solve Example 3.1 if ethylene obeys the principle of corresponding states.

Solution

From Example 3.1

$$T_r = 1.239, \quad P_r = 0.198, \quad \omega = 0.089.$$

The use of Eqn (D-2) in Appendix D gives

$$\tilde{V}_r^{(0)} = 6.0522 \quad \text{and} \quad \tilde{V}_r^{(R)} = 6.0776.$$

The values of $Z^{(0)}$ and $Z^{(R)}$ are calculated from Eqns (D-6) and (D-7), respectively, as

$$Z^{(0)} = 0.9654 \quad \text{and} \quad Z^{(R)} = 0.9695.$$

From Eqn (D-8)

$$Z^{(1)} = \frac{0.9695 - 0.9654}{0.3978} = 0.0103.$$

The compressibility factor is calculated from Eqn (3.5-3) as

$$Z = 0.9654 + (0.089)(0.0103) = 0.9663,$$

which is almost the same as the Z value obtained in Example 3.1. Therefore, $\tilde{V} = 2811 \text{ cm}^3/\text{mol}$.

3.6 Departure Functions

Internal energy, enthalpy, and entropy are all state functions. When a system goes from an initial state of (T_1, P_1) to a final state of (T_2, P_2) , calculations of ΔU , ΔH , and ΔS are independent of the path chosen. Such changes can be easily calculated if one chooses the path shown in Fig. 3.8.

For an ideal gas, the changes in enthalpy and entropy are given by

$$\Delta \tilde{H}^{\text{IG}} = \int_{T_1}^{T_2} \tilde{C}_P^* dT, \quad (3.6-1)$$

$$\Delta \tilde{S}^{\text{IG}} = \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right). \quad (3.6-2)$$

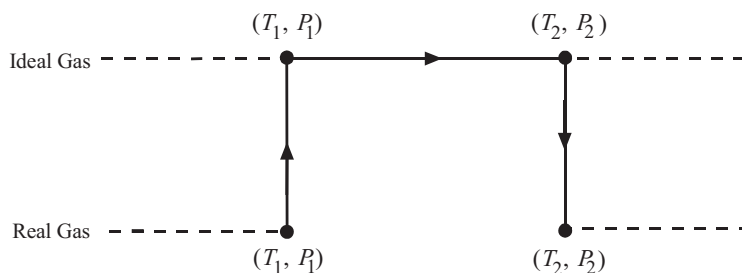


Figure 3.8: The path followed in the solution of thermodynamic problems.

Using the path shown in Fig. 3.8, the changes in enthalpy and entropy are given by

$$\Delta \tilde{H} = -(\tilde{H} - \tilde{H}^{\text{IG}})_{T_1, P_1} + \int_{T_1}^{T_2} \tilde{C}_P^* dT + (\tilde{H} - \tilde{H}^{\text{IG}})_{T_2, P_2}, \quad (3.6-3)$$

$$\Delta \tilde{S} = -(\tilde{S} - \tilde{S}^{\text{IG}})_{T_1, P_1} + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right) + (\tilde{S} - \tilde{S}^{\text{IG}})_{T_2, P_2}. \quad (3.6-4)$$

The term *departure function* is defined as the change in property in going from the real state to the ideal gas state or vice versa at constant temperature and pressure, i.e. $(\psi - \psi^{\text{IG}})_{T, P}$.

In a similar fashion, the change in internal energy is given by

$$\Delta \tilde{U} = -(\tilde{U} - \tilde{U}^{\text{IG}})_{T_1, P_1} + \int_{T_1}^{T_2} \tilde{C}_V^* dT + (\tilde{U} - \tilde{U}^{\text{IG}})_{T_2, P_2}. \quad (3.6-5)$$

The departure function for internal energy, $(\tilde{U} - \tilde{U}^{\text{IG}})_{T, P}$, is calculated indirectly from the departure function for enthalpy as follows. From the definition of enthalpy

$$\tilde{H} = \tilde{U} + P\tilde{V} = \tilde{U} + ZRT. \quad (3.6-6)$$

For an ideal gas, Eqn (3.6-6) reduces to

$$\tilde{H}^{\text{IG}} = \tilde{U}^{\text{IG}} + RT. \quad (3.6-7)$$

Subtraction of Eqn (3.6-7) from Eqn (3.6-6) gives

$$(\tilde{U} - \tilde{U}^{\text{IG}})_{T, P} = (\tilde{H} - \tilde{H}^{\text{IG}})_{T, P} + (1 - Z)RT. \quad (3.6-8)$$

3.6.1 Departure Functions from the Cubic Equations of State

Comparison of Eqn (3.6-3) with either Eqn (3.3-13) or Eqn (3.3-20) gives the departure function for enthalpy as

$$\begin{aligned} (\tilde{H} - \tilde{H}^{IG})_{T,P} &= \int_0^P \left[\tilde{V} - T \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP \\ &= \int_\infty^{\tilde{V}} \left[\tilde{V} \left(\frac{\partial P}{\partial \tilde{V}} \right)_T + T \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} \right] d\tilde{V}. \end{aligned} \quad (3.6-9)$$

On the other hand, comparison of Eqn (3.6-4) with Eqn (3.4-12) gives the departure function for entropy as

$$(\tilde{S} - \tilde{S}^{IG})_{T,P} = \int_0^P \left[\frac{R}{P} - \left(\frac{\partial \tilde{V}}{\partial T} \right)_P \right] dP. \quad (3.6-10)$$

The use of Eqns (3.3-16) and (3.3-18) in Eqn (3.6-10) also expresses the departure function for entropy in the form

$$(\tilde{S} - \tilde{S}^{IG})_{T,P} = \int_\infty^{\tilde{V}} \left[\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} - \frac{R}{\tilde{V}} \right] d\tilde{V} + R \ln Z. \quad (3.6-11)$$

Departure functions for various equations of state are given below.

van der Waals equation of state: The use of the van der Waals equation of state in Eqns (3.6-9) and (3.6-11) leads to

$$\frac{(\tilde{H} - \tilde{H}^{IG})_{T,P}}{RT} = Z - 1 - \frac{A}{Z}, \quad (3.6-12)$$

$$\frac{(\tilde{S} - \tilde{S}^{IG})_{T,P}}{R} = \ln(Z - B), \quad (3.6-13)$$

where the dimensionless parameters A and B are given in Table 3.2. The use of Eqn (3.6-12) in Eqn (3.6-8) gives

$$\frac{(\tilde{U} - \tilde{U}^{IG})_{T,P}}{RT} = -\frac{A}{Z}. \quad (3.6-14)$$

Redlich-Kwong equation of state: The use of the Redlich-Kwong equation of state in Eqns (3.6-9) and (3.6-11) leads to

$$\frac{(\tilde{H} - \tilde{H}^{IG})_{T,P}}{RT} = Z - 1 - \frac{3}{2} \frac{A}{B} \ln \left(1 + \frac{B}{Z} \right), \quad (3.6-15)$$

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}}{R} = \ln(Z - B) - \frac{1}{2} \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right), \quad (3.6-16)$$

where the dimensionless parameters A and B are given in Table 3.2. The use of Eqn (3.6-15) in Eqn (3.6-8) gives

$$\frac{(\tilde{U} - \tilde{U}^{\text{IG}})_{T,P}}{RT} = -\frac{3}{2} \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right). \quad (3.6-17)$$

Peng-Robinson equation of state: The use of the Peng-Robinson equation of state in Eqns (3.6-9) and (3.6-11) leads to

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}}{RT} = Z - 1 + \frac{1}{RT} \left[\frac{T (da/dT) - a}{\sqrt{8}b} \right] \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (3.6-18)$$

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}}{R} = \ln(Z - B) + \frac{1}{R} \left(\frac{da/dT}{\sqrt{8}b} \right) \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (3.6-19)$$

where the dimensionless parameters A and B are given in Table 3.2, and the derivative term da/dT is given by Eqn (3.2-19). Substitution of Eqn (3.6-18) into Eqn (3.6-8) gives

$$\frac{(\tilde{U} - \tilde{U}^{\text{IG}})_{T,P}}{RT} = \left[\frac{T (da/dT) - a}{\sqrt{8}bRT} \right] \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right]. \quad (3.6-20)$$

Equations (3.6-18)–(3.6-20) can be further simplified with the help of Eqn (3.2-19). The resulting equations are given as

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}}{RT} = Z - 1 - \frac{A(1 + \Gamma)}{\sqrt{8}B} \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (3.6-21)$$

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}}{R} = \ln(Z - B) - \frac{A\Gamma}{\sqrt{8}B} \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (3.6-22)$$

$$\frac{(\tilde{U} - \tilde{U}^{\text{IG}})_{T,P}}{RT} = -\frac{A(1 + \Gamma)}{\sqrt{8}B} \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (3.6-23)$$

where Γ is defined by Eqn (3.2-20).

Example 3.19 Isopentane is to be compressed from 3 bar and 500 K to 20 bar in a reversible isothermal compressor. Determine the work required per mole of isopentane as well as the heat interaction of the compressor with the surroundings. Assume that isopentane obeys the Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 461.0 \text{ K}, \quad P_c = 33.8 \text{ bar}, \quad \omega = 0.227.$$

System: Compressor.

The first law for a steady-state flow system is given by

$$\Delta \tilde{H} = \tilde{Q} + \tilde{W}_s. \quad (1)$$

For a reversible and isothermal process, the heat interaction can be calculated from the second law as

$$\tilde{Q} = T \Delta \tilde{S}. \quad (2)$$

The values of reduced temperature, reduced pressure, and α at the compressor inlet (state 1) and outlet (state 2) are calculated as

State	T_r	P_r	α
1	1.085	0.089	0.942
2	1.085	0.592	0.942

The values of A , B , and Z , and the departure functions at the compressor inlet and outlet are given in the following table:

State	P (bar)	A	B	Z	Γ	$\tilde{H} - \tilde{H}^{\text{IG}}$ (J/mol)	$\tilde{S} - \tilde{S}^{\text{IG}}$ (J/mol K)
1	3	0.033	6.382×10^{-3}	0.973	0.763	-359.2	-0.496
2	20	0.217	0.042	0.812	0.763	-2647	-3.787

The change in enthalpy is calculated from Eqn (3.6-3) as

$$\Delta \tilde{H} = -(\tilde{H} - \tilde{H}^{\text{IG}})_{T_1, P_1} + (\tilde{H} - \tilde{H}^{\text{IG}})_{T_2, P_2} = 359.2 - 2647 = -2287.8 \text{ J/mol}.$$

The change in entropy is calculated from Eqn (3.6-4) as

$$\begin{aligned} \Delta \tilde{S} &= -(\tilde{S} - \tilde{S}^{\text{IG}})_{T_1, P_1} - R \ln \left(\frac{P_2}{P_1} \right) + (\tilde{S} - \tilde{S}^{\text{IG}})_{T_2, P_2} \\ &= 0.496 - (8.314) \ln \left(\frac{20}{3} \right) - 3.787 = -19.064 \text{ J/mol K} \end{aligned}$$

The use of Eqn (2) gives the heat interaction as

$$\tilde{Q} = (500) (-19.064) = -9532 \text{ J/mol.}$$

The work done can be calculated from Eqn (1) as

$$\tilde{W}_s = \Delta\tilde{H} - \tilde{Q} = -2287.8 + 9532 = 7244.2 \text{ J/mol.}$$

Example 3.20 A piston-cylinder assembly initially contains 0.1 m^3 of ethylene at 393 K and 35 bar. The pressure of the gas is increased to 205 bar by a reversible isothermal compression. Calculate the heat and work interactions with the surroundings by assuming

- (a) Ethylene is an ideal gas,
- (b) Ethylene obeys the Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 282.5 \text{ K}, \quad P_c = 50.6 \text{ bar}, \quad \omega = 0.089$$

System: Ethylene in the piston-cylinder assembly.

- (a) Since the internal energy of an ideal gas depends only on temperature, $\Delta U = 0$ and the first law simplifies to

$$Q = -W. \quad (1)$$

The work done on the gas can be determined from

$$W = - \int_{V_1}^{V_2} P \, dV = - \int_{V_1}^{V_2} \frac{nRT}{V} \, dV = -nRT \ln \left(\frac{V_2}{V_1} \right) = -nRT \ln \left(\frac{P_1}{P_2} \right). \quad (2)$$

The number of moles of ethylene is

$$n = \frac{P_1 V_1}{RT_1} = \frac{(35)(0.1)}{(8.314 \times 10^{-2})(393)} = 0.107 \text{ kmol.}$$

Substitution of the values into Eqn (2) gives

$$W = -(0.107)(8.314)(393) \ln \left(\frac{35}{205} \right) = 618 \text{ kJ.}$$

From Eqn (1), $Q = -68$ kJ. (b) For a reversible isothermal process, the second law states that

$$Q = T \Delta S. \quad (3)$$

Therefore, once Q and ΔU are calculated, the work done on the gas is calculated from the first law, i.e.

$$W = \Delta U - Q. \quad (4)$$

Determination of internal energy and entropy values at the initial (state 1) and final (state 2) states requires reduced temperature, reduced pressure, and α values to be known:

State	T_r	P_r	α
1	1.391	0.692	0.825
2	1.391	4.051	0.825

The values of A , B , Z , and Γ , and the departure functions at the initial and final states are given in the table below:

State	P (bar)	A	B	Z	Γ	$\tilde{U} - \tilde{U}^{\text{IG}}$ (kJ/kmol)	$\tilde{S} - \tilde{S}^{\text{IG}}$ (kJ/kmol K)
1	35	0.135	0.039	0.908	0.662	-775	-1.953
2	205	0.790	0.227	0.740	0.662	-4611	-10.222

The number of moles of ethylene is

$$n = \frac{P_1 V_1}{RT_1 Z_1} = \frac{(35)(0.1)}{(8.314 \times 10^{-2})(393)(0.908)} = 0.118 \text{ kmol.}$$

Since the process is isothermal, i.e. $T_1 = T_2 = T$, Eqn (3.6-5) gives

$$\Delta \tilde{U} = -(\tilde{U} - \tilde{U}^{\text{IG}})_{T,P_1} + (\tilde{U} - \tilde{U}^{\text{IG}})_{T,P_2} = 775 - 4611 = -3836 \text{ kJ/kmol.}$$

Thus, the total change in internal energy is

$$\Delta U = n\Delta\tilde{U} = (0.118)(-3836) = -452.6 \text{ kJ.}$$

The change in entropy is calculated from Eqn (3.6-4) as

$$\begin{aligned}\Delta\tilde{S} &= -(\tilde{S} - \tilde{S}^{\text{IG}})_{T_1, P_1} - R \ln \left(\frac{P_2}{P_1} \right) + (\tilde{S} - \tilde{S}^{\text{IG}})_{T_2, P_2} \\ &= 1.953 - (8.314) \ln \left(\frac{205}{35} \right) - 10.222 = -22.965 \text{ kJ/kmol K.}\end{aligned}$$

The amount of heat exchange is

$$Q = T\Delta S = (393) [(0.118)(-22.965)] = -1065 \text{ kJ.}$$

The work done on the gas can be calculated from Eqn (4) as

$$W = -452.6 + 1065 = 612.4 \text{ kJ.}$$

3.6.2 Departure Functions from the Principle of Corresponding States

Departure functions for enthalpy and entropy are given as

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}}{RT_c} = \frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}^{(0)}}{RT_c} + \omega \frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}^{(1)}}{RT_c}, \quad (3.6-24)$$

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T, P}}{R} = \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T, P}^{(0)}}{R} + \omega \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T, P}^{(1)}}{R}. \quad (3.6-25)$$

Appendix D gives the equations developed by Lee and Kesler (1975) to calculate the terms on the right-hand sides of Eqns (3.6-24) and (3.6-25).

Example 3.21 Solve Example 3.13 if propane obeys the principle of corresponding states.

Solution

Note that

$$\tilde{W}_s = \Delta\tilde{H}. \quad (1)$$

From Appendix A

$$T_c = 369.9 \text{ K}, \quad P_c = 42.5 \text{ bar}, \quad \omega = 0.153.$$

From the equations given in Appendix D, the enthalpy departures are tabulated below. Therefore, the use of Eqn (3.6-24) gives

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T_1, P_1}}{RT_c} = -0.159 + (0.153)(-0.194) = -0.189,$$

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T_2, P_2}}{RT_c} = -0.471 + (0.153)(-0.527) = -0.552.$$

Parameters	State 1	State 2
	($T_r = 0.873$, $P_r = 0.118$)	($T_r = 0.927$, $P_r = 0.353$)
$\tilde{V}_r^{(0)}$	6.9658	2.1836
$Z^{(0)}$	0.9385	0.8311
$\tilde{V}_r^{(R)}$	6.8739	2.1142
$Z^{(R)}$	0.9261	0.8047
$Z^{(1)}$	-0.0311	-0.0664
$(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}^{(0)} / RT_c$	-0.159	-0.471
$(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}^{(R)} / RT_c$	-0.236	-0.681
$(\tilde{H} - \tilde{H}^{\text{IG}})_{T, P}^{(1)} / RT_c$	-0.194	-0.527

The change in enthalpy can be calculated from Eqn (3.6-3). From Example 3.13

$$\int_{323}^{343} \tilde{C}_p^* dT = 1612 \text{ J/mol.}$$

Thus,

$$\Delta \tilde{H} = (0.189)(8.314)(369.9) + 1612$$

$$-(0.552)(8.314)(369.9) = 495.7 \text{ J/mol.}$$

From Eqn (1), the work done on the system is 495.7 J/mol.

3.7 Which Equation of State to Use?

In the low density region, i.e. when either pressure is very low or temperature is very high, gas behavior can be described by the ideal gas equation of state. Outside of this region, one might think of how to choose the “best” equation of state among various alternatives. In other words, similar to the queen in Snow White who asked, “Mirror, mirror on the wall, who’s the fairest of them all?” (Martin, 1979).

The truncated form of the virial equation of state, Eqn (3.1-9), can be used only for the vapor phase for pressures up to approximately 15 bar.

For computational purposes, an equation of state representing both vapor and liquid phases would be ideal. In that respect, cubic equations of state have the advantage. However, no cubic equation of state can provide the *PVT* behavior in all regions of interest. For example, they usually fail near the critical region.

Chapter 4 of the book “*The Properties of Gases and Liquids*” (Poling *et al.* 2004) provides the guidelines for selecting an equation of state for *PVT* properties of pure substances.

Problems

Problems Related to Section 3.1

- 3.1** Calculate the second virial coefficient for *n*-butane at 375 K. The experimental value is $-426.5 \text{ cm}^3/\text{mol}$ (Gupta and Eubank, 1997). Also estimate the density of *n*-butane at 375 K and 5 bar.

(Answer: $-430.8 \text{ cm}^3/\text{mol}$, 0.01 g/cm^3)

- 3.2** Various correlations are available in the literature to calculate the second virial coefficient for nonpolar fluids besides Eqn (3.1-10). For example, Meng *et al.* (2004) modified the well-known Tsonopoulos (1979) correlation as

$$B = \frac{RT_c}{P_c} \left[f^{(0)} + \omega f^{(1)} \right], \quad (1)$$

where

$$f^{(0)} = 0.13356 - \frac{0.30252}{T_r^3} - \frac{0.15668}{T_r^2} - \frac{0.00724}{T_r^3} - \frac{0.00022}{T_r^8}, \quad (2)$$

$$f^{(1)} = 0.17404 - \frac{0.15581}{T_r} + \frac{0.38183}{T_r^2} - \frac{0.44044}{T_r^3} - \frac{0.00541}{T_r^8}. \quad (3)$$

Calculate the second virial coefficients for methane, ethane, and carbon dioxide at 300 K using Eqn (1).

(Answer: $-42.098 \text{ cm}^3/\text{mol}$, $-181.483 \text{ cm}^3/\text{mol}$, $-121.869 \text{ cm}^3/\text{mol}$)

3.3 Saturated isobutane vapor at 320 K is contained in a piston-cylinder assembly. It is expanded to 3 bar by a reversible isothermal process. At 320 K, the vapor pressure of isobutane is 6.243 bar. Using the virial equation of state, Eqn (3.1-9),

- Estimate the molar volume of isobutane at the initial and final states.
- Determine the work done during the expansion process.

(Answer: (a) $3704 \text{ cm}^3/\text{mol}$, $8310.7 \text{ cm}^3/\text{mol}$ and (b) -1949.7 J/mol)

3.4 If a gas obeys the virial equation of state, Eqn (3.1-9), use Eqn (1) of Problem 2.1 and show that the Joule-Thomson coefficient is given by

$$\mu = \frac{T \left(\frac{dB}{dT} \right) - B}{\tilde{C}_P^* - T \left(\frac{d^2B}{dT^2} \right) P} \quad (1)$$

Substitute Eqn (3.1-10) into Eqn (1) to obtain

$$\mu = \frac{\frac{T_c}{P_c} \left[-0.083 + \frac{1.0972}{T_r^{1.6}} + \omega \left(-0.139 + \frac{0.8944}{T_r^{4.2}} \right) \right]}{\frac{\tilde{C}_P^*}{R} + P_r \left(\frac{1.75552}{T_r^{2.6}} + \omega \frac{3.75648}{T_r^{5.2}} \right)}.$$

3.5 Analytical calculation of the roots of a quadratic equation

$$x^3 + px^2 + qx + r = 0 \quad (1)$$

is as follows.

Let the terms M and N be defined as

$$M = \frac{3q - p^2}{9} \quad \text{and} \quad N = \frac{9pq - 27r - 2p^3}{54}. \quad (2)$$

If p , q , and r are real and if $\Delta = M^3 + N^2$ is the discriminant, then

- $\Delta > 0$; one root is real and two are complex conjugate,
- $\Delta = 0$; all roots are real and at least two are equal,

- $\Delta < 0$; all roots are real and unequal.

Case (i) Solutions for $\Delta > 0$

In this case the roots are given by

$$x_1 = S + T - \frac{1}{3}p, \quad (3)$$

$$x_2 = -\frac{1}{2}(S + T) - \frac{1}{3}p + \frac{1}{2}i\sqrt{3}(S - T), \quad (4)$$

$$x_3 = -\frac{1}{2}(S + T) - \frac{1}{3}p - \frac{1}{2}i\sqrt{3}(S - T), \quad (5)$$

where

$$S = \sqrt[3]{N + \sqrt{\Delta}} \quad \text{and} \quad T = \sqrt[3]{N - \sqrt{\Delta}}. \quad (6)$$

Case (ii) Solutions for $\Delta < 0$

The roots are given by

$$x_i = \pm 2\sqrt{-M} \cos \left[\frac{\theta}{3} + (i - 1)120^\circ \right] - \frac{1}{3}p \quad i = 1, 2, 3, \quad (7)$$

where

$$\theta = \arccos \sqrt{\frac{N^2}{(-M)^3}} \quad (\theta \text{ is in degrees}). \quad (8)$$

In Eqn (7) the upper sign applies if N is positive, while the lower sign applies if N is negative.

If methanol obeys the van der Waals equation of state, predict the density of saturated methanol vapor at 11 bar and 413 K analytically.

(Answer: 0.01106 g/cm³)

3.6 Estimate the molar volumes of saturated liquid and vapor for n -octane at 293 K using:

- Redlich-Kwong equation of state.
- Soave-Redlich-Kwong equation of state.
- Peng-Robinson equation of state.

At 293 K, the vapor pressure of n -octane is 1.387 bar.

(Answer: (a) 196.9 cm³/mol, 15,250 cm³/mol, (b) 189.6 cm³/mol, 14,560 cm³/mol, and (c) 168.6 cm³/mol, 14,540 cm³/mol)

- 3.7** The maximum pressure, P , that a spherical tank can withstand is dependent on the tank radius, R , the wall thickness, t_w , and the maximum tensile strength of the material, σ . The equation proposed by [Strelzhoﬀ and Pan \(1968\)](#) states that

$$P = \begin{cases} \frac{2t_w\sigma}{R + 0.2t_w} & \text{if } P < 0.665\sigma, \\ \frac{2\sigma \left(\frac{t_w}{R} + 1 \right)^2 - 2\sigma}{\left(\frac{t_w}{R} + 1 \right)^2 + 2} & \text{if } P > 0.665\sigma, \end{cases}$$

where P is the internal gauge pressure.

Estimate the maximum amount of ethylene that can be stored in a spherical tank at 300 K. The tank is made of stainless steel 316, has a radius of 1.5 m, and has a wall thickness of 13 mm. Ethylene is represented by the Peng-Robinson equation of state. For stainless steel 316, $\sigma = 586$ MPa.

(Answer: 4476 kg)

- 3.8** A rigid tank contains propane at 350 K and an unknown pressure. When the tank is cooled to 303 K, the vapor starts condensing. Estimate the initial pressure in the tank using the Peng-Robinson equation of state. At 303 K, the vapor pressure of propane is 10.84 bar.

(Answer: 13.234 bar)

- 3.9** A 0.8 m^3 rigid tank contains 2 kg of acetylene at 200 K. Determine the state and pressure of acetylene in the tank using the Peng-Robinson equation of state. The vapor pressure of acetylene at 200 K is 1.905 bar.

(Answer: 1.544 bar)

- 3.10** A 0.05 m^3 piston-cylinder device contains 7 mol of isobutane at 350 K. It is compressed isothermally until the volume becomes 0.01 m^3 . The vapor pressure of isobutane is given as a function of temperature as

$$\ln P^{\text{vap}} = -9.416723 \ln T - \frac{4577.132}{T} + 73.51351 + 1.514279 \times 10^{-5} T^2,$$

where P^{vap} is in kPa and T is in K. Estimate the initial and final pressures using the Peng-Robinson equation of state.

(Answer: 3.812 bar, 12.469 bar)

- 3.11** Suppose that a rigid cylinder contains propylene at 380 K and 150 bar. The contents of the cylinder are cooled to 285 K and part of the liquid that

condensed at the bottom of the cylinder is removed. If the cylinder temperature is again increased to 380 K, will the pressure be greater than, equal to, or less than 150 bar? Explain clearly.

(Answer: Less than 150 bar)

- 3.12** A piston-cylinder assembly contains 3 kg of acetone with 60% quality. The system is heated at constant pressure until all the acetone vaporizes. Calculate the work done if the temperature at the final state is 400 K. Assume acetone is represented by the Peng-Robinson equation of state. The vapor pressure of acetone is given as a function of temperature as

$$\ln P^{\text{vap}} = 10.0311 - \frac{2940.46}{T - 35.93},$$

where P^{vap} is in bar and T is in K.

(Answer: $-58,598$ J)

- 3.13** From Fig. 3.3a, conclude that the isotherm satisfies the following equation in the two-phase region:

$$P^{\text{vap}}(\tilde{V}^{\text{V}} - \tilde{V}^{\text{L}}) = \int_{\tilde{V}^{\text{L}}}^{\tilde{V}^{\text{V}}} P \, d\tilde{V}. \quad (1)$$

- (a) If a substance is represented by the Redlich-Kwong equation of state, show that Eqn (1) takes the form

$$P^{\text{vap}}(\tilde{V}^{\text{V}} - \tilde{V}^{\text{L}}) = RT \ln \left(\frac{\tilde{V}^{\text{V}} - b}{\tilde{V}^{\text{L}} - b} \right) - \frac{a}{b\sqrt{T}} \ln \left[\frac{\tilde{V}^{\text{V}}(\tilde{V}^{\text{L}} + b)}{\tilde{V}^{\text{L}}(\tilde{V}^{\text{V}} + b)} \right]. \quad (2)$$

- (b) Express Eqn (2) in terms of dimensionless quantities to obtain

$$Z^{\text{V}} - Z^{\text{L}} - \ln \left(\frac{Z^{\text{V}} - B}{Z^{\text{L}} - B} \right) + \frac{A}{B} \ln \left[\frac{Z^{\text{V}}(Z^{\text{L}} + B)}{Z^{\text{L}}(Z^{\text{V}} + B)} \right] = 0. \quad (3)$$

Note that Eqn (3) can be used to estimate vapor pressures of pure substances at any given temperature.

- (c) Using a trial-and-error procedure, estimate the vapor pressure of propane at 350 K if it obeys the Redlich-Kwong equation of state.

(Answer: (c) 30.9 bar)

Problems Related to Section 3.2

- 3.14** A rigid container is completely filled with 2 kg of mercury at 293 K and 1 bar and sealed. Estimate the amount of heat transferred so as to increase the pressure to 100 bar. The following data are provided for mercury:

$$\rho = 13.6 \text{ g/cm}^3, \quad \widehat{C}_P = 0.139 \text{ J/g K},$$

$$\beta = 1.82 \times 10^{-4} \text{ K}^{-1}, \quad \kappa = 3.95 \times 10^{-6} \text{ bar}^{-1}.$$

Hint: In this case $\widehat{C}_V \neq \widehat{C}_P$.

(Answer: 507.9 J)

- 3.15** A piston-cylinder assembly contains 5 mol of carbon monoxide at 300 K. The gas is compressed from 3 L to 1 L by a reversible isothermal process. Calculate the heat removed from the system if carbon monoxide is represented by the van der Waals equation of state.

(Answer: −15,620 J)

- 3.16** One mole of carbon dioxide undergoes the following changes in a series of nonflow processes: (i) From an initial state of 300 K and 4 bar (state 1), it is compressed to 415 K and 15 bar (state 2); (ii) It is then cooled to 325 K at a constant pressure of 15 bar (state 3). For the overall process, the work done on the gas is 320 J. Estimate the heat interaction with the surroundings assuming carbon dioxide obeys the Redlich-Kwong equation of state.

(Answer: 184 J)

- 3.17** A vertical cylinder fitted with a frictionless piston contains 1 mol of acetylene at 330 K and 20 bar. The piston-cylinder assembly is well insulated and a mechanical lock prevents the piston from rising. The piston has a mass of 500 kg and a cross-sectional area of 0.02 m². The ambient pressure is 1 bar. The mechanical lock is now released. If acetylene obeys the Peng-Robinson equation of state, estimate the gas temperature when the system reaches equilibrium. Take $g = 10 \text{ m/s}^2$.

(Answer: 267.8 K)

- 3.18** Consider a closed system undergoing a reversible adiabatic expansion from state 1 to state 2.

(a) Starting with

$$\left(\frac{\partial T}{\partial \widetilde{V}} \right)_{\widetilde{S}} \left(\frac{\partial \widetilde{V}}{\partial \widetilde{S}} \right)_T \left(\frac{\partial \widetilde{S}}{\partial T} \right)_{\widetilde{V}} = -1 \quad (1)$$

show that

$$\left(\frac{\partial T}{\partial \tilde{V}}\right)_{\tilde{s}} = -\frac{T}{\tilde{C}_V} \left(\frac{\partial P}{\partial T}\right)_{\tilde{V}}. \quad (2)$$

- (b) If an ideal gas with constant heat capacity undergoes an isentropic process, show that Eqn (2) reduces to

$$\frac{T_2}{T_1} = \left(\frac{\tilde{V}_1}{\tilde{V}_2}\right)^{\gamma-1}, \quad (3)$$

where $\gamma = \tilde{C}_P^*/\tilde{C}_V^*$.

Problems Related to Section 3.3

- 3.19** A gas at T_1 and P_1 is throttled to a pressure P_2 . It is represented by the virial equation of state, Eqn (3.1-9), with the parameter B dependent on temperature in the form

$$B = \alpha + \beta T,$$

where α and β are known constants. If $\tilde{C}_P^*$ is assumed constant, show that the outlet temperature is given by

$$T_2 = T_1 + \frac{\alpha (P_1 - P_2)}{\tilde{C}_P^*}.$$

- 3.20** If a gas is represented by the Peng-Robinson equation of state, use Eqn (3.3-10) and show that

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = \frac{R}{P} \left\{ \frac{\frac{1}{Z-B} + \frac{A\Gamma}{Z(Z+B) + B(Z-B)}}{\frac{1}{(Z-B)^2} - \frac{2A(Z+B)}{[Z(Z+B) + B(Z-B)]^2}} \right\},$$

where Γ is defined by Eqn (3.2-20).

- 3.21** Calculate the change in molar enthalpy when isobutane gas is compressed from 350 K and 10 bar to 450 K and 90 bar in a steady-flow compressor. Use the Peng-Robinson equation of state.

(Answer: 2431 J/mol)

- 3.22** It is proposed to preheat a gas from 303 K to 398 K in a heat exchanger. The pressure of the gas at the inlet is 35 bar and the pressure drop in the heat exchanger may be considered negligible. Estimate the amount of heat required in J/mol using the following data

$$Z = 1 + \frac{aP}{\sqrt{T}} \quad \text{with } a = -0.02 \text{ K}^{1/2}/\text{bar},$$

$$\tilde{C}_P^* = 11.7 + 0.03T,$$

where $\tilde{C}_P^*$ is in J/mol K, and T is in K.

(Answer: 2103 J/mol)

- 3.23** A vertical cylinder fitted with a frictionless piston contains 0.15 m^3 of nitrogen at 350 K. The piston has a mass of 250 kg and a cross-sectional area of 0.1 m^2 . The piston and cylinder do not conduct heat, but heat can be added to the gas by a heating coil. The ambient pressure is 1 bar and the acceleration of gravity is 10 m/s^2 . Assume that nitrogen is described by the Redlich-Kwong equation of state with $\tilde{C}_V^* = 2.5R$. If 2 kJ of energy is supplied to nitrogen through a heating coil, estimate the final temperature and pressure for the following two cases:

- (a) Stops placed at the initial equilibrium position of the piston prevent it from rising.
- (b) Piston moves freely within the cylinder.

(Answer: (a) 365 K, 1.32 bar and (b) 360.6 K, 1.25 bar)

Problems Related to Section 3.4

- 3.24** Using the van der Waals equation of state show that

$$\left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} = \frac{R}{\tilde{V} - b}. \quad (1)$$

Substitute Eqn (1) and $\tilde{C}_V^* = \tilde{C}_P^* - R$ into Eqn (3.4-11) to obtain

$$\Delta \tilde{S} = R \ln \left[\frac{T_1(\tilde{V}_2 - b)}{T_2(\tilde{V}_1 - b)} \right] + \int_{T_1}^{T_2} \frac{\tilde{C}_P^*}{T} dT. \quad (2)$$

Show that the substitution of

$$\tilde{V} = \frac{ZRT}{P} \quad \text{and} \quad b = \frac{BRT}{P} \quad (3)$$

into Eqn (2) leads to Eqn (3.4-13).

- 3.25** A piston-cylinder assembly initially contains 5 mol of *n*-pentane at 500 K and 3 bar. The gas is compressed isothermally to 30 bar with a work input of 10 kJ. During this process, heat transfer takes place with the surroundings at a temperature of 298 K. Is this process reversible, irreversible, or impossible? Assume that *n*-pentane obeys the Peng-Robinson equation of state.

(Answer: Impossible)

- 3.26** Steam enters an adiabatic turbine at 70 bar and 873 K and leaves at a pressure of 4 bar. Estimate the maximum amount of work that can be delivered by the turbine if steam obeys the Peng-Robinson equation of state.

(Answer: −15, 190 J/mol)

- 3.27** A chemical plant is in need of ethylene at 325 K and 15 bar. Since the available ethylene is at 480 K and 50 bar, two schemes have been suggested to decrease the temperature and pressure of ethylene to the desired values.

In Scheme I, ethylene is first throttled from 50 bar to 15 bar and then sent to the heat exchanger to decrease its temperature to 325 K. In Scheme II, ethylene is first expanded in a reversible and adiabatic turbine to 15 bar and then sent to the heat exchanger. If ethylene obeys the Peng-Robinson equation of state, determine the heat transfer and work per unit mass of ethylene in each of the suggested schemes.

(Answer: Scheme I: $\hat{W} = 0$, $\hat{Q} = -279.6$ J/g; Scheme II: $\hat{W} = -148.2$ J/g, $\hat{Q} = -131.4$ J/g)

Problem Related to Section 3.5

- 3.28** Estimate the volume of a cylinder holding 150 mol of methane at 300 K and 40 bar using the principle of corresponding states.

(Answer: 0.0876 m³)

Problems Related to Section 3.6

- 3.29** If a gas is represented by the virial equation of state, show that the enthalpy and entropy departures are given by

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}}{RT} = P_r \left[\frac{0.083}{T_r} - \frac{1.0972}{T_r^{2.6}} + \omega \left(\frac{0.139}{T_r} - \frac{0.8944}{T_r^{5.2}} \right) \right], \quad (1)$$

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}}{R} = -P_r \left(\frac{0.6752}{T_r^{2.6}} + \omega \frac{0.7224}{T_r^{5.2}} \right). \quad (2)$$

Ammonia at 350 K and 15 bar enters a throttling valve and leaves it at 1 bar. Estimate the exit temperature using the virial equation of state.

(Answer: 331.13 K)

- 3.30** Calculate the departure functions for internal energy, enthalpy, and entropy for propane at 400 K and 80 bar using

- (a) van der Waals equation of state.
- (b) Redlich-Kwong equation of state.
- (c) Peng-Robinson equation of state.

(Answer: (a) -5164 J/mol, -7035 J/mol, -12.597 J/mol K, (b) -6956 J/mol, -8941 J/mol, -17.245 J/mol K, and (c) -7645 J/mol, -9673 J/mol, -18.861 J/mol K)

- 3.31** A gas is represented by the following equation of state:

$$Z = 1 + \left(a - \frac{b}{T} \right) \frac{P}{RT},$$

where $a = 15 \times 10^{-6} \text{ m}^3/\text{mol}$ and $b = 0.06 \text{ m}^3 \text{ K}/\text{mol}$.

- (a) Show that the departure functions for enthalpy and entropy take the form

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}}{RT} = A - 2B \quad \text{and} \quad \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}}{R} = -B,$$

where the dimensionless parameters A and B are defined by

$$A = \frac{aP}{RT} \quad \text{and} \quad B = \frac{bP}{RT^2}.$$

- (b) The gas at 573 K and 10 bar expands reversibly and adiabatically in a steady-flow turbine to 1 bar. Calculate the work output of the turbine. Take $\tilde{C}_P^* = 3.5R$.

(Answer: (b) -7916.7 J/mol)

- 3.32** A well-insulated cylindrical tank of volume 0.1 m^3 initially contains oxygen at 50 bar and 300 K. A relief valve is opened and oxygen is allowed to escape at a constant flow rate of 8 mol/min. If oxygen is represented by the Peng-Robinson equation of state, estimate the temperature and pressure of oxygen in the tank after 10 min.

Hint: Assume that the oxygen remaining in the tank undergoes a reversible and adiabatic expansion. (Why?)

(Answer: 247.6 K, 24.5 bar)

- 3.33** A steady-flow turbine compresses ethane from 300 K and 1 bar to 350 K and 50 bar. Use Eqns (3.6-3) and (3.6-4) to determine the changes in enthalpy and entropy, respectively, if ethane obeys the Peng-Robinson equation of state.

(Answer: 101.7 J/mol, -29.415 J/mol K)

- 3.34** Use Eqn (3.2-9) and show that the departure function for heat capacity at constant volume is given by

$$(\tilde{C}_V - \tilde{C}_V^*)_{T,P} = T \int_{\infty}^{\tilde{V}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V}. \quad (1)$$

Use Eqn (1) in Eqn (2.2-49) and show that the departure function for heat capacity at constant pressure is given by

$$(\tilde{C}_P - \tilde{C}_P^*)_{T,P} = T \int_{\infty}^{\tilde{V}} \left(\frac{\partial^2 P}{\partial T^2} \right)_{\tilde{V}} d\tilde{V} - T \frac{(\partial P / \partial T)_{\tilde{V}}^2}{(\partial P / \partial \tilde{V})_T} - R. \quad (2)$$

- 3.35** The departure function for molar heat capacity at constant volume is defined by

$$(\tilde{C}_V - \tilde{C}_V^*)_{T,P} = \frac{\partial}{\partial T} \left[(\tilde{U} - \tilde{U}^{\text{IG}})_{T,P} \right]_{\tilde{V}}. \quad (1)$$

- (a) If a gas is represented by the Peng-Robinson equation of state, show that the use of Eqn (3.6-20) in Eqn (1) leads to

$$\frac{(\tilde{C}_V - \tilde{C}_V^*)_{T,P}}{R} = \frac{A\Gamma(1 + \Gamma)}{2\sqrt{8}B} \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right], \quad (2)$$

where Γ is defined by Eqn (3.2-20).

(b) Combine Eqn (2) with Eqns (2.2-51) and (3.2-12) to obtain

$$\begin{aligned} \frac{(\tilde{C}_P - \tilde{C}_P^*)_{T,P}}{R} = & \frac{A\Gamma(1 + \Gamma)}{2\sqrt{8}B} \ln \left[\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right] \\ & + \frac{\left[\frac{1}{Z - B} + \frac{A\Gamma}{Z(Z + B) + B(Z - B)} \right]^2}{\frac{1}{(Z - B)^2} - \frac{2A(Z + B)}{[Z(Z + B) + B(Z - B)]^2}} - 1. \quad (3) \end{aligned}$$

(c) Calculate the departure functions for molar heat capacities at constant volume and pressure for ethylene gas at 300 K and 150 bar if it obeys the Peng-Robinson equation of state.

(Answer: (c) 6.470 J/mol K, 49.506 J/mol K)

3.36 In a chemical refinery, a certain process requires 12 mol/s of process steam at 2 bar with not less than 97% quality and not more than 15 K superheat. It is proposed to throttle the available steam at 1173 K and 20 bar through an adiabatic valve and then cool it to the required conditions for the process steam. Estimate the minimum duty of heat exchanger using the principle of corresponding states. The vapor pressure of steam is given as a function of temperature as

$$\ln P^{\text{vap}} = 11.6834 - \frac{3816.44}{T - 46.13},$$

where P^{vap} is in bar and T is in K.

(Answer: 394,296 W)

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Chapter 4

Equilibrium and Phase Stability in One-Component Systems

The purpose of this chapter is to provide answers to the following questions: (i) under what conditions are the two phases of a pure substance in equilibrium with each other? and (ii) at a given temperature and pressure, in which form does a pure substance exist—solid, liquid, or gas?

4.1 Equilibrium Criteria for Closed Systems

For a closed system, the second law of thermodynamics, Eqn (1.6-8), states that

$$T \, dS - \delta Q \geq 0. \quad (4.1-1)$$

From the first law of thermodynamics

$$dU = \delta Q - P \, dV. \quad (4.1-2)$$

Elimination of δQ between Eqns (4.1-1) and (4.1-2) leads to

$$T \, dS - dU - P \, dV \geq 0. \quad (4.1-3)$$

4.1.1 Condition of Maximum Entropy

Under the conditions of constant internal energy and volume, Eqn (4.1-3) simplifies to

$$dS_{U,V} \geq 0, \quad (4.1-4)$$

where the subscripts U and V imply that these quantities are kept constant during a process. Under these conditions, any spontaneous process tends to increase the

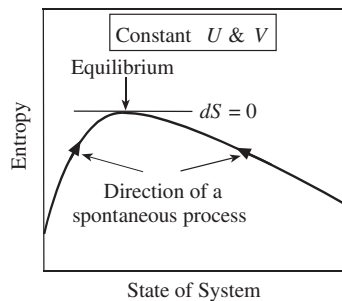


Figure 4.1: The equilibrium condition for constant internal energy and volume.

entropy of a system as shown in Fig. 4.1. When the system is at equilibrium, entropy reaches its maximum value and this condition is mathematically expressed as

$$dS_{U,V} = 0 \quad \text{At equilibrium.} \quad (4.1-5)$$

4.1.2 Condition of Minimum Internal Energy

When entropy and volume are kept constant, Eqn (4.1-3) simplifies to

$$dU_{S,V} \leq 0. \quad (4.1-6)$$

Therefore, the internal energy of a system decreases as a result of a spontaneous process under the conditions of constant entropy and volume. This phenomenon can be explained as follows. Spontaneous processes lead to an increase in entropy. In order to keep entropy constant, heat must be transferred to the surroundings with a concomitant decrease in internal energy. When the system is at equilibrium, internal energy reaches its minimum value and this condition is mathematically expressed as

$$dU_{S,V} = 0 \quad \text{At equilibrium.} \quad (4.1-7)$$

4.1.3 Condition of Minimum Helmholtz Energy

In differential form, Helmholtz energy is given by

$$dA = d(U - TS) = dU - S dT - T dS. \quad (4.1-8)$$

The use of Eqn (4.1-8) in Eqn (4.1-3) gives

$$dA + S dT + P dV \leq 0. \quad (4.1-9)$$

When temperature and volume are kept constant, Eqn (4.1-9) simplifies to

$$dA_{T,V} \leq 0. \quad (4.1-10)$$

Therefore, the Helmholtz energy of a system decreases as a result of a spontaneous process under the conditions of constant temperature and volume. When the system is at equilibrium, Helmholtz energy reaches its minimum value and this condition is mathematically expressed as

$$dA_{T,V} = 0 \quad \text{At equilibrium.} \quad (4.1-11)$$

4.1.4 Condition of Minimum Gibbs Energy

In differential form, Gibbs energy is given by

$$dG = d(H - TS) = d(U + PV - TS) = dU + P dV + V dP - T dS - S dT. \quad (4.1-12)$$

The use of Eqn (4.1-12) in Eqn (4.1-3) gives

$$dG + S dT - V dP \leq 0. \quad (4.1-13)$$

When temperature and pressure are kept constant, Eqn (4.1-13) simplifies to

$$dG_{T,P} \leq 0. \quad (4.1-14)$$

Under these conditions, any spontaneous process tends to decrease the Gibbs energy of a system as shown in Fig. 4.2. When the system is at equilibrium, Gibbs energy reaches its minimum value and this condition is mathematically expressed as

$$dG_{T,P} = 0 \quad \text{At equilibrium.} \quad (4.1-15)$$

Since Gibbs energy describes equilibrium at conditions of constant temperature and pressure, it is the most commonly used thermodynamic function in chemical engineering applications.

4.2 Equilibrium Criteria for Open Systems

To develop the conditions for equilibrium, it is first necessary to develop the combined first and second laws of thermodynamics for an open system. The first law of thermodynamics for an open system, Eqn (1.5-3), is given by

$$\tilde{H}_{in} dn_{in} - \tilde{H}_{out} dn_{out} + \delta Q + \delta W = d(n\tilde{U})_{sys} \quad (4.2-1)$$

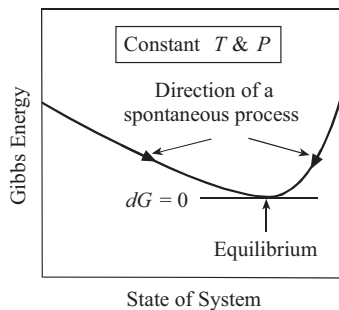


Figure 4.2: The equilibrium condition for constant temperature and pressure.

in which changes in kinetic and potential energies are considered negligible. On the other hand, the entropy balance, Eqn (1.6-6), is written as

$$\delta S_{\text{gen}} = d(n\tilde{S})_{\text{sys}} + \tilde{S}_{\text{out}} dn_{\text{out}} - \tilde{S}_{\text{in}} dn_{\text{in}} - \frac{\delta Q}{T_{\text{sys}}}. \quad (4.2-2)$$

To combine the first and the second law expressions, we have to restrict our attention to processes that are reversible, i.e. $\delta S_{\text{gen}} = 0$. Mass can be added and withdrawn reversibly only if the properties of the entering and leaving mass are identical to those of the system, i.e.

$$T_{\text{in}} = T_{\text{out}} = T_{\text{sys}}, \quad P_{\text{in}} = P_{\text{out}} = P_{\text{sys}}, \quad \tilde{H}_{\text{in}} = \tilde{H}_{\text{out}} = \tilde{H}_{\text{sys}}, \quad \tilde{S}_{\text{in}} = \tilde{S}_{\text{out}} = \tilde{S}_{\text{sys}}. \quad (4.2-3)$$

Under these conditions, Eqns (4.2-1) and (4.2-2) become

$$\tilde{H}_{\text{sys}}(dn_{\text{in}} - dn_{\text{out}}) + \delta Q + \delta W = dU_{\text{sys}}, \quad (4.2-4)$$

$$0 = dS_{\text{sys}} + \tilde{S}_{\text{sys}}(dn_{\text{out}} - dn_{\text{in}}) - \frac{\delta Q}{T_{\text{sys}}}. \quad (4.2-5)$$

For a one-component system under consideration, the mass balance is expressed in the form

$$dn_{\text{in}} - dn_{\text{out}} = dn_{\text{sys}}. \quad (4.2-6)$$

Elimination of δQ between Eqns (4.2-4) and (4.2-5) and the use of Eqn (4.2-6) lead to

$$dU_{\text{sys}} = T_{\text{sys}} dS_{\text{sys}} + \delta W + (\tilde{H}_{\text{sys}} - T_{\text{sys}}\tilde{S}_{\text{sys}})dn_{\text{sys}}. \quad (4.2-7)$$

If the only work is that of the expansion type of work, i.e. $-P_{\text{sys}} dV_{\text{sys}}$, Eqn (4.2-7) takes the form

$$dU = T dS - P dV + (\tilde{H} - T\tilde{S})dn, \quad (4.2-8)$$

where the subscript “sys” is dropped with the understanding that all properties belong to the system. The coefficient of dn is nothing more than the Gibbs energy per mole, $\tilde{G}$, for a pure substance.¹ Thus, Eqn (4.2-8) is expressed as

$$dU = T dS - P dV + \tilde{G} dn. \quad (4.2-9)$$

Consider an isolated system containing α - and β -phases of a pure substance. Note that although the overall system is an isolated one, each phase may be considered an open system, i.e. there may be an exchange of mass and/or energy between α - and β -phases. Since the system is isolated, then the total values of U , V , and n are all constant, i.e.

$$\begin{aligned} U = U_\alpha + U_\beta = \text{constant} &\Rightarrow dU_\alpha = -dU_\beta, \\ V = V_\alpha + V_\beta = \text{constant} &\Rightarrow dV_\alpha = -dV_\beta, \\ n = n_\alpha + n_\beta = \text{constant} &\Rightarrow dn_\alpha = -dn_\beta. \end{aligned} \quad (4.2-10)$$

The equilibrium criterion for an isolated system is given by Eqn (4.1-5), i.e.

$$dS = dS_\alpha + dS_\beta = 0. \quad (4.2-11)$$

The total differential of entropy can be obtained from Eqn (4.2-9) as

$$dS = \frac{1}{T} dU + \frac{P}{T} dV - \frac{\tilde{G}}{T} dn, \quad (4.2-12)$$

so that

$$dS_\alpha = \frac{1}{T_\alpha} dU_\alpha + \frac{P_\alpha}{T_\alpha} dV_\alpha - \frac{\tilde{G}_\alpha}{T_\alpha} dn_\alpha, \quad (4.2-13)$$

$$dS_\beta = \frac{1}{T_\beta} dU_\beta + \frac{P_\beta}{T_\beta} dV_\beta - \frac{\tilde{G}_\beta}{T_\beta} dn_\beta. \quad (4.2-14)$$

Substitution of Eqns (4.2-13) and (4.2-14) into Eqn (4.2-11) and making use of the relationships given by Eqn (4.2-10) lead to

$$dS = \left(\frac{1}{T_\alpha} - \frac{1}{T_\beta} \right) dU_\alpha + \left(\frac{P_\alpha}{T_\alpha} - \frac{P_\beta}{T_\beta} \right) dV_\alpha - \left(\frac{\tilde{G}_\alpha}{T_\alpha} - \frac{\tilde{G}_\beta}{T_\beta} \right) dn_\alpha = 0, \quad (4.2-15)$$

which can only be satisfied if the terms in parentheses in Eqn (4.2-15) are all equal to zero. Therefore, the following conditions hold at equilibrium:

$$\begin{aligned} T_\alpha &= T_\beta && \text{Thermal equilibrium} \\ P_\alpha &= P_\beta && \text{Mechanical equilibrium} \\ \tilde{G}_\alpha &= \tilde{G}_\beta && \text{Chemical equilibrium.} \end{aligned} \quad (4.2-16)$$

¹Historically, for a single component system, Gibbs energy per mole has been called the *chemical potential* and designated by μ , i.e. $\tilde{G} = \mu$.

Note that a temperature gradient implies the performance of a thermal work, i.e. heat engine or heat pump, and a pressure gradient implies performance of mechanical work, i.e. expansion/contraction work. The gradient of Gibbs energy, on the other hand, implies performance of a chemical work.

4.3 Phase Stability

Why does water exist as a liquid when $0 < T < 100^\circ\text{C}$ under atmospheric conditions at sea level? Why does CO_2 exist as a gas under room conditions? To answer such questions one should realize that any spontaneous process taking place at constant temperature and pressure leads to a decrease in the Gibbs energy of a system. The Gibbs energy reaches its minimum value when the system is at equilibrium. In other words, at a given temperature and pressure, a pure substance is *stable* in a phase in which it has the minimum Gibbs energy.²

In differential form, Gibbs energy of a pure substance is expressed as

$$d\tilde{G} = \tilde{V} dP - \tilde{S} dT. \quad (4.3-1)$$

Therefore, variation of Gibbs energy with pressure and temperature is essential for understanding phase stability.

4.3.1 Change of Gibbs Energy with Pressure

Variation of Gibbs energy with pressure at constant temperature is

$$\left(\frac{\partial \tilde{G}}{\partial P} \right)_T = \tilde{V}. \quad (4.3-2)$$

Since molar volume is always greater than zero, Eqn (4.3-2) implies that Gibbs energy increases with increasing pressure under isothermal conditions.

For an ideal gas, integration of Eqn (4.3-2) at constant temperature gives

$$\Delta \tilde{G}^{\text{IG}} = \int_{P_1}^{P_2} \frac{RT}{P} dP = RT \ln \left(\frac{P_2}{P_1} \right). \quad (4.3-3)$$

²Since $G = H - TS$, the value of G is dependent on the terms “ H ” and “ TS .” At high temperatures the term “ TS ” dominates and, as a result, the phase with the highest value of entropy is the most stable. At low temperatures, the term “ H ” dominates and the phase with the lowest value of enthalpy is the most stable.

Thus, Gibbs energy of an ideal gas increases exponentially with increasing pressure.

The molar volume of liquids and solids is almost independent of pressure. In this case, integration of Eqn (4.3-2) at constant temperature leads to

$$\Delta \tilde{G}^C = \tilde{V}^C \Delta P, \quad (4.3-4)$$

where the superscript “C” indicates the condensed phase, i.e. either liquid or solid. Equation (4.3-4) indicates a linear variation of Gibbs energy with pressure, with a slope being equal to $\tilde{V}^C$.

If m is the slope of $\tilde{G}$ versus P curve at any given temperature, then the above discussion implies that

$$m^V \gg m^L > m^S. \quad (4.3-5)$$

While m^L and m^S are almost constant, m^V changes exponentially with pressure. Figure 4.3a indicates the variation of $\tilde{G}$ as a function of pressure for gas, liquid, and solid phases.

As shown in Fig. 4.3b, when $P < P^{\text{vap}}$, the gas phase has the lowest Gibbs energy and the substance exists as a gas since it is most stable in this phase. At $P = P^{\text{vap}}$, molar Gibbs energies of the gas and liquid phases are equal to each other or, in other words, the gas and liquid phases are in equilibrium. At this point the temperature and pressure are called the saturation temperature (or boiling temperature) and saturation pressure (or vapor pressure), respectively. When $P^{\text{vap}} < P < P_m$, the liquid phase has the minimum Gibbs energy and the substance exists as a liquid. At $P = P_m$, molar Gibbs energies of the liquid and solid phases are equal to each

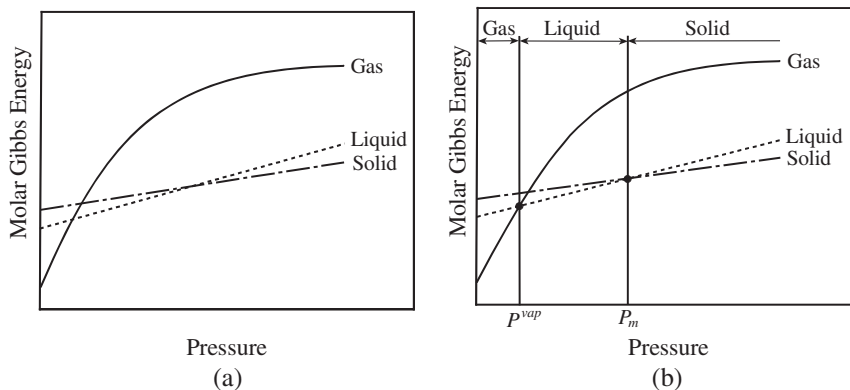


Figure 4.3: Variation of $\tilde{G}$ as a function of P at constant T and stable equilibrium states.

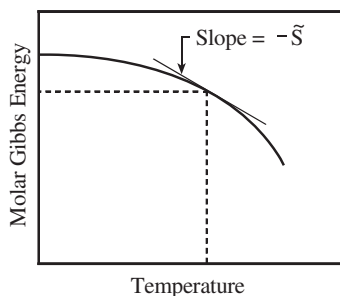


Figure 4.4: Variation of $\tilde{G}$ as a function of T at constant P .

other or, in other words, the liquid and solid phases are in equilibrium. At this point the temperature and pressure are called the melting (or freezing) temperature and melting pressure, respectively. When $P > P_m$, the solid phase is the most stable phase since it has the lowest Gibbs energy.

4.3.2 Change of Gibbs Energy with Temperature

Variation of Gibbs energy with temperature at constant pressure is given by

$$\left(\frac{\partial \tilde{G}}{\partial T} \right)_P = -\tilde{S}. \quad (4.3-6)$$

Since the entropy is always positive, then the slope of $\tilde{G}$ versus T is negative. Moreover, since $\tilde{S}$ increases with temperature for all substances, then $\tilde{G}$ is a concave³ function of T as shown in Fig. 4.4.

Variation of Gibbs energies of different phases as a function of temperature at constant pressure is shown in Fig. 4.5a. In plotting such a graph one should remember that $\tilde{S}^V > \tilde{S}^L > \tilde{S}^S$. As shown in Fig. 4.5b, solid, having the lowest Gibbs energy, is the most stable phase when $T < T_m$. At $T = T_m$ (melting or freezing temperature), solid is in equilibrium with liquid and their molar Gibbs energies are equal to each other. When $T_m < T < T_b$, the liquid phase is the stable phase. At $T = T_b$ (boiling or saturation temperature), the liquid and gas phases are in equilibrium. The gas phase becomes the most stable phase for $T > T_b$.

³A function $f(x)$ is *concave* if the chord joining any two points lies entirely below the curve.

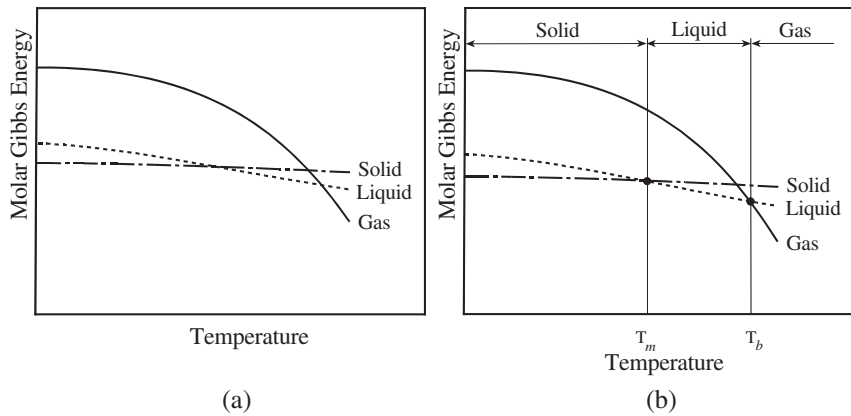


Figure 4.5: Variation of $\tilde{G}$ for different phases as a function of T at constant P and stable equilibrium states.

Example 4.1 The following values are for saturated steam at 508.15 K. Show that the liquid and vapor states of water are in equilibrium.

T (K)	P^{vap} (MPa)	$\hat{H}^L$ (kJ/kg)	$\hat{H}^V$ (kJ/kg)	$\hat{S}^L$ (kJ/kg K)	$\hat{S}^V$ (kJ/kg K)
508.15	3.06	1013.62	2804.2	2.6558	6.1791

Solution

When liquid and vapor phases of water are in equilibrium, then the number of degrees of freedom is calculated from Eqn (1.1-4) as

$$\mathcal{F} = \mathcal{C} + 2 - \mathcal{P} = 1 + 2 - 2 = 1.$$

Therefore, either temperature or pressure has to be specified to fix the state of the system. For saturated steam at 508.15 K, the corresponding vapor pressure (or saturation pressure) is 3.06 MPa. Under the condition of equilibrium, Gibbs energies of liquid and vapor must be equal to each other:

$$\begin{aligned}\hat{G}^L &= \hat{H}^L - T\hat{S}^L = 1013.62 - (508.15)(2.6558) = -335.9248 \text{ kJ/kg}, \\ \hat{G}^V &= \hat{H}^V - T\hat{S}^V = 2804.2 - (508.15)(6.1791) = -335.7097 \text{ kJ/kg}.\end{aligned}$$

Problems

Problem Related to Section 4.1

4.1 When the gravitational force is appreciable, Eqn (5) of Problem 2.11 indicates that the change in Gibbs energy is expressed as

$$dG = V dP - S dT + mg dz. \quad (1)$$

- (a) Since Gibbs energy reaches its minimum value under the condition of equilibrium, i.e. $dG = 0$, conclude that

$$dT = 0 \Rightarrow T \text{ is uniform} \quad (2)$$

and

$$V dP + mg dz = 0 \Rightarrow \frac{dP}{dz} = -\rho g, \quad (3)$$

which is the hydrostatic pressure distribution. Keep in mind that the distance z is measured in the direction opposite to gravity.

- (b) Since density, ρ , is constant for liquids, show that the integration of Eqn (3) gives

$$P = -\rho gz + C, \quad (4)$$

where C is an integration constant.

- (c) Density changes as a function of temperature and pressure in the case of gases. Assuming ideal gas behavior, i.e.

$$\rho = \frac{PM}{RT}, \quad (5)$$

show that the integration of Eqn (3) leads to

$$\ln P = -\frac{Mgz}{RT} + C, \quad (6)$$

where M is the molecular weight and C is an integration constant.

- (d) Once the pressure in an oil reservoir decreases to low values, oil flow to the well practically stops. *Enhanced oil recovery* (EOR) is a technique used to recover oil that is left in the reservoir. One of the EOR techniques is to inject high-pressure carbon dioxide into the reservoir. When CO_2 is mixed with oil, the resulting liquid has a lower viscosity, i.e. thinner, and becomes less sticky. This process facilitates the flow of oil from the rock to the well. Suppose that CO_2 is injected into a well 4 km deep at a pressure of 140 bar. If the temperature within the well is uniform at 310 K, estimate the pressure at the bottom of the well.

(Answer: (d) 273.4 bar)

Problems Related to Section 4.2

4.2 Complete the following table without using the steam table:

T (°C)	$\hat{H}^L$ (kJ/kg)	$\hat{H}^V$ (kJ/kg)	$\hat{S}^L$ (kJ/kg K)	$\hat{S}^V$ (kJ/kg K)
–	567.69	2727.3	1.6870	6.9777
205	875.04	–	2.3780	6.3952
340	1594.2	2622.0	–	5.3357

Use T (K) = T (°C) + 273.15.

(Answer: $T = 135^\circ\text{C}$, $\hat{H}^V = 2795.9$ kJ/kg, $\hat{S}^L = 3.6594$ kJ/kg K).

4.3 When the gravitational force is appreciable, the combined first and second laws of thermodynamics for a closed system lead to

$$dU = T dS - P dV + mg dz \quad (1)$$

as shown in Problem 2.11.

(a) In the case of an open system, show that the combined first and second laws of thermodynamics give

$$dU = T dS - P dV + mg dz + (\tilde{G} + Mgz)dn, \quad (2)$$

where m and M represent the mass and the molecular weight, respectively.

(b) Show that the change in Gibbs energy is given by

$$dG = V dP - S dT + mg dz + (\tilde{G} + Mgz)dn. \quad (3)$$

(c) Under equilibrium conditions show that

$$dT = 0, \quad (4)$$

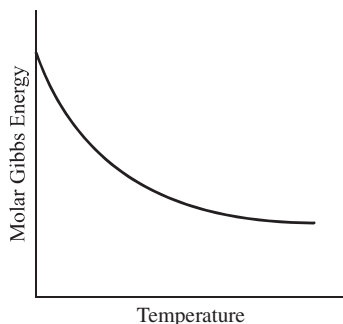
$$V dP + mg dz = 0, \quad (5)$$

$$\tilde{G} + Mgz = 0. \quad (6)$$

(d) Differentiate Eqn (6) under constant temperature and conclude that Eqns (5) and (6) both lead to hydrostatic pressure distribution as given by Eqn (3) in Problem 4.1.

Problems Related to Section 4.3

- 4.4** A function $f(x)$ is *convex* if the chord joining any two points lies entirely above the curve. Explain why Gibbs energy cannot be a convex function of temperature at constant pressure. In other words, show that the following plot of $\tilde{G}$ versus T at constant pressure is impossible.



- 4.5** An element X exists in nature in two different solid forms, X_α and X_β . The standard Gibbs energy of formations of X_α and X_β at 298 K and atmospheric pressure are 510 J/mol and 485 J/mol, respectively.
- Which form of X is the more stable at 298 K? Explain.
 - The molar volumes of X_α and X_β are 25 cm³/mol and 30 cm³/mol, respectively. Keeping the temperature constant at 298 K, does the increase in pressure make X_α more stable than X_β ? If so, at what pressure will the transition occur?
 - The standard molar entropies of X_α and X_β are 23.6 J/mol K and 34.8 J/mol K, respectively. Under atmospheric pressure, does the increase in temperature make X_α more stable than X_β ? If so, at what temperature will the transition occur?

(Answer: (a) X_β , (b) 51 bar and (c) No)

- 4.6** Tin exists in two solid forms, white and gray tin. While white tin has a body-centered tetragonal structure, the structure of gray tin is diamond cubic. The enthalpy and entropy changes associated with the transformation of gray tin to white tin at 298 K and atmospheric pressure are 2090 J/mol and 7.3 J/mol K, respectively.

- (a) Which form of tin is the more stable at 298 K?
 (b) Under atmospheric pressure, estimate the temperature at which white and gray tin coexist in equilibrium. Indicate your assumptions.

(Answer: (b) 286.3 K)

- 4.7 Phase transitions of SiO_2 under high pressure are of interest to earth scientists. Atake et al. (2000) reported that the enthalpy and entropy changes during α -quartz to coesite (polymorph of quartz) transition at 298 K and atmospheric pressure are 3400 J/mol and -2.16 J/mol K, respectively. The molar volume of α -quartz is $22.688 \times 10^{-6} \text{ m}^3/\text{mol}$ and the molar volume of coesite is $20.6 \times 10^{-6} \text{ m}^3/\text{mol}$. At 298 K, what is the pressure at which these two forms of SiO_2 will be in equilibrium?

(Answer: 1.94 GPa)

- 4.8 The vapor pressure of water as a function of temperature is given by

$$\ln P_{\text{H}_2\text{O}}^{\text{vap}} = 13.8365 - \frac{6.6075 \times 10^3}{T} + \frac{10.1855 \times 10^5}{T^2} - \frac{17.7039 \times 10^7}{T^3},$$

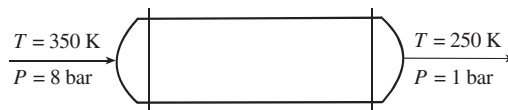
$$300 \leq T \leq 600,$$

where $P_{\text{H}_2\text{O}}^{\text{vap}}$ is in bar and T is in K.

- (a) Estimate the minimum pressure at which water exists as a liquid at 350 K.
 (b) Estimate the maximum pressure at which water exists as a vapor at 500 K.

(Answer: (a) 0.425 bar and (b) 26.56 bar)

- 4.9 A pure substance X flows through a heat exchanger under steady conditions as shown in the figure below. The inlet and outlet streams are single-phase and the ideal gas heat capacity is $\tilde{C}_p^* = 11.5R$.



Estimate the amount of heat added to or removed from substance X using the following information provided from the solutions of the Peng-Robinson equation of state at the inlet and outlet conditions:

	Z	$\left(\tilde{H} - \tilde{H}^{\text{IG}}\right)_{T,P} / RT$	$\left(\tilde{S} - \tilde{S}^{\text{IG}}\right)_{T,P} / R$
Inlet	0.0334	−5.868	−6.088
	0.0895	−3.107	−3.465
	0.8573	−0.410	−0.275
Outlet	0.0043	−10.451	−9.988
	0.0334	−2.673	−4.033
	0.9588	−0.106	−0.066

(Answer: −30,090 J/mol)

Reference

Atake, T., I. Noriko, H. Kawaji, K. Matsuzaka and M. Akaogi, 2000, *J. Chem. Thermodyn.*, **32**, 217-227.

Chapter 5

Fugacity of a Pure Component

This chapter introduces a new concept called fugacity, which is the basis for establishing equilibrium criteria between different phases of a pure substance. After showing how to calculate this quantity for gases, liquids, and solids, the variation of temperature with pressure (or vice versa) during phase transformation of pure substances will be discussed.

5.1 Molar Gibbs Energy of a Pure Ideal Gas

For a pure substance, molar Gibbs energy is given in differential form as

$$d\tilde{G} = \tilde{V} dP - \tilde{S} dT. \quad (5.1-1)$$

The ideal gas equation of state relates the molar volume to pressure as

$$\tilde{V}^{\text{IG}} = \frac{RT}{P}. \quad (5.1-2)$$

Substitution of Eqn (5.1-2) into Eqn (5.1-1) leads to

$$d\tilde{G}^{\text{IG}} = \frac{RT}{P} dP - \tilde{S} dT = RT d \ln P - \tilde{S} dT. \quad (5.1-3)$$

At constant temperature, Eqn (5.1-3) reduces to

$$d\tilde{G}^{\text{IG}} = RT d \ln P \quad \text{at constant } T. \quad (5.1-4)$$

Integration of Eqn (5.1-4) at constant temperature from the standard state pressure, P° , to the state of interest at which the pressure is P gives

$$\ln \left(\frac{P}{P^\circ} \right) = \frac{\tilde{G}^{\text{IG}}(T, P) - \tilde{G}^{\text{IG}}(T, P^\circ)}{RT}. \quad (5.1-5)$$

5.2 Definition of Fugacity and Fugacity Coefficient

Equation (5.1-5) is valid only for pure, ideal gases. In order to generalize it, in 1901 Gilbert Newton Lewis¹ introduced a function f_i , called the *fugacity of pure component i* , in terms of the departure function for molar Gibbs energy as

$$\ln \left[\frac{f_i(T, P)}{P} \right] = \frac{(\tilde{G}_i - \tilde{G}_i^{\text{IG}})_{T, P}}{RT} = \frac{(\tilde{H}_i - \tilde{H}_i^{\text{IG}})_{T, P}}{RT} - \frac{(\tilde{S}_i - \tilde{S}_i^{\text{IG}})_{T, P}}{R} \quad (5.2-1)$$

subject to the following constraint:

$$f_i/P \rightarrow 1 \quad \text{as } P \rightarrow 0 \text{ (ideal gas state),} \quad (5.2-2)$$

which applies to solid, liquid, and gas.

The *fugacity coefficient of pure component i* , ϕ_i , is defined as the ratio of the fugacity to the pressure, i.e.

$$\phi_i = \frac{f_i}{P}. \quad (5.2-3)$$

It is a measure of the deviation of the system from ideal gas behavior, i.e. $\phi_i = 1$ for an ideal gas. Note that both fugacity and the fugacity coefficient are intensive properties of the system and are a function of any two other intensive properties, such as temperature and pressure, for a single-phase pure material. Fugacity coefficient may be greater or less than unity as shown in Fig. 5.1.

Addition of Eqns (5.1-5) and (5.2-1) leads to an alternative expression for the fugacity of pure component i in the form

$$\tilde{G}_i(T, P) = \tilde{G}_i^{\text{IG}}(T, P^o) - RT \ln P^o + RT \ln f_i(T, P) \quad (5.2-4)$$

or

$$\tilde{G}_i(T, P) = \lambda_i(T) + RT \ln f_i(T, P), \quad (5.2-5)$$

where $\lambda_i(T)$ is the molar Gibbs energy of pure component i at unit fugacity, and is defined by

$$\lambda_i(T) = \tilde{G}_i^{\text{IG}}(T, P^o) - RT \ln P^o. \quad (5.2-6)$$

Differentiation of Eqn (5.2-5) at constant temperature gives

$$d\tilde{G}_i = RT d \ln f_i \quad \text{at constant } T. \quad (5.2-7)$$

When two phases of a pure substance are in equilibrium, we have

$$T_\alpha = T_\beta, \quad P_\alpha = P_\beta, \quad \tilde{G}_i^\alpha = \tilde{G}_i^\beta. \quad (5.2-8)$$

¹Professor of physical chemistry and the dean of the College of Chemistry at University of California, Berkeley.

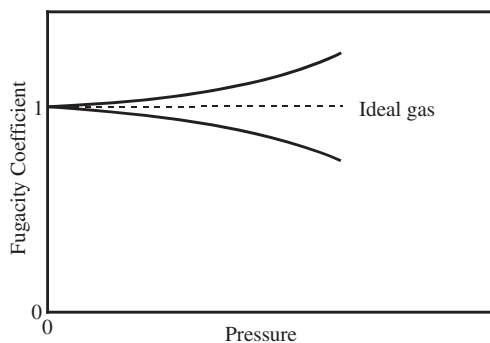


Figure 5.1: Variation of the fugacity coefficient with pressure.

Letting $T_\alpha = T_\beta = T$ and $P_\alpha = P_\beta = P$, equality of molar Gibbs energies leads to

$$\lambda_i(T) + RT \ln f_i^\alpha(T, P) = \lambda_i(T) + RT \ln f_i^\beta(T, P) \Rightarrow f_i^\alpha(T, P) = f_i^\beta(T, P) \quad (5.2-9)$$

indicating that fugacities of a pure component must be equal to each other under the conditions of equilibrium.

The word fugacity comes from the same root as “fugitive.” It can be thought of as a degree of “discomfort.” The higher the fugacity, the more uncomfortable a molecule is in a given phase. To ease the measure of discomfort, a molecule escapes to another phase in which its fugacity is lower. For this reason, fugacity is also called “escaping tendency,” the measure of which is molar Gibbs energy. All spontaneous physical and/or chemical processes take place in the direction of decreasing fugacity and molar Gibbs energy. For example, the driving force for mass transfer is the difference in fugacity.

At constant temperature, Eqn (5.1-1) reduces to

$$d\tilde{G}_i = \tilde{V}_i dP. \quad (5.2-10)$$

Substitution of Eqn (5.2-10) into Eqn (5.2-7) gives

$$RT d \ln f_i = \tilde{V}_i dP. \quad (5.2-11)$$

Subtraction of the term $RT d \ln P$ from both sides of Eqn (5.2-11) and integration from an ideal gas state to any state lead to

$$\ln \left(\frac{f_i}{P} \right) = \ln \phi_i = \int_0^P \left(\frac{\tilde{V}_i}{RT} - \frac{1}{P} \right) dP, \quad T = \text{constant}, \quad (5.2-12)$$

which is applicable to gases, liquids, and solids. Fugacity (or fugacity coefficient) can be calculated from either Eqn (5.2-1) or Eqn (5.2-12).

5.3 Fugacity of a Pure Gas

Fugacity of a pure gas at a given temperature and pressure can be calculated from Eqn (5.2-12), i.e.

$$\ln \phi_i^V = \int_0^P \left(\frac{\tilde{V}_i}{RT} - \frac{1}{P} \right) dP, \quad T = \text{constant} \quad (5.3-1)$$

or

$$\ln \phi_i^V = \int_0^P \left(\frac{Z_i^V - 1}{P} \right) dP, \quad T = \text{constant}. \quad (5.3-2)$$

At low pressures, gases behave ideally, i.e. $Z_i^V = 1$, and Eqn (5.3-2) leads to

$$\ln \phi_i^V = 0 \Rightarrow \phi_i^V = 1 \text{ or } f_i^V(T, P) = P \quad (5.3-3)$$

indicating that for a pure gas at low pressures the fugacity coefficient is unity and the fugacity is equal to the total pressure.

5.3.1 Fugacity from Volume-Explicit Equations of State

Substitution of $\tilde{V}_i = \tilde{V}_i(P, T)$ into Eqn (5.3-1) and integration of the resulting equation either analytically or numerically give the fugacity of a gas at a specified temperature and pressure.

Example 5.1 Calculate the fugacity coefficient of ammonia at 350 K and 50 bar if the equation of state is given as

$$P(\tilde{V} - b) = RT,$$

where $b = 37.3 \text{ cm}^3/\text{mol}$.

Solution

From the given equation of state

$$\tilde{V}_{\text{NH}_3} = b + \frac{RT}{P}. \quad (1)$$

Substitution of Eqn (1) into Eqn (5.3-1) gives

$$\ln \phi_{\text{NH}_3} = \int_0^{50} \left(\frac{b}{RT} + \frac{1}{P} - \frac{1}{P} \right) dP = \frac{bP}{RT} \Big|_0^{50} = \frac{(37.3)(50)}{(83.14)(350)} = 0.064.$$

Therefore,

$$\phi_{\text{NH}_3} = 1.066.$$

5.3.2 Fugacity from Property Tables

If property tables are available, then fugacity can be calculated directly without using an equation of state as shown in the following example.

Example 5.2 Calculate the fugacity of steam at 873 K and 50 bar using the following data:

T (K)	P (bar)	$\hat{H}$ (kJ/kg)	$\hat{S}$ (kJ/kg K)
873	50	3666.5	7.2589
873	0.1	3705.4	10.1608

Solution

Let the values of temperature and pressure at states 1 and 2 be (873 K, 0.1 bar) and (873 K, 50 bar), respectively. From Eqn (5.2-5)

$$\tilde{G}_1 = \lambda(873 \text{ K}) + RT \ln f_1, \quad (1)$$

$$\tilde{G}_2 = \lambda(873 \text{ K}) + RT \ln f_2. \quad (2)$$

Subtraction of Eqn (1) from Eqn (2) results in

$$\tilde{G}_2 - \tilde{G}_1 = RT \ln \left(\frac{f_2}{f_1} \right). \quad (3)$$

Note that Eqn (3) can also be obtained from Eqn (5.2-1). Since $\tilde{G} = \tilde{H} - T\tilde{S}$, Eqn (3) becomes

$$(\tilde{H}_2 - \tilde{H}_1) - T(\tilde{S}_2 - \tilde{S}_1) = RT \ln \left(\frac{f_2}{f_1} \right) \quad (4)$$

or

$$M [(\hat{H}_2 - \hat{H}_1) - T(\hat{S}_2 - \hat{S}_1)] = RT \ln \left(\frac{f_2}{f_1} \right), \quad (5)$$

where M is the molecular weight of water. At 873 K and 0.1 bar, it is plausible to assume that steam behaves as an ideal gas, i.e.

$$f_1 = P_1 = 0.1 \text{ bar}.$$

Substitution of the numerical values into Eqn (5) gives

$$18 [(3666.5 - 3705.4) - (873)(7.2589 - 10.1608)] = (8.314)(873) \ln \left(\frac{f_2}{0.1} \right).$$

The solution yields

$$f_2 = 48.6 \text{ bar}.$$

5.3.3 Fugacity from the Virial Equation of State

At low to moderate pressures, it is possible to truncate the virial equation of state after the second term as given by Eqn (3.1-9), i.e.

$$Z_i^V = 1 + \frac{B_i P}{RT}, \quad (5.3-4)$$

where B_i , the second virial coefficient for pure i , is calculated from Eqn (3.1-10). Substitution of Eqn (5.3-4) into Eqn (5.3-2) and integration give

$$\ln \phi_i^V = \frac{B_i P}{RT}. \quad (5.3-5)$$

5.3.4 Fugacity from Cubic Equations of State

The cubic equations of state are pressure-explicit, i.e. $P = P(\tilde{V}, T)$, and the use of Eqn (5.3-1) is not practical for this case. Therefore, it is useful to have an equation relating fugacity to an integral over volume rather than pressure.

For this purpose, the starting point is the equation of state for a real gas, i.e.

$$P\tilde{V}_i = Z_i^V RT. \quad (5.3-6)$$

At constant temperature, differentiation of Eqn (5.3-6) gives

$$\frac{d\tilde{V}_i}{\tilde{V}_i} + \frac{dP}{P} = \frac{dZ_i^V}{Z_i^V}. \quad (5.3-7)$$

Table 5.1: The term Θ_i^V in Eqn (5.3-9).^a

Equation of State	Θ_i^V	
van der Waals	$\frac{A_i}{Z_i^V}$	(A)
Redlich-Kwong	$\frac{A_i}{B_i} \ln \left(1 + \frac{B_i}{Z_i^V} \right)$	(B)
Soave-Redlich-Kwong	$\frac{A_i}{B_i} \ln \left(1 + \frac{B_i}{Z_i^V} \right)$	(C)
Peng-Robinson	$\frac{A_i}{\sqrt{8}B_i} \ln \left[\frac{Z_i^V + (1 + \sqrt{2})B_i}{Z_i^V + (1 - \sqrt{2})B_i} \right]$	(D)

^a The dimensionless parameters A_i and B_i can be calculated from the expressions given in Table 3.2.

Substitution of Eqns (5.3-6) and (5.3-7) into Eqn (5.3-2) yields

$$\ln \phi_i^V = Z_i^V - 1 - \ln Z_i^V + \frac{1}{RT} \int_{\tilde{V}_i}^{\infty} \left(P - \frac{RT}{\tilde{V}_i} \right) d\tilde{V}_i, \quad T = \text{constant}. \quad (5.3-8)$$

Substitution of the cubic equation of state into Eqn (5.3-8) and integration lead to the following general expression for calculating fugacity coefficients in the vapor phase:

$$\ln \phi_i^V = Z_i^V - 1 - \ln(Z_i^V - B_i) - \Theta_i^V, \quad (5.3-9)$$

where the term Θ_i^V is given in Table 5.1 depending on the equation of state.

Equation (5.3-9) can also be obtained by substituting departure function expressions developed in Section 3.6.1 into Eqn (5.2-1).

Example 5.3 Calculate the fugacity coefficient of acetylene at 373 K and 1, 10, and 50 bar if it obeys

- (a) van der Waals equation of state.
- (b) Redlich-Kwong equation of state.
- (c) Soave-Redlich-Kwong equation of state.
- (d) Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 308.3 \text{ K}, \quad P_c = 61.4 \text{ bar}, \quad \omega = 0.190.$$

The reduced temperature is

$$T_r = \frac{T}{T_c} = \frac{373}{308.3} = 1.210.$$

The values of the reduced pressures are

P (bar)	1	10	50
P_r	0.016	0.163	0.814

- (a) For the van der Waals equation of state, the values of A , B , Z , and ϕ are given in the following table:

P (bar)	A	B	Z	ϕ
1	4.61×10^{-3}	1.653×10^{-3}	0.997	0.997
10	0.047	0.017	0.969	0.970
50	0.235	0.084	0.830	0.852

- (b) For the Redlich-Kwong equation of state, the values of A , B , Z , and ϕ are given in the following table:

P (bar)	A	B	Z	ϕ
1	4.247×10^{-3}	1.146×10^{-3}	0.997	0.997
10	0.043	0.012	0.968	0.969
50	0.216	0.058	0.833	0.850

- (c) For the Soave-Redlich-Kwong equation of state, the values of α , A , B , Z , and ϕ are given in the following table:

P (bar)	α	A	B	Z	ϕ
1	0.853	3.985×10^{-3}	1.146×10^{-3}	0.997	0.997
10	0.853	0.041	0.012	0.971	0.971
50	0.853	0.203	0.058	0.850	0.863

- (d) For the Peng-Robinson equation of state, the values of α , A , B , Z , and ϕ are given in the following table:

P (bar)	α	A	B	Z	ϕ
1	0.873	4.362×10^{-3}	1.029×10^{-3}	0.997	0.997
10	0.873	0.044	0.010	0.966	0.966
50	0.873	0.222	0.052	0.829	0.843

5.3.5 Fugacity from the Principle of Corresponding States

In terms of reduced temperature and pressure, Eqn (5.2-1) takes the form

$$\ln \phi_i^V = \frac{1}{T_r} \frac{(\tilde{H}_i - \tilde{H}_i^{\text{IG}})_{T,P}}{RT_c} - \frac{(\tilde{S}_i - \tilde{S}_i^{\text{IG}})_{T,P}}{R} \quad (5.3-10)$$

in which the enthalpy and entropy departure functions are calculated from Eqns (3.6-24) and (3.6-25), respectively.

5.4 Fugacity of a Pure Liquid

When liquid and vapor phases of a pure component i are in equilibrium with each other, the number of degrees of freedom is unity. In other words, the equilibrium state can be determined by specifying one independent intensive variable, for example, temperature. At any given temperature, the corresponding pressure is the vapor pressure (or saturation pressure), P_i^{vap} , as shown in Fig. 5.2.

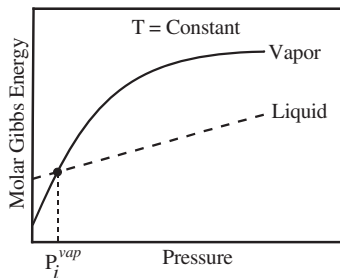


Figure 5.2: Variation of Gibbs energy with pressure at constant temperature.

The vapor and liquid phases are in equilibrium with each other at the point of intersection of $\tilde{G}_i$ versus P curves in Fig. 5.2. Thus,

$$\tilde{G}_i^V(T, P_i^{\text{vap}}) = \tilde{G}_i^L(T, P_i^{\text{vap}}) \Rightarrow f_i^V(T, P_i^{\text{vap}}) = f_i^L(T, P_i^{\text{vap}}). \quad (5.4-1)$$

In other words, there is no variation in fugacity during a phase change, i.e. from liquid to vapor or vice versa.

Using Eqn (5.2-5), molar Gibbs energy of a pure liquid at any temperature and pressure is expressed as

$$\tilde{G}_i^L(T, P) = \lambda_i(T) + RT \ln f_i^L(T, P). \quad (5.4-2)$$

On the other hand, molar Gibbs energy of a pure liquid at the same temperature but at the corresponding vapor pressure is given by

$$\tilde{G}_i^L(T, P_i^{\text{vap}}) = \lambda_i(T) + RT \ln f_i^L(T, P_i^{\text{vap}}). \quad (5.4-3)$$

Subtraction of Eqn (5.4-3) from Eqn (5.4-2) gives

$$\tilde{G}_i^L(T, P) - \tilde{G}_i^L(T, P_i^{\text{vap}}) = RT \ln \left[\frac{f_i^L(T, P)}{f_i^L(T, P_i^{\text{vap}})} \right]. \quad (5.4-4)$$

The use of Eqn (5.4-1) in Eqn (5.4-4) leads to

$$\tilde{G}_i^L(T, P) - \tilde{G}_i^L(T, P_i^{\text{vap}}) = RT \ln \left[\frac{f_i^L(T, P)}{f_i^V(T, P_i^{\text{vap}})} \right]. \quad (5.4-5)$$

At constant temperature, integration of Eqn (5.2-10) from P_i^{vap} to P yields

$$\tilde{G}_i^L(T, P) - \tilde{G}_i^L(T, P_i^{\text{vap}}) = \tilde{V}_i^L(P - P_i^{\text{vap}}) \quad (5.4-6)$$

Table 5.2: Variation of the Poynting correction factor as a function of pressure and temperature when $\tilde{V}^L = 80 \text{ cm}^3/\text{mol}$.

$T \text{ (K)}$	$P - P^{\text{vap}}$ (bar)	Poynting Correction Factor
298	1	1.003
298	100	1.381
298	1000	25.254
300	50	1.174
100	50	1.618
50	50	2.618

in which the molar volume of liquid is considered constant. Substitution of Eqn (5.4-6) into Eqn (5.4-5) and rearrangement result in

$$f_i^L(T, P) = f_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right] \quad (5.4-7)$$

or

$$f_i^L(T, P) = P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right]. \quad (5.4-8)$$

The exponential term in Eqns (5.4-7) and (5.4-8) is commonly referred to as the *Poynting correction factor* (PCF). It accounts for the compression of the liquid to a pressure P greater than its vapor pressure, P_i^{vap} . Table 5.2 indicates the variation in Poynting correction factor as a function of pressure. Examination of the values clearly reveals that the Poynting correction factor is almost unity at low pressures but may become large at extremely high pressures or low temperatures.

Example 5.4 Specify the conditions under which the fugacity of liquid is almost equal to the vapor pressure at a given temperature, i.e. $f_i^L(T, P) \simeq P_i^{\text{vap}}$.

Solution

When pressure is not very high, the Poynting correction factor is unity and Eqn (5.4-8) simplifies to

$$f_i^L(T, P) = P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}). \quad (1)$$

When the vapor pressure is not very high, the vapor phase may be considered an ideal gas and $\phi_i^V(T, P_i^{\text{vap}}) = 1$. Thus, Eqn (1) becomes

$$f_i^L(T, P) = P_i^{\text{vap}}. \quad (2)$$

Comment: Using steam tables, [Rittmann *et al.* \(1982\)](#) expressed the fugacity of saturated water and saturation pressure of water as

$$\ln f_{\text{H}_2\text{O}}^{\text{sat}} = 8.9672 - \frac{1138.26}{T} - \frac{10.6317 \times 10^5}{T^2} + \frac{8.8730 \times 10^7}{T^3},$$

$$\ln P_{\text{H}_2\text{O}}^{\text{vap}} = 13.8365 - \frac{6.6075 \times 10^3}{T} + \frac{10.1855 \times 10^5}{T^2} - \frac{17.7039 \times 10^7}{T^3},$$

where $f_{\text{H}_2\text{O}}^{\text{sat}}$ and $P_{\text{H}_2\text{O}}^{\text{vap}}$ are in bar, and T is in K. These equations are valid when

$$300 \leq T \leq 600.$$

The calculated values of $P_{\text{H}_2\text{O}}^{\text{vap}}$ and $f_{\text{H}_2\text{O}}^{\text{sat}}$ are given in the following table:

T (K)	$P_{\text{H}_2\text{O}}^{\text{vap}}$ (bar)	$f_{\text{H}_2\text{O}}^{\text{sat}}$ (bar)
300	0.032	0.035
350	0.425	0.413
400	2.503	2.392
450	9.395	8.760
500	26.560	23.496

Thus, at higher values of vapor pressure, the difference between vapor pressure and fugacity becomes appreciable.

Evaluation of liquid fugacities requires molar volumes of saturated liquids to be known. Molar volumes of various liquids as a function of temperature are given in [Table 5.3](#).

If experimental data are unavailable, molar volumes of saturated liquids can be estimated by the *modified Rackett equation* ([Spencer and Danner, 1972](#))

$$\tilde{V}^{\text{L}} = \frac{RT_{\text{c}}}{P_{\text{c}}} Z_{\text{RA}}^{1+(1-T_{\text{r}})^{2/7}}, \quad (5.4-9)$$

where Z_{RA} is the *Rackett parameter*, given by

$$Z_{\text{RA}} = 0.29056 - 0.08775\omega. \quad (5.4-10)$$

Table 5.3: Molar volumes of various liquids (Holmes and van Winkle, 1970).

Substance	T (K)	$\tilde{V}^L$ (cm ³ /mol)	Substance	T (K)	$\tilde{V}^L$ (cm ³ /mol)
Acetone	228.15	67.380	<i>n</i> -Heptane	273.15	143.045
	273.15	71.483		323.15	152.303
	323.15	76.826		373.15	163.619
Acetonitrile	273.15	51.092	<i>n</i> -Hexane	273.15	127.301
	303.15	53.214		323.15	136.388
	355.15	57.4		373.15	148.211
Benzene	273.15	86.783	Methanol	273.15	39.556
	323.15	92.263		373.15	44.874
	373.15	98.537		473.15	57.939
<i>n</i> -Butanol	273.15	89.873	Methyl ethyl ketone	273.15	87.3
	343.15	97.8		333.15	94.5
	413.15	108.7		373.15	100.0
Carbon tetrachloride	293.15	96.518	<i>n</i> -Octane	273.15	158.970
	353.15	104.192		333.15	170.630
	413.15	114.379		393.15	185.182
Chloroform	273.15	78.218	<i>n</i> -Pentane	273.15	111.8
	303.15	81.185		333.15	122.9
	333.15	84.5		373.15	131.4
Cyclohexane	288.15	107.470	<i>n</i> -Propanol	293.15	74.785
	306.30	109.841		343.15	78.962
	352.35	116.630		393.15	84.515
Ethanol	273.15	57.141	Toluene	303.15	107.415
	323.15	60.356		353.15	113.717
	373.15	64.371		400.00	120.879
Ethyl acetate	273.15	95.3	Water	277.13	18.060
	323.15	102.1		323.15	18.278
	373.15	110.5		373.15	18.844

Example 5.5 Estimate the molar volume of liquid carbon dioxide at 274 K and 70 bar using the Soave-Redlich-Kwong equation of state, the Peng-Robinson equation of state, and the Rackett equation.

Solution

From Appendix A

$$T_c = 304.2 \text{ K}, \quad P_c = 73.8 \text{ bar}, \quad \omega = 0.239.$$

The values of the reduced temperature and pressure are

$$T_r = \frac{274}{304.2} = 0.901 \quad \text{and} \quad P_r = \frac{70}{73.8} = 0.949.$$

The molar volume is calculated from

$$\tilde{V}^L = \frac{Z^L RT}{P}.$$

For the Soave-Redlich-Kwong and Peng-Robinson equations of state the results are given in the following table:

Equation of State	A	B	Z^L	$\tilde{V}^L$ (cm ³ /mol)
Soave-Redlich-Kwong	0.544	0.091	0.160	52.1
Peng-Robinson	0.575	0.082	0.142	46.2

The Rackett parameter is

$$Z_{RA} = 0.29056 - (0.08775)(0.239) = 0.270.$$

The use of Eqn (5.4-9) gives

$$\tilde{V}^L = \frac{(83.14)(304.2)}{73.8} (0.270)^{1+(1-0.901)^{2/7}} = 47.1 \text{ cm}^3/\text{mol}.$$

Comment: In general, the Soave-Redlich-Kwong equation of state overpredicts the liquid molar volume by 10–35%.

Example 5.6 Calculate the fugacity of benzene at 320 K and 100 bar. The vapor pressure of benzene at 320 K is 0.308 bar.

Solution

From Appendix A

$$T_c = 562 \text{ K}, \quad P_c = 48.9 \text{ bar}, \quad \omega = 0.212.$$

Since $T < T_c$ and $P > P^{\text{vap}}$, benzene exists as a liquid at the given temperature and pressure. Application of Eqn (5.4-8) gives

$$f_{\text{benzene}}^L(320 \text{ K}, 100 \text{ bar}) = 0.308 \phi_{\text{benzene}}^V(320 \text{ K}, 0.308 \text{ bar}) \times \exp \left[\frac{\tilde{V}_{\text{benzene}}^L (100 - 0.308)}{(83.14)(320)} \right]. \quad (1)$$

Since the vapor pressure is low, vapor may be considered an ideal gas, i.e.

$$\phi_{\text{benzene}}^V(320 \text{ K}, 0.308 \text{ bar}) \simeq 1.$$

From Table 5.3

$$\tilde{V}_{\text{benzene}}^L \simeq 92.263 \text{ cm}^3/\text{mol}.$$

Substitution of the numerical values into Eqn (1) gives the liquid fugacity as

$$f_{\text{benzene}}^L(320 \text{ K}, 100 \text{ bar}) = 0.308 \exp \left[\frac{(92.263)(100 - 0.308)}{(83.14)(320)} \right] = 0.435 \text{ bar}.$$

If liquid fugacity is known at (T, P_1) , the value of the liquid fugacity at (T, P_2) can be calculated from Eqn (5.4-8) as

$$f_i^L(T, P_2) = f_i^L(T, P_1) \exp \left[\frac{\tilde{V}_i^L(P_2 - P_1)}{RT} \right]. \quad (5.4-11)$$

5.4.1 Fugacity from Cubic Equations of State

Since cubic equations of state can describe pure substances in both the vapor and liquid phases, Eqn (5.3-9) is also applicable to liquids, i.e.

$$\ln \phi_i^L = Z_i^L - 1 - \ln(Z_i^L - B_i) - \Theta_i^L, \quad (5.4-12)$$

where the term Θ_i^L is given in Table 5.4 depending on the equation of state.

Table 5.4: The term Θ_i^L in Eqn (5.4-12).^a

Equation of State	Θ_i^L	
van der Waals	$\frac{A_i}{Z_i^L}$	(A)
Redlich-Kwong	$\frac{A_i}{B_i} \ln \left(1 + \frac{B_i}{Z_i^L} \right)$	(B)
Soave-Redlich-Kwong	$\frac{A_i}{B_i} \ln \left(1 + \frac{B_i}{Z_i^L} \right)$	(C)
Peng-Robinson	$\frac{A_i}{\sqrt{8}B_i} \ln \left[\frac{Z_i^L + (1 + \sqrt{2})B_i}{Z_i^L + (1 - \sqrt{2})B_i} \right]$	(D)

^a The dimensionless parameters A_i and B_i can be calculated from the expressions given in Table 3.2.

Example 5.7 Estimate the vapor pressure of ethanol at 313 K by using the condition of chemical equilibrium. Assume that ethanol is represented by the Peng-Robinson equation of state.

Solution

From Appendix A

$$T_c = 514 \text{ K}, \quad P_c = 63 \text{ bar}, \quad \omega = 0.644.$$

The condition of equilibrium is given by Eqn (5.4-1), i.e.

$$f_{\text{ethanol}}^V(T, P_{\text{ethanol}}^{\text{vap}}) = f_{\text{ethanol}}^L(T, P_{\text{ethanol}}^{\text{vap}}). \quad (1)$$

Dividing Eqn (1) by $P_{\text{ethanol}}^{\text{vap}}$ gives

$$\phi_{\text{ethanol}}^V(T, P_{\text{ethanol}}^{\text{vap}}) = \phi_{\text{ethanol}}^L(T, P_{\text{ethanol}}^{\text{vap}}). \quad (2)$$

An algorithm to estimate the vapor pressure at a given temperature is given as follows:

1. Obtain T_c , P_c , and ω from Appendix A.
2. Determine the reduced temperature.
3. Use Eqn (3.1-16) to determine the term α .
4. Assume $P_{\text{ethanol}}^{\text{vap}}$ and determine the reduced pressure.
5. Determine the dimensionless parameters A and B (Table 3.2).
6. Evaluate the parameters p , q , and r appearing in Eqn (3.1-18) (Table 3.2).
7. Solve Eqn (3.1-18), i.e. $Z^3 + pZ^2 + qZ + r = 0$. The largest and the smallest roots correspond to Z^V and Z^L , respectively.
8. Use Z^V in Eqn (5.3-9) and calculate ϕ_{ethanol}^V . Use Z^L in Eqn (5.4-12) and calculate ϕ_{ethanol}^L . Repeat steps (4)–(8) until $\phi_{\text{ethanol}}^V = \phi_{\text{ethanol}}^L$.

The reduced temperature is

$$T_r = \frac{T}{T_c} = \frac{313}{514} = 0.609.$$

The term α is calculated from Eqn (3.1-16) as

$$\alpha = \left[1 + (0.37464 + (1.54226)(0.644) - (0.26992)(0.644)^2) \times (1 - \sqrt{0.609}) \right]^2 = 1.628.$$

The results of the iterative procedure are given below:

$P_{\text{ethanol}}^{\text{vap}}$ (bar)	$A \times 10^3$	$B \times 10^4$	$Z^L \times 10^4$	Z^V	ϕ^L	ϕ^V
0.1	3.186	2.028	2.379	0.997	1.877	0.997
0.2	6.372	4.056	4.759	0.994	0.939	0.994
0.1888	6.015	3.829	4.492	0.994	0.994	0.994

Therefore, the vapor pressure of ethanol at 313 K is 0.1888 bar (or 141.6 mmHg). In other words, ethanol exists in liquid form at 313 K when $P > 0.1888$ bar.

5.5 Fugacity of a Pure Solid

When solid and vapor phases of a pure component i are in equilibrium with each other, the number of degrees of freedom is unity. In other words, the equilibrium state can be determined by specifying one independent intensive variable, for example temperature. At any given temperature, the corresponding pressure is the sublimation pressure (or vapor pressure), P_i^{sub} , as shown in Fig. 5.3.

The vapor and solid phases are in equilibrium with each other at the point of intersection of $\tilde{G}_i$ versus P curves in Fig. 5.3. Thus,

$$\tilde{G}_i^V(T, P_i^{\text{sub}}) = \tilde{G}_i^S(T, P_i^{\text{sub}}) \Rightarrow f_i^V(T, P_i^{\text{sub}}) = f_i^S(T, P_i^{\text{sub}}). \quad (5.5-1)$$

Using Eqn (5.2-5), molar Gibbs energy of a pure solid at any temperature and pressure is expressed as

$$\tilde{G}_i^S(T, P) = \lambda_i(T) + RT \ln f_i^S(T, P). \quad (5.5-2)$$

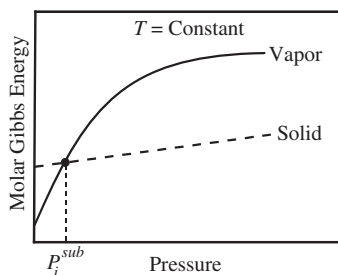


Figure 5.3: Variation of Gibbs energy with pressure at constant temperature.

On the other hand, molar Gibbs energy of a pure solid at the same temperature but at the corresponding sublimation pressure is given by

$$\tilde{G}_i^S(T, P_i^{\text{sub}}) = \lambda_i(T) + RT \ln f_i^S(T, P_i^{\text{sub}}). \quad (5.5-3)$$

Subtraction of Eqn (5.5-3) from Eqn (5.5-2) gives

$$\tilde{G}_i^S(T, P) - \tilde{G}_i^S(T, P_i^{\text{sub}}) = RT \ln \left[\frac{f_i^S(T, P)}{f_i^S(T, P_i^{\text{sub}})} \right]. \quad (5.5-4)$$

The use of Eqn (5.5-1) in Eqn (5.5-4) leads to

$$\tilde{G}_i^S(T, P) - \tilde{G}_i^S(T, P_i^{\text{sub}}) = RT \ln \left[\frac{f_i^S(T, P)}{f_i^V(T, P_i^{\text{sub}})} \right]. \quad (5.5-5)$$

At constant temperature, integration of Eqn (5.2-10) from P_i^{sub} to P yields

$$\tilde{G}_i^S(T, P) - \tilde{G}_i^S(T, P_i^{\text{sub}}) = \tilde{V}_i^S(P - P_i^{\text{sub}}). \quad (5.5-6)$$

Substitution of Eqn (5.5-6) into Eqn (5.5-5) and rearrangement result in

$$f_i^S(T, P) = P_i^{\text{sub}} \phi_i^V(T, P_i^{\text{sub}}) \exp \left[\frac{\tilde{V}_i^S(P - P_i^{\text{sub}})}{RT} \right]. \quad (5.5-7)$$

Sublimation pressure of solids is normally low. Therefore, ideal gas behavior for the gas phase over the pure solid can be assumed, i.e. $\phi_i^V(T, P_i^{\text{sub}}) \simeq 1$, and Eqn (5.5-7) simplifies to

$$f_i^S(T, P) = P_i^{\text{sub}} \exp \left[\frac{\tilde{V}_i^S(P - P_i^{\text{sub}})}{RT} \right]. \quad (5.5-8)$$

If solid fugacity is known at (T, P_1) , the value of solid fugacity at (T, P_2) can be calculated from Eqn (5.5-7) as

$$f_i^S(T, P_2) = f_i^S(T, P_1) \exp \left[\frac{\tilde{V}_i^S(P_2 - P_1)}{RT} \right]. \quad (5.5-9)$$

Example 5.8 Calculate the fugacity of ice at 250 K and 100 bar using the following data:

$$\ln P_{\text{ice}}^{\text{sub}} (\text{bar}) = -\frac{2378.6}{T} + 51.06 \ln T - 0.225T + 1.62 \times 10^{-4}T^2 - 233.45,$$

$$\tilde{V}_{\text{ice}} (\text{cm}^3/\text{mol}) = 19.655 + 0.0022364(T - 273.15).$$

Solution

At 250 K

$$P_{\text{ice}}^{\text{sub}} = 7.743 \times 10^{-4} \text{ bar} \quad \text{and} \quad \tilde{V}_{\text{ice}} = 19.603 \text{ cm}^3/\text{mol}.$$

The use of Eqn (5.5-8) gives

$$f_{\text{ice}} = 7.743 \times 10^{-4} \exp \left[\frac{(19.603)(100)}{(83.14)(250)} \right] = 8.51 \times 10^{-4} \text{ bar}.$$

5.6 Phase Transitions and Equilibrium Criteria

The states of a pure substance can be represented as a surface in a three-dimensional $P\hat{V}T$ diagram as shown in Fig. 5.4. For convenience, however, it is generally preferred to plot two-dimensional diagrams, such as pressure versus specific (or molar) volume and pressure versus temperature. The pressure-temperature diagram of a pure substance includes three curves representing liquid-vapor, solid-vapor, and liquid-solid equilibrium as shown in Fig. 5.5. Such plots are also referred to as the *phase diagrams*.

When $P < P_c$, a substance in the gaseous state is called either a *gas* ($T > T_c$) or a *vapor* ($T < T_c$). Under isothermal conditions, while a vapor can be liquefied by exerting pressure, a gas cannot be liquefied no matter what pressure is applied to it.

The *triple point* is the only point on the phase diagram where the solid, liquid, and vapor phases coexist in equilibrium.² In other words, it is the intersection of the liquid-vapor (vapor pressure curve), solid-liquid (fusion or melting curve), and solid-vapor (sublimation pressure curve) coexistence curves. Note that the number of degrees of freedom, $\mathcal{F}$, is zero at the triple point. Triple point temperatures and pressures of some substances are given in Table 5.5. If the triple point pressure is less than 1 atm, as in the case of water, it is possible to have all three phases of a substance under atmospheric pressure depending on temperature. If the triple point pressure is higher than 1 atm, as in the case of carbon dioxide, then a substance cannot exist in the liquid form under atmospheric pressure, and the transition from the solid to vapor form, i.e. sublimation, takes place with an increase in temperature.

At temperatures and pressures higher than the critical values, substances exist in the fluid (or supercritical) region and are called *supercritical fluids*. They possess

²In general, the triple point is the point of intersection of three different phases. If a substance exists in different forms of solid, e.g. graphite and diamond for carbon, it can have more than one triple point.

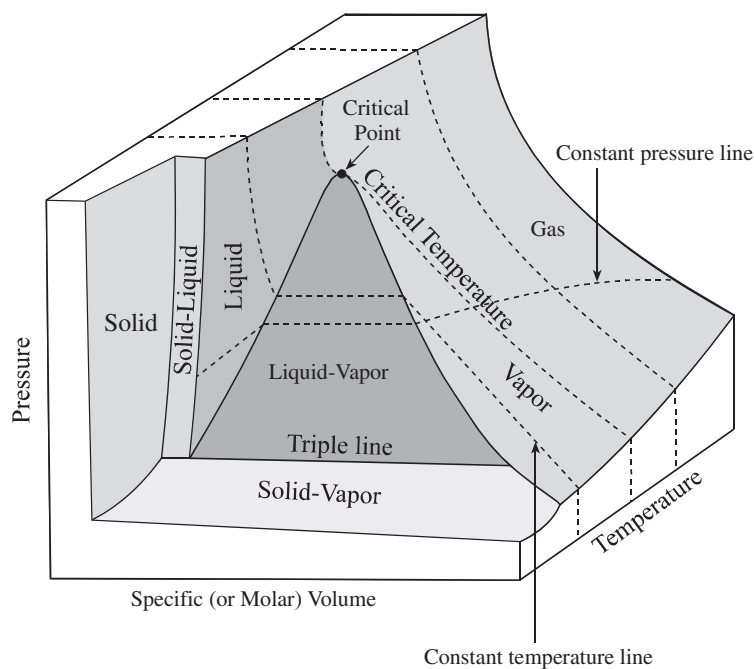
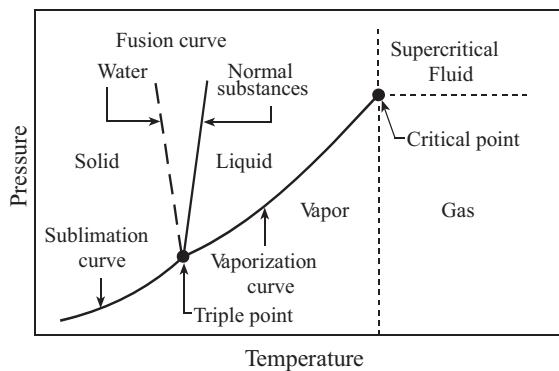
Figure 5.4: The $P\hat{V}T$ surface of a pure substance.

Figure 5.5: Temperature-pressure diagram.

both the gaseous properties (viscosity, diffusivity, surface tension) of being able to diffuse into substances easily, and the liquid property (density) of being able to dissolve substances.

The slope of the vaporization curve gives the rate of change of vapor pressure of liquid with temperature. The slope of the sublimation curve gives the rate of change

Table 5.5: Normal boiling point, normal melting point, and triple point temperature and pressure for various substances.^a

Substance	Temperature (°C) at $P = 1$ atm		Triple point	
	Boiling Point	Melting (Freezing) Point	T (°C)	P (atm)
Ammonia	−33.33	−77.74	−77.74	0.06
Bromine	58.75	−7.25	7.25	0.058
Carbon dioxide	−78.45	−56.57	−56.57	5.1172
Naphthalene	217.99	80.15	80.27	0.0099
Sulfur dioxide	−10.05	−75.55	−75.47	0.0165
Water	100	0	0.01	0.006

^a Compiled from Korea Thermophysical Properties Data Bank (www.thermo.org/research/kdb).

of sublimation (or vapor) pressure of solid with temperature. Finally, the slope of the fusion (or melting) curve gives the rate of change of the melting (or freezing) pressure of solid with temperature. While the fusion curve has a positive slope for most substances, the slope becomes negative for water. Since the fusion curve generally has a very steep slope, the triple point temperature for most substances is close to their melting (or freezing) temperature at atmospheric pressure, known as *normal melting (or freezing) point*.

5.6.1 Clapeyron Equation

The starting point for obtaining quantitative information on the slopes of these equilibrium curves, i.e. dP/dT , is the fact that molar Gibbs energies of the phases are equal to each other

$$\tilde{G}_\alpha = \tilde{G}_\beta, \quad (5.6-1)$$

where subscripts α and β represent different phases. The differential form of Eqn (5.6-1) is

$$d\tilde{G}_\alpha = d\tilde{G}_\beta. \quad (5.6-2)$$

The use of Eqn (5.1-1) in Eqn (5.6-2) leads to

$$\tilde{V}_\alpha dP - \tilde{S}_\alpha dT = \tilde{V}_\beta dP - \tilde{S}_\beta dT. \quad (5.6-3)$$

Rearrangement of Eqn (5.6-3) yields

$$\left(\frac{dP}{dT}\right)_{\text{eq}} = \frac{\tilde{S}_\alpha - \tilde{S}_\beta}{\tilde{V}_\alpha - \tilde{V}_\beta}, \quad (5.6-4)$$

where the subscript “eq” indicates that the path of integration is on the vaporization curve, the fusion curve, or the sublimation curve.

The entropy difference in Eqn (5.6-4) must be expressed in terms of measurable quantities. For this purpose, the use of the identity $\tilde{G} = \tilde{H} - T\tilde{S}$ in Eqn (5.6-1) leads to

$$\tilde{H}_\alpha - T\tilde{S}_\alpha = \tilde{H}_\beta - T\tilde{S}_\beta \Rightarrow \tilde{S}_\alpha - \tilde{S}_\beta = \frac{\tilde{H}_\alpha - \tilde{H}_\beta}{T}. \quad (5.6-5)$$

Substitution of Eqn (5.6-5) into Eqn (5.6-4) gives

$$\left(\frac{dP}{dT}\right)_{\text{eq}} = \frac{\tilde{H}_\alpha - \tilde{H}_\beta}{T(\tilde{V}_\alpha - \tilde{V}_\beta)}, \quad (5.6-6)$$

which is known as the *Clapeyron*³ equation.

5.6.2 Vapor-Liquid Equilibrium

The amount of heat to convert saturated liquid to saturated vapor is called the enthalpy change on vaporization or simply the *heat of vaporization*, $\Delta\tilde{H}^{\text{vap}}$. If the α -phase is considered to be vapor and the β -phase to be liquid, then

$$\tilde{H}_\alpha - \tilde{H}_\beta = \tilde{H}^{\text{V}} - \tilde{H}^{\text{L}} = \Delta\tilde{H}^{\text{vap}} \quad (5.6-7)$$

and

$$\tilde{V}_\alpha - \tilde{V}_\beta = \tilde{V}^{\text{V}} - \tilde{V}^{\text{L}} \simeq \tilde{V}^{\text{V}} = \frac{RT}{P^{\text{vap}}} \quad \text{if the vapor is an ideal gas.} \quad (5.6-8)$$

Substitution of Eqns (5.6-7) and (5.6-8) into Eqn (5.6-6) results in

$$\frac{dP^{\text{vap}}}{P^{\text{vap}}} = \frac{\Delta\tilde{H}^{\text{vap}}}{R} \frac{dT}{T^2} \quad (5.6-9)$$

or

$$\frac{d \ln P^{\text{vap}}}{dT} = \frac{\Delta\tilde{H}^{\text{vap}}}{RT^2}, \quad (5.6-10)$$

which is referred to as the *Clausius*⁴-*Clapeyron equation*. Integration of Eqn (5.6-10) depends on whether $\Delta\tilde{H}^{\text{vap}}$ is constant or changes with temperature.

$\Delta\tilde{H}^{\text{vap}}$ is independent of temperature: When $\Delta\tilde{H}^{\text{vap}}$ is constant, Eqn (5.6-10) can be rearranged as

$$d \ln P^{\text{vap}} = -\frac{\Delta\tilde{H}^{\text{vap}}}{R} d\left(\frac{1}{T}\right). \quad (5.6-11)$$

³Benoit Paul Emile Clapeyron, French engineer and physicist.

⁴Rudolf Julius Emanuel Clausius, German physicist and mathematician.

Integration of Eqn (5.6-11) gives

$$\ln P^{\text{vap}} = -\frac{\Delta\tilde{H}^{\text{vap}}}{RT} + C, \quad (5.6-12)$$

where C is an integration constant. Equation (5.6-12) indicates that a plot of $\ln P^{\text{vap}}$ versus $1/T$ is a straight line with a slope equal to $(-\Delta\tilde{H}^{\text{vap}}/R)$. Between any two states 1 and 2 on the vapor pressure curve, Eqn (5.6-12) yields

$$\ln \left(\frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} \right) = -\frac{\Delta\tilde{H}^{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right), \quad (5.6-13)$$

which provides accurate estimates of vapor pressures only over small temperature ranges.

$\Delta\tilde{H}^{\text{vap}}$ varies with temperature: The variation of $\Delta\tilde{H}^{\text{vap}}$ with temperature is expressed in the form

$$\frac{d\Delta\tilde{H}^{\text{vap}}}{dT} = \frac{d\tilde{H}^{\text{V}}}{dT} - \frac{d\tilde{H}^{\text{L}}}{dT} = \tilde{C}_P^{\text{V}} - \tilde{C}_P^{\text{L}} = \Delta\tilde{C}_P^{\text{VL}}. \quad (5.6-14)$$

Integration of Eqn (5.6-14) gives

$$\Delta\tilde{H}^{\text{vap}} = \int_{T_{\text{ref}}}^T \Delta\tilde{C}_P^{\text{VL}} dT + \Delta\tilde{H}_{\text{ref}}^{\text{vap}}, \quad (5.6-15)$$

where $\Delta\tilde{H}_{\text{ref}}^{\text{vap}}$ is the value of $\Delta\tilde{H}^{\text{vap}}$ at the reference temperature, T_{ref} . Once $\Delta\tilde{H}^{\text{vap}}$ is expressed as a function of temperature from Eqn (5.6-15), substitution of the resulting expression into Eqn (5.6-10) and integration yield the relationship between vapor pressure and temperature.

If $\Delta\tilde{C}_P^{\text{VL}}$ is independent of temperature, then Eqn (5.6-15) reduces to

$$\Delta\tilde{H}^{\text{vap}} = \Delta\tilde{C}_P^{\text{VL}}(T - T_{\text{ref}}) + \Delta\tilde{H}_{\text{ref}}^{\text{vap}}. \quad (5.6-16)$$

Substitution of Eqn (5.6-16) into Eqn (5.6-10) and integration yield

$$\ln P^{\text{vap}} = \left(\frac{\Delta\tilde{C}_P^{\text{VL}}}{R} \right) \ln T - \left(\frac{\Delta\tilde{H}_{\text{ref}}^{\text{vap}} - \Delta\tilde{C}_P^{\text{VL}}T_{\text{ref}}}{R} \right) \frac{1}{T} + C, \quad (5.6-17)$$

where C is an integration constant.

In the literature, among the various equations relating vapor pressure to temperature, the most widely used is the *Antoine equation*, expressed in the form

$$\ln P^{\text{vap}} = A - \frac{B}{T + C}. \quad (5.6-18)$$

The constants A , B , and C are given in Appendix C. Note that when $C = 0$, Eqn (5.6-18) reduces to Eqn (5.6-12).

Example 5.9 Calculate the heat of vaporization of n -heptanol [$\text{CH}_3(\text{CH}_2)_6\text{OH}$] at the normal boiling point.

Solution

From Appendix C

$$\ln P^{\text{vap}} = 8.6866 - \frac{2626.42}{T - 146.6}. \quad (1)$$

At the normal boiling point, the vapor pressure of liquid equals 1 atm (~ 1.013 bar). Therefore, the normal boiling point can be calculated from the Antoine equation as

$$\ln 1.013 = 8.6866 - \frac{2626.42}{T - 146.6} \Rightarrow T = 449.4 \text{ K}.$$

The heat of vaporization can be determined from Eqn (5.6-10). Differentiation of Eqn (1) with respect to temperature gives

$$\frac{d \ln P^{\text{vap}}}{dT} = \frac{2626.42}{(T - 146.6)^2}. \quad (2)$$

Thus, comparison of Eqn (2) with Eqn (5.6-10) results in

$$\frac{\Delta \tilde{H}^{\text{vap}}}{RT^2} = \frac{2626.42}{(T - 146.6)^2}$$

or

$$\Delta \tilde{H}^{\text{vap}} = \frac{2626.42 RT^2}{(T - 146.6)^2} = \frac{(2626.42)(8.314)(449.4)^2}{(449.4 - 146.6)^2} = 48,098 \text{ J/mol}.$$

5.6.3 Solid-Liquid Equilibrium

The amount of heat required to melt a solid is called the *heat of fusion*, $\Delta \tilde{H}^{\text{fus}}$. If the α -phase is considered to be liquid and the β -phase to be solid, then

$$\tilde{H}_\alpha - \tilde{H}_\beta = \tilde{H}^{\text{L}} - \tilde{H}^{\text{S}} = \Delta \tilde{H}^{\text{fus}} > 0. \quad (5.6-19)$$

The volume change on melting, $\Delta \tilde{V}^{\text{fus}}$, is given by

$$\Delta \tilde{V}^{\text{fus}} = \tilde{V}^{\text{L}} - \tilde{V}^{\text{S}}. \quad (5.6-20)$$

Therefore, the slope of the solid-liquid equilibrium curve, i.e. Eqn (5.6-6), becomes

$$\frac{dP}{dT_m} = \frac{\Delta \tilde{H}^{\text{fus}}}{T_m \Delta \tilde{V}^{\text{fus}}}, \quad (5.6-21)$$

where T_m is the melting temperature. Since $\Delta \tilde{H}^{\text{fus}} > 0$, the slope of the solid-liquid equilibrium curve is either positive or negative depending on the sign of $\Delta \tilde{V}^{\text{fus}}$. For almost all substances $\Delta \tilde{V}^{\text{fus}} > 0$ and thus the slope of the solid-liquid equilibrium curve has a positive slope. Water is an exception as the density of ice, 917 kg/m^3 , is less than that of liquid water, 1000 kg/m^3 . In this case

$$\Delta \tilde{V}^{\text{fus}} = \left(\frac{1}{1000} - \frac{1}{917} \right) 18 = -1.63 \times 10^{-3} \text{ m}^3/\text{kmol}$$

and the slope of the solid-liquid equilibrium curve is negative.

Integration of Eqn (5.6-21) depends on whether $\Delta \tilde{H}^{\text{fus}}$ is constant or changes with temperature.

$\Delta \tilde{H}^{\text{fus}}$ is independent of temperature: When $\Delta \tilde{H}^{\text{fus}}$ is constant, integration of Eqn (5.6-21) gives

$$P = \left(\frac{\Delta \tilde{H}^{\text{fus}}}{\Delta \tilde{V}^{\text{fus}}} \right) \ln T_m + C, \quad (5.6-22)$$

where C is an integration constant. Between any two states 1 and 2 on the solid-liquid equilibrium curve, Eqn (5.6-22) yields

$$P_2 - P_1 = \left(\frac{\Delta \tilde{H}^{\text{fus}}}{\Delta \tilde{V}^{\text{fus}}} \right) \ln \left(\frac{T_{m_2}}{T_{m_1}} \right) = \left(\frac{\Delta \hat{H}^{\text{fus}}}{\Delta \hat{V}^{\text{fus}}} \right) \ln \left(\frac{T_{m_2}}{T_{m_1}} \right). \quad (5.6-23)$$

Example 5.10 Explain quantitatively why a 55 kg ice skater glides on solid ice but not on dry ice (solid CO_2) under atmospheric conditions? For water $\Delta \tilde{H}^{\text{fus}} = 6012 \text{ kJ/kmol}$.

Solution

The melting temperature of water at 1 atm ($\sim 1.013 \text{ bar}$) is 0°C , which is also known as the normal melting (or freezing) point. If the length and the thickness of the skating blade are 150 mm and 3 mm, respectively, then the pressure exerted by the weight of the skater is

$$P = \frac{mg}{A} = \frac{(55)(9.8)}{(150 \times 10^{-3})(3 \times 10^{-3})} = 1.198 \times 10^6 \text{ Pa} = 1.198 \times 10^3 \text{ kPa}.$$

As a result of the increase in pressure, the melting temperature can be calculated from Eqn (5.6-23) as

$$\begin{aligned} T_{m_2} &= T_{m_1} \exp \left[\left(\frac{\Delta \tilde{V}^{\text{fus}}}{\Delta \tilde{H}^{\text{fus}}} \right) (P_2 - P_1) \right] \\ &= 273 \exp \left[\left(\frac{-1.63 \times 10^{-3}}{6012} \right) (1.198 \times 10^3 - 101.3) \right] = 272.92 \text{ K.} \end{aligned}$$

The decrease in the melting temperature leads to liquefaction of water, which in turn acts as a lubricating layer during skating. In the case of dry ice, on the other hand, dP/dT_m is positive and the melting temperature increases with increasing pressure. As a result, carbon dioxide remains as a solid.

Comment: Melting under pressure and refreezing again with a release of pressure is known as “regelation.” Another typical example of this phenomenon is the movement of a weighted wire through an ice block without cutting it into two pieces. The applied pressure lowers the melting temperature of ice and a film of water forms. As the wire moves downward, the release of pressure on the meltwater causes refreezing of the melted portion.

$\Delta \tilde{H}^{\text{fus}}$ varies with temperature: The variation of $\Delta \tilde{H}^{\text{fus}}$ with temperature is expressed in the form

$$\frac{d\Delta \tilde{H}^{\text{fus}}}{dT} = \frac{d\tilde{H}^{\text{L}}}{dT} - \frac{d\tilde{H}^{\text{S}}}{dT} = \tilde{C}_P^{\text{L}} - \tilde{C}_P^{\text{S}} = \Delta \tilde{C}_P^{\text{LS}}. \quad (5.6-24)$$

Integration of Eqn (5.6-24) gives

$$\Delta \tilde{H}^{\text{fus}} = \int_{T_{\text{ref}}}^{T_m} \Delta \tilde{C}_P^{\text{LS}} dT + \Delta \tilde{H}_{\text{ref}}^{\text{fus}}, \quad (5.6-25)$$

where $\Delta \tilde{H}_{\text{ref}}^{\text{fus}}$ is the value of $\Delta \tilde{H}^{\text{fus}}$ at the reference temperature, T_{ref} . Once $\Delta \tilde{H}^{\text{fus}}$ is expressed as a function of melting temperature from Eqn (5.6-25), substitution of the resulting expression into Eqn (5.6-21) and integration yield the relationship between pressure and melting temperature on the solid-liquid equilibrium curve.

If $\Delta \tilde{C}_P^{\text{LS}}$ is independent of temperature, then Eqn (5.6-25) reduces to

$$\Delta \tilde{H}^{\text{fus}} = \Delta \tilde{C}_P^{\text{LS}} (T_m - T_{\text{ref}}) + \Delta \tilde{H}_{\text{ref}}^{\text{fus}}. \quad (5.6-26)$$

Substitution of Eqn (5.6-26) into Eqn (5.6-21) and integration yield

$$P = \frac{\Delta \tilde{C}_P^{\text{LS}}}{\Delta \tilde{V}^{\text{fus}}} T_m + \left(\frac{\Delta \tilde{H}_{\text{ref}}^{\text{fus}} - \Delta \tilde{C}_P^{\text{LS}} T_{\text{ref}}}{\Delta \tilde{V}^{\text{fus}}} \right) \ln T_m + C, \quad (5.6-27)$$

where C is an integration constant.

Example 5.11 The normal melting point of lead (atomic weight = 207) is 600.65 K and the enthalpy of fusion is 4812 J/mol. Estimate the melting temperatures of lead at 50, 100, 150, and 200 bar using the following data:

$$\rho^L = 10,215 \text{ kg/m}^3, \quad \rho^S = 11,350 \text{ kg/m}^3,$$

$$\tilde{C}_P^L = 32.4 - 3.1 \times 10^{-3}T, \quad \tilde{C}_P^S = 23.6 + 9.75 \times 10^{-3}T,$$

where $\tilde{C}_P$ is in J/mol K and T is in K.

Solution

The difference between the heat capacities of liquid and solid is

$$\Delta\tilde{C}_P^{LS} = 8.8 - 12.85 \times 10^{-3}T. \quad (1)$$

Taking $T_{\text{ref}} = 600.65 \text{ K}$ and $\Delta\tilde{H}_{\text{ref}}^{\text{fus}} = 4812 \text{ J/mol}$, substitution of Eqn (1) into Eqn (5.6-25) gives

$$\Delta\tilde{H}^{\text{fus}} = 8.8T - 6.425 \times 10^{-3}T^2 + 1844.3. \quad (2)$$

The volume change on melting, $\Delta\tilde{V}^{\text{fus}}$, is given by

$$\begin{aligned} \Delta\tilde{V}^{\text{fus}} &= \left(\frac{1}{10,215} - \frac{1}{11,350} \right) 207 = 2.0264 \times 10^{-3} \text{ m}^3/\text{kmol} \\ &= 2.0264 \times 10^{-6} \text{ m}^3/\text{mol}. \end{aligned} \quad (3)$$

Thus, substitution of Eqns (2) and (3) into Eqn (5.6-21) and integration result in

$$\begin{aligned} &(P - 1.013) \times 10^5 \\ &= \frac{1}{2.0264 \times 10^{-6}} \int_{600.65}^{T_m} \left(\frac{8.8T - 6.425 \times 10^{-3}T^2 + 1844.3}{T} \right) dT. \end{aligned}$$

Simplification gives

$$P = 43.427T_m + 9101.4 \ln T_m - 1.5853 \times 10^{-2}T_m^2 - 78,594. \quad (4)$$

Melting temperatures calculated at different pressures are given in the table below:

P (bar)	50	100	150	200
T_m (K)	601.9	603.1	604.4	605.7

5.6.4 Solid-Vapor Equilibrium

The amount of heat required to convert solid to vapor is called the *heat of sublimation*, $\Delta\tilde{H}^{\text{sub}}$. In general, heat of sublimation is indirectly determined from heat of vaporization and heat of fusion as

$$\Delta\tilde{H}^{\text{sub}} = \tilde{H}^{\text{V}} - \tilde{H}^{\text{S}} = (\tilde{H}^{\text{V}} - \tilde{H}^{\text{L}}) + (\tilde{H}^{\text{L}} - \tilde{H}^{\text{S}}) = \Delta\tilde{H}^{\text{vap}} + \Delta\tilde{H}^{\text{fus}}. \quad (5.6-28)$$

The volume change on sublimation is given by

$$\Delta\tilde{V}^{\text{sub}} = \tilde{V}^{\text{V}} - \tilde{V}^{\text{S}} \simeq \tilde{V}^{\text{V}} = \frac{RT}{P^{\text{sub}}} \quad \text{if the vapor is an ideal gas.} \quad (5.6-29)$$

Thus, the slope of the solid-vapor equilibrium line, i.e. Eqn (5.6-6), becomes

$$\frac{d \ln P^{\text{sub}}}{dT} = \frac{\Delta\tilde{H}^{\text{sub}}}{RT^2}, \quad (5.6-30)$$

which is similar to Eqn (5.6-10). Since $\Delta\tilde{H}^{\text{sub}} > \Delta\tilde{H}^{\text{vap}}$, the slope of the sublimation curve is steeper than that of the vaporization curve.

If $\Delta\tilde{H}^{\text{sub}}$ is independent of temperature, integration of Eqn (5.6-30) gives

$$\ln P^{\text{sub}} = -\frac{\Delta\tilde{H}^{\text{sub}}}{RT} + C. \quad (5.6-31)$$

The constant of integration in Eqn (5.6-31), C , can be evaluated using the triple point data as

$$C = \frac{\Delta\tilde{H}^{\text{sub}}}{RT_{\text{t}}} + \ln P_{\text{t}}, \quad (5.6-32)$$

where P_{t} and T_{t} represent triple point pressure and temperature, respectively. Substitution of Eqn (5.6-32) into Eqn (5.6-31) gives

$$\ln \left(\frac{P^{\text{sub}}}{P_{\text{t}}} \right) = -\frac{\Delta\tilde{H}^{\text{sub}}}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{t}}} \right). \quad (5.6-33)$$

5.7 Analysis of Phase Diagrams Using Stability Criteria

It was stated in Section 4.3 that a substance, at a given temperature and pressure, is most stable in a phase in which its Gibbs energy is minimum. For example, under normal conditions, i.e. 298 K and 1 bar (0.1 MPa), the difference between the Gibbs energies of graphite and diamond is given as

$$\tilde{G}_{\text{diamond}}(298 \text{ K}, 10^5 \text{ Pa}) - \tilde{G}_{\text{graphite}}(298 \text{ K}, 10^5 \text{ Pa}) = 2870 \text{ J/mol}.$$

Since $\tilde{G}_{\text{graphite}} < \tilde{G}_{\text{diamond}}$, graphite is the more stable phase under these conditions. If it is desired to produce diamond from graphite at room temperature, what should be the minimum pressure imposed for such a transformation?

The starting point for answering such a question is the fact that spontaneous transition from graphite to diamond takes place when the two phases are in equilibrium with each other, i.e.

$$\tilde{G}_{\text{diamond}}(298 \text{ K}, P) = \tilde{G}_{\text{graphite}}(298 \text{ K}, P). \quad (5.7-1)$$

At constant temperature, Eqn (5.1-1) gives the variation of Gibbs energy with pressure as

$$d\tilde{G} = \tilde{V} dP. \quad (5.7-2)$$

Therefore,

$$\tilde{G}_{\text{diamond}}(298 \text{ K}, P) = \tilde{G}_{\text{diamond}}(298 \text{ K}, 10^5 \text{ Pa}) + \tilde{V}_{\text{diamond}}(P - 10^5), \quad (5.7-3)$$

$$\tilde{G}_{\text{graphite}}(298 \text{ K}, P) = \tilde{G}_{\text{graphite}}(298 \text{ K}, 10^5 \text{ Pa}) + \tilde{V}_{\text{graphite}}(P - 10^5). \quad (5.7-4)$$

Substitution of Eqns (5.7-3) and (5.7-4) into Eqn (5.7-1) and simplification lead to

$$P = 10^5 + \frac{\tilde{G}_{\text{diamond}}(298 \text{ K}, 10^5 \text{ Pa}) - \tilde{G}_{\text{graphite}}(298 \text{ K}, 10^5 \text{ Pa})}{\tilde{V}_{\text{graphite}} - \tilde{V}_{\text{diamond}}}. \quad (5.7-5)$$

The densities of graphite and diamond are approximately 2300 and 3530 kg/m³, respectively. Thus, substitution of the numerical values into Eqn (5.7-5) gives

$$P = 10^5 + \frac{2870 \times 10^3}{\left(\frac{1}{2300} - \frac{1}{3530}\right)} = 1.58 \times 10^9 \text{ Pa}.$$

Thus, at room temperature, graphite can be converted into diamond when $P \geq 1580 \text{ MPa}$.

5.7.1 Gibbs-Helmholtz Equation

In some cases, it is much more convenient to determine the variation of $\tilde{G}/T$ with temperature under constant pressure. Mathematically speaking, we are interested in

$$\frac{\partial}{\partial T} \left(\frac{\tilde{G}}{T} \right)_P = \frac{1}{T} \left(\frac{\partial \tilde{G}}{\partial T} \right)_P - \frac{\tilde{G}}{T^2} \quad (5.7-6)$$

From Eqn (5.1-1)

$$\left(\frac{\partial \tilde{G}}{\partial T} \right)_P = -\tilde{S}. \quad (5.7-7)$$

Substitution of Eqn (5.7-7) into Eqn (5.7-6) gives

$$\frac{\partial}{\partial T} \left(\frac{\tilde{G}}{T} \right)_P = -\frac{\tilde{H}}{T^2}, \quad (5.7-8)$$

which is known as the *Gibbs-Helmholtz equation*.

Example 5.12 Estimate the normal melting point of lead using the following data:

P (bar)	T (K)	Phase	$\tilde{H}$ (J/mol)	$\tilde{S}$ (J/mol K)
1.013	400	Solid	2770	72.8
1.013	1200	Liquid	31,160	113.1

The heat capacities at constant pressure are given by

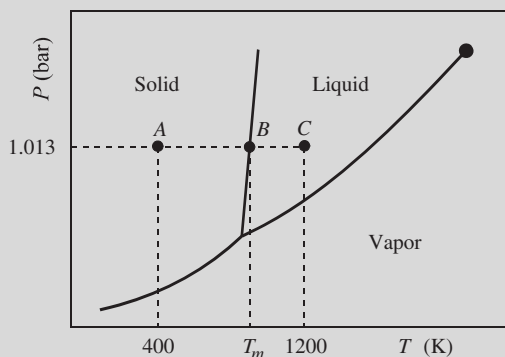
$$\tilde{C}_P^S = 23.6 + 9.75 \times 10^{-3}T \quad \text{and} \quad \tilde{C}_P^L = 32.4 - 3.1 \times 10^{-3}T$$

where $\tilde{C}_P$ is in J/mol K and T is in K.

Solution

A representative phase diagram of lead is shown below. At the normal melting point, T_m , solid and liquid phases are in equilibrium with each other (state B), i.e.

$$\frac{\tilde{G}^S|_{T_m}}{T_m} = \frac{\tilde{G}^L|_{T_m}}{T_m}. \quad (1)$$



At constant pressure, Eqn (5.7-8) is expressed in the form

$$d \left(\frac{\tilde{G}}{T} \right) = -\frac{\tilde{H}}{T^2} dT. \quad (2)$$

Integration of Eqn (2) requires the enthalpy to be expressed as a function of temperature, i.e.

$$\frac{d\tilde{H}}{dT} = \tilde{C}_P. \quad (3)$$

For the solid phase

$$\frac{d\tilde{H}^S}{dT} = 23.6 + 9.75 \times 10^{-3}T \Rightarrow \tilde{H}^S = 23.6T + 4.875 \times 10^{-3}T^2 + C_1.$$

From the value of enthalpy at 400 K, the integration constant C_1 is evaluated as -7450 . Thus

$$\tilde{H}^S = 23.6T + 4.875 \times 10^{-3}T^2 - 7450. \quad (4)$$

Substitution of Eqn (4) into Eqn (2) and integration from 400 K to T_m give

$$\frac{\tilde{G}^S|_{T_m}}{T_m} = 96.1 - \frac{7450}{T_m} - 23.6 \ln T_m - 4.875 \times 10^{-3}T_m. \quad (5)$$

For the liquid phase

$$\frac{d\tilde{H}^L}{dT} = 32.4 - 3.1 \times 10^{-3}T \Rightarrow \tilde{H}^L = 32.4T - 1.55 \times 10^{-3}T^2 + C_2.$$

Since the value of enthalpy at 1200 K is known, the integration constant C_2 is evaluated as -5488 . Thus,

$$\tilde{H}^L = 32.4T - 1.55 \times 10^{-3}T^2 - 5488. \quad (6)$$

Substitution of Eqn (6) into Eqn (2) and integration from T_m to 1200 K give

$$\frac{\tilde{G}^L|_{T_m}}{T_m} = 145.3 - \frac{5488}{T_m} - 32.4 \ln T_m + 1.55 \times 10^{-3}T_m. \quad (7)$$

Substitution of Eqns (5) and (7) into Eqn (1) and simplification lead to

$$49.2 + \frac{1962}{T_m} - 8.8 \ln T_m + 6.425 \times 10^{-3}T_m = 0. \quad (8)$$

The solution of Eqn (8) by MATHCAD[®] gives $T_m = 602.3$ K.

Comment: The actual value of the normal melting point of lead is 600.65 K as given in Example 5.11.

5.8 Variation of Fugacity with Pressure and Temperature

Variation of fugacity with pressure under isothermal conditions can be easily obtained from Eqn (5.2-7) as

$$\left(\frac{\partial \ln f_i}{\partial P} \right)_T = \frac{1}{RT} \underbrace{\left(\frac{\partial \tilde{G}_i}{\partial P} \right)_T}_{\tilde{V}_i} = \frac{\tilde{V}_i}{RT}. \quad (5.8-1)$$

On the other hand, differentiation of Eqn (5.2-1) with respect to temperature by keeping pressure constant gives

$$\left(\frac{\partial \ln f_i}{\partial T} \right)_P = \frac{1}{R} \frac{\partial}{\partial T} \left(\frac{\tilde{G}_i - \tilde{G}_i^{\text{IG}}}{T} \right)_P. \quad (5.8-2)$$

The use of the Gibbs-Helmholtz equation, Eqn (5.7-8), in Eqn (5.8-2) yields

$$\left(\frac{\partial \ln f_i}{\partial T} \right)_P = -\frac{\tilde{H}_i - \tilde{H}_i^{\text{IG}}}{RT^2}. \quad (5.8-3)$$

Note that the term $(\tilde{H}_i - \tilde{H}_i^{\text{IG}})$ is simply the departure function for molar enthalpy.

Variation in $\ln f_i$ is expressed in the form of an exact differential as

$$d \ln f_i = \left(\frac{\partial \ln f_i}{\partial P} \right)_T dP + \left(\frac{\partial \ln f_i}{\partial T} \right)_P dT. \quad (5.8-4)$$

Substitution of Eqns (5.8-1) and (5.8-3) into Eqn (5.8-4) gives

$$d \ln f_i = \left(\frac{\tilde{V}_i}{RT} \right) dP - \left(\frac{\tilde{H}_i - \tilde{H}_i^{\text{IG}}}{RT^2} \right) dT. \quad (5.8-5)$$

Example 5.13 Use Eqn (5.8-5) to express the fugacity of pure solid in terms of the fugacity of pure liquid.

Solution

For a liquid, Eqn (5.8-5) takes the form

$$d \ln f_i^{\text{L}} = \left(\frac{\tilde{V}_i^{\text{L}}}{RT} \right) dP - \left(\frac{\tilde{H}_i^{\text{L}} - \tilde{H}_i^{\text{IG}}}{RT^2} \right) dT. \quad (1)$$

For a solid, on the other hand, Eqn (5.8-5) becomes

$$d \ln f_i^{\text{S}} = \left(\frac{\tilde{V}_i^{\text{S}}}{RT} \right) dP - \left(\frac{\tilde{H}_i^{\text{S}} - \tilde{H}_i^{\text{IG}}}{RT^2} \right) dT. \quad (2)$$

Subtraction of Eqn (1) from Eqn (2) gives

$$d \ln \left(\frac{f_i^S}{f_i^L} \right) = -\frac{\Delta \tilde{V}_i^{\text{fus}}}{RT} dP + \frac{\Delta \tilde{H}_i^{\text{fus}}}{RT^2} dT. \quad (3)$$

Integration of Eqn (3) from the normal melting point, T_{m_i} , to T , and from 1 atm (~ 1 bar) to P , under the assumption that the heat of fusion is a constant leads to

$$\ln \left(\frac{f_i^S}{f_i^L} \right) = -\frac{\Delta \tilde{V}_i^{\text{fus}}(P - 1)}{RT} + \frac{\Delta \tilde{H}_i^{\text{fus}}}{RT_{m_i}} \left(1 - \frac{T_{m_i}}{T} \right). \quad (4)$$

Since the difference between the liquid and solid molar volumes is usually negligible, it is plausible to neglect the first term on the right-hand side of Eqn (4) without loss of accuracy. Hence, Eqn (4) simplifies to

$$\ln \left(\frac{f_i^S}{f_i^L} \right) = \frac{\Delta \tilde{H}_i^{\text{fus}}}{RT_{m_i}} \left(1 - \frac{T_{m_i}}{T} \right). \quad (5)$$

Comment: Equation (5) will be derived in Section 12.1 using a different approach.

Problems

Problems Related to Section 5.3

5.1 Derive Eqn (5.3-9) by substituting the van der Waals equation of state into Eqn (5.3-8).

5.2 Use Eqn (5.3-2) and calculate the fugacity coefficient of dimethyl ether at 420 K and 50 bar using the following data:

P (bar)	1	5	10	15	20	25	30	35	40	45	50
Z	0.994	0.969	0.937	0.905	0.871	0.836	0.799	0.761	0.721	0.678	0.631

(Answer: 0.72)

5.3 Calculate the fugacity of propylene at 400 K and 50 bar if it obeys

- (a) van der Waals equation of state.
- (b) Redlich-Kwong equation of state.
- (c) Peng-Robinson equation of state.

(Answer: (a) 37.207 bar, (b) 36.597 bar, and (c) 35.741 bar)

5.4 Calculate the fugacity of propane at 390 K and 30 bar if it obeys

- (a) Soave-Redlich-Kwong equation of state.
- (b) Peng-Robinson equation of state.

(Answer: (a) 24.141 bar and (b) 23.558 bar)

5.5 Using the following data for the fugacity coefficient of isobutane as a function of pressure at 500 K, estimate its molar volume at 500 K and 75 bar.

P (bar)	$\phi_{C_4H_{10}}$	P (bar)	$\phi_{C_4H_{10}}$	P (bar)	$\phi_{C_4H_{10}}$
0	1.000	40	0.803	80	0.653
10	0.947	50	0.761	90	0.624
20	0.896	60	0.722	100	0.599
30	0.848	70	0.686		

(Answer: 349.3 cm³/mol)

5.6 In Example 5.2, Eqn (3) indicates that the change in Gibbs energy under isothermal conditions is given by

$$\tilde{G}_2 - \tilde{G}_1 = RT \ln \left(\frac{f_2}{f_1} \right). \quad (1)$$

Combine Eqn (1) with Eqn (2.1-5) to show that

$$(\tilde{W}_s)_{\text{rev}} = RT \ln \left(\frac{f_2}{f_1} \right), \quad (2)$$

which can be used to estimate the work interaction of a steady-flow system under reversible and isothermal conditions.

Carbon dioxide is compressed in a reversible isothermal steady-state flow process from 3 bar and 350 K to 70 bar. Determine the work of compression per mole of carbon dioxide using the Peng-Robinson equation of state.

(Answer: 8509 J/mol)

Problems Related to Section 5.4

5.7 Estimate the fugacity of 1-octene at 480 K and 12 bar if it obeys the virial equation of state. The vapor pressure of 1-octene at 480 K is 6.791 bar.

(Answer: 5.919 bar)

5.8 Estimate the pressure at which the fugacity of liquid ethanol at 293 K is 5% higher than the fugacity of liquid at its saturation pressure.

(Answer: 23.5 bar)

- 5.9** Use the condition of vapor-liquid equilibrium and estimate the temperature of saturated liquid water at 4.5 bar by using the Peng-Robinson equation of state. Compare the result with the one obtained from the following equation:

$$\ln P_{\text{H}_2\text{O}}^{\text{vap}} = 13.8365 - \frac{6.6075 \times 10^3}{T} + \frac{10.1855 \times 10^5}{T^2} - \frac{17.7039 \times 10^7}{T^3},$$

$$300 \leq T \leq 600$$

in which $P_{\text{H}_2\text{O}}^{\text{vap}}$ is in bar and T is in K.

(Answer: 421.54 K)

- 5.10** Estimate the vapor pressure of ethylene at 240 K using the van der Waals equation of state.

(Answer: 25.5 bar)

- 5.11** Estimate the vapor pressure of liquid methane at 150 K using the Redlich-Kwong equation of state.

(Answer: 10.1 bar)

Problem Related to Section 5.5

- 5.12** Pure A exists in solid-vapor equilibrium in a rigid container at 150 K and 10 bar. Calculate the fugacity of pure solid if the vapor phase is described by the virial equation of state, i.e.

$$Z = 1 + \frac{BP}{RT},$$

where $B = 45 \times 10^{-5} \text{ m}^3/\text{mol}$.

(Answer: 14.35 bar)

Problems Related to Section 5.6

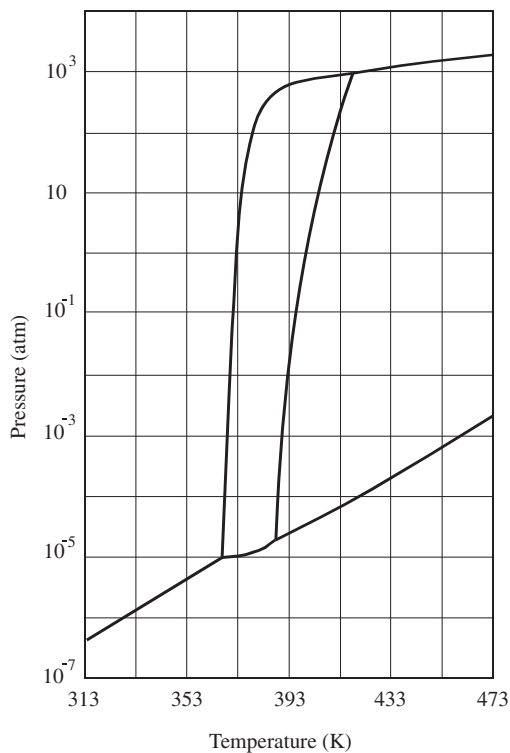
- 5.13** Consider transformation of saturated liquid to saturated vapor at constant temperature and pressure by the addition of heat.

- Show the process on a P - $\tilde{V}$ diagram.
- Show the process on a P - T diagram.
- Which plot is more informative for the given process? Explain.

- 5.14** A portion of the phase diagram of sulfur is shown below. There are two solid phases of sulfur, rhombic and monoclinic. The standard Gibbs energy of formation of rhombic sulfur is 0 kJ/mol and that of monoclinic sulfur is 0.33 kJ/mol at 298 K.

- (a) On the phase diagram, label the four regions and indicate the triple points.
- (b) Is it possible to completely melt rhombic sulfur at 473 K? If no, explain the reason clearly. If yes, what pressure should be imposed on rhombic sulfur?
- (c) It is required to convert rhombic sulfur to monoclinic sulfur and your friend suggests the following options:
- (i) Heat rhombic sulfur at 1 atm.
 - (ii) Heat rhombic sulfur at 1×10^{-6} atm.
 - (iii) Heat rhombic sulfur at 9×10^3 atm.
 - (iv) Decrease the pressure on rhombic sulfur at 338 K.
 - (v) Decrease the pressure on rhombic sulfur at 433 K.
 - (vi) Decrease the pressure on rhombic sulfur at 373 K.

Which one(s) of these processes is possible?



- 5.15** *Trouton's rule* (Trouton, 1883) states that the entropy of vaporization of liquids at their normal boiling point is constant, i.e.

$$\Delta \tilde{S}^{\text{vap}} = \frac{\Delta \tilde{H}^{\text{vap}}}{T} = 88 \text{ J/mol K.} \quad (1)$$

It should be kept in mind that Trouton's rule is an empirical relation and is not valid for hydrogen-bonded liquids.

- (a) Show that the combination of Trouton's rule with the Clausius-Clapeyron equation leads to

$$P_2^{\text{vap}} = P_1^{\text{vap}} \left(\frac{T_2}{T_1} \right)^{10.6}. \quad (2)$$

- (b) The normal boiling point of acetone is 56.5 °C. Calculate the vapor pressure of acetone at various temperatures using Eqn (2) and compare the values with the ones given in the literature.

- 5.16** An improved version of Trouton's rule, known as the *Trouton-Hildebrand-Everett rule*, is given by

$$\Delta \tilde{S}^{\text{vap}} = \frac{\Delta \tilde{H}^{\text{vap}}}{T} = R (4 + \ln T). \quad (1)$$

The vapor pressure of chloroform at 323.15 K is 0.703 bar. Use Eqn (1) to estimate the normal boiling point of chloroform. The experimental value is 334.3 K.

(Answer: 335.2 K)

- 5.17** As stated in Problem 3.13, an isotherm representing the cubic equation of state satisfies the following equation:

$$P^{\text{vap}}(\tilde{V}^{\text{V}} - \tilde{V}^{\text{L}}) = \int_{\tilde{V}^{\text{L}}}^{\tilde{V}^{\text{V}}} P \, d\tilde{V}. \quad (1)$$

- (a) Differentiate Eqn (1) with respect to temperature to obtain

$$\frac{dP^{\text{vap}}}{dT}(\tilde{V}^{\text{V}} - \tilde{V}^{\text{L}}) = \int_{\tilde{V}^{\text{L}}}^{\tilde{V}^{\text{V}}} \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} d\tilde{V}. \quad (2)$$

Hint: Use Leibnitz's rule to differentiate the right-hand side of Eqn (1).

- (b) Combine Eqn (2) with the Clapeyron equation and show that

$$\Delta \tilde{H}^{\text{vap}} = T \int_{\tilde{V}^{\text{L}}}^{\tilde{V}^{\text{V}}} \left(\frac{\partial P}{\partial T} \right)_{\tilde{V}} d\tilde{V}. \quad (3)$$

- (c) For the van der Waals equation of state, show that the integration of the right-hand side of Eqn (3) leads to

$$\frac{\Delta \tilde{H}^{\text{vap}}}{RT} = \ln \left(\frac{Z^{\text{V}} - B}{Z^{\text{L}} - B} \right). \quad (4)$$

- (d) For the Redlich-Kwong equation of state, show that the integration of the right-hand side of Eqn (3) leads to

$$\frac{\Delta \tilde{H}^{\text{vap}}}{RT} = \ln \left(\frac{Z^{\text{V}} - B}{Z^{\text{L}} - B} \right) + \frac{A}{2B} \ln \left[\frac{Z^{\text{V}}(Z^{\text{L}} + B)}{Z^{\text{L}}(Z^{\text{V}} + B)} \right]. \quad (5)$$

- (e) Estimate $\Delta \tilde{H}^{\text{vap}}$ for saturated benzene at 298.15 K if it obeys the Redlich-Kwong equation of state.

(Answer: (e) 32.65 kJ/mol)

5.18 Consider vapor-liquid equilibrium of a pure component.

- (a) Use $P\tilde{V} = ZRT$ and show that Eqn (5.6-6) takes the form

$$\frac{dP^{\text{vap}}}{P^{\text{vap}}} = \frac{\Delta \tilde{H}^{\text{vap}}}{R \Delta Z^{\text{vap}}} \frac{dT}{T^2}, \quad (1)$$

where $\Delta Z^{\text{vap}} = Z^{\text{V}} - Z^{\text{L}}$.

- (b) Assume $\Delta \tilde{H}^{\text{vap}} / \Delta Z^{\text{vap}}$ to be a constant and show that the integration of Eqn (1) from the critical pressure, P_c , to any pressure, P^{vap} , leads to

$$\log \left(\frac{P^{\text{vap}}}{P_c} \right) = \frac{\Delta \tilde{H}^{\text{vap}}}{2.303R \Delta Z^{\text{vap}} T_c} \left(1 - \frac{1}{T_r} \right). \quad (2)$$

- (c) Use Eqn (3.1-14) and show that

$$\frac{\Delta \tilde{H}^{\text{vap}}}{2.303R \Delta Z^{\text{vap}} T_c} = \frac{7}{3} (\omega + 1). \quad (3)$$

- (d) Combine Eqns (2) and (3) to get

$$\log \left(\frac{P^{\text{vap}}}{P_c} \right) = \frac{7}{3} (1 + \omega) \left(1 - \frac{1}{T_r} \right), \quad (4)$$

which can be used to estimate vapor pressures of liquids when $T_r \geq 0.6$.

- (e) Estimate the vapor pressure of carbon dioxide at 273.16 K using Eqn (4) and compare it with the literature value of 34.86 bar (Bignell and Bean, 1988).

5.19 Calculate the normal boiling point of *n*-propanol by the following two methods.

- Use vapor pressure data in the form of the Antoine equation.
- One way of estimating chemical and physical properties of pure compounds is the so-called *group contribution method*, which is based on the molecular structure of a compound. The property in question is estimated by the additive contributions of the functional groups constituting the molecule. To estimate normal boiling points, T_{nb} , of pure compounds, [Stein and Brown \(1994\)](#) proposed the following formula:

$$T_{nb}(K) = 198.2 + \sum_i n_i g_i, \quad (1)$$

where n_i is the number of groups of type i in the molecule and g_i is the contribution of each group. The normal boiling point predicted by Eqn (1) must be corrected by the following equations:

$$\begin{aligned} &T_{nb}(\text{corrected}) \\ &= \begin{cases} 1.5577T_{nb} - 94.84 - 7.705 \times 10^{-4}T_{nb}^2, & T_{nb} \leq 700 \text{ K}, \\ 0.4791T_{nb} + 282.7, & T_{nb} > 700 \text{ K}. \end{cases} \quad (2) \end{aligned}$$

Take the g_i values of 21.98, 24.22, and 88.46 for the CH_3 , CH_2 , and primary-OH groups.

(Answer: (a) 370.35 K and (b) 363.14 K)

5.20 As shown in Problem 4.1, the hydrostatic pressure distribution in fluids can be calculated from the equation

$$\frac{dP}{dz} = -\rho g, \quad (1)$$

where the distance z is measured in the direction opposite to gravity. The highest point on the earth's surface is the top of Mount Everest, located in the Himalayas on the border of Nepal and China. It is approximately 8900 m above sea level. Combine Eqn (1) with the Clausius-Clapeyron equation, Eqn (5.6-10), to estimate the boiling temperature of water at the top of Mount Everest. The following data are provided:

- The average rate of decrease in air temperature with altitude is 6.5 K/km.
- The air temperature at sea level is 293 K.
- The heat of vaporization of water is 40,800 J/mol and it may be considered independent of temperature.

- The molecular weight of air is 29 g/mol.

(Answer: 342.9 K)

- 5.21** Based on Eqn (5.6-12), it is possible to express the vapor pressure of substances as a function of temperature in the form

$$\log P_r^{\text{vap}} = a - \frac{b}{T_r}, \quad (1)$$

which is valid from the triple point to the critical point. For monatomic gases (Ar, Kr, Xe), [Pitzer *et al.* \(1955\)](#) plotted $\log P_r^{\text{vap}}$ versus $1/T_r$ and noted that $\log P_r^{\text{vap}}$ takes the value of -1 at $T_r = 0.7$ (or $1/T_r = 1.43$).

- (a) Using the following data for *n*-pentane, plot $\log P_r^{\text{vap}}$ versus $1/T_r$.

T (K)	298	323	348	373	398	423	448
P^{vap} (bar)	0.6833	1.5923	3.2340	5.9447	9.9109	15.5454	23.0773

- (b) Using the method of least squares show that the data are represented by

$$\log P_r^{\text{vap}} = 2.883 - \frac{2.894}{T_r}. \quad (2)$$

- (c) Use Eqn (3.1-14) and calculate the acentric factor for *n*-pentane.

- 5.22** Under atmospheric pressure, the heat of fusion and the melting temperature of *p*-xylene are 17,127 J/mol and 286.4 K, respectively. Determine the entropy change on fusion.

(Answer: 59.8 J/mol K)

- 5.23** The vapor pressure of *n*-heptanol is 84.398 kPa at 443.15 K and the normal boiling point is 448 K. Estimate the heat of vaporization of *n*-heptanol.

(Answer: 62,208 J/mol)

- 5.24** The vapor pressure of methyl sulfoxide is correlated by [Douglas \(1948\)](#) as a function of temperature in the form

$$\log P^{\text{vap}} = 26.5 - \frac{3539.32}{T} - 6 \log T, \quad 293.15 \leq T \leq 323.15,$$

where P^{vap} is in mmHg and T is in K. Estimate the heat of vaporization of methyl sulfoxide at 298.15 K.

(Answer: 52.9 kJ/mol)

- 5.25** Uracil ($\text{C}_4\text{H}_4\text{N}_2\text{O}_2$) is an organic base of the pyrimidine family. The sublimation pressure of uracil as a function of temperature is given below (Brunetti *et al.*, 2000):

T (K)	P (Pa)	T (K)	P (Pa)
397.0	4.365×10^{-2}	428.0	5.370×10^{-1}
404.0	8.128×10^{-2}	431.0	7.244×10^{-1}
409.0	1.260×10^{-1}	435.5	1.047
416.0	2.188×10^{-1}	440.0	1.479
419.5	2.951×10^{-1}	443.0	1.950
424.0	4.266×10^{-1}		

Calculate the heat of sublimation of uracil.

(Answer: 119.2 kJ/mol)

- 5.26** The following data are available for iodine:

$$\rho_{\text{liquid}} = 4000 \text{ kg/m}^3, \quad \rho_{\text{solid}} = 4930 \text{ kg/m}^3,$$

$$\ln P_{\text{liquid}}^{\text{vap}} = 47.83 - \frac{7381}{T} - 5.18 \ln T,$$

$$\ln P_{\text{solid}}^{\text{vap}} = 34.16 - \frac{8240}{T} - 2.51 \ln T,$$

where P^{vap} is in atm and T is in K.

- Calculate the values of temperature and pressure at the triple point and plot a representative phase diagram for iodine.
- When iodine vapor, initially at 0.06 atm and 430 K, is cooled at constant pressure, determine the temperature at which the condensed phase first appears? What is this condensed phase?
- When iodine vapor, initially at 0.21 atm and 430 K, is cooled at constant pressure, determine the temperature at which the condensed phase first appears? What is this condensed phase?
- Calculate the heat of fusion of iodine at the triple point.
- Calculate the normal boiling point of iodine.

(Answer: (a) 385.5 K, 0.115 atm; (b) 372.6 K; (c) 403 K; (d) 15,700 J/mol; and (e) 459 K)

- 5.27** Using the following data provided for *n*-hexanol ($\text{C}_6\text{H}_{14}\text{O}$)

- Normal boiling point = 430 K.
- Normal melting point = 226 K.
- $\Delta \tilde{H}^{\text{vap}} = 44.5 \text{ kJ/mol}$.

- $\Delta \tilde{H}^{\text{fus}} = 15.48 \text{ kJ/mol}$.

estimate its vapor pressure at

(a) 200 K.

(b) 415 K.

(Answer: (a) 2.08×10^{-7} atm and (b) 0.638 atm)

5.28 The vapor pressure of phenol as a function of temperature is given by

$$\ln P^{\text{vap}} = -11.04 \ln T - \frac{10,000}{T} + 93.275 + 4.32 \times 10^{-6} T^2,$$

where P^{vap} is in kPa and T is in K.

(a) Calculate the heat of vaporization at the triple point temperature of 314.06 K.

(b) If the heat of fusion is 11,514 J/mol, show that the sublimation pressure of phenol is expressed as

$$\ln P^{\text{sub}} = 31.34 - \frac{8181}{T},$$

where P^{sub} is in Pa and T is in K.

(Answer: (a) 56,501 J/mol)

5.29 The sublimation pressure of solid benzene is 299 Pa at 243 K. The vapor pressure of liquid benzene is 6022 Pa at 283 K. If the triple point of benzene is at 278.5 K and 4786 Pa, estimate its heat of fusion.

(Answer: 10.5 kJ/mol)

5.30 One gram of copper is placed in a rigid tank having a volume of 1 m^3 . Once the air is evacuated, the tank is sealed and the temperature is increased to 1600 K. The melting point of copper is 1356 K and the vapor pressure of liquid copper is given by

$$\ln P^{\text{vap}} = -\frac{39,129}{T} + 14.203,$$

where P^{vap} is in atm and T is in K.

(a) How many phases are present within the tank?

(b) Estimate the heat of vaporization of copper.

(Answer: (a) 2 (liquid and vapor) and (b) 325.3 kJ/mol)

- 5.31** Using the Peng-Robinson equation of state, estimate the maximum amount of propane that can be stored as vapor in a 0.1 m^3 tank at 250 K.
(Answer: 0.491 kg)
- 5.32** What should be the minimum pressure inside the container of a disposable butane lighter at a temperature of 298 K? Assume that *n*-butane is represented by the Peng-Robinson equation of state.
(Answer: 2.43 bar)
- 5.33** At what temperature does the pure methane liquefy under atmospheric conditions? Assume that methane is represented by the Peng-Robinson equation of state.
(Answer: 111.55 K)
- 5.34** Propylene is to be transported through a pipeline as a liquid. If the maximum temperature expected in the summer is around 330 K, estimate the minimum pressure that should be imposed in the pipeline. Assume that propylene is represented by the Peng-Robinson equation of state.
(Answer: 23.6 bar)
- 5.35** Determine the state of methanol, i.e. liquid or vapor, under the following conditions:
- (a) $T = 550 \text{ K}$ and $P = 2000 \text{ bar}$.
 - (b) $T = 550 \text{ K}$ and $P = 1 \text{ bar}$.
 - (c) $T = 350 \text{ K}$ and $P = 2 \text{ bar}$.
 - (d) $T = 350 \text{ K}$ and $P = 1 \text{ bar}$.
- 5.36** A piston-cylinder assembly initially contains 4 mol of substance X in vapor phase at 0.8 bar and 373 K. The vapor is compressed isothermally and reversibly until the first droplet of liquid forms. Calculate the heat and work interactions with the surroundings if the vapor obeys the virial equation of state with $B = -1887 \text{ cm}^3/\text{mol}$.
The normal boiling temperature and the heat of vaporization of substance X are 354.8 K and 29,750 J/mol, respectively.
(Answer: $Q = -9032 \text{ J}$ and $W = 9032 \text{ J}$).
- 5.37** When common gas-liquid-solid transitions take place, while temperature remains constant during vaporization, condensation, and sublimation, volume and entropy show abrupt change. Such transformations are called “first-order

transitions.” Noting that

$$\left(\frac{\partial G}{\partial P}\right)_T = V \quad \text{and} \quad \left(\frac{\partial G}{\partial T}\right)_P = -S,$$

the derivatives of Gibbs energy are discontinuous for a first-order transition. Since $H = G + TS$, enthalpy is also discontinuous at the transition temperature. A “second-order transition,” on the other hand, is defined as one in which the first derivative of Gibbs energy is continuous but the second derivative is discontinuous at the transition temperature. In other words, quantities such as volume, entropy, and enthalpy remain unchanged during phase change. Transition of a paramagnetic material into a ferromagnetic one or vice versa is an example of a second-order transition.

How does the heat capacity at constant pressure vary during the first-order and second-order transitions?

Problem Related to Section 5.8

5.38 Lee and Edmister (1971) proposed the following empirical equation for predicting the fugacity of pure liquid hydrocarbons:

$$\begin{aligned} \ln \left(\frac{f}{P} \right) = & A_1 + \frac{A_2}{T_r} + A_3 \ln T_r + A_4 T_r^2 + A_5 T_r^6 \\ & + \left(\frac{A_6}{T_r} + A_7 \ln T_r + A_8 T_r^2 \right) P_r + A_9 T_r^3 P_r^2 \\ & - \ln P_r + \omega \left[(1 - T_r) \left(A_{10} + \frac{A_{11}}{T_r} \right) + A_{12} \frac{P_r}{T_r} + A_{13} T_r^3 P_r^2 \right], \end{aligned}$$

where

$$\begin{aligned} A_1 &= 6.32873, & A_6 &= -0.018706, & A_{11} &= -11.201, \\ A_2 &= -8.45167, & A_7 &= -0.286517, & A_{12} &= -0.05044, \\ A_3 &= -6.90287, & A_8 &= 0.18940, & A_{13} &= 0.002255 \\ A_4 &= 1.87895, & A_9 &= -0.002584, \\ A_5 &= -0.33448, & A_{10} &= 8.7015, \end{aligned}$$

Estimate the specific enthalpy of liquid *n*-pentane at 433 K and 18.5 bar relative to that of the ideal gas at the same temperature.

(Answer: -264.5 J/g)

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Chapter 6

Thermodynamics of Mixtures

A mixture is a gaseous, liquid, or solid phase containing more than one component. The chemical process industries mainly deal with the separation of mixtures, and their properties are needed in design calculations. Mixture properties, however, cannot be determined from the weighted average of the properties of the pure components comprising the mixture. The purpose of this chapter is to show how to calculate the mixture properties.

6.1 Equations of State for Mixtures

6.1.1 Virial Equation of State

At low to moderate pressures (up to ~ 15 bar), the virial equation of state, truncated after the second term, for a gas mixture consisting of k species is given by

$$Z_{\text{mix}} = 1 + \frac{B_{\text{mix}}P}{RT}, \quad (6.1-1)$$

where the second virial coefficient for the mixture, B_{mix} , is defined as

$$B_{\text{mix}} = \sum_{i=1}^k \sum_{j=1}^k y_i y_j B_{ij} \quad (6.1-2)$$

in which y_i represents the mole fraction of species i . When $i = j$, the terms B_{ii} and B_{jj} represent second virial coefficients corresponding to the pure components i and j , respectively. When $i \neq j$, second virial cross coefficients are symmetric, i.e. $B_{ij} = B_{ji}$. In engineering calculations involving normal (nonpolar and a few

slightly polar) fluids, second virial cross coefficients, B_{ij} , can be estimated from Eqns (3.1.10)–(3.1.12) with the following modifications of T_c , P_c , and ω :

$$T_{cij} = (1 - k_{ij}^*) \sqrt{T_{ci} T_{cj}} \quad P_{cij} = \frac{4T_{cij}}{(\tilde{V}_{ci}^{1/3} + \tilde{V}_{cj}^{1/3})^3} \left(\frac{P_{ci} \tilde{V}_{ci}}{T_{ci}} + \frac{P_{cj} \tilde{V}_{cj}}{T_{cj}} \right)$$

$$\omega_{ij} = \frac{\omega_i + \omega_j}{2}, \quad (6.1-3)$$

where $\tilde{V}_{ci}$ and $\tilde{V}_{cj}$ are the critical molar volumes of components i and j , respectively.¹ The values of the binary interaction parameter,² k_{ij}^* , have been reported by Chueh and Prausnitz (1967a), Tsonopoulos (1979), Nishiumi *et al.* (1988), and Meng *et al.* (2007). For binary mixtures of chemically similar molecules, i.e. similar size, shape, and chemical nature, k_{ij}^* may be considered zero. Chueh and Prausnitz (1967b) proposed the following equation to estimate k_{ij}^* :

$$k_{ij}^* = 1 - 8 \left(\frac{\tilde{V}_{ci}^{1/6} \tilde{V}_{cj}^{1/6}}{\tilde{V}_{ci}^{1/3} + \tilde{V}_{cj}^{1/3}} \right)^3. \quad (6.1-4)$$

In the absence of reliable data, the second virial cross coefficient, B_{ij} , may be assumed to be the arithmetic mean of B_{ii} and B_{jj} , i.e. $B_{ij} \simeq (B_{ii} + B_{jj})/2$.

Example 6.1 A rigid tank contains 3 mol of ethane (1) at 373 K and 4 bar. Estimate the pressure in the tank when 5 mol of *n*-butane (2) is added isothermally to the tank. The mixture is described by the virial equation of state and the following critical properties are provided:

Component	$\tilde{V}_c$ (cm ³ /mol)
Ethane	145.5
<i>n</i> -Butane	255.0

Solution

From Appendix A

¹In the absence of data, critical molar volume can be estimated from the following formula:

$$\tilde{V}_c = \frac{RT_c}{P_c} (0.290 - 0.085\omega).$$

²Keep in mind that interaction parameters have no theoretical basis; they are simply adjustable parameters to fit experimental data.

Component	T_c (K)	P_c (bar)	ω
Ethane	305.3	49	0.099
<i>n</i> -Butane	425.0	38	0.199

The final pressure in the tank can be estimated from Eqn (6.1-1), i.e.

$$(Z_{\text{mix}})_f = \frac{P_f(\tilde{V}_{\text{mix}})_f}{RT} = 1 + \frac{(B_{\text{mix}})_f P_f}{RT} \implies P_f = \frac{RT}{(\tilde{V}_{\text{mix}})_f - (B_{\text{mix}})_f}, \quad (1)$$

where the subscript “f” represents the final state. The term B_{mix} is calculated from Eqn (6.1-2) as

$$B_{\text{mix}} = y_1^2 B_{11} + 2y_1 y_2 B_{12} + y_2^2 B_{22}. \quad (2)$$

The total tank volume, V , can be calculated from the initial conditions, i.e. when the tank is initially filled with ethane. The use of the virial equation of state gives

$$Z_{\text{initial}} = \frac{P_{\text{initial}} V}{n_{\text{initial}} RT} = 1 + \frac{B_{11} P_{\text{initial}}}{RT} \implies V = n_{\text{initial}} \left(\frac{RT}{P_{\text{initial}}} + B_{11} \right). \quad (3)$$

Thus, the molar volume of the mixture at the final state is

$$(\tilde{V}_{\text{mix}})_f = \frac{V}{n_f}. \quad (4)$$

The second virial coefficients for pure components, i.e. B_{11} and B_{22} , are calculated from Eqns (3.1.10)–(3.1.12). The results are given in the table below:

Component	T_r	$B^{(0)}$	$B^{(1)}$	B_{ii} (cm ³ /mol)
Ethane	1.222	−0.223	0.065	−112.3
<i>n</i> -Butane	0.878	−0.437	−0.159	−435.7

To calculate the second virial cross coefficient, B_{12} , it is first necessary to calculate the parameters defined by Eqns (6.1-3) and (6.1-4):

$$k_{12}^* = 1 - 8 \left[\frac{(145.5)^{1/6} (255)^{1/6}}{145.5^{1/3} + 255^{1/3}} \right]^3 = 0.013,$$

$$T_{c12} = (1 - 0.013) \sqrt{(305.3)(425)} = 355.5 \text{ K},$$

$$\omega_{12} = \frac{0.099 + 0.199}{2} = 0.149,$$

$$P_{c12} = \frac{(4)(355.5)}{(145.5^{1/3} + 255^{1/3})^3} \left[\frac{(49)(145.5)}{305.3} + \frac{(38)(255)}{425} \right] = 42.04 \text{ bar}.$$

The parameters $B_{12}^{(0)}$ and $B_{12}^{(1)}$ are calculated from Eqns (3.1-11) and (3.1-12), respectively, as

$$B_{12}^{(0)} = 0.083 - \frac{0.422}{(373/355.5)^{1.6}} = -0.308,$$

$$B_{12}^{(1)} = 0.139 - \frac{0.172}{(373/355.5)^{4.2}} = -1.57 \times 10^{-3}.$$

Finally, the second virial cross coefficient B_{12} is calculated from Eqn (3.1-10) as

$$B_{12} = \frac{(83.14)(355.5)}{42.04} [-0.308 + (0.149)(-1.57 \times 10^{-3})]$$

$$= -216.7 \text{ cm}^3/\text{mol}.$$

The term B_{mix} is calculated from Eqn (2) as

$$B_{\text{mix}} = \left(\frac{3}{8}\right)^2 (-112.3) + 2 \left(\frac{3}{8}\right) \left(\frac{5}{8}\right) (-216.7) + \left(\frac{5}{8}\right)^2 (-435.7)$$

$$= -287.6 \text{ cm}^3/\text{mol}.$$

The tank volume is calculated from Eqn (3) as

$$V = (3) \left[\frac{(83.14)(373)}{4} - 112.3 \right] = 22,922 \text{ cm}^3$$

The use of Eqn (4) gives

$$(\tilde{V}_{\text{mix}})_f = \frac{22,922}{8} = 2865.3 \text{ cm}^3/\text{mol}.$$

Substitution of the numerical values into Eqn (1) results in

$$P_f = \frac{(83.14)(373)}{2865.3 + 287.6} = 9.8 \text{ bar}.$$

Example 6.2 A gas mixture containing 90 mol% methane (1) and 10% ethane (2) is irreversibly compressed from 1 bar and 310 K to 30 bar and 350 K in an adiabatic compressor. If the gas mixture is represented by the virial equation of

state, estimate the work required per mole of gas mixture. The virial coefficients are expressed as a function of temperature in the form

$$B_{ij} = \alpha_{ij} - \frac{\beta_{ij}}{T} - \frac{\gamma_{ij}}{T^2}$$

in which the parameters are given by Estela-Urbe *et al.* (2003) as

$$\begin{aligned} \alpha_{11} &= 42.19 & \alpha_{12} &= 100.19 & \alpha_{22} &= 89.75, \\ \beta_{11} &= 17.13 \times 10^3 & \beta_{12} &= 52.82 \times 10^3 & \beta_{22} &= 49.80 \times 10^3, \\ \gamma_{11} &= 24.57 \times 10^5 & \gamma_{12} &= 12.63 \times 10^5 & \gamma_{22} &= 96.19 \times 10^5 \end{aligned}$$

with B is in cm^3/mol and T is in K.

Solution

From Appendix B

$$\begin{aligned} \tilde{C}_{P_1}^* &= 36.155 - 0.511 \times 10^{-1}T + 2.215 \times 10^{-4}T^2 \\ &\quad - 1.824 \times 10^{-7}T^3 + 4.899 \times 10^{-11}T^4, \\ \tilde{C}_{P_2}^* &= 33.313 - 0.111 \times 10^{-1}T + 3.566 \times 10^{-4}T^2 \\ &\quad - 3.762 \times 10^{-7}T^3 + 11.983 \times 10^{-11}T^4. \end{aligned}$$

System: Compressor

For a steady-state flow system, the first law gives

$$\tilde{W}_s = \Delta \tilde{H}. \quad (1)$$

The change in enthalpy can be determined from Eqn (3.3-13), i.e.

$$\begin{aligned} \Delta \tilde{H} &= \int_{P_1}^0 \left[\tilde{V}_{\text{mix}} - T \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial T} \right)_P \right]_{T_1} dP + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT \\ &\quad + \int_0^{P_2} \left[\tilde{V}_{\text{mix}} - T \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial T} \right)_P \right]_{T_2} dP. \end{aligned} \quad (2)$$

The use of the virial equation of state, Eqn (6.1-1), leads to

$$Z_{\text{mix}} = \frac{P \tilde{V}_{\text{mix}}}{RT} = 1 + \frac{B_{\text{mix}} P}{RT} \Rightarrow \tilde{V}_{\text{mix}} = \frac{RT}{P} + B_{\text{mix}}. \quad (3)$$

The partial derivative appearing in Eqn (2) is

$$\left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial T} \right)_P = \frac{R}{P} + \frac{dB_{\text{mix}}}{dT} \quad (4)$$

and the integrand in Eqn (2) becomes

$$\tilde{V}_{\text{mix}} - T \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial T} \right)_P = B_{\text{mix}} - T \frac{dB_{\text{mix}}}{dT}. \quad (5)$$

Therefore, it is first necessary to express B_{mix} as a function of temperature. From Eqn (6.1-2)

$$B_{\text{mix}} = y_1^2 B_{11} + 2y_1 y_2 B_{12} + y_2^2 B_{22}. \quad (6)$$

Expression of virial coefficients in terms of temperature leads to

$$B_{\text{mix}} = \varphi_1 - \frac{\varphi_2}{T} - \frac{\varphi_3}{T^2}, \quad (7)$$

where

$$\varphi_1 = y_1^2 \alpha_{11} + 2y_1 y_2 \alpha_{12} + y_2^2 \alpha_{22}, \quad (8)$$

$$\varphi_2 = y_1^2 \beta_{11} + 2y_1 y_2 \beta_{12} + y_2^2 \beta_{22}, \quad (9)$$

$$\varphi_3 = y_1^2 \gamma_{11} + 2y_1 y_2 \gamma_{12} + y_2^2 \gamma_{22}. \quad (10)$$

The use of Eqn (7) in Eqn (5) gives

$$\tilde{V}_{\text{mix}} - T \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial T} \right)_P = \varphi_1 - \frac{2\varphi_2}{T} - \frac{3\varphi_3}{T^2}. \quad (11)$$

Substitution of Eqn (11) into Eqn (2) and integration give

$$\Delta \tilde{H} = \varphi_1 (P_2 - P_1) - 2\varphi_2 \left(\frac{P_2}{T_2} - \frac{P_1}{T_1} \right) - 3\varphi_3 \left(\frac{P_2}{T_2^2} - \frac{P_1}{T_1^2} \right) + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT. \quad (12)$$

The numerical values of φ_1 , φ_2 , and φ_3 are calculated from Eqns (8–10) as

$$\begin{aligned} \varphi_1 &= 53.106 \text{ cm}^3/\text{mol} \quad \varphi_2 = 2.388 \times 10^4 \text{ cm}^3 \text{ K/mol} \\ \varphi_3 &= 2.314 \times 10^6 \text{ cm}^3 \text{ K}^2/\text{mol}. \end{aligned}$$

The molar heat capacity of an ideal gas mixture is the mole fraction-weighted sum of the pure molar heat capacities, i.e.

$$\begin{aligned} \tilde{C}_{P_{\text{mix}}}^* &= y_1 \tilde{C}_{P_1}^* + y_2 \tilde{C}_{P_2}^* = 35.871 - 0.471 \times 10^{-1} T + 2.350 \times 10^{-4} T^2 \\ &\quad - 2.018 \times 10^{-7} T^3 + 5.607 \times 10^{-11} T^4. \end{aligned} \quad (13)$$

Thus, the numerical value of the integral in Eqn (12) is

$$\int_{310}^{350} \tilde{C}_{P_{\text{mix}}}^* dT = 1574 \text{ J/mol}.$$

Substitution of the numerical values into Eqn (12) yields

$$\begin{aligned}\Delta\tilde{H} = & 53.106 \times 10^{-6}(30 - 1) \times 10^5 - 2(2.388 \times 10^{-2}) \left(\frac{30}{350} - \frac{1}{310} \right) \\ & \times 10^5 - 3(2.314) \left(\frac{30}{350^2} - \frac{1}{310^2} \right) \times 10^5 + 1574 = 1171.3 \text{ J/mol.}\end{aligned}$$

6.1.2 Cubic Equations of State

The cubic equations of state given in Table 3.1 are also valid for gas and liquid mixtures. Table 6.1 summarizes these equations for mixtures. The practical question to be asked at this stage is how to evaluate the mixture parameters, i.e. a_{mix} and b_{mix} , once pure component parameters are calculated.

According to the van der Waals (or quadratic) mixing rules,³ the mixture parameters are given by the following composition-dependent expressions:

$$a_{\text{mix}} = \sum_{i=1}^k \sum_{j=1}^k x_i x_j a_{ij}, \quad (6.1-5)$$

$$b_{\text{mix}} = \sum_{i=1}^k \sum_{j=1}^k x_i x_j \left(\frac{b_i + b_j}{2} \right) = \sum_{i=1}^k x_i b_i, \quad (6.1-6)$$

where x_i represents the mole fraction of species i . In Eqns (6.1-5) and (6.1-6), the terms a_{ii} (or a_{jj}) and b_i are the pure component parameters. When $i \neq j$, the term a_{ij} is called the *unlike interaction parameter* and is related to the pure component parameters as

$$a_{ij} = \begin{cases} \sqrt{a_{ii} a_{jj}} & \text{VDW} \\ (1 - k_{ij}) \sqrt{a_{ii} a_{jj}} & \text{RK, SRK, PR} \end{cases} \quad (6.1-7)$$

where k_{ij} is the adjustable binary interaction parameter with a symmetric property, i.e. $k_{ij} = k_{ji}$.

The parameter α_i in the Soave-Redlich-Kwong (SRK) and Peng-Robinson (PR) equations of state is defined by

$$\alpha_i = \begin{cases} [1 + (0.480 + 1.574\omega_i - 0.176\omega_i^2)(1 - \sqrt{T_{ri}})]^2 & \text{SRK,} \\ [1 + (0.37464 + 1.54226\omega_i - 0.26992\omega_i^2)(1 - \sqrt{T_{ri}})]^2 & \text{PR.} \end{cases} \quad (6.1-8)$$

³Equations (6.1-5) and (6.1-6) are not the only equations used to calculate the parameters for the mixtures. Ghosh (1999) and Twu *et al.* (2002) summarized various mixing rules used in the literature. See also Orbey and Sandler (1998).

Table 6.1: Cubic equations of state for mixtures.^a

Equation	P	a_{ii}	b_i
VDW	$\frac{RT}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} - \frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}^2}$	$\frac{27}{64} \left(\frac{R^2 T_{c_i}^2}{P_{c_i}} \right)$	$\frac{1}{8} \left(\frac{RT_{c_i}}{P_{c_i}} \right)$
RK	$\frac{RT}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} - \frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}(\tilde{V}_{\text{mix}} + b_{\text{mix}})\sqrt{T}}$	$0.42748 \left(\frac{R^2 T_{c_i}^{2.5}}{P_{c_i}} \right)$	$0.08664 \left(\frac{RT_{c_i}}{P_{c_i}} \right)$
SRK	$\frac{RT}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} - \frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}(\tilde{V}_{\text{mix}} + b_{\text{mix}})}$	$0.42748 \left(\frac{R^2 T_{c_i}^2}{P_{c_i}} \right) \alpha_i$	$0.08664 \left(\frac{RT_{c_i}}{P_{c_i}} \right)$
PR	$\frac{RT}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} - \frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}(\tilde{V}_{\text{mix}} + b_{\text{mix}}) + b_{\text{mix}}(\tilde{V}_{\text{mix}} - b_{\text{mix}})}$	$0.45724 \left(\frac{R^2 T_{c_i}^2}{P_{c_i}} \right) \alpha_i$	$0.07780 \left(\frac{RT_{c_i}}{P_{c_i}} \right)$

^aVDW—van der Waals, RK—Redlich-Kwong, SRK—Soave-Redlich-Kwong, and PR—Peng-Robinson.

On the other hand, the compressibility factor for the mixture, Z_{mix} , is obtained from

$$Z_{\text{mix}}^3 + pZ_{\text{mix}}^2 + qZ_{\text{mix}} + r = 0, \quad (6.1-9)$$

where the coefficients p , q , and r are given in Table 6.2. The dimensionless parameters for the mixture are defined by

$$A_{\text{mix}} = \sum_{i=1}^k \sum_{j=1}^k x_i x_j A_{ij}, \quad (6.1-10)$$

Table 6.2: Parameters in Eqn (6.1-9).

Equation	p	q	r
VDW	$-1 - B_{\text{mix}}$	A_{mix}	$-A_{\text{mix}}B_{\text{mix}}$
RK	-1	$A_{\text{mix}} - B_{\text{mix}} - B_{\text{mix}}^2$	$-A_{\text{mix}}B_{\text{mix}}$
SRK	-1	$A_{\text{mix}} - B_{\text{mix}} - B_{\text{mix}}^2$	$-A_{\text{mix}}B_{\text{mix}}$
PR	$-1 + B_{\text{mix}}$	$A_{\text{mix}} - 2B_{\text{mix}} - 3B_{\text{mix}}^2$	$-A_{\text{mix}}B_{\text{mix}} + B_{\text{mix}}^2 + B_{\text{mix}}^3$

Table 6.3: Dimensionless parameters A_{ij} and B_i .

Equation	A_{ii}	A_{ij}	B_i
van der Waals	$\frac{27}{64} \left(\frac{P_r}{T_r^2} \right)_i$	$\sqrt{A_{ii}A_{jj}}$	$\frac{1}{8} \left(\frac{P_r}{T_r} \right)_i$
Redlich-Kwong	$0.42748 \left(\frac{P_r}{T_r^{2.5}} \right)_i$	$(1 - k_{ij})\sqrt{A_{ii}A_{jj}}$	$0.08664 \left(\frac{P_r}{T_r} \right)_i$
Soave-Redlich-Kwong	$0.42748 \left(\frac{P_r}{T_r^2} \right)_i \alpha_i$	$(1 - k_{ij})\sqrt{A_{ii}A_{jj}}$	$0.08664 \left(\frac{P_r}{T_r} \right)_i$
Peng-Robinson	$0.45724 \left(\frac{P_r}{T_r^2} \right)_i \alpha_i$	$(1 - k_{ij})\sqrt{A_{ii}A_{jj}}$	$0.07780 \left(\frac{P_r}{T_r} \right)_i$

$$B_{\text{mix}} = \sum_{i=1}^k x_i B_i \quad (6.1-11)$$

in which the terms A_{ij} and B_i are defined in Table 6.3.

Example 6.3 Calculate the molar volume of a gas mixture consisting of 50 mol% propane (1), 20% *n*-butane (2), and 30% *n*-pentane (3) at 393 K and 40 bar using

- van der Waals equation of state.
- Peng-Robinson equation of state ($k_{12} = 0.003$, $k_{13} = 0.027$, $k_{23} = 0.017$).

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Propane	369.9	42.5	0.153
<i>n</i> -Butane	425.0	38.0	0.199
<i>n</i> -Pentane	469.8	33.6	0.251

The molar volume of the mixture is calculated from

$$\tilde{V}_{\text{mix}} = \frac{Z_{\text{mix}}^V RT}{P}.$$

Therefore, it is first necessary to estimate the compressibility factor for the

mixture, Z_{mix}^V . The reduced temperature and pressure values are

Component	T_r	P_r
Propane	1.062	0.941
<i>n</i> -Butane	0.925	1.053
<i>n</i> -Pentane	0.837	1.190

(a) Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 0.352 \quad A_{12} = A_{21} = 0.427 \quad B_1 = 0.111, \\ A_{22} &= 0.519 \quad A_{13} = A_{31} = 0.502 \quad B_2 = 0.142, \\ A_{33} &= 0.717 \quad A_{23} = A_{32} = 0.610 \quad B_3 = 0.178. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}} = 0.482 \quad \text{and} \quad B_{\text{mix}} = 0.137.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}}^3 - 1.137Z_{\text{mix}}^2 + 0.482Z_{\text{mix}} - 0.066 = 0 \quad \Rightarrow \quad Z_{\text{mix}}^V = 0.26.$$

The molar volume of the mixture is

$$\tilde{V}_{\text{mix}} = \frac{Z_{\text{mix}}^V RT}{P} = \frac{(0.26)(83.14)(393)}{40} = 212.4 \text{ cm}^3/\text{mol}.$$

(b) The use of Eqn (6.1-8) gives the parameter α as

Component	α
Propane	0.963
<i>n</i> -Butane	1.052
<i>n</i> -Pentane	1.131

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 0.367 \quad A_{12} = A_{21} = 0.465 \quad B_1 = 0.069, \\ A_{22} &= 0.592 \quad A_{13} = A_{31} = 0.553 \quad B_2 = 0.089, \\ A_{33} &= 0.878 \quad A_{23} = A_{32} = 0.709 \quad B_3 = 0.111. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}} = 0.538 \quad \text{and} \quad B_{\text{mix}} = 0.086.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}}^3 - 0.914Z_{\text{mix}}^2 + 0.344Z_{\text{mix}} - 0.038 = 0 \quad \Rightarrow \quad Z_{\text{mix}}^V = 0.179.$$

The molar volume of the mixture is

$$\tilde{V}_{\text{mix}} = \frac{Z_{\text{mix}}^V RT}{P} = \frac{(0.179)(83.14)(393)}{40} = 146.2 \text{ cm}^3/\text{mol}.$$

6.1.2.1 Calculation of the change in internal energy

The equations developed in Section 3.2 can be extended to mixtures with the following modifications.

van der Waals equation of state: For mixtures, Eqn (3.2-16) becomes

$$\Delta \tilde{U} = R \left(\frac{A_{\text{mix}_1} T_1}{Z_{\text{mix}_1}} - \frac{A_{\text{mix}_2} T_2}{Z_{\text{mix}_2}} \right) + \int_{T_1}^{T_2} \tilde{C}_{V_{\text{mix}}}^* dT, \quad (6.1-12)$$

where

$$\tilde{C}_{V_{\text{mix}}}^* = \sum_{i=1}^k x_i \tilde{C}_{V_i}^*. \quad (6.1-13)$$

Redlich-Kwong equation of state: For mixtures, Eqn (3.2-17) becomes

$$\begin{aligned} \Delta \tilde{U} = \frac{3R}{2} \left[\frac{A_{\text{mix}_1} T_1}{B_{\text{mix}_1}} \ln \left(1 + \frac{B_{\text{mix}_1}}{Z_{\text{mix}_1}} \right) - \frac{A_{\text{mix}_2} T_2}{B_{\text{mix}_2}} \ln \left(1 + \frac{B_{\text{mix}_2}}{Z_{\text{mix}_2}} \right) \right] \\ + \int_{T_1}^{T_2} \tilde{C}_{V_{\text{mix}}}^* dT. \end{aligned} \quad (6.1-14)$$

Peng-Robinson equation of state: For mixtures, Eqn (3.2.18) becomes

$$\begin{aligned} \Delta \tilde{U} = \frac{T_2 (da_{\text{mix}}/dT)_{T_2} - a_{\text{mix}_2}}{\sqrt{8}b_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}_2} + (1 + \sqrt{2}) B_{\text{mix}_2}}{Z_{\text{mix}_2} + (1 - \sqrt{2}) B_{\text{mix}_2}} \right] \\ - \frac{T_1 (da_{\text{mix}}/dT)_{T_1} - a_{\text{mix}_1}}{\sqrt{8}b_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}_1} + (1 + \sqrt{2}) B_{\text{mix}_1}}{Z_{\text{mix}_1} + (1 - \sqrt{2}) B_{\text{mix}_1}} \right] \\ + \int_{T_1}^{T_2} \tilde{C}_{V_{\text{mix}}}^* dT, \end{aligned} \quad (6.1-15)$$

where

$$\frac{da_{\text{mix}}}{dT} = -\frac{1}{2T} \sum_{i=1}^k \sum_{j=1}^k x_i x_j a_{ij} (\Gamma_i + \Gamma_j) \quad (6.1-16)$$

in which

$$\Gamma_i = (0.37464 + 1.54226 \omega_i - 0.26992 \omega_i^2) \sqrt{\frac{T_{r_i}}{\alpha_i}}. \quad (6.1-17)$$

Substitution of Eqn (6.1-16) into Eqn (6.1-15) gives

$$\begin{aligned} \Delta \tilde{U} = & \frac{RT_2 \Omega_{\text{mix}_2}}{\sqrt{8} B_{\text{mix}_2}} \ln \left[\frac{Z_{\text{mix}_2} + (1 + \sqrt{2}) B_{\text{mix}_2}}{Z_{\text{mix}_2} + (1 - \sqrt{2}) B_{\text{mix}_2}} \right] - \frac{RT_1 \Omega_{\text{mix}_1}}{\sqrt{8} B_{\text{mix}_1}} \\ & \times \ln \left[\frac{Z_{\text{mix}_1} + (1 + \sqrt{2}) B_{\text{mix}_1}}{Z_{\text{mix}_1} + (1 - \sqrt{2}) B_{\text{mix}_1}} \right] + \int_{T_1}^{T_2} \tilde{C}_{V_{\text{mix}}}^* dT, \end{aligned} \quad (6.1-18)$$

where

$$\Omega_{\text{mix}} = -\frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k x_i x_j A_{ij} (\Gamma_i + \Gamma_j) - A_{\text{mix}}. \quad (6.1-19)$$

6.1.2.2 Calculation of the change in enthalpy

The equations developed in Section 3.3 can be extended to mixtures with the following modifications.

van der Waals equation of state: For mixtures, Eqn (3.3-21) becomes

$$\Delta \tilde{H} = RT_2 \left(Z_{\text{mix}_2} - 1 - \frac{A_{\text{mix}_2}}{Z_{\text{mix}_2}} \right) - RT_1 \left(Z_{\text{mix}_1} - 1 - \frac{A_{\text{mix}_1}}{Z_{\text{mix}_1}} \right) + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT \quad (6.1-20)$$

where

$$\tilde{C}_{P_{\text{mix}}}^* = \sum_{i=1}^k x_i \tilde{C}_{P_i}^* \quad (6.1-21)$$

Redlich-Kwong equation of state: For mixtures, Eqn (3.3-22) becomes

$$\begin{aligned} \Delta \tilde{H} = & RT_2 \left[Z_{\text{mix}_2} - 1 - \frac{3 A_{\text{mix}_2}}{2 B_{\text{mix}_2}} \ln \left(1 + \frac{B_{\text{mix}_2}}{Z_{\text{mix}_2}} \right) \right] \\ & - RT_1 \left[Z_{\text{mix}_1} - 1 - \frac{3 A_{\text{mix}_1}}{2 B_{\text{mix}_1}} \ln \left(1 + \frac{B_{\text{mix}_1}}{Z_{\text{mix}_1}} \right) \right] \\ & + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT. \end{aligned} \quad (6.1-22)$$

Peng-Robinson equation of state: For mixtures, Eqn (3.3-24) becomes

$$\begin{aligned} \Delta \tilde{H} = & RT_2 \left\{ Z_{\text{mix}_2} - 1 + \frac{\Omega_{\text{mix}_2}}{\sqrt{8}B_{\text{mix}_2}} \ln \left[\frac{Z_{\text{mix}_2} + (1 + \sqrt{2}) B_{\text{mix}_2}}{Z_{\text{mix}_2} + (1 - \sqrt{2}) B_{\text{mix}_2}} \right] \right\} \\ & - RT_1 \left\{ Z_{\text{mix}_1} - 1 + \frac{\Omega_{\text{mix}_1}}{\sqrt{8}B_{\text{mix}_1}} \ln \left[\frac{Z_{\text{mix}_1} + (1 + \sqrt{2}) B_{\text{mix}_1}}{Z_{\text{mix}_1} + (1 - \sqrt{2}) B_{\text{mix}_1}} \right] \right\} \\ & + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT, \end{aligned} \quad (6.1-23)$$

where Ω_{mix} is defined by Eqn (6.1-19).

6.1.2.3 Calculation of the change in entropy

The equations developed in Section 3.4 can be extended to mixtures with the following modifications.

van der Waals equation of state: For mixtures, Eqn (3.4-13) becomes

$$\Delta \tilde{S} = R \ln \left(\frac{Z_{\text{mix}_2} - B_{\text{mix}_2}}{Z_{\text{mix}_1} - B_{\text{mix}_1}} \right) + \int_{T_1}^{T_2} \frac{\tilde{C}_{P_{\text{mix}}}^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right). \quad (6.1-24)$$

Redlich-Kwong equation of state: For mixtures, Eqn (3.4-14) becomes

$$\begin{aligned} \Delta \tilde{S} = & R \ln \left(\frac{Z_{\text{mix}_2} - B_{\text{mix}_2}}{Z_{\text{mix}_1} - B_{\text{mix}_1}} \right) - \frac{R}{2} \left[\frac{A_{\text{mix}_2}}{B_{\text{mix}_2}} \ln \left(1 + \frac{B_{\text{mix}_2}}{Z_{\text{mix}_2}} \right) \right. \\ & \left. - \frac{A_{\text{mix}_1}}{B_{\text{mix}_1}} \ln \left(1 + \frac{B_{\text{mix}_1}}{Z_{\text{mix}_1}} \right) \right] + \int_{T_1}^{T_2} \frac{\tilde{C}_{P_{\text{mix}}}^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right). \end{aligned} \quad (6.1-25)$$

Peng-Robinson equation of state: For mixtures, Eqn (3.4-16) becomes

$$\begin{aligned} \Delta \tilde{S} = & R \ln \left(\frac{Z_{\text{mix}_2} - B_{\text{mix}_2}}{Z_{\text{mix}_1} - B_{\text{mix}_1}} \right) + \frac{R\Theta_{\text{mix}_2}}{\sqrt{8}B_{\text{mix}_2}} \ln \left[\frac{Z_{\text{mix}_2} + (1 + \sqrt{2}) B_{\text{mix}_2}}{Z_{\text{mix}_2} + (1 - \sqrt{2}) B_{\text{mix}_2}} \right] \\ & - \frac{R\Theta_{\text{mix}_1}}{\sqrt{8}B_{\text{mix}_1}} \ln \left[\frac{Z_{\text{mix}_1} + (1 + \sqrt{2}) B_{\text{mix}_1}}{Z_{\text{mix}_1} + (1 - \sqrt{2}) B_{\text{mix}_1}} \right] \\ & + \int_{T_1}^{T_2} \frac{\tilde{C}_{P_{\text{mix}}}^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right), \end{aligned} \quad (6.1-26)$$

where Θ_{mix} is defined by

$$\Theta_{\text{mix}} = \Omega_{\text{mix}} + A_{\text{mix}}. \quad (6.1-27)$$

Example 6.4 A gas mixture consisting of 50 mol% propane (1), 20% *n*-butane (2), and 30% *n*-pentane (3) is compressed irreversibly from 300 K and 1 bar to 500 K and 40 bar by an adiabatic compressor. Estimate the work input per mole of gas mixture passing through the compressor if the mixture obeys the Peng-Robinson equation of state with $k_{12} = 0.003$, $k_{13} = 0.027$, and $k_{23} = 0.017$.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Propane	369.9	42.5	0.153
<i>n</i> -Butane	425.0	38.0	0.199
<i>n</i> -Pentane	469.8	33.6	0.251

From Appendix B

$$\begin{aligned} \tilde{C}_{P_1}^* &= 29.595 + 0.838 \times 10^{-1}T + 3.256 \times 10^{-4}T^2 \\ &\quad - 3.958 \times 10^{-7}T^3 + 13.129 \times 10^{-11}T^4, \\ \tilde{C}_{P_2}^* &= 24.258 + 2.335 \times 10^{-1}T + 1.279 \times 10^{-4}T^2 \\ &\quad - 2.437 \times 10^{-7}T^3 + 8.552 \times 10^{-11}T^4, \\ \tilde{C}_{P_3}^* &= 33.780 + 2.485 \times 10^{-1}T + 2.535 \times 10^{-4}T^2 \\ &\quad - 3.838 \times 10^{-7}T^3 + 12.977 \times 10^{-11}T^4. \end{aligned}$$

System: Compressor

This is a steady-state flow process and from the first law

$$\Delta \tilde{H} = \underbrace{\tilde{Q}}_0 + \tilde{W}_s \quad (1)$$

Therefore, the work input to the compressor is equal to the change in enthalpy, which can be determined from Eqn (6.1-23).

State 1 ($T_1 = 300$ K, $P_1 = 1$ bar)

The values of reduced temperature, reduced pressure, and α are

Component	T_r	P_r	α
Propane	0.811	0.024	1.124
<i>n</i> -Butane	0.706	0.026	1.226
<i>n</i> -Pentane	0.639	0.030	1.322

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 0.019 \quad A_{12} = A_{21} = 0.023 \quad B_1 = 2.302 \times 10^{-3}, \\ A_{22} &= 0.029 \quad A_{13} = A_{31} = 0.028 \quad B_2 = 2.865 \times 10^{-3}, \\ A_{33} &= 0.044 \quad A_{23} = A_{32} = 0.035 \quad B_3 = 3.653 \times 10^{-3}. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}_1} = 0.027 \quad B_{\text{mix}_1} = 2.82 \times 10^{-3}.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}_1}^3 - 0.997Z_{\text{mix}_1}^2 + 0.021Z_{\text{mix}_1} - 6.817 \times 10^{-5} = 0 \Rightarrow Z_{\text{mix}_1} = 0.976.$$

The parameter Γ is calculated from Eqn (6.1-17) as

$$\Gamma_1 = 0.513 \quad \Gamma_2 = 0.509 \quad \Gamma_3 = 0.518.$$

The use of Eqn (6.1-19) gives

$$\Omega_{\text{mix}_1} = -0.041.$$

State 2 ($T_2 = 500\text{K}$, $P_2 = 40\text{ bar}$)

The values of reduced temperature, reduced pressure, and α are

Component	T_r	P_r	α
Propane	1.352	0.941	0.813
<i>n</i> -Butane	1.176	1.053	0.890
<i>n</i> -Pentane	1.064	1.190	0.953

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 0.191 \quad A_{12} = A_{21} = 0.243 \quad B_1 = 0.054, \\ A_{22} &= 0.310 \quad A_{13} = A_{31} = 0.288 \quad B_2 = 0.070, \\ A_{33} &= 0.459 \quad A_{23} = A_{32} = 0.371 \quad B_3 = 0.087. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}_2} = 0.281 \quad B_{\text{mix}_2} = 0.067.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}_2}^3 - 0.933Z_{\text{mix}_2}^2 + 0.134Z_{\text{mix}_2} - 0.014 = 0 \Rightarrow Z_{\text{mix}_2} = 0.785.$$

The parameter Γ is calculated from Eqn (6.1-17) as

$$\Gamma_1 = 0.779 \quad \Gamma_2 = 0.771 \quad \Gamma_3 = 0.787.$$

The use of Eqn (6.1-19) gives

$$\Omega_{\text{mix}_2} = -0.5$$

The heat capacity of the mixture can be calculated as

$$\begin{aligned}\tilde{C}_{P_{\text{mix}}}^* &= y_1 \tilde{C}_{P_1}^* + y_2 \tilde{C}_{P_2}^* + y_3 \tilde{C}_{P_3}^* = 29.783 + 1.631 \times 10^{-1}T \\ &\quad + 2.644 \times 10^{-4}T^2 - 3.618 \times 10^{-7}T^3 + 12.168 \times 10^{-11}T^4\end{aligned}$$

so that

$$\int_{300}^{500} \tilde{C}_{P_{\text{mix}}}^* dT = 23,428 \text{ J/mol.}$$

Substitution of the numerical values into Eqn (6.1-23) gives

$$\Delta \tilde{H} = -3343 + 164.3 + 23,428 = 20,249 \text{ J/mol.}$$

6.1.3 Departure Functions

The departure functions developed for pure components in Section 3.6 can be extended to mixtures. For mixtures, Eqns (3.6-3)–(3.6-5) take the form

$$\Delta \tilde{H} = -(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_1, P_1} + \int_{T_1}^{T_2} \tilde{C}_{P_{\text{mix}}}^* dT + (\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_2, P_2}, \quad (6.1-28)$$

$$\Delta \tilde{S} = -(\tilde{S}_{\text{mix}} - \tilde{S}^{\text{IGM}})_{T_1, P_1} + \int_{T_1}^{T_2} \frac{\tilde{C}_{P_{\text{mix}}}^*}{T} dT - R \ln \left(\frac{P_2}{P_1} \right) + (\tilde{S}_{\text{mix}} - \tilde{S}^{\text{IGM}})_{T_2, P_2}, \quad (6.1-29)$$

$$\Delta \tilde{U} = -(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T_1, P_1} + \int_{T_1}^{T_2} \tilde{C}_{V_{\text{mix}}}^* dT + (\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T_2, P_2}, \quad (6.1-30)$$

where the departure functions are given by the following equations depending on the type of the equation of state.

van der Waals equation of state: For mixtures, Eqns (3.6-12)–(3.6-14) become

$$\frac{(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T, P}}{RT} = Z_{\text{mix}} - 1 - \frac{A_{\text{mix}}}{Z_{\text{mix}}}, \quad (6.1-31)$$

$$\frac{(\tilde{S}_{\text{mix}} - \tilde{S}^{\text{IGM}})_{T, P}}{R} = \ln(Z_{\text{mix}} - B_{\text{mix}}), \quad (6.1-32)$$

$$\frac{(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T,P}}{RT} = -\frac{A_{\text{mix}}}{Z_{\text{mix}}}. \quad (6.1-33)$$

Redlich-Kwong equation of state: For mixtures, Eqns (3.6-15)–(3.6-17) become

$$\frac{(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T,P}}{RT} = Z_{\text{mix}} - 1 - \frac{3}{2} \frac{A_{\text{mix}}}{B_{\text{mix}}} \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right), \quad (6.1-34)$$

$$\frac{(\tilde{S}_{\text{mix}} - \tilde{S}^{\text{IGM}})_{T,P}}{R} = \ln(Z_{\text{mix}} - B_{\text{mix}}) - \frac{1}{2} \frac{A_{\text{mix}}}{B_{\text{mix}}} \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right), \quad (6.1-35)$$

$$\frac{(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T,P}}{RT} = -\frac{3}{2} \frac{A_{\text{mix}}}{B_{\text{mix}}} \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right). \quad (6.1-36)$$

Peng-Robinson equation of state: For mixtures, Eqns (3.6-18)–(3.6-20) become

$$\frac{(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T,P}}{RT} = Z_{\text{mix}} - 1 + \frac{\Omega_{\text{mix}}}{\sqrt{8}B_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}} + (1 + \sqrt{2})B_{\text{mix}}}{Z_{\text{mix}} + (1 - \sqrt{2})B_{\text{mix}}} \right], \quad (6.1-37)$$

$$\frac{(\tilde{S}_{\text{mix}} - \tilde{S}^{\text{IGM}})_{T,P}}{R} = \ln(Z_{\text{mix}} - B_{\text{mix}}) + \frac{\Theta_{\text{mix}}}{\sqrt{8}B_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}} + (1 + \sqrt{2})B_{\text{mix}}}{Z_{\text{mix}} + (1 - \sqrt{2})B_{\text{mix}}} \right], \quad (6.1-38)$$

$$\frac{(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T,P}}{RT} = \frac{\Omega_{\text{mix}}}{\sqrt{8}B_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}} + (1 + \sqrt{2})B_{\text{mix}}}{Z_{\text{mix}} + (1 - \sqrt{2})B_{\text{mix}}} \right], \quad (6.1-39)$$

where Ω_{mix} and Θ_{mix} are given by Eqns (6.1-19) and (6.1-27), respectively.

Example 6.5 Solve Example 6.2 using departure functions if the gas mixture obeys the Redlich-Kwong equation of state.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)
Methane	190.6	46.1
Ethane	305.3	49.0

For a steady-state flow system, the first law gives

$$\tilde{W}_s = \Delta \tilde{H} \quad (1)$$

The change in enthalpy can be calculated from Eqn (6.1-28).

State 1 ($T_1 = 310$ K, $P_1 = 1$ bar)

The values of reduced temperature and reduced pressure are

Component	T_r	P_r
Methane	1.626	0.022
Ethane	1.015	0.020

Using the equations given in Table 6.3 and taking k_{ij} to be equal to zero, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 2.790 \times 10^{-3} & A_{12} = A_{21} &= 4.794 \times 10^{-3} & B_1 &= 1.172 \times 10^{-3}, \\ A_{22} &= 8.237 \times 10^{-3} & & & B_2 &= 1.707 \times 10^{-3}. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}_1} = 3.205 \times 10^{-3} \quad B_{\text{mix}_1} = 1.225 \times 10^{-3}.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}_1}^3 - Z_{\text{mix}_1}^2 + 1.978 \times 10^{-3} Z_{\text{mix}_1} - 3.926 \times 10^{-6} = 0 \Rightarrow Z_{\text{mix}_1} = 0.998.$$

The enthalpy departure function is calculated from Eqn (6.1-34) as

$$\begin{aligned} (\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_1, P_1} &= (8.314)(310) \left[0.998 - 1 - \frac{3}{2} \left(\frac{3.205 \times 10^{-3}}{1.225 \times 10^{-3}} \right) \right. \\ &\quad \left. \times \ln \left(1 + \frac{1.225 \times 10^{-3}}{0.998} \right) \right] = -17.6 \text{ J/mol}. \end{aligned}$$

State 2 ($T_2 = 350$ K, $P_2 = 30$ bar)

The values of reduced temperature and reduced pressure are

Component	T_r	P_r
Methane	1.836	0.651
Ethane	1.146	0.612

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$\begin{aligned} A_{11} &= 0.061 & A_{12} = A_{21} &= 0.106 & B_1 &= 0.031, \\ A_{22} &= 0.186 & & & B_2 &= 0.046. \end{aligned}$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}_2} = 0.07 \quad B_{\text{mix}_2} = 0.033.$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}_2}^3 - Z_{\text{mix}_2}^2 + 0.036Z_{\text{mix}_2} - 2.31 \times 10^{-3} = 0 \Rightarrow Z_{\text{mix}_2} = 0.965.$$

The enthalpy departure function is calculated from Eqn (6.1-34) as

$$\begin{aligned} (\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_2, P_2} &= (8.314)(350) \left[0.965 - 1 - \frac{3}{2} \left(\frac{0.07}{0.033} \right) \right. \\ &\quad \left. \times \ln \left(1 + \frac{0.033}{0.965} \right) \right] = -413.2 \text{ J/mol.} \end{aligned}$$

From Example 6.2

$$\int_{310}^{350} \tilde{C}_{P_{\text{mix}}}^* dT = 1574 \text{ J/mol.}$$

Substitution of the numerical values into Eqn (6.1-28) leads to

$$\Delta \tilde{H} = 17.6 + 1574 - 413.2 = 1178.4 \text{ J/mol.}$$

Example 6.6 A carbon dioxide stream at 10 bar and 350 K and a methane stream at 5 bar and 270 K are continuously fed to an adiabatic mixer in the mole ratio 1:3. If the exit gas mixture is at 3 bar, estimate its temperature. Assume that the pure gases as well as the mixture obey the Redlich-Kwong equation of state.

Solution

From Appendix A, the critical constants are given below:

Component	T_c (K)	P_c (bar)
Carbon dioxide	304.2	73.8
Methane	190.6	46.1

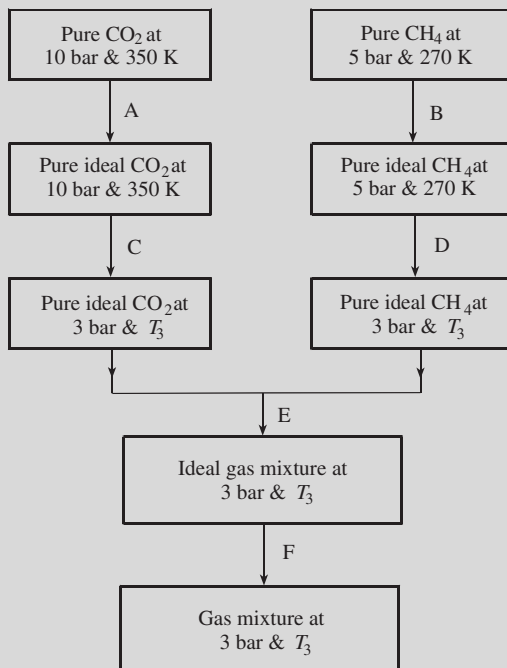
From Appendix B

$$\begin{aligned} \tilde{C}_{P_{\text{CO}_2}}^* &= 29.268 - 0.224 \times 10^{-1}T + 2.653 \times 10^{-4}T^2 - 4.153 \times 10^{-7}T^3 \\ &\quad + 20.057 \times 10^{-11}T^4, \\ \tilde{C}_{P_{\text{CH}_4}}^* &= 36.155 - 0.511 \times 10^{-1}T + 2.215 \times 10^{-4}T^2 - 1.824 \times 10^{-7}T^3 \\ &\quad + 4.899 \times 10^{-11}T^4. \end{aligned}$$

For a steady-state flow system, the first law gives

$$\Delta H = \underbrace{Q}_0 + \underbrace{W_s}_0 = 0. \quad (1)$$

The process path for calculating enthalpy change is shown in the figure below:



Taking 1 mol of CO₂ as a basis, the enthalpy change is expressed as

$$\Delta H = \Delta \tilde{H}_A + 3\Delta \tilde{H}_B + \Delta \tilde{H}_C + 3\Delta \tilde{H}_D + 4\Delta \tilde{H}_E + 4\Delta \tilde{H}_F. \quad (2)$$

Considering the following states

State	T (K)	P (bar)
1	350	10
2	270	5
3	T_3	3

Eqn (2) is expressed in the form

$$\begin{aligned} \Delta H = & -(\tilde{H}_{\text{CO}_2} - \tilde{H}_{\text{CO}_2}^{\text{IG}})_{T_1, P_1} - 3(\tilde{H}_{\text{CH}_4} - \tilde{H}_{\text{CH}_4}^{\text{IG}})_{T_2, P_2} \\ & + \int_{350}^{T_3} \tilde{C}_{P_{\text{CO}_2}}^* dT + 3 \int_{270}^{T_3} \tilde{C}_{P_{\text{CH}_4}}^* dT + 4(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_3, P_3}. \end{aligned} \quad (3)$$

Since there is no interaction between ideal gas molecules, mixing of ideal gases produces no change in enthalpy. Thus, $\Delta \tilde{H}_E = 0$. Substitution of Eqn (3) into Eqn (1) gives

$$\begin{aligned} & -(\tilde{H}_{\text{CO}_2} - \tilde{H}_{\text{CO}_2}^{\text{IG}})_{T_1, P_1} - 3(\tilde{H}_{\text{CH}_4} - \tilde{H}_{\text{CH}_4}^{\text{IG}})_{T_2, P_2} + \int_{350}^{T_3} \tilde{C}_{P_{\text{CO}_2}}^* dT \\ & + 3 \int_{270}^{T_3} \tilde{C}_{P_{\text{CH}_4}}^* dT + 4(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_3, P_3} = 0. \end{aligned} \quad (4)$$

Departure function for pure CO₂: At 350 K and 10 bar

$$A = 0.041 \quad B = 0.010 \quad Z = 0.969.$$

The use of Eqn (3.6-15) gives

$$(\tilde{H}_{\text{CO}_2} - \tilde{H}_{\text{CO}_2}^{\text{IG}})_{T_1, P_1} = -274 \text{ J/mol}. \quad (5)$$

Departure function for pure CH₄: At 270 K and 5 bar

$$A = 0.019 \quad B = 6.634 \times 10^{-3} \quad Z = 0.987.$$

The use of Eqn (3.6-15) gives

$$(\tilde{H}_{\text{CH}_4} - \tilde{H}_{\text{CH}_4}^{\text{IG}})_{T_2, P_2} = -93.8 \text{ J/mol}. \quad (6)$$

The remaining terms in Eqn (4) cannot be calculated since T_3 is not known. Therefore, a trial-and-error procedure will be used to estimate T_3 .

Assuming $T_3 = 288.84 \text{ K}$, the values of the integrals in Eqn (4) are

$$\int_{350}^{288.84} \tilde{C}_{P_{\text{CO}_2}}^* dT = -2308 \text{ J/mol}, \quad (7)$$

$$\int_{270}^{288.84} \tilde{C}_{P_{\text{CH}_4}}^* dT = 668.7 \text{ J/mol}. \quad (8)$$

Departure function for the mixture: Taking k_{12} to be equal to zero, the dimensionless parameters and the compressibility factor become

$$A_{\text{mix}} = 0.012 \quad B_{\text{mix}} = 3.718 \times 10^{-3} \quad Z_{\text{mix}} = 0.992.$$

The use of Eqn (6.1-34) gives

$$(\tilde{H}_{\text{mix}} - \tilde{H}^{\text{IGM}})_{T_3, P_3} = -62.7 \text{ J/mol}. \quad (9)$$

If the assumed temperature is correct, then the summation of the terms on the left-hand side of Eqn (4) should be zero. Substitution of Eqns (5)–(9) into the left-hand side of Eqn (4) gives

$$274 + 3(93.8) - 2308 + 3(668.7) - 4(62.7) = 2.7,$$

which is close enough to zero.

6.1.4 Which Equation of State to Use?

The two central questions in solving problems encountered in real life are: (i) Which equation of state should be used for a given mixture? (ii) Which mixing rule should be employed in the calculations? Unfortunately, there is no clearcut recipe for answering such questions. Simulation packages given in Appendix G provide decision trees for making the proper selection by taking into account various factors, some of which are given below:

- Nature of the chemical species (Polar/Nonpolar, Electrolyte/Nonelectrolyte).
- Real or pseudo-components.⁴
- Value of pressure (less than or greater than 10 bar).
- Availability of interaction parameters.

The Peng-Robinson and the Soave-Redlich-Kwong equations of state are mostly preferred in the natural-gas, petroleum, and petrochemical industries (Prausnitz and Tavares, 2004). For more details on the subject, the reader should refer to Chen and Mathias (2002), Poling *et al.* (2004), de Hemptinne and Behar (2006), and Edwards (2008).

6.2 Partial Molar Property

Consider two separate beakers containing water of volumes 20 cm³ and 50 cm³. If the contents of these two beakers are mixed, the total volume of water will be 70 cm³. Since volumes of a pure component are additive, this is an expected result.

When 20 cm³ of water is mixed with 50 cm³ of ethanol, the total volume of the solution turns out to be 67 cm³, less than the sum of pure liquid volumes. The reason for this is the fact that water and ethanol molecules interact with each other and

⁴In complex mixtures, such as crude oil, it is not possible to identify each component making up the mixture. In that case, some components are lumped into a so-called *pseudo-component*. Reduction in the number of components makes the mixture amenable to modeling. The criteria used in lumping and the estimation of pseudo-component properties are beyond the scope of this text.

pack differently than with like molecules.⁵ If 1 and 2 represent water and ethanol, respectively, then

$$V_{\text{mix}} \neq n_1 \tilde{V}_1 + n_2 \tilde{V}_2 \quad (6.2-1)$$

in which $\tilde{V}_1$ and $\tilde{V}_2$ represent the molar volumes of pure 1 and 2, respectively.

In engineering calculations, however, we want to have an equation similar to Eqn (6.2-1) to calculate the total volume of a mixture. In other words, we have to come up with a quantity $\bar{V}_i$ so that the additive property holds, i.e.

$$V_{\text{mix}} = n_1 \bar{V}_1 + n_2 \bar{V}_2 \quad (6.2-2)$$

or, in general,

$$V_{\text{mix}} = \sum_{i=1}^k n_i \bar{V}_i. \quad (6.2-3)$$

The term $\bar{V}_i$ is called the *partial molar volume of species i* and is defined by

$$\bar{V}_i = \left(\frac{\partial V_{\text{mix}}}{\partial n_i} \right)_{T, P, n_j \neq i}. \quad (6.2-4)$$

The subscripts on the partial derivative mean that pressure, temperature, and number of moles of all components other than i are kept constant.

In general, a *partial molar property*, $\bar{\varphi}_i$, is the rate at which an extensive property of the entire mixture, φ_{mix} , changes with the number of moles of component i in the mixture when temperature, pressure, and number of moles of all components other than i are kept constant, i.e.

$$\bar{\varphi}_i = \left(\frac{\partial \varphi_{\text{mix}}}{\partial n_i} \right)_{T, P, n_j \neq i}. \quad (6.2-5)$$

It should be kept in mind that $\bar{\varphi}_i$ is a *property of the mixture and not simply a property of component i*. The partial molar quantity is an intensive property that is generally dependent on the composition of the mixture as well as temperature and pressure.

The total property of the mixture is expressed in terms of the partial molar properties as

$$\varphi_{\text{mix}} = \sum_{i=1}^k n_i \bar{\varphi}_i \quad (6.2-6)$$

⁵Consider a box of volume V_1 filled with small metal balls and another box of volume V_2 filled with rice. If the contents of these two boxes are mixed, the final volume will be less than $V_1 + V_2$.

or dividing each term by the total number of moles gives

$$\tilde{\varphi}_{\text{mix}} = \sum_{i=1}^k x_i \bar{\varphi}_i. \quad (6.2-7)$$

Example 6.7 Pecar and Dolecek (2005) experimentally determined the densities and the partial molar volumes of a binary mixture of ethanol (1) and water (2) as a function of temperature and pressure. In a mixture with an ethanol mole fraction of 0.35 at 298 K and 1 bar, the mixture density and the partial molar volume of ethanol are determined as 891.9 kg/m³ and 57.31 cm³/mol, respectively. Calculate the partial molar volume of water.

Solution

From Eqn (6.2-7) we can write

$$\tilde{V}_{\text{mix}} = x_1 \bar{V}_1 + x_2 \bar{V}_2. \quad (1)$$

Note that the molar volume of the mixture is given by

$$\tilde{V}_{\text{mix}} = \frac{M_{\text{mix}}}{\rho_{\text{mix}}} \quad (2)$$

in which M_{mix} and ρ_{mix} represent the molecular weight and the density of the mixture, respectively. The molecular weight of the mixture is given by

$$M_{\text{mix}} = x_1 M_1 + x_2 M_2 = (0.35)(46.07) + (0.65)(18.02) = 27.84 \text{ g/mol.}$$

Therefore, the molar volume of the mixture is calculated from Eqn (2) as

$$\tilde{V}_{\text{mix}} = \frac{27.84}{0.8919} = 31.21 \text{ cm}^3/\text{mol.}$$

The partial molar volume of water is then determined from Eqn (1) as

$$31.21 = (0.35)(57.31) + 0.65 \bar{V}_2 \Rightarrow \bar{V}_2 = 17.16 \text{ cm}^3/\text{mol.}$$

Consider the definition of enthalpy, i.e.

$$H = U + PV. \quad (6.2-8)$$

Differentiation of Eqn (6.2-8) with respect to n_i by keeping temperature, pressure,

and number of moles of all components other than i constant gives

$$\left(\frac{\partial H}{\partial n_i}\right)_{T,P,n_{j \neq i}} = \left(\frac{\partial U}{\partial n_i}\right)_{T,P,n_{j \neq i}} + P \left(\frac{\partial V}{\partial n_i}\right)_{T,P,n_{j \neq i}}. \quad (6.2-9)$$

Application of the definition of a partial molar quantity leads to

$$\overline{H}_i = \overline{U}_i + P\overline{V}_i. \quad (6.2-10)$$

Similarly, we can write

$$\overline{A}_i = \overline{U}_i - T\overline{S}_i, \quad (6.2-11)$$

$$\overline{G}_i = \overline{H}_i - T\overline{S}_i. \quad (6.2-12)$$

6.2.1 Determination of Partial Molar Properties

For simplicity let us consider the determination of partial molar volumes in a binary mixture. In general, partial molar volumes can be determined by two different approaches.

Method of tangent slope: In this method, a known amount of a compound, say one mole of component 1, is placed in a graduated cylinder. By keeping temperature and pressure constant throughout the experiment, the volume of the mixture is recorded by adding increasing amounts of component 2. Then either the total volume of the mixture, V_{mix} , is plotted as a function of n_2 or an equation is fitted to the volume data in the form $V_{\text{mix}} = \tilde{V}_1 + \alpha n_2 + \beta n_2^{1.5} + \gamma n_2^2 + \dots$, in which $\tilde{V}_1$ is the molar volume of pure 1.

At any given value of n_2 , the slope of the tangent to the curve gives the partial molar volume of component 2, i.e.

$$\text{Slope} = \left(\frac{\partial V_{\text{mix}}}{\partial n_2}\right)_{T,P,n_1} = \overline{V}_2. \quad (6.2-13)$$

If an equation is fitted, then differentiation of the equation with respect to n_2 gives the partial molar volume of component 2 as

$$\overline{V}_2 = \left(\frac{\partial V_{\text{mix}}}{\partial n_2}\right)_{T,P,n_1} = \alpha + 1.5\beta n_2^{0.5} + 2\gamma n_2 + \dots \quad (6.2-14)$$

Once $\overline{V}_2$ is determined, the partial molar volume of component 1 can be obtained from Eqn (6.2-6) as

$$\overline{V}_1 = \frac{V_{\text{mix}} - n_2 \overline{V}_2}{n_1}. \quad (6.2-15)$$

Example 6.8 When NaCl is dissolved in water at 298 K, the volume of the resulting mixture is represented as a function of molality,⁶ m , in the form

$$V_{\text{mix}} = 1003 + 16.62m + 1.77m^{1.5} + 0.12m^2, \quad (1)$$

where V_{mix} is in cm^3 . For a 0.01 m solution of NaCl, calculate:

- (a) The total solution volume.
- (b) The partial molar volumes of NaCl and water.

Solution

- (a) From Eqn (1)

$$V_{\text{mix}} = 1003 + 16.62(0.01) + 1.77(0.01)^{1.5} + 0.12(0.01)^2 = 1003.2 \text{ cm}^3.$$

- (b) Let the subscripts 1 and 2 designate H_2O and NaCl, respectively. Choosing 1 kg of water as a basis, $m = n_2$. Thus, differentiation of Eqn (1) with respect to m gives the partial molar volume of NaCl as

$$\begin{aligned} \bar{V}_2 &= 16.62 + 2.655m^{0.5} + 0.24m \\ &= 16.62 + 2.655(0.01)^{0.5} + 0.24(0.01) \\ &= 16.89 \text{ cm}^3/\text{mol}. \end{aligned}$$

From Eqn (6.2-15)

$$\bar{V}_1 = \frac{V_{\text{mix}} - n_2 \bar{V}_2}{n_1} = \frac{1003.2 - (0.01)(16.89)}{1000/18} = 18.05 \text{ cm}^3/\text{mol}.$$

Method of tangent intercepts: The total volume of a binary mixture is given by

$$V_{\text{mix}} = n \tilde{V}_{\text{mix}} = (n_1 + n_2) \tilde{V}_{\text{mix}}. \quad (6.2-16)$$

The partial molar volume of component 1 is

$$\bar{V}_1 = \left(\frac{\partial V_{\text{mix}}}{\partial n_1} \right)_{T,P,n_2} = \left\{ \frac{\partial}{\partial n_1} [(n_1 + n_2) \tilde{V}_{\text{mix}}] \right\}_{T,P,n_2}. \quad (6.2-17)$$

⁶Molality is the number of moles of solute dissolved in 1 kg of solvent.

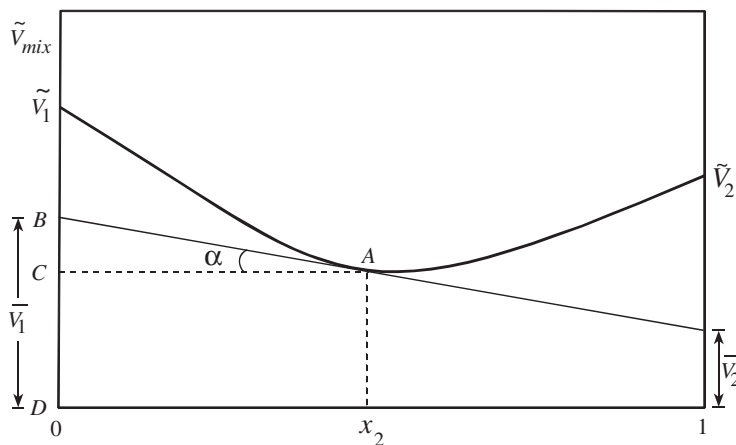


Figure 6.1: Graphical determination of partial molar volumes.

Carrying out the differentiation gives

$$\bar{V}_1 = \tilde{V}_{\text{mix}} + n \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial n_1} \right)_{T,P,n_2}. \quad (6.2-18)$$

By the application of the chain rule, Eqn (6.2-18) becomes

$$\bar{V}_1 = \tilde{V}_{\text{mix}} + n \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2} \left(\frac{\partial x_2}{\partial n_1} \right)_{T,P,n_2}. \quad (6.2-19)$$

Since $x_2 = n_2/(n_1 + n_2)$, then

$$\left(\frac{\partial x_2}{\partial n_1} \right)_{T,P,n_2} = -\frac{x_2}{n} \quad (6.2-20)$$

so that Eqn (6.2-19) takes the form

$$\bar{V}_1 = \tilde{V}_{\text{mix}} - x_2 \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2} \quad (6.2-21)$$

Suppose that $\tilde{V}_{\text{mix}}$ varies as a function of x_2 as shown in Fig. 6.1. At any given value of x_2 , let us draw a tangent to the curve. Note that the slope of the tangent line is given by

$$\text{Slope} = \frac{\partial \tilde{V}_{\text{mix}}}{\partial x_2} = -\tan \alpha = -\frac{\overline{BC}}{\overline{AC}}. \quad (6.2-22)$$

Substitution of Eqn (6.2-22) into Eqn (6.2-21) leads to

$$\bar{V}_1 = \tilde{V}_{\text{mix}} - x_2 \left(\frac{\partial \tilde{V}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2} = \overline{CD} - \overline{AC} \left(-\frac{\overline{BC}}{\overline{AC}} \right) = \overline{CD} + \overline{BC} = \overline{BD}. \quad (6.2-23)$$

Thus, the intercept of the tangent line at $x_2 = 0$ is the partial molar volume of component 1. Similarly, the intercept of the tangent line at $x_2 = 1$ gives the partial molar volume of component 2.

For a binary system,⁷ Eqn (6.2-21) can be generalized as

$$\bar{\varphi}_1 = \tilde{\varphi}_{\text{mix}} - x_2 \left(\frac{d\tilde{\varphi}_{\text{mix}}}{dx_2} \right) \quad \text{constant } T \text{ \& } P \quad (6.2-24)$$

and

$$\bar{\varphi}_2 = \tilde{\varphi}_{\text{mix}} - x_1 \left(\frac{d\tilde{\varphi}_{\text{mix}}}{dx_1} \right) \quad \text{constant } T \text{ \& } P. \quad (6.2-25)$$

Since $x_1 + x_2 = 1$ for a binary system, then

$$\frac{d\tilde{\varphi}_{\text{mix}}}{dx_1} = -\frac{d\tilde{\varphi}_{\text{mix}}}{dx_2}. \quad (6.2-26)$$

Thus, depending on the functional form of $\tilde{\varphi}_{\text{mix}}$, the derivatives in Eqns (6.2-24) and (6.2-25) can be interchanged according to Eqn (6.2-26).

In the limiting cases of the mole fractions, i.e. $x_i \rightarrow 0$ or $x_i \rightarrow 1$, the partial molar properties become

$$\lim_{x_i \rightarrow 0} \bar{\varphi}_i = \bar{\varphi}_i^\infty \quad \text{and} \quad \lim_{x_i \rightarrow 1} \bar{\varphi}_i = \tilde{\varphi}_i, \quad (6.2-27)$$

where $\bar{\varphi}_i^\infty$ is called the *partial molar property of component i at infinite dilution*.

Example 6.9 It is required to prepare 1 L of a solution containing 10 wt.% of component 1 and 90% of component 2. The stock solution available in the laboratory has the composition of 60 mol % component 1 and 40% of component 2. Determine the volumes of the stock solution and pure component 2 that must be mixed.

Data: The molar volume of the mixture (in cm^3/mol) is experimentally determined as

$$\tilde{V}_{\text{mix}} = 60x_1 + 110x_2 + 5x_1x_2.$$

The molecular weights of components 1 and 2 are 40 and 60 g/mol, respectively.

⁷See Problem 6.10 for the generalization of Eqns (6.2-24) and (6.2-25) for a k -component system.

Solution

Let X and Y be the volumes (in cm^3) of the stock solution and pure component 2, respectively, that must be mixed to produce 1 L of desired mixture. The conservation statements for the number of moles of components 1 and 2 are expressed as

$$\left(\frac{X}{\tilde{V}_{\text{mix}}^{\text{stock}}} \right) (0.6) = \left(\frac{1000}{\tilde{V}_{\text{mix}}^{\text{final}}} \right) x_1^{\text{final}}, \quad (1)$$

$$\left(\frac{X}{\tilde{V}_{\text{mix}}^{\text{stock}}} \right) (1 - 0.6) + \frac{Y}{\tilde{V}_2} = \left(\frac{1000}{\tilde{V}_{\text{mix}}^{\text{final}}} \right) (1 - x_1^{\text{final}}). \quad (2)$$

Determination of the molar volumes of the stock and final solutions requires the mole fractions to be known. In a binary mixture, the mole fraction, x , is related to the weight fraction, ω , as

$$x_1 = \frac{\frac{\omega_1}{M_1}}{\frac{\omega_1}{M_1} + \frac{\omega_2}{M_2}},$$

where M_1 and M_2 are the molecular weights of components 1 and 2, respectively. Therefore, the mole fraction of component 1 in the final solution is

$$x_1^{\text{final}} = \frac{\frac{0.1}{40}}{\frac{0.1}{40} + \frac{0.9}{60}} = 0.143.$$

The molar volume of the final solution is

$$\begin{aligned} \tilde{V}_{\text{mix}}^{\text{final}} &= (60)(0.143) + (110)(1 - 0.143) + (5)(0.143)(1 - 0.143) \\ &= 103.46 \text{ cm}^3/\text{mol}. \end{aligned}$$

The molar volume of the stock solution is

$$\tilde{V}_{\text{mix}}^{\text{stock}} = (60)(0.6) + (110)(1 - 0.6) + (5)(0.6)(1 - 0.6) = 81.2 \text{ cm}^3/\text{mol}.$$

The molar volume of pure component 2 is

$$\tilde{V}_2 = \lim_{x_2 \rightarrow 1} \tilde{V}_{\text{mix}} = 110 \text{ cm}^3/\text{mol}.$$

Substitution of the numerical values into Eqn (1) gives

$$X = 187.1 \text{ cm}^3.$$

Then the use of Eqn (2) results in

$$Y = 809.8 \text{ cm}^3.$$

Note that $X + Y = 187 + 809.8 = 996.8 \text{ cm}^3 < 1000 \text{ cm}^3$, because of volume increase upon mixing.

Example 6.10 The molar enthalpy of a binary liquid mixture of 1 and 2 at constant temperature and pressure is given by

$$\tilde{H}_{\text{mix}} = 400x_1 + 250x_2 + x_1x_2(15x_1 + 45x_2),$$

where $\tilde{H}_{\text{mix}}$ is in J/mol. Calculate

- The molar enthalpies of pure components 1 and 2.
- The partial molar enthalpies of components 1 and 2 at $x_1 = 0.35$.
- The partial molar enthalpies of components 1 and 2 at infinite dilution.

Solution

- Molar enthalpies of pure components are

$$\tilde{H}_1 = \lim_{x_1 \rightarrow 1} \tilde{H}_{\text{mix}} = 400 \text{ J/mol} \quad \tilde{H}_2 = \lim_{x_2 \rightarrow 1} \tilde{H}_{\text{mix}} = 250 \text{ J/mol}.$$

- From Eqn (6.2-24)

$$\bar{H}_1 = \tilde{H}_{\text{mix}} - x_2 \left(\frac{d\tilde{H}_{\text{mix}}}{dx_2} \right). \quad (1)$$

The derivative of $\tilde{H}_{\text{mix}}$ with respect to x_2 is

$$\begin{aligned} \frac{d\tilde{H}_{\text{mix}}}{dx_2} &= -400 + 250 + (1 - 2x_2)(15x_1 + 45x_2) + x_1x_2(-15 + 45) \\ &= -150 + 15x_1 + 45x_2 - 90x_2^2. \end{aligned} \quad (2)$$

Thus, Eqn (1) becomes

$$\begin{aligned} \bar{H}_1 &= 400x_1 + 250x_2 + x_1x_2(15x_1 + 45x_2) \\ &\quad - x_2(-150 + 15x_1 + 45x_2 - 90x_2^2) = 400 - 15x_2^2 + 60x_2^3. \end{aligned} \quad (3)$$

The partial molar enthalpy of component 2 can be calculated from Eqn (6.2-25)

$$\bar{H}_2 = \tilde{H}_{\text{mix}} - x_1 \left(\frac{d\tilde{H}_{\text{mix}}}{dx_1} \right). \quad (4)$$

Since

$$\frac{d\tilde{H}_{\text{mix}}}{dx_1} = -\frac{d\tilde{H}_{\text{mix}}}{dx_2} \quad (5)$$

Eqn (4) becomes

$$\bar{H}_2 = \tilde{H}_{\text{mix}} + x_1 \left(\frac{d\tilde{H}_{\text{mix}}}{dx_2} \right) \quad (6)$$

or

$$\begin{aligned} \bar{H}_2 = 400x_1 + 250x_2 + x_1x_2(15x_1 + 45x_2) + x_1(-150 + 15x_1 \\ + 45x_2 - 90x_2^2) = 250 + 75x_1^2 - 60x_1^3. \end{aligned} \quad (7)$$

When $x_1 = 0.35$

$$\bar{H}_1 = 410.14 \text{ J/mol} \quad \text{and} \quad \bar{H}_2 = 256.62 \text{ J/mol}.$$

(c) The partial molar enthalpies of components 1 and 2 at infinite dilution are

$$\begin{aligned} \bar{H}_1^\infty &= \lim_{x_1 \rightarrow 0} \bar{H}_1 = 400 - 15 + 60 = 445 \text{ J/mol}, \\ \bar{H}_2^\infty &= \lim_{x_2 \rightarrow 0} \bar{H}_2 = 250 + 75 - 60 = 265 \text{ J/mol}. \end{aligned}$$

6.2.2 Homogeneous Functions and Partial Molar Properties

A function $f(z_1, z_2)$ is said to be *homogeneous of degree n* if

$$f(\lambda z_1, \lambda z_2) = \lambda^n f(z_1, z_2) \quad (6.2-28)$$

for all λ . For a homogeneous function of degree n , *Euler's theorem* states that

$$nf(z_1, z_2, \dots, z_k) = \sum_{i=1}^k \left(\frac{\partial f}{\partial z_i} \right)_{z_j \neq i} z_i. \quad (6.2-29)$$

The extensive properties in thermodynamics can be regarded as homogeneous functions of degree “1”. Therefore, for every extensive property we can write

$$f(z_1, z_2, \dots, z_k) = \sum_{i=1}^k \left(\frac{\partial f}{\partial z_i} \right)_{z_j \neq i} z_i \quad \text{When } f \text{ is an extensive property.} \quad (6.2-30)$$

On the other hand, the intensive properties are homogeneous functions of degree “0” and can be expressed as

$$0 = \sum_{i=1}^k \left(\frac{\partial f}{\partial z_i} \right)_{z_j \neq i} z_i \quad \text{When } f \text{ is an intensive property.} \quad (6.2-31)$$

Any extensive thermodynamic property of a mixture, φ_{mix} , is dependent on temperature, pressure, and the number of moles of each component present in the mixture, i.e.

$$\varphi_{\text{mix}} = \varphi_{\text{mix}}(T, P, n_1, n_2, \dots, n_k). \quad (6.2-32)$$

At constant temperature and pressure, application of Eqn (6.2-30) gives

$$\varphi_{\text{mix}} = \sum_{i=1}^k \left(\frac{\partial \varphi_{\text{mix}}}{\partial n_i} \right)_{T, P, n_j \neq i} n_i \quad (6.2-33)$$

or

$$\varphi_{\text{mix}} = \sum_{i=1}^k \bar{\varphi}_i n_i, \quad (6.2-34)$$

which is identical with Eqn (6.2-6).

6.3 Property Changes on Mixing

When two non-identical liquids are mixed, the total property of the mixture, φ_{mix} , is different from the sum of the properties of the pure liquids, i.e.

$$\varphi_{\text{mix}} \neq \sum_{i=1}^k n_i \tilde{\varphi}_i. \quad (6.3-1)$$

The inequality sign in Eqn (6.3-1) can be removed by the introduction of a “correction factor”, $\Delta\varphi_{\text{mix}}$, so that

$$\varphi_{\text{mix}} = \sum_{i=1}^k n_i \tilde{\varphi}_i + \Delta\varphi_{\text{mix}}. \quad (6.3-2)$$

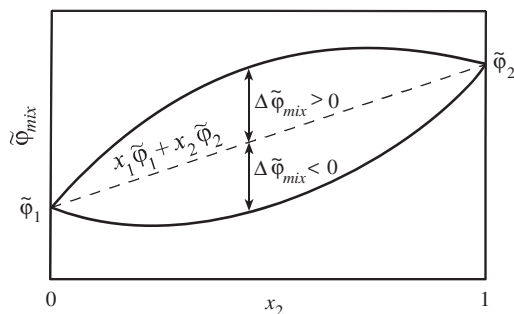


Figure 6.2: Property change on mixing for a binary system.

The term $\Delta\varphi_{\text{mix}}$ is called *isothermal property change on mixing*. In other words, it represents the difference between the actual property of the mixture and the total property of the unmixed pure components at the same temperature and pressure.

Dividing each term in Eqn (6.3-2) by the total number of moles gives

$$\tilde{\varphi}_{\text{mix}} = \sum_{i=1}^k x_i \tilde{\varphi}_i + \Delta\tilde{\varphi}_{\text{mix}}. \quad (6.3-3)$$

Rearrangement of Eqn (6.3-3) gives

$$\Delta\tilde{\varphi}_{\text{mix}} = \tilde{\varphi}_{\text{mix}} - \sum_{i=1}^k x_i \tilde{\varphi}_i = \sum_{i=1}^k x_i (\bar{\varphi}_i - \tilde{\varphi}_i). \quad (6.3-4)$$

As shown in Fig. 6.2, property change on mixing may take positive or negative values depending on the interactions between molecules. Note that the dotted line represents the case when $\Delta\tilde{\varphi}_{\text{mix}}$ is zero.

Among the property changes on mixing, chemical engineers are particularly interested in the *isothermal volume change on mixing*, $\Delta\tilde{V}_{\text{mix}}$, and the *isothermal enthalpy change on mixing* (or simply the *heat of mixing*), $\Delta\tilde{H}_{\text{mix}}$, both of which have to be determined experimentally.

When $\Delta V_{\text{mix}} > 0$, the interactions between unlike molecules are weaker than those between like molecules. On the other hand, if the interactions between unlike molecules are stronger than those between like molecules, then molecules pack themselves more tightly so as $\Delta V_{\text{mix}} < 0$. As shown in Fig. 6.3, depending on the nature of interactions between species in a mixture, the following cases may be encountered in practice:

- $\Delta\tilde{V}_{\text{mix}}$ is positive over the entire range of composition (Fig. 6.3a).

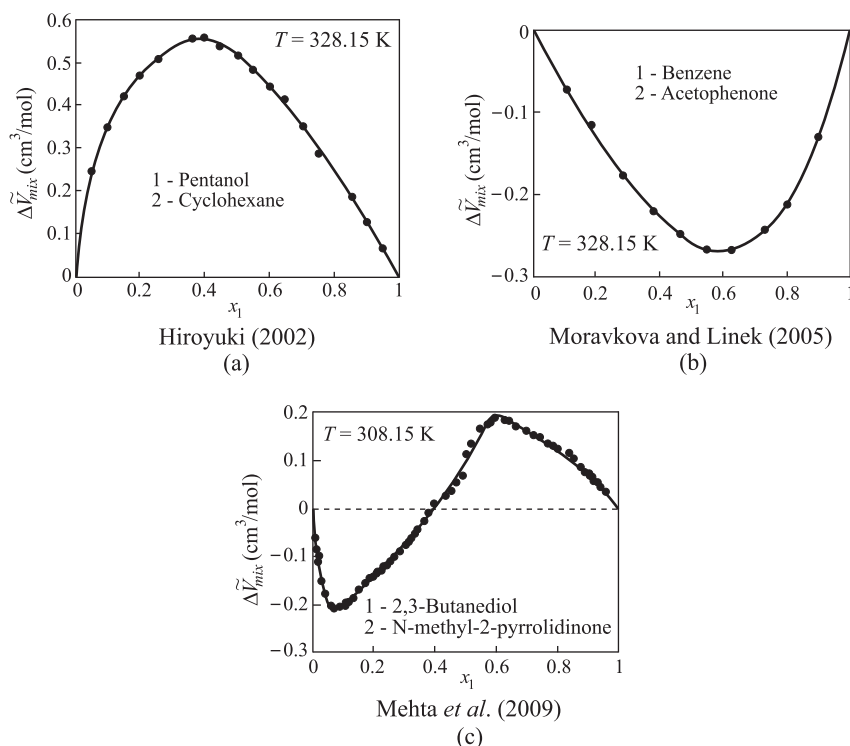


Figure 6.3: Variation of $\Delta \tilde{V}_{\text{mix}}$ as a function of composition.

- $\Delta \tilde{V}_{\text{mix}}$ is negative over the entire range of composition (Fig. 6.3b).
- $\Delta \tilde{V}_{\text{mix}}$ exhibits an S-shaped dependence (Fig. 6.3c).

In the case of heat of mixing, note that the enthalpy change is associated with the change in bond energies. If pure components A and B are mixed to form a mixture, it is necessary to break A–A and B–B bonds to create A–B bonds as shown in Fig. 6.4. Bond breaking requires or absorbs energy, i.e. an endothermic process. On the other hand, bond formation releases energy, i.e. an exothermic process.

When A–B bonds are stronger than A–A and B–B bonds, components in the mixture are more stable than pure components and $\Delta H_{\text{mix}} < 0$. On the other hand, when A–A and B–B bonds are stronger than A–B bonds, pure components are more stable than the components in the mixture and $\Delta H_{\text{mix}} > 0$. As in the case of ΔV_{mix} , variation of ΔH_{mix} with composition may exhibit the following cases:

- $\Delta \tilde{H}_{\text{mix}}$ is positive over the entire range of composition (Fig. 6.5a).
- $\Delta \tilde{H}_{\text{mix}}$ is negative over the entire range of composition (Fig. 6.5b).

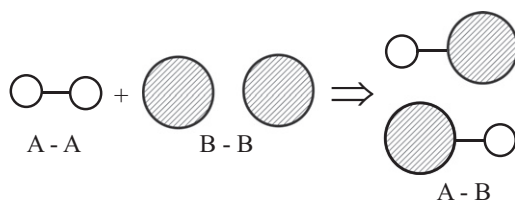


Figure 6.4: Formation of a binary mixture of species A and B.

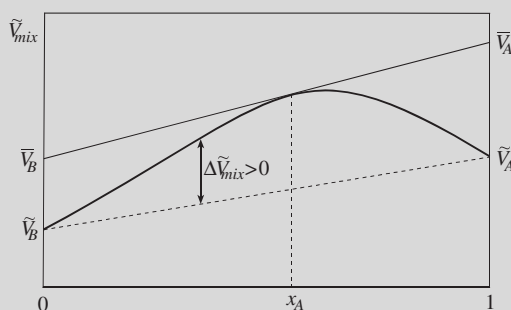
- $\Delta \tilde{H}_{\text{mix}}$ exhibits an S-shaped dependence (Fig. 6.5c).

Example 6.11 When 50 cm³ of liquid A is mixed with 60 cm³ of liquid B, the resulting solution has a volume of 113 cm³. What can be said about the relative magnitudes of $\bar{V}_A$ and $\tilde{V}_A$, as well as $\bar{V}_B$ and $\tilde{V}_B$?

Solution

Since $\Delta V_{\text{mix}} = 113 - (50 + 60) = 3 \text{ cm}^3 > 0$, a $\tilde{V}_{\text{mix}}$ versus composition diagram can be qualitatively drawn as shown in the figure below. From the graph one can easily conclude that

$$\bar{V}_A > \tilde{V}_A \quad \text{and} \quad \bar{V}_B > \tilde{V}_B.$$



6.3.1 Determination of the Volume Change on Mixing

Once the mixture is prepared by specifying the mole fractions of each species, its density as well as the densities of the pure components are measured by a densitometer. Then the volume change on mixing is determined from the following

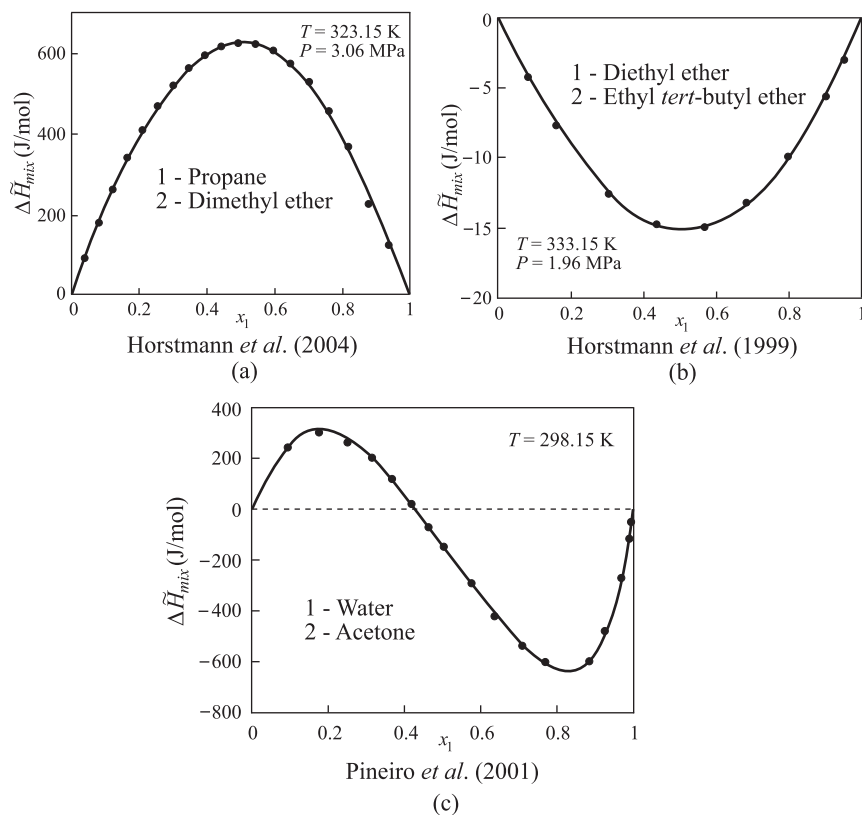


Figure 6.5: Variation of $\Delta \tilde{H}_{mix}$ as a function of composition.

equation:

$$\Delta \tilde{V}_{mix} = \frac{\sum_{i=1}^k x_i M_i}{\rho_{mix}} - \sum_{i=1}^k \frac{x_i M_i}{\rho_i} = \sum_{i=1}^k x_i M_i \left(\frac{1}{\rho_{mix}} - \frac{1}{\rho_i} \right), \quad (6.3-5)$$

where M_i is the molecular weight of species i and ρ_i is the density of pure i .

Example 6.12 The following data are reported by Grenner *et al.* (2006) for a binary mixture of cyclohexanamine (1) and *n*-heptane (2) at 303.15 K under atmospheric pressure:

x_1	0.0000	0.3006	0.5000	0.7996	1.0000
$\rho_{\text{mix}} \text{ (g/cm}^3\text{)}$	0.67532	0.71971	0.75383	0.81268	0.85820

The molecular weights of cyclohexanamine and *n*-heptane are 99.17 and 100.20 g/mol, respectively. Determine the volume change on mixing as a function of composition.

Solution

From the given data, pure component densities for cyclohexanamine and *n*-heptane are 0.85820 and 0.67532 g/cm³, respectively. The volume change on mixing can be calculated from Eqn (6.3-5), i.e.

$$\Delta \tilde{V}_{\text{mix}} = x_1 M_1 \left(\frac{1}{\rho_{\text{mix}}} - \frac{1}{\rho_1} \right) + x_2 M_2 \left(\frac{1}{\rho_{\text{mix}}} - \frac{1}{\rho_2} \right).$$

The calculated values are given below:

x_1	0.3006	0.5000	0.7996
$\Delta \tilde{V}_{\text{mix}} \text{ (cm}^3\text{/mol)}$	0.284	0.273	0.150

6.3.2 Determination of the Heat of Mixing

The heat of mixing data can be obtained by the use of an isothermal flow calorimeter as shown in Fig. 6.6. Note that the flow calorimeter is simply a steady-state mixing device. Pure fluid 1 and pure fluid 2, both at a temperature T , enter the calorimeter at molar flow rates of $\dot{n}_1$ and $\dot{n}_2$, respectively. They are mixed thoroughly by a stirrer

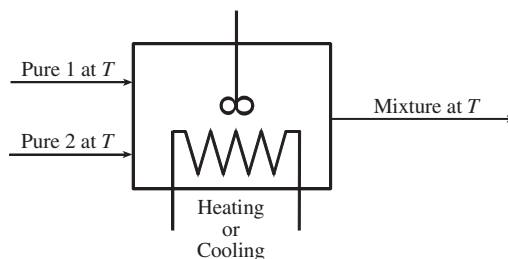


Figure 6.6: An isothermal flow calorimeter.

to maintain uniform temperature within the calorimeter. Heat is added or removed through the coils placed in the calorimeter so as to keep the temperature of the mixture equal to that of the entering streams. Taking the contents of the calorimeter to be the system, the mass and energy balances are

$$\dot{n}_1 + \dot{n}_2 = \dot{n}_{\text{mix}} \quad (6.3-6)$$

and

$$\dot{n}_{\text{mix}} \tilde{H}_{\text{mix}} - \dot{n}_1 \tilde{H}_1 - \dot{n}_2 \tilde{H}_2 = \dot{Q} + \dot{W}_s. \quad (6.3-7)$$

Assuming $\dot{W}_s$ to be negligible and dividing Eqn (6.3-7) by $\dot{n}_{\text{mix}}$ give

$$\underbrace{\tilde{H}_{\text{mix}} - (x_1 \tilde{H}_1 + x_2 \tilde{H}_2)}_{\Delta \tilde{H}_{\text{mix}}} = \frac{\dot{Q}}{\dot{n}_1 + \dot{n}_2}. \quad (6.3-8)$$

Therefore, once the molar flow rates of the pure streams are fixed, one can determine the heat of mixing from Eqn (6.3-8) by measuring the heat transfer rate, $\dot{Q}$. Carrying out experiments at different flow ratios of the two streams yields the heat of mixing as a function of composition.

Experimental determination of the heat of mixing enables one to calculate the molar enthalpy of the mixture from the following equation:

$$\tilde{H}_{\text{mix}} = \sum_{i=1}^k x_i \tilde{H}_i + \Delta \tilde{H}_{\text{mix}}. \quad (6.3-9)$$

Thus, it is possible to plot the enthalpy of the mixture as a function of composition. Such plots, also known as *enthalpy-concentration diagrams*, are very useful in determining energy balances.

Example 6.13 An isothermal mixer at 294 K is fed with a pure sulfuric acid stream of flow rate 160 kg/min and an aqueous solution stream containing 38 wt.% sulfuric acid of flow rate 150 kg/min. Estimate the rate of heat that must be added or removed so as to keep the temperature of the exit stream at 294 K under steady conditions.

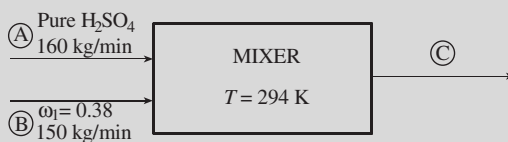
The heat of mixing data for the sulfuric acid (1) and water (2) mixture at 294 K is represented by Ross (1952) as

$$\Delta \tilde{H}_{\text{mix}} \text{ (kJ/mol)} = x_1 x_2 [-53.531 + 20.869(x_1 - x_2)].$$

The pure component enthalpies for sulfuric acid and water are 1.596 and 1.591 kJ/mol, respectively.

Solution

The flow diagram of the mixing process is given below:



The heat that must be removed or added to the mixer can be determined from the steady-state energy balance as

$$\Delta \dot{H} = \dot{Q} + \underbrace{\dot{W}_s}_{\sim 0} \quad (1)$$

or

$$\dot{Q} = \dot{n}_C \tilde{H}_C - \dot{n}_A \tilde{H}_A - \dot{n}_B \tilde{H}_B. \quad (2)$$

The molar enthalpies from streams B and C can be calculated from

$$\tilde{H}_{\text{mix}} = x_1 \tilde{H}_1 + x_2 \tilde{H}_2 + \Delta \tilde{H}_{\text{mix}} \quad (3)$$

$$= 1.596x_1 + 1.591x_2 + x_1x_2 \times [-53.531 + 20.869(x_1 - x_2)]. \quad (4)$$

To calculate the weight fraction of sulfuric acid in stream C, first it is necessary to write the overall material balance as

$$\dot{m}_A + \dot{m}_B = \dot{m}_C \Rightarrow \dot{m}_C = 160 + 150 = 310 \text{ kg/min.} \quad (5)$$

On the other hand, the material balance for sulfuric acid is

$$\dot{m}_A + \dot{m}_B(\omega_1)_B = \dot{m}_C(\omega_1)_C \Rightarrow (\omega_1)_C = \frac{160 + (150)(0.38)}{310} = 0.7. \quad (6)$$

The use of Eqn (4) requires mole fractions of components to be known. The relationship between the weight and mole fractions is given by

$$x_1 = \frac{\frac{\omega_1}{M_1}}{\frac{\omega_1}{M_1} + \frac{\omega_2}{M_2}}. \quad (7)$$

Thus, the mole fractions of sulfuric acid in streams B and C are

$$(x_1)_B = \frac{\frac{0.38}{98}}{\frac{0.38}{98} + \frac{1 - 0.38}{18}} = 0.1 \quad (x_1)_C = \frac{\frac{0.7}{98}}{\frac{0.7}{98} + \frac{1 - 0.7}{18}} = 0.3.$$

The use of Eqn (4) yields

$$\text{Stream A} \left\{ \begin{array}{l} \tilde{H}_A = 1.596 \text{ kJ/mol}, \end{array} \right.$$

$$\text{Stream B} \left\{ \begin{array}{l} \tilde{H}_B = (1.596)(0.1) + (1.591)(0.9) + (0.1)(0.9) [-53.531 + (20.869)(0.1 - 0.9)] \\ = -4.729 \text{ kJ/mol}, \end{array} \right.$$

$$\text{Stream C} \left\{ \begin{array}{l} \tilde{H}_C = (1.596)(0.3) + (1.591)(0.7) + (0.3)(0.7) [-53.531 + (20.869)(0.3 - 0.7)] \\ = -11.402 \text{ kJ/mol}. \end{array} \right.$$

To determine the molar flow rates of streams B and C, we have to calculate their molecular weights:

$$M_B = (0.1)(98) + (0.9)(18) = 26 \text{ g/mol} = 26 \times 10^{-3} \text{ kg/mol},$$

$$M_C = (0.3)(98) + (0.7)(18) = 42 \text{ g/mol} = 42 \times 10^{-3} \text{ kg/mol}.$$

Substitution of the numerical values into Eqn (2) gives

$$\begin{aligned} \dot{Q} &= \left(\frac{310}{42 \times 10^{-3}} \right) (-11.402) - \left(\frac{160}{98 \times 10^{-3}} \right) (1.596) - \left(\frac{150}{26 \times 10^{-3}} \right) \\ &\quad \times (-4.729) = -59,512 \text{ kJ/min} \end{aligned}$$

indicating that heat must be removed from the system.

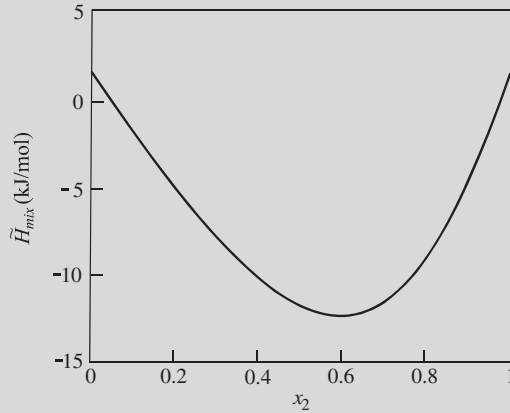
Example 6.14 Using the data given in Example 6.13 first plot the enthalpy-concentration diagram for sulfuric acid (1) and water (2) mixtures at 294 K. Then prove that it is always safer to add acid to water rather than water to acid.

Solution

The molar enthalpy of the mixture is given by

$$\tilde{H}_{\text{mix}} \text{ (kJ/mol)} = 1.596x_1 + 1.591x_2 + x_1x_2 [-53.531 + 20.869(x_1 - x_2)]. \quad (1)$$

The plot of the molar enthalpy of the mixture, $\tilde{H}_{\text{mix}}$, versus mole fraction of water, x_2 , is shown below:



Using Eqns (6.2-24) and (6.2-25), the partial molar enthalpies are obtained as a function of composition in the form

$$\bar{H}_1 = 1.596 - 53.531x_2^2 + 62.607x_1x_2^2 - 20.869x_2^3, \quad (2)$$

$$\bar{H}_2 = 1.591 - 53.531x_1^2 - 62.607x_1^2x_2 + 20.869x_1^3. \quad (3)$$

The partial molar enthalpies at infinite dilution are

$$\bar{H}_1^\infty = \lim_{x_1 \rightarrow 0} \bar{H}_1 = 1.596 - 53.531 - 20.869 = -72.804 \text{ kJ/mol},$$

$$\bar{H}_2^\infty = \lim_{x_2 \rightarrow 0} \bar{H}_2 = 1.591 - 53.531 + 20.869 = -31.071 \text{ kJ/mol}.$$

The heat of mixing is given by

$$\Delta \tilde{H}_{\text{mix}} = x_1(\bar{H}_1 - \tilde{H}_1) + x_2(\bar{H}_2 - \tilde{H}_2). \quad (4)$$

Now, let us consider two cases:

Process A (Adding a small amount of acid to a very large amount of water)

If 1 g of H_2SO_4 is added to 1000 g of H_2O , then

$$x_1 = \frac{\frac{1}{98}}{\frac{1}{98} + \frac{1000}{18}} = 1.84 \times 10^{-4}.$$

Since $x_1 \ll x_2$, then $\bar{H}_1 = \bar{H}_1^\infty$ and $\bar{H}_2 \simeq \tilde{H}_2$. Thus, Eqn (4) simplifies to

$$\begin{aligned} \Delta \tilde{H}_{\text{mix}}|_{\text{Process A}} &= x_1(\bar{H}_1^\infty - \tilde{H}_1) = (1.84 \times 10^{-4}) \\ &\times (-72,804 - 1596) = -13.7 \text{ J/mol}. \end{aligned} \quad (3)$$

Process B (Adding a small amount of water to a very large amount of acid)
If 1 g of H₂O is added to 1000 g of H₂SO₄, then

$$x_2 = \frac{\frac{1}{18}}{\frac{1000}{98} + \frac{1}{18}} = 5.42 \times 10^{-3}.$$

Since $x_2 \ll x_1$, then $\bar{H}_1 \simeq \tilde{H}_1$ and $\bar{H}_2 = \bar{H}_2^\infty$. Thus, Eqn (4) simplifies to

$$\begin{aligned} \Delta \tilde{H}_{\text{mix}}|_{\text{Process B}} &= x_2(\bar{H}_2^\infty - \tilde{H}_2) = (5.42 \times 10^{-3})(-31,071 - 1591) \\ &= -177 \text{ J/mol.} \end{aligned}$$

Note that mixing is exothermic for both processes. However, more heat is released when water is added to acid. This may lead to violent boiling of the solution and splashing of the acid. Therefore, acid should be slowly added to water to prevent boiling.

Once property change on mixing, $\Delta \tilde{\varphi}_{\text{mix}}$, is determined experimentally as a function of mole fraction, the next step is to fit the data and to express property change on mixing as a function of composition. For a binary system, for example, such a relationship is expressed by the following equation:

$$\Delta \tilde{\varphi}_{\text{mix}} = x_1 x_2 f(x_1, x_2). \quad (6.3-10)$$

The term $x_1 x_2$ originates from the fact that $\Delta \tilde{\varphi}_{\text{mix}}$ becomes zero for a pure component, i.e. $x_1 = 1$ (or $x_2 = 0$) and $x_2 = 1$ (or $x_1 = 0$). In the literature, it is customary to express $f(x_1, x_2)$ in polynomial form as

$$\frac{\Delta \tilde{\varphi}_{\text{mix}}}{x_1 x_2} = \sum_{i=1}^N A_i (x_1 - x_2)^{i-1} = A_1 + A_2(x_1 - x_2) + A_3(x_1 - x_2)^2 + \cdots \quad (6.3-11)$$

which is known as the Redlich-Kister type expansion.

Example 6.15 Conti *et al.* (1997) reported the following experimental values of $\Delta \tilde{H}_{\text{mix}}$ for a binary system of chloroform (1) and cyclohexane (2) at 298 K:

x_1	$\Delta\tilde{H}_{\text{mix}}$ (J/mol)	x_1	$\Delta\tilde{H}_{\text{mix}}$ (J/mol)	x_1	$\Delta\tilde{H}_{\text{mix}}$ (J/mol)
0.0563	150.8	0.2666	520.2	0.5740	618.1
0.0563	147.4	0.3661	609.6	0.6690	567.7
0.1643	375.2	0.4732	630.9	0.7587	478.3
0.2520	500.1	0.4732	644.3	0.8435	351.5
0.2666	519.0	0.5740	618.7	0.9238	199.7

Fit the data to the equation of the form

$$\Delta\tilde{H}_{\text{mix}} = x_1x_2 \sum_{i=1}^4 A_i(x_1 - x_2)^{i-1} \quad (1)$$

and evaluate the coefficients A_1 , A_2 , A_3 , and A_4 .

Solution

The coefficients can be estimated by the method of least squares. Note that Eqn (1) can be rearranged in the form of

$$\frac{\Delta\tilde{H}_{\text{mix}}}{x_1x_2} = A_1 + A_2(x_1 - x_2) + A_3(x_1 - x_2)^2 + A_4(x_1 - x_2)^3 \quad (2)$$

or

$$y = A_1 + A_2z + A_3z^2 + A_4z^3, \quad (3)$$

where

$$y = \frac{\Delta\tilde{H}_{\text{mix}}}{x_1x_2} \quad \text{and} \quad z = x_1 - x_2. \quad (4)$$

The coefficients A_i must be chosen such that the sum of the squares of the deviations, S , given by

$$S = \sum_{i=1}^N [y_i - (A_1 + A_2z_i + A_3z_i^2 + A_4z_i^3)]^2 \quad (5)$$

is minimum. Note that N represents the number of data points. Minimization of S is accomplished by differentiating Eqn (5) with respect to A_i ($i = 1, 2, 3, 4$),

and setting the derivatives equal to zero. The result is

$$A_1 N + A_2 \sum_{i=1}^N z_i + A_3 \sum_{i=1}^N z_i^2 + A_4 \sum_{i=1}^N z_i^3 = \sum_{i=1}^N y_i, \quad (6)$$

$$A_1 \sum_{i=1}^N z_i + A_2 \sum_{i=1}^N z_i^2 + A_3 \sum_{i=1}^N z_i^3 + A_4 \sum_{i=1}^N z_i^4 = \sum_{i=1}^N z_i y_i, \quad (7)$$

$$A_1 \sum_{i=1}^N z_i^2 + A_2 \sum_{i=1}^N z_i^3 + A_3 \sum_{i=1}^N z_i^4 + A_4 \sum_{i=1}^N z_i^5 = \sum_{i=1}^N z_i^2 y_i, \quad (8)$$

$$A_1 \sum_{i=1}^N z_i^3 + A_2 \sum_{i=1}^N z_i^4 + A_3 \sum_{i=1}^N z_i^5 + A_4 \sum_{i=1}^N z_i^6 = \sum_{i=1}^N z_i^3 y_i. \quad (9)$$

Equations (6)–(9) can be solved for A_i using the matrix algebra, i.e.

$$\begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{bmatrix} = \begin{bmatrix} N & \sum_{i=1}^N z_i & \sum_{i=1}^N z_i^2 & \sum_{i=1}^N z_i^3 \\ \sum_{i=1}^N z_i & \sum_{i=1}^N z_i^2 & \sum_{i=1}^N z_i^3 & \sum_{i=1}^N z_i^4 \\ \sum_{i=1}^N z_i^2 & \sum_{i=1}^N z_i^3 & \sum_{i=1}^N z_i^4 & \sum_{i=1}^N z_i^5 \\ \sum_{i=1}^N z_i^3 & \sum_{i=1}^N z_i^4 & \sum_{i=1}^N z_i^5 & \sum_{i=1}^N z_i^6 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^N y_i \\ \sum_{i=1}^N z_i y_i \\ \sum_{i=1}^N z_i^2 y_i \\ \sum_{i=1}^N z_i^3 y_i \end{bmatrix} \quad (10)$$

or

$$\begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{bmatrix} = \begin{bmatrix} 15 & -1.565 & 4.401 & -0.928 \\ -1.565 & 4.401 & -0.928 & 2.429 \\ 4.401 & -0.928 & 2.429 & -0.681 \\ -0.928 & 2.429 & -0.681 & 1.601 \end{bmatrix}^{-1} \begin{bmatrix} 3.979 \times 10^4 \\ -4.389 \times 10^3 \\ 1.206 \times 10^4 \\ -2.597 \times 10^3 \end{bmatrix}$$

$$= \begin{bmatrix} 2.546 \times 10^3 \\ -125.091 \\ 361.22 \\ 197.836 \end{bmatrix}.$$

Comment: The correlation coefficient⁸ for the third-order polynomial fit is 0.979.

6.3.3 Determination of Partial Molar Properties from $\Delta\tilde{\varphi}_{\text{mix}}$

One of the practical questions to ask is how to determine partial molar properties once a property change on mixing is determined as a function of composition. For

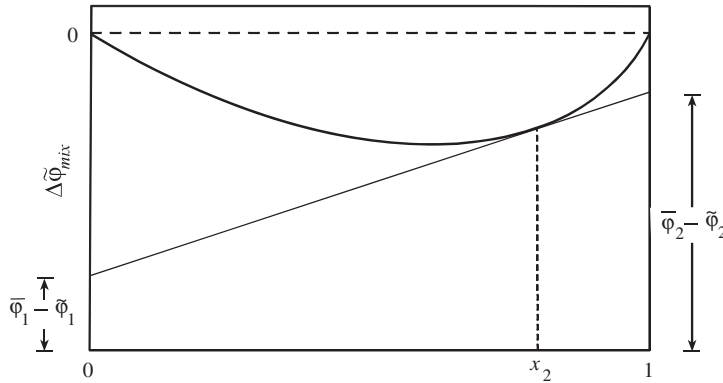


Figure 6.7: Determination of partial molar properties from a property change on mixing.

a binary mixture, the molar property of a mixture is given by

$$\tilde{\varphi}_{\text{mix}} = (x_1 \tilde{\varphi}_1 + x_2 \tilde{\varphi}_2) + \Delta \tilde{\varphi}_{\text{mix}}. \quad (6.3-12)$$

On the other hand, Eqn (6.2-24) gives the partial molar property of component 1 as

$$\bar{\varphi}_1 = \tilde{\varphi}_{\text{mix}} - x_2 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2}. \quad (6.3-13)$$

Substitution of Eqn (6.3-12) into Eqn (6.3-13) gives

$$\bar{\varphi}_1 = x_1 \tilde{\varphi}_1 + x_2 \tilde{\varphi}_2 + \Delta \tilde{\varphi}_{\text{mix}} - x_2 \left[-\tilde{\varphi}_1 + \tilde{\varphi}_2 + \left(\frac{\partial \Delta \tilde{\varphi}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2} \right]. \quad (6.3-14)$$

Simplification of Eqn (6.3-14) leads to

$$\bar{\varphi}_1 - \tilde{\varphi}_1 = \Delta \tilde{\varphi}_{\text{mix}} - x_2 \left(\frac{\partial \Delta \tilde{\varphi}_{\text{mix}}}{\partial x_2} \right)_{T,P,n_2}. \quad (6.3-15)$$

Since Eqns (6.2-24) and (6.3-15) have similar forms, graphical interpretation of Eqn (6.3-15) is the same as that of Eqn (6.2-24) as shown in Fig. 6.7.

6.4 The Gibbs-Duhem Equation

The Gibbs-Duhem equation gives a relationship between the partial molar properties of different components in a mixture. For any extensive thermodynamic property, the total property of the mixture is expressed as

$$\varphi_{\text{mix}} = \sum_{i=1}^k n_i \bar{\varphi}_i. \quad (6.4-1)$$

Differentiation of Eqn (6.4-1) gives

$$d\varphi_{\text{mix}} = \sum_{i=1}^k n_i d\bar{\varphi}_i + \sum_{i=1}^k \bar{\varphi}_i dn_i. \quad (6.4-2)$$

On the other hand, the total property of the mixture, φ_{mix} , is a function of temperature, pressure, and the number of moles of each component present in the mixture, i.e.

$$\varphi_{\text{mix}} = \varphi_{\text{mix}}(T, P, n_1, n_2, \dots, n_k). \quad (6.4-3)$$

The total differential of φ_{mix} is

$$d\varphi_{\text{mix}} = \left(\frac{\partial \varphi_{\text{mix}}}{\partial T} \right)_{P, n_j} dT + \left(\frac{\partial \varphi_{\text{mix}}}{\partial P} \right)_{T, n_j} dP + \sum_{i=1}^k \underbrace{\left(\frac{\partial \varphi_{\text{mix}}}{\partial n_i} \right)_{T, P, n_{j \neq i}}}_{\bar{\varphi}_i} dn_i. \quad (6.4-4)$$

At constant temperature and pressure Eqn (6.4-4) simplifies to

$$d\varphi_{\text{mix}} = \sum_{i=1}^k \bar{\varphi}_i dn_i. \quad (6.4-5)$$

The use of Eqn (6.4-5) in Eqn (6.4-2) leads to

$$\sum_{i=1}^k n_i d\bar{\varphi}_i = 0 \quad (6.4-6)$$

or dividing each term by the total number of moles gives

$$\sum_{i=1}^k x_i d\bar{\varphi}_i = 0 \quad \text{constant } T \text{ \& } P, \quad (6.4-7)$$

which is known as the Gibbs-Duhem equation under the conditions of constant temperature and pressure.

For simplicity let us consider a binary mixture and exploit the practical uses of the Gibbs-Duhem equation. For a binary mixture, Eqn (6.4-7) simplifies to

$$x_1 d\bar{\varphi}_1 + x_2 d\bar{\varphi}_2 = 0. \quad (6.4-8)$$

Differentiation of Eqn (6.4-8) with respect to x_1 results in

$$x_1 \frac{d\bar{\varphi}_1}{dx_1} + x_2 \frac{d\bar{\varphi}_2}{dx_1} = 0 \quad \text{constant } T \text{ \& } P. \quad (6.4-9)$$

The Gibbs-Duhem equation, Eqn (6.4-9), can be used for the following purposes:

- If the partial molar quantity of one of the species is known, the partial molar quantity of the other species can be calculated from Eqn (6.4-9).
- If the partial molar quantities of both species are experimentally determined, the expressions for $\bar{\varphi}_1$ and $\bar{\varphi}_2$ should satisfy the Gibbs-Duhem equation. Therefore, the Gibbs-Duhem equation is quite useful for checking whether the experimental data are thermodynamically consistent or not.

Example 6.16 If the partial molar volume of component 1 in a binary mixture at constant T and P is given by

$$\bar{V}_1 = \tilde{V}_1 + \beta x_2^2,$$

where $\tilde{V}_1$ is the molar volume of pure 1, find the corresponding equation for $\bar{V}_2$.

Solution

For $\varphi = V$, the Gibbs-Duhem equation, Eqn (6.4-9), is written as

$$x_1 \frac{d\bar{V}_1}{dx_1} + x_2 \frac{d\bar{V}_2}{dx_1} = 0 \quad \text{constant } T \text{ \& } P. \quad (1)$$

The term $d\bar{V}_1/dx_1$ becomes

$$\frac{d\bar{V}_1}{dx_1} = \frac{d}{dx_1} [\tilde{V}_1 + \beta(1 - x_1)^2] = -2\beta(1 - x_1) = -2\beta x_2. \quad (2)$$

It is also possible to evaluate the same term as

$$\frac{d\bar{V}_1}{dx_1} = -\frac{d\bar{V}_1}{dx_2} = -\frac{d}{dx_2} (\tilde{V}_1 + \beta x_2^2) = -2\beta x_2.$$

Substitution of Eqn (2) into Eqn (1) gives

$$\frac{d\bar{V}_2}{dx_1} = 2\beta x_1. \quad (3)$$

Integration of Eqn (3) leads to

$$\bar{V}_2 = \beta x_1^2 + C. \quad (4)$$

The constant C is evaluated by using the fact that as $x_1 \rightarrow 0$, $\bar{V}_2 \rightarrow \tilde{V}_2$. This gives $C = \tilde{V}_2$ and the partial molar volume of species 2 is expressed in the form

$$\bar{V}_2 = \tilde{V}_2 + \beta x_1^2. \quad (5)$$

The generalized form of Eqn (6.4-9) for a k -component system is given in Problem 6.29.

Problems

Problems related to Section 6.1

- 6.1** Estimate the molar volume of a binary mixture containing 30 mol% propane (1) and 70% n -butane (2) at 350 K and 3 bar. The mixture is represented by the virial equation state, and the critical molar volumes of propane and n -butane are $2.00 \times 10^{-4} \text{ m}^3/\text{mol}$ and $2.55 \times 10^{-4} \text{ m}^3/\text{mol}$, respectively.

(Answer: $9.27 \times 10^{-3} \text{ m}^3/\text{mol}$)

- 6.2** A binary gaseous mixture at T_1 and P_1 enters a throttling valve and leaves it at P_2 . The mixture is represented by the virial equation of state with the virial coefficients depending on temperature in the form

$$B_{ij} = \alpha_{ij} + \beta_{ij}T.$$

Show that the exit temperature, T_2 , is given by

$$T_2 = T_1 + \frac{\alpha_{\text{mix}}(P_1 - P_2)}{\tilde{C}_{P_{\text{mix}}}^*},$$

where

$$\begin{aligned}\alpha_{\text{mix}} &= y_1^2 \alpha_{11} + 2y_1 y_2 \alpha_{12} + y_2^2 \alpha_{22}, \\ \tilde{C}_{P_{\text{mix}}}^* &= y_1 \tilde{C}_{P_1}^* + y_2 \tilde{C}_{P_2}^*\end{aligned}$$

in which y_1 and y_2 represent mole fractions of components 1 and 2, respectively.

- 6.3** A rigid cylinder holds a gas mixture containing 75 mol% propane (1) and 25% n -butane (2) at 300 K and 50 bar. Calculate the molar volume of the gas mixture ($k_{12} = 0.003$) using

- Redlich-Kwong equation of state.
- Peng-Robinson equation of state.

(Answer: (a) $99.53 \text{ cm}^3/\text{mol}$ and (b) $86.02 \text{ cm}^3/\text{mol}$)

- 6.4** It is required to store 10 kg of ethane (1), 25 kg of ethylene (2), and 40 kg of propane (3) at 400 K and 35 bar. Estimate the volume of the tank if the mixture

obeys the Peng-Robinson equation of state. Take $k_{12} = 0.010$, $k_{13} = 0.001$, and $k_{23} = 0$.

(Answer: 1.712 m³)

- 6.5** A rigid tank is divided into two parts by a rigid partition. One side contains 2 kmol of methane at 300 K and 100 bar, and the other side contains 3 kmol of nitrogen at 300 K and 100 bar. The partition is removed and the gases mix with each other. If the temperature is kept constant at 300 K, estimate the final pressure. The system is represented by the van der Waals equation of state.

(Answer: 101.1 bar)

- 6.6** The term $(\partial \tilde{V}/\partial T)_P$ is needed in the calculation of the Joule-Thomson coefficient (see Problem 2.1). If a gas mixture is represented by the Peng-Robinson equation of state, then this term is calculated indirectly by the use of Eqn (2.2-50), i.e.

$$\left(\frac{\partial \tilde{V}}{\partial T}\right)_P = -\frac{(\partial P/\partial T)_{\tilde{V}}}{(\partial P/\partial \tilde{V})_T}. \quad (1)$$

- (a) Show that

$$\left(\frac{\partial P}{\partial T}\right)_{\tilde{V}} = \frac{P}{T} \left[\frac{1}{Z_{\text{mix}} - B_{\text{mix}}} - \frac{\Theta_{\text{mix}}}{Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})} \right], \quad (2)$$

$$\left(\frac{\partial P}{\partial \tilde{V}}\right)_T = \frac{P^2}{RT} \left\{ -\frac{1}{(Z_{\text{mix}} - B_{\text{mix}})^2} + \frac{2A_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}})}{[Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})]^2} \right\}, \quad (3)$$

where Θ_{mix} is defined by Eqn (6.1-27).

- (b) For a binary gas mixture consisting of 75 mol% propane (1) and 25% *n*-butane (2) at 310 K and 10 bar, calculate the value of $(\partial \tilde{V}/\partial T)_P$. Take $k_{12} = 0.003$.

(Answer: (b) 12.383 cm³/mol K)

- 6.7** A gas mixture consisting of 65 mol % methane (1) and 35% isobutane (2) is initially at 400 K and 60 bar. Estimate the temperature of the gas mixture if it is throttled to 5 bar. Use the Peng-Robinson equation of state with $k_{12} = 0.026$.

(Answer: 372.6 K)

6.8 This problem is the extension of Problem 3.35 to mixtures. For mixtures, the departure function for molar heat capacity at constant volume is defined by

$$(\tilde{C}_{V_{\text{mix}}} - \tilde{C}_V^{\text{IGM}})_{T,P} = \frac{\partial}{\partial T} \left[(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}})_{T,P} \right]_{\tilde{V}_{\text{mix}}} \quad (1)$$

- (a) For a gas mixture represented by the Peng-Robinson equation of state, show that the use of Eqn (6.1-39) in Eqn (1) leads to

$$\frac{(\tilde{C}_{V_{\text{mix}}} - \tilde{C}_V^{\text{IGM}})_{T,P}}{R} = \frac{\psi_{\text{mix}}}{\sqrt{8}B_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}} + (1 + \sqrt{2})B_{\text{mix}}}{Z_{\text{mix}} + (1 - \sqrt{2})B_{\text{mix}}} \right], \quad (2)$$

where

$$\psi_{\text{mix}} = \frac{1}{4} \sum_{i=1}^k \sum_{j=1}^k x_i x_j A_{ij} (\Gamma_i + 2\Gamma_i \Gamma_j + \Gamma_j) \quad (3)$$

in which the term Γ_i is defined by Eqn (6.1-17).

- (b) Combine Eqn (2) with Eqns (2.2-51) and (3.2-12) to obtain

$$\begin{aligned} \frac{(\tilde{C}_{P_{\text{mix}}} - \tilde{C}_P^{\text{IGM}})_{T,P}}{R} &= \frac{\psi_{\text{mix}}}{\sqrt{8}B_{\text{mix}}} \ln \left[\frac{Z_{\text{mix}} + (1 + \sqrt{2})B_{\text{mix}}}{Z_{\text{mix}} + (1 - \sqrt{2})B_{\text{mix}}} \right], \\ &+ \frac{\left[\frac{1}{Z_{\text{mix}} - B_{\text{mix}}} - \frac{\Theta_{\text{mix}}}{Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})} \right]^2}{\frac{1}{(Z_{\text{mix}} - B_{\text{mix}})^2} - \frac{2A_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}})}{[Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})]^2}} - 1, \end{aligned} \quad (4)$$

where Θ_{mix} is defined by Eqn (6.1-27). Note that the molar heat capacity of an ideal gas mixture is the mole fraction weighted average of the pure component values, i.e.

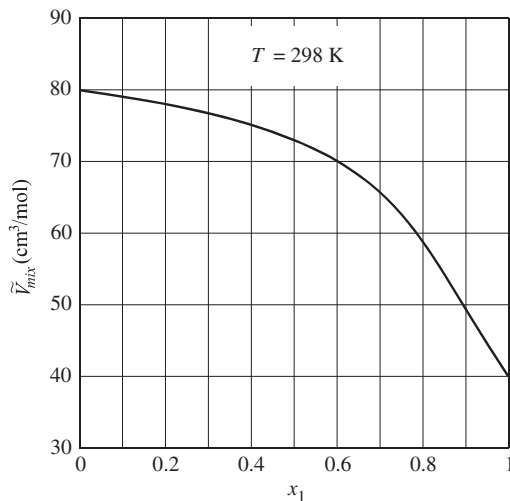
$$\tilde{C}_P^{\text{IGM}} = \sum_{i=1}^k y_i \tilde{C}_{P_i}^* \quad (5)$$

- (c) For a binary gas mixture consisting of 75 mol% propane (1) and 25% *n*-butane (2) at 310 K and 10 bar, calculate the departure functions for molar heat capacities at constant volume and pressure. The mixture obeys the Peng-Robinson equation of state with $k_{12} = 0.003$.

(Answer: (c) 0.775 J/mol K and 10.173 J/mol K)

Problems related to Section 6.2

6.9 It is required to prepare 35 L of a mixture containing 60 mol% component 1 and 40% component 2 at 298 K. The molar volume of the mixture as a function of composition is shown below.



- (a) Estimate the volumes of pure components that must be mixed for this purpose.
- (b) Are the like interactions stronger or weaker than the unlike interactions?

(Answer: (a) $V_1 = 12 \text{ L}$, $V_2 = 16 \text{ L}$)

6.10 Consider a k -component mixture in which n represents the total number of moles of material.

- (a) Show that the substitution of

$$\varphi_{\text{mix}} = n\tilde{\varphi}_{\text{mix}} \quad (1)$$

into Eqn (6.2-5) gives

$$\bar{\varphi}_i = \tilde{\varphi}_{\text{mix}} + n \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial n_i} \right)_{T, P, n_{j \neq i}}. \quad (2)$$

- (b) Use the chain rule of differentiation and express Eqn (2) as

$$\bar{\varphi}_i = \tilde{\varphi}_{\text{mix}} + n \sum_{\substack{r=1 \\ r \neq i}}^k \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_r} \right)_{T, P, x_{j \neq i, r}} \left(\frac{\partial x_r}{\partial n_i} \right), \quad (3)$$

where the subscript $x_{j \neq i, r}$ indicates that mole fractions of all components other than i and r are kept constant.

(c) Noting that

$$\frac{\partial x_r}{\partial n_i} = -\frac{n_r}{n^2} \quad (4)$$

show that Eqn (3) takes the form

$$\bar{\varphi}_i = \tilde{\varphi}_{\text{mix}} - \sum_{\substack{r \neq i \\ r=1}}^k x_r \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_r} \right)_{T, P, x_{j \neq i, r}}. \quad (5)$$

(d) For a ternary system show that

$$\bar{\varphi}_1 = \tilde{\varphi}_{\text{mix}} - x_2 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_2} \right)_{x_3} - x_3 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_3} \right)_{x_2}, \quad (6)$$

$$\bar{\varphi}_2 = \tilde{\varphi}_{\text{mix}} - x_1 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_1} \right)_{x_3} - x_3 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_3} \right)_{x_1}, \quad (7)$$

$$\bar{\varphi}_3 = \tilde{\varphi}_{\text{mix}} - x_1 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_1} \right)_{x_2} - x_2 \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial x_2} \right)_{x_1}. \quad (8)$$

6.11 For a ternary system of components 1, 2, and 3, the molar volume of the mixture is given by

$$\tilde{V}_{\text{mix}} = \tilde{V}_1 x_1 + \tilde{V}_2 x_2 + \tilde{V}_3 x_3 + A x_1 x_2 x_3,$$

where A is a function of temperature, and the terms $\tilde{V}_1$, $\tilde{V}_2$, and $\tilde{V}_3$ represent the molar volumes of pure components 1, 2, and 3, respectively. Use Eqns (6), (7), and (8) of Problem 6.10 and show that the partial molar volumes of components 1, 2, and 3 are expressed as a function of composition by the following equations:

$$\begin{aligned} \bar{V}_1 &= \tilde{V}_1 + A x_2 x_3 (1 - 2x_1), \\ \bar{V}_2 &= \tilde{V}_2 + A x_1 x_3 (1 - 2x_2), \\ \bar{V}_3 &= \tilde{V}_3 + A x_1 x_2 (1 - 2x_3). \end{aligned}$$

6.12 The molar enthalpy of a binary liquid mixture of A and B is represented by

$$\tilde{H}_{\text{mix}} = (\alpha_1 + \beta_1 x_A) x_A + (\alpha_2 + \beta_2 x_B) x_B,$$

where α_1 , α_2 , β_1 , and β_2 are dependent on temperature. Show that

$$\begin{aligned} \bar{H}_A &= (\alpha_1 - \beta_2) + 2(\beta_1 + \beta_2) x_A - (\beta_1 + \beta_2) x_A^2, \\ \bar{H}_B &= (\alpha_2 + \beta_2) - (\beta_1 + \beta_2) x_A^2. \end{aligned}$$

- 6.13** To prevent freezing of water in a car radiator, ethylene glycol (MW = 62.07g/mol) is added as an antifreeze. Ray and Nemethy (1973) provided the following data for the mixture of water (1) and ethylene glycol (2) at 298 K:

$$\rho_{\text{mix}} = 0.99722 + 0.42114x_2 - 0.72022x_2^2 + 0.62036x_2^3 - 0.20911x_2^4,$$

where ρ_{mix} is in g/cm³.

- Calculate the molar volumes of pure water and ethylene glycol.
- If 1 L of ethylene glycol is added to 5 L of water, what will be the total volume of the resulting mixture?
- Determine the partial molar volumes of water and ethylene glycol at $x_2 = 0.2$.

(Answer: (a) 18.07 cm³/mol and 55.95 cm³/mol, (b) 5980 cm³, and (c) 17.37 cm³/mol and 57.45 cm³/mol)

- 6.14** Partial molar volume of species i can be calculated from Eqn (6.2-4) once the equation of state is specified.

- Since cubic equations of state are pressure-explicit, show that it is more convenient to express Eqn (6.2-4) in the form

$$\bar{V}_i = - \frac{\left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}}}{\left(\frac{\partial P}{\partial V_{\text{mix}}} \right)_{T, n_j}}.$$

- Use the Redlich-Kwong equation of state to obtain

$$\bar{V}_i = \frac{RT}{P} \left[\frac{\frac{1}{Z_{\text{mix}} - B_{\text{mix}}} \left(1 + \frac{B_i}{Z_{\text{mix}} - B_{\text{mix}}} \right) - \frac{2 \sum_{j=1}^k x_j A_{ij} - \frac{A_{\text{mix}} B_i}{Z_{\text{mix}} + B_{\text{mix}}}}{Z_{\text{mix}} (Z_{\text{mix}} + B_{\text{mix}})}}{\frac{1}{(Z_{\text{mix}} - B_{\text{mix}})^2} - \frac{A_{\text{mix}} (2Z_{\text{mix}} + B_{\text{mix}})}{Z_{\text{mix}}^2 (Z_{\text{mix}} + B_{\text{mix}})^2}} \right].$$

- A ternary gas mixture containing 45 mol% methane (1), 35% ethane (2), and 20% carbon dioxide (3) is at 350 K and 25 bar. Estimate the partial molar volumes of each component if the mixture obeys the Redlich-Kwong equation of state. Take $k_{ij} = 0$.

(Answer: 1146 cm³/mol, 1035 cm³/mol, 1072 cm³/mol)

- 6.15** For a mixture obeying the Peng-Robinson equation of state, use Eqn (6.4) of Problem 6.14 and show that the partial molar volume of species *i* is given by

$$\bar{V}_i = \frac{RT}{P} \left[\frac{1}{Z_{\text{mix}} - B_{\text{mix}}} \left(1 + \frac{B_i}{Z_{\text{mix}} - B_{\text{mix}}} \right) - \frac{2 \sum_{j=1}^k x_j A_{ij} - \frac{2A_{\text{mix}}B_i(Z_{\text{mix}} - B_{\text{mix}})}{Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})}}{Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})} \right] \times \left[\frac{1}{(Z_{\text{mix}} - B_{\text{mix}})^2} - \frac{2A_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}})}{[Z_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}}) + B_{\text{mix}}(Z_{\text{mix}} - B_{\text{mix}})]^2} \right].$$

Consider a binary mixture consisting of 75 mol% ethane (1) and 25% *n*-decane (2) at 444 K and 138 bar. Estimate the partial molar volume of ethane using the Peng-Robinson equation of state. Take $k_{ij} = 0$.

(Answer: 153.4 cm³/mol).

Problems related to Section 6.3

- 6.16** For a binary mixture of components 1 and 2, the molar enthalpy of the mixture (in J/mol) is given by

$$\tilde{H}_{\text{mix}} = 750x_1 + 380x_2 + 10x_1(5x_1 + 17x_2).$$

- (a) Show that the partial molar enthalpies of components 1 and 2 are expressed as

$$\bar{H}_1 = 920 - 240x_1 + 120x_1^2 \quad \text{and} \quad \bar{H}_2 = 380 + 120x_1^2.$$

- (b) Show that

$$\Delta\tilde{H}_{\text{mix}} = 120x_1x_2.$$

- (c) Calculate $\bar{H}_1$, $\bar{H}_2$, and $\Delta\tilde{H}_{\text{mix}}$ at $x_1 = 0.4$.

(Answer: (c) 843.2 J/mol, 399.2 J/mol, 28.8 J/mol)

- 6.17** Using the following data reported by Moravkova and Linek (2005) for a binary mixture of benzene (1) and acetophenone (2) at 25 °C, determine the volume

change on mixing (in cm^3/mol) at the given mole fractions of benzene. The molecular weights of benzene and acetophenone are 78.1 and 120.15 g/mol, respectively.

x_1	$\rho_{\text{mix}} (\text{g}/\text{cm}^3)$	x_1	$\rho_{\text{mix}} (\text{g}/\text{cm}^3)$	x_1	$\rho_{\text{mix}} (\text{g}/\text{cm}^3)$
0.00000	1.02311	0.37918	0.97658	0.72894	0.92384
0.10571	1.01102	0.46261	0.96498	0.80033	0.91156
0.18602	1.00134	0.54838	0.95249	0.89527	0.89423
0.28547	0.98894	0.62477	0.94071	1.00000	0.87364

- 6.18** An isothermal mixer at 300 K is fed with a stream containing 35 mol% A of flow rate 2 mol/s and a stream containing 80 mol% A of flow rate 1 mol/s. Calculate the heat that must be removed or added per mole of solution leaving the mixer under steady conditions so as to keep the outlet stream temperature constant at 300 K. The heat of mixing data for a binary liquid mixture of A and B at 300 K are represented by the following equation:

$$\Delta \tilde{H}_{\text{mix}} = x_A x_B (10x_A + 30x_B),$$

where $\Delta \tilde{H}_{\text{mix}}$ is in J/mol.

(Answer: 0.765 J/mol of solution leaving the mixer)

- 6.19** Suppose that the pure component enthalpies of A and B in Problem 6.18 are 180 and 250 J/mol, respectively.

- Calculate the partial molar enthalpies of components A and B at infinite dilution.
- Plot $\tilde{H}_{\text{mix}}$ versus x_A and evaluate $\overline{H}_A^\infty$ and $\overline{H}_B^\infty$ graphically.

(Answer: (a) 210 J/mol and 260 J/mol)

- 6.20** The enthalpy of solution (or heat of solution) is defined as the change in enthalpy that results when one mole of solute (component 1) is dissolved in a solvent (component 2). [Ogawa *et al.* \(1997\)](#) reported the following data for the enthalpy of solution, ΔH_{sol} , for benzene (1) in cyclohexane (2) at 298.15 K:

m_1 (mol/kg)	ΔH_{sol} (kJ/mol of benzene)
0.0912	3.15
0.1529	3.16
0.1544	3.16
0.1569	3.13

where m_1 is the molality of benzene. Determine the heat of mixing as a

function of the mole fraction of benzene.

(Answer: 24 J/mol, 40.1 J/mol, 40.5 J/mol, 40.8 J/mol)

6.21 Constantinescu and Wichterle (2002) reported the following experimental values of $\Delta \tilde{V}_{\text{mix}}$ for a binary system of ethanol (1) and methyl propanoate (2) at 298.15 K:

x_1	$\Delta \tilde{V}_{\text{mix}}$ (cm ³ /mol)	x_1	$\Delta \tilde{V}_{\text{mix}}$ (cm ³ /mol)	x_1	$\Delta \tilde{V}_{\text{mix}}$ (cm ³ /mol)
0.0718	0.04714	0.3976	0.12064	0.7065	0.08995
0.1378	0.07003	0.4838	0.12086	0.7531	0.08079
0.2216	0.10068	0.5411	0.11743	0.8285	0.06207
0.2947	0.10946	0.5873	0.10664	0.9115	0.03438
0.3441	0.11500	0.6496	0.09741		

Fit the data to the following equation

$$\Delta \tilde{V}_{\text{mix}} = x_1 x_2 \sum_{i=1}^3 A_i (x_1 - x_2)^{i-1} \quad (1)$$

and evaluate the coefficients A_1 , A_2 , and A_3 .

(Answer: $A_1 = 0.4660$, $A_2 = -0.1352$, $A_3 = 0.1226$).

6.22 Serbanovic *et al.* (2006) reported the following data for the densities of a binary mixture of methanol (1) and benzene (2) at 298.15 K:

x_1	ρ_{mix} (g/cm ³)	x_1	ρ_{mix} (g/cm ³)	x_1	ρ_{mix} (g/cm ³)
0.0000	0.873582	0.2030	0.864285	0.5503	0.842600
0.0516	0.871283	0.2607	0.861329	0.5810	0.840082
0.0768	0.870172	0.2978	0.859308	0.6499	0.833959
0.1011	0.869074	0.3312	0.857433	0.6993	0.829097
0.1241	0.868029	0.3497	0.856357	0.8000	0.817709
0.1501	0.866825	0.4019	0.853164	0.8992	0.804020
0.1989	0.864485	0.4493	0.850021	1.0000	0.786694
0.2004	0.864414	0.4990	0.846501		

(a) Fit the data to the following equation:

$$\Delta \tilde{V}_{\text{mix}} = x_1 x_2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} \quad (1)$$

and show that the coefficients are given as

$$A_1 = -0.0199 \quad A_2 = -0.1342 \quad A_3 = 0.1740 \quad A_4 = -0.1906.$$

- (b) Use Eqn (6.3-15) and show that the partial molar volumes of components 1 and 2 are expressed as a function of composition in the form

$$\bar{V}_1 = \tilde{V}_1 + x_2^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} + 2x_1 x_2^2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1}, \quad (2)$$

$$\bar{V}_2 = \tilde{V}_2 + x_1^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} - 2x_1^2 x_2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1} \quad (3)$$

and conclude that

$$\bar{V}_1^\infty = \tilde{V}_1 + \sum_{i=1}^4 (-1)^{i-1} A_i \quad \text{and} \quad \bar{V}_2^\infty = \tilde{V}_2 + \sum_{i=1}^4 A_i. \quad (4)$$

Also calculate the numerical values of $\bar{V}_1^\infty$ and $\bar{V}_2^\infty$.

(Answer: (b) 41.206 cm³/mol and 89.243 cm³/mol)

- 6.23** The molar volume of a mixture containing methanol (1) and n-hexane (2) at 298.15 K is given by Orge *et al.* (1997) as

$$\tilde{V}_{\text{mix}} \text{ (cm}^3\text{/mol)} = 40.74x_1 + 131.55x_2 + x_1x_2 \sum_{i=1}^3 A_i (x_1 - x_2)^{i-1}$$

with

$$A_1 = 2.0741 \quad A_2 = 0.3195 \quad A_3 = 1.7733.$$

- (a) Determine the volumes of pure methanol and n-hexane required to form 600 cm³ of a solution containing 30 mol % methanol.
 (b) Determine the total volume of the mixture when 5 mol of methanol are mixed with 3 mol of n-hexane. What would the volume be if the mixture behaved ideally?

(Answer: (a) 70 cm³ and 527.3 cm³ and (b) 602.6 cm³, 598.4 cm³)

- 6.24** Lepori *et al.* (2002) determined volume change on mixing for a binary mixture of perfluorohexane (C₆F₁₄) and n-octane at 298.15 K using a vibrating-tube densimeter. The experimental results are fitted to an equation of the form

$$\Delta \tilde{V}_{\text{mix}} \text{ (cm}^3\text{/mol)} = x_1 x_2 [16.00 + 3.67(x_1 - x_2) + 11.70(x_1 - x_2)^2],$$

where subscripts 1 and 2 represent perfluorohexane and n-octane, respectively. The molar volumes of pure components are reported as

$$\tilde{V}_1 = 201.64 \text{ cm}^3\text{/mol} \quad \tilde{V}_2 = 163.50 \text{ cm}^3\text{/mol}.$$

- (a) Estimate the total volume of the mixture when 500 cm³ of perfluorohexane is mixed with 800 cm³ of *n*-octane.
 (b) Calculate the partial molar volume of each component at the resulting mixture composition.
 (c) Calculate the partial molar volume of each component at infinite dilution.

(Answer: (a) 1326 cm³, (b) 207.53 cm³/mol and 165.92 cm³/mol, and (c) 225.67 cm³/mol and 194.87 cm³/mol)

6.25 The heat of mixing data for a binary mixture of tetramethyl-1,3-butanediamine (TMBD) and *n*-heptane at 298.15 K were correlated by Dahmani *et al.* (2002) in the form

$$\Delta\tilde{H}_{\text{mix}} \text{ (J/mol)} = x_1x_2 [1186.86 - 338.28(x_1 - x_2)],$$

where subscripts 1 and 2 represent TMBD and *n*-heptane, respectively.

- (a) Plot $\Delta\tilde{H}_{\text{mix}}$ versus x_1 and estimate the difference between the partial molar and pure component enthalpies of TMBD and *n*-heptane at $x_1 = 0.4$ graphically.
 (b) Repeat the calculations by using Eqn (6.3-15).

(Answer: (b) $\bar{H}_1 - \tilde{H}_1 = 354.2 \text{ J/mol}$, $\bar{H}_2 - \tilde{H}_2 = 265.7 \text{ J/mol}$)

6.26 The heat of mixing data for water (1) and tetrahydrofuran (2) mixtures at 298.15 K are reported by Kiyohara and Benson (1977) as follows:

x_1	$\Delta\tilde{H}_{\text{mix}} \text{ (J/mol)}$	x_1	$\Delta\tilde{H}_{\text{mix}} \text{ (J/mol)}$	x_1	$\Delta\tilde{H}_{\text{mix}} \text{ (J/mol)}$
0.04	151.83	0.38	23.22	0.70	-578.39
0.06	208.51	0.40	-14.21	0.80	-706.59
0.10	283.04	0.42	-52.31	0.90	-708.48
0.20	289.35	0.50	-210.49	0.94	-581.55
0.30	160.86	0.58	-367.94	0.96	-455.06
0.34	94.26	0.62	-443.05	0.98	-266.58

- (a) Plot $\Delta\tilde{H}_{\text{mix}}$ versus x_1 and determine $\bar{H}_1 - \tilde{H}_1$ and $\bar{H}_2 - \tilde{H}_2$ graphically at $x_1 = 0.5$.
 (b) How much heat would be released (or absorbed) upon mixing 6 mol of water with 14 mol of tetrahydrofuran at a constant temperature of 298.15 K.

(Answer: (a) $\bar{H}_1 - \tilde{H}_1 = -1110 \text{ J/mol}$, $\bar{H}_2 - \tilde{H}_2 = 705 \text{ J/mol}$ and (b) 3217 J)

Problems related to Section 6.4

6.27 After carrying out experiments on volume change on mixing of binary systems, your friend proposes the following expression to express the molar volume of a mixture as a function of composition:

$$\tilde{V}_{\text{mix}} = (\alpha_1 + \beta_1 x_1^2)x_1 + (\alpha_2 + \beta_2 x_2^2)x_2, \quad (1)$$

where α_i and β_i are constants. Since Eqn (1) has the form

$$\tilde{V}_{\text{mix}} = \bar{V}_1 x_1 + \bar{V}_2 x_2, \quad (2)$$

your friend compares Eqns (1) and (2) and suggests that

$$\bar{V}_1 = \alpha_1 + \beta_1 x_1^2 \quad \text{and} \quad \bar{V}_2 = \alpha_2 + \beta_2 x_2^2.$$

Do you agree? Why?

6.28 Pure components 1 and 2 have the molar enthalpies of 200 J/mol and 350 J/mol, respectively. When these two components are mixed at constant temperature and pressure, the partial molar enthalpy of component 1 is reported as

$$\bar{H}_1 \text{ (J/mol)} = 200 + 40x_2^2. \quad (1)$$

Your boss wants you to express the molar enthalpy of the mixture as a function of composition.

(a) First use the Gibbs-Duhem equation, i.e.

$$x_1 \frac{d\bar{H}_1}{dx_1} + x_2 \frac{d\bar{H}_2}{dx_1} = 0 \quad (2)$$

to obtain the partial molar enthalpy of component 2 as

$$\bar{H}_2 = 350 + 40x_1^2. \quad (3)$$

Then use

$$\tilde{H}_{\text{mix}} = x_1 \bar{H}_1 + x_2 \bar{H}_2 \quad (4)$$

to show that

$$\tilde{H}_{\text{mix}} = 350 - 150x_1 + 40x_1x_2. \quad (5)$$

(b) Rearrange Eqn (6.2-24) for enthalpy as

$$\frac{d\tilde{H}_{\text{mix}}}{dx_2} - \frac{1}{x_2} \tilde{H}_{\text{mix}} = -\frac{\bar{H}_1}{x_2}, \quad (6)$$

which is a linear equation with an integrating factor of $1/x_2$. Multiply Eqn (6) by an integrating factor to transform it into

$$\frac{d}{dx_2} \left(\frac{\tilde{H}_{\text{mix}}}{x_2} \right) = -\frac{\bar{H}_1}{x_2^2}. \quad (7)$$

Integrate Eqn (7) to obtain the solution as

$$\tilde{H}_{\text{mix}} = -x_2 \int \frac{\bar{H}_1}{x_2^2} dx_2 + Cx_2, \quad (8)$$

where C is an integration constant. Show that the substitution of Eqn (1) into Eqn (8) and integration also lead to Eqn (5).

6.29 The generalized form of the Gibbs-Duhem equation can be derived as follows:

(a) Use $\varphi_{\text{mix}} = n\tilde{\varphi}_{\text{mix}}$ in Eqn (6.4-4) to obtain

$$d\varphi_{\text{mix}} = n \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial T} \right)_{P, n_j} dT + n \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial P} \right)_{T, n_j} dP + \sum_{i=1}^k \bar{\varphi}_i dn_i. \quad (1)$$

(b) Show that the combination of Eqn (1) with Eqn (6.4-2) gives

$$- \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial T} \right)_{P, x_j} dT - \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial P} \right)_{T, x_j} dP + \sum_{i=1}^k x_i d\bar{\varphi}_i = 0, \quad (2)$$

which is known as the *generalized Gibbs-Duhem equation*.

(c) Use

$$\bar{\varphi}_i = \bar{\varphi}_i(T, P, x_1, x_2, \dots, x_{k-1}) \quad (3)$$

to obtain

$$d\bar{\varphi}_i = \left(\frac{\partial \bar{\varphi}_i}{\partial T} \right)_{P, x_j} dT + \left(\frac{\partial \bar{\varphi}_i}{\partial P} \right)_{T, x_j} dP + \sum_{m=1}^{k-1} \left(\frac{\partial \bar{\varphi}_i}{\partial x_m} \right)_{T, P} dx_m. \quad (4)$$

(d) Show that the substitution of Eqn (4) into Eqn (2) leads to

$$\begin{aligned} & - \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial T} \right)_{P, x_j} dT - \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial P} \right)_{T, x_j} dP + \sum_{i=1}^k x_i \left(\frac{\partial \bar{\varphi}_i}{\partial T} \right)_{P, x_j} dT \\ & + \sum_{i=1}^k x_i \left(\frac{\partial \bar{\varphi}_i}{\partial P} \right)_{T, x_j} dP + \sum_{i=1}^k x_i \sum_{m=1}^{k-1} \left(\frac{\partial \bar{\varphi}_i}{\partial x_m} \right)_{T, P} dx_m = 0. \end{aligned} \quad (5)$$

(e) Note that

$$\begin{aligned} \sum_{i=1}^k x_i \left(\frac{\partial \bar{\varphi}_i}{\partial T} \right)_{P, x_j} dT &= \sum_{i=1}^k \left[\frac{\partial (x_i \bar{\varphi}_i)}{\partial T} \right]_{P, x_j} dT \\ &= \left(\frac{\partial}{\partial T} \sum_{i=1}^k x_i \bar{\varphi}_i \right)_{P, x_j} dT = \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial T} \right)_{P, x_j} dT. \end{aligned} \quad (6)$$

Similarly,

$$\sum_{i=1}^k x_i \left(\frac{\partial \bar{\varphi}_i}{\partial P} \right)_{T, x_j} dP = \left(\frac{\partial \tilde{\varphi}_{\text{mix}}}{\partial P} \right)_{T, x_j} dP. \quad (7)$$

(f) Use Eqns (6) and (7) in Eqn (5) and conclude that

$$\sum_{i=1}^k x_i \sum_{m=1}^{k-1} \left(\frac{\partial \bar{\varphi}_i}{\partial x_m} \right)_{T, P} dx_m = 0, \quad (8)$$

which is the generalized form of Eqn (6.4-9) for a k -component system.

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Chapter 7

Fugacity of a Component in a Mixture

When two phases of a pure component are in equilibrium, fugacities in each phase must be the same. The purpose of this chapter is to replace a pure component with a multicomponent mixture and investigate the conditions under which two phases of a multicomponent mixture are in equilibrium with each other. For this purpose, first the fugacity definition given in Chapter 5 will be extended to multicomponent mixtures; then equations to calculate fugacities of components making up the mixture will be developed.

7.1 Fundamental Equations for a Multicomponent Mixture

The combined law for a single-phase, single-component open system, Eqn (4.2-9), is given as

$$dU = T dS - P dV + \tilde{G} dn. \quad (7.1-1)$$

By definition, Gibbs energy is expressed as

$$G = U + PV - TS. \quad (7.1-2)$$

In differential form, Eqn (7.1-2) becomes

$$dG = dU + P dV + V dP - T dS - S dT. \quad (7.1-3)$$

Substitution of Eqn (7.1-1) into Eqn (7.1-3) leads to

$$dG = V dP - S dT + \tilde{G} dn \quad (7.1-4)$$

indicating that the Gibbs energy is dependent on pressure, temperature, and number of moles, i.e.

$$G = G(P, T, n). \quad (7.1-5)$$

The total differential of G is

$$dG = \left(\frac{\partial G}{\partial P} \right)_{T,n} dP + \left(\frac{\partial G}{\partial T} \right)_{P,n} dT + \left(\frac{\partial G}{\partial n} \right)_{P,T} dn. \quad (7.1-6)$$

Comparison of Eqn (7.1-6) with Eqn (7.1-4) yields

$$V = \left(\frac{\partial G}{\partial P} \right)_{T,n}, \quad (7.1-7)$$

$$S = - \left(\frac{\partial G}{\partial T} \right)_{P,n}, \quad (7.1-8)$$

$$\tilde{G} = \left(\frac{\partial G}{\partial n} \right)_{P,T}. \quad (7.1-9)$$

In the case of a single-phase, multicomponent open system, Gibbs energy of the phase not only depends on pressure and temperature, but also on the number of moles of each component existing in the phase, i.e.

$$G = G(P, T, n_1, n_2, \dots, n_k). \quad (7.1-10)$$

Therefore, the total differential of G is expressed in the form

$$dG = \left(\frac{\partial G}{\partial P} \right)_{T,n_j} dP + \left(\frac{\partial G}{\partial T} \right)_{P,n_j} dT + \sum_{i=1}^k \left(\frac{\partial G}{\partial n_i} \right)_{P,T,n_{j \neq i}} dn_i. \quad (7.1-11)$$

The subscript n_j in the first two terms of Eqn (7.1-11) implies that the number of moles of each component is held constant during differentiation. For a constant number of moles, Eqns (7.1-7) and (7.1-8) are still valid. On the other hand, the partial molar Gibbs energy is defined by

$$\bar{G}_i = \left(\frac{\partial G}{\partial n_i} \right)_{P,T,n_{j \neq i}}. \quad (7.1-12)$$

Thus, Eqn (7.1-11) becomes

$$dG = V dP - S dT + \sum_{i=1}^k \bar{G}_i dn_i. \quad (7.1-13)$$

Historically, for a multicomponent system, the partial molar Gibbs energy has been called the *chemical potential* and designated by μ_i , i.e. $\bar{G}_i = \mu_i$.

The enthalpy is related to the Gibbs energy by

$$H = G + TS. \quad (7.1-14)$$

Differentiation of Eqn (7.1-14) gives

$$dH = dG + T dS + S dT. \quad (7.1-15)$$

Substitution of Eqn (7.1-13) into Eqn (7.1-15) yields

$$dH = T dS + V dP + \sum_{i=1}^k \bar{G}_i dn_i. \quad (7.1-16)$$

The internal energy is related to the enthalpy by

$$U = H - PV. \quad (7.1-17)$$

Differentiation of Eqn (7.1-17) gives

$$dU = dH - P dV - V dP. \quad (7.1-18)$$

The use of Eqn (7.1-16) in Eqn (7.1-18) results in

$$dU = T dS - P dV + \sum_{i=1}^k \bar{G}_i dn_i. \quad (7.1-19)$$

The Helmholtz energy is related to the internal energy by

$$A = U - TS. \quad (7.1-20)$$

Differentiation of Eqn (7.1-20) gives

$$dA = dU - T dS - S dT. \quad (7.1-21)$$

Substitution of Eqn (7.1-19) into Eqn (7.1-21) yields

$$dA = -P dV - S dT + \sum_{i=1}^k \bar{G}_i dn_i. \quad (7.1-22)$$

Equations (7.1-13), (7.1-16), (7.1-19), and (7.1-22) are the fundamental equations for a single-phase, multicomponent open system. From these equations, partial molar Gibbs energy can be expressed in different forms as

$$\bar{G}_i = \left(\frac{\partial G}{\partial n_i} \right)_{T,P,n_{j \neq i}} = \left(\frac{\partial H}{\partial n_i} \right)_{P,S,n_{j \neq i}} = \left(\frac{\partial U}{\partial n_i} \right)_{V,S,n_{j \neq i}} = \left(\frac{\partial A}{\partial n_i} \right)_{T,V,n_{j \neq i}}. \quad (7.1-23)$$

Example 7.1 Develop the Gibbs-Duhem equation for the Gibbs energy by the application of Euler's theorem.

Solution

Equation (7.1-19) indicates that the internal energy of a single-phase, multicomponent system depends on S , V , and the number of moles of each component, i.e.

$$U = U(S, V, n_1, n_2, \dots, n_k). \quad (1)$$

Application of Euler's theorem, Eqn (6.2-30), gives

$$U = \left(\frac{\partial U}{\partial S} \right)_{V, n_j} S + \left(\frac{\partial U}{\partial V} \right)_{S, n_j} V + \sum_{i=1}^k \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_{j \neq i}} n_i. \quad (2)$$

From Eqn (7.1-19)

$$\left(\frac{\partial U}{\partial S} \right)_{V, n_j} = T, \quad \left(\frac{\partial U}{\partial V} \right)_{S, n_j} = -P, \quad \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_{j \neq i}} = \bar{G}_i. \quad (3)$$

Thus, Eqn (2) takes the form

$$U = TS - PV + \sum_{i=1}^k \bar{G}_i n_i. \quad (4)$$

Differentiation of Eqn (4) yields

$$dU = T dS + S dT - P dV - V dP + \sum_{i=1}^k \bar{G}_i dn_i + \sum_{i=1}^k n_i d\bar{G}_i. \quad (5)$$

Subtraction of Eqn (7.1-19) from Eqn (5) results in

$$S dT - V dP + \sum_{i=1}^k n_i d\bar{G}_i = 0, \quad (6)$$

which is the most general form of the Gibbs-Duhem equation. It provides the relationship between temperature, pressure, and the partial molar Gibbs energy. At constant temperature and pressure, Eqn (6) reduces to

$$\sum_{i=1}^k n_i d\bar{G}_i = 0 \quad \text{constant } T \text{ \& } P. \quad (7)$$

Dividing Eqn (7) by the total number of moles gives

$$\sum_{i=1}^k x_i d\bar{G}_i = 0 \quad \text{constant } T \text{ \& } P. \quad (8)$$

Comment: For a single-component system, the Gibbs-Duhem equation, Eqn (6), reduces to

$$S dT - V dP + n d\tilde{G} = 0. \quad (9)$$

Dividing Eqn (9) by the number of moles, n , leads to

$$d\tilde{G} = \tilde{V} dP - \tilde{S} dT, \quad (10)$$

which is identical with Eqn (2.2-15).

7.1.1 Criteria for Chemical Equilibrium

Consider an isolated multicomponent system containing α - and β -phases. Note that although the overall system is an isolated one each phase may be considered an open system, i.e. there may be an exchange of mass and/or energy between α - and β -phases.

In the case of no chemical reaction, the total values of internal energy, volume, and number of moles of species i can be expressed as

$$\begin{aligned} U &= U_\alpha + U_\beta = \text{constant} \Rightarrow dU_\alpha = -dU_\beta, \\ V &= V_\alpha + V_\beta = \text{constant} \Rightarrow dV_\alpha = -dV_\beta, \\ n_i &= n_{i\alpha} + n_{i\beta} = \text{constant} \Rightarrow dn_{i\alpha} = -dn_{i\beta}. \end{aligned} \quad (7.1-24)$$

Suppose that an arbitrary process causes changes in the entropies of the α - and β -phases. Such changes can be expressed from Eqn (7.1-19) as

$$dS_\alpha = \frac{1}{T_\alpha} dU_\alpha + \frac{P_\alpha}{T_\alpha} dV_\alpha - \sum_{i=1}^k \frac{\bar{G}_i^\alpha}{T_\alpha} dn_{i\alpha}, \quad (7.1-25)$$

$$dS_\beta = \frac{1}{T_\beta} dU_\beta + \frac{P_\beta}{T_\beta} dV_\beta - \sum_{i=1}^k \frac{\bar{G}_i^\beta}{T_\beta} dn_{i\beta}. \quad (7.1-26)$$

Summation of Eqns (7.1-25) and (7.1-26) and making use of the constraints given by Eqn (7.1-24) lead to

$$dS = \left(\frac{1}{T_\alpha} - \frac{1}{T_\beta} \right) dU_\alpha + \left(\frac{P_\alpha}{T_\alpha} - \frac{P_\beta}{T_\beta} \right) dV_\alpha - \sum_{i=1}^k \left(\frac{\bar{G}_i^\alpha}{T_\alpha} - \frac{\bar{G}_i^\beta}{T_\beta} \right) dn_{i\alpha}. \quad (7.1-27)$$

The condition of equilibrium is satisfied, i.e. $dS = 0$, if the terms in parentheses in Eqn (7.1-27) are all equal to zero. Therefore, at equilibrium we have the following conditions:

$$\begin{aligned} T_\alpha &= T_\beta, & \text{Thermal equilibrium,} \\ P_\alpha &= P_\beta, & \text{Mechanical equilibrium,} \\ \bar{G}_i^\alpha &= \bar{G}_i^\beta, & \text{Chemical equilibrium.} \end{aligned} \quad (7.1-28)$$

Equality of partial molar Gibbs energies implies that there are no chemical driving forces to move species from one phase to another.

7.1.2 The Gibbs Phase Rule

The Gibbs phase rule is given by Eqn (1.1-4), i.e. $\mathcal{P} + \mathcal{F} = \mathcal{C} + 2$. We are now in a position to derive this equation theoretically. The number of degrees of freedom, $\mathcal{F}$, is defined by

$$\begin{aligned} \mathcal{F} &= [\text{Total \# of independent intensive variables}] \\ &\quad - [\text{\# of independent equations relating these variables}]. \end{aligned} \quad (7.1-29)$$

Consider a composite system containing $\mathcal{P}$ phases and $\mathcal{C}$ components. Assume that each component exists in each phase. Hence,

$$\begin{aligned} \text{For each phase} &: (\mathcal{C} - 1) \quad \text{independent mole fractions} \\ \text{Total mole fractions: } &\mathcal{P}(\mathcal{C} - 1) \end{aligned} \quad (7.1-30)$$

At equilibrium, temperature and pressure must be uniform throughout the phases. Therefore, the total number of independent intensive variables is $2 + \mathcal{P}(\mathcal{C} - 1)$. The condition of chemical equilibrium is designated by the following equations:

$$\text{At equilibrium} \quad \bar{G}_i^1 = \bar{G}_i^2 = \dots = \bar{G}_i^{\mathcal{P}}, \quad i = 1, 2, \dots, \mathcal{C}. \quad (7.1-31)$$

Under these circumstances, we have

$$\begin{aligned} \text{For each species} &: (\mathcal{P} - 1) \quad \text{independent equations} \\ \text{Total \# of independent equations: } &\mathcal{C}(\mathcal{P} - 1) \end{aligned} \quad (7.1-32)$$

Substitution of Eqns (7.1-30) and (7.1-32) into Eqn (7.1-29) gives

$$\mathcal{F} = 2 + \mathcal{P}(\mathcal{C} - 1) - \mathcal{C}(\mathcal{P} - 1) \Rightarrow \mathcal{P} + \mathcal{F} = \mathcal{C} + 2. \quad (7.1-33)$$

Note that Eqn (7.1-33) applies only to systems at equilibrium when there is no chemical reaction taking place.

7.2 Partial Molar Gibbs Energy of an Ideal Gas Mixture

An ideal gas mixture is a gas mixture with a density so low that its molecules do not appreciably interact. The equation of state for an ideal gas mixture containing k species is given by

$$PV^{\text{IGM}} = \underbrace{(n_1 + n_2 + \cdots + n_k)}_n RT \quad (7.2-1)$$

in which the superscript “IGM” stands for “ideal gas mixture.” In an ideal gas mixture, the partial pressure of species i , p_i , is expressed as

$$p_i = \frac{n_i RT}{V} \Rightarrow p_i = Py_i. \quad (7.2-2)$$

In other words, the partial pressure of species i in the ideal gas mixture is the pressure exerted by n_i moles of pure species i at the same volume and temperature.

Any total property in an ideal gas mixture is the sum of individual species making up the mixture, each evaluated at the mixture temperature and its own partial pressure. Therefore, we can write

$$U^{\text{IGM}} = \sum_{i=1}^k n_i \tilde{U}_i^{\text{IG}}(T) \Rightarrow \bar{U}_i^{\text{IGM}} = \tilde{U}_i^{\text{IG}}, \quad (7.2-3)$$

$$H^{\text{IGM}} = \sum_{i=1}^k n_i \tilde{H}_i^{\text{IG}}(T) \Rightarrow \bar{H}_i^{\text{IGM}} = \tilde{H}_i^{\text{IG}}, \quad (7.2-4)$$

$$S^{\text{IGM}} = \sum_{i=1}^k n_i \tilde{S}_i^{\text{IG}}(T, p_i). \quad (7.2-5)$$

Equations (7.2-3) and (7.2-4) imply that both the internal energy change on mixing and the enthalpy change on mixing are zero for an ideal gas mixture.

From Eqn (2.2-41), the change in the entropy of a pure ideal gas with pressure, keeping temperature constant, is given by

$$d\tilde{S}_i^{\text{IG}} = -\frac{R}{P}dP \quad \text{constant } T. \quad (7.2-6)$$

Integration at constant temperature from P to p_i yields

$$\tilde{S}_i^{\text{IG}}(T, p_i) = \tilde{S}_i^{\text{IG}}(T, P) - R \ln \left(\frac{p_i}{P} \right) = \tilde{S}_i^{\text{IG}}(T, P) - R \ln y_i. \quad (7.2-7)$$

Substitution of Eqn (7.2-7) into Eqn (7.2-5) gives

$$S^{\text{IGM}} = \sum_{i=1}^k n_i \tilde{S}_i^{\text{IG}}(T, P) - R \underbrace{\sum_{i=1}^k n_i \ln y_i}_{\Delta S_{\text{mix}}^{\text{IGM}}}. \quad (7.2-8)$$

The Gibbs energy of an ideal gas mixture is

$$G^{\text{IGM}} = H^{\text{IGM}} - TS^{\text{IGM}}. \quad (7.2-9)$$

Substitution of Eqns (7.2-4) and (7.2-8) into Eqn (7.2-9) gives

$$G^{\text{IGM}} = \sum_{i=1}^k n_i \tilde{G}_i^{\text{IG}}(T, P) + RT \sum_{i=1}^k n_i \ln y_i. \quad (7.2-10)$$

The partial molar Gibbs energy of species i in an ideal gas mixture can be calculated as

$$\bar{G}_i^{\text{IGM}}(T, P, y_i) = \left(\frac{\partial G^{\text{IGM}}}{\partial n_i} \right)_{T, P, n_{j \neq i}} = \tilde{G}_i^{\text{IG}}(T, P) + RT \ln y_i. \quad (7.2-11)$$

It is possible to rearrange Eqn (7.2-11) as

$$\ln \left(\frac{p_i}{P} \right) = \frac{\bar{G}_i^{\text{IGM}}(T, P, y_i) - \tilde{G}_i^{\text{IG}}(T, P)}{RT}. \quad (7.2-12)$$

7.3 Fugacity of a Component in a Mixture

Lewis extended the definition introduced by Eqn (7.2-12) to multicomponent systems, and the fugacity of a component i in a mixture, $\hat{f}_i$, is expressed in the form

$$\ln \left[\frac{\widehat{f}_i(T, P, x_i)}{Px_i} \right] = \frac{\overline{G}_i(T, P, x_i) - \overline{G}_i^{\text{IGM}}(T, P, x_i)}{RT} \quad (7.3-1)$$

subject to the following constraint:

$$\widehat{f}_i/(x_i P) \rightarrow 1 \quad \text{as } P \rightarrow 0 \text{ (ideal gas state),} \quad (7.3-2)$$

which applies to solid, liquid, and gas.

The fugacity coefficient of a component i in a mixture, $\widehat{\phi}_i$, is defined by

$$\widehat{\phi}_i = \frac{\widehat{f}_i}{x_i P}, \quad (7.3-3)$$

where the symbol $\widehat{}$ is used to represent properties in the mixture. Although they look similar to partial properties, their definitions are not the same.

Combination of Eqns (7.3-1), (7.2-12), and (5.1-5) leads to an alternative expression for the fugacity of component i in a mixture as

$$\overline{G}_i = \widetilde{G}_i^{\text{IG}}(T, P^o) - RT \ln P^o + RT \ln \widehat{f}_i \quad (7.3-4)$$

or

$$\overline{G}_i = \lambda_i(T) + RT \ln \widehat{f}_i, \quad (7.3-5)$$

where $\lambda_i(T)$ is the molar Gibbs energy of pure component i at unit fugacity, and is defined by Eqn (5.2-6). Differentiation of Eqn (7.3-5) by keeping temperature and the number of moles of each component constant leads to

$$d\overline{G}_i = RT d \ln \widehat{f}_i \quad \text{constant } T \text{ \& } n_j. \quad (7.3-6)$$

When α - and β -phases of a multicomponent mixture are in equilibrium with each other, the condition of equilibrium is represented by Eqn (7.1-28), i.e.

$$T_\alpha = T_\beta, \quad P_\alpha = P_\beta, \quad \overline{G}_i^\alpha = \overline{G}_i^\beta. \quad (7.3-7)$$

Letting $T_\alpha = T_\beta = T$ and $P_\alpha = P_\beta = P$, equality of partial molar Gibbs energies leads to

$$\lambda_i(T) + RT \ln \widehat{f}_i^\alpha(T, P, x_i^\alpha) = \lambda_i(T) + RT \ln \widehat{f}_i^\beta(T, P, x_i^\beta) \quad (7.3-8)$$

or

$$\widehat{f}_i^\alpha(T, P, x_i^\alpha) = \widehat{f}_i^\beta(T, P, x_i^\beta). \quad (7.3-9)$$

Thus, at equilibrium, fugacities of each component are equal in all phases.

Differentiation of Eqn (7.3-6) with respect to pressure gives

$$\left(\frac{\partial \bar{G}_i}{\partial P}\right)_{T,n_j} = RT \left(\frac{\partial \ln \hat{f}_i}{\partial P}\right)_{T,n_j}. \quad (7.3-10)$$

The left-hand side of Eqn (7.3-10) is expressed as

$$\left(\frac{\partial \bar{G}_i}{\partial P}\right)_{T,n_j} = \frac{\partial}{\partial P} \left[\left(\frac{\partial G_{\text{mix}}}{\partial n_i}\right)_{T,P,n_j \neq i} \right]_{T,n_j}. \quad (7.3-11)$$

Changing the order of differentiation leads to

$$\left(\frac{\partial \bar{G}_i}{\partial P}\right)_{T,n_j} = \frac{\partial}{\partial n_i} \left[\left(\frac{\partial G_{\text{mix}}}{\partial P}\right)_{T,n_j} \right]_{T,P,n_j \neq i} = \left(\frac{\partial V_{\text{mix}}}{\partial n_i}\right)_{T,P,n_j \neq i} = \bar{V}_i. \quad (7.3-12)$$

Therefore, at constant temperature and number of moles, Eqn (7.3-10) is expressed as

$$RT \, d \ln \hat{f}_i = \bar{V}_i \, dP. \quad (7.3-13)$$

Subtraction of the term $RT \, d \ln(x_i P)$ from both sides of Eqn (7.3-13) and integration result in

$$\ln \left(\frac{\hat{f}_i}{x_i P} \right) = \ln \hat{\phi}_i = \int_0^P \left(\frac{\bar{V}_i}{RT} - \frac{1}{P} \right) dP \quad \text{constant } T \text{ \& } n_j. \quad (7.3-14)$$

Noting that $P\bar{V}_i = \bar{Z}_i RT$, it is also possible to express Eqn (7.3-14) in the form

$$\ln \hat{\phi}_i = \int_0^P \left(\frac{\bar{Z}_i - 1}{P} \right) dP \quad \text{constant } T \text{ \& } n_j. \quad (7.3-15)$$

The fugacity of a pure component can be calculated from Eqn (5.2-12) as

$$\ln \left(\frac{f_i}{P} \right) = \int_0^P \left(\frac{\tilde{V}_i}{RT} - \frac{1}{P} \right) dP \quad \text{constant } T. \quad (7.3-16)$$

Subtraction of Eqn (7.3-16) from Eqn (7.3-14) yields

$$\ln \left(\frac{\hat{f}_i}{x_i f_i} \right) = \frac{1}{RT} \int_0^P (\bar{V}_i - \tilde{V}_i) dP \quad \text{constant } T \text{ \& } n_j. \quad (7.3-17)$$

Equations (7.3-14), (7.3-15), and (7.3-17) can be used to calculate the fugacity of component i in a mixture, and all equations are valid for gases, liquids, and solids. Note that the evaluation of integrals in these equations requires either the equation of state or the volumetric data in tabular form to be known.

7.4 Ideal Mixture¹

A mixture, either gaseous or liquid, is said to be an *ideal mixture* if it has the following properties:

- (i) It is homogeneous.
- (ii) The sizes of the molecules are equal.
- (iii) The forces between unlike molecules of the mixture are the same as the forces between like molecules.

It follows from properties (ii) and (iii) that

$$\bar{V}_i^{\text{IM}}(T, P, x_i) = \tilde{V}_i(T, P). \quad (7.4-1)$$

On the other hand, property (iii) implies that

$$\bar{H}_i^{\text{IM}}(T, P, x_i) = \tilde{H}_i(T, P). \quad (7.4-2)$$

Since $\bar{U}_i = \bar{H}_i - P\bar{V}_i$, then Eqns (7.4-1) and (7.4-2) indicate that

$$\bar{U}_i^{\text{IM}}(T, P, x_i) = \tilde{U}_i(T, P). \quad (7.4-3)$$

Mixtures of two similar substances usually form an ideal mixture, i.e. benzene and toluene, *o*-xylene and *m*-xylene, butane and isobutane, *n*-butanol and *n*-pentanol, *n*-pentane and *n*-hexane.

Substitution of Eqn (7.4-1) into Eqn (7.3-17) yields

$$\ln \left(\frac{\hat{f}_i^{\text{IM}}}{x_i f_i} \right) = 0 \quad (7.4-4)$$

or

$$\hat{f}_i^{\text{IM}}(T, P, x_i) = x_i f_i(T, P) \quad (7.4-5)$$

indicating that the fugacity of component *i* in an ideal mixture can be calculated by multiplying the pure component fugacity by its mole fraction. This relationship is known as the *Lewis-Randall rule*. Pure component fugacities, on the other hand, can be calculated by using one of the methods given in Chapter 5.

The use of Eqn (7.4-5) in Eqn (7.3-3) gives

$$\hat{\phi}_i^{\text{IM}}(T, P, x_i) = \phi_i(T, P) \quad (7.4-6)$$

¹In the literature, *ideal mixture* is also referred to as *ideal solution*.

indicating that the fugacity coefficient of i in an ideal mixture is independent of the mixture composition.

Although the simplicity of the Lewis-Randall rule makes it an attractive tool in the calculation of mixture fugacities, one should remember the limitations of using it. The Lewis-Randall rule is applicable under the following conditions:

- Component i is present in large excess, i.e. $x_i > 0.9$.
- Components are similar.
- Gas phase is nearly ideal, i.e. pressure is low.

7.5 Calculation of Component Fugacities in a Gas Mixture

7.5.1 Fugacity of a Component in an Ideal Gas Mixture

Using Eqn (7.2-1), the partial molar volume of an ideal gas mixture is calculated as

$$\bar{V}_i = \left(\frac{\partial V^{\text{IGM}}}{\partial n_i} \right)_{T, P, n_{j \neq i}} = \frac{RT}{P}. \quad (7.5-1)$$

Substitution of Eqn (7.5-1) into Eqn (7.3-14) gives

$$\ln \hat{\phi}_i^{\text{IGM}} = 0 \Rightarrow \hat{\phi}_i^{\text{IGM}} = 1 \text{ or } \hat{f}_i^{\text{IGM}} = y_i P = p_i. \quad (7.5-2)$$

Therefore, for an ideal gas mixture, the fugacity coefficient of component i is unity and the fugacity of component i is equal to its partial pressure. At high pressures and low temperatures, the fugacity coefficient of a component in a gas mixture deviates from unity, indicating nonideal behavior.

7.5.2 Fugacity of a Component in an Ideal Mixture

In this case the Lewis-Randall rule applies and from Eqns (7.4-5) and (7.4-6)

$$\hat{f}_i^{\text{IM}}(T, P, y_i) = y_i f_i(T, P) \text{ and } \hat{\phi}_i^{\text{IM}}(T, P, y_i) = \phi_i(T, P). \quad (7.5-3)$$

Fugacities of pure components are determined by one of the methods described in Section 5.3.

It is important not to confuse *ideal gas mixture* with *ideal mixture*.² An ideal gas mixture always behaves as an ideal mixture. However, an ideal mixture does

²There is no interaction between the molecules of an ideal gas mixture. In the case of an ideal mixture, however, the molecules interact with each other, the interaction between like molecules being almost equal to the interaction between unlike molecules.

not necessarily behave as an ideal gas mixture. Note that the compressibility factor for an ideal mixture is given by

$$Z^{\text{IM}} = \frac{P\tilde{V}^{\text{IM}}}{RT} = \frac{P}{RT} \sum_{i=1}^k y_i \tilde{V}_i = \sum_{i=1}^k y_i \left(\frac{P\tilde{V}_i}{RT} \right) = \sum_{i=1}^k y_i Z_i. \quad (7.5-4)$$

On the other hand, $Z^{\text{IGM}} = 1.0$.

Example 7.2 Consider an ideal binary mixture of 35 mol% methane (1) and 65% ethane (2) at 373.15 K and 30 bar. Determine the fugacities of methane and ethane in the mixture using the Peng-Robinson equation of state.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Methane	190.6	46.1	0.011
Ethane	305.3	49.0	0.099

The Lewis-Randall rule requires pure component fugacities. The use of the Peng-Robinson equation of state, Eqn (5.3-9), yields the following values:

Component	α	A	B	Z^{V}	ϕ	f (bar)
Methane	0.712	0.055	0.026	0.973	0.972	29.16
Ethane	0.892	0.167	0.039	0.870	0.879	26.37

Thus, the fugacities of components in the mixture are

$$\begin{aligned} \hat{f}_1 &= (0.35)(29.16) = 10.21 \text{ bar}, \\ \hat{f}_2 &= (0.65)(26.37) = 17.14 \text{ bar}. \end{aligned}$$

7.5.3 Fugacity from the Virial Equation of State

At low to moderate pressures, the virial equation of state of a multicomponent gas mixture, truncated after the second term, is given by

$$Z_{\text{mix}} = 1 + \frac{B_{\text{mix}}P}{RT}, \quad (7.5-5)$$

where the second virial coefficient for the mixture, B_{mix} , is defined by Eqn (6.1-2). The use of Eqn (7.5-5) in Eqn (7.3-15) gives the fugacity coefficient of species i in a mixture as³

$$\ln \hat{\phi}_i^V = \left(2 \sum_{j=1}^k y_j B_{ij} - B_{\text{mix}} \right) \frac{P}{RT}. \quad (7.5-6)$$

An alternative derivation of Eqn (7.5-6) for a binary system is given in Problem 7.4.

Example 7.3 For a mixture of n -butane (1) and n -hexane (2) at 373 K, determine the fugacity coefficients at 1, 10, and 15 bar when $y_1 = 0.35$. The virial coefficients at 373 K are given as

$$\begin{aligned} B_{11} &= -422 \text{ cm}^3/\text{mol}, \\ B_{22} &= -103.1 \text{ cm}^3/\text{mol}, \\ B_{12} &= B_{21} = -650 \text{ cm}^3/\text{mol}. \end{aligned}$$

Solution

For a binary system, Eqn (6.1-2) yields

$$B_{\text{mix}} = y_1^2 B_{11} + 2y_1 y_2 B_{12} + y_2^2 B_{22}. \quad (1)$$

Substitution of Eqn (1) into Eqn (7.5-6) leads to

$$\hat{\phi}_1 = \exp \left\{ \frac{P}{RT} [B_{11} + y_2^2 (2B_{12} - B_{11} - B_{22})] \right\}, \quad (2)$$

$$\hat{\phi}_2 = \exp \left\{ \frac{P}{RT} [B_{22} + y_1^2 (2B_{12} - B_{11} - B_{22})] \right\}. \quad (3)$$

Fugacity coefficients of n -butane (1) and n -hexane (2), calculated from Eqns (2) and (3), are tabulated below:

P (bar)	$\hat{\phi}_1$	$\hat{\phi}_2$
1	0.976	0.994
10	0.785	0.938
15	0.696	0.909

³See Problem 7.2.

7.5.4 Fugacity from Cubic Equations of State

Since the cubic equations of state are pressure-explicit, i.e.

$$P = P(T, V_{\text{mix}}, n_1, n_2, \dots, n_k), \quad (7.5-7)$$

then it is useful to have an equation relating the fugacity of component i in a mixture to an integral over volume rather than pressure. To convert Eqn (7.3-14) to the form amenable to the cubic equations of state, note that the total differential of Eqn (7.5-7) is

$$dP = \left(\frac{\partial P}{\partial T} \right)_{V_{\text{mix}}, n_j} dT + \left(\frac{\partial P}{\partial V_{\text{mix}}} \right)_{T, n_j} dV_{\text{mix}} + \sum_{i=1}^k \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} dn_i. \quad (7.5-8)$$

Differentiation of Eqn (7.5-8) with respect to n_i by keeping T , P , and the number of moles of each component other than i constant gives

$$\left(\frac{\partial P}{\partial V_{\text{mix}}} \right)_{T, n_j} \underbrace{\left(\frac{\partial V_{\text{mix}}}{\partial n_i} \right)_{T, P, n_{j \neq i}}}_{\bar{V}_i} + \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} = 0 \quad (7.5-9)$$

or

$$\bar{V}_i = - \frac{\left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}}}{\left(\frac{\partial P}{\partial V_{\text{mix}}} \right)_{T, n_j}} = - \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} \left(\frac{\partial V_{\text{mix}}}{\partial P} \right)_{T, n_j}. \quad (7.5-10)$$

On the other hand, integration of the equation of state, $P\tilde{V}_{\text{mix}} = Z_{\text{mix}}RT$, at constant temperature gives

$$\frac{dP}{P} + \frac{d\tilde{V}_{\text{mix}}}{\tilde{V}_{\text{mix}}} = \frac{dZ_{\text{mix}}}{Z_{\text{mix}}}. \quad (7.5-11)$$

Substitution of Eqns (7.5-10) and (7.5-11) into Eqn (7.3-14) yields

$$\ln \hat{\phi}_i = \frac{1}{RT} \int_{\tilde{V}_{\text{mix}}}^{\infty} \left[n \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} - \frac{RT}{\tilde{V}_{\text{mix}}} \right] d\tilde{V}_{\text{mix}} - \ln Z_{\text{mix}}. \quad (7.5-12)$$

van der Waals equation of state: Substitution of the van der Waals equation of state into Eqn (7.5-12) gives the fugacity coefficient of species i in a mixture as

$$\ln \hat{\phi}_i^V = \frac{B_i}{Z_{\text{mix}}^V - B_{\text{mix}}} - \ln (Z_{\text{mix}}^V - B_{\text{mix}}) - \frac{2 \sum_{j=1}^k y_j A_{ij}}{Z_{\text{mix}}^V}. \quad (7.5-13)$$

Example 7.4 Show how one can obtain Eqn (7.5-13) by the substitution of the van der Waals equation of state into Eqn (7.5-12).

Solution

Since $\tilde{V} = V/n$, the van der Waals equation of state takes the form

$$\begin{aligned}
 P &= \frac{nRT}{V_{\text{mix}} - nb_{\text{mix}}} - \frac{n^2 a_{\text{mix}}}{V_{\text{mix}}^2} \\
 &= \frac{nRT}{V_{\text{mix}} - n \sum_{i=1}^k y_i b_i} - \frac{n^2 \sum_{i=1}^k \sum_{j=1}^k y_i y_j a_{ij}}{V_{\text{mix}}^2}, \\
 &= \frac{nRT}{V_{\text{mix}} - \sum_{i=1}^k n_i b_i} - \frac{\sum_{i=1}^k \sum_{j=1}^k n_i n_j a_{ij}}{V_{\text{mix}}^2}. \quad (1)
 \end{aligned}$$

Differentiation of Eqn (1) with respect to n_i gives

$$\begin{aligned}
 \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} &= \frac{RT}{V_{\text{mix}} - \sum_{i=1}^k n_i b_i} + \frac{nRT b_i}{\left(V_{\text{mix}} - \sum_{i=1}^k n_i b_i \right)^2} \\
 &\quad - \frac{1}{V_{\text{mix}}^2} \frac{\partial}{\partial n_i} \left(\sum_{i=1}^k \sum_{j=1}^k n_i n_j a_{ij} \right)_{T, V_{\text{mix}}, n_{j \neq i}}. \quad (2)
 \end{aligned}$$

For the differentiation of the third term on the right-hand side of Eqn (2), let us carry out the differentiation for a binary mixture of components 1 and 2, i.e.

$$\begin{aligned}
 \frac{\partial}{\partial n_1} \left(\sum_{i=1}^2 \sum_{j=1}^2 n_i n_j a_{ij} \right)_{T, V_{\text{mix}}, n_2} &= \left[\frac{\partial}{\partial n_1} (n_1^2 a_{11} + 2n_1 n_2 a_{12} + n_2^2 a_{22}) \right]_{T, V_{\text{mix}}, n_2} \\
 &= 2(n_1 a_{11} + n_2 a_{12}) = 2 \sum_{j=1}^2 n_j a_{1j}. \quad (3)
 \end{aligned}$$

It is possible to generalize Eqn (3) as

$$\frac{\partial}{\partial n_i} \left(\sum_{i=1}^k \sum_{j=1}^k n_i n_j a_{ij} \right)_{T, V_{\text{mix}}, n_{j \neq i}} = 2 \sum_{j=1}^k n_j a_{ij}. \quad (4)$$

Substitution of Eqn (4) into Eqn (2) gives

$$\left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} = \frac{RT}{V_{\text{mix}} - nb_{\text{mix}}} + \frac{nRTb_i}{(V_{\text{mix}} - nb_{\text{mix}})^2} - \frac{2n \sum_{j=1}^k y_j a_{ij}}{V_{\text{mix}}^2}. \quad (5)$$

Multiplication of Eqn (5) by the total number of moles, n , results in

$$n \left(\frac{\partial P}{\partial n_i} \right)_{T, V_{\text{mix}}, n_{j \neq i}} = \frac{RT}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} + \frac{RTb_i}{(\tilde{V}_{\text{mix}} - b_{\text{mix}})^2} - \frac{2 \sum_{j=1}^k y_j a_{ij}}{\tilde{V}_{\text{mix}}^2}. \quad (6)$$

Substitution of Eqn (6) into Eqn (7.5-12) and integration lead to

$$\ln \hat{\phi}_i^V = \ln \left(\frac{\tilde{V}_{\text{mix}}}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} \right) + \frac{b_i}{\tilde{V}_{\text{mix}} - b_{\text{mix}}} - \frac{2 \sum_{j=1}^k y_j a_{ij}}{\tilde{V}_{\text{mix}} RT} - \ln Z_{\text{mix}}^V. \quad (7)$$

Finally, the use of dimensionless parameters reduces Eqn (7) to Eqn (7.5-13).

An alternative derivation of Eqn (7.5-13) is presented in Problem 7.6.

Redlich-Kwong equation of state: The use of the Redlich-Kwong equation of state in Eqn (7.5-12) leads to the following expression for the fugacity coefficient of species i in a mixture:

$$\begin{aligned} \ln \hat{\phi}_i^V &= \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^V - 1) - \ln (Z_{\text{mix}}^V - B_{\text{mix}}) \\ &\quad - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k y_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}^V} \right). \end{aligned} \quad (7.5-14)$$

For an alternative derivation of this equation, see Problem 7.7.

Example 7.5 Calculate the fugacity coefficients of carbon dioxide (1) in a mixture containing 85 mol% *n*-butane (2) at 444 K as a function of pressure. The system is represented by the Redlich-Kwong equation of state with $k_{12} = 0.18$.

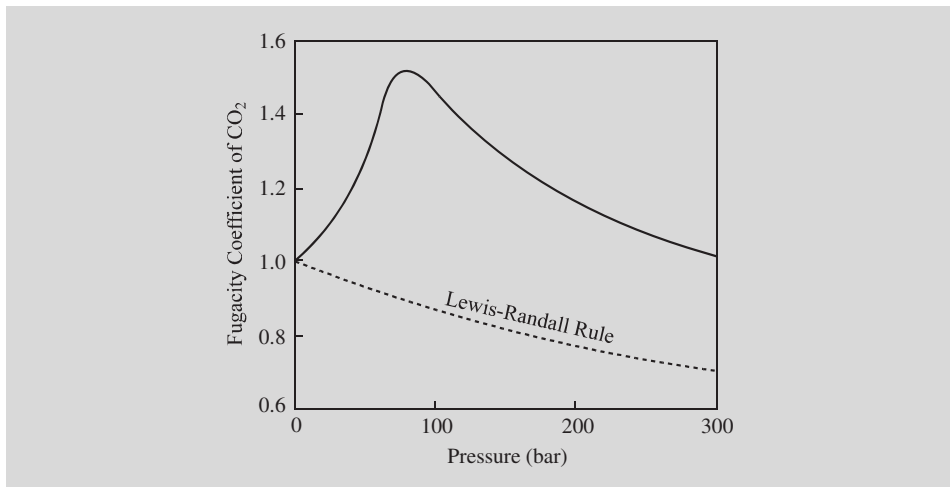
Solution

From Appendix A

Component	T_c (K)	P_c (bar)
Carbon dioxide	304.2	73.8
<i>n</i> -Butane	425.0	38.0

Once the dimensionless parameters, A_{mix} and B_{mix} , and the compressibility factor for the vapor mixture, Z_{mix}^V , are calculated using the equations given in Section 6.1.2, the fugacity coefficient of carbon dioxide in the mixture, $\hat{\phi}_{\text{CO}_2}^V$, can be determined from Eqn (7.5-14). On the other hand, if the mixture behaves ideally, then the Lewis-Randall rule applies and the fugacity coefficient of CO_2 in the mixture is equal to the fugacity coefficient of pure CO_2 , $\phi_{\text{CO}_2}^V$. The fugacity coefficient of pure CO_2 is calculated by using Eqn (5.3-9). The results are presented in tabular form and also plotted as shown below. As can be seen from the figure, the fugacity coefficient of CO_2 is strongly dependent on pressure with a maximum point. An ideal mixture assumption fails at all pressures, since the mole fraction of carbon dioxide is small. For more details on this problem see [Chueh and Prausnitz \(1967\)](#).

P (bar)	A_{mix}	B_{mix}	Z_{mix}^V	$\hat{\phi}_{\text{CO}_2}^V$	$\phi_{\text{CO}_2}^V$
1	8.333×10^{-3}	1.976×10^{-3}	0.994	1.002	0.999
10	0.083	0.020	0.934	1.021	0.986
50	0.417	0.099	0.602	1.250	0.931
60	0.500	0.119	0.505	1.392	0.918
70	0.583	0.138	0.453	1.498	0.905
80	0.667	0.158	0.445	1.519	0.892
90	0.750	0.178	0.456	1.498	0.880
100	0.833	0.198	0.474	1.462	0.868
150	1.250	0.296	0.593	1.277	0.813
200	1.667	0.395	0.721	1.151	0.767
250	2.083	0.494	0.847	1.069	0.730
300	2.500	0.593	0.970	1.015	0.701



Soave-Redlich-Kwong equation of state: The use of the Soave-Redlich-Kwong equation of state in Eqn (7.5-12) leads to the following expression for the fugacity coefficient of species i in a mixture:

$$\ln \hat{\phi}_i^V = \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^V - 1) - \ln (Z_{\text{mix}}^V - B_{\text{mix}}) - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k y_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}^V} \right). \quad (7.5-15)$$

In terms of dimensionless parameters, Eqn (7.5-15) is identical with the one given for the Redlich-Kwong equation of state, Eqn (7.5-14).

Peng-Robinson equation of state: The use of the Peng-Robinson equation of state in Eqn (7.5-12) leads to the following expression for the fugacity coefficient of species i in a mixture:

$$\ln \hat{\phi}_i^V = \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^V - 1) - \ln (Z_{\text{mix}}^V - B_{\text{mix}}) - \frac{A_{\text{mix}}}{\sqrt{8} B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k y_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left[\frac{Z_{\text{mix}}^V + (1 + \sqrt{2}) B_{\text{mix}}}{Z_{\text{mix}}^V + (1 - \sqrt{2}) B_{\text{mix}}} \right]. \quad (7.5-16)$$

Example 7.6 Determine the fugacity coefficients of methane (1) and ethane (2) in a 35 mol% methane, 65% ethane mixture at 373.15 K and 30 bar using the Peng-Robinson equation of state. Take $k_{12} = -0.003$.

Solution

The properties at the critical point as well as the acentric factors were given in Example 7.2. The values of the reduced temperature and pressure are

Component	T_r	P_r
Methane	1.958	0.651
Ethane	1.222	0.612

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$A_{11} = 0.055, A_{12} = A_{21} = 0.096, B_1 = 0.026, \\ A_{22} = 0.167, B_2 = 0.039.$$

The dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}} = 0.121 \quad \text{and} \quad B_{\text{mix}} = 0.034$$

Therefore, Eqn (6.1-9) becomes

$$Z_{\text{mix}}^3 - 0.966Z_{\text{mix}}^2 + 0.050Z_{\text{mix}} - 0.003 = 0 \Rightarrow Z_{\text{mix}}^V = 0.915.$$

Therefore, the fugacity coefficients of components are calculated from Eqn (7.5-16) as

$$\hat{\phi}_1 = 0.986, \quad \hat{\phi}_2 = 0.883.$$

Comment: The fugacities of components in the mixture are

$$\hat{f}_1 = (0.986)(30)(0.35) = 10.35 \text{ bar}, \\ \hat{f}_2 = (0.883)(30)(0.65) = 17.22 \text{ bar}.$$

Comparison of these values with the ones calculated in Example 7.2 indicates that the use of the Lewis-Randall rule is satisfactory for this specific case.

7.6 Calculation of Component Fugacities in a Liquid Mixture

Cubic equations of state can also describe $P - \tilde{V} - T$ relations of liquid mixtures. However, estimation of mixture parameters is not a straightforward task. Depending on the components making up the mixture, different mixing rules may be used.

When van der Waals mixing rules, i.e. Eqns (6.1-5) and (6.1-6), are used, the equations given in Section 7.5.4 are also applicable for liquid mixtures.

van der Waals equation of state:

$$\ln \hat{\phi}_i^L = \frac{B_i}{Z_{\text{mix}}^L - B_{\text{mix}}} - \ln (Z_{\text{mix}}^L - B_{\text{mix}}) - \frac{2 \sum_{j=1}^k x_j A_{ij}}{Z_{\text{mix}}^L}. \quad (7.6-1)$$

Redlich-Kwong equation of state:

$$\begin{aligned} \ln \hat{\phi}_i^L = & \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^L - 1) - \ln (Z_{\text{mix}}^L - B_{\text{mix}}) \\ & - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k x_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}^L} \right). \end{aligned} \quad (7.6-2)$$

Soave-Redlich-Kwong equation of state:

$$\begin{aligned} \ln \hat{\phi}_i^L = & \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^L - 1) - \ln (Z_{\text{mix}}^L - B_{\text{mix}}) \\ & - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k x_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}^L} \right). \end{aligned} \quad (7.6-3)$$

Peng-Robinson equation of state:

$$\begin{aligned} \ln \hat{\phi}_i^L = & \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^L - 1) - \ln (Z_{\text{mix}}^L - B_{\text{mix}}) - \frac{A_{\text{mix}}}{\sqrt{8} B_{\text{mix}}} \\ & \times \left(\frac{2 \sum_{j=1}^k x_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left[\frac{Z_{\text{mix}}^L + (1 + \sqrt{2}) B_{\text{mix}}}{Z_{\text{mix}}^L + (1 - \sqrt{2}) B_{\text{mix}}} \right]. \end{aligned} \quad (7.6-4)$$

Example 7.7 Shibata and Sandler (1989) reported the following data for the coexisting vapor and liquid phases of nitrogen (1) and cyclohexane (2) mixture:

$$T = 366.4 \text{ K}, \quad P = 138.76 \text{ bar}, \quad x_1 = 0.1286, \quad y_1 = 0.9721.$$

The system is represented by the Peng-Robinson equation of state with $k_{12} = 0.109$.

- Show that the vapor and liquid phases are indeed in equilibrium with each other.
- Estimate the densities of vapor and liquid phases.

Solution

From Appendix A

Component	M (g/mol)	T_c (K)	P_c (bar)	ω
Nitrogen	28.01	126.2	34.0	0.039
Cyclohexane	84.16	554.0	40.7	0.212

- The condition of equilibrium is given by

$$\hat{f}_i^V = \hat{f}_i^L \Rightarrow y_i \hat{\phi}_i^V = x_i \hat{\phi}_i^L. \quad (1)$$

Using the equations given in Table 6.3, the dimensionless parameters are calculated as

$$A_{11} = 0.107, \quad A_{12} = A_{21} = 0.620, \quad B_1 = 0.109, \\ A_{22} = 4.541, \quad B_2 = 0.401.$$

Vapor phase: Using $y_1 = 0.9721$ and $y_2 = 0.0279$, the dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}} = \sum_{i=1}^2 \sum_{j=1}^2 y_i y_j A_{ij} = 0.138 \quad \text{and} \quad B_{\text{mix}} = \sum_{i=1}^2 y_i B_i = 0.118.$$

The compressibility factor for the vapor phase is calculated from Eqn (6.1-9) as

$$Z_{\text{mix}}^V = 1.019.$$

Substitution of the numerical values into Eqn (7.5-16) leads to

$$\hat{\phi}_1^V = 1.021, \quad \hat{\phi}_2^V = 0.493.$$

Liquid phase: Using $x_1 = 0.1286$ and $x_2 = 0.8714$, the dimensionless parameters for the mixture are calculated from Eqns (6.1-10) and (6.1-11) as

$$A_{\text{mix}} = \sum_{i=1}^2 \sum_{j=1}^2 x_i x_j A_{ij} = 3.589 \quad \text{and} \quad B_{\text{mix}} = \sum_{i=1}^2 x_i B_i = 0.364.$$

The compressibility factor for the liquid phase is calculated from Eqn (6.1-9) as

$$Z_{\text{mix}}^{\text{L}} = 0.471.$$

Substitution of the numerical values into Eqn (7.6-4) leads to

$$\hat{\phi}_1^{\text{L}} = 7.623, \quad \hat{\phi}_2^{\text{L}} = 0.017.$$

The use of Eqn (1) gives

$$\begin{aligned} y_1 \hat{\phi}_1^{\text{V}} &= (0.9721)(1.021) = 0.993, & x_1 \hat{\phi}_1^{\text{L}} &= (0.1286)(7.623) = 0.980, \\ y_2 \hat{\phi}_2^{\text{V}} &= (0.0279)(0.493) = 0.014, & x_2 \hat{\phi}_2^{\text{L}} &= (0.8714)(0.017) = 0.015. \end{aligned}$$

Since $y_i \hat{\phi}_i^{\text{V}} \simeq x_i \hat{\phi}_i^{\text{L}}$, vapor and liquid phases are in equilibrium with each other.

(b) The molar volume of the vapor phase is

$$\tilde{V}_{\text{mix}}^{\text{V}} = \frac{Z_{\text{mix}}^{\text{V}} RT}{P} = \frac{(1.019)(83.14)(366.4)}{138.76} = 223.7 \text{ cm}^3/\text{mol}.$$

The molecular weight of the vapor mixture is

$$M^{\text{V}} = (0.9721)(28.01) + (0.0279)(84.16) = 29.577 \text{ g/mol}.$$

The density of the vapor is

$$\rho^{\text{V}} = \frac{29.577}{223.7} = 0.132 \text{ g/cm}^3.$$

The molar volume of the liquid phase is

$$\tilde{V}_{\text{mix}}^{\text{L}} = \frac{Z_{\text{mix}}^{\text{L}} RT}{P} = \frac{(0.471)(83.14)(366.4)}{138.76} = 103.4 \text{ cm}^3/\text{mol}.$$

The molecular weight of the liquid mixture is

$$M^{\text{L}} = (0.1286)(28.01) + (0.8714)(84.16) = 76.939 \text{ g/mol}.$$

The density of the liquid is

$$\rho^{\text{L}} = \frac{76.939}{103.4} = 0.744 \text{ g/cm}^3.$$

In most applications, however, liquid mixtures cannot be expressed by equations of state. In that case, component fugacities in liquid mixtures are determined by using activity coefficients as will be explained in the next chapter.

7.7 Variation of Component Fugacity with Pressure and Temperature

The variation of component fugacity with pressure by keeping temperature and composition constant can be easily obtained from Eqn (7.3-13) as

$$\left(\frac{\partial \ln \hat{f}_i}{\partial P} \right)_{T, x_j} = \frac{\bar{V}_i}{RT}. \quad (7.7-1)$$

On the other hand, differentiation of Eqn (7.3-1) with respect to temperature by keeping pressure and composition constant gives

$$\left(\frac{\partial \ln \hat{f}_i}{\partial T} \right)_{P, x_j} = \frac{1}{R} \frac{\partial}{\partial T} \left(\frac{\bar{G}_i}{T} \right)_{P, x_j} - \frac{1}{R} \frac{\partial}{\partial T} \left(\frac{\bar{G}_i^{\text{IGM}}}{T} \right)_{P, x_j}. \quad (7.7-2)$$

The Gibbs-Helmholtz equation developed for pure components, Eqn (5.7-8), is also valid for partial molar properties, i.e.

$$\frac{\partial}{\partial T} \left(\frac{\bar{G}_i}{T} \right)_{P, x_j} = -\frac{\bar{H}_i}{T^2}. \quad (7.7-3)$$

Therefore, Eqn (7.7-2) reduces to

$$\left(\frac{\partial \ln \hat{f}_i}{\partial T} \right)_{P, x_j} = -\frac{\bar{H}_i - \bar{H}_i^{\text{IGM}}}{RT^2} \quad (7.7-4)$$

or

$$\left(\frac{\partial \ln \hat{f}_i}{\partial T} \right)_{P, x_j} = -\frac{\bar{H}_i - \tilde{H}_i^{\text{IG}}}{RT^2} \quad (7.7-5)$$

Variation in $\ln \hat{f}_i$ is expressed in the form of an exact differential as

$$d \ln \hat{f}_i = \left(\frac{\partial \ln \hat{f}_i}{\partial P} \right)_{T, x_j} dP + \left(\frac{\partial \ln \hat{f}_i}{\partial T} \right)_{P, x_j} dT. \quad (7.7-6)$$

Substitution of Eqns (7.7-1) and (7.7-5) into Eqn (7.7-6) gives

$$d \ln \hat{f}_i = \left(\frac{\bar{V}_i}{RT} \right) dP - \left(\frac{\bar{H}_i - \tilde{H}_i^{\text{IG}}}{RT^2} \right) dT. \quad (7.7-7)$$

Note that Eqn (7.7-7) is analogous to Eqn (5.8-5) in which molar quantities are replaced by partial molar quantities.

7.8 The Use of Fugacity in Phase Equilibrium Calculations

When two phases of a pure component are in equilibrium with each other, compositions of the phases are the same. For example, when water vapor is in equilibrium with liquid water at 373 K under atmospheric pressure, there is no difference in compositions of phases. In the case of multicomponent mixtures, on the other hand, compositions of the phases that are in equilibrium with each other are not the same. Estimation of the compositions of components in each phase is of practical interest to chemical engineers.

7.8.1 Vapor-Liquid Equilibrium

When pure liquid A evaporates into vapor B, evaporation takes place as long as

$$f_A^{\text{L}}(T, P) > \hat{f}_A^{\text{V}}(T, P, y_A) \quad \text{Mass transfer from liquid to vapor.} \quad (7.8-1)$$

In writing Eqn (7.8-1) it is implicitly assumed that the solubility of B in the liquid is negligible, i.e. liquid exists as pure A. The mass transfer of A to the vapor phase stops when the system reaches equilibrium, i.e.

$$f_A^{\text{L}}(T, P) = \hat{f}_A^{\text{V}}(T, P, y_A^*) \quad \text{Condition of equilibrium.} \quad (7.8-2)$$

Under these conditions vapor is saturated with A, y_A^* being the corresponding mole fraction. In other words, when vapor is saturated with A, it contains the maximum amount of A at a given temperature and pressure. The use of Eqns (5.4-8) and (7.3-3) for the left- and right-hand sides of Eqn (7.8-2), respectively, results in

$$P_A^{\text{vap}} \phi_A^{\text{V}}(T, P_A^{\text{vap}}) \exp \left[\frac{\tilde{V}_A^{\text{L}}(P - P_A^{\text{vap}})}{RT} \right] = y_A^* P \hat{\phi}_A^{\text{V}}(T, P, y_A^*). \quad (7.8-3)$$

If pressure is not very high, the following simplifying assumptions are plausible:

- Poynting correction factor is unity.
- The vapor of A at T and P_A^{vap} is ideal, i.e. $\phi_A^V(T, P_A^{\text{vap}}) = 1$.
- Vapor is an ideal gas mixture, i.e. $\hat{\phi}_A^V(T, P, y_A^*) = 1$.

Under these circumstances, Eqn (7.8-3) simplifies to

$$P_A^{\text{vap}} = y_A^* P \quad (7.8-4)$$

indicating that vapor is saturated with A when its partial pressure equals vapor pressure at the given temperature.

For example, vapor pressure of liquid water at 298 K is 3.169 kPa. Air at 298 K and 101.3 kPa is saturated with water vapor (or relative humidity is 100%) when the partial pressure of water vapor equals 3.169 kPa. Under these conditions, air holds the maximum water with a mole fraction of

$$y_{\text{H}_2\text{O}}^* = \frac{P_{\text{H}_2\text{O}}^{\text{vap}}}{P} = \frac{3.169}{101.3} = 0.0313.$$

Example 7.8 Three liters of toluene accidentally spills on the laboratory floor. Determine the fraction of toluene vaporized over a long period of time, say 24 h, if the volume of the laboratory is 180 m³. Assume that the laboratory temperature remains constant at 298 K.

Solution

From Appendix A

$$T_c = 593 \text{ K}, \quad P_c = 41 \text{ bar}, \quad \omega = 0.263.$$

From Appendix C

$$\ln P^{\text{vap}} = 9.3935 - \frac{3096.52}{T - 53.67}.$$

The toluene spilled on the floor will vaporize into air as long as

$$f_{\text{toluene}}^L(T, P) > \hat{f}_{\text{toluene}}^V(T, P, y_{\text{toluene}}). \quad (1)$$

In writing Eqn (1) it is implicitly assumed that air has a negligible solubility in liquid toluene so that the liquid phase is pure toluene. Under equilibrium conditions, i.e. when air is saturated with toluene, vaporization stops and we have

$$f_{\text{toluene}}^L(298 \text{ K}, 1.013 \text{ bar}) = \hat{f}_{\text{toluene}}^V(298 \text{ K}, 1.013 \text{ bar}, y_{\text{toluene}}^*), \quad (2)$$

where y_{toluene}^* represents the mole fraction of toluene in air saturated with toluene at 298 K and atmospheric pressure, i.e. 1.013 bar. Using the same assumptions in simplifying Eqn (7.8-3), Eqn (2) reduces to

$$P_{\text{toluene}}^{\text{vap}} = P y_{\text{toluene}}^* \Rightarrow y_{\text{toluene}}^* = \frac{P_{\text{toluene}}^{\text{vap}}}{P}. \quad (3)$$

At 298 K, the vapor pressure of toluene is

$$P_{\text{toluene}}^{\text{vap}} = \exp \left(9.3935 - \frac{3096.52}{298 - 53.67} \right) = 0.038 \text{ bar}.$$

Substitution of numerical values into Eqn (3) gives

$$y_{\text{toluene}}^* = \frac{0.038}{1.013} = 3.75 \times 10^{-2}.$$

In other words, liquid toluene vaporizes until its mole fraction in the vapor phase reaches 3.75×10^{-2} .

Now it is necessary to determine whether the spilled toluene causes air saturated with toluene or not. From Table 5.3, the molar volume of liquid toluene is approximately 107 cm³/mol. Therefore, the number of moles of toluene is

$$n_{\text{toluene}} = \frac{V_{\text{bottle}}}{\tilde{V}_{\text{L}}} = \frac{3000}{107} = 28.04 \text{ mol}.$$

The number of moles of air is

$$n_{\text{air}} = \frac{PV_{\text{lab}}}{RT} = \frac{(1.013)(180)}{(8.314 \times 10^{-5})(298)} = 7359.6 \text{ mol}.$$

If all toluene were evaporated, then its mole fraction would be

$$y_{\text{toluene}} = \frac{28.04}{28.04 + 7359.6} = 3.8 \times 10^{-3},$$

which is less than the saturation value of 3.75×10^{-2} . Therefore, all toluene is evaporated.

It should be kept in mind that Eqn (7.8-3) simplifies to Eqn (7.8-4) as long as pressure is low. When pressure is moderate or high, then it is necessary to use Eqn (7.8-3) as will be shown in the next example.

Example 7.9 A cylinder contains propylene (1) and nitrogen (2) at 298 K and 18 bar. Estimate the vapor phase composition if the solubility of nitrogen in liquid propylene is negligible. Assume that the vapor phase is represented by the virial equation of state. Critical molar volumes of propylene and nitrogen are $1.81 \times 10^{-4} \text{ m}^3/\text{mol}$ and $0.895 \times 10^{-4} \text{ m}^3/\text{mol}$, respectively. The vapor pressure of propylene at 298 K is 11.53 bar.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Propylene	365.2	46	0.144
Nitrogen	126.2	34	0.039

Since the solubility of nitrogen in liquid propylene is negligible, the liquid phase consists of pure propylene and the condition of equilibrium is given by Eqn (7.8-2) as

$$f_1^L(T, P) = \hat{f}_1^V(T, P, y_1^*) \quad (1)$$

or, from Eqn (7.8-3),

$$P_1^{\text{vap}} \phi_1^V(T, P_1^{\text{vap}}) \exp \left[\frac{\tilde{V}_1^L(P - P_1^{\text{vap}})}{RT} \right] = y_1^* P \hat{\phi}_1^V(T, P, y_1^*). \quad (2)$$

Calculation of the fugacity coefficients requires virial coefficients, B_{11} , B_{22} , and B_{12} , to be calculated.

Calculation of second virial coefficients for pure components: From Eqns (3.1-10)–(3.1-13)

$$\begin{aligned}
 B_{11} &= \frac{(83.14)(365.2)}{46} \left\{ \left[0.083 - \frac{0.422}{(298/365.2)^{1.6}} \right] \right. \\
 &\quad \left. + (0.144) \left[0.139 - \frac{0.172}{(298/365.2)^{4.2}} \right] \right\} = -356 \text{ cm}^3/\text{mol}, \\
 B_{22} &= \frac{(83.14)(126.2)}{34} \left\{ \left[0.083 - \frac{0.422}{(298/126.2)^{1.6}} \right] \right. \\
 &\quad \left. + (0.039) \left[0.139 - \frac{0.172}{(298/126.2)^{4.2}} \right] \right\} = -5.7 \text{ cm}^3/\text{mol}.
 \end{aligned}$$

Calculation of second virial cross coefficient: The parameters defined by Eqns (6.1-3) and (6.1-4) are calculated as

$$\begin{aligned}
 k_{12}^* &= 1 - \frac{8\sqrt{\tilde{V}_{c1}\tilde{V}_{c2}}}{(\tilde{V}_{c1}^{1/3} + \tilde{V}_{c2}^{1/3})^3} \\
 &= 1 - \frac{8\sqrt{(1.81 \times 10^{-4})(0.895 \times 10^{-4})}}{\left[(1.81 \times 10^{-4})^{1/3} + (0.895 \times 10^{-4})^{1/3}\right]^3} = 0.020, \\
 T_{c12} &= (1 - k_{12}^*)\sqrt{T_{c1}T_{c2}} = (1 - 0.020)\sqrt{(365.2)(126.2)} = 210.4 \text{ K}, \\
 \omega_{12} &= \frac{\omega_1 + \omega_2}{2} = \frac{0.144 + 0.039}{2} = 0.092, \\
 P_{c12} &= \frac{(4)(210.4)}{\left[(1.81 \times 10^{-4})^{1/3} + (0.895 \times 10^{-4})^{1/3}\right]^3} \\
 &\quad \times \left[\frac{(46)(1.81 \times 10^{-4})}{365.2} + \frac{(34)(0.895 \times 10^{-4})}{126.2} \right] \\
 &= 37.98 \text{ bar}.
 \end{aligned}$$

The parameters $B_{12}^{(0)}$ and $B_{12}^{(1)}$ are calculated from Eqns (3.1-11) and (3.1-12), respectively, as

$$\begin{aligned}
 B_{12}^{(0)} &= 0.083 - \frac{0.422}{(298/210.4)^{1.6}} = -0.159, \\
 B_{12}^{(1)} &= 0.139 - \frac{0.172}{(298/210.4)^{4.2}} = 0.099.
 \end{aligned}$$

Finally, the second virial cross coefficient B_{12} is calculated from Eqn (3.1-10) as

$$B_{12} = \frac{(83.14)(210.4)}{37.98} [-0.159 + (0.092)(0.099)] = -69.04 \text{ cm}^3/\text{mol}.$$

Calculation of fugacity coefficients: The fugacity coefficient of pure propylene vapor is calculated from Eqn (5.3-5) as

$$\phi_1^V(T, P_1^{\text{vap}}) = \exp\left(\frac{B_{11}P_1^{\text{vap}}}{RT}\right) = \exp\left[\frac{(-356)(11.53)}{(83.14)(298)}\right] = 0.847.$$

The fugacity coefficient of propylene in the vapor phase is calculated from Eqn (7.5-6) as

$$\begin{aligned}\hat{\phi}_1^V(T, P, y_1^*) &= \exp \left\{ \frac{P}{RT} [B_{11} + (1 - y_1^*)^2 (2B_{12} - B_{11} - B_{22})] \right\} \\ &= \exp \left(\frac{(18)}{(83.14)(298)} \left\{ -356 + (1 - y_1^*)^2 [(2)(-69.04) \right. \right. \\ &\quad \left. \left. + 356 + 5.7] \right\} \right) \\ &= \exp [0.259 + 0.162(1 - y_1^*)^2].\end{aligned}\quad (3)$$

The molar volume of liquid propylene is calculated from the Rackett equation, Eqn (5.4-9), as

$$\begin{aligned}Z_{RA} &= 0.29056 - (0.08775)(0.144) = 0.278, \\ \tilde{V}_1^L &= \frac{(83.14)(365.2)}{46} (0.278)^{1+[1-(298/365.2)]^2/7} = 83.3 \text{ cm}^3/\text{mol}.\end{aligned}$$

Substitution of the numerical values into Eqn (2) gives

$$\begin{aligned}(11.53)(0.847) \exp \left[\frac{(83.3)(18 - 11.53)}{(83.14)(298)} \right] \\ = y_1^*(18) \exp [0.259 + 0.162(1 - y_1^*)^2].\end{aligned}\quad (4)$$

The solution of Eqn (4) by MATHCAD[®] gives $y_1^* = 0.404$.

Condensation is the reverse of vaporization. Let us calculate the temperature at which condensation starts when the vapor composition and pressure are known. Suppose that air at 298 K and 101.3 kPa has a relative humidity (RH) of 80%. The partial pressure of water vapor, $p_{\text{H}_2\text{O}}$, in air is

$$p_{\text{H}_2\text{O}} = (\text{RH})P_{\text{H}_2\text{O}}^{\text{vap}} = (0.80)(3.169) = 2.535 \text{ kPa}.$$

Condensation, or mass transfer from the vapor to the liquid, takes place if

$$\hat{f}_{\text{H}_2\text{O}}^V(T, P, y_{\text{H}_2\text{O}}) > f_{\text{H}_2\text{O}}^L(T, P) \Rightarrow p_{\text{H}_2\text{O}} > P_{\text{H}_2\text{O}}^{\text{vap}}.$$

In writing the above equation, it is implicitly assumed that the solubility of air in water is negligible. In our case since $p_{\text{H}_2\text{O}} < P_{\text{H}_2\text{O}}^{\text{vap}}$, no condensation occurs. As the

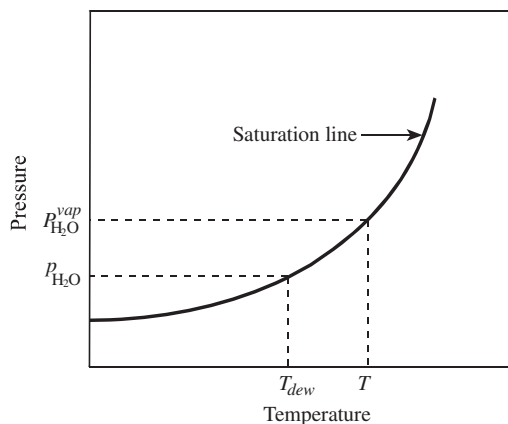


Figure 7.1: Dew point temperature of an air-water system.

temperature is decreased, the amount of water vapor in air does not change. The first liquid droplet appears when the temperature drops to T^* , at which point⁴

$$p_{\text{H}_2\text{O}} = P_{\text{H}_2\text{O}}^{\text{vap}} = 2.535 \text{ kPa.}$$

In other words, air becomes saturated with water vapor. The use of the Antoine equation gives the dew point temperature as

$$T^* = 46.13 + \frac{3816.44}{11.6834 - \ln(2.535/100)} = 294.6 \text{ K (21.6}^\circ\text{C)}.$$

Figure 7.1 illustrates the dew point temperature of an air-water system.

Example 7.10 A piston-cylinder arrangement holds a gas mixture of 20 mol% chloroform (1) and 80% nitrogen (2) at 340 K and 1.013 bar. The system is cooled under constant pressure. Estimate the temperature at which condensation of chloroform takes place.

Solution

From Appendix C

$$\ln P_1^{\text{vap}} = 8.3530 - \frac{2696.79}{T - 46.16}.$$

⁴This is known as the dew point temperature. Such calculations will be covered in detail in Chapter 9.

Let T^* be the temperature at which the first liquid droplet of chloroform appears. If the solubility of nitrogen in liquid chloroform is assumed negligible, i.e. the liquid phase is pure chloroform, then the condition of equilibrium is expressed as

$$f_1^L(T^*, 1.013 \text{ bar}) = \hat{f}_1^V(T^*, 1.013 \text{ bar}, y_1) \quad (1)$$

or

$$\begin{aligned} P_1^{\text{vap}}(T^*) \phi_1^V [T^*, P_1^{\text{vap}}(T^*)] \exp \left\{ \frac{\tilde{V}_1^L [1.013 - P_1^{\text{vap}}(T^*)]}{RT^*} \right\} \\ = (1.013) y_1 \hat{\phi}_1^V(T^*, 1.013 \text{ bar}, y_1). \end{aligned} \quad (2)$$

Since the pressure is atmospheric, the following simplifying assumptions are plausible:

- Poynting correction factor is unity.
- Vapor phase is an ideal gas mixture, i.e. $\hat{\phi}_1^V(T^*, 1.013 \text{ bar}, y_1) \simeq 1$.
- $\phi_1^V [T^*, P_1^{\text{vap}}(T^*)] \simeq 1$.

Therefore, Eqn (2) simplifies to

$$P_1^{\text{vap}}(T^*) = 1.013 y_1 = (1.013)(0.2) = 0.203 \text{ bar}.$$

The temperature, T^* , is calculated from the Antoine equation as

$$T^* = 46.16 + \frac{2696.79}{8.3530 - \ln(0.203)} = 317.3 \text{ K}.$$

Comment: Since the pressure is atmospheric, simplifying assumptions lead to a straightforward calculation of T^* . In the case of high pressures, one should use a trial-and-error procedure to estimate T^* , i.e. assume T^* and check whether Eqn (2) is satisfied.

7.8.2 Solid-Liquid Equilibrium

Let us add a small amount of sugar to a cup filled with tea. If we stir the tea in the cup, the sugar will disappear. This means the tea is unsaturated with sugar or, in other words, the fugacity of the solid sugar is greater than the fugacity of the sugar in the tea, i.e.

$$f_{\text{sugar}}^S(T, P) > \hat{f}_{\text{sugar}}^L(T, P, x_{\text{sugar}}) \quad \text{Mass transfer from solid to liquid.} \quad (7.8-5)$$

If we keep continue adding sugar to the tea at constant temperature, a point comes where the liquid is saturated with sugar

$$f_{\text{sugar}}^S(T, P) = \hat{f}_{\text{sugar}}^L(T, P, x_{\text{sugar}}^*) \quad \text{Condition of equilibrium.} \quad (7.8-6)$$

The saturated condition, i.e. $x_{\text{sugar}} = x_{\text{sugar}}^*$, represents the maximum amount of sugar that can be present in the tea at the given temperature and pressure. After this point, any additional sugar will not dissolve and will appear as a solid phase within the cup. Solid-liquid equilibrium will be discussed in detail in Chapter 12.

7.8.3 Solid-Vapor Equilibrium

Consider sublimation of species A into vapor B. Sublimation takes place as long as

$$f_A^S(T, P) > \hat{f}_A^V(T, P, y_A) \quad \text{Mass transfer from solid to vapor.} \quad (7.8-7)$$

In writing Eqn (7.8-7) it is implicitly assumed that the solid phase is pure A, i.e. solubility of B in the solid phase is negligible. Sublimation stops when vapor B is saturated with A, i.e. $y_A = y_A^*$, at the given temperature and pressure. Under these conditions

$$f_A^S(T, P) = \hat{f}_A^V(T, P, y_A^*) \quad \text{Condition of equilibrium.} \quad (7.8-8)$$

The use of Eqns (5.5-8) and (7.3-3) for the left- and right-hand sides, respectively, of Eqn (7.8-8) leads to

$$P_A^{\text{sub}} \exp \left[\frac{\tilde{V}_A^S (P - P_A^{\text{sub}})}{RT} \right] = y_A^* P \hat{\phi}_A^V(T, P, y_A^*) \quad (7.8-9)$$

or

$$y_A^* = \left(\frac{P_A^{\text{sub}}}{P} \right) \mathcal{E}, \quad (7.8-10)$$

where the *enhancement factor*, $\mathcal{E}$, is defined by

$$\mathcal{E} = \frac{\exp \left[\frac{\tilde{V}_A^S (P - P_A^{\text{sub}})}{RT} \right]}{\hat{\phi}_A^V}. \quad (7.8-11)$$

As stated in Section 5.6, supercritical fluids are at a temperature and pressure greater than or equal to the critical temperature and pressure of fluids. At high pressures, supercritical fluids possess solvent properties that are superior to those of liquid solvents. As a result, they may be an alternative to replace toxic and environmentally hazardous solvents used in separation and or purification processes. Being nontoxic, inexpensive, and nonflammable, and having a low critical temperature, carbon dioxide is one of the most widely used supercritical fluids for extraction purposes. The design of extraction processes involving supercritical solvents requires the understanding of solid-vapor equilibrium.

Example 7.11 It is required to estimate the solubility of naphthalene (2) in carbon dioxide (1) at 118.6 bar and 328 K for a supercritical extraction. Assume that the vapor phase is represented by the Peng-Robinson equation of state with $k_{12} = 0.08246$. For naphthalene, the sublimation pressure at 328 K and the molar volume are 15.9×10^{-4} bar and $111.9 \text{ cm}^3/\text{mol}$, respectively.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Carbon dioxide	304.2	73.8	0.239
Naphthalene	748.0	41.0	0.302

Note that the given temperature is less than the triple point temperature of naphthalene, which is 353.42 K from Table 5.5. Therefore, when the naphthalene content of vapor exceeds its solubility limit, solid naphthalene is formed by desublimation. If the solid phase consists of pure naphthalene, the condition of phase equilibrium states that the fugacity of pure solid naphthalene is equal to the fugacity of naphthalene in the gas phase. Thus, Eqn (7.8-9) is expressed as

$$P_2^{\text{sub}} \exp \left[\frac{\tilde{V}_2^{\text{S}}(P - P_2^{\text{sub}})}{RT} \right] = y_2^* P \hat{\phi}_2^{\text{V}}(T, P, y_2^*) \quad (1)$$

or

$$\underbrace{15.9 \times 10^{-4} \exp \left[\frac{(111.9)(118.6 - 15.9 \times 10^{-4})}{(83.14)(328)} \right]}_{2.587 \times 10^{-3}} = y_2^* P \hat{\phi}_2^{\text{V}}(T, P, y_2^*). \quad (2)$$

Since the fugacity coefficient of component 2 is dependent on the mole fraction, determination of y_2^* from Eqn (2) requires a trial-and-error solution. Once the value of y_2^* is assumed, $\hat{\phi}_2^{\text{V}}$ is determined from Eqn (7.5-16). If the assumed

y_2^* value is correct, then the right-hand side (RHS) of Eqn (2) should be equal to 2.587×10^{-3} . The results of the iterations are summarized in the following table:

$y_2^* \times 10^3$	A_{mix}	B_{mix}	Z_{mix}^V	$\hat{\phi}_2^V$	RHS of Eqn (3)
7.00	0.622	0.119	0.383	2.877×10^{-3}	2.388×10^{-3}
7.95	0.626	0.119	0.379	2.732×10^{-3}	2.576×10^{-3}
8.01	0.626	0.119	0.379	2.723×10^{-3}	2.587×10^{-3}

Therefore, the mole fraction of naphthalene in CO_2 at 118.6 bar and 328 K is 8.01×10^{-3} . The experimental value reported by [McHugh and Paulaitis \(1980\)](#) is 12.29×10^{-3} .

Comment: When the mole fraction of one of the components is very small, i.e. $y_1 \simeq 1$ (or $y_2 \simeq 0$), the following simplifying assumptions are plausible:

$$A_{\text{mix}} = A_{11}, \quad B_{\text{mix}} = B_1, \quad Z_{\text{mix}}^V = Z_1^V \quad (3)$$

and the fugacity coefficient of component 2 takes the form

$$\begin{aligned} \ln (\hat{\phi}_2^V)_{\text{approx}} &= \frac{B_2}{B_1} (Z_1^V - 1) - \ln (Z_1^V - B_1) - \frac{A_{11}}{\sqrt{8}B_1} \left(\frac{2A_{12}}{A_{11}} - \frac{B_2}{B_1} \right) \\ &\quad \times \ln \left[\frac{Z_1^V + (1 + \sqrt{2})B_1}{Z_1^V + (1 - \sqrt{2})B_1} \right], \end{aligned} \quad (4)$$

which is independent of composition. Thus, Eqn (2) becomes

$$2.587 \times 10^{-3} = y_2^* P (\hat{\phi}_2^V)_{\text{approx}}. \quad (5)$$

The numerical values of the parameters in the Peng-Robinson equation of state are

$$\begin{aligned} A_{11} &= 0.597, \quad B_1 = 0.116, \\ A_{12} &= 2.372, \quad B_2 = 0.513. \end{aligned}$$

For pure CO_2 , Eqn (6.1-9) becomes

$$Z_1^3 - 0.884Z_1^2 + 0.325Z_1 - 0.054 = 0 \quad \Rightarrow \quad Z_1^V = 0.414.$$

The approximate value of the fugacity coefficient of component 2 is calculated from Eqn (4) as

$$(\hat{\phi}_2^V)_{\text{approx}} = 4.195 \times 10^{-3}.$$

Substitution of the numerical values into Eqn (5) gives

$$2.587 \times 10^{-3} = y_2^*(118.6)(4.195 \times 10^{-3}) \Rightarrow y_2^* = 5.2 \times 10^{-3}.$$

As far as the order of magnitude is concerned, this value is not very far from the previously calculated one.

In the above examples, our analysis is restricted to cases in which equilibrium takes place between a pure component in one phase and a multicomponent mixture in another phase. Obviously, this is not necessarily the case in general.

Problems

Problem Related to Section 7.2

7.1 Derive Eqn (7.2-11).

Problems Related to Section 7.5

7.2 Derive Eqn (7.5-6) as follows:

- (a) By definition, the partial molar compressibility factor of component i is given by

$$\bar{Z}_i = \left[\frac{\partial(nZ_{\text{mix}})}{\partial n_i} \right]_{T,P,n_{j \neq i}}. \quad (1)$$

Show that the substitution of Eqn (7.5-5) into Eqn (1) and rearrangement give

$$\bar{Z}_i = \left[\frac{\partial}{\partial n_i} \left(n + \frac{n^2 B_{\text{mix}} P}{nRT} \right) \right]_{T,P,n_{j \neq i}}. \quad (2)$$

Carry out the differentiation to obtain

$$\bar{Z}_i = 1 - \frac{B_{\text{mix}} P}{RT} + \frac{P}{nRT} \left(\frac{\partial}{\partial n_i} \sum_{i=1}^k \sum_{j=1}^k n_i n_j B_{ij} \right)_{T,P,n_{j \neq i}}. \quad (3)$$

- (b) To differentiate the third term on the right-hand side of Eqn (3), first consider a binary mixture of components 1 and 2, and show that

$$\left(\frac{\partial}{\partial n_1} \sum_{i=1}^2 \sum_{j=1}^2 n_i n_j B_{ij} \right)_{T,P,n_2} = 2(n_1 B_{11} + n_2 B_{12}) = 2 \sum_{j=1}^2 n_j B_{1j}. \quad (4)$$

Generalize Eqn (4) as

$$\left(\frac{\partial}{\partial n_i} \sum_{i=1}^k \sum_{j=1}^k n_i n_j B_{ij} \right)_{T,P,n_{j \neq i}} = 2 \sum_{j=1}^k n_j B_{ij}. \quad (5)$$

Therefore, Eqn (3) takes the form

$$\bar{Z}_i = 1 + \frac{P}{RT} \left(2 \sum_{j=1}^k y_j B_{ij} - B_{\text{mix}} \right). \quad (6)$$

- (c) Show that the substitution of Eqn (6) into Eqn (7.3-15) leads to Eqn (7.5-6).

7.3 Derive an expression to represent the overall fugacity coefficient of the mixture, ϕ_{mix} , as follows:

- (a) Start with

$$\tilde{G}_{\text{mix}} = \sum_{i=1}^k x_i \bar{G}_i \quad (1)$$

and show that

$$\tilde{G}_{\text{mix}} = \sum_{i=1}^k x_i \lambda_i + RT \sum_{i=1}^k x_i \ln \hat{f}_i. \quad (2)$$

- (b) The fugacity of the mixture, f_{mix} , is defined by

$$\tilde{G}_{\text{mix}} = \lambda_{\text{mix}} + RT \ln f_{\text{mix}} \quad (3)$$

with the following constraint:

$$\phi_{\text{mix}} = f_{\text{mix}}/P \rightarrow 1 \quad \text{as } P \rightarrow 0. \quad (4)$$

Equate Eqns (2) and (3) to get

$$\lambda_{\text{mix}} + RT \ln f_{\text{mix}} = \sum_{i=1}^k x_i \lambda_i + RT \sum_{i=1}^k x_i \ln \hat{f}_i. \quad (5)$$

Since Eqn (5) is valid at all pressures, evaluate Eqn (5) as $P \rightarrow 0$ to conclude that

$$\lambda_{\text{mix}} = \sum_{i=1}^k x_i \lambda_i + \sum_{i=1}^k x_i \ln x_i. \quad (6)$$

(c) Show that the use of Eqn (6) in Eqn (5) leads to

$$\ln \phi_{\text{mix}} = \sum_{i=1}^k x_i \ln \hat{\phi}_i \quad (7)$$

or

$$\phi_{\text{mix}} = \prod_{i=1}^k \hat{\phi}_i^{x_i}. \quad (8)$$

Note that the form of Eqn (3) is similar to the one given for a pure component, Eqn (5.2-6). Therefore, the pure component fugacity coefficient equations given in Chapter 5 can also be used to express the overall fugacity coefficient of mixtures.

7.4 Equation (7) of Problem 7.3 suggests that the term $\ln \hat{\phi}_i$ can be interpreted as the partial molar property of $\ln \phi_{\text{mix}}$, i.e.

$$\ln \hat{\phi}_i = \left[\frac{\partial}{\partial n_i} (n \ln \phi_{\text{mix}}) \right]_{T, P, n_{j \neq i}}. \quad (1)$$

Therefore, for a binary mixture, show that the application of Eqns (6.2-24) and (6.2-25) gives

$$\ln \hat{\phi}_1 = \ln \phi_{\text{mix}} - x_2 \frac{d \ln \phi_{\text{mix}}}{dx_2}, \quad (2)$$

$$\ln \hat{\phi}_2 = \ln \phi_{\text{mix}} - x_1 \frac{d \ln \phi_{\text{mix}}}{dx_1}. \quad (3)$$

(a) Note that Eqn (5.3-5) can be extended to mixtures as

$$\ln \phi_{\text{mix}} = \frac{B_{\text{mix}} P}{RT}. \quad (4)$$

(b) Use Eqn (4) in Eqns (2) and (3) to obtain

$$\ln \hat{\phi}_1 = \frac{P}{RT} [B_{11} + y_2^2 (2B_{12} - B_{11} - B_{22})], \quad (5)$$

$$\ln \hat{\phi}_2 = \frac{P}{RT} [B_{22} + y_1^2 (2B_{12} - B_{11} - B_{22})]. \quad (6)$$

7.5 In the case of a pressure-explicit equation of state, the use of Eqn (1) in Problem 7.4, i.e.

$$\ln \hat{\phi}_i = \left[\frac{\partial}{\partial n_i} (n \ln \phi_{\text{mix}}) \right]_{T, P, n_{j \neq i}} \quad (1)$$

is not practical in the calculation of component fugacities. An alternative expression can be developed as follows:

(a) In general

$$n \ln \phi_{\text{mix}} = f(T, P, V_{\text{mix}}, n_1, n_2, \dots, n_k). \quad (2)$$

In Eqn (1), temperature, pressure, and the number of moles of each component other than i are kept constant during differentiation. Therefore, Eqn (2) can be expressed as

$$n \ln \phi_{\text{mix}} = g(V_{\text{mix}}, n_i). \quad (3)$$

(b) First express the total differential of Eqn (3) as

$$\begin{aligned} d(n \ln \phi_{\text{mix}}) &= \left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial V_{\text{mix}}} \right]_{T, P, n_j} dV_{\text{mix}} \\ &+ \left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial n_i} \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} dn_i \end{aligned} \quad (4)$$

and then differentiate Eqn (4) with respect to n_i , by keeping T , P , and $n_{j \neq i}$ constant, to obtain

$$\begin{aligned} \left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial n_i} \right]_{T, P, n_{j \neq i}} &= \left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial V_{\text{mix}}} \right]_{T, P, n_j} \underbrace{\left(\frac{\partial V_{\text{mix}}}{\partial n_i} \right)_{T, P, n_{j \neq i}}}_{\bar{V}_i} \\ &+ \left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial n_i} \right]_{T, P, V_{\text{mix}}, n_{j \neq i}}. \end{aligned} \quad (5)$$

(c) Note that Eqn (5.3-1) can be extended to mixtures as

$$\ln \phi_{\text{mix}} = \int_0^P \left(\frac{\tilde{V}_{\text{mix}}}{RT} - \frac{1}{P} \right) dP \quad \text{constant } T \text{ \& } n_j. \quad (6)$$

Hence, conclude that

$$\left[\frac{\partial (n \ln \phi_{\text{mix}})}{\partial V_{\text{mix}}} \right]_{T, P, n_j} = 0. \quad (7)$$

(d) Combine Eqns (1), (5), and (7) to obtain

$$\ln \hat{\phi}_i = \left[\frac{\partial(n \ln \phi_{\text{mix}})}{\partial n_i} \right]_{T, P, V_{\text{mix}}, n_{j \neq i}}. \quad (8)$$

7.6 From Eqn (5.3-9), the overall fugacity coefficient of the mixture obeying the van der Waals equation of state is given by

$$\ln \phi_{\text{mix}} = Z_{\text{mix}} - 1 - \ln(Z_{\text{mix}} - B_{\text{mix}}) - \frac{A_{\text{mix}}}{Z_{\text{mix}}}. \quad (1)$$

(a) Show that the substitution of Eqn (1) into Eqn (8) of Problem 7.5 and rearrangement give

$$\begin{aligned} \ln \hat{\phi}_i = & \left[\frac{\partial(n Z_{\text{mix}})}{\partial n_i} \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} - 1 - \left[\frac{\partial}{\partial n_i} \left(\frac{n^2 A_{\text{mix}}}{n Z_{\text{mix}}} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} \\ & - \ln(Z_{\text{mix}} - B_{\text{mix}}) - \frac{n}{Z_{\text{mix}} - B_{\text{mix}}} \left\{ \left[\frac{\partial}{\partial n_i} \left(\frac{n Z_{\text{mix}}}{n} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} \right. \\ & \left. - \left[\frac{\partial}{\partial n_i} \left(\frac{n B_{\text{mix}}}{n} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} \right\}. \end{aligned} \quad (2)$$

(b) Show that

$$\left[\frac{\partial(n Z_{\text{mix}})}{\partial n_i} \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} = 0, \quad (3)$$

$$\left[\frac{\partial}{\partial n_i} \left(\frac{n^2 A_{\text{mix}}}{n Z_{\text{mix}}} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} = \frac{2 \sum_{j=1}^k x_j A_{ij}}{Z_{\text{mix}}}, \quad (4)$$

$$\left[\frac{\partial}{\partial n_i} \left(\frac{n Z_{\text{mix}}}{n} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} = -\frac{Z_{\text{mix}}}{n}, \quad (5)$$

$$\left[\frac{\partial}{\partial n_i} \left(\frac{n B_{\text{mix}}}{n} \right) \right]_{T, P, V_{\text{mix}}, n_{j \neq i}} = \frac{B_i - B_{\text{mix}}}{n}. \quad (6)$$

Substitute Eqns (3)–(6) into Eqn (2) to obtain Eqn (7.5-13).

7.7 From Eqn (5.3-9), the overall fugacity coefficient of the mixture obeying the Redlich-Kwong equation of state is given by

$$\ln \phi_{\text{mix}} = Z_{\text{mix}} - 1 - \ln(Z_{\text{mix}} - B_{\text{mix}}) - \frac{A_{\text{mix}}}{B_{\text{mix}}} \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right). \quad (1)$$

- (a) Show that the substitution of Eqn (1) into Eqn (8) of Problem 7.5 and rearrangement give

$$\begin{aligned} \ln \hat{\phi}_i = & \left[\frac{\partial(nZ_{\text{mix}})}{\partial n_i} \right]_{T,P,V_{\text{mix}},n_{j \neq i}} - 1 - \ln(Z_{\text{mix}} - B_{\text{mix}}) \\ & - \frac{n}{Z_{\text{mix}} - B_{\text{mix}}} \left\{ \left[\frac{\partial}{\partial n_i} \left(\frac{nZ_{\text{mix}}}{n} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}} \right. \\ & \left. - \left[\frac{\partial}{\partial n_i} \left(\frac{nB_{\text{mix}}}{n} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}} \right\} \\ & - \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right) \left[\frac{\partial}{\partial n_i} \left(\frac{n^2 A_{\text{mix}}}{nB_{\text{mix}}} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}} \\ & - \frac{nA_{\text{mix}}Z_{\text{mix}}}{B_{\text{mix}}(Z_{\text{mix}} + B_{\text{mix}})} \left[\frac{\partial}{\partial n_i} \left(\frac{nB_{\text{mix}}}{nZ_{\text{mix}}} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}}. \quad (2) \end{aligned}$$

- (b) Show that

$$\left[\frac{\partial}{\partial n_i} \left(\frac{n^2 A_{\text{mix}}}{nB_{\text{mix}}} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}} = \frac{2 \sum_{j=1}^k x_j A_{ij}}{B_{\text{mix}}} - \frac{A_{\text{mix}} B_i}{B_{\text{mix}}^2}, \quad (3)$$

$$\left[\frac{\partial}{\partial n_i} \left(\frac{nB_{\text{mix}}}{nZ_{\text{mix}}} \right) \right]_{T,P,V_{\text{mix}},n_{j \neq i}} = \frac{B_i}{nZ_{\text{mix}}}. \quad (4)$$

Substitute Eqns (3) and (4) together with Eqns (3), (5), and (6) of Problem 7.6 into Eqn (2) to obtain

$$\begin{aligned} \ln \hat{\phi}_i = & \frac{B_i}{Z_{\text{mix}} - B_{\text{mix}}} - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{B_i}{Z_{\text{mix}} + B_{\text{mix}}} \right) - \ln(Z_{\text{mix}} - B_{\text{mix}}) \\ & - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k x_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}} \right). \quad (5) \end{aligned}$$

- (c) Combining the first two terms on the right-hand side of Eqn (5) and making use of the Redlich-Kwong equation of state, i.e.

$$Z_{\text{mix}}^3 - Z_{\text{mix}}^2 + (A_{\text{mix}} - B_{\text{mix}} - B_{\text{mix}}^2)Z_{\text{mix}} - A_{\text{mix}}B_{\text{mix}} = 0 \quad (6)$$

show that Eqn (5) reduces to Eqn (7.5-14).

7.8 Twenty-eight moles of methane (1), 16 mol of ethane (2), and 36 mol of carbon dioxide (3) are mixed in a rigid tank at 320 K and 10 bar. Calculate the fugacities of each component in the gas mixture using

- (a) The Lewis-Randall rule if pure gases obey the Peng-Robinson equation of state.
- (b) The virial equation of state with

$$B_{11} = -35.17 \text{ cm}^3/\text{mol}, \quad B_{22} = -159.42 \text{ cm}^3/\text{mol}, \quad B_{33} = -104.54 \text{ cm}^3/\text{mol}, \\ B_{12} = -76.71 \text{ cm}^3/\text{mol}, \quad B_{13} = -54.02 \text{ cm}^3/\text{mol}, \quad B_{23} = -105.65 \text{ cm}^3/\text{mol}.$$

(Answer: (a) $\hat{f}_1 = 3.440 \text{ bar}$, $\hat{f}_2 = 1.867 \text{ bar}$, $\hat{f}_3 = 4.305 \text{ bar}$ and
(b) $\hat{f}_1 = 3.467 \text{ bar}$, $\hat{f}_2 = 1.902 \text{ bar}$, $\hat{f}_3 = 4.341 \text{ bar}$)

7.9 A ternary mixture consisting of 50 mol% methane (1), 30% ethane (2), and 20% propane (3) is at 380 K and 200 bar. The mixture is represented by the van der Waals equation of state.

- (a) Estimate the fugacities of each component in the mixture.
- (b) Repeat the calculations for an ideal mixture.

(Answer: (a) $\hat{f}_1 = 92.132 \text{ bar}$, $\hat{f}_2 = 31.860 \text{ bar}$, $\hat{f}_3 = 14.875 \text{ bar}$ and
(b) $\hat{f}_1 = 86.362 \text{ bar}$, $\hat{f}_2 = 31.209 \text{ bar}$, $\hat{f}_3 = 13.572 \text{ bar}$)

7.10 A gaseous mixture consisting of 15 mol% methane (1), 25% ethylene (2), 20% ethane (3), and 40% propane (4) is at 400 K and 150 bar. Calculate the component fugacities in the mixture using the Peng-Robinson equation of state with

$$k_{12} = 0.022, \quad k_{13} = -0.003, \quad k_{14} = 0.016, \\ k_{23} = 0.010, \quad k_{24} = 0, \quad k_{34} = 0.001.$$

(Answer: $\hat{f}_1 = 25.935 \text{ bar}$, $\hat{f}_2 = 27.508 \text{ bar}$, $\hat{f}_3 = 19.058 \text{ bar}$, $\hat{f}_4 = 24.077 \text{ bar}$)

7.11 A rigid tank is divided into two parts by a rigid semi-permeable membrane through which carbon dioxide can pass. One side contains a gaseous mixture consisting of 70 mol% carbon dioxide (1) and 30% methane (2) at 323 K and 60 bar. The other side contains pure carbon dioxide at 323 K. Estimate the pressure of pure carbon dioxide so as to stop diffusion of carbon dioxide through the membrane if the gas phase obeys the Peng-Robinson equation of state with $k_{12} = 0.09$.

(Answer: 38.3 bar)

7.12 This is the continuation of Problem 4.3 for a multicomponent system.

- (a) For a multicomponent system, show that Eqn (3) of Problem 4.3 takes the form

$$dG = V dP - S dT + mg dz + \sum_{i=1}^k (\bar{G}_i + M_i g z) dn_i. \quad (1)$$

- (b) Under equilibrium conditions, i.e. $dG = 0$, conclude that

$$dT = 0 \Rightarrow T \text{ is uniform}, \quad (2)$$

$$V dP + mg dz = 0 \Rightarrow \frac{dP}{dz} = -\rho g, \quad (3)$$

$$\bar{G}_i + M_i g z = 0. \quad (4)$$

Keep in mind that the distance z is measured in the direction opposite to gravity.

- (c) Differentiate Eqn (4) by keeping temperature constant to obtain

$$d\bar{G}_i = RT d \ln \hat{f}_i = -M_i g dz, \quad (5)$$

which is known as the *Gibbs sedimentation expression* (Firoozabadi, 1999).

- (d) Show that the integration of Eqn (5) gives

$$\ln \hat{f}_i = -\frac{M_i g z}{RT} + C, \quad (6)$$

where C is an integration constant. If the temperature is uniform in a well of depth H , show that Eqn (6) takes the form

$$\frac{\hat{f}_{i,\text{bottom}}}{\hat{f}_{i,\text{top}}} = \exp\left(\frac{M_i g H}{RT}\right). \quad (7)$$

- (e) The composition at the top of a natural gas well of depth 3 km is 90 mol% methane (1) and 10% ethane (2). The temperature within the well is uniform at 310 K and the pressure at the top of the well is 120 bar. It is required to estimate the composition and the pressure at the bottom of the well.

- (i) Assuming natural gas to be an ideal gas mixture, show that Eqn (7) reduces to

$$\left(\frac{y_1}{y_2}\right)_{\text{bottom}} = \left(\frac{y_1}{y_2}\right)_{\text{top}} \exp\left[\frac{(M_1 - M_2)gH}{RT}\right]. \quad (8)$$

First use Eqn (8) with $y_1 + y_2 = 1$ and estimate the composition at the bottom of the well. Then calculate the pressure at the bottom of the well.

- (ii) Assuming natural gas to be an ideal mixture, show that Eqn (7) takes the form

$$y_{1,\text{bottom}} = y_{1,\text{top}} \left(\frac{f_{1,\text{top}}}{f_{1,\text{bottom}}} \right) \exp \left(\frac{M_1 g H}{RT} \right), \quad (9)$$

$$y_{2,\text{bottom}} = y_{2,\text{top}} \left(\frac{f_{2,\text{top}}}{f_{2,\text{bottom}}} \right) \exp \left(\frac{M_2 g H}{RT} \right). \quad (10)$$

First assume the pressure at the bottom of the well and calculate pure component fugacities using the Peng-Robinson equation of state. Then determine the mole fractions using Eqns (9) and (10). Repeat this procedure until $y_1 + y_2 = 1$.

(Answer: (e) (i) $y_1 = 0.8847$, $P = 146.5$ bar; (ii) $y_1 = 0.873$, $P = 155$ bar)

Problems Related to Section 7.8

- 7.13** Since the sublimation pressure of solids is very small, its determination by experimental techniques is often difficult. One way of determining the sublimation pressure is to use the solubility data. Mendes *et al.* (1999) reported that the solubility of β -carotene (2) in carbon dioxide (1) at 313.15 K and 201 bar is $y_2 = 13 \times 10^{-8}$. Using the following data for β -carotene

$$T_c = 1177 \text{ K}, \quad P_c = 12.4 \text{ bar}, \quad \omega = 1.04, \quad \tilde{V}^S = 536.5 \text{ cm}^3/\text{mol}$$

and the Peng-Robinson equation of state with $k_{12} = 0.1013$, estimate the sublimation pressure of β -carotene at 313.15 K. Mendes *et al.* reported the sublimation pressure as 1.72×10^{-19} bar.

(Answer: 1.68×10^{-19} bar)

- 7.14** Determine the solubility of salicylic acid (2) in supercritical carbon dioxide (1) at 313 K and 192.5 bar. Assume that the vapor phase is represented by the Peng-Robinson equation of state with $k_{12} = 0.233$. For salicylic acid

$$T_c = 864.79 \text{ K}, \quad P_c = 50.24 \text{ bar}, \quad \omega = 0.785.$$

At 313 K, the sublimation pressure and the molar volume of solid salicylic acid are 2.86×10^{-6} bar and $95.70 \text{ cm}^3/\text{mol}$, respectively. The experimental value reported by Gurdial and Foster (1991) is $y_2 = 3.64 \times 10^{-4}$.

(Answer: 3.52×10^{-4})

7.15 A natural gas mainly consists of methane. Cryogenic processes are used in natural gas plants for the separation of methane from heavier alkanes. The presence of carbon dioxide, due to its relatively high triple point, in a methane stream causes freezing, leading to clogging of pipes and other operational problems. Consider a mixture of methane (1) and carbon dioxide (2) in which the mole fraction of CO_2 is 0.1. If the pressure is 50 bar, estimate the temperature at which carbon dioxide freezes out using the Peng-Robinson equation of state. The following data are provided by [ZareNezhad and Eggeman \(2006\)](#):

$$\rho_2^S = 1562 \text{ kg/m}^3,$$

$$P_2^{\text{sub}} = 9.44 \times 10^6 \exp\left(-\frac{3108.2}{T}\right),$$

where P_2^{sub} is in bar and T is in K. The interaction parameter is given as a function of temperature in the form

$$k_{12} = 0.0998 + \frac{5.4835}{T} - \frac{36.134}{T^2}.$$

Repeat the calculations when $y_2 = 0.01$. For more details on the problem, see [Eggeman and Chafin \(2005\)](#).

(Answer: 189.8 K, 152 K)

7.16 Consider vapor-liquid equilibrium between nitrogen (1) and water (2) in a rigid container at low temperature. The vapor phase nonideality is represented by the virial equation of state. Note that the following assumptions are plausible for this problem:

- Solubility of nitrogen in liquid water is negligible, i.e. $x_2 \simeq 1$.
- Since the temperature is low, then the vapor pressure of water is also low, i.e. $y_2 \ll y_1$.

Under these circumstances, show that the equality of water fugacities in the liquid and vapor phases leads to

$$y_2 = \frac{P_2^{\text{vap}}}{P} \exp\left\{\frac{[(\tilde{V}_2^L + B_{11}) - 2B_{12}]P}{RT}\right\}.$$

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Chapter 8

Excess Mixture Properties and Activity Coefficients

The behavior of most liquid and solid mixtures cannot be represented by the cubic equations of state. For this reason, it is necessary to define another quantity, called the *activity coefficient*, to express fugacities of components in liquid and solid mixtures. In the literature, various activity coefficient models are used in phase equilibrium calculations. The purpose of this chapter is to introduce such models and to show how to evaluate the parameters appearing in these models.

8.1 Property Changes on Mixing for an Ideal Mixture

Property change on mixing per mole, $\Delta\tilde{\varphi}_{\text{mix}}$, is defined by Eqn (6.3-4), i.e.

$$\Delta\tilde{\varphi}_{\text{mix}} = \sum_{i=1}^k x_i(\bar{\varphi}_i - \tilde{\varphi}_i). \quad (8.1-1)$$

For an ideal mixture, Eqn (8.1-1) takes the form

$$\Delta\tilde{\varphi}_{\text{mix}}^{IM} = \sum_{i=1}^k x_i(\bar{\varphi}_i^{IM} - \tilde{\varphi}_i). \quad (8.1-2)$$

The properties of an ideal mixture are given in Section 7.4 as

$$\bar{V}_i^{IM} = \tilde{V}_i \quad \text{and} \quad \bar{H}_i^{IM} = \tilde{H}_i. \quad (8.1-3)$$

The use of Eqn (8.1-3) in Eqn (8.1-2) gives

$$\Delta\tilde{V}_{\text{mix}}^{IM} = 0 \quad \text{and} \quad \Delta\tilde{H}_{\text{mix}}^{IM} = 0, \quad (8.1-4)$$

indicating that volume and enthalpy changes on mixing are both zero. For an ideal mixture this is an expected result because not only are the sizes of the molecules equal to each other, but also the interactions between unlike molecules are equal to those between like molecules.

To obtain an expression for the Gibbs energy change on mixing for an ideal mixture, it is necessary to relate $\overline{G}_i^{IM}$ to $\tilde{G}_i$. The partial molar Gibbs energy of component i in an ideal mixture is given by

$$\overline{G}_i^{IM} = \lambda_i(T) + RT \ln \hat{f}_i^{IM} = \lambda_i(T) + RT \ln(x_i f_i). \quad (8.1-5)$$

The molar Gibbs energy of pure i is expressed by Eqn (5.2-5), i.e.

$$\tilde{G}_i = \lambda_i(T) + RT \ln f_i. \quad (8.1-6)$$

Subtraction of Eqn (8.1-6) from Eqn (8.1-5) leads to

$$\overline{G}_i^{IM} - \tilde{G}_i = RT \ln x_i. \quad (8.1-7)$$

Thus, the use of Eqn (8.1-7) in Eqn (8.1-2) results in

$$\Delta \tilde{G}_{\text{mix}}^{IM} = RT \sum_{i=1}^k x_i \ln x_i. \quad (8.1-8)$$

For a binary mixture of components 1 and 2, a representative plot of $\Delta \tilde{G}_{\text{mix}}^{IM}$ versus x_1 is shown in Fig. 8.1. Since $x_i < 1$, it follows from Eqn (8.1-8) that $\Delta \tilde{G}_{\text{mix}}^{IM} < 0$. In other words, the Gibbs energy of an ideal mixture is always less than the summation of the Gibbs energies of the unmixed pure components. Thus, upon mixing at a specified temperature and pressure, components 1 and 2 form a **stable**¹ ideal mixture.

To obtain an expression for the entropy change on mixing for an ideal mixture, note that

$$\Delta \tilde{G}_{\text{mix}}^{IM} = \underbrace{\Delta \tilde{H}_{\text{mix}}^{IM}}_0 - T \Delta \tilde{S}_{\text{mix}}^{IM}. \quad (8.1-9)$$

Substitution of Eqn (8.1-8) into Eqn (8.1-9) yields

$$\Delta \tilde{S}_{\text{mix}}^{IM} = -R \sum_{i=1}^k x_i \ln x_i. \quad (8.1-10)$$

For a binary mixture of components 1 and 2, a representative plot of $\Delta \tilde{S}_{\text{mix}}^{IM}$ versus x_1 is also shown in Fig. 8.1. Mixing increases disorder and thus $\Delta \tilde{S}_{\text{mix}}^{IM} > 0$.

¹In other words, components 1 and 2 are completely miscible in each other. Miscibility of components will be discussed in Chapter 11 in detail.

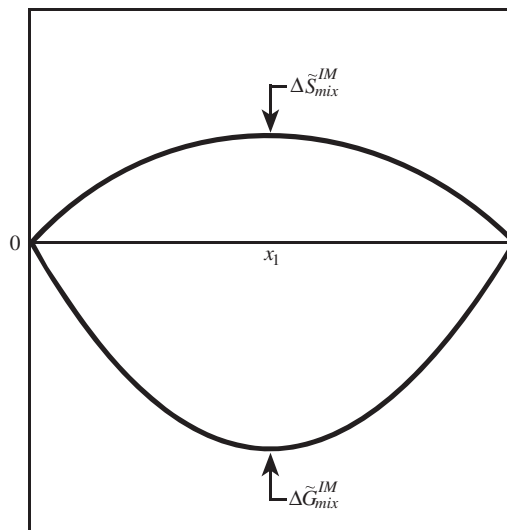


Figure 8.1: Representative plots of $\Delta\tilde{G}_{\text{mix}}^{\text{IM}}$ and $\Delta\tilde{S}_{\text{mix}}^{\text{IM}}$ for a binary system.

8.2 Excess Properties

In the literature, it is customary to split the property change on mixing into two terms as

$$\Delta\varphi_{\text{mix}} = \Delta\varphi_{\text{mix}}^{\text{IM}} + \varphi^{\text{ex}}, \quad (8.2-1)$$

where φ^{ex} is called the *excess property*. The use of Eqn (8.2-1) in Eqn (6.3-2) leads to

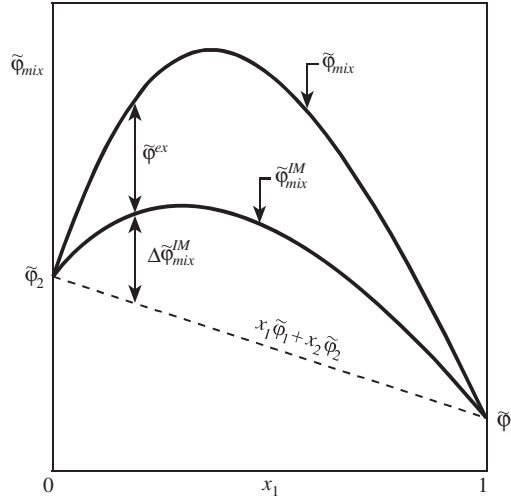
$$\varphi_{\text{mix}} = \underbrace{\sum_{i=1}^k n_i \tilde{\varphi}_i}_{\varphi_{\text{mix}}^{\text{IM}}} + \Delta\varphi_{\text{mix}}^{\text{IM}} + \varphi^{\text{ex}}. \quad (8.2-2)$$

Thus, an excess property is the difference between the actual property and the property for an ideal mixture at the same temperature, pressure, and composition of the mixture, i.e.

$$\varphi^{\text{ex}}(T, P, x_i) = \varphi_{\text{mix}}(T, P, x_i) - \varphi_{\text{mix}}^{\text{IM}}(T, P, x_i). \quad (8.2-3)$$

By definition, excess properties are zero for an ideal mixture. They are only defined for mixtures² and have no physical meaning for pure fluids.

²This is the reason why an excess property is designated as φ^{ex} and not as $\varphi_{\text{mix}}^{\text{ex}}$.

Figure 8.2: Representative plot of $\tilde{\varphi}_{mix}$ for a binary system.

Dividing each term in Eqn (8.2-2) by the total number of moles gives the molar property of the mixture as

$$\tilde{\varphi}_{mix} = \underbrace{\sum_{i=1}^k x_i \tilde{\varphi}_i}_{\tilde{\varphi}_{mix}^{IM}} + \tilde{\varphi}^{ex}. \quad (8.2-4)$$

For a binary mixture, a representative plot of $\tilde{\varphi}_{mix}$ as a function of composition is shown in Fig. 8.2.

The relationship between the excess property and the property change on mixing can be found from Eqn (8.2-1) as

$$\tilde{V}^{ex} = \Delta\tilde{V}_{mix} - \underbrace{\Delta\tilde{V}_{mix}^{IM}}_0 = \Delta\tilde{V}_{mix}, \quad (8.2-5)$$

$$\tilde{H}^{ex} = \Delta\tilde{H}_{mix} - \underbrace{\Delta\tilde{H}_{mix}^{IM}}_0 = \Delta\tilde{H}_{mix}, \quad (8.2-6)$$

$$\tilde{G}^{ex} = \Delta\tilde{G}_{mix} - \Delta\tilde{G}_{mix}^{IM} = \Delta\tilde{G}_{mix} - RT \sum_{i=1}^k x_i \ln x_i, \quad (8.2-7)$$

$$\tilde{S}^{ex} = \Delta\tilde{S}_{mix} - \Delta\tilde{S}_{mix}^{IM} = \Delta\tilde{S}_{mix} + R \sum_{i=1}^k x_i \ln x_i. \quad (8.2-8)$$

Partial molar excess functions are defined in a manner analogous to that used for partial molar properties, i.e.

$$\bar{\varphi}_i^{\text{ex}} = \left(\frac{\partial \varphi^{\text{ex}}}{\partial n_i} \right)_{T,P,n_{j \neq i}}. \quad (8.2-9)$$

Thus, substitution of Eqn (8.2-3) into Eqn (8.2-9) gives

$$\begin{aligned} \bar{\varphi}_i^{\text{ex}} &= \left(\frac{\partial \varphi_{\text{mix}}}{\partial n_i} \right)_{T,P,n_{j \neq i}} - \left(\frac{\partial \varphi_{\text{mix}}^{\text{IM}}}{\partial n_i} \right)_{T,P,n_{j \neq i}} \\ &= \bar{\varphi}_i - \bar{\varphi}_i^{\text{IM}}. \end{aligned} \quad (8.2-10)$$

8.2.1 Relations Between Excess Properties

For a single phase, multicomponent open system the differential change in the Gibbs energy is given by Eqn (7.1-13), i.e.

$$dG = VdP - SdT + \sum_{i=1}^k \bar{G}_i dn_i. \quad (8.2-11)$$

In terms of the excess functions, Eqn (8.2-11) is expressed as

$$dG^{\text{ex}} = V^{\text{ex}}dP - S^{\text{ex}}dT + \sum_{i=1}^k \bar{G}_i^{\text{ex}} dn_i. \quad (8.2-12)$$

Dividing each term by RT yields

$$\frac{1}{RT}dG^{\text{ex}} = \frac{V^{\text{ex}}}{RT}dP - \frac{S^{\text{ex}}}{RT}dT + \sum_{i=1}^k \frac{\bar{G}_i^{\text{ex}}}{RT}dn_i. \quad (8.2-13)$$

The use of the identity

$$d\left(\frac{G^{\text{ex}}}{RT}\right) = \frac{1}{RT}dG^{\text{ex}} - \frac{G^{\text{ex}}}{RT^2}dT \quad (8.2-14)$$

in Eqn (8.2-13) leads to

$$d\left(\frac{G^{\text{ex}}}{RT}\right) = \frac{V^{\text{ex}}}{RT}dP - \frac{H^{\text{ex}}}{RT^2}dT + \sum_{i=1}^k \frac{\bar{G}_i^{\text{ex}}}{RT}dn_i, \quad (8.2-15)$$

which is known as the *fundamental excess-property relation*.

Differentiation of Eqn (8.2-15) with respect to pressure by keeping temperature and the mole numbers of all components constant gives

$$\left[\frac{\partial(G^{\text{ex}}/RT)}{\partial P} \right]_{T, n_j} = \frac{V^{\text{ex}}}{RT} \quad (8.2-16)$$

or

$$\left(\frac{\partial \tilde{G}^{\text{ex}}}{\partial P} \right)_{T, x_j} = \tilde{V}^{\text{ex}}. \quad (8.2-17)$$

Differentiation of Eqn (8.2-15) with respect to temperature by keeping pressure and the mole numbers of all components constant gives

$$\left[\frac{\partial(G^{\text{ex}}/RT)}{\partial T} \right]_{P, n_j} = -\frac{H^{\text{ex}}}{RT^2} \quad (8.2-18)$$

or

$$\left[\frac{\partial}{\partial T} \left(\frac{\tilde{G}^{\text{ex}}}{T} \right) \right]_{P, x_j} = -\frac{\tilde{H}^{\text{ex}}}{T^2}, \quad (8.2-19)$$

which is simply the extension of the Gibbs-Helmholtz equation, Eqn (5.7-8), to excess quantities.

Once molar excess Gibbs energy is known, molar excess volume (or volume change on mixing) and molar excess enthalpy (or enthalpy change on mixing) can be evaluated by using Eqns (8.2-17) and (8.2-19), respectively. On the other hand, $\tilde{U}^{\text{ex}}$ and $\tilde{S}^{\text{ex}}$ can be calculated from the following equations:

$$\tilde{U}^{\text{ex}} = \tilde{H}^{\text{ex}} - P\tilde{V}^{\text{ex}}, \quad (8.2-20)$$

$$\tilde{S}^{\text{ex}} = \frac{\tilde{H}^{\text{ex}}}{T} - \frac{\tilde{G}^{\text{ex}}}{T}. \quad (8.2-21)$$

Example 8.1 A stream of liquid 1 flowing at 5 mol/s and a stream of liquid 2 flowing at 3 mol/s, both at 298 K, are mixed in a steady flow process. For this mixture the molar excess Gibbs energy is given by

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2$$

in which the parameter A is given as a function of temperature in the form

$$A = 14.8 - \frac{1000}{T} - 2 \ln T,$$

where T is in K.

- (a) How much heat must be added to or removed from the mixer so as to maintain the mixing process isothermal?
- (b) Suppose that the entering temperature of liquid 2 is increased from 298 K to 330 K and the mixer is adiabatic. Determine the temperature of the stream leaving the mixer under these conditions. Take pure component heat capacities as $\tilde{C}_{P_1} = 35 \text{ J/mol K}$ and $\tilde{C}_{P_2} = 50 \text{ J/mol K}$.

Solution

(a) Considering the mixer as a system, the mass and energy balances are

$$\dot{n}_1 + \dot{n}_2 = \dot{n}_{\text{mix}}, \quad (1)$$

and

$$\Delta \dot{H} = \dot{Q} + \underbrace{\dot{W}_s}_{\sim 0} \Rightarrow \dot{Q} = \dot{n}_{\text{mix}} \tilde{H}_{\text{mix}} - (\dot{n}_1 \tilde{H}_1 + \dot{n}_2 \tilde{H}_2). \quad (2)$$

The molar enthalpy of the mixture is given by

$$\tilde{H}_{\text{mix}} = x_1 \tilde{H}_1 + x_2 \tilde{H}_2 + \Delta \tilde{H}_{\text{mix}}. \quad (3)$$

Substitution of Eqns (1) and (2) into Eqn (3) and using the relationship

$$\dot{n}_{\text{mix}} x_i = \dot{n}_i \quad i = 1, 2 \quad (4)$$

yield

$$\begin{aligned} \dot{Q} &= \dot{n}_1 \tilde{H}_1 + \dot{n}_2 \tilde{H}_2 + (\dot{n}_1 + \dot{n}_2) \Delta \tilde{H}_{\text{mix}} - (\dot{n}_1 \tilde{H}_1 + \dot{n}_2 \tilde{H}_2) \\ &= (\dot{n}_1 + \dot{n}_2) \Delta \tilde{H}_{\text{mix}} = (\dot{n}_1 + \dot{n}_2) \tilde{H}^{\text{ex}}. \end{aligned} \quad (5)$$

The quantity $\tilde{H}^{\text{ex}}$ can be determined from Eqn (8.2-19) as

$$\tilde{H}^{\text{ex}} = -RT^2 x_1 x_2 \frac{dA}{dT} = R x_1 x_2 (2T - 1000). \quad (6)$$

Substitution of the numerical values gives

$$\tilde{H}^{\text{ex}} = (8.314) \left(\frac{5}{8} \right) \left(\frac{3}{8} \right) [(2)(298) - 1000] = -787.2 \text{ J/mol}.$$

Thus, the amount of heat that must be removed from the system is calculated as

$$\dot{Q} = (8)(-787.2) = -6298 \text{ W}.$$

(b) Since the mixer is adiabatic, i.e. $\dot{Q} = 0$, Eqn (2) becomes

$$\Delta \dot{H} = \dot{n}_{\text{mix}} \tilde{H}_{\text{mix}}|_T - (\dot{n}_1 \tilde{H}_1|_{298} + \dot{n}_2 \tilde{H}_2|_{330}) = 0, \quad (7)$$

where T represents the temperature of the stream leaving the mixer. The molar enthalpy of the mixture, Eqn (3), is given by

$$\tilde{H}_{\text{mix}}|_T = x_1 \tilde{H}_1|_T + x_2 \tilde{H}_2|_T + \Delta \tilde{H}_{\text{mix}}|_T. \quad (8)$$

Substitution of Eqns (1) and (8) into Eqn (7) and rearrangement lead to

$$\dot{n}_1 (\tilde{H}_1|_T - \tilde{H}_1|_{298}) + \dot{n}_2 (\tilde{H}_2|_T - \tilde{H}_2|_{330}) + (\dot{n}_1 + \dot{n}_2) \Delta \tilde{H}_{\text{mix}}|_T = 0. \quad (9)$$

Since $\Delta \tilde{H} = \tilde{C}_P \Delta T$ (assuming constant $\tilde{C}_P$) and $\Delta \tilde{H}_{\text{mix}} = \tilde{H}^{\text{ex}}$, Eqn (9) becomes

$$\dot{n}_1 \tilde{C}_{P1} (T - 298) + \dot{n}_2 \tilde{C}_{P2} (T - 330) + (\dot{n}_1 + \dot{n}_2) R x_1 x_2 (2T - 1000) = 0$$

or

$$(5)(35)(T - 298) + (3)(50)(T - 330) + (8)(8.314) \left(\frac{5}{8}\right) \left(\frac{3}{8}\right) (2T - 1000) = 0.$$

Solving for temperature gives $T = 329.2$ K.

8.3 Activity and Activity Coefficient

In a multicomponent mixture, the *activity* of component i , $\hat{a}_i$, is defined as

$$\hat{a}_i(T, P, x_i) = \frac{\hat{f}_i(T, P, x_i)}{\hat{f}_i^o(T^o, P^o, x_i^o)}, \quad (8.3-1)$$

where the superscript o implies the standard state. Like fugacity, activity is a measure of escaping tendency. Among the various standard states used in the literature, the most commonly used standard state is the pure component, i.e. $x_i^o = 1$, at the same temperature, pressure, and phase as the mixture. Under these conditions, Eqn (8.3-1) becomes

$$\hat{a}_i(T, P, x_i) = \frac{\hat{f}_i(T, P, x_i)}{f_i(T, P)}. \quad (8.3-2)$$

For an ideal mixture, application of the Lewis-Randall rule simplifies Eqn (8.3-2) to

$$\hat{a}_i^{IM} = x_i. \quad (8.3-3)$$

The *activity coefficient*, γ , is defined to account for the deviation from an ideal mixture behavior, i.e.

$$\hat{a}_i = \gamma_i x_i. \quad (8.3-4)$$

Comparison of Eqns (8.3-2) and (8.3-4) leads to

$$\gamma_i(T, P, x_i) = \frac{\hat{f}_i(T, P, x_i)}{\hat{f}_i^{IM}(T, P, x_i)} = \frac{\hat{f}_i(T, P, x_i)}{x_i f_i(T, P)}. \quad (8.3-5)$$

From Eqn (8.3-5), it is obvious that $\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$.

For a binary liquid mixture, representative plots of fugacity of component 1, $\hat{f}_1$, versus the mole fraction of 1, x_1 , are shown in Fig. 8.3 by solid curves. When the Lewis-Randall rule is applicable, fugacity of component 1 varies linearly with mole fraction as shown by a dashed straight line. When $x_i \rightarrow 1$, the solid curve coincides with the dashed straight line.

Since $\hat{f}_1 > \hat{f}_1^{IM}$ in Fig. 8.3a, the activity coefficient is greater than unity when $0 < x_1 < 1$. The higher the activity coefficient (or activity), the higher the escaping tendency of a component from the liquid phase. Thus, $\gamma_1 > 1$ can be regarded as a form of repulsion between unlike molecules in a liquid mixture. In other words, the like interactions are stronger than the unlike interactions. In Fig. 8.3b, on the other hand, $\hat{f}_1 < \hat{f}_1^{IM}$ and the activity coefficient is less than unity when $0 < x_1 < 1$. In this case, component 1 prefers to remain in the liquid phase since the unlike interactions are stronger than the like interactions.

8.3.1 Relationship Between Activity Coefficients and Excess Properties

Using Eqn (8.2-10), partial molar excess Gibbs energy is expressed as

$$\bar{G}_i^{\text{ex}} = \bar{G}_i - \bar{G}_i^{IM} = \lambda_i + RT \ln \hat{f}_i - (\lambda_i + RT \ln \hat{f}_i^{IM}) = RT \ln \left(\frac{\hat{f}_i}{\hat{f}_i^{IM}} \right). \quad (8.3-6)$$

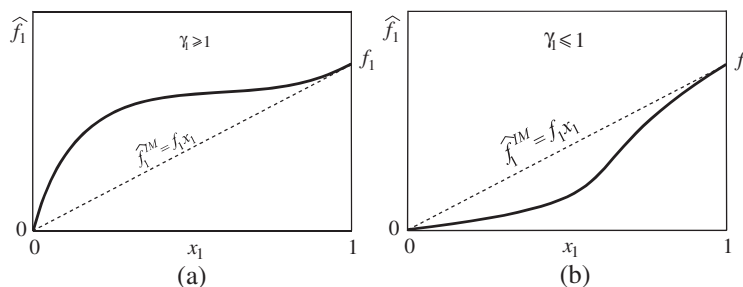


Figure 8.3: Variation of fugacity of component 1 in a liquid mixture with composition.

Substitution of Eqn (8.3-5) into Eqn (8.3-6) gives

$$\bar{G}_i^{\text{ex}} = RT \ln \gamma_i. \quad (8.3-7)$$

The molar excess Gibbs energy is given by

$$\tilde{G}^{\text{ex}} = \sum_{i=1}^k x_i \bar{G}_i^{\text{ex}}. \quad (8.3-8)$$

Substitution of Eqn (8.3-7) into Eqn (8.3-8) gives

$$\tilde{G}^{\text{ex}} = RT \sum_{i=1}^k x_i \ln \gamma_i. \quad (8.3-9)$$

Activity coefficients are dependent on temperature, pressure, and composition of the mixture. The limiting values of the activity coefficients are given as

$$\gamma_i = \begin{cases} 1 & x_i \rightarrow 1 \\ \gamma_i^\infty & x_i \rightarrow 0 \end{cases}, \quad (8.3-10)$$

where γ_i^∞ is the *activity coefficient of component i at infinite dilution*. It reflects the behavior of a single solute molecule completely surrounded by solvent.

In the literature, it is customary to plot either γ_i or $\ln \gamma_i$ versus composition under the conditions of constant temperature and pressure. Figure 8.4 shows how activity coefficients vary with composition in a binary mixture of ethanol (1) and 2,2,4-trimethylpentane (2) at 333.15 K (Hiaki *et al.*, 1994).

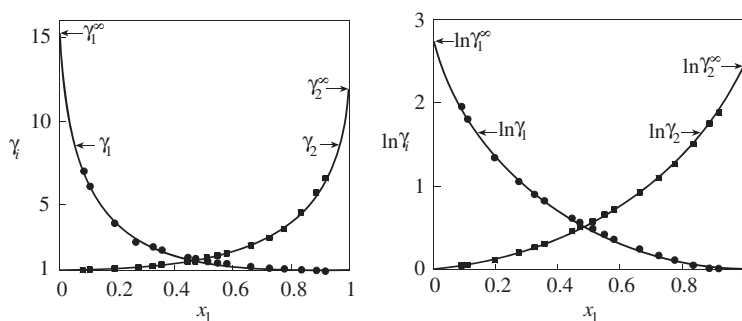


Figure 8.4: Activity coefficient-liquid composition diagram for ethanol (1) and 2,2,4-trimethylpentane (2) at 333.15 K (Hiaki *et al.*, 1994).

8.3.2 Estimation of Excess Properties

For liquid mixtures, experimental determinations of volume change on mixing and heat of mixing were explained in Section 6.3. Since $\Delta \tilde{V}_{\text{mix}} = \tilde{V}^{\text{ex}}$ and $\Delta \tilde{H}_{\text{mix}} = \tilde{H}^{\text{ex}}$, then experimental determinations of $\tilde{V}^{\text{ex}}$ and $\tilde{H}^{\text{ex}}$ are rather straightforward.

According to Eqn (8.3-9), evaluation of molar excess Gibbs energy requires experimental determination of activity coefficients. Activity coefficients are calculated from the experimental vapor-liquid equilibrium (VLE) measurements. At equilibrium, fugacities of each component in the vapor and liquid phases must be equal to each other, i.e.

$$\hat{f}_i^V(T, P, y_i) = \hat{f}_i^L(T, P, x_i) \quad (8.3-11)$$

or

$$\begin{aligned} \hat{\phi}_i^V(T, P, y_i)y_iP &= f_i^L(T, P)\gamma_i(T, P, x_i)x_i \\ &= P_i^{\text{vap}}\phi_i^V(T, P_i^{\text{vap}})\exp\left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT}\right]\gamma_i(T, P, x_i)x_i. \end{aligned} \quad (8.3-12)$$

When pressure is not very high, the Poynting correction factor is almost unity and the vapor phase can be considered an ideal gas mixture, i.e. $\hat{\phi}_i^V(T, P, y_i) = 1$ and $\phi_i^V(T, P_i^{\text{vap}}) = 1$. In other words, nonidealities associated with the vapor phase are neglected³ and the liquid phase nonideality is represented by the activity coefficient.

With such simplifications, Eqn (8.3-12) reduces to

$$\gamma_i = \frac{y_iP}{x_iP_i^{\text{vap}}}. \quad (8.3-13)$$

The procedure for expressing molar excess Gibbs energy as a function of composition can be outlined as follows:

- At a specified temperature, let the vapor and liquid phases of a multicomponent mixture reach equilibrium and record the pressure (isothermal data); or, at a specified pressure, let the vapor and liquid phases of a multicomponent mixture reach equilibrium and record the temperature (isobaric data),
- Take small samples from the liquid and vapor phases and determine x_i and y_i ,
- Using the Antoine equation calculate P_i^{vap} at the specified (or recorded) temperature,

³It should be kept in mind that the vapor phase nonideality cannot be neglected at all times, especially when pressure is moderate or high. See Example 8.7 and Problem 8.10 for further details.

- Use Eqn (8.3-13) and calculate γ_i values,
- Calculate $\tilde{G}^{\text{ex}}$ from Eqn (8.3-9),
- Develop a mathematical equation for expressing $\tilde{G}^{\text{ex}}$ as a function of x_i . Note that the expression for $\tilde{G}^{\text{ex}}$ should satisfy the fact that

$$\tilde{G}^{\text{ex}} = 0 \quad \text{when} \quad x_i = 1 \quad (i = 1, 2, \dots, k). \quad (8.3-14)$$

For a binary mixture, $\tilde{G}^{\text{ex}}/RT$ is generally expressed⁴ as a function of composition using a Redlich-Kister type expansion, i.e.

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \sum_{i=1}^n A_i (x_1 - x_2)^{i-1}. \quad (8.3-15)$$

Example 8.2 Huang *et al.* (2004) reported the following vapor-liquid equilibrium data for tetraethyl orthosilicate (1)–ethanol (2) mixtures at 400 mmHg

$T(\text{K})$	x_1	y_1	$T(\text{K})$	x_1	y_1
336.5	0.0000	0.0000	378.2	0.9061	0.2211
337.9	0.1622	0.0144	398.7	0.9668	0.4953
339.5	0.2721	0.0205	410.0	0.9892	0.7370
344.4	0.5132	0.0352	419.2	1.0000	1.0000
360.8	0.8252	0.1170			

The vapor pressures of pure components are given by the Antoine equation as

$$\ln P_1^{\text{vap}} = 16.784 - \frac{3942.79}{T - 53.797} \quad \text{and} \quad \ln P_2^{\text{vap}} = 20.421 - \frac{4657.84}{T - 13.722},$$

where P is in mmHg and T is in K. Calculate the activity coefficients and express $\tilde{G}^{\text{ex}}/RT$ as a function of composition.

⁴In the literature, the term $\tilde{G}^{\text{ex}}/RT$ is sometimes expressed in the form

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \sum_{i=0}^n A_i L_i(x_1),$$

where L_i 's are the Legendre polynomials given by

$$L_i(x_1) = \frac{(2i-1)(2x_1-1)L_{i-1}(x_1) - (i-1)L_{i-2}(x_1)}{i}$$

in which

$$L_0(x_1) = 1 \quad \text{and} \quad L_1(x_1) = 2x_1 - 1.$$

Solution

At each temperature, vapor pressures are calculated from

$$P_1^{\text{vap}} = \exp\left(16.784 - \frac{3942.79}{T - 53.797}\right) \quad P_2^{\text{vap}} = \exp\left(20.421 - \frac{4657.84}{T - 13.722}\right).$$

The activity coefficients are then calculated by using Eqn (8.3-13), i.e.

$$\gamma_1 = \frac{y_1 P}{x_1 P_1^{\text{vap}}} \quad \text{and} \quad \gamma_2 = \frac{y_2 P}{x_2 P_2^{\text{vap}}}.$$

Finally, the term $\tilde{G}^{\text{ex}}/RT$ is determined from Eqn (8.3-9) as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2.$$

The results are tabulated below:

$T(\text{K})$	x_1	y_1	$P_1^{\text{vap}}(\text{mmHg})$	$P_2^{\text{vap}}(\text{mmHg})$	γ_1	γ_2	$\tilde{G}^{\text{ex}}/RT$
336.5	0.0000	0.0000	17.1	399.6	—	1.000	0
337.9	0.1622	0.0144	18.3	425.3	1.942	1.106	0.192
339.5	0.2721	0.0205	19.8	456.4	1.525	1.179	0.235
344.4	0.5132	0.0352	24.9	564.1	1.100	1.405	0.214
360.8	0.8252	0.1170	51.5	1098.0	1.102	1.841	0.186
378.2	0.9061	0.2211	102.5	2083.0	0.952	1.593	−0.001
398.7	0.9668	0.4953	211.1	4113.0	0.971	1.478	−0.016
410.0	0.9892	0.7370	303.4	5808.0	0.982	1.677	−0.012
419.2	1.0000	1.0000	400.9	7583.4	1.000	—	0

Using the procedure outlined in Example 6.15, $\tilde{G}^{\text{ex}}/RT$ is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1}$$

with $A_1 = 0.914$, $A_2 = 0.532$, $A_3 = 1.957$, $A_4 = -1.918$, and $A_5 = -2.889$.

Sometimes an equation relating molar excess Gibbs energy to the composition in the form

$$\frac{\tilde{G}^{\text{ex}}}{RT} = f(x_1, x_2, \dots, x_k) \quad (8.3-16)$$

is given and it is required to calculate the activity coefficients. For this purpose, the use of Eqn (8.3-7) in Eqn (8.2-15) results in

$$d\left(\frac{G^{\text{ex}}}{RT}\right) = \frac{V^{\text{ex}}}{RT} dP - \frac{H^{\text{ex}}}{RT^2} dT + \sum_{i=1}^k \ln \gamma_i dn_i. \quad (8.3-17)$$

It follows from Eqn (8.3-17) that

$$\ln \gamma_i = \left[\frac{\partial(G^{\text{ex}}/RT)}{\partial n_i} \right]_{T,P,n_{j \neq i}}, \quad (8.3-18)$$

implying that $\ln \gamma_i$ can be interpreted as the partial molar property of G^{ex}/RT .

Examination of Eqns (8.2-16), (8.2-18), and (8.3-18) reveals that excess Gibbs energy can be regarded as a generating function to estimate other excess properties as shown in Fig. 8.5.

For a binary mixture, taking $\tilde{\varphi}_{\text{mix}} = \tilde{G}^{\text{ex}}/RT$ and $\bar{\varphi}_i = \ln \gamma_i$, the use of Eqns (6.2-24) and (6.2-25) leads to the following equations for the evaluation of activity coefficients:

$$\ln \gamma_1 = \frac{\tilde{G}^{\text{ex}}}{RT} - x_2 \frac{d}{dx_2} \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right) \quad \text{constant } T \text{ \& } P. \quad (8.3-19)$$

$$\ln \gamma_2 = \frac{\tilde{G}^{\text{ex}}}{RT} - x_1 \frac{d}{dx_1} \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right) \quad \text{constant } T \text{ \& } P. \quad (8.3-20)$$

In other words, intercepts of the tangent to the $\tilde{G}^{\text{ex}}/RT$ versus x_1 (or x_2) curve at $x_1 = 1$ and $x_1 = 0$ give $\ln \gamma_1$ and $\ln \gamma_2$, respectively.

When $\tilde{G}^{\text{ex}}/RT$ is expressed by Eqn (8.3-15), Eqns (8.3-19) and (8.3-20) yield

$$\ln \gamma_1 = x_2^2 \sum_{i=1}^n A_i (x_1 - x_2)^{i-1} + 2x_1 x_2^2 \sum_{i=1}^{n-1} i A_{i+1} (x_1 - x_2)^{i-1}, \quad (8.3-21)$$

$$\ln \gamma_2 = x_1^2 \sum_{i=1}^n A_i (x_1 - x_2)^{i-1} - 2x_1^2 x_2 \sum_{i=1}^{n-1} i A_{i+1} (x_1 - x_2)^{i-1}. \quad (8.3-22)$$

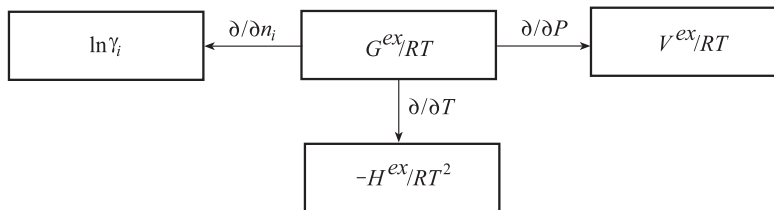


Figure 8.5: Partial differentiation of G^{ex}/RT leads to various excess quantities.

8.3.3 Variation of Activity Coefficients with Temperature

It follows from Eqn (8.2-19) that

$$\left[\frac{\partial}{\partial T} \left(\frac{\bar{G}_i^{\text{ex}}}{RT} \right) \right]_{P, x_j} = -\frac{\bar{H}_i^{\text{ex}}}{RT^2}. \quad (8.3-23)$$

Substitution of Eqn (8.3-7) into Eqn (8.3-23) gives

$$\left(\frac{\partial \ln \gamma_i}{\partial T} \right)_{P, x_j} = -\frac{\bar{H}_i^{\text{ex}}}{RT^2}. \quad (8.3-24)$$

For small temperature changes, partial molar excess enthalpy, $\bar{H}_i^{\text{ex}}$, may be considered constant. In this case, integration of Eqn (8.3-24) at constant pressure and composition gives

$$\gamma_i(T_2, P, x_i) = \gamma_i(T_1, P, x_i) \exp \left[\frac{\bar{H}_i^{\text{ex}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right]. \quad (8.3-25)$$

Partial molar excess enthalpy is calculated from

$$\bar{H}_i^{\text{ex}} = \bar{H}_i - \tilde{H}_i. \quad (8.3-26)$$

Once heat of mixing is determined experimentally, partial molar excess enthalpy can be easily determined as explained in Problem 8.5. Note that when $\bar{H}_i^{\text{ex}}$ is of the order of 10^4 J/mol or greater, the exponential term in Eqn (8.3-25) is different from unity.

8.3.4 Variation of Activity Coefficients with Pressure

It follows from Eqn (8.2-17) that

$$\left(\frac{\partial \bar{G}_i^{\text{ex}}}{\partial P} \right)_{T, x_j} = \bar{V}_i^{\text{ex}}. \quad (8.3-27)$$

Substitution of Eqn (8.3-7) into Eqn (8.3-27) gives

$$\left(\frac{\partial \ln \gamma_i}{\partial P} \right)_{T, x_j} = \frac{\bar{V}_i^{\text{ex}}}{RT}. \quad (8.3-28)$$

The partial molar excess volume of component i in the mixture, $\bar{V}_i^{\text{ex}}$, is usually pressure independent. As a result, integration of Eqn (8.3-28) at constant temperature and composition gives

$$\gamma_i(T, P_2, x_i) = \gamma_i(T, P_1, x_i) \exp \left[\frac{\bar{V}_i^{\text{ex}} (P_2 - P_1)}{RT} \right]. \quad (8.3-29)$$

Partial molar excess volume is calculated from

$$\bar{V}_i^{\text{ex}} = \bar{V}_i - \tilde{V}_i. \quad (8.3-30)$$

Example 8.3 Using the data given in Problem 6.22, estimate partial molar excess volumes for a mixture of methanol (1) and benzene (2) at 298 K. Show that the exponential correction term in Eqn (8.3-29) is almost unity.

Solution Partial molar volumes of methanol and benzene can be determined from Eqns (2) and (3) of Problem 6.22, respectively, as

$$\bar{V}_1 = \tilde{V}_1 + x_2^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} + 2x_1 x_2^2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1}, \quad (1)$$

$$\bar{V}_2 = \tilde{V}_2 + x_1^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} - 2x_1^2 x_2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1}. \quad (2)$$

The use of Eqns (1) and (2) in Eqn (8.3-30) gives partial molar excess volumes as

$$\bar{V}_1^{\text{ex}} = x_2^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} + 2x_1 x_2^2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1}, \quad (3)$$

$$\bar{V}_2^{\text{ex}} = x_1^2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1} - 2x_1^2 x_2 \sum_{i=1}^3 i A_{i+1} (x_1 - x_2)^{i-1}, \quad (4)$$

where the coefficients A_i are given as

$$A_1 = -0.0199 \quad A_2 = -0.1342 \quad A_3 = 0.1740 \quad A_4 = -0.1906.$$

The calculated values of $\bar{V}_1^{\text{ex}}$ and $\bar{V}_2^{\text{ex}}$ as a function of composition are given in the following table:

x_1	$\bar{V}_1^{\text{ex}} (\text{cm}^3/\text{mol})$	$\bar{V}_2^{\text{ex}} (\text{cm}^3/\text{mol})$	x_1	$\bar{V}_1^{\text{ex}} (\text{cm}^3/\text{mol})$	$\bar{V}_2^{\text{ex}} (\text{cm}^3/\text{mol})$
0.0	0.479	0	0.6	-0.023	0.010
0.1	0.114	0.017	0.7	-0.016	-0.003
0.2	-0.035	0.042	0.8	-0.012	-0.017
0.3	-0.071	0.053	0.9	-0.005	-0.056
0.4	-0.060	0.046	1.0	0	-0.171
0.5	-0.039	0.029			

Note that $\bar{V}_i^{\text{ex}}$ is of the order of $\pm 10^{-1} \text{ cm}^3/\text{mol}$. Taking $P_2 - P_1 = 500 \text{ bar}$, the exponential term in Eqn (8.3-29) becomes

$$\exp \left[\frac{\bar{V}_i^{\text{ex}} (P_2 - P_1)}{RT} \right] = \begin{cases} \exp \left[\frac{(0.1)(500)}{(83.14)(298)} \right] = 1.002, \\ \exp \left[\frac{(-0.1)(500)}{(83.14)(298)} \right] = 0.998. \end{cases}$$

Comment: Assuming activity coefficients to be independent of pressure is plausible even at high pressures.

8.4 Binary Activity Coefficient Models

Activity coefficient models can be broadly classified as follows:

- **Empirical models:** The molar excess Gibbs energy is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 f(x_1, x_2),$$

and activity coefficients are determined from Eqns (8.3-19) and (8.3-20). The parameters appearing in these equations are estimated with the help of experimental data. Some examples are two- and three-suffix Margules equations, and the van Laar equation.

- **Local composition models:** These models assume that the bulk (or mean) concentration of species in solution is very different from the local concentration. Hence, local concentrations are dependent on the interactions between molecules and a radial distribution function. Some examples are Wilson (Wilson, 1964), NRTL—NonRandom Two-Liquid (Renon and Prausnitz, 1968), and UNIQUAC—UNIversal QUAsi-Chemical (Abrams and Prausnitz, 1975).
- **Group contribution models:** These models are based on the molecular structure. The activity coefficients are estimated by the additive contributions of the

functional groups comprising the molecule. Some examples are UNIFAC—**UN**ified **A**ctivity **C**oefficient (Fredenslund *et al.*, 1975), Modified UNIFAC—Dortmund (Weidlich and Gmehling, 1987), and ASOG—**A**nalytical **S**olutions of **G**roups (Kojima and Tochigi, 1979).

Coverage of all activity coefficient models available in the literature is beyond the scope of this text. In the following section, some of the most commonly used activity coefficient models for binary liquid mixtures will be briefly given.

8.4.1 Two-Suffix⁵ (One-Constant) Margules Equation

This is the simplest empirical model, in which molar excess Gibbs energy is proposed in the form

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2. \quad (8.4-1)$$

In other words, $i = 1$ in the Redlich-Kister expansion, Eqn (8.3-15). Substitution of Eqn (8.4-1) into Eqns (8.3-19) and (8.3-20) gives the activity coefficients as

$$\ln \gamma_1 = Ax_2^2, \quad (8.4-2)$$

$$\ln \gamma_2 = Ax_1^2. \quad (8.4-3)$$

Since the function representing $\tilde{G}^{\text{ex}}$ is symmetric, then the activity coefficients are mirror images of each other. Figure 8.6 shows the representation of the activity coefficients of a binary system consisting of 3-methylpentane (1) and tetrahydrofuran (2) by the two-suffix Margules equation.

In general, the parameter A is dependent on temperature and pressure, and related to the logarithm of the activity coefficients at infinite dilution as

$$A = \ln \gamma_1^\infty = \ln \gamma_2^\infty. \quad (8.4-4)$$

When A is positive, activity coefficients are greater than unity, and like interactions are much greater than unlike interactions. On the other hand, when A is negative, activity coefficients are less than unity, and unlike interactions are stronger than like interactions.

The two-suffix Margules equation applies best to mixtures of simple molecules of similar size, shape, and chemical nature. For example, argon-oxygen (liquid mixture in cryogenic applications) and benzene-cyclohexane mixtures can be described by this model.

⁵Two-suffix implies that the expansion for $\tilde{G}^{\text{ex}}$ is quadratic in mole fraction.

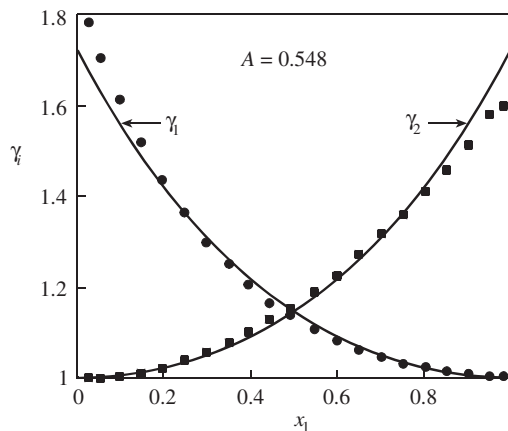


Figure 8.6: The two-suffix Margules equation representing the system 3-methylpentane (1) and tetrahydrofuran (2) at 101.3 kPa (Loras *et al.*, 1999).

Example 8.4 Derive Eqns (8.4-2) and (8.4-3).

Solution

Equation (8.4-1) can be expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = A \left(\frac{n_1}{n} \right) \left(\frac{n_2}{n} \right), \quad (1)$$

where $n = n_1 + n_2$ is the total number of moles. The total excess Gibbs energy, G^{ex} , is

$$\frac{G^{\text{ex}}}{RT} = n \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right) = A \left(\frac{n_1 n_2}{n_1 + n_2} \right). \quad (2)$$

From Eqn (8.3-18)

$$\ln \gamma_1 = \left[\frac{\partial (G^{\text{ex}}/RT)}{\partial n_1} \right]_{T,P,n_2} = A \frac{n_2^2}{(n_1 + n_2)^2} = Ax_2^2, \quad (3)$$

$$\ln \gamma_2 = \left[\frac{\partial (G^{\text{ex}}/RT)}{\partial n_2} \right]_{T,P,n_1} = A \frac{n_1^2}{(n_1 + n_2)^2} = Ax_1^2. \quad (4)$$

Alternate solution: Substitution of Eqn (8.4-1) into Eqns (8.3-19) and (8.3-20) leads to

$$\ln \gamma_1 = Ax_1 x_2 - Ax_2(-x_2 + x_1) = Ax_2^2, \quad (5)$$

$$\ln \gamma_2 = Ax_1 x_2 - Ax_1(x_2 - x_1) = Ax_1^2. \quad (6)$$

Example 8.5 Mara *et al.* (1997) obtained the following set of vapor-liquid equilibrium data for the 1-propanol (1) and 2-pentanol (2) system at 313.5 K:

x_1	y_1	$P(\text{kPa})$
0.2022	0.4332	3.189
0.4030	0.6784	4.154
0.6026	0.8282	5.097
0.8015	0.9278	6.026

If the two-suffix Margules equation represents this system, calculate the parameter A. The vapor pressures of 1-propanol and 2-pentanol at 313.5 K are 6.970 kPa and 2.261 kPa, respectively.

Solution

The parameter A in the two-suffix Margules equation is given by

$$A = \frac{\tilde{G}^{\text{ex}}/RT}{x_1 x_2}. \quad (1)$$

Therefore, it is necessary to calculate $\tilde{G}^{\text{ex}}/RT$ as a function of composition from Eqn (8.3-9), i.e.

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2. \quad (2)$$

The activity coefficients, on the other hand, are calculated by using Eqn (8.3-13), i.e.

$$\gamma_1 = \frac{y_1 P}{x_1 P_1^{\text{vap}}} \quad \text{and} \quad \gamma_2 = \frac{y_2 P}{x_2 P_2^{\text{vap}}}. \quad (3)$$

The results are given in the following table:

x_1	y_1	γ_1	γ_2	$(\tilde{G}^{\text{ex}}/RT) \times 10^3$	A
0.2022	0.4332	0.980	1.002	-2.403	-0.015
0.4030	0.6784	1.003	0.990	-4.863	-0.020
0.6026	0.8282	1.005	0.975	-7.204	-0.030
0.8015	0.9278	1.001	0.969	-5.527	-0.035

Taking the arithmetic average of A gives

$$A = - \left(\frac{0.015 + 0.020 + 0.030 + 0.035}{4} \right) = -0.025.$$

Alternate solution: The parameter A can also be determined by defining an “objective function”. For example, an objective function, F, with respect to molar excess Gibbs energy is defined as

$$F = \sum_{i=1}^4 \left(\frac{\tilde{G}^{\text{ex}}}{RT} - Ax_1x_2 \right)_i^2.$$

The value of A that minimizes the sum of the squares of the deviations is given by

$$\frac{d}{dA} \sum_{i=1}^4 \left(\frac{\tilde{G}^{\text{ex}}}{RT} - Ax_1x_2 \right)_i^2 = 0$$

or

$$A = \frac{\sum_{i=1}^4 \left(\frac{\tilde{G}^{\text{ex}}}{RT} x_1x_2 \right)_i}{\sum_{i=1}^4 (x_1x_2)_i^2}.$$

Substitution of the numerical values yields

$$A = -0.025.$$

8.4.2 Three-Suffix⁶ (Two-Constant) Margules Equation

In this empirical model the expression for molar excess Gibbs energy is given by⁷

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1x_2 [A + B(x_1 - x_2)]. \quad (8.4-5)$$

In other words, $i = 2$ in the Redlich-Kister expansion, Eqn (8.3-15). The use of Eqn (8.4-5) in Eqns (8.3-19) and (8.3-20) gives the activity coefficients as

$$\ln \gamma_1 = x_2^2(A + 3B - 4Bx_2), \quad (8.4-6)$$

$$\ln \gamma_2 = x_1^2(A - 3B + 4Bx_1). \quad (8.4-7)$$

⁶Three-suffix implies that the expansion for $\tilde{G}^{\text{ex}}$ is third-order in mole fraction.

⁷For an alternate expression of molar excess Gibbs energy, see Problem 8.12.

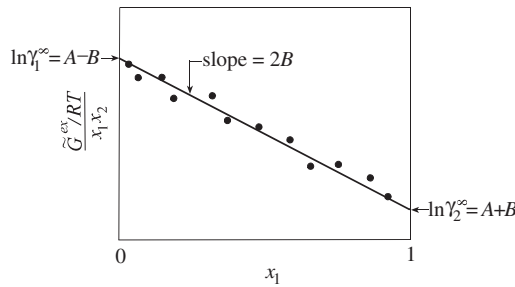


Figure 8.7: A binary liquid mixture represented by the three-suffix Margules equation.

Linearization of Eqn (8.4-5) gives

$$\frac{\tilde{G}^{\text{ex}}/RT}{x_1 x_2} = 2Bx_1 + (A - B). \quad (8.4-8)$$

so that the parameters A and B can be determined graphically⁸, i.e. a plot of $(\tilde{G}^{\text{ex}}/RT)/x_1 x_2$ versus x_1 gives a straight line with a slope of $2B$ and an intercept of $(A - B)$ as shown in Fig. 8.7.

The infinite dilution activity coefficients can be obtained from Eqn (8.4-6) as $x_1 \rightarrow 0$ and from Eqn (8.4-7) as $x_2 \rightarrow 0$. The resulting expressions are

$$\ln \gamma_1^\infty = A - B \quad \text{and} \quad \ln \gamma_2^\infty = A + B. \quad (8.4-9)$$

Thus, the parameters A and B are expressed in terms of the activity coefficients at infinite dilution as

$$A = \ln \sqrt{\gamma_1^\infty \gamma_2^\infty} \quad \text{and} \quad B = \ln \sqrt{\frac{\gamma_2^\infty}{\gamma_1^\infty}}. \quad (8.4-10)$$

⁸Graphical determination of parameters A and B is also possible by linearizing Eqn (8.4-6), i.e.

$$\frac{\ln \gamma_1}{x_2^2} = -4Bx_2 + (A + 3B)$$

or by linearizing Eqn (8.4-7), i.e.

$$\frac{\ln \gamma_2}{x_1^2} = 4Bx_1 + (A - 3B).$$

For a more thorough discussion on the subject, see Shacham *et al.* (1993).

Example 8.6 Gabaldon *et al.* (1996) obtained the following set of data for the water (1) and n-propanol (2) system at 0.6 bar:

x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.103	3.107	1.011	0.557	1.642	1.443
0.231	2.603	1.051	0.686	1.385	1.925
0.294	2.397	1.083	0.796	1.203	2.850
0.361	2.212	1.158	0.869	1.099	4.294
0.487	1.814	1.306			

Estimate the Margules parameters A and B , and calculate the infinite dilution activity coefficients.

Solution

The molar excess Gibbs energy is calculated from Eqn (8.3-9), i.e.

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2,$$

and the values are given in the table below:

x_1	$\tilde{G}^{\text{ex}}/RT$	x_1	$\tilde{G}^{\text{ex}}/RT$
0.103	0.127	0.557	0.439
0.231	0.259	0.686	0.429
0.294	0.313	0.796	0.361
0.361	0.380	0.869	0.273
0.487	0.427		

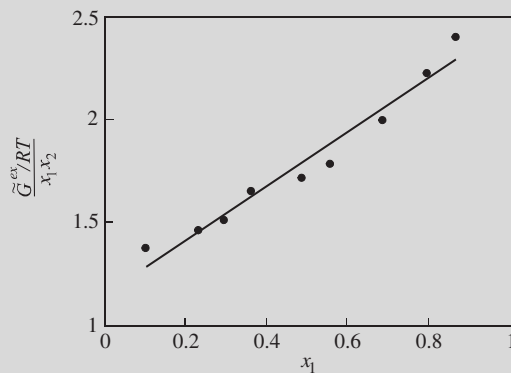
The plot of $(\tilde{G}^{\text{ex}}/RT)/x_1x_2$ versus x_1 is shown below. The data fit a straight line with a correlation coefficient⁹ of 0.979. The slope and the intercept of the best straight line passing through the data points are 1.309 and 1.150, respectively. Therefore, the parameters are

$$A = 1.805 \quad \text{and} \quad B = 0.655.$$

The use of Eqn (8.4-9) gives the infinite dilution activity coefficients as

$$\gamma_1^\infty = 3.158 \quad \text{and} \quad \gamma_2^\infty = 11.705.$$

⁹A correlation coefficient measures the degree to which two variables are directly related. Its value changes from -1 to 1 , -1 and 1 being perfect negative and positive correlations, respectively.



Alternate solution: The parameters A and B can also be determined by defining an objective function, F , as

$$F = \sum_{i=1}^9 \left\{ \frac{\tilde{G}^{\text{ex}}}{RT} - x_1 x_2 [A + B(x_1 - x_2)] \right\}_i^2.$$

The values of A and B that minimize the sum of the squares of the deviations are given by

$$\frac{\partial}{\partial A} \sum_{i=1}^9 \left\{ \frac{\tilde{G}^{\text{ex}}}{RT} - x_1 x_2 [A + B(x_1 - x_2)] \right\}_i^2 = 0, \quad (1)$$

$$\frac{\partial}{\partial B} \sum_{i=1}^9 \left\{ \frac{\tilde{G}^{\text{ex}}}{RT} - x_1 x_2 [A + B(x_1 - x_2)] \right\}_i^2 = 0. \quad (2)$$

Once the differentiations are carried out, Eqns (1) and (2) take the form

$$A \sum_{i=1}^9 (x_1 x_2)_i + B \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)]_i = \sum_{i=1}^9 \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right)_i, \quad (3)$$

$$A \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)]_i + B \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)^2]_i = \sum_{i=1}^9 \left(\frac{\tilde{G}^{\text{ex}}}{RT} x_1 x_2 \right)_i. \quad (4)$$

Using matrix algebra, the parameters A and B can be determined from Eqns (3) and (4) as

$$\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^9 (x_1 x_2)_i & \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)]_i \\ \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)]_i & \sum_{i=1}^9 [x_1 x_2 (x_1 - x_2)^2]_i \end{pmatrix}^{-1} \\ \times \begin{pmatrix} \sum_{i=1}^9 \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right)_i \\ \sum_{i=1}^9 \left(\frac{\tilde{G}^{\text{ex}}}{RT} x_1 x_2 \right)_i \end{pmatrix} = \begin{pmatrix} 1.696 & -0.037 \\ -0.037 & 0.315 \end{pmatrix}^{-1} \begin{pmatrix} 3.008 \\ 0.139 \end{pmatrix} = \begin{pmatrix} 1.788 \\ 0.651 \end{pmatrix}$$

which are slightly lower than the previously calculated values.

The parameters A and B can be estimated from a single set of activity coefficient data as

$$A = \frac{\ln \gamma_2}{x_1^2} \left(2x_1 - \frac{1}{2} \right) + \frac{\ln \gamma_1}{x_2^2} \left(\frac{3}{2} - 2x_1 \right) \quad \text{and} \quad B = \frac{1}{2} \left(\frac{\ln \gamma_1}{x_2^2} - \frac{\ln \gamma_2}{x_1^2} \right). \quad (8.4-11)$$

Example 8.7 The following vapor-liquid equilibrium data are provided for a binary mixture of ethanol (1) and acetone (2) by [Campbell *et al.* \(1987\)](#):

$T(\text{K})$	$P(\text{kPa})$	x_1	y_1
397.7	652.9	0.342	0.305

At 397.7 K

$$P_1^{\text{vap}} = 487.1 \text{ kPa} \quad P_2^{\text{vap}} = 665.3 \text{ kPa} \\ \tilde{V}_1^L = 67.4 \text{ cm}^3/\text{mol} \quad \tilde{V}_2^L = 88.3 \text{ cm}^3/\text{mol}.$$

The vapor phase obeys the virial equation of state with the following parameters:

$$B_{11} = -562.3 \text{ cm}^3/\text{mol} \quad B_{12} = -519.6 \text{ cm}^3/\text{mol} \quad B_{22} = -692.9 \text{ cm}^3/\text{mol}.$$

If the liquid mixture is represented by the three-suffix Margules equation, estimate the parameters A and B .

Solution

Once the activity coefficients, γ_1 and γ_2 , are calculated at the given experimental data point, the parameters A and B can be determined from Eqn (8.4-11). The starting point for the calculation of activity coefficients is Eqn (8.3-12), i.e.

$$\begin{aligned}\widehat{\phi}_i^V(T, P, y_i)y_iP &= f_i^L(T, P)\gamma_i(T, P, x_i)x_i \\ &= P_i^{\text{vap}}\phi_i^V(T, P_i^{\text{vap}})\exp\left[\frac{\widetilde{V}_i^L(P - P_i^{\text{vap}})}{RT}\right]\gamma_i(T, P, x_i)x_i \quad i = 1, 2. \quad (1)\end{aligned}$$

Since pressure is high, simplifying assumptions of this equation leading to Eqn (8.3-13) are no longer valid. Therefore, activity coefficients are calculated from Eqn (1) as

$$\gamma_i = \frac{\widehat{\phi}_i^V(T, P, y_i)y_iP}{P_i^{\text{vap}}\phi_i^V(T, P_i^{\text{vap}})x_i\exp\left[\widetilde{V}_i^L(P - P_i^{\text{vap}})/RT\right]}. \quad (2)$$

From Eqn (5.3-5)

$$\begin{aligned}\phi_1^V(T, P_1^{\text{vap}}) &= \exp\left(\frac{B_{11}P_1^{\text{vap}}}{RT}\right) = \exp\left[\frac{(-562.3)(487.1)}{(8314)(397.7)}\right] = 0.921, \\ \phi_2^V(T, P_2^{\text{vap}}) &= \exp\left(\frac{B_{22}P_2^{\text{vap}}}{RT}\right) = \exp\left[\frac{(-692.9)(665.3)}{(8314)(397.7)}\right] = 0.870.\end{aligned}$$

From Eqn (7.5-6)

$$\begin{aligned}\widehat{\phi}_1^V &= \exp\left\{\frac{P}{RT}[B_{11} + y_2^2(2B_{12} - B_{11} - B_{22})]\right\} \\ &= \exp\left(\frac{652.9}{(8314)(397.7)}\{-562.3 + (0.695)^2[2(-519.6) \right. \\ &\quad \left. + 562.3 + 692.9]\}\right) = 0.914, \\ \widehat{\phi}_2^V &= \exp\left\{\frac{P}{RT}[B_{22} + y_1^2(2B_{12} - B_{11} - B_{22})]\right\} \\ &= \exp\left(\frac{652.9}{(8314)(397.7)}\{-692.9 + (0.305)^2[2(-519.6) \right. \\ &\quad \left. + 562.3 + 692.9]\}\right) = 0.876.\end{aligned}$$

The Poynting correction factors for ethanol and acetone are

$$\exp \left[\frac{\tilde{V}_1^L (P - P_1^{\text{vap}})}{RT} \right] = \exp \left[\frac{(67.4)(652.9 - 487.1)}{(8314)(397.7)} \right] = 1.003,$$

$$\exp \left[\frac{\tilde{V}_2^L (P - P_2^{\text{vap}})}{RT} \right] = \exp \left[\frac{(88.3)(652.9 - 665.3)}{(8314)(397.7)} \right] = 1.000.$$

Substitution of the numerical values into Eqn (2) gives the activity coefficients as

$$\gamma_1 = \frac{(0.914)(0.305)(652.9)}{(487.1)(0.921)(0.342)(1.003)} = 1.183,$$

$$\gamma_2 = \frac{(0.876)(0.695)(652.9)}{(665.3)(0.870)(0.658)(1.000)} = 1.044.$$

The use of Eqn (8.4-11) gives the parameters as

$$A = \frac{\ln(1.044)}{(0.342)^2} \left[2(0.342) - \frac{1}{2} \right] + \frac{\ln(1.183)}{(0.658)^2} \left[\frac{3}{2} - 2(0.342) \right] = 0.384,$$

$$B = \frac{1}{2} \left[\frac{\ln(1.183)}{(0.658)^2} - \frac{\ln(1.044)}{(0.342)^2} \right] = 0.010.$$

Comment: One should be cautious in using parameters based on a single set of experimental data.

The three-suffix Margules equation can be used to model moderately nonideal mixtures.

8.4.3 van Laar Equation

The van Laar equation, like the three-suffix Margules equation, is another two-parameter empirical model in which molar excess Gibbs energy is represented by¹⁰

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \left(\frac{AB}{Ax_1 + Bx_2} \right). \quad (8.4-12)$$

The use of Eqn (8.4-12) in Eqns (8.3-19) and (8.3-20) gives the activity coefficients as

$$\ln \gamma_1 = \frac{A}{\left(1 + \frac{A x_1}{B x_2} \right)^2}, \quad (8.4-13)$$

¹⁰See Problem 8.15 for the development of Eqn (8.4-12).

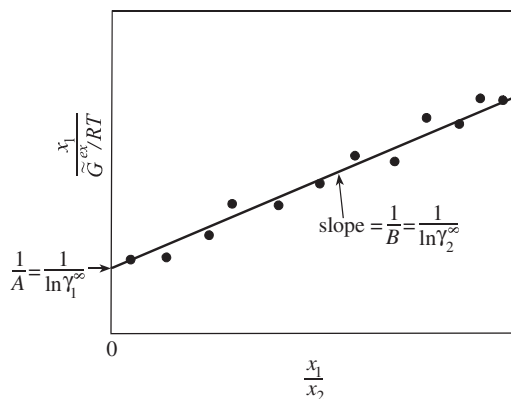


Figure 8.8: A binary liquid mixture represented by the van Laar equation.

$$\ln \gamma_2 = \frac{B}{\left(1 + \frac{B x_2}{A x_1}\right)^2}. \quad (8.4-14)$$

Linearization of Eqn (8.4-12) gives

$$\frac{x_1}{\tilde{G}^{\text{ex}}/RT} = \frac{1}{B} \frac{x_1}{x_2} + \frac{1}{A}, \quad (8.4-15)$$

so that the parameters A and B can be determined graphically¹¹, i.e. a plot of $x_1/(\tilde{G}^{\text{ex}}/RT)$ versus x_1/x_2 gives a straight line with a slope of $1/B$ and an intercept of $1/A$ as shown in Fig. 8.8.

The parameters A and B can also be estimated from a single set of activity coefficient data as

$$\begin{aligned} A &= \ln \gamma_1^\infty = \ln \gamma_1 \left(1 + \frac{x_2 \ln \gamma_2}{x_1 \ln \gamma_1}\right)^2 \quad \text{and} \\ B &= \ln \gamma_2^\infty = \ln \gamma_2 \left(1 + \frac{x_1 \ln \gamma_1}{x_2 \ln \gamma_2}\right)^2. \end{aligned} \quad (8.4-16)$$

¹¹Graphical determination of parameters A and B is also possible by linearizing Eqn (8.4-13), i.e.

$$\frac{1}{\sqrt{\ln \gamma_1}} = \frac{\sqrt{A}}{B} \frac{x_1}{x_2} + \frac{1}{\sqrt{A}}$$

or by linearizing Eqn (8.4-14), i.e.

$$\frac{1}{\sqrt{\ln \gamma_2}} = \frac{\sqrt{B}}{A} \frac{x_2}{x_1} + \frac{1}{\sqrt{B}}.$$

For a more thorough discussion on the subject, see Shacham *et al.* (1993).

The van Laar model is adequate for moderately nonideal mixtures, but cannot be used for highly nonideal mixtures.

In most cases, the parameters A and B are difficult to estimate as a result of the lack of reliable experimental data. For nonpolar binary mixtures, the van Laar equation can be reduced to a one-parameter form by replacing the ratio A/B by the ratio of molar liquid volumes (Poling *et al.*, 2004), i.e.

$$\frac{A}{B} = \frac{\tilde{V}_1^L}{\tilde{V}_2^L}. \quad (8.4-17)$$

Such simplification, however, cannot be used if one of the components is polar. Molar liquid volumes can be calculated by the use of the modified Rackett equation, Eqn (5.4-9).

Example 8.8 For a binary mixture of carbon tetrachloride (1) and *n*-propanol (2), Carley and Bertelsen (1949) obtained the following data from VLE experiments under atmospheric pressure:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.058	2.90	1.03	0.441	1.64	1.29	0.818	1.10	2.73
0.141	2.51	1.06	0.522	1.50	1.40	0.851	1.07	3.05
0.300	2.00	1.15	0.683	1.24	1.84	0.950	1.013	5.54

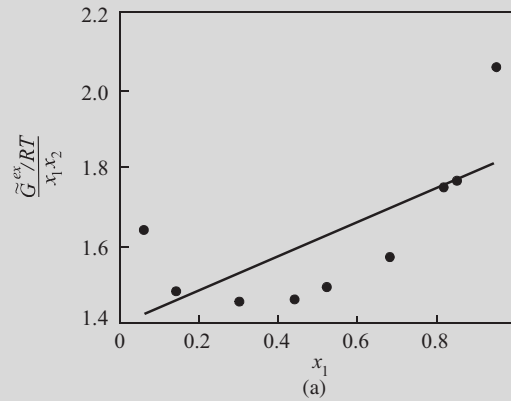
Which model, three-suffix Margules or van Laar, better represents the activity coefficients of this system?

Solution

The molar excess Gibbs energy is calculated from Eqn (8.3-9), i.e.

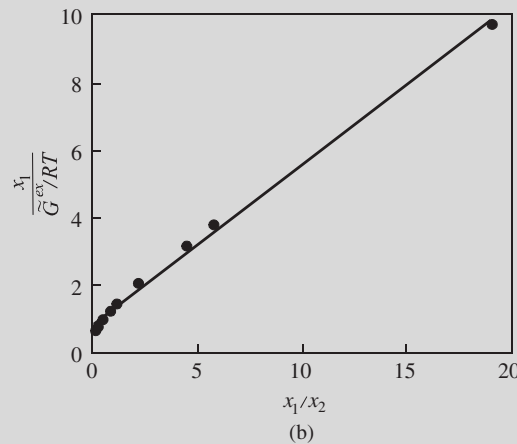
$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2.$$

If the system is represented by the three-suffix Margules equation, then the plot of $(\tilde{G}^{\text{ex}}/RT)/x_1x_2$ versus x_1 should yield a straight line. In Figure (a), the correlation coefficient of the best straight line passing through the data points is 0.698. The scatter of the data indicates that the representation of this binary system by the three-suffix Margules equation is rather poor.



On the other hand, if the activity coefficients are represented by the van Laar model, then the plot of $x_1/(\tilde{G}^{\text{ex}}/RT)$ versus x_1/x_2 should yield a straight line. As shown in Figure (b), the result is a straight line with a correlation coefficient of 0.998, indicating that the system is best represented by the van Laar model. The slope and the intercept of the straight line are 0.473 and 0.852, respectively. Therefore, the parameters are

$$A = 1.174 \quad \text{and} \quad B = 2.114.$$



8.4.4 Wilson Equation

The Wilson equation is based on a local composition model and the molar excess Gibbs energy is given by

$$\frac{\tilde{G}^{\text{ex}}}{RT} = -[x_1 \ln(x_1 + \Lambda_{12}x_2) + x_2 \ln(x_2 + \Lambda_{21}x_1)]. \quad (8.4-18)$$

The use of Eqn (8.4-18) in Eqns (8.3-19) and (8.3-20) gives the activity coefficients as

$$\ln \gamma_1 = - \left[\ln (x_1 + \Lambda_{12}x_2) + x_2 \left(\frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} - \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} \right) \right], \quad (8.4-19)$$

$$\ln \gamma_2 = - \left[\ln (x_2 + \Lambda_{21}x_1) + x_1 \left(\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right) \right]. \quad (8.4-20)$$

The parameters Λ_{12} and Λ_{21} can be evaluated as follows¹²: In terms of the infinite dilution activity coefficients, Eqns (8.4-19) and (8.4-20) reduce to

$$\ln \gamma_1^\infty = - \ln \Lambda_{12} + 1 - \Lambda_{21}, \quad (8.4-21)$$

$$\ln \gamma_2^\infty = - \ln \Lambda_{21} + 1 - \Lambda_{12}. \quad (8.4-22)$$

Equations (8.4-21) and (8.4-22) can be reduced to a single equation of the form

$$\Lambda_{12} = \frac{1}{\gamma_1^\infty} \exp \left[1 - \frac{1}{\gamma_2^\infty} \exp (1 - \Lambda_{12}) \right], \quad (8.4-23)$$

so that Λ_{12} can be determined. Then the parameter Λ_{21} is calculated from

$$\Lambda_{21} = 1 - \ln (\Lambda_{12} \gamma_1^\infty). \quad (8.4-24)$$

The evaluation of Λ_{12} and Λ_{21} from Eqns (8.4-23) and (8.4-24) is not straightforward when one of the infinite dilution activity coefficients (or both) is less than unity. In that case the solution of Eqn (8.4-23) yields multiple roots. It should be kept in mind that the Wilson parameters cannot be negative (Why?).

Example 8.9 For a binary system of thiophene (1) and *n*-hexane (2), Sapei *et al.* (2006) reported $\gamma_1^\infty = 1.82$ and $\gamma_2^\infty = 2.49$ at 338.15 K. If this system is represented by the Wilson equation, evaluate the parameters Λ_{12} and Λ_{21} , and express activity coefficients as a function of mole fractions.

Solution

The use of Eqn (8.4-23) gives

$$\Lambda_{12} = \frac{1}{1.82} \exp \left[1 - \frac{1}{2.49} \exp (1 - \Lambda_{12}) \right] \Rightarrow \Lambda_{12} \simeq 1.$$

Then the parameter Λ_{21} can be estimated from Eqn (8.4-24) as

$$\Lambda_{21} = 1 - \ln (1.82) = 0.4.$$

¹²See also Problem 8.19 for evaluation of Wilson parameters.

Thus, Eqns (8.4-19) and (8.4-20) take the form

$$\ln \gamma_1 = x_2 \left(1 - \frac{0.4}{x_2 + 0.4x_1} \right),$$

$$\ln \gamma_2 = - \left[\ln(x_2 + 0.4x_1) + x_1 \left(1 - \frac{0.4}{x_2 + 0.4x_1} \right) \right].$$

The numerical values of the activity coefficients are given in the table below:

x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.1	1.677	1.004	0.6	1.162	1.248
0.2	1.547	1.019	0.7	1.098	1.387
0.3	1.431	1.046	0.8	1.047	1.599
0.4	1.329	1.089	0.9	1.013	1.933
0.5	1.239	1.153			

In the literature, the parameters Λ_{12} and Λ_{21} are usually expressed as

$$\Lambda_{12} = \frac{\tilde{V}_2^L}{\tilde{V}_1^L} \exp \left(-\frac{\lambda_{12} - \lambda_{11}}{RT} \right) \quad \text{and} \quad \Lambda_{21} = \frac{\tilde{V}_1^L}{\tilde{V}_2^L} \exp \left(-\frac{\lambda_{21} - \lambda_{22}}{RT} \right), \quad (8.4-25)$$

where λ_{ij} 's are the energy parameters associated with the interaction between molecules i and j . In most applications, the parameters $(\lambda_{12} - \lambda_{11})$ and $(\lambda_{21} - \lambda_{22})$ are considered independent of temperature. The molar volume of pure liquid i , $\tilde{V}_i^L$, can be estimated by the modified Rackett equation, Eqn (5.4-9).

The Wilson equation can be used when the components in the liquid phase are completely miscible over the whole composition range. For example, it works well for mixtures of highly polar compounds, i.e. alcohol and water, and mixtures of hydrocarbons.

8.4.5 NRTL (Nonrandom Two-Liquid) Equation

The NRTL equation, which is based on a local composition model, is proposed in the form

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \left(\frac{\tau_{21} G_{21}}{x_1 + G_{21} x_2} + \frac{\tau_{12} G_{12}}{x_2 + G_{12} x_1} \right), \quad (8.4-26)$$

where

$$G_{12} = \exp(-\alpha \tau_{12}) \quad \text{and} \quad G_{21} = \exp(-\alpha \tau_{21}). \quad (8.4-27)$$

The parameters τ_{12} and τ_{21} are expressed in terms of the energy parameters, g_{ij} , in the form

$$\tau_{12} = \frac{g_{12} - g_{22}}{RT} \quad \text{and} \quad \tau_{21} = \frac{g_{21} - g_{11}}{RT}. \quad (8.4-28)$$

The use of Eqn (8.4-26) in Eqns (8.3-19) and (8.3-20) leads to the following activity coefficients:

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + G_{21}x_2} \right)^2 + \frac{\tau_{12}G_{12}}{(x_2 + G_{12}x_1)^2} \right], \quad (8.4-29)$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + G_{12}x_1} \right)^2 + \frac{\tau_{21}G_{21}}{(x_1 + G_{21}x_2)^2} \right]. \quad (8.4-30)$$

The use of the NRTL equation requires two temperature-dependent parameters, τ_{12} and τ_{21} , in addition to a nonrandomness parameter¹³, α . Comparison with the experimental VLE data indicates that α values change between 0.2 and 0.47. Thus, in the absence of information, it is generally recommended to take $\alpha = 0.3$ in the calculations.

At infinite dilution, and when α is estimated, Eqns (8.4-29) and (8.4-30) can be reduced to a single equation when τ_{21} from

$$\tau_{21} = \ln \gamma_1^\infty - \tau_{12} \exp(-\alpha \tau_{12}) \quad (8.4-31)$$

is substituted into

$$\tau_{12} = \ln \gamma_2^\infty - \tau_{21} \exp(-\alpha \tau_{21}). \quad (8.4-32)$$

The NRTL equation can be used for highly nonideal systems as well as for partially miscible systems. Among the various activity coefficient models discussed above, the Wilson model usually provides the best combination of ease of use and accuracy, as long as there is no miscibility problem. If limited solubility is a problem, then the NRTL model should be used.

8.5 Regular Mixture

As defined in Section 7.4, an ideal mixture is one in which molecules are of the same size and shape, and the interactions between unlike molecules are, more or less, the same as those between like molecules. Thus, molar excess Gibbs energy, like all other excess properties, is zero for an ideal mixture. A *regular mixture*, on the other hand, is defined as the one in which excess volume and excess entropy are both zero. Since $G^{\text{ex}} = H^{\text{ex}} - TS^{\text{ex}} = U^{\text{ex}} + PV^{\text{ex}} - TS^{\text{ex}}$, it follows that

$$G^{\text{ex}} = H^{\text{ex}} = U^{\text{ex}} \quad \text{Regular mixture.} \quad (8.5-1)$$

“Regular mixture” approximation relaxes one of the assumptions of an “ideal mixture”, i.e. interactions between unlike molecules are different from those between

¹³It takes into account the nonrandom distribution of molecules of “1” around a molecule of “2”.

Table 8.1: Molar liquid volumes and solubility parameters for various solvents at 298 K.

Solvent	$\tilde{V}^L (\text{cm}^3/\text{mol})$	$\delta (\text{J}/\text{cm}^3)^{1/2}$
Benzene	89	18.7
Carbon tetrachloride	97	18.0
Cyclohexane	109	16.8
Ethyl benzene	123	18.0
<i>n</i> -Hexane	132	14.9
<i>n</i> -Octane	164	15.3
<i>n</i> -Pentane	116	14.4
Toluene	107	18.3
Water	18	48.0

like molecules. For a binary “regular mixture”, the Hildebrand-Scatchard theory gives the activity coefficients as¹⁴

$$RT \ln \gamma_1 = \tilde{V}_1^L \Phi_2^2 (\delta_1 - \delta_2)^2, \quad (8.5-2)$$

$$RT \ln \gamma_2 = \tilde{V}_2^L \Phi_1^2 (\delta_1 - \delta_2)^2. \quad (8.5-3)$$

The terms Φ_1 and Φ_2 represent the volume fractions of components 1 and 2, respectively:

$$\Phi_1 = \frac{x_1 \tilde{V}_1^L}{x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L} \quad \text{and} \quad \Phi_2 = \frac{x_2 \tilde{V}_2^L}{x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L}. \quad (8.5-4)$$

The terms δ_1 and δ_2 represent *solubility parameters* and their values are given by Barton (1975, 1991). As shown in Problem 8.23, solubility parameters can be estimated by using the following formula:

$$\delta_i = \sqrt{\frac{\Delta \tilde{H}_i^{\text{vap}} - RT}{\tilde{V}_i^L}}. \quad (8.5-5)$$

It follows from Eqns (8.5-2), (8.5-3), and (8.5-5) that only pure component experimental data (liquid density and enthalpy of vaporization) are sufficient to estimate activity coefficients.

Solubility parameters of various liquids are given in Table 8.1¹⁵. The smaller the difference in solubility parameters, the higher the solubility of substances in

¹⁴See Problem 8.23 for the development of Eqns (8.5-2) and (8.5-3).

¹⁵Traditionally, solubility parameters are reported in $(\text{cal}/\text{cm}^3)^{1/2}$. Note that

$$(\text{cal}/\text{cm}^3)^{1/2} \simeq 2.045 (\text{J}/\text{cm}^3)^{1/2}.$$

each other. For example, solubility parameters of water and toluene are very different from each other. As a result, solubility of water in toluene is very low. When solubility parameters are equal to each other, Eqn (11) of Problem 8.23 gives $\tilde{G}^{\text{ex}} = 0$, implying ideal mixture behavior, i.e. $\gamma_i = 1$.

The squared terms on the right-hand sides of Eqns (8.5-2) and (8.5-3) imply that

$$\ln \gamma_i > 0 \Rightarrow \gamma_i > 1. \quad (8.5-6)$$

Therefore, the regular mixture theory is not able to predict the cases when $\gamma_i < 1$. Moreover, at constant composition, the right-hand sides of Eqns (8.5-2) and (8.5-3) are constant. Thus, $\ln \gamma_i$ is proportional to $1/T$. In the case of insufficient experimental data, such a relationship can be used to estimate the variation in activity coefficient with temperature at constant composition.

If the parameters A and B in the van Laar equations are defined as

$$A = \frac{\tilde{V}_1^L}{RT}(\delta_1 - \delta_2)^2 \quad \text{and} \quad B = \frac{\tilde{V}_2^L}{RT}(\delta_1 - \delta_2)^2, \quad (8.5-7)$$

then Eqns (8.4-13) and (8.4-14) reduce to Eqns (8.5-2) and (8.5-3), respectively.

Example 8.10 A binary liquid mixture consists of 40 mol% *n*-pentane (1) and 60% benzene (2). Calculate the activity coefficients at 318 K by using the regular mixture theory.

Solution

From Table 8.1

$$\begin{aligned} \tilde{V}_1^L &= 116 \text{ cm}^3/\text{mol} \quad \text{and} \quad \tilde{V}_2^L = 89 \text{ cm}^3/\text{mol}, \\ \delta_1 &= 14.4 \text{ (J/cm}^3\text{)}^{1/2} \quad \text{and} \quad \delta_2 = 18.7 \text{ (J/cm}^3\text{)}^{1/2}. \end{aligned}$$

The volume fractions of *n*-pentane and benzene are calculated from Eqn (8.5-4) as

$$\begin{aligned} \Phi_1 &= \frac{(0.4)(116)}{(0.4)(116) + (0.6)(89)} = 0.465, \\ \Phi_2 &= \frac{(0.6)(89)}{(0.4)(116) + (0.6)(89)} = 0.535. \end{aligned}$$

The use of Eqns (8.5-2) and (8.5-3) yields

$$\begin{aligned} \gamma_1 &= \exp \left[\frac{(116)(0.535)^2(14.4 - 18.7)^2}{(8.314)(318)} \right] = 1.261, \\ \gamma_2 &= \exp \left[\frac{(89)(0.465)^2(14.4 - 18.7)^2}{(8.314)(318)} \right] = 1.144. \end{aligned}$$

Comment: It is implicitly assumed that the liquid volumes are independent of temperature.

In general, predictions of “regular mixture” approximation are good for mixtures of nonpolar molecules.

8.6 UNIFAC

If experimental data are unavailable, a group contribution technique such as UNIFAC can be used to estimate activity coefficients of nonelectrolytes in a liquid mixture. In the UNIFAC model, the activity coefficient is calculated from the sum of the *combinatorial activity coefficient*, γ_i^C , and the *residual activity coefficient*, γ_i^R , in the form

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R. \quad (8.6-1)$$

The combinational part, which is dependent on the surface area and volume of each molecule, is calculated from

$$\ln \gamma_i^C = \ln \left(\frac{\Phi_i}{x_i} \right) + 5q_i \ln \left(\frac{\theta_i}{\Phi_i} \right) + l_i - \frac{\Phi_i}{x_i} \sum_j x_j l_j \quad (8.6-2)$$

in which x_i represents the mole fraction of component i and the summations are over all components, including component i . The terms Φ_i (volume fraction), θ_i (surface area fraction), and l_i are defined by

$$\Phi_i = \frac{r_i x_i}{\sum_j r_j x_j} \quad \theta_i = \frac{q_i x_i}{\sum_j q_j x_j} \quad l_i = 1 - r_i + 5(r_i - q_i). \quad (8.6-3)$$

In Eqn (8.6-3), the molecular volume, r_i , and the molecular surface area, q_i , are calculated as

$$r_i = \sum_k v_k^{(i)} R_k \quad \text{and} \quad q_i = \sum_k v_k^{(i)} Q_k, \quad (8.6-4)$$

where $v_k^{(i)}$ is the number of k groups present in component i . Values of R and Q for some selected groups and subgroups are given in Table 8.2.

The residual part of the activity coefficient describes the intermolecular forces and is calculated from

$$\ln \gamma_i^R = \sum_k v_k^{(i)} \left[\ln \Gamma_k - \ln \Gamma_k^{(i)} \right], \quad (8.6-5)$$

Table 8.2: UNIFAC group volume and surface area parameters for selected groups (Hansen *et al.*, 1991).

Group	Subgroup	R	Q	Example
CH_x	CH_3	0.9011	0.848	n -Pentane = $2\text{CH}_3 + 3\text{CH}_2$
	CH_2	0.6744	0.540	n -Butane = $2\text{CH}_3 + 2\text{CH}_2$
	CH	0.4469	0.228	2,3-Dimethylbutane = $4\text{CH}_3 + 2\text{CH}$
	C	0.2195	0.000	2,2,3,3-Tetramethylbutane = $2\text{C} + 6\text{CH}_3$
$\text{C}=\text{C}$	$\text{CH}_2=\text{CH}$	1.3454	1.176	1-Hexene = $1\text{CH}_3 + 3\text{CH}_2 + 1\text{CH}_2=\text{CH}$
	$\text{CH}=\text{CH}$	1.1167	0.867	2-Hexene = $2\text{CH}_3 + 2\text{CH}_2 + 1\text{CH}=\text{CH}$
	$\text{CH}_2=\text{C}$	1.1173	0.988	3-Methyl-1-Butene = $2\text{CH}_3 + 1\text{CH} + 1\text{CH}_2=\text{C}$
	$\text{CH}=\text{C}$	0.8886	0.676	2-Methyl-2-Butene = $3\text{CH}_3 + 1\text{CH}=\text{C}$
	$\text{C}=\text{C}$	0.6605	0.485	2,3-Dimethylbutene = $4\text{CH}_3 + 1\text{C}=\text{C}$
ACH^1	ACH	0.5313	0.400	Benzene = 6ACH
	AC	0.3652	0.120	Styrene = $1\text{CH}_2=\text{CH} + 5\text{ACH} + 1\text{AC}$
OH	OH	1.0000	1.200	Ethanol = $1\text{CH}_3 + 1\text{CH}_2 + 1\text{OH}$
CH_3OH	CH_3OH	1.4311	1.432	Methanol = $1\text{CH}_3\text{OH}$
H_2O	H_2O	0.9200	1.400	Water = $1\text{H}_2\text{O}$
ACOH^1	ACOH	0.8952	0.680	Phenol = $5\text{ACH} + 1\text{ACOH}$
CH_2CO	CH_3CO	1.6724	1.488	Methyl ethyl ketone = $1\text{CH}_3 + 1\text{CH}_2 + 1\text{CH}_3\text{CO}$
	CH_2CO	1.4457	1.180	Diethyl ketone = $2\text{CH}_3 + 1\text{CH}_2 + 1\text{CH}_2\text{CO}$
COOH	COOH	1.3013	1.224	Acetic acid = $1\text{CH}_3 + 1\text{COOH}$
	HCOOH	1.5280	1.532	Formic acid = 1HCOOH
CCl_3	CHCl_3	2.8700	2.410	Chloroform = 1CHCl_3
	CCl_3	2.6401	2.184	1,1,1-Trichloroethane = $1\text{CH}_3 + 1\text{CCl}_3$

¹ "A" stands for aromatic.

where Γ_k is the contribution of functional group k to the residual activity coefficient, and $\Gamma_k^{(i)}$ is the contribution of group k in the pure fluid i at the same temperature and pressure as the mixture. The term $\Gamma_k^{(i)}$ is needed in Eqn (8.6-5) to satisfy the condition that $\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$. These quantities are calculated by the following expression:

$$\ln \Gamma_k (\text{or } \ln \Gamma_k^{(i)}) = Q_k \left[1 - \ln \left(\sum_m \Theta_m \Psi_{mk} \right) - \sum_m \frac{\Theta_m \Psi_{km}}{\sum_n \Theta_n \Psi_{nm}} \right], \quad (8.6-6)$$

where the summations are over all different groups. The area fraction of group m , Θ_m , is defined by

$$\Theta_m = \frac{Q_m X_m}{\sum_n Q_n X_n}, \quad (8.6-7)$$

Table 8.3: UNIFAC group-group interaction parameters for selected groups (Poling *et al.*, 2004).

m ↓	n →	$a_{mn}(\text{K})$					
		CH _x	C = C	ACH	OH	H ₂ O	CH ₂ CO
CH _x		0.0	86.02	61.13	986.5	1318	476.4
C = C		−35.36	0.0	38.81	524.1	270.6	182.6
ACH		−11.12	3.446	0.0	636.1	903.8	25.77
OH		156.4	457.0	89.60	0.0	353.5	84.00
H ₂ O		300	496.1	362.3	−229.1	0.0	−195.4
CH ₂ CO		26.76	42.92	140.1	164.5	472.5	0.0

where X_m is the mole fraction of the group m in the mixture

$$X_m = \frac{\sum_j v_m^{(j)} x_j}{\sum_j \sum_n v_n^{(j)} x_j}. \quad (8.6-8)$$

The group interaction parameter, Ψ_{mn} , is given by

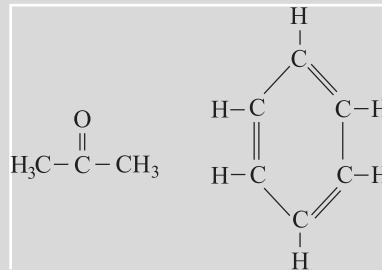
$$\Psi_{mn} = \exp\left(-\frac{a_{mn}}{T}\right) \quad (8.6-9)$$

in which $a_{mn} \neq a_{nm}$. Table 8.3 gives values of a_{mn} for interactions between six representative main groups.

Example 8.11 Estimate the activity coefficients of a binary liquid consisting of 73 mol% acetone (1) and 27% benzene (2) at 332.4 K.

Solution

The structural formulas of acetone and benzene are shown below.



Note that while acetone has one CH₃ group and one CH₃CO group, benzene has six ACH groups. From Table 8.2

Acetone ($i = 1$) :	1CH ₃ ($k = A$)	$R = 0.9011$	$Q = 0.848$
	1CH ₃ CO ($k = B$)	$R = 1.6724$	$Q = 1.488$
Benzene ($i = 2$) :	6ACH ($k = C$)	$R = 0.5313$	$Q = 0.400$.

Thus, $v_k^{(i)}$ values are

$$\begin{aligned} v_A^{(1)} &= 1 \quad v_B^{(1)} = 1 \quad v_C^{(1)} = 0 \\ v_A^{(2)} &= 0 \quad v_B^{(2)} = 0 \quad v_C^{(2)} = 6. \end{aligned}$$

• **Combinatorial Activity Coefficient**

The molecular volumes and surface areas are calculated from Eqn (8.6-4) as

$$\begin{aligned} r_1 &= (1)(0.9011) + (1)(1.6724) = 2.5735, \\ r_2 &= (6)(0.5313) = 3.1878, \\ q_1 &= (1)(0.848) + (1)(1.488) = 2.336, \\ q_2 &= (6)(0.400) = 2.400. \end{aligned}$$

The volume and surface area fractions as well as the parameter l_i are calculated from Eqn (8.6-3) as

$$\begin{aligned} \Phi_1 &= \frac{(2.5735)(0.73)}{(2.5735)(0.73) + (3.1878)(0.27)} = 0.6858, \\ \Phi_2 &= \frac{(3.1878)(0.27)}{(2.5735)(0.73) + (3.1878)(0.27)} = 0.3142, \\ \theta_1 &= \frac{(2.336)(0.73)}{(2.336)(0.73) + (2.400)(0.27)} = 0.7246, \\ \theta_2 &= \frac{(2.400)(0.27)}{(2.336)(0.73) + (2.400)(0.27)} = 0.2754, \\ l_1 &= 1 - 2.5735 + (5)(2.5735 - 2.336) = -0.3860, \\ l_2 &= 1 - 3.1878 + (5)(3.1878 - 2.400) = 1.7512. \end{aligned}$$

Thus, the combinatorial part of the activity coefficients is calculated from Eqn (8.6-2) as

$$\begin{aligned} \ln \gamma_1^C &= \ln \left(\frac{0.6858}{0.73} \right) + (5)(2.336) \ln \left(\frac{0.7246}{0.6858} \right) - 0.3860 \\ &\quad - \frac{0.6858}{0.73} [(0.73)(-0.386) + (0.27)(1.7512)] = 0.01486, \\ \ln \gamma_2^C &= \ln \left(\frac{0.3142}{0.27} \right) + (5)(2.400) \ln \left(\frac{0.2754}{0.3142} \right) + 1.7512 \\ &\quad - \frac{0.3142}{0.27} [(0.73)(-0.386) + (0.27)(1.7512)] = 0.09883. \end{aligned}$$

• Residual Activity Coefficient

Interaction parameters should be considered between the main groups. The subgroups CH_3 ($k = A$), CH_3CO ($k = B$), and ACH ($k = C$) belong to CH_x , CH_2CO , and ACH main groups, respectively. Hence, from Table 8.3

$$\begin{array}{lll} a_{AA} = 0 & a_{AB} = 476.4 & a_{AC} = 61.13 \\ a_{BA} = 26.76 & a_{BB} = 0 & a_{BC} = 140.1 \\ a_{CA} = -11.12 & a_{CB} = 25.77 & a_{CC} = 0. \end{array}$$

From Eqn (8.6-9)

$$\begin{array}{lll} \Psi_{AA} = 1 & \Psi_{AB} = 0.2385 & \Psi_{AC} = 0.8320 \\ \Psi_{BA} = 0.9226 & \Psi_{BB} = 1 & \Psi_{BC} = 0.6561 \\ \Psi_{CA} = 1.0340 & \Psi_{CB} = 0.9254 & \Psi_{CC} = 1. \end{array}$$

For pure acetone ($i = 1$), the mole fractions of groups CH_3 and CH_3CO are

$$X_A^{(1)} = \frac{v_A^{(1)}}{v_A^{(1)} + v_B^{(1)}} = \frac{1}{1 + 1} = 0.5 \quad \text{and} \quad X_B^{(1)} = 0.5.$$

The group area fractions in pure acetone are calculated from Eqn (8.6-7) as

$$\begin{aligned} \Theta_A^{(1)} &= \frac{(0.848)(0.5)}{(0.848)(0.5) + (1.488)(0.5)} = 0.363, \\ \Theta_B^{(1)} &= \frac{(1.488)(0.5)}{(0.848)(0.5) + (1.488)(0.5)} = 0.637. \end{aligned}$$

Contributions of groups CH_3 and CH_3CO in pure acetone are calculated from Eqn (8.6-6) as

$$\begin{aligned} \ln \Gamma_A^{(1)} &= 0.848 \left\{ 1 - \ln [(0.363)(1) + (0.637)(0.9226)] \right. \\ &\quad \left. - \left[\frac{(0.363)(1)}{(0.363)(1) + (0.637)(0.9226)} + \frac{(0.637)(0.2385)}{(0.363)(0.2385) + (0.637)(1)} \right] \right\} \\ &= 0.389, \\ \ln \Gamma_B^{(1)} &= 1.488 \left\{ 1 - \ln [(0.363)(0.2385) + (0.637)(1)] \right. \\ &\quad \left. - \left[\frac{(0.363)(0.9226)}{(0.363)(1) + (0.637)(0.9226)} + \frac{(0.637)(1)}{(0.363)(0.2385) + (0.637)(1)} \right] \right\} \\ &= 0.135. \end{aligned}$$

Since only one main group exists in benzene

$$\ln \Gamma_C^{(2)} = 0.$$

Now, contributions of the functional groups to the residual activity coefficient, Γ_k , should be calculated. The use of Eqn (8.6-8) gives the group mole fractions in solution as

$$\begin{aligned} X_A &= \frac{v_A^{(1)}x_1 + v_A^{(2)}x_2}{\left[v_A^{(1)} + v_B^{(1)} + v_C^{(1)}\right]x_1 + \left[v_A^{(2)} + v_B^{(2)} + v_C^{(2)}\right]x_2} \\ &= \frac{(1)(0.73)}{(2)(0.73) + (6)(0.27)} = 0.237, \\ X_B &= \frac{(1)(0.73)}{(2)(0.73) + (6)(0.27)} = 0.237, \\ X_C &= \frac{(6)(0.27)}{(2)(0.73) + (6)(0.27)} = 0.526. \end{aligned}$$

The area fractions are calculated from Eqn (8.6-7) as

$$\begin{aligned} \Theta_A &= \frac{Q_A X_A}{Q_A X_A + Q_B X_B + Q_C X_C} \\ &= \frac{(0.848)(0.237)}{(0.848)(0.237) + (1.488)(0.237) + (0.400)(0.526)} = 0.263, \\ \Theta_B &= \frac{(1.488)(0.237)}{(0.848)(0.237) + (1.488)(0.237) + (0.400)(0.526)} = 0.462, \\ \Theta_C &= \frac{(0.400)(0.526)}{(0.848)(0.237) + (1.488)(0.237) + (0.400)(0.526)} = 0.275. \end{aligned}$$

Substitution of the numerical values into Eqn (8.6-6) gives

$$\begin{aligned} \ln \Gamma_A &= 0.848 \left\{ 1 - \ln [(0.263)(1) + (0.462)(0.9226) + (0.275)(1.0340)] \right. \\ &\quad - \left[\frac{(0.263)(1)}{(0.263)(1) + (0.462)(0.9226) + (0.275)(1.0340)} \right. \\ &\quad + \frac{(0.462)(0.2385)}{(0.263)(0.2385) + (0.462)(1) + (0.275)(0.9254)} \\ &\quad \left. \left. + \frac{(0.275)(0.8320)}{(0.263)(0.8320) + (0.462)(0.6561) + (0.275)(1)} \right] \right\} = 0.2782. \end{aligned}$$

Similarly,

$$\ln \Gamma_B = 0.2692,$$

$$\ln \Gamma_C = 0.0216.$$

From Eqn (8.6-5)

$$\ln \gamma_1^R = (1)(0.2782 - 0.389) + (1)(0.2692 - 0.135) = 0.0234,$$

$$\ln \gamma_2^R = (6)(0.0216 - 0) = 0.1296.$$

Activity coefficients can be calculated from Eqn (8.6-1) as

$$\ln \gamma_1 = 0.01486 + 0.0234 = 0.03826 \implies \gamma_1 = 1.039,$$

$$\ln \gamma_2 = 0.09883 + 0.1296 = 0.22843 \implies \gamma_2 = 1.257.$$

8.7 Infinite Dilution Activity Coefficients

The use of the activity coefficient models discussed in Section 8.4 requires the parameters to be known. The parameters appearing in these models, on the other hand, are closely related to the infinite-dilution activity coefficients as shown in Table 8.4. Once the infinite-dilution activity coefficients are known, then parameters can be easily determined. Moreover, the design of processes involving the separation of dilute contaminants (or hazardous substances) from water and/or a waste stream requires the infinite dilution activity coefficient to be known.

Sherman *et al.* (1996) summarized various direct and indirect experimental measurement methods for the estimation of infinite-dilution activity coefficients. On the other hand, the values of activity coefficients at infinite dilution are tabulated by Tiegs *et al.* (1987).

Once the experimental VLE data are obtained, infinite-dilution activity coefficients can be estimated as follows: The use of Eqn (8.3-13) gives the activity coefficients as a function of composition. Then extrapolation of the experimental data to infinite dilution, i.e. $x_i \rightarrow 0$, gives infinite-dilution activity coefficients, γ_i^∞ .

Extrapolation to infinite dilution can be carried out by using various techniques, one of which is the *Lagrange three-point formula*. The Lagrange interpolating polynomial is the polynomial of degree $n - 1$ that passes through n points $y_1 = f(x_1)$, $y_2 = f(x_2)$, $\dots$, $y_n = f(x_n)$. It is given by

$$P(x) = \sum_{j=1}^n P_j(x), \quad (8.7-1)$$

where

$$P_j(x) = y_j \prod_{\substack{k=1 \\ k \neq j}}^n \frac{x - x_k}{x_j - x_k}. \quad (8.7-2)$$

For $n = 3$, substitution of Eqn (8.7-2) into Eqn (8.7-1) leads to

$$\begin{aligned} P(x) = & y_1 \left[\frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)} \right] + y_2 \left[\frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)} \right] \\ & + y_3 \left[\frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)} \right]. \end{aligned} \quad (8.7-3)$$

When $x = 0$, Eqn (8.7-3) reduces to

$$\begin{aligned} P(0) = & y_1 \left[\frac{x_2 x_3}{(x_1 - x_2)(x_1 - x_3)} \right] \\ & + y_2 \left[\frac{x_1 x_3}{(x_2 - x_1)(x_2 - x_3)} \right] + y_3 \left[\frac{x_1 x_2}{(x_3 - x_1)(x_3 - x_2)} \right]. \end{aligned} \quad (8.7-4)$$

Therefore, infinite-dilution activity coefficients are calculated as

$$\gamma_1^\infty = \frac{x_1^{(2)} x_1^{(3)} \gamma_1^{(1)}}{[x_1^{(1)} - x_1^{(2)}][x_1^{(1)} - x_1^{(3)}]} + \frac{x_1^{(1)} x_1^{(3)} \gamma_1^{(2)}}{[x_1^{(2)} - x_1^{(1)}][x_1^{(2)} - x_1^{(3)}]} + \frac{x_1^{(1)} x_1^{(2)} \gamma_1^{(3)}}{[x_1^{(3)} - x_1^{(1)}][x_1^{(3)} - x_1^{(2)}]}. \quad (8.7-5)$$

$$\gamma_2^\infty = \frac{x_2^{(2)} x_2^{(3)} \gamma_2^{(1)}}{[x_2^{(1)} - x_2^{(2)}][x_2^{(1)} - x_2^{(3)}]} + \frac{x_2^{(1)} x_2^{(3)} \gamma_2^{(2)}}{[x_2^{(2)} - x_2^{(1)}][x_2^{(2)} - x_2^{(3)}]} + \frac{x_2^{(1)} x_2^{(2)} \gamma_2^{(3)}}{[x_2^{(3)} - x_2^{(1)}][x_2^{(3)} - x_2^{(2)}]}. \quad (8.7-6)$$

Table 8.4: Infinite dilution activity coefficients for various models.

Model	Infinite Dilution Activity Coefficients
Two-Suffix Margules	$\gamma_1^\infty = \gamma_2^\infty = \exp A$
Three-Suffix Margules	$\gamma_1^\infty = \exp(A - B)$ $\gamma_2^\infty = \exp(A + B)$
van Laar	$\gamma_1^\infty = \exp A$ $\gamma_2^\infty = \exp B$
Wilson	$\gamma_1^\infty = \exp(-\ln \Lambda_{12} + 1 - \Lambda_{21})$ $\gamma_2^\infty = \exp(-\ln \Lambda_{21} + 1 - \Lambda_{12})$
NRTL	$\gamma_1^\infty = \exp[\tau_{21} + \tau_{12} \exp(-\alpha \tau_{12})]$ $\gamma_2^\infty = \exp[\tau_{12} + \tau_{21} \exp(-\alpha \tau_{21})]$
Hildebrand-Scatchard	$\gamma_1^\infty = \exp \left[\frac{\tilde{V}_1^L}{RT} (\delta_1 - \delta_2)^2 \right]$ $\gamma_2^\infty = \exp \left[\frac{\tilde{V}_2^L}{RT} (\delta_1 - \delta_2)^2 \right]$

Example 8.12 Alonso *et al.* (2004) reported the following isothermal VLE data for a binary mixture of methanol (1) and *n*-hexane (2) at 313.15 K:

- Determine the infinite-dilution activity coefficients by using the Lagrange three-point formula.
- Determine the parameters in the Wilson equation and estimate the vapor phase composition for the given x_1 values.

$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$	x_1	y_1
37.313	0.0000	0.0000	68.943	0.3449	0.4848	68.748	0.7757	0.4840
61.555	0.0386	0.4060	68.948	0.4065	0.4849	68.616	0.7986	0.4859
64.676	0.0544	0.4390	68.967	0.4163	0.4848	68.353	0.8259	0.4890
66.110	0.0667	0.4530	68.960	0.5004	0.4821	67.935	0.8489	0.4931
67.822	0.1027	0.4698	68.969	0.5008	0.4821	67.284	0.8711	0.4992
68.539	0.1856	0.4768	68.962	0.5987	0.4795	66.016	0.8968	0.5112
68.664	0.2050	0.4779	68.960	0.6043	0.4795	63.747	0.9224	0.5336
68.759	0.2273	0.4794	68.948	0.6517	0.4796	59.703	0.9472	0.5764
68.831	0.2524	0.4811	68.927	0.7008	0.4800	50.569	0.9757	0.6925
68.878	0.2770	0.4826	68.902	0.7256	0.4810	35.464	1.0000	1.0000
68.912	0.3008	0.4839	68.850	0.7516	0.4824			

Solution

(a) It is first necessary to calculate the activity coefficients from Eqn (8.3-13), i.e.

$$\gamma_1 = \frac{y_1 P}{x_1 P_1^{\text{vap}}} \quad \text{and} \quad \gamma_2 = \frac{y_2 P}{x_2 P_2^{\text{vap}}}. \quad (1)$$

For example, let us calculate γ_1 and γ_2 values for the following data set:

$P(\text{kPa})$	x_1	y_1
68.759	0.2273	0.4794

From the given data, $P_1^{\text{vap}} = 35.464$ kPa and $P_2^{\text{vap}} = 37.313$ kPa at 313.15 K. The activity coefficients are

$$\gamma_1 = \frac{y_1 P}{x_1 P_1^{\text{vap}}} = \frac{(0.4794)(68.759)}{(0.2273)(35.464)} = 4.089,$$

$$\gamma_2 = \frac{y_2 P}{x_2 P_2^{\text{vap}}} = \frac{(1 - 0.4794)(68.759)}{(1 - 0.2273)(37.313)} = 1.242.$$

The calculated values of the activity coefficients as a function of composition are given in the table below:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.0386	18.256	1.019	0.3449	2.733	1.453	0.7516	1.246	3.845
0.0544	14.717	1.028	0.4065	2.319	1.604	0.7757	1.210	4.239
0.0667	12.661	1.038	0.4163	2.265	1.631	0.7986	1.177	4.694
0.1027	8.748	1.074	0.5004	1.873	1.916	0.8259	1.141	5.377
0.1856	4.965	1.180	0.5008	1.872	1.918	0.8489	1.113	6.108
0.2050	4.514	1.209	0.5987	1.557	2.397	0.8711	1.087	7.006
0.2273	4.089	1.242	0.6043	1.543	2.431	0.8968	1.061	8.380
0.2524	3.699	1.280	0.6517	1.431	2.761	0.9224	1.040	10.268
0.2770	3.384	1.321	0.7008	1.331	3.210	0.9472	1.024	12.837
0.3008	3.126	1.363	0.7256	1.288	3.493	0.9757	1.012	17.150

The values of the liquid phase mole fractions and the activity coefficients to be used in Eqns (8.7-5) and (8.7-6) are given below:

i	$x_1^{(i)}$	$\gamma_1^{(i)}$	$x_2^{(i)}$	$\gamma_2^{(i)}$
1	0.0386	18.256	0.0243	17.150
2	0.0544	14.717	0.0528	12.837
3	0.0667	12.661	0.0776	10.268

Therefore, the infinite-dilution activity coefficients are calculated as

$$\begin{aligned}
 \gamma_1^\infty &= \frac{(0.0544)(0.0667)(18.256)}{(0.0386 - 0.0544)(0.0386 - 0.0667)} \\
 &\quad + \frac{(0.0386)(0.0667)(14.717)}{(0.0544 - 0.0386)(0.0544 - 0.0667)} \\
 &\quad + \frac{(0.0386)(0.0544)(12.661)}{(0.0667 - 0.0386)(0.0667 - 0.0544)} = 31.149, \\
 \gamma_2^\infty &= \frac{(0.0528)(0.0776)(17.150)}{(0.0243 - 0.0528)(0.0243 - 0.0776)} \\
 &\quad + \frac{(0.0243)(0.0776)(12.837)}{(0.0528 - 0.0243)(0.0528 - 0.0776)} \\
 &\quad + \frac{(0.0243)(0.0528)(10.268)}{(0.0776 - 0.0243)(0.0776 - 0.0528)} = 21.977.
 \end{aligned}$$

(b) The use of Eqn (8.4-23) gives

$$\Lambda_{12} = \frac{1}{31.149} \exp \left[1 - \frac{1}{21.977} \exp(1 - \Lambda_{12}) \right] \Rightarrow \Lambda_{12} = 0.078.$$

The parameter Λ_{21} can be estimated from Eqn (8.4-24) as

$$\Lambda_{21} = 1 - \ln [(0.078)(31.149)] = 0.112.$$

Once γ_1 values are calculated from Eqn (8.4-19) at each x_1 value, the gas phase composition is determined from

$$y_1^{\text{calc}} = \frac{\gamma_1 x_1 P_1^{\text{vap}}}{P},$$

and the results are tabulated below:

x_1	γ_1	y_1^{exp}	y_1^{calc}	x_1	γ_1	y_1^{exp}	y_1^{calc}	x_1	γ_1	y_1^{exp}	y_1^{calc}
0.0386	15.239	0.4060	0.3389	0.3449	2.585	0.4848	0.4586	0.7516	1.223	0.4824	0.4736
0.0544	12.413	0.4390	0.3703	0.4065	2.205	0.4849	0.4609	0.7757	1.189	0.4840	0.4757
0.0667	10.810	0.4530	0.3868	0.4163	2.154	0.4848	0.4612	0.7986	1.159	0.4859	0.4782
0.1027	7.775	0.4698	0.4176	0.5004	1.802	0.4821	0.4637	0.8259	1.125	0.4890	0.4822
0.1856	4.644	0.4768	0.4460	0.5008	1.801	0.4821	0.4637	0.8489	1.100	0.4931	0.4873
0.2050	4.237	0.4779	0.4486	0.5987	1.516	0.4795	0.4666	0.8711	1.077	0.4992	0.4945
0.2273	3.848	0.4794	0.4511	0.6043	1.502	0.4795	0.4668	0.8968	1.054	0.5112	0.5075
0.2524	3.486	0.4811	0.4533	0.6517	1.398	0.4796	0.4685	0.9224	1.033	0.5336	0.5302
0.2770	3.191	0.4826	0.4550	0.7008	1.305	0.4800	0.4706	0.9472	1.017	0.5764	0.5723
0.3008	2.949	0.4839	0.4564	0.7256	1.264	0.4810	0.4719	0.9757	1.004	0.6925	0.6872

An alternative approach for determining infinite dilution activity coefficients is to use Redlich-Kister type expansion in expressing $\tilde{G}^{\text{ex}}/RT$ as a function of composition, i.e. Eqn (8.3-15). Then it follows from Eqns (8.3-21) and (8.3-22) that

$$\ln \gamma_1^\infty = \sum_{i=1}^n (-1)^{i-1} A_i \quad \text{and} \quad \ln \gamma_2^\infty = \sum_{i=1}^n A_i. \quad (8.7-7)$$

Example 8.13 Using the result obtained in Example 8.12, i.e. $\Lambda_{12} = 0.078$ and $\Lambda_{21} = 0.112$, estimate infinite dilution activity coefficients using Eqn (8.7-7).

Solution

The term $\tilde{G}^{\text{ex}}/RT$ can be determined from Eqn (8.3-9), i.e.

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2. \quad (1)$$

The calculated values are given below:

x_1	$\tilde{G}^{ex}/RT$	x_1	$\tilde{G}^{ex}/RT$	x_1	$\tilde{G}^{ex}/RT$
0.0386	0.117	0.3449	0.559	0.7516	0.469
0.0544	0.159	0.4065	0.588	0.7757	0.442
0.0667	0.188	0.4163	0.591	0.7986	0.413
0.1027	0.266	0.5004	0.603	0.8259	0.375
0.1856	0.405	0.5008	0.602	0.8489	0.339
0.2050	0.430	0.5987	0.581	0.8711	0.302
0.2273	0.457	0.6043	0.579	0.8968	0.254
0.2524	0.485	0.6517	0.554	0.9224	0.201
0.2770	0.508	0.7008	0.518	0.9472	0.144
0.3008	0.529	0.7256	0.495	0.9757	0.071

Using the procedure outlined in Example 6.15, $\tilde{G}^{ex}/RT$ is expressed as

$$\frac{\tilde{G}^{ex}}{RT} = x_1 x_2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1}$$

with $A_1 = 2.415$, $A_2 = -0.039$, $A_3 = 0.388$, $A_4 = -0.091$, and $A_5 = 0.414$. The use of Eqn (8.7-7) gives the infinite dilution activity coefficients as

$$\begin{aligned} \ln \gamma_1^\infty &= 2.415 + 0.039 + 0.388 + 0.091 + 0.414 \\ &= 3.347 \Rightarrow \gamma_1^\infty = 28.417, \\ \ln \gamma_2^\infty &= 2.415 - 0.039 + 0.388 - 0.091 + 0.414 \\ &= 3.087 \Rightarrow \gamma_2^\infty = 21.911. \end{aligned}$$

8.8 Testing Consistency of Experimental Data

On a molar basis, Eqn (8.3-17) becomes

$$d\left(\frac{\tilde{G}^{ex}}{RT}\right) = \frac{\tilde{V}^{ex}}{RT} dP - \frac{\tilde{H}^{ex}}{RT^2} dT + \sum_{i=1}^k \ln \gamma_i dx_i. \quad (8.8-1)$$

Substitution of Eqn (8.3-9) into the left-hand side of Eqn (8.8-1) and rearrangement give

$$\sum_{i=1}^k x_i d \ln \gamma_i = \frac{\tilde{V}^{ex}}{RT} dP - \frac{\tilde{H}^{ex}}{RT^2} dT, \quad (8.8-2)$$

which is known as the *Gibbs-Duhem equation for excess properties*. In the case of constant temperature and pressure, Eqn (8.8-2) takes the form

$$\sum_{i=1}^k x_i d \ln \gamma_i = 0 \quad \text{constant } T \text{ \& } P. \quad (8.8-3)$$

For a binary mixture, Eqn (8.8-3) reduces to

$$x_1 \left(\frac{d \ln \gamma_1}{dx_1} \right) + x_2 \left(\frac{d \ln \gamma_2}{dx_1} \right) = 0 \quad \text{constant } T \text{ \& } P. \quad (8.8-4)$$

Equation (8.8-4) is used to check whether the activity coefficients obtained from the experimental data are thermodynamically consistent or not.

Example 8.14 In a binary liquid system, the activity coefficients of the components are proposed in the form

$$\begin{aligned} \ln \gamma_1 &= Ax_1x_2 + x_1^2, \\ \ln \gamma_2 &= Ax_1x_2 + x_2^2. \end{aligned}$$

Are these equations thermodynamically consistent?

Solution

The given activity coefficient expressions should satisfy the Gibbs-Duhem equation, Eqn (8.8-4), i.e.

$$x_1 \left(\frac{d \ln \gamma_1}{dx_1} \right) + x_2 \underbrace{\left(\frac{d \ln \gamma_2}{dx_1} \right)}_{-\frac{d \ln \gamma_2}{dx_2}} = 0. \quad (1)$$

The derivatives of the activity coefficients are

$$\frac{d \ln \gamma_1}{dx_1} = A + 2(1 - A)x_1, \quad (2)$$

$$\frac{d \ln \gamma_2}{dx_2} = A + 2(1 - A)x_2. \quad (3)$$

Therefore, the left-hand side of Eqn (1) becomes

$$\begin{aligned} x_1 \frac{d \ln \gamma_1}{dx_1} - x_2 \frac{d \ln \gamma_2}{dx_2} &= x_1 [A + 2(1 - A)x_1] - x_2 [A + 2(1 - A)x_2] \\ &= (2 - A)(x_1 - x_2), \end{aligned} \quad (4)$$

which is different from zero. Therefore, the given equations are thermodynamically inconsistent.

Comment: In cases in which the Gibbs-Duhem equation is satisfied, it is also necessary to check whether $\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$.

It is important to keep in mind that Eqn (8.8-4) is valid only when both temperature and pressure remain constant. On the other hand, VLE measurements are made at either constant temperature or constant pressure¹⁶. For this purpose, the integrated form of Eqn (8.8-1) is used. For a binary system, Eqn (8.8-1) takes the form

$$d\left(\frac{\tilde{G}^{\text{ex}}}{RT}\right) = \frac{\tilde{V}^{\text{ex}}}{RT}dP - \frac{\tilde{H}^{\text{ex}}}{RT^2}dT + \ln\left(\frac{\gamma_1}{\gamma_2}\right)dx_1 \quad (8.8-5)$$

or

$$\frac{d}{dx_1}\left(\frac{\tilde{G}^{\text{ex}}}{RT}\right) = \frac{\tilde{V}^{\text{ex}}}{RT}\frac{dP}{dx_1} - \frac{\tilde{H}^{\text{ex}}}{RT^2}\frac{dT}{dx_1} + \ln\left(\frac{\gamma_1}{\gamma_2}\right). \quad (8.8-6)$$

Integration of Eqn (8.8-6) from $x_1 = 0$ to $x_1 = 1$ gives

$$\underbrace{\left.\frac{\tilde{G}^{\text{ex}}}{RT}\right|_{x_1=1} - \left.\frac{\tilde{G}^{\text{ex}}}{RT}\right|_{x_1=0}}_0 = \int_{P_2^{\text{vap}}}^{P_1^{\text{vap}}} \frac{\tilde{V}^{\text{ex}}}{RT}dP - \int_{T_2^{\text{sat}}}^{T_1^{\text{sat}}} \frac{\tilde{H}^{\text{ex}}}{RT^2}dT + \int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right)dx_1 \quad (8.8-7)$$

or

$$\int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right)dx_1 = - \int_{P_2^{\text{vap}}}^{P_1^{\text{vap}}} \frac{\tilde{V}^{\text{ex}}}{RT}dP + \int_{T_2^{\text{sat}}}^{T_1^{\text{sat}}} \frac{\tilde{H}^{\text{ex}}}{RT^2}dT. \quad (8.8-8)$$

8.8.1 Isothermal VLE Data

When the VLE measurements are made under constant temperature, Eqn (8.8-8) simplifies to

$$\int_0^1 \ln\left(\frac{\gamma_1}{\gamma_2}\right)dx_1 = - \int_{P_2^{\text{vap}}}^{P_1^{\text{vap}}} \frac{\tilde{V}^{\text{ex}}}{RT}dP. \quad (8.8-9)$$

¹⁶For a binary system, Gibbs phase rule indicates that $\mathcal{F} = 2$. Therefore, when temperature (or pressure) is fixed, compositions of the liquid and vapor phases cannot be changed without varying pressure (or temperature).

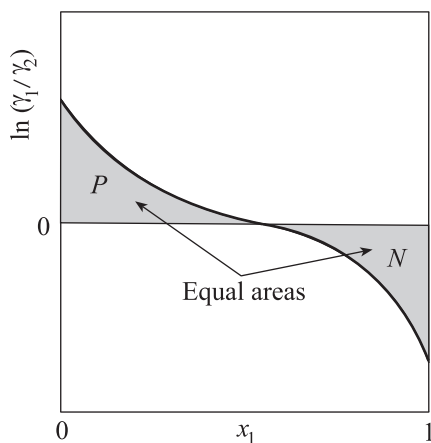


Figure 8.9: Area test for the thermodynamic consistency of isothermal VLE data.

Excess volume (or volume change on mixing) may be considered negligible for most systems. Under these conditions, the right-hand side of Eqn (8.8-9) is almost zero, i.e.

$$\int_0^1 \ln \left(\frac{\gamma_1}{\gamma_2} \right) dx_1 = 0. \quad (8.8-10)$$

Therefore, when the data are plotted in the form of $\ln(\gamma_1/\gamma_2)$ versus x_1 , as shown in Fig. 8.9, the areas above (P) and below (N) the x_1 -axis must be equal. In the literature, this is known as the *area test* of Redlich and Kister (1948).

As a result of errors involved in experiments, it is obvious that Eqn (8.8-10) can never be satisfied exactly. In the literature, the following criteria are used for consistent data:

$$A = 100 \left| \int_0^1 \ln \left(\frac{\gamma_1}{\gamma_2} \right) dx_1 \right| < 3 \quad (8.8-11)$$

or

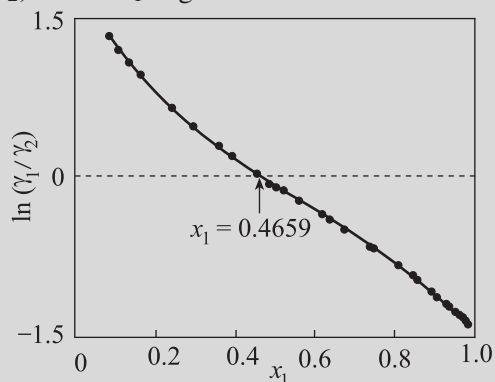
$$D = 100 \left[\frac{\left| \int_0^1 \ln \left(\frac{\gamma_1}{\gamma_2} \right) dx_1 \right|}{P + |N|} \right] < 10. \quad (8.8-12)$$

Example 8.15 The following isothermal VLE data are reported by Hiaki *et al.* (1998) for a binary system of 2-methyl-2-propanol (1) and octane (2) at 343.15 K. Check whether the experimental data are consistent.

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.0834	3.8957	1.0297	0.5039	1.3684	1.5106	0.8537	1.0395	2.7352
0.1061	3.4608	1.0448	0.5209	1.3493	1.5339	0.8920	1.0228	3.0081
0.1329	3.0925	1.0539	0.5598	1.2915	1.6061	0.9061	1.0169	3.1844
0.1610	2.8056	1.0626	0.6175	1.2184	1.7332	0.9285	1.0119	3.3567
0.2411	2.2293	1.1540	0.6372	1.2058	1.7932	0.9353	1.0106	3.4333
0.2939	1.9356	1.1962	0.6756	1.1626	1.8955	0.9533	1.0088	3.6277
0.3591	1.6969	1.2765	0.7372	1.1084	2.1265	0.9608	1.0070	3.7223
0.3896	1.6053	1.3185	0.7468	1.1011	2.1688	0.9684	1.0063	3.8161
0.4553	1.4540	1.4211	0.8083	1.0621	2.4505	0.9763	1.0053	3.8714
0.4872	1.3917	1.4831	0.8453	1.0435	2.6489	0.9839	1.0034	4.0228

Solution

The plot of $\ln(\gamma_1/\gamma_2)$ versus x_1 is given below.



Fitting a fifth-order polynomial to the data points yields

$$\ln \left(\frac{\gamma_1}{\gamma_2} \right) = 1.7576 - 5.6602x_1 + 4.6434x_1^2 + 0.5137x_1^3 - 4.8185x_1^4 + 2.1238x_1^5.$$

Thus,

$$\int_0^1 (1.7576 - 5.6602x_1 + 4.6434x_1^2 + 0.5137x_1^3 - 4.8185x_1^4 + 2.1238x_1^5) dx_1 = -6.0083 \times 10^{-3}.$$

From Eqn (8.8-11)

$$A = (100)(6.0083 \times 10^{-3}) = 0.6 < 3.$$

Note that $\ln(\gamma_1/\gamma_2) = 0$ when $x_1 = 0.4659$. Therefore, the positive and negative areas are

$$P = \int_0^{0.4659} (1.7576 - 5.6602x_1 + 4.6434x_1^2 + 0.5137x_1^3 - 4.8185x_1^4 + 2.1238x_1^5) dx_1 = 0.3496,$$

$$N = \int_{0.4659}^1 (1.7576 - 5.6602x_1 + 4.6434x_1^2 + 0.5137x_1^3 - 4.8185x_1^4 + 2.1238x_1^5) dx_1 = -0.3556.$$

The use of Eqn (8.8-12) gives

$$D = 100 \left(\frac{6.0083 \times 10^{-3}}{0.3496 + 0.3556} \right) = 0.852 < 10.$$

Therefore, the data are consistent.

8.8.2 Isobaric VLE Data

When the VLE measurements are carried out under constant pressure, Eqn (8.8-8) simplifies to

$$\int_0^1 \ln \left(\frac{\gamma_1}{\gamma_2} \right) dx_1 = \int_{T_2^{sat}}^{T_1^{sat}} \frac{\tilde{H}^{ex}}{RT^2} dT. \quad (8.8-13)$$

Herington (1951) evaluated the right-hand side of Eqn (8.8-13) as follows:

$$\begin{aligned} \int_{T_2^{sat}}^{T_1^{sat}} \frac{\tilde{H}^{ex}}{RT^2} dT &= - \int_{1/T_2^{sat}}^{1/T_1^{sat}} \frac{\tilde{H}^{ex}}{R} d \left(\frac{1}{T} \right), \\ &= \frac{\langle \tilde{H}^{ex} \rangle}{R} \left| \frac{1}{T_1^{sat}} - \frac{1}{T_2^{sat}} \right|, \\ &= \frac{\langle \Delta \tilde{H}_{mix} \rangle}{R} \left| \frac{1}{T_1^{sat}} - \frac{1}{T_2^{sat}} \right|, \end{aligned} \quad (8.8-14)$$

where $\langle \Delta \tilde{H}_{mix} \rangle$ is the average value of the heat of mixing in the total concentration range. By examining various binary systems, Herington proposed the following empirical criterion for thermodynamic consistency:

$$|D - J| < 10, \quad (8.8-15)$$

where D is defined by Eqn (8.8-12) and J is given by

$$J = 150 \left(\frac{T_{\max} - T_{\min}}{T_{\min}} \right) \quad (8.8-16)$$

in which $T_{\min}$ and $T_{\max}$ are the minimum and maximum temperatures (in K) in the range $0 \leq x_1 \leq 1$, respectively.

Example 8.16 The following isobaric VLE data are reported by Huang *et al.* (2004) for a binary system of tetraethyl orthosilicate (1) and ethanol (2) at 24 kPa. Check whether the experimental data are consistent.

T (K)	x_1	γ_1	γ_2	T (K)	x_1	γ_1	γ_2
319.5	0.0000	—	1.0000	350.5	0.8875	1.2140	1.7640
319.8	0.0430	5.9456	1.0239	355.8	0.9163	1.1772	1.7894
320.3	0.0661	3.9699	1.0280	358.5	0.9281	1.1395	1.8098
321.0	0.1080	2.5875	1.0333	366.0	0.9534	1.0995	1.8151
321.8	0.1623	2.3458	1.0525	371.7	0.9681	1.0859	1.8210
323.1	0.2224	1.8888	1.0607	375.0	0.9749	1.0575	1.8499
323.9	0.3402	1.5448	1.1949	376.6	0.9782	1.0563	1.8701
325.6	0.4322	1.3788	1.2712	382.0	0.9872	1.0553	1.8758
327.0	0.5064	1.3360	1.3577	384.9	0.9911	1.0436	1.9020
329.0	0.5931	1.3287	1.4806	387.5	0.9942	1.0339	1.9363
331.1	0.7131	1.3196	1.6928	390.7	0.9987	1.0215	3.8249
338.5	0.7875	1.3113	1.7300	393.4	1.0000	1.0000	—
342.0	0.8227	1.2878	1.7333				

Solution

The minimum and maximum temperatures are

$$T_{\min} = 319.5 \text{ K} \quad T_{\max} = 393.4 \text{ K}.$$

The use of Eqn (8.8-16) gives

$$J = 150 \left(\frac{393.4 - 319.5}{319.5} \right) = 34.7.$$

Fitting a fifth-order polynomial to the data points yields

$$\ln \left(\frac{\gamma_1}{\gamma_2} \right) = 2.413 - 20.043x_1 + 84.010x_1^2 - 182.492x_1^3 \\ + 186.021x_1^4 - 70.733x_1^5.$$

Thus,

$$\int_0^1 (2.413 - 20.043x_1 + 84.010x_1^2 - 182.492x_1^3 + 186.021x_1^4 - 70.733x_1^5) dx_1 = 0.187.$$

Note that $\ln(\gamma_1/\gamma_2) = 0$ when $x_1 = 0.499$. Therefore, the positive and negative areas are

$$\begin{aligned} P &= \int_0^{0.499} (2.413 - 20.043x_1 + 84.010x_1^2 - 182.492x_1^3 + 186.021x_1^4 - 70.733x_1^5) dx_1 = 0.329, \\ N &= \int_{0.499}^1 (2.413 - 20.043x_1 + 84.010x_1^2 - 182.492x_1^3 + 186.021x_1^4 - 70.733x_1^5) dx_1 = -0.141. \end{aligned}$$

The use of Eqn (8.8-12) gives

$$D = 100 \left(\frac{0.187}{0.329 + 0.141} \right) = 39.8.$$

From Eqn (8.8-15)

$$|D - J| = 39.8 - 34.7 = 5.1 < 10.$$

Therefore, the data are consistent.

One should keep in mind that even if the data satisfy the aforementioned tests, the reliability of experimental data set is still in question. Various other thermodynamic consistency tests are also available in the literature (Van Ness, 1995; Wisniak, 1993, 1994). As stated by Prausnitz *et al.* (1999), “The literature is rich with articles on testing for thermodynamic consistency because it is much easier to test someone else’s data than to obtain one’s own in the laboratory. Much (but by no means all) of this literature is obscured by excessive use of statistics. It has been said that *patriotism is the last refuge of a scoundrel*. Similarly, we might say that *statistics is the last refuge of a poor experimentalist* or, in a more positive way, that *a gram of good data is worth more than a ton of consistency tests*.”.

8.9 Concluding Remarks

As will be shown in the following chapters, calculation of activity coefficients is of critical importance in the design of separation equipment. The activity coefficient models mentioned in Sections 8.4, 8.5, and 8.6 are not the only ones available in the literature. Since there is no universal equation that can represent all types of liquid mixtures, there is an abundance of activity coefficient models. This brings up the question of how to choose the “best” activity coefficient model for a given system. Furthermore, the aforementioned models are all limited to binary mixtures. How one can calculate the activity coefficients for multicomponent mixtures?

8.9.1 Which Activity Coefficient Model to Use?

Simulation packages given in Appendix G provide guidelines in choosing the appropriate activity coefficient model. For moderately nonideal and polar solutions, the NRTL, Wilson, and van Laar models can be used. When the system is highly nonideal and polar, the NRTL model is preferred. For more details on the subject, the reader should refer to Carlson (1996), Zygula *et al.* (2001), Chen and Mathias (2002), Suppes (2002), and Poling *et al.* (2004).

8.9.2 Activity Coefficients for Multicomponent Mixtures

Most of the published data related to vapor-liquid, liquid-liquid, and solid-liquid equilibria are for binary systems. In that respect, efforts have been directed to extend available binary models to multicomponent systems so that the resulting equations require only the data for the binary subsystems. For example, the Wilson model for a multicomponent system is given by

$$\ln \gamma_i = 1 - \ln \left(\sum_{j=1}^k \Lambda_{ij} x_j \right) - \sum_{m=1}^k \left(\frac{\Lambda_{mi} x_m}{\sum_{j=1}^k \Lambda_{mj} x_j} \right), \quad (8.9-1)$$

where $\Lambda_{ij} = 1$ for $i = j$. In the case of a ternary system, Eqn (8.9-1) takes the form

$$\begin{aligned} \ln \gamma_i = 1 - \ln (\Lambda_{i1} x_1 + \Lambda_{i2} x_2 + \Lambda_{i3} x_3) - & \frac{\Lambda_{1i} x_1}{x_1 + \Lambda_{12} x_2 + \Lambda_{13} x_3} \\ - \frac{\Lambda_{2i} x_2}{\Lambda_{21} x_1 + x_2 + \Lambda_{23} x_3} - & \frac{\Lambda_{3i} x_3}{\Lambda_{31} x_1 + \Lambda_{32} x_2 + x_3}. \end{aligned} \quad (8.9-2)$$

Example 8.17 Kirss *et al.* (2005) reported the following binary Wilson parameters as a function of temperature:

$$\begin{aligned}\Lambda_{12} &= \exp\left(1.70016 - \frac{728.735}{T}\right) & \Lambda_{21} &= \exp\left(-1.03432 + \frac{155.701}{T}\right), \\ \Lambda_{13} &= \exp\left(2.17687 - \frac{839.958}{T}\right) & \Lambda_{31} &= \exp\left(-0.86126 + \frac{195.303}{T}\right), \\ \Lambda_{23} &= \exp\left(0.05676 - \frac{314.903}{T}\right) & \Lambda_{32} &= \exp\left(3.58544 - \frac{1774.377}{T}\right),\end{aligned}$$

where 1, 2, and 3 represent methylbutyl ketone, nonane, and cyclohexanol, respectively. For a ternary system, the experimental data at 408.88 K and 80 kPa indicate the liquid phase composition as

$$x_1 = 0.128 \quad x_2 = 0.256 \quad x_3 = 0.616.$$

- Determine the activity coefficients in a ternary mixture of methylbutyl ketone, nonane, and cyclohexanol.
- Estimate the vapor phase composition in equilibrium with the liquid.

Data: The vapor pressures of the components are given by

$$\begin{aligned}\ln P_1^{\text{vap}} &= 14.005 - \frac{3104.454}{T - 69.962}, \\ \ln P_2^{\text{vap}} &= 13.8546 - \frac{3224.816}{T - 74.824}, \\ \ln P_3^{\text{vap}} &= 13.7219 - \frac{2778.058}{T - 128.724},\end{aligned}$$

where T is in K and P^{vap} is in kPa.

Solution

- At 408.88 K, the binary Wilson parameters are

$$\begin{aligned}\Lambda_{12} &= 0.921 & \Lambda_{21} &= 0.520, \\ \Lambda_{13} &= 1.130 & \Lambda_{31} &= 0.681, \\ \Lambda_{23} &= 0.490 & \Lambda_{32} &= 0.470.\end{aligned}$$

Substitution of the numerical values into Eqn (8.9-2) gives the activity coefficients as

$$\begin{aligned}
\gamma_1 &= \exp \left\{ 1 - \ln [0.128 + (0.921)(0.256) + (1.130)(0.616)] \right. \\
&\quad - \frac{0.128}{0.128 + (0.921)(0.256) + (1.130)(0.616)} \\
&\quad - \frac{(0.520)(0.256)}{(0.520)(0.128) + 0.256 + (0.490)(0.616)} \\
&\quad \left. - \frac{(0.681)(0.616)}{(0.681)(0.128) + (0.470)(0.256) + 0.616} \right\} = 1.104, \\
\gamma_2 &= \exp \left\{ 1 - \ln [(0.520)(0.128) + 0.256 + (0.490)(0.616)] \right. \\
&\quad - \frac{(0.921)(0.128)}{0.128 + (0.921)(0.256) + (1.130)(0.616)} \\
&\quad - \frac{0.256}{(0.520)(0.128) + 0.256 + (0.490)(0.616)} \\
&\quad \left. - \frac{(0.470)(0.616)}{(0.681)(0.128) + (0.470)(0.256) + 0.616} \right\} = 1.819, \\
\gamma_3 &= \exp \left\{ 1 - \ln [(0.681)(0.128) + (0.470)(0.256) + 0.616] \right. \\
&\quad - \frac{(1.130)(0.128)}{0.128 + (0.921)(0.256) + (1.130)(0.616)} \\
&\quad - \frac{(0.490)(0.256)}{(0.520)(0.128) + 0.256 + (0.490)(0.616)} \\
&\quad \left. - \frac{0.616}{(0.681)(0.128) + (0.470)(0.256) + 0.616} \right\} = 1.115.
\end{aligned}$$

(b) At 408.88 K, the vapor pressures are

$$P_1^{\text{vap}} = 127.116 \text{ kPa} \quad P_2^{\text{vap}} = 66.758 \text{ kPa} \quad P_3^{\text{vap}} = 44.961 \text{ kPa}.$$

The use of Eqn (8.3-13) gives the vapor phase composition as

$$\begin{aligned}
y_1 &= \frac{\gamma_1 x_1 P_1^{\text{vap}}}{P} = \frac{(1.104)(0.128)(127.116)}{80} = 0.225, \\
y_2 &= \frac{\gamma_2 x_2 P_2^{\text{vap}}}{P} = \frac{(1.819)(0.256)(66.758)}{80} = 0.389, \\
y_3 &= \frac{\gamma_3 x_3 P_3^{\text{vap}}}{P} = \frac{(1.115)(0.616)(44.961)}{80} = 0.386.
\end{aligned}$$

Problems

Problem related to Section 8.1

- 8.1** A beaker contains 20 mol of pure water at 298 K. Calculate the change in Gibbs energy when 0.05 mol of solid X is dissolved in water under isothermal conditions. Assume that the mixture behaves as an ideal mixture and $\tilde{G}_X = 300 \text{ J/mol}$.

(Answer: -851.2 J)

Problems related to Section 8.2

- 8.2** A binary liquid mixture of components 1 and 2 is to be heated at constant pressure from T_1 to T_2 . The molar excess Gibbs energy of the mixture is represented by

$$\tilde{G}^{\text{ex}} = Ax_1x_2$$

in which the parameter A is given as a function of temperature in the form

$$A = \alpha + \frac{\beta}{T}.$$

Under isobaric conditions, show that the change in molar entropy of the system is given by

$$\tilde{S}_{\text{sys}}(T_2, P) - \tilde{S}_{\text{sys}}(T_1, P) = \tilde{C}_{P_{\text{mix}}} \ln \left(\frac{T_2}{T_1} \right) + \beta x_1x_2 \left(\frac{1}{T_2^2} - \frac{1}{T_1^2} \right),$$

where

$$\tilde{C}_{P_{\text{mix}}} = x_1\tilde{C}_{P_1} + x_2\tilde{C}_{P_2}.$$

- 8.3** A stream of liquid 1 at 300 K with a flow rate of 2 mol/s and a stream of liquid 2 at 340 K with a flow rate of 3 mol/s are mixed in an adiabatic mixer in a steady flow process. The molar excess Gibbs energy of the mixture is represented by

$$\tilde{G}^{\text{ex}} = 1500x_1x_2.$$

- (a) Estimate the temperature of the stream exiting the mixer if the heat capacities for pure components 1 and 2 are 40 J/mol K and 60 J/mol K, respectively.

(b) Determine the rate of entropy change as a result of the mixing process.

(Answer: (a) 320.8 K and (b) 22.9 W/K)

8.4 Excess molar heat capacity at constant pressure is defined by

$$\tilde{C}_P^{\text{ex}} = \left(\frac{\partial \tilde{H}^{\text{ex}}}{\partial T} \right)_{P, x_j} = T \left(\frac{\partial \tilde{S}^{\text{ex}}}{\partial T} \right)_{P, x_j}. \quad (1)$$

(a) Show that the substitution of Eqn (8.2-19) into Eqn (1) gives

$$\tilde{C}_P^{\text{ex}} = -T \left(\frac{\partial^2 \tilde{G}^{\text{ex}}}{\partial T^2} \right)_{P, x_j}. \quad (2)$$

(b) If $\tilde{C}_P^{\text{ex}}$ is represented by

$$\frac{\tilde{C}_P^{\text{ex}}}{R} = a + bT + cT^2 \quad (3)$$

then show that, at constant pressure and composition, molar excess enthalpy and molar excess Gibbs energy are expressed as

$$\frac{\tilde{H}^{\text{ex}}}{R} = aT + \frac{b}{2}T^2 + \frac{c}{3}T^3 + I_1 \text{ and } \frac{\tilde{G}^{\text{ex}}}{RT} = -a \ln T - \frac{b}{2}T - \frac{c}{6}T^2 + \frac{I_1}{T} + I_2, \quad (4)$$

where I_1 and I_2 are integration constants.

Problems related to Section 8.3

8.5 For a binary mixture of ethanol (1) and cyclohexane (2), the following heat of mixing data are reported by Lien *et al.* (2004) at 348.15 K:

x_1	$\Delta \tilde{H}_{\text{mix}} \text{ (J/mol)}$	x_1	$\Delta \tilde{H}_{\text{mix}} \text{ (J/mol)}$
0.102	704	0.550	1198
0.151	840	0.600	1147
0.198	941	0.650	1076
0.250	1036	0.700	978
0.300	1110	0.750	858
0.350	1167	0.800	717
0.401	1205	0.849	559
0.450	1222	0.900	391
0.500	1224	0.949	226

- (a) Fit the data to an equation of the form

$$\Delta \tilde{H}_{\text{mix}} = x_1 x_2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1} \quad (1)$$

and show that $A_1 = 4925$, $A_2 = -327$, $A_3 = -550$, $A_4 = -2643$, and $A_5 = 3608$.

- (b) Use Eqn (6.3-15) and show that the partial molar excess enthalpy of components 1 and 2 are given in the form

$$\bar{H}_1^{\text{ex}} = x_2^2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1} + 2x_1 x_2^2 \sum_{i=1}^4 i A_{i+1} (x_1 - x_2)^{i-1}. \quad (2)$$

$$\bar{H}_2^{\text{ex}} = x_1^2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1} - 2x_1^2 x_2 \sum_{i=1}^4 i A_{i+1} (x_1 - x_2)^{i-1}. \quad (3)$$

- (c) Evaluate $\bar{H}_1^{\text{ex}}$ and $\bar{H}_2^{\text{ex}}$ when $x_1 = 0.55$.

- (d) Show that

$$\bar{H}_1^{\text{ex},\infty} = \sum_{i=1}^5 (-1)^{i-1} A_i \quad \text{and} \quad \bar{H}_2^{\text{ex},\infty} = \sum_{i=1}^5 A_i \quad (4)$$

and evaluate $\bar{H}_1^{\text{ex},\infty}$ and $\bar{H}_2^{\text{ex},\infty}$.

(Answer: (c) $\bar{H}_1^{\text{ex}} = 877.3$ J/mol, $\bar{H}_2^{\text{ex}} = 1614$ J/mol and (d) $\bar{H}_1^{\text{ex},\infty} = 10,953$ J/mol, $\bar{H}_2^{\text{ex},\infty} = 5013$ J/mol)

8.6 Fenclova *et al.* (2002) reported the following data for a binary system of trichloromethane (1) and tetrahydropyran (2) at 333.15 K. Plot $\tilde{G}^{\text{ex}}/RT$ versus x_1 and evaluate the activity coefficients at $x_1 = 0.4$ graphically.

x_1	$\tilde{G}^{\text{ex}}$ (J/mol)	x_1	$\tilde{G}^{\text{ex}}$ (J/mol)	x_1	$\tilde{G}^{\text{ex}}$ (J/mol)
0.0000	0	0.4242	-788.0	0.7631	-593.3
0.0298	-83.1	0.5308	-814.6	0.8453	-425.2
0.0736	-203.0	0.6244	-769.5	0.9458	-161.5
0.1363	-357.6	0.6825	-720.2	0.9767	-69.0
0.2150	-525.2	0.6903	-703.3	1.0000	0
0.3149	-685.5	0.7101	-669.7		

(Answer: $\gamma_1 = 0.63$, $\gamma_2 = 0.85$)

8.7 For a ternary system represented by the two-suffix Margules model, the molar excess Gibbs energy is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = A_{12}x_1x_2 + A_{13}x_1x_3 + A_{23}x_2x_3.$$

Use Eqn (8.3-18) and show that the activity coefficients are given by

$$\begin{aligned}\ln \gamma_1 &= A_{12}x_2^2 + A_{13}x_3^2 + (A_{12} + A_{13} - A_{23})x_2x_3, \\ \ln \gamma_2 &= A_{12}x_1^2 + A_{23}x_3^2 + (A_{12} + A_{23} - A_{13})x_1x_3, \\ \ln \gamma_3 &= A_{13}x_1^2 + A_{23}x_2^2 + (A_{13} + A_{23} - A_{12})x_1x_2.\end{aligned}$$

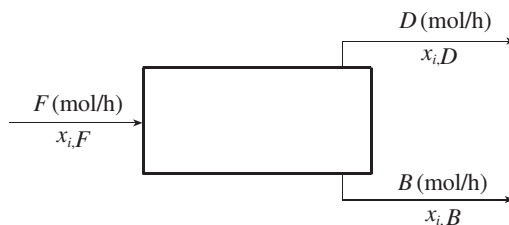
8.8 For a binary system of components 1 and 2, the molar excess Gibbs energy is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1x_2 (Ax_2 + Bx_1 - Dx_1x_2).$$

Show that the activity coefficients are expressed as

$$\begin{aligned}\ln \gamma_1 &= x_2^2 [A + 2(B - A - D)x_1 + 3Dx_1^2], \\ \ln \gamma_2 &= x_1^2 [B + 2(A - B - D)x_2 + 3Dx_2^2].\end{aligned}$$

8.9 Consider the separation of a multicomponent mixture as shown in the figure below.



(a) Show that the minimum amount of work required for this separation process taking place at constant temperature and pressure is given by

$$\begin{aligned}\frac{W}{F} &= RT \left[\frac{D}{F} \sum_{i=1}^k x_{i,D} \ln(x_{i,D} \gamma_{i,D}) + \frac{B}{F} \sum_{i=1}^k x_{i,B} \ln(x_{i,B} \gamma_{i,B}) \right. \\ &\quad \left. - \sum_{i=1}^k x_{i,F} \ln(x_{i,F} \gamma_{i,F}) \right].\end{aligned}$$

- (b) Show that the minimum work required for the complete separation of an equimolar ideal binary mixture into its pure components at constant temperature and pressure is given by

$$\frac{W}{F} = 0.693 RT.$$

8.10 In the development of Eqn (8.3-13), all nonidealities associated with the vapor phase are neglected and the Poynting correction factor is considered unity. Consider a binary mixture and assume that the vapor phase nonidealities are represented by the virial equation of state.

- (a) Show that Eqn (8.3-12) takes the form

$$\gamma_1 = \frac{y_1 P}{x_1 P_1^{\text{vap}}} \exp \left[\frac{(B_{11} - \tilde{V}_1^L)(P - P_1^{\text{vap}}) + y_2^2(2B_{12} - B_{11} - B_{22})P}{RT} \right]. \quad (1)$$

$$\gamma_2 = \frac{y_2 P}{x_2 P_2^{\text{vap}}} \exp \left[\frac{(B_{22} - \tilde{V}_2^L)(P - P_2^{\text{vap}}) + y_1^2(2B_{12} - B_{11} - B_{22})P}{RT} \right]. \quad (2)$$

- (b) For a binary mixture of *p*-dioxane (1) and acetonitrile (2), the following isothermal VLE data are reported by Davollo *et al.* (1981) at 298.15 K:

x_1	y_1	$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$
0.0000	0.0000	11.87	0.5170	0.3002	9.14	0.8988	0.7262	6.05
0.1125	0.0640	11.33	0.5506	0.3235	8.91	0.9300	0.7965	5.69
0.2000	0.1125	10.90	0.6470	0.4013	8.27	1.0000	1.0000	4.83
0.3070	0.1715	10.30	0.7465	0.4971	7.52			
0.4005	0.2272	9.81	0.8130	0.5812	6.94			

With the help of the following data

$$\begin{aligned} \tilde{V}_1^L &= 85.78 \text{ cm}^3/\text{mol} & \tilde{V}_2^L &= 52.85 \text{ cm}^3/\text{mol}, \\ B_{11} &= -1948 \text{ cm}^3/\text{mol} & B_{22} &= -3722 \text{ cm}^3/\text{mol} \\ B_{12} &= -1499 \text{ cm}^3/\text{mol}. \end{aligned}$$

calculate the activity coefficients using Eqns (1) and (2).

- (c) Fit the data to the following equation

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \sum_{i=1}^4 A_i (x_1 - x_2)^{i-1},$$

and show that $A_1 = 0.414$, $A_2 = 0.080$, $A_3 = 0.051$, and $A_4 = -0.043$.

Problems related to Section 8.4

8.11 Loras *et al.* (1999) reported the following VLE data for the system of 3-methylpentane (1) and methyl 1,1-dimethylpropyl ether (2) at 101.3 kPa:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.026	1.176	1.000	0.370	1.082	1.026	0.722	1.014	1.100
0.061	1.192	1.000	0.432	1.063	1.036	0.765	1.011	1.110
0.126	1.167	1.002	0.488	1.053	1.042	0.811	1.006	1.132
0.181	1.132	1.002	0.508	1.050	1.041	0.858	1.001	1.144
0.235	1.126	1.008	0.565	1.039	1.052	0.905	1.000	1.168
0.289	1.100	1.013	0.620	1.029	1.063	0.950	0.998	1.192
0.329	1.091	1.021	0.674	1.022	1.085	0.978	0.999	1.198

If the system is represented by the two-suffix Margules equation, estimate the parameter A that minimizes the objective function

$$F = \sum_{i=1}^{21} \left(\frac{\tilde{G}^{ex}}{RT} - A x_1 x_2 \right)_i^2.$$

(Answer: $A = 0.183$)

8.12 For the three-suffix Margules equation, molar excess Gibbs energy is sometimes expressed as

$$\frac{\tilde{G}^{ex}}{RT} = x_1 x_2 (A_{21} x_1 + A_{12} x_2). \quad (1)$$

(a) Compare Eqn (1) with Eqn (8.4-5) and conclude that

$$A_{12} = A - B \quad \text{and} \quad A_{21} = A + B. \quad (2)$$

(b) Show that

$$\ln \gamma_1 = x_2^2 [A_{12} + 2(A_{21} - A_{12})x_1], \quad (3)$$

$$\ln \gamma_2 = x_1^2 [A_{21} + 2(A_{12} - A_{21})x_2]. \quad (4)$$

Note that when $A_{12} = A_{21}$, Eqns (3) and (4) are equivalent to the two-suffix Margules equations.

8.13 A binary mixture of components 1 and 2 is represented by the three-suffix Margules equation with temperature dependent parameters expressed in the form

$$A = a_1 + \frac{a_2}{T} \quad \text{and} \quad B = b_1 + \frac{b_2}{T}. \quad (1)$$

(a) Show that the infinite-dilution activity coefficients are expressed as

$$\ln \gamma_1^\infty = (a_1 - b_1) + \frac{a_2 - b_2}{T}, \quad (2)$$

$$\ln \gamma_2^\infty = (a_1 + b_1) + \frac{a_2 + b_2}{T}. \quad (3)$$

(b) Show that the heat of mixing is given by

$$\frac{\Delta \tilde{H}_{\text{mix}}}{R} = x_1 x_2 [(a_2 + b_2)x_1 + (a_2 - b_2)x_2]. \quad (4)$$

(c) For a mixture of mono-ethanol amine (1) and water (2), [Kundu and Bandyopadhyay \(2007\)](#) reported the following data:

$$a_1 = 1.531 \quad a_2 = -828.4 \text{ K} \quad b_1 = 0.253 \quad b_2 = -13.84 \text{ K}.$$

Estimate γ_1^∞ and γ_2^∞ at 313 K.

(Answer: (c) $\gamma_1^\infty = 0.266$, $\gamma_2^\infty = 0.404$)

8.14 For a binary mixture of di-isopropyl ether (1) and benzene (2), the following isothermal VLE data are reported by [Villamanan *et al.* \(2006\)](#) at 313.15 K:

x_1	y_1	$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$	x_1	y_1	$P(\text{kPa})$
0.0000	0.0000	24.398	0.3992	0.5040	30.794	0.6485	0.7231	33.642
0.0504	0.0878	25.442	0.4489	0.5503	31.400	0.6991	0.7646	34.184
0.1011	0.1652	26.387	0.4501	0.5514	31.405	0.7493	0.8052	34.698
0.1498	0.2318	27.221	0.4991	0.5954	31.988	0.7978	0.8438	35.174
0.1994	0.2936	28.019	0.5003	0.5965	31.995	0.8508	0.8854	35.707
0.2497	0.3515	28.770	0.5490	0.6391	32.563	0.9078	0.9296	36.253
0.2998	0.4054	29.481	0.5499	0.6399	32.565	0.9640	0.9727	36.790
0.3500	0.4563	30.155	0.5986	0.6814	33.113	1.0000	1.0000	37.126
0.3992	0.5039	30.785	0.6005	0.6830	33.127			

(a) With the help of the following data

$$\begin{aligned} \tilde{V}_1^L &= 145 \text{ cm}^3/\text{mol} & \tilde{V}_2^L &= 91 \text{ cm}^3/\text{mol}, \\ B_{11} &= -1687.8 \text{ cm}^3/\text{mol} & B_{22} &= -1310.5 \text{ cm}^3/\text{mol} \\ B_{12} &= -1701.0 \text{ cm}^3/\text{mol}. \end{aligned}$$

calculate the activity coefficients using Eqns (1) and (2) of Problem 8.10.

(b) If the activity coefficients are represented by the three-suffix Margules equation, plot $(\tilde{G}^{\text{ex}}/RT)/x_1 x_2$ versus x_1 and show that the Margules parameters are given by

$$A = 0.165 \quad \text{and} \quad B = -0.044.$$

- (c) Show that the linearization of Eqn (8.4-6), i.e. a plot of $\ln \gamma_1/x_2^2$ versus x_2 , yields

$$A = 0.162 \quad \text{and} \quad B = -0.035.$$

- (d) Show that the linearization of Eqn (8.4-7), i.e. a plot of $\ln \gamma_2/x_1^2$ versus x_1 , yields

$$A = 0.158 \quad \text{and} \quad B = -0.060.$$

- (e) Villamanan *et al.* (2006) defined an objective function as

$$F = \sum_{i=1}^{24} \left(\frac{P - P_{calc}}{P} \right)_i^2,$$

where

$$P_{calc} = P_1^{\text{vap}} x_1 \gamma_1 + P_2^{\text{vap}} x_2 \gamma_2.$$

By minimizing this objective function, they evaluated the parameters A and B as

$$A = 0.171 \quad \text{and} \quad B = -0.060.$$

Compare the predictions of activity coefficients using the parameters obtained from four different approaches.

8.15 The molar excess Gibbs energy expression for the van Laar model, Eqn (8.4-12), is developed as follows:

- (a) Assuming $\tilde{S}^{\text{ex}} = \tilde{V}^{\text{ex}} = 0$, show that

$$\tilde{G}^{\text{ex}} = \tilde{U}^{\text{ex}}. \quad (1)$$

- (b) By definition

$$\tilde{U}^{\text{ex}} = \tilde{U}_{\text{mix}} - \sum_{i=1}^k x_i \tilde{U}_i. \quad (2)$$

Rearrange Eqn (2) in the form

$$\begin{aligned} \tilde{U}^{\text{ex}} = & \left(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}} \right) + \left(\tilde{U}^{\text{IGM}} - \sum_{i=1}^k x_i \tilde{U}_i^{\text{IG}} \right) \\ & - \sum_{i=1}^k x_i \left(\tilde{U}_i - \tilde{U}_i^{\text{IG}} \right). \end{aligned} \quad (3)$$

Note that the second term on the right-hand side of Eqn (3) is zero. For the van der Waals equation of state, Eqns (6.1-33) and (3.6-14) give

$$\tilde{U}_{\text{mix}} - \tilde{U}^{IGM} = -\frac{A_{\text{mix}}RT}{Z_{\text{mix}}} = -\frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}}, \quad (4)$$

$$\tilde{U}_i - \tilde{U}_i^{IG} = -\frac{A_iRT}{Z_i} = -\frac{a_i}{\tilde{V}_i}. \quad (5)$$

Substitute Eqns (4) and (5) into Eqn (3) to obtain

$$\tilde{U}^{\text{ex}} = \tilde{G}^{\text{ex}} = -\frac{a_{\text{mix}}}{\tilde{V}_{\text{mix}}} + \sum_{i=1}^k \frac{x_i a_i}{\tilde{V}_i}. \quad (6)$$

(c) Assuming incompressible liquids, i.e.

$$\tilde{V}_i = b_i \quad \text{and} \quad \tilde{V}_{\text{mix}} = \sum_{i=1}^k x_i b_i, \quad (7)$$

show that Eqn (6) takes the form

$$\tilde{G}^{\text{ex}} = \sum_{i=1}^k \frac{x_i a_i}{b_i} - \frac{\sum_{i=1}^k \sum_{j=1}^k x_i x_j a_{ij}}{\sum_{i=1}^k x_i b_i}, \quad (8)$$

where

$$a_{ij} = \sqrt{a_{ii}a_{jj}}. \quad (9)$$

(d) Show that the rearrangement of Eqn (8) gives

$$\tilde{G}^{\text{ex}} = \frac{\sum_{i=1}^k \sum_{j=1}^k x_i x_j \left(\frac{a_i b_j}{b_i} - a_{ij} \right)}{\sum_{i=1}^k x_i b_i}. \quad (10)$$

(e) For a binary mixture show that Eqn (10) becomes

$$\tilde{G}^{\text{ex}} = \frac{x_1 x_2 b_1 b_2}{x_1 b_1 + x_2 b_2} \left(\frac{\sqrt{a_1}}{b_1} - \frac{\sqrt{a_2}}{b_2} \right)^2. \quad (11)$$

(f) Define parameters A and B as

$$A = \frac{b_1}{RT} \left(\frac{\sqrt{a_1}}{b_1} - \frac{\sqrt{a_2}}{b_2} \right)^2 \quad B = \frac{b_2}{RT} \left(\frac{\sqrt{a_1}}{b_1} - \frac{\sqrt{a_2}}{b_2} \right)^2 \quad (12)$$

and show that Eqn (11) reduces to

$$\tilde{G}^{\text{ex}} = x_1 x_2 \left(\frac{AB}{Ax_1 + Bx_2} \right). \quad (13)$$

which is Eqn (8.4-12).

8.16 When $A = B$, show that the van Laar and the two-suffix Margules models become identical.

8.17 For a binary mixture of ethanol (1) and benzene (2) at 343 K, the van Laar parameters are $A = 1.946$ and $B = 1.610$.

- Calculate the activity coefficients for a mixture containing 30 mol % ethanol.
- Calculate the activity coefficients at infinite dilution.
- Using the values of the activity coefficients at infinite dilution, determine the parameters of the three-suffix Margules equation.
- Use the Margules equation and calculate the activity coefficients for a mixture containing 30 mol % ethanol.

(Answer: (a) $\gamma_1 = 2.327$, $\gamma_2 = 1.206$ (b) $\gamma_1^\infty = 7$, $\gamma_2^\infty = 5$ and (c) $A = 1.778$, $B = -0.168$ d) $\gamma_1 = 2.351$, $\gamma_2 = 1.206$)

8.18 For a binary mixture of ethanol (1) and water (2), the following vapor-liquid equilibrium data have been reported by [Pemberton and Mash \(1978\)](#) at 303.15 K:

$P(\text{bar})$	x_1	y_1	$P(\text{bar})$	x_1	y_1
0.07329	0.10991	0.4743	0.09999	0.63434	0.7329
0.08723	0.24688	0.5907	0.10341	0.80840	0.8337
0.09303	0.38655	0.6404	0.10445	0.92444	0.9284
0.09663	0.50492	0.6797			

Evaluate the parameters of the van Laar activity coefficient model graphically by plotting $x_1/(\tilde{G}^{\text{ex}}/RT)$ versus x_1/x_2 .

(Answer: $A = 1.785$, $B = 0.916$)

8.19 If the activity coefficients are represented by the Wilson equations, for an equimolar mixture show that

$$\ln \gamma_1^* = - \left\{ \ln [0.5 (1 + \Lambda_{12})] + \left(\frac{\Lambda_{21}}{1 + \Lambda_{21}} - \frac{\Lambda_{12}}{1 + \Lambda_{12}} \right) \right\}, \quad (1)$$

$$\ln \gamma_2^* = - \left\{ \ln [0.5 (1 + \Lambda_{21})] + \left(\frac{\Lambda_{12}}{1 + \Lambda_{12}} - \frac{\Lambda_{21}}{1 + \Lambda_{21}} \right) \right\}, \quad (2)$$

where γ_1^* and γ_2^* represent the activity coefficients at $x_1 = x_2 = 0.5$. Add Eqns (1) and (2) to get

$$\Lambda_{21} = \frac{4}{\gamma_1^* \gamma_2^* (1 + \Lambda_{12})} - 1. \quad (3)$$

Show that the substitution Eqn (3) into Eqn (1) leads to

$$\ln \left[\frac{\gamma_1^* (1 + \Lambda_{12})}{2} \right] = \frac{\Lambda_{12}}{1 + \Lambda_{12}} + \frac{\gamma_1^* \gamma_2^* (1 + \Lambda_{12})}{4} - 1. \quad (4)$$

The value of Λ_{12} is obtained from the solution of this transcendental equation. Once Λ_{12} is determined, Λ_{21} can be evaluated from Eqn (3).

8.20 Prasad *et al.* (2007) reported the following vapor-liquid equilibrium data for a binary mixture of trichloroethylene (1) and *p*-cresol (2) at 95.23 kPa:

$T(K)$	x_1	y_1	$T(K)$	x_1	y_1	$T(K)$	x_1	y_1
472.85	0.0000	0.0000	386.55	0.3002	0.9647	365.25	0.7003	0.9931
441.15	0.0512	0.6229	378.75	0.3936	0.9782	362.55	0.8130	0.9954
422.05	0.1002	0.8128	372.45	0.5002	0.9861	360.75	0.9004	0.9970
392.85	0.2450	0.9500	367.65	0.6210	0.9910	358.35	1.0000	1.0000

The vapor pressures of pure components are given by the Antoine equation as

$$\ln P_1^{\text{vap}} = 14.1654 - \frac{3028.13}{T - 43.15} \quad \ln P_2^{\text{vap}} = 14.1815 - \frac{3479.39}{T - 111.30},$$

where P is in kPa and T is in K. The densities of trichloroethylene and *p*-cresol are 1464 and 1154 kg/m³, respectively.

- Calculate the activity coefficients as a function of composition.
- Use the equations given in Problem 8.19 to evaluate the Wilson parameters. Note that Prasad *et al.* (2007) evaluated the Wilson parameters as

$$\frac{\lambda_{12} - \lambda_{11}}{R} = -103.56 \text{ K} \quad \text{and} \quad \frac{\lambda_{21} - \lambda_{22}}{R} = 630.97 \text{ K}$$

by minimizing the objective function defined as

$$F = \sum_{i=1}^{10} \left(\frac{P - P_{\text{calc}}}{P} \right)_i^2.$$

(Answer: (b) $(\lambda_{12} - \lambda_{11})/R = -77.1$ K, $(\lambda_{21} - \lambda_{22})/R = 590.4$ K)

8.21 For a binary system of ethanol (1) and methyl butanoate (2) at 346.3 K, [Constantinescu and Wichterle \(2002\)](#) reported the following NRTL parameters:

$$\tau_{12} = 0.356 \quad \tau_{21} = 0.460 \quad \alpha = -1.0914.$$

Estimate the infinite dilution activity coefficients.

(Answer: $\gamma_1^\infty = 2.678$ $\gamma_2^\infty = 3.053$)

8.22 Note that the molar excess enthalpy can be calculated from Eqn (8.2-19), i.e.

$$\frac{\tilde{H}^{\text{ex}}}{R} = -T^2 \left[\frac{\partial}{\partial T} \left(\frac{\tilde{G}^{\text{ex}}}{RT} \right) \right]_{P, x_j}. \quad (1)$$

(a) For the NRTL model, let the parameters τ_{12} and τ_{21} be dependent on temperature in the form

$$\tau_{12} = a_{12} + \frac{b_{12}}{T} \quad \text{and} \quad \tau_{21} = a_{21} + \frac{b_{21}}{T}. \quad (2)$$

Show that the substitution of Eqn (8.4-26) into Eqn (1) leads to

$$\frac{\tilde{H}^{\text{ex}}}{R} = x_1 x_2 \left\{ \frac{G_{21} b_{21} [x_1 (1 - \alpha \tau_{21}) + G_{21} x_2]}{(x_1 + G_{21} x_2)^2} + \frac{G_{12} b_{12} [x_2 (1 - \alpha \tau_{12}) + G_{12} x_1]}{(x_2 + G_{12} x_1)^2} \right\} \quad (3)$$

For a binary system of 2-methylamino ethanol (1) and water (2), [Mundhwa and Henni \(2007\)](#) reported the following parameters:

$$\tau_{12} = -67.695 - \frac{159.767}{T} \quad \tau_{21} = 20.123 - \frac{1265.64}{T}$$

where T is in K. If the nonrandomness parameter is $\alpha = 0.019$, calculate the molar excess enthalpy at 298.15 K as a function of composition and compare with the following experimental data:

x_1	$\tilde{H}^{\text{ex}}(\text{J/mol})$	x_1	$\tilde{H}^{\text{ex}}(\text{J/mol})$	x_1	$\tilde{H}^{\text{ex}}(\text{J/mol})$
0.0302	-530	0.1992	-2243	0.5017	-2508
0.0514	-902	0.3021	-2671	0.5992	-2148
0.0695	-1163	0.3471	-2750	0.6960	-1707
0.1002	-1480	0.3996	-2732	0.7988	-1219
0.1485	-1919	0.4471	-2667	0.8939	-688

- (b) For a binary system represented by the Wilson equation, show that the substitution of Eqn (8.4-18) into Eqn (1) leads to

$$\tilde{H}^{\text{ex}} = x_1 x_2 \left[\frac{(\lambda_{12} - \lambda_{11})\Lambda_{12}}{x_1 + \Lambda_{12}x_2} + \frac{(\lambda_{21} - \lambda_{22})\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right]. \quad (4)$$

For a binary system of formamide (1) and ethanol (2), Zarei and Iloukhani (2003) reported the following parameters:

$$\begin{aligned} \lambda_{12} - \lambda_{11} &= 2733.86 \text{ J/mol} & \lambda_{21} - \lambda_{22} &= 1271.70 \text{ J/mol} \\ \tilde{V}_1^L &= 44.87 \text{ cm}^3/\text{mol} & \tilde{V}_2^L &= 58.35 \text{ cm}^3/\text{mol}. \end{aligned}$$

Calculate the molar excess enthalpy at 298.15 K as a function of composition and compare with the following experimental data:

x_1	$\tilde{H}^{\text{ex}}(\text{J/mol})$	x_1	$\tilde{H}^{\text{ex}}(\text{J/mol})$
0.0880	250.1	0.5457	619.3
0.1808	426.1	0.6065	589.4
0.2731	542.1	0.7001	516.2
0.3620	606.9	0.8001	397.5
0.4524	632.7	0.9014	223.7

Problems related to Section 8.5

8.23 The activity coefficient expressions for a regular mixture, Eqns (8.5-2) and (8.5-3), are developed as follows:

- (a) Assuming $\tilde{S}^{\text{ex}} = \tilde{V}^{\text{ex}} = 0$, follow the procedure outlined in Problem 8.15 and show that

$$\tilde{G}^{\text{ex}} = \tilde{U}^{\text{ex}} = \left(\tilde{U}_{\text{mix}} - \tilde{U}^{\text{IGM}} \right) - \sum_{i=1}^k x_i \left(\tilde{U}_i - \tilde{U}_i^{\text{IG}} \right). \quad (1)$$

- (b) In the van Laar model, the terms on the right-hand side of Eqn (1) are evaluated with the help of the van der Waals equation of state. The Scatchard-Hildebrand theory, on the other hand, introduces the concept of *cohesive energy density*, C_{ii} . Cohesive energy density (in J/cm³) is the change in internal energy of species i in going from the liquid to the ideal gas state, i.e.

$$C_{ii} = \frac{\tilde{U}_i^{IG} - \tilde{U}_i}{\tilde{V}_i^L}. \quad (2)$$

In other words, it is the energy required to vaporize a liquid per unit liquid volume.

- (c) For the mixtures, extension of Eqn (2) is given by

$$C_{\text{mix}} = \frac{\tilde{U}_i^{IGM} - \tilde{U}_{\text{mix}}}{\tilde{V}_{\text{mix}}^L}, \quad (3)$$

where C_{mix} is defined by

$$C_{\text{mix}} = \sum_{i=1}^k \sum_{j=1}^k \Phi_i \Phi_j C_{ij} \quad (4)$$

in which

$$C_{ij} = \sqrt{C_{ii} C_{jj}}. \quad (5)$$

- (d) Since $\tilde{V}^{\text{ex}} = 0$, conclude that

$$\tilde{V}_{\text{mix}}^L = x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L. \quad (6)$$

- (e) For a binary mixture, show that the substitution of Eqns (2)–(6) into Eqn (1) leads to

$$\begin{aligned} \tilde{G}^{\text{ex}} = & - \left(\Phi_1^2 C_{11} + 2 \Phi_1 \Phi_2 \sqrt{C_{11} C_{22}} + \Phi_2^2 C_{22} \right) \left(x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L \right) \\ & + x_1 \tilde{V}_1^L C_{11} + x_2 \tilde{V}_2^L C_{22}. \end{aligned} \quad (7)$$

- (f) The relationship between the volume and mole fractions is given by

$$\Phi_1 = \frac{x_1 \tilde{V}_1^L}{x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L} \quad \Phi_2 = \frac{x_2 \tilde{V}_2^L}{x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L}. \quad (8)$$

Show that the use of Eqn (8) in Eqn (7) results in

$$\tilde{G}^{\text{ex}} = \left(x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L \right) \Phi_1 \Phi_2 \left(\sqrt{C_{11}} - \sqrt{C_{22}} \right)^2 \quad (9)$$

- (g) The solubility parameter, δ_i , is defined as the square root of the cohesive energy density, i.e.

$$\delta_i = \sqrt{C_{ii}} \quad (10)$$

so that Eqn (9) takes the form

$$\tilde{G}^{\text{ex}} = (x_1 \tilde{V}_1^L + x_2 \tilde{V}_2^L) \Phi_1 \Phi_2 (\delta_1 - \delta_2)^2 \quad (11)$$

Use Eqn (11) in Eqns (8.3-19) and (8.3-20) to obtain Eqns (8.5-2) and (8.5-3).

- (h) At temperatures well below the critical temperature, Eqn (2) can be expressed as

$$C_{ii} = \frac{\Delta \tilde{H}_i^{\text{vap}} - RT}{\tilde{V}_i^L} \quad (12)$$

Using Eqns (10) and (12), estimate the solubility parameter and compare it with the value given in Table 8.1. Enthalpy of vaporization of benzene at 298 K is 33.83 kJ/mol.

(Answer: (h) $18.6 \text{ (J/cm}^3\text{)}^{1/2}$)

- 8.24** The solubility of solid biphenyl (2) in liquid pyridine (1) at 297.3 K is reported by Choi and McLaughlin (1983) as $x_2 = 0.3742$. Using the following data, estimate the activity coefficient of biphenyl in the liquid mixture.

$$\begin{aligned} \tilde{V}_1^L &= 84.87 \text{ cm}^3/\text{mol} & \delta_1 &= 20.259 \text{ (J/cm}^3\text{)}^{1/2}, \\ \tilde{V}_2^L &= 155.16 \text{ cm}^3/\text{mol} & \delta_2 &= 19.304 \text{ (J/cm}^3\text{)}^{1/2}. \end{aligned}$$

(Answer: $\gamma_2 = 1.013$)

- 8.25** When $\tilde{V}_1^L = \tilde{V}_2^L = \tilde{V}^L$, show that the activity coefficient expressions for a regular mixture reduce to those for the two-suffix Margules equations with

$$A = \frac{\tilde{V}^L (\delta_1 - \delta_2)^2}{RT}.$$

Problems related to Section 8.7

- 8.26** For a binary mixture of ethanol (1) and methylcyclohexane (2), Russinyol *et al.* (2004) reported the following isobaric VLE data under atmospheric pressure:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.008	12.130	1.000	0.358	2.153	1.395	0.754	1.149	3.051
0.024	10.650	1.002	0.399	1.948	1.483	0.797	1.105	3.495
0.040	9.413	1.007	0.440	1.784	1.580	0.849	1.061	4.227
0.086	6.824	1.028	0.523	1.537	1.814	0.899	1.029	5.239
0.143	4.876	1.074	0.549	1.476	1.901	0.948	1.008	6.705
0.197	3.743	1.133	0.607	1.359	2.128	0.976	1.002	7.864
0.221	3.378	1.164	0.660	1.272	2.387			
0.307	2.486	1.299	0.718	1.192	2.757			

- (a) Use the Lagrange three-point formula and show that the activity coefficients at infinite dilution are given as

$$\gamma_1^\infty = 12.961 \quad \gamma_2^\infty = 8.605.$$

- (b) Calculate molar excess Gibbs energy as a function of composition, fit the data to the following equation

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 \sum_{i=1}^5 A_i (x_1 - x_2)^{i-1},$$

and show that $A_1 = 2.046$, $A_2 = -0.184$, $A_3 = 0.356$, $A_4 = 0.018$, and $A_5 = -0.023$.

8.27 The activity coefficient of species i in a mixture is defined by Eqn (8.3-5) as

$$\gamma_i = \frac{\hat{\phi}_i^L(T, P, x_i)}{\phi_i^L(T, P)}. \quad (1)$$

Taking the limit as $x_i \rightarrow 0$, Eqn (1) becomes

$$\gamma_i^\infty = \lim_{x_i \rightarrow 0} \left[\frac{\hat{\phi}_i^L(T, P, x_i)}{\phi_i^L(T, P)} \right]. \quad (2)$$

- (a) Consider a binary system of species 1 and 2. If the fugacity coefficients can be calculated by the Redlich-Kwong equation of state, show that

$$\begin{aligned} \ln \gamma_1^\infty = & \frac{B_1}{B_2} (Z_2^L - 1) - (Z_1^L - 1) + \ln \left(\frac{Z_1^L - B_1}{Z_2^L - B_2} \right) + \frac{A_{11}}{B_1} \\ & \times \ln \left(1 + \frac{B_1}{Z_1^L} \right) - \frac{A_{22}}{B_2} \left[2(1 - k_{12}) \sqrt{\frac{A_{11}}{A_{22}}} - \frac{B_1}{B_2} \right] \\ & \times \ln \left(1 + \frac{B_2}{Z_2^L} \right), \end{aligned} \quad (3)$$

$$\begin{aligned}
\ln \gamma_2^\infty = & \frac{B_2}{B_1} (Z_1^L - 1) - (Z_2^L - 1) + \ln \left(\frac{Z_2^L - B_2}{Z_1^L - B_1} \right) + \frac{A_{22}}{B_2} \\
& \times \ln \left(1 + \frac{B_2}{Z_2^L} \right) - \frac{A_{11}}{B_1} \left[2(1 - k_{12}) \sqrt{\frac{A_{22}}{A_{11}} - \frac{B_2}{B_1}} \right] \\
& \times \ln \left(1 + \frac{B_1}{Z_1^L} \right). \quad (4)
\end{aligned}$$

- (b) For a binary system of benzene (1) and cyclohexane (2) at 310.9 K and 1.013 bar, the infinite dilution activity coefficient for benzene is reported as $\gamma_1^\infty = 1.48$ (Twu and Coon, 1995). Estimate the binary interaction parameter, k_{12} , using Eqn (3).
- (c) Assuming k_{12} to be independent of temperature, estimate the infinite dilution activity coefficient for cyclohexane at 314.9 K and 1.013 bar.
- (Answer: (b) 0.0268 and (c) $\gamma_2^\infty = 1.58$)

Problems related to Section 8.8

8.28 For a binary system, the molar excess Gibbs energy is given by

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2^2. \quad (1)$$

- (a) Show that the activity coefficients are given by

$$\ln \gamma_1 = Ax_2^2(x_2 - x_1), \quad (2)$$

$$\ln \gamma_2 = 2Ax_1^2x_2. \quad (3)$$

- (b) Show that Eqns (2) and (3) satisfy the Gibbs-Duhem equation, Eqn (8.8-4).

8.29 Mirmehrabi *et al.* (2006) proposed the following equation to express the molar excess Gibbs energy of a binary system:

$$\frac{\tilde{G}^{\text{ex}}}{RT} = A(x_1x_2)^\beta.$$

- (a) Show that

$$\ln \gamma_1 = Ax_2^\beta \left[\beta x_1^{\beta-1} + (1 - 2\beta)x_1^\beta \right]. \quad (1)$$

$$\ln \gamma_2 = Ax_1^\beta \left[\beta x_2^{\beta-1} + (1 - 2\beta)x_2^\beta \right]. \quad (2)$$

- (b) Show that Eqns (1) and (2) satisfy the Gibbs-Duhem equation, Eqn (8.8-4).

8.30 For a binary system of ethanol (1) and water (2) at 298.15 K, Hansen and Miller (1954) expressed the activity coefficient of ethanol as a function of composition as

$$\ln \gamma_1 = (1 - x_1^2) (1.531 - 1.289x_1 - 0.207e^{-11x_1}).$$

- (a) Show that the activity coefficient of water is given by

$$\ln \gamma_2 = x_1^2 \left\{ 2.1755 - 1.289x_1 - 0.207e^{-11x_1} - \frac{0.0188}{x_1^2} [1 - (1 + 11x_1)e^{-11x_1}] \right\}.$$

- (b) Calculate the infinite dilution activity coefficients.

(Answer: (b) $\gamma_1^\infty = 3.758$, $\gamma_2^\infty = 2.381$)

8.31 One of the tests used to verify the consistency of VLE data is the *infinite dilution test*. For a binary system of components 1 and 2, start with Eqn (8.3-10) to obtain

$$\frac{\tilde{G}^{\text{ex}}/RT}{x_1x_2} = \frac{\ln \gamma_1}{x_2} + \frac{\ln \gamma_2}{x_1}. \quad (1)$$

- (a) Show that

$$\left(\frac{\tilde{G}^{\text{ex}}/RT}{x_1x_2} \right)_{x_1=0} = \ln \gamma_1^\infty \quad \text{and} \quad \left(\frac{\tilde{G}^{\text{ex}}/RT}{x_1x_2} \right)_{x_1=1} = \ln \gamma_2^\infty. \quad (2)$$

- (b) In the infinite dilution test, it is required to plot $\tilde{G}^{\text{ex}}/(RTx_1x_2)$ versus x_1 , $\ln \gamma_1$ versus x_1 , and $\ln \gamma_2$ versus x_1 on the same graph. If the data are consistent, the values of $\tilde{G}^{\text{ex}}/(RTx_1x_2)$ at $x_1 = 0$ and $x_1 = 1$ should approach $\ln \gamma_1$ and $\ln \gamma_2$, respectively, as closely as possible. This obviously raises the question of how to quantify the closeness? In the literature it is suggested that consistent data should satisfy

$$I_1 = 100 \left| \frac{\left(\frac{\tilde{G}^{\text{ex}}/RT}{x_1 x_2} \right)_{x_1=0} - \ln \gamma_1^\infty}{\ln \gamma_1^\infty} \right| < 30 \quad \text{and}$$

$$I_2 = 100 \left| \frac{\left(\frac{\tilde{G}^{\text{ex}}/RT}{x_1 x_2} \right)_{x_1=1} - \ln \gamma_2^\infty}{\ln \gamma_2^\infty} \right| < 30. \quad (3)$$

For a binary system of 2-butanol (1) and octane (2), [Hiaki *et al.* \(1998\)](#) calculated the following activity coefficients from VLE data at 343.15 K:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.0425	4.9534	1.0072	0.4941	1.4304	1.4723	0.7891	1.0821	2.4214
0.0868	3.8434	1.0295	0.5314	1.3628	1.5459	0.8228	1.0602	2.6276
0.1678	2.8271	1.0664	0.5882	1.2779	1.6779	0.8475	1.0482	2.7930
0.2263	2.3529	1.1175	0.6140	1.2440	1.7423	0.8674	1.0380	2.9680
0.2598	2.1642	1.1509	0.6502	1.2017	1.8522	0.8858	1.0270	3.1351
0.3061	1.9524	1.1987	0.6613	1.1884	1.8992	0.9397	1.0130	3.5843
0.3420	1.8159	1.2405	0.6883	1.1655	1.9743	0.9539	1.0105	3.7849
0.4130	1.6103	1.3411	0.7384	1.1190	2.1756			
0.4487	1.5232	1.3960	0.7603	1.0988	2.2990			

Plot $\tilde{G}^{\text{ex}}/(RTx_1x_2)$ versus x_1 , $\ln \gamma_1$ versus x_1 , and $\ln \gamma_2$ versus x_1 on the same graph and calculate I_1 and I_2 .

(Answer: (b) $I_1 = 2.7$, $I_2 = 13.3$)

8.32 [Kyyny *et al.* \(2001\)](#) reported the following isothermal VLE data for a binary system of 2-butanol (1) and 2,4,4-trimethyl-1-pentene (2) at 360 K:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.0511	3.69	1.01	0.3905	1.54	1.25	0.8235	1.03	2.42
0.0533	3.63	1.01	0.4579	1.41	1.34	0.8595	0.98	2.83
0.0587	3.42	1.01	0.5150	1.31	1.43	0.8877	0.98	2.95
0.0929	3.15	1.02	0.5831	1.23	1.56	0.9140	0.97	3.18
0.1288	2.76	1.03	0.6454	1.17	1.69	0.9322	0.97	3.44
0.1792	2.39	1.06	0.7008	1.11	1.85	0.9517	0.97	3.73
0.2322	2.07	1.10	0.7529	1.09	2.00	0.9675	0.98	3.86
0.2814	1.87	1.14	0.7815	1.06	2.13	0.9807	0.99	3.93
0.3330	1.69	1.19	0.7875	1.06	2.20			

(a) Fit the data to a fifth-order polynomial and show that

$$\ln \left(\frac{\gamma_1}{\gamma_2} \right) = 1.4280 - 2.6508x_1 - 8.7773x_1^2 + 31.9029x_1^3 - 38.5755x_1^4 + 15.2042x_1^5.$$

(b) Show that the experimental data are consistent.

(Answer: (b) $A = 2.9$, $D = 4.4$)

Problem related to Section 8.9

8.33 The NRTL model for a multicomponent system is given by

$$\ln \gamma_i = \frac{\sum_{j=1}^k \tau_{ji} G_{ji} x_j}{\sum_{m=1}^k \tau_{mi} x_m} + \sum_{j=1}^k \frac{G_{ij} x_j}{\sum_{m=1}^k G_{mj} x_m} \left(\tau_{ij} - \frac{\sum_{r=1}^k \tau_{rj} G_{rj} x_r}{\sum_{m=1}^k G_{mj} x_m} \right),$$

where

$$\tau_{ij} = \frac{g_{ij} - g_{jj}}{RT} \quad G_{ij} = \exp(-\alpha_{ij} \tau_{ij}) \quad \alpha_{ij} = \alpha_{ji}.$$

Merzougui *et al.* (2011) reported the following data for a ternary mixture of water (1), methanol (2), and dichloromethane (3) at 293.15 K:

$i - j$	$g_{ij} - g_{jj}(\text{cal/mol})$	$g_{ji} - g_{ii}(\text{cal/mol})$	α_{ij}
1 - 2	238.5741	566.9698	0.2
1 - 3	749.2058	1498.0961	0.2
2 - 3	-359.8070	1108.6181	0.2

Calculate the activity coefficients when $x_1 = 0.1$, $x_2 = 0.3$, and $x_3 = 0.6$.

(Answer: $\gamma_1 = 2.170$, $\gamma_2 = 1.783$, $\gamma_3 = 2.794$)

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Chapter 9

Vapor-Liquid Equilibrium

Distillation is one of the most commonly used separation processes in the chemical process industries. It is based on the differences in boiling points of components in the mixture. A typical distillation column is shown in Fig. 9.1. The feed enters the distillation column in any state (from subcooled liquid to superheated vapor). The parts of the distillation column above and below the feed point are called the *rectifying* and *stripping sections*, respectively. The vapor stream from the top of the distillation column is sent to a condenser. Part of the liquid is taken as the overhead product, while the rest is returned to the column as “reflux.” The overhead product is rich in more volatile component(s). The ratio of the reflux flow rate to the overhead product flow rate is called the *reflux ratio*. Part of the liquid stream from the bottom of the distillation column is taken as the bottoms product, which is rich in less volatile component(s). The rest is vaporized in a reboiler and returned to the column.

The trays placed in the column are used to bring the vapor and liquid streams, which flow in countercurrent direction, into contact with each other. The trays may be of various types, i.e. sieve, bubble-cap, valve, etc. In a sieve tray column, for example, liquid flows across the tray through channels separated by baffles and vapor flows from the bottom, up through holes in the tray. When the two streams mix with each other, if the component fugacities are not the same in the vapor and liquid phases, the resulting driving force causes mass transfer to take place. While the more volatile component transfers from the liquid to the vapor phase, the less volatile component transfers from the vapor to the liquid phase. The system reaches equilibrium, or, in other words, mass transfer between the phases stops, when component fugacities in the vapor and liquid phases are equal to each other. Once the system reaches equilibrium, the two phases leave the tray as two separate streams. The combination of these stages, i.e. mixing, reaching equilibrium, and separation, is called an *equilibrium stage*. The purpose of this chapter is to provide

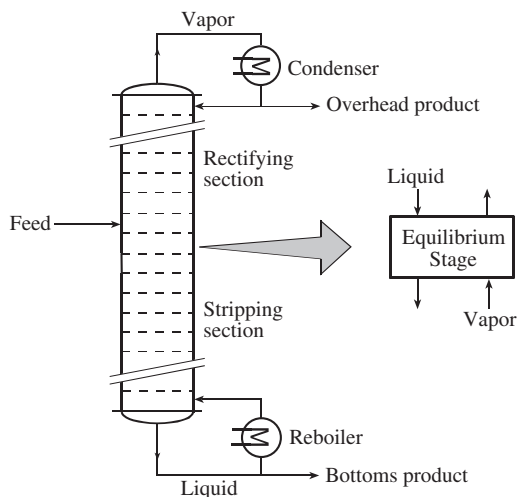


Figure 9.1: A distillation column.

the necessary tools to determine the compositions of the liquid and vapor streams leaving each equilibrium stage.

9.1 Vapor-Liquid Equilibrium Calculations

When vapor is in equilibrium with liquid, besides temperature and pressure of the vapor and liquid phases, fugacities of each component in the vapor and liquid phases must be equal to each other, i.e.

$$\hat{f}_i^V(T, P, y_i) = \hat{f}_i^L(T, P, x_i) \quad i = 1, 2, \dots, k. \quad (9.1-1)$$

There are mainly two approaches for the modeling of vapor-liquid equilibrium: In the first approach, also known as the γ - ϕ model, fugacity of component i in the vapor phase is expressed in terms of the fugacity coefficient, and the fugacity of component i in the liquid phase is expressed in terms of the activity coefficient, i.e.

$$y_i P \hat{\phi}_i^V(T, P, y_i) = x_i \gamma_i(T, P, x_i) f_i^L(T, P). \quad (9.1-2)$$

The use of Eqn (5.4-8) to express the fugacity of pure component i in the liquid phase reduces Eqn (9.1-2) to

$$y_i P \hat{\phi}_i^V(T, P, y_i) = x_i \gamma_i(T, P, x_i) P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right]. \quad (9.1-3)$$

The fugacity coefficient of component i , $\hat{\phi}_i^V$, and the fugacity coefficient of pure i , ϕ_i^V , are calculated from the equation of state.

In the second approach, also known as the ϕ - ϕ model, fugacities of component i in the vapor and liquid phases are expressed in terms of the fugacity coefficients so that Eqn (9.1-1) takes the form

$$y_i \hat{\phi}_i^V(T, P, y_i) = x_i \hat{\phi}_i^L(T, P, x_i) \quad (9.1-4)$$

in which fugacity coefficients are calculated either by using a single equation of state for both vapor and liquid phases or by using different equations of state for vapor and liquid phases.

This chapter mainly focuses on the use of the first approach, i.e. Eqn (9.1-3). Note that the use of Eqn (9.1-3) requires the vapor pressures of pure components, P_i^{vap} , to be known. The most commonly used vapor pressure equation is that of Antoine, given by Eqn (5.6-18), i.e.

$$\ln P_i^{\text{vap}} = A_i - \frac{B_i}{T + C_i} \quad (9.1-5)$$

Appendix C provides the parameters A_i , B_i , and C_i for various substances.

Once the equilibrium data are obtained either experimentally or theoretically, the results are presented either in tabular form or as a *vapor-liquid equilibrium* (VLE) diagram. Such equilibrium diagrams may be in different forms as presented below.

Pressure-composition diagram: To construct a pressure-composition diagram, compositions of the liquid and vapor phases are experimentally determined as a function of pressure at a specified temperature. Liquid phase composition, x_i , versus pressure, P , and vapor phase composition, y_i , versus pressure, P , curves are called *bubble point* and *dew point* curves, respectively. For a binary system, a typical pressure-composition diagram is shown in Fig. 9.2a.

The region between the bubble point and dew point curves represents the two-phase, i.e. vapor and liquid, region in which the number of degrees of freedom, $\mathcal{F}$, is two. When liquid and vapor phases are in equilibrium, their temperatures and pressures must be equal to each other. Thus, any horizontal line in the region between dew point and bubble point curves joins the vapor and liquid phases that are in equilibrium. Such a line is called the *tie line*.

The region above the bubble point curve represents the subcooled liquid. The region below the dew point curve represents the superheated vapor. In these two regions $\mathcal{F} = 3$.

At a specified temperature, examination of Fig. 9.2a indicates that the vapor pressures of pure components 1 and 2 are 17.82 kPa and 1.84 kPa, respectively. Since $P_1^{\text{vap}} > P_2^{\text{vap}}$, pure component 1 boils at a lower temperature as compared to component 2. In other words, component 1 is more volatile.

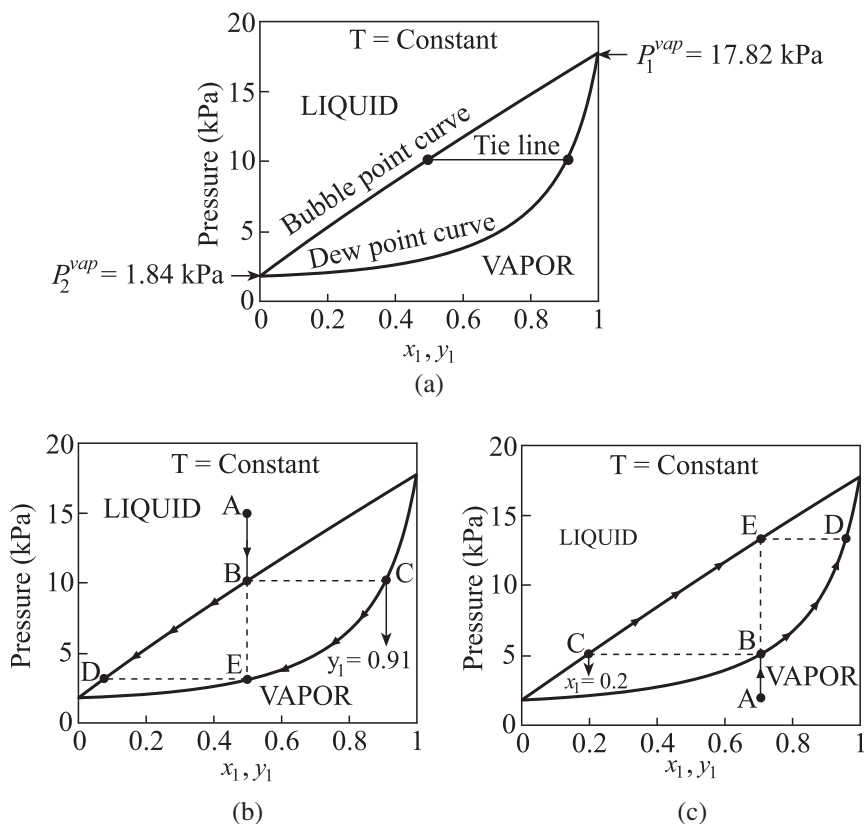


Figure 9.2: Pressure-composition diagram for a binary system.

Suppose that a liquid mixture of $x_1 = 0.5$ is initially at a pressure of 15 kPa (point A) as shown in Fig. 9.2b. Let us decrease pressure while keeping the temperature constant. When the vertical line passing through the point A intersects the bubble point curve (point B), the liquid mixture starts to boil with the corresponding pressure of 10 kPa, known as the *bubble point*¹ pressure. The dashed horizontal line passing through the point B, i.e. tie line, intersects the dew point curve at point C. Thus, the composition of the first bubble of vapor is $y_1 = 0.91$. Since component 1 is more volatile, the vapor is rich in component 1. This, in turn, increases the amount of component 2 (the less volatile component) in the liquid. As the boiling continues, while the composition of the liquid moves along $\overline{BD}$ on the bubble point curve, the composition of the vapor moves along $\overline{CE}$ on the dew point curve. The tie line $\overline{DE}$ represents the equilibrium between the last liquid droplet (point D)

¹Bubble point is the state at which the saturated liquid starts to boil.

having a composition of $x_1 = 0.08$ and vapor (point E). The pressure at point D (or point E) is 3.1 kPa. Further decrease of pressure leads to superheated vapor with composition $y_1 = 0.5$.

At a given temperature, pure liquid boils at a single pressure, i.e. when its vapor pressure equals the ambient pressure. On the other hand, for a multicomponent liquid at a specified temperature and initial composition, boiling takes place over a range of pressure (10–3.1 kPa in the above example) due to the varying composition of the liquid phase.

Now consider a vapor mixture at a pressure of 2 kPa with a composition of $y_1 = 0.7$ (point A) as shown in Fig. 9.2c. Let us increase pressure while keeping the temperature constant. When the vertical line passing through the point A intersects the dew point curve (point B), the vapor mixture starts to condense with the corresponding pressure of 5 kPa, known as the *dew point² pressure*. The dashed horizontal line passing through the point B , i.e. tie line, intersects the bubble point curve at point C . Thus, the composition of the first liquid droplet is around $x_1 = 0.2$. Since component 2 is less volatile, the liquid is rich in component 2. This, in turn, increases the amount of component 1 (the more volatile component) in the vapor. As the condensation continues, while the composition of the vapor moves along $\overline{BD}$ on the dew point curve, the composition of the liquid moves along $\overline{CE}$ on the bubble point curve. The tie line $\overline{DE}$ represents the equilibrium between the last vapor molecule (point D) having a composition of $y_1 = 0.96$ and liquid (point E). The pressure at point D (or point E) is 13.3 kPa. Further increase of pressure leads to subcooled liquid with composition $x_1 = 0.7$.

At a given temperature, pure vapor condenses at a single pressure. On the other hand, for a multicomponent vapor at a specified temperature and initial composition, condensation takes place over a range of pressure (5–13.3 kPa in the above example) due to the varying composition of the vapor phase.

Temperature-composition diagram: To construct a temperature-composition diagram, compositions of the liquid and vapor phases are experimentally determined as a function of temperature at a specified pressure. Liquid phase composition, x_i , versus temperature, T , and vapor phase composition, y_i , versus temperature, T , curves are called *bubble point* and *dew point* curves, respectively. For a binary system, a typical temperature-composition diagram is shown in Fig. 9.3a.

The region between the bubble point and dew point curves represents the two-phase region. The horizontal line within the two-phase region joining the vapor and liquid phases that are in equilibrium is called the *tie line*. The region below the bubble point curve represents the subcooled liquid. The region above the dew point curve represents the superheated vapor.

²Dew point is the state at which the saturated vapor starts to condense.

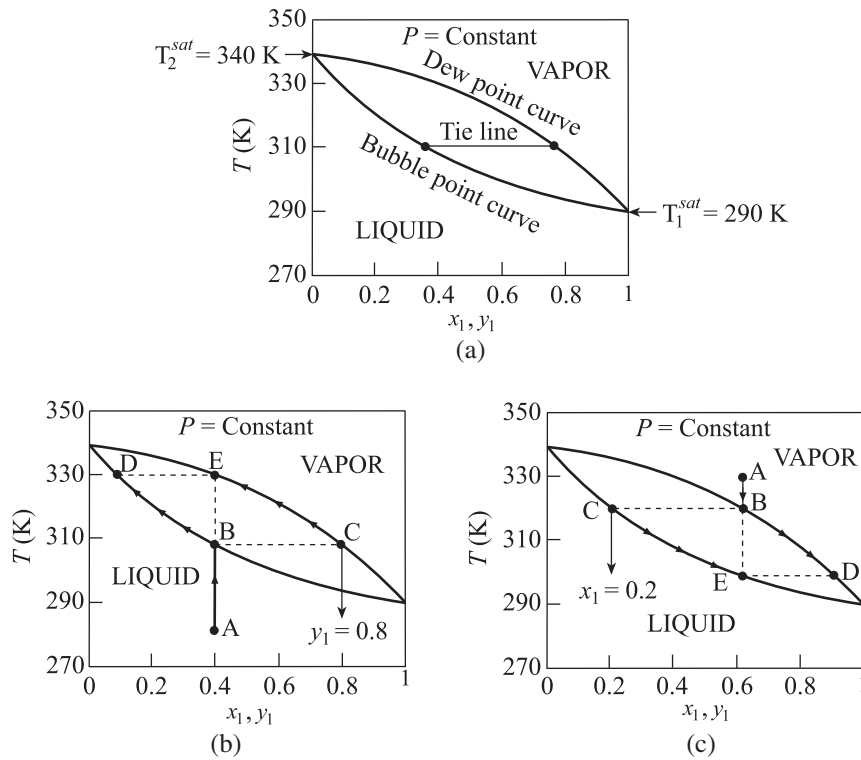


Figure 9.3: Temperature-composition diagram for a binary system.

At a specified pressure, examination of Fig. 9.3a indicates that the saturation (or boiling point) temperatures of pure components 1 and 2 are 290 K and 340 K, respectively. Since $T_1^{\text{sat}} < T_2^{\text{sat}}$, component 1 is more volatile.

Suppose that a liquid mixture of $x_1 = 0.4$ is initially at a temperature of 280 K (point A) as shown in Fig. 9.3b. Let us increase temperature while keeping the pressure constant. When the vertical line passing through the point A intersects the bubble point curve (point B), the liquid mixture starts to boil with the corresponding temperature of 308 K, known as the *bubble point temperature*. The tie line passing through the point B intersects the dew point curve at point C. Thus, the composition of the first bubble of vapor is $y_1 = 0.8$. Since component 1 is more volatile, the vapor is rich in it. This, in turn, increases the amount of component 2 (the less volatile component) in the liquid. As the boiling continues, while the composition of the liquid moves along $\overline{BD}$ on the bubble point curve,³ the composition of the vapor

³As liquid becomes rich in the less volatile component, its boiling point temperature increases.

moves along $\overline{CE}$ on the dew point curve. The tie line $\overline{DE}$ represents the equilibrium between the last liquid droplet (point D) having a composition of $x_1 = 0.09$ and vapor (point E). The temperature at point D (or point E) is 330 K. Further increase of temperature leads to superheated vapor with composition $y_1 = 0.4$.

At a given pressure, pure liquid boils at a single temperature. On the other hand, for a multicomponent liquid at a specified pressure and initial composition, boiling takes place over a range of temperature (308–330 K in the above example) due to the varying composition of the liquid phase.

Now consider a vapor mixture at a temperature of 330 K with a composition of $y_1 = 0.62$ (point A) as shown in Fig. 9.3c. Let us decrease temperature while keeping the pressure constant. When the vertical line passing through the point A intersects the dew point curve (point B), the vapor mixture starts to condense with the corresponding temperature of 320 K, known as the *dew point temperature*. The tie line passing through the point B intersects the bubble point curve at point C . Thus, the composition of the first liquid droplet is $x_1 = 0.2$. Since component 2 is less volatile, the liquid is rich in it. This, in turn, increases the amount of component 1 (the more volatile component) in the vapor. As the condensation continues, while the composition of the vapor moves along $\overline{BD}$ on the dew point curve,⁴ the composition of the liquid moves along $\overline{CE}$ on the bubble point curve. The tie line $\overline{DE}$ represents the equilibrium between the last vapor molecule (point D) having a composition of $y_1 = 0.91$ and liquid (point E). The temperature at point D (or point E) is 299 K. Further decrease of temperature leads to subcooled liquid with composition $x_1 = 0.62$.

At a given pressure, pure vapor condenses at a single temperature. On the other hand, for a multicomponent vapor at a specified pressure and initial composition, condensation takes place over a range of temperature (320–295 K in the above example) due to the varying composition of the vapor phase.

x-y diagram: In this diagram, vapor phase composition, y , is plotted versus the liquid phase composition, x , as shown in Fig. 9.4. Usually, a “45° line,” i.e. $x = y$, is included on the diagram for reference. These diagrams are typically made at constant pressure, and so each point represents a different temperature.

9.1.1 Types of Vapor-Liquid Equilibrium Calculations

To solve a well-posed vapor-liquid equilibrium problem, the number of equations must be equal to the number of unknowns. The number of degrees of freedom, $\mathcal{F}$, is

$$\mathcal{F} = \left[\begin{array}{c} \text{Total \# of independent} \\ \text{intensive variables} \end{array} \right] - \left[\begin{array}{c} \text{\# of independent equations} \\ \text{relating these variables} \end{array} \right]. \quad (9.1-6)$$

⁴As vapor becomes rich in the more volatile component, it condenses at a lower temperature.

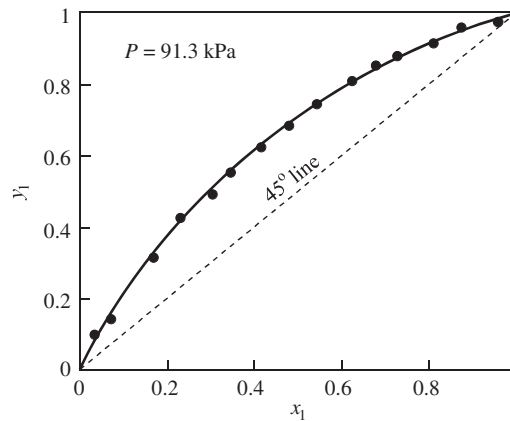


Figure 9.4: x - y diagram for a binary mixture of 1-propanol (1) and 1-butanol (2) (Mohsen-Nia and Memarzadeh, 2010).

For a k -component system at equilibrium, if each component is present in each phase, the total number of independent intensive variables is expressed as

$$\begin{aligned} \text{Variables} &= T + P + (x_1, x_2, \dots, x_{k-1}) + (y_1, y_2, \dots, y_{k-1}) \\ &= 2 + 2(k - 1) = 2k. \end{aligned} \quad (9.1-7)$$

The equations relating x_i to y_i (or vice versa) are given by either Eqn (9.1-2) or Eqn (9.1-4). Therefore, the total number of independent equations is k and Eqn (9.1-6) becomes

$$\mathcal{F} = 2k - k = k, \quad (9.1-8)$$

which indicates that k variables must be specified so that equations at equilibrium, i.e. Eqn (9.1-2), are used for determining the remaining unknown variables.

Vapor-liquid equilibrium calculations can be divided into five types depending on the specified set of variables:

- *Bubble point pressure calculation:* Calculate pressure and vapor phase composition, given temperature and liquid phase composition.
- *Dew point pressure calculation:* Calculate pressure and liquid phase composition, given temperature and vapor phase composition.
- *Bubble point temperature calculation:* Calculate temperature and vapor phase composition, given pressure and liquid phase composition.
- *Dew point temperature calculation:* Calculate temperature and liquid phase composition, given pressure and vapor phase composition.
- *Flash calculation:* Calculate amounts and compositions of liquid and vapor phases, given overall composition and amount of feed stream besides temperature and pressure within the flash chamber.

Once temperature is specified, vapor pressures of species making up the mixture can be easily determined. Thus, bubble point and dew point pressure calculations are easiest to perform. On the other hand, if pressure is specified, bubble point and dew point temperature calculations rely on a trial-and-error procedure since vapor pressures of species cannot be determined *a priori*.

9.2 Raoult's Law

Let us simplify Eqn (9.1-3) with the following assumptions:

- *Assumption 1:* Vapor is an ideal gas mixture.

For an ideal gas mixture, fugacity of component i in the mixture is equal to its partial pressure, i.e. $\hat{\phi}_i^V = 1$. Thus, Eqn (9.1-3) simplifies to

$$y_i P = x_i \gamma_i(T, P, x_i) P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right]. \quad (9.2-1)$$

- *Assumption 2:* Liquid is an ideal mixture.

Forces between like and unlike molecules do not differ and the Lewis-Randall rule applies, i.e. $\hat{f}_i = f_i x_i$ and $\gamma_i = 1$. Thus, Eqn (9.2-1) reduces to

$$y_i P = x_i P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right]. \quad (9.2-2)$$

- *Assumption 3:* The Poynting correction factor is equal to unity.

When pressure is not very high, Eqn (9.2-2) simplifies to

$$y_i P = x_i P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \quad (9.2-3)$$

- *Assumption 4:* The saturated vapor of i at T and P_i^{vap} is ideal.

This assumption implies that fugacity of pure vapor is equal to the vapor pressure, i.e. $\phi_i^V(T, P_i^{\text{vap}}) = 1$, and Eqn (9.2-3) takes the final form of

$$y_i P = x_i P_i^{\text{vap}}. \quad (9.2-4)$$

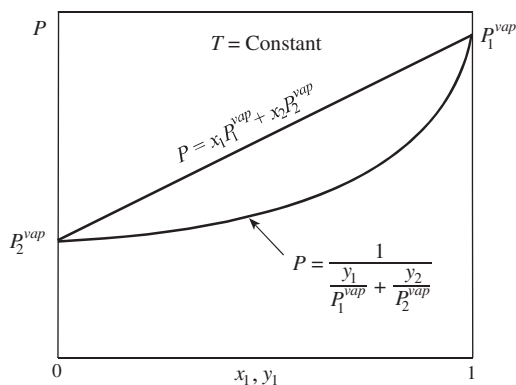


Figure 9.5: Pressure-composition diagram of a system obeying Raoult's law.

which is known as *Raoult's law*.⁵ It simply states that the partial pressure of each component in the vapor phase is equal to its mole fraction in the liquid phase times its pure-component vapor pressure.

Summation of Eqn (9.2-4) over all components present in the system yields

$$\sum_{i=1}^k y_i P = \sum_{i=1}^k x_i P_i^{\text{vap}} \Rightarrow P \underbrace{\sum_{i=1}^k y_i}_1 = \sum_{i=1}^k x_i P_i^{\text{vap}}. \quad (9.2-5)$$

Therefore, the total pressure (or bubble point pressure) is given by

$$P = \sum_{i=1}^k x_i P_i^{\text{vap}}. \quad (9.2-6)$$

For a binary system, Eqn (9.2-6) reduces to

$$P = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} = P_2^{\text{vap}} + (P_1^{\text{vap}} - P_2^{\text{vap}}) x_1, \quad (9.2-7)$$

which indicates that the equilibrium total pressure changes linearly with the liquid phase mole fraction at constant temperature as shown in Fig. 9.5.

It is also possible to rearrange Eqn (9.2-4) as

$$x_i = \frac{y_i P}{P_i^{\text{vap}}}. \quad (9.2-8)$$

⁵Named after French chemist François Marie Raoult.

Summation of Eqn (9.2-8) over all components present in the system gives

$$\sum_{i=1}^k x_i = \sum_{i=1}^k \frac{y_i P}{P_i^{\text{vap}}} \Rightarrow 1 = P \sum_{i=1}^k \frac{y_i}{P_i^{\text{vap}}}. \quad (9.2-9)$$

Therefore, the total pressure (or dew point pressure) is given by

$$P = \frac{1}{\sum_{i=1}^k \frac{y_i}{P_i^{\text{vap}}}}. \quad (9.2-10)$$

For a binary system, Eqn (9.2-10) reduces to

$$P = \frac{1}{\frac{y_1}{P_1^{\text{vap}}} + \frac{y_2}{P_2^{\text{vap}}}}. \quad (9.2-11)$$

Figure 9.5 also shows the variation of system pressure as a function of the vapor phase mole fraction at constant temperature. Note that Fig. 9.5 is nothing more than the pressure-composition diagram for a system obeying Raoult's law.

The system pressure is calculated from Eqn (9.2-6) when the liquid phase composition is known. On the other hand, the system pressure is calculated from Eqn (9.2-10) if the vapor phase composition is known.

For a binary system obeying Raoult's law, the temperature-composition diagram can be prepared as follows:

- Specify pressure P .
- Calculate T_1^{sat} and T_2^{sat} at the specified P from

$$T_i^{\text{sat}} = \frac{B_i}{A_i - \ln P} - C_i.$$

- Pick temperature values between T_1^{sat} and T_2^{sat} .
- Calculate P_1^{vap} and P_2^{vap} .
- Calculate x_1 from Eqn (9.2-7), i.e.

$$x_1 = \frac{P - P_2^{\text{vap}}}{P_1^{\text{vap}} - P_2^{\text{vap}}}.$$

- Calculate y_1 from Eqn (9.2-8), i.e.

$$y_1 = \frac{x_1 P_1^{\text{vap}}}{P}.$$

Example 9.1 Prepare temperature-composition diagram for a binary mixture of *n*-hexane (1) and *n*-heptane (2) at 1.013 bar. Compare your results with the following experimental data given by Jan *et al.* (1994):

<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁	<i>T</i> (K)	<i>x</i> ₁	<i>y</i> ₁
371.47	0.0000	0.0000	356.37	0.3953	0.6236	345.55	0.8160	0.9240
367.18	0.0964	0.1986	352.93	0.5123	0.7270	343.74	0.8991	0.9637
363.52	0.1904	0.3600	350.27	0.6187	0.8074	341.81	1.0000	1.0000
359.11	0.3146	0.5258	348.13	0.7012	0.8615			

Solution

From Appendix C, the vapor pressures are given as

$$\ln P_1^{\text{vap}} = 9.2164 - \frac{2697.55}{T - 48.78},$$

$$\ln P_2^{\text{vap}} = 9.2535 - \frac{2911.32}{T - 56.51}.$$

At 1.013 bar, the saturation temperatures are

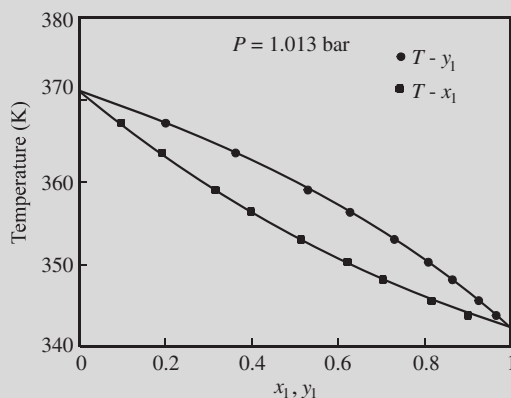
$$T_1^{\text{sat}} = \frac{2697.55}{9.2164 - \ln 1.013} + 48.78 = 341.9 \text{ K},$$

$$T_2^{\text{sat}} = \frac{2911.32}{9.2535 - \ln 1.013} + 56.51 = 371.6 \text{ K}.$$

By picking temperature values between 341.9 K and 371.6 K, the calculated values of *x*₁ and *y*₁ are tabulated as follows:

<i>T</i> (K)	<i>P</i> ₁ ^{vap} (bar)	<i>P</i> ₂ ^{vap} (bar)	<i>x</i> ₁	<i>y</i> ₁	<i>T</i> (K)	<i>P</i> ₁ ^{vap} (bar)	<i>P</i> ₂ ^{vap} (bar)	<i>x</i> ₁	<i>y</i> ₁
343	1.049	0.403	0.944	0.978	359	1.683	0.690	0.325	0.540
345	1.116	0.433	0.849	0.936	361	1.780	0.735	0.266	0.467
347	1.186	0.464	0.760	0.890	363	1.880	0.782	0.210	0.390
349	1.260	0.497	0.676	0.841	365	1.985	0.832	0.157	0.307
351	1.337	0.531	0.598	0.789	367	2.095	0.884	0.106	0.220
353	1.418	0.568	0.524	0.733	369	2.209	0.939	0.058	0.127
355	1.503	0.607	0.454	0.673	371	2.327	0.996	0.013	0.029
357	1.591	0.647	0.388	0.609					

The plot of Txy diagram is shown in the figure below. The points represent the experimental data given by Jan *et al.* (1994).



9.3 VLE Calculations When Raoult's Law is Applicable

Raoult's law is applicable for mixtures composed of chemically similar components at low pressure.

9.3.1 Bubble Point Pressure Calculation

At a given temperature, it is required to calculate the pressure at which a liquid of known composition first begins to boil (see Fig. 9.6a). The bubble point pressure calculation is straightforward and is given below:

1. Calculate vapor pressures of pure components from Eqn (9.1-5).
2. Calculate the bubble point pressure from Eqn (9.2-6).
3. Determine the composition of the vapor phase from Eqn (9.2-4), i.e.

$$y_i = \frac{x_i P_i^{\text{vap}}}{P}. \quad (9.3-1)$$

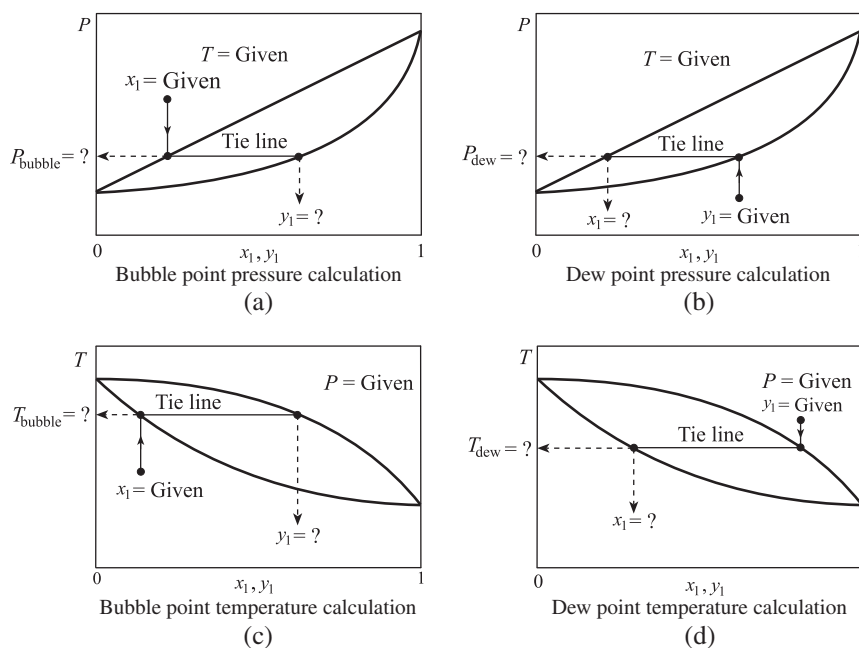


Figure 9.6: Bubble point and dew point calculations.

Example 9.2 Estimate the bubble point pressure of a liquid mixture of 20 mol% *n*-butane (1), 25% *n*-pentane (2), 35% *n*-hexane (3), and 20% *n*-heptane (4) at 290 K. Also calculate the composition of the first bubble of vapor.

Solution

From Appendix C, the vapor pressures are given in the form

$$\begin{aligned}\ln P_1^{\text{vap}} &= 9.0580 - \frac{2154.90}{T - 34.42}, \\ \ln P_2^{\text{vap}} &= 9.2131 - \frac{2477.07}{T - 39.94}, \\ \ln P_3^{\text{vap}} &= 9.2164 - \frac{2697.55}{T - 48.78}, \\ \ln P_4^{\text{vap}} &= 9.2535 - \frac{2911.32}{T - 56.51}.\end{aligned}$$

At 290 K, the vapor pressures are calculated as

$$P_1^{\text{vap}} = 1.871 \text{ bar} \quad P_2^{\text{vap}} = 0.500 \text{ bar} \quad P_3^{\text{vap}} = 0.140 \text{ bar} \quad P_4^{\text{vap}} = 0.040 \text{ bar}.$$

The use of Eqn (9.2-6) gives the bubble point pressure as

$$\begin{aligned} P_{\text{bubble}} &= x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} + x_3 P_3^{\text{vap}} + x_4 P_4^{\text{vap}} \\ &= (0.20)(1.871) + (0.25)(0.500) + (0.35)(0.140) \\ &\quad + (0.20)(0.040) = 0.556 \text{ bar}. \end{aligned}$$

Composition of the first bubble of vapor can be calculated from Eqn (9.3-1), i.e.

$$\begin{aligned} y_1 &= \frac{(0.20)(1.871)}{0.556} = 0.673, \\ y_2 &= \frac{(0.25)(0.500)}{0.556} = 0.225, \\ y_3 &= \frac{(0.35)(0.140)}{0.556} = 0.088, \\ y_4 &= \frac{(0.20)(0.040)}{0.556} = 0.014. \end{aligned}$$

Note that $\sum y_i = 1.0$. Sometimes, as a result of roundoff errors, the summation of mole fractions may not be exactly equal to unity.

Comments:

- When Raoult's law is applicable

$$(P_i^{\text{vap}})_{\text{least volatile}} < P_{\text{bubble}} < (P_i^{\text{vap}})_{\text{most volatile}}.$$

- The vapor phase is rich in *n*-butane, the component with the highest vapor pressure.
- The problem can also be stated as follows: "A vapor mixture of 20 mol% *n*-butane (1), 25% *n*-pentane (2), 35% *n*-hexane (3), and 20% *n*-heptane (4) at 290 K is compressed isothermally. Estimate the minimum pressure at which the vapor mixture is completely condensed."

9.3.2 Dew Point Pressure Calculation

At a given temperature, it is required to calculate the pressure at which a vapor of known composition first begins to condense (see Fig. 9.6b). The dew point pressure calculation is also straightforward and is given below:

- Calculate vapor pressures of pure components from Eqn (9.1-5).
- Calculate the dew point pressure from Eqn (9.2-10).
- Determine the composition of the liquid phase from Eqn (9.2-4), i.e.

$$x_i = \frac{y_i P}{P_i^{\text{vap}}}. \quad (9.3-2)$$

Example 9.3 Estimate the dew point pressure of a vapor mixture of 20 mol% *n*-butane (1), 25% *n*-pentane (2), 35% *n*-hexane (3), and 20% *n*-heptane (4) at 290 K. Also calculate the composition of the first drop of liquid.

Solution

From Example 9.2, the vapor pressures at 290 K are

$$P_1^{\text{vap}} = 1.871 \text{ bar} \quad P_2^{\text{vap}} = 0.500 \text{ bar} \quad P_3^{\text{vap}} = 0.140 \text{ bar} \quad P_4^{\text{vap}} = 0.040 \text{ bar}.$$

The use of Eqn (9.2-10) gives the dew point pressure as

$$P_{\text{dew}} = \frac{1}{\frac{0.20}{1.871} + \frac{0.25}{0.500} + \frac{0.35}{0.140} + \frac{0.20}{0.040}} = 0.123 \text{ bar}.$$

Composition of the first drop of liquid can be calculated from Eqn (9.3-2), i.e.

$$\begin{aligned} x_1 &= \frac{(0.20)(0.123)}{1.871} = 0.013, \\ x_2 &= \frac{(0.25)(0.123)}{0.500} = 0.062, \\ x_3 &= \frac{(0.35)(0.123)}{0.140} = 0.308, \\ x_4 &= \frac{(0.20)(0.123)}{0.040} = 0.615. \end{aligned}$$

As a result of roundoff errors $\sum x_i = 0.998$.

Comments:

- When Raoult's law is applicable

$$(P_i^{\text{vap}})_{\text{least volatile}} < P_{\text{dew}} < (P_i^{\text{vap}})_{\text{most volatile}}.$$

- The liquid phase is rich in *n*-heptane, the component with the lowest vapor pressure.
- The problem can also be stated in the following ways:
 - (a) A vapor mixture of 20 mol% *n*-butane (1), 25% *n*-pentane (2), 35% *n*-hexane (3), and 20% *n*-heptane (4) at 290 K is compressed isothermally until condensation occurs. Estimate the pressure at which condensation starts and determine the composition of the first liquid droplet.
 - (b) A liquid mixture of 20 mol% *n*-butane (1), 25% *n*-pentane (2), 35% *n*-hexane (3), and 20% *n*-heptane (4) is depressurized isothermally at 290 K. What is the maximum pressure for the complete evaporation of the mixture?

9.3.3 Bubble Point Temperature Calculation

At a given pressure, it is required to calculate the temperature at which a liquid of known composition first begins to boil (see Fig. 9.6c). Since temperature is an unknown quantity, vapor pressures of pure liquids cannot be determined *a priori* and the following trial-and-error procedure should be followed:

1. Assume T and determine vapor pressures of pure components from Eqn (9.1-5). When Raoult's law is applicable, $(T_i^{\text{sat}})_{\text{min}} < T_{\text{bubble}} < (T_i^{\text{sat}})_{\text{max}}$.
2. Use Eqn (9.2-6) and calculate P_{calc} .
3. Compare P_{calc} with the specified P . If $P_{\text{calc}} > P$, assumed T is too high; if $P_{\text{calc}} < P$, assumed T is too low. When the calculated pressure matches the specified pressure, mole fractions in the vapor phase are calculated from

$$y_i = \frac{x_i P_i^{\text{vap}}}{P}. \quad (9.3-3)$$

9.3.4 Dew Point Temperature Calculation

At a given pressure, it is required to calculate the temperature at which a vapor of known composition first begins to condense (see Fig. 9.6d). Since temperature is an unknown quantity, we have to use the following trial-and-error-procedure:

1. Assume T and determine vapor pressures of pure components from Eqn (9.1-5). When Raoult's law is applicable, $(T_i^{\text{sat}})_{\min} < T_{\text{dew}} < (T_i^{\text{sat}})_{\max}$.
2. Use Eqn (9.2-10) and calculate P_{calc} .
3. Compare P_{calc} with the specified P . If $P_{\text{calc}} > P$, assumed T is too high; if $P_{\text{calc}} < P$, assumed T is too low. When the calculated pressure matches the specified pressure, mole fractions in the liquid phase are calculated from

$$x_i = \frac{y_i P}{P_i^{\text{vap}}} \quad (9.3-4)$$

Example 9.4 Estimate the bubble point and dew point temperatures of a mixture consisting of 50% *n*-pentane (1), 45% *n*-hexane (2), and 5% *n*-heptane (3) at 0.8 bar.

Solution

The boiling point temperatures of pure components at 0.8 bar are

$$\begin{aligned} T_1^{\text{sat}} &= 39.94 + \frac{2477.07}{9.2131 - \ln 0.8} = 302.4 \text{ K}, \\ T_2^{\text{sat}} &= 48.78 + \frac{2697.55}{9.2164 - \ln 0.8} = 334.6 \text{ K}, \\ T_3^{\text{sat}} &= 56.51 + \frac{2911.32}{9.2535 - \ln 0.8} = 363.7 \text{ K}. \end{aligned}$$

Therefore, our initial estimates for the bubble and the dew point temperatures should lie between 302.4 K and 363.7 K. Let us choose the mole fraction weighted average of the pure component boiling points as an initial estimate, i.e.

$$T = (0.50)(302.4) + (0.45)(334.6) + (0.05)(363.7) = 320 \text{ K}.$$

Bubble point temperature: In this case we know the total pressure, 0.8 bar, and the liquid phase composition, $x_1 = 0.50$, $x_2 = 0.45$, and $x_3 = 0.05$. At the correct value of temperature, the pressure calculated from

$$P_{\text{calc}} = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} + x_3 P_3^{\text{vap}}$$

must be equal to 0.8. The results of the iterative procedure are shown in the table below:

Trial #	T (K)	P_1^{vap} (bar)	P_2^{vap} (bar)	P_3^{vap} (bar)	P_{calc} (bar)
1	320.00	1.445	0.482	0.166	0.948
2	310.00	1.042	0.329	0.107	0.674
3	314.91	1.227	0.399	0.134	0.800

Therefore, the bubble point temperature is approximately 315 K. Composition of the first bubble of vapor can be calculated from

$$y_i = \frac{x_i P_i^{\text{vap}}}{P}.$$

Substitution of the values gives the vapor phase mole fractions as

$$\begin{aligned} y_1 &= \frac{(0.50)(1.227)}{0.8} = 0.767, \\ y_2 &= \frac{(0.45)(0.399)}{0.8} = 0.224, \\ y_3 &= \frac{(0.05)(0.134)}{0.8} = 0.008. \end{aligned}$$

As a result of roundoff errors, $\Sigma y_i = 0.999$.

Dew point temperature: In this case we know the total pressure, 0.8 bar, and the vapor phase composition, $y_1 = 0.50$, $y_2 = 0.45$, and $y_3 = 0.05$. At the correct value of temperature, the pressure calculated from

$$P_{\text{calc}} = \frac{1}{\frac{y_1}{P_1^{\text{vap}}} + \frac{y_2}{P_2^{\text{vap}}} + \frac{y_3}{P_3^{\text{vap}}}}$$

must be equal to 0.8. The results of the iterative procedure are shown in the table below:

Trial #	T (K)	P_1^{vap} (bar)	P_2^{vap} (bar)	P_3^{vap} (bar)	P_{calc} (bar)
1	320.00	1.445	0.482	0.166	0.633
2	330.00	1.961	0.687	0.249	0.900
3	326.57	1.770	0.610	0.217	0.800

Therefore, the dew point temperature is approximately 327 K. Composition of the first drop of liquid can be calculated from

$$x_i = \frac{y_i P}{P_i^{\text{vap}}}.$$

Substitution of the values gives the liquid phase mole fractions as

$$\begin{aligned} x_1 &= \frac{(0.50)(0.8)}{1.770} = 0.226, \\ x_2 &= \frac{(0.45)(0.8)}{0.610} = 0.590, \\ x_3 &= \frac{(0.05)(0.8)}{0.217} = 0.184. \end{aligned}$$

Note that $\sum x_i = 1.0$. The dew point and bubble point temperatures differ by $327 - 315 = 12$ K for this mixture.

Comment: Problem 9.5 discusses the use of the Newton-Raphson method for the calculation of bubble point and dew point temperatures.

9.3.5 Flash Calculation

A typical example to an equilibrium stage is a flash vaporization unit, where a liquid feed is brought to conditions such that a liquid phase and a vapor phase will develop and approach equilibrium in a vessel, which is commonly called a *flash drum*. The overhead vapor product and bottoms liquid product leave the flash drum as separate streams and they are assumed to be in equilibrium with each other. In this way, more-volatile components can be separated from less-volatile components. Separation of liquid feed into vapor and liquid phases requires a sudden decrease of the pressure

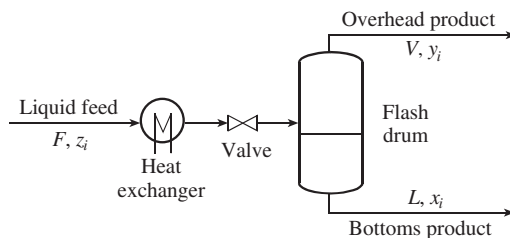


Figure 9.7: A flash vaporization unit.

of the feed stream as it enters the flash drum. This is accomplished by a throttling valve. A typical flash vaporization unit is shown in Fig. 9.7.

At a given temperature (or pressure) and feed composition, the specification of the flash drum pressure (or temperature) is not arbitrary. The pressure-composition and temperature-composition diagrams of a binary mixture obeying Raoult's law are given in Fig. 9.8. At a specified feed composition (z_1) and flash drum temperature (T), the drum pressure (P) should lie between the bubble point and dew point pressures in order to have a two-phase mixture within the drum. On the other hand, at a specified feed composition (z_1) and flash drum pressure (P), the drum temperature (T) should lie between the bubble point and dew point temperatures in order to have a two-phase mixture within the drum.

Example 9.5 A liquid mixture containing 65 mol% benzene (1) and 35% toluene (2) is flowing at 80 kmol/h. It is flashed across a valve and into a flash drum held at 363 K. Determine the limiting values of the drum pressure in order to have a two-phase mixture within the flash drum.

Solution

From Appendix C, the vapor pressures are expressed in the form

$$\ln P_1^{\text{vap}} = 9.2806 - \frac{2788.51}{T - 52.36} \quad \text{and} \quad \ln P_2^{\text{vap}} = 9.3935 - \frac{3096.52}{T - 53.67}.$$

At 363 K, the vapor pressures are

$$P_1^{\text{vap}} = 1.355 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 0.540 \text{ bar}.$$

The pressure of the system should be between the bubble point and dew point pressures in order to have a two-phase mixture within the flash drum. The bubble

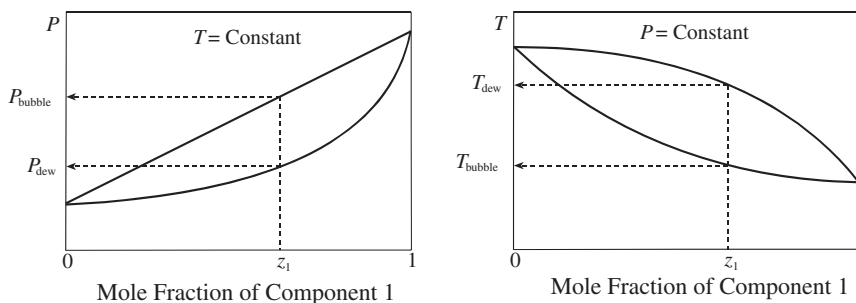


Figure 9.8: Extremum values of pressure and temperature to ensure a two-phase mixture within the flash drum.

point pressure, P_{bubble} , is

$$P_{\text{bubble}} = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} = (0.65)(1.355) + (0.35)(0.540) = 1.070 \text{ bar.}$$

On the other hand, the dew point pressure, P_{dew} , is

$$P_{\text{dew}} = \frac{1}{\frac{y_1}{P_1^{\text{vap}}} + \frac{y_2}{P_2^{\text{vap}}}} = \frac{1}{\frac{0.65}{1.355} + \frac{0.35}{0.540}} = 0.887 \text{ bar.}$$

To ensure the presence of both vapor and liquid phases within the drum

$$0.887 \text{ bar} < P_{\text{system}} < 1.070 \text{ bar.}$$

The overall and component material balances around the flash drum are given as

$$F = V + L, \quad (9.3-5)$$

$$Fz_i = Vy_i + Lx_i. \quad (9.3-6)$$

Simultaneous solution of Eqns (9.3-5) and (9.3-6) yields the fraction vaporized as

$$\frac{V}{F} = \frac{z_i - x_i}{y_i - x_i}. \quad (9.3-7)$$

Once V/F is determined, then the use of Eqn (9.3-5) gives

$$\frac{L}{F} = 1 - \frac{V}{F} = \frac{y_i - z_i}{y_i - x_i}. \quad (9.3-8)$$

From Eqns (9.3-7) and (9.3-8)

$$\frac{V}{L} = \frac{z_i - x_i}{y_i - z_i} \Rightarrow L(z_i - x_i) = V(y_i - z_i). \quad (9.3-9)$$

For a binary system at specified temperature and pressure, the locations of the feed, vapor, and liquid streams on the Txy diagram are shown in Fig. 9.9. For this system, Eqn (9.3-9) becomes

$$L(z_1 - x_1) = V(y_1 - z_1) \Rightarrow \overline{LLF} = \overline{VFV}. \quad (9.3-10)$$

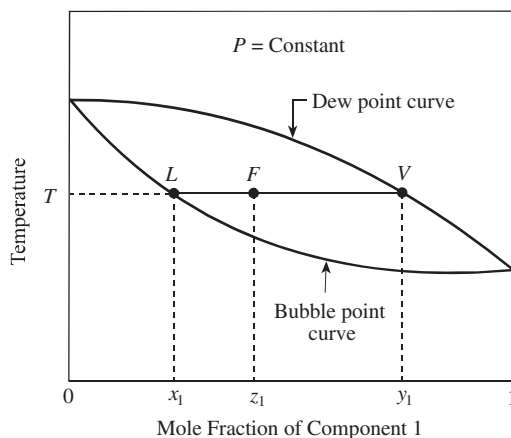


Figure 9.9: Location of the feed, liquid, and vapor streams.

On the other hand, consider two masses, m_A and m_B , that are in equilibrium on a lever as shown in Fig. 9.10. From physics

$$m_A d_A = m_B d_B. \quad (9.3-11)$$

In other words, masses are inversely proportional to their distances from the fulcrum.

Equation (9.3-10) is completely analogous to Eqn (9.3-11) with the feed point F acting as a fulcrum. For this reason, Eqns (9.3-7)–(9.3-9) are also referred to as the *lever rule* and can be used to determine the amounts of vapor and liquid streams. Note that Eqns (9.3-7) and (9.3-8) are also expressed in the form

$$\frac{V}{F} = \frac{\overline{LF}}{\overline{LV}} \quad L \text{ is the fulcrum,} \quad (9.3-12)$$

$$\frac{L}{F} = \frac{\overline{FV}}{\overline{LV}} \quad V \text{ is the fulcrum.} \quad (9.3-13)$$

The use of the lever rule is not only limited to Txy diagrams. It can also be used for Pxy diagrams.

In phase equilibrium calculations, mole fractions in the vapor and liquid phases are related to each other through the K -value⁶ defined as

$$K_i = \frac{y_i}{x_i}. \quad (9.3-14)$$

⁶ K -value is sometimes called *distribution coefficient* or *partition coefficient*.

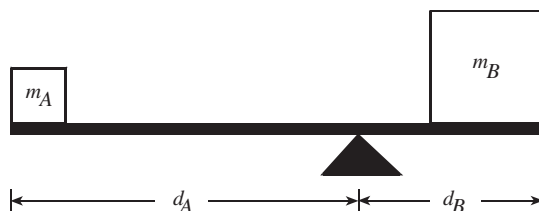


Figure 9.10: Static equilibrium on a lever.

The use of Raoult's law, Eqn (9.2-4), in Eqn (9.3-14) leads to

$$K_i(T, P) = \frac{y_i}{x_i} = \frac{P_i^{\text{vap}}(T)}{P}. \quad (9.3-15)$$

Once temperature and pressure are specified, K_i values can be easily calculated from Eqn (9.3-15). When $T_r \geq 0.6$, Eqn (4) of Problem 5.18 may be used for the estimation of vapor pressures. The use of such an equation in Eqn (9.3-15) leads to

$$\log K_i = \frac{7}{3}(1 + \omega_i) \left(1 - \frac{1}{T_{r_i}} \right) - \log P_{r_i} \quad T_{r_i} \geq 0.6. \quad (9.3-16)$$

In the literature, monographs are available for the estimation of K -values of certain substances. For example, DePriester (1953) presented K -values for light hydrocarbons as a function of temperature and pressure. A representative DePriester chart in SI units is shown in Fig. 9.11. This chart may be used for various purposes:

- *Determination of the K -value of a specified hydrocarbon at a given temperature and pressure:* Draw a straight line joining the given temperature and pressure. Read the K -value at the intersection of this straight line and the specified hydrocarbon's curve.
- *Determination of the boiling point temperature of a specified hydrocarbon at a given pressure:* Since $K = 1$ for a pure substance, draw a straight line joining the given pressure and $K = 1$ on the specified hydrocarbon's curve. Extension of this straight line to the temperature axis gives the boiling point temperature.
- *Determination of the vapor pressure at a given temperature:* Draw a straight line joining the given temperature and $K = 1$ on the specified hydrocarbon's curve. Extension of this straight line to the pressure axis gives the vapor pressure.

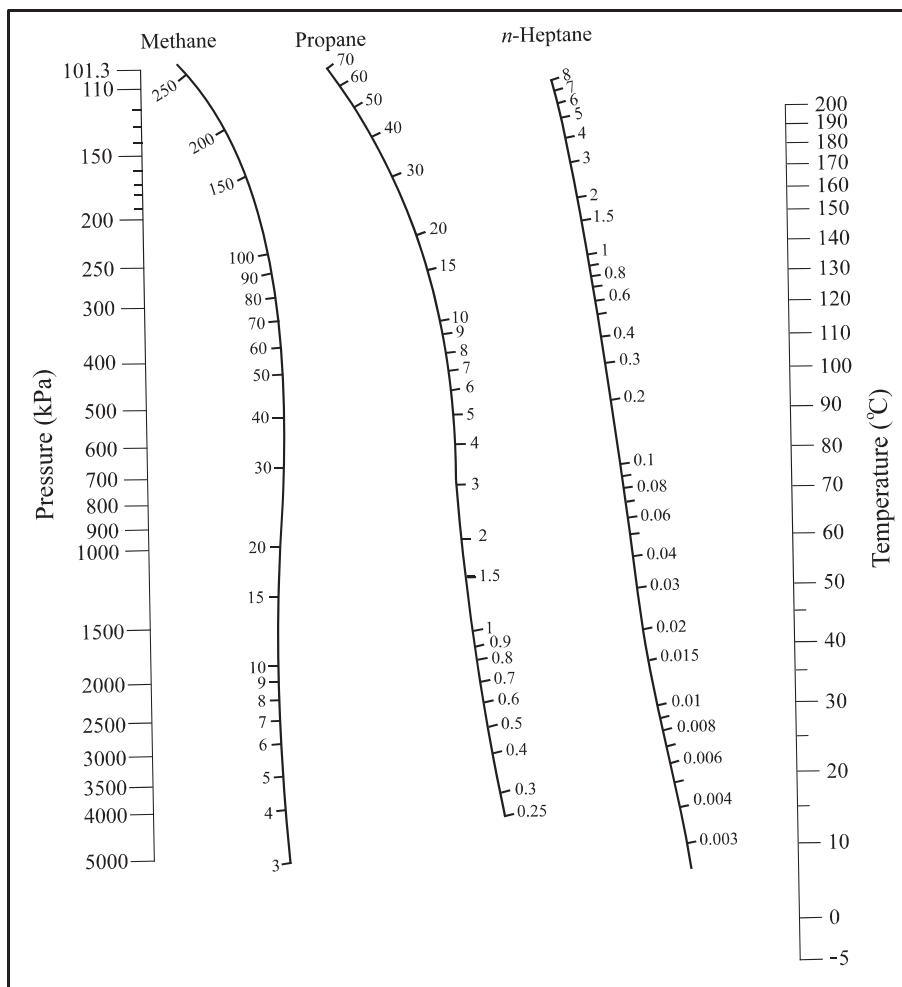


Figure 9.11: A representative DePriester chart. (Reproduced with permission from Dadyburjor (1978). Copyright 1978 by the American Institute of Chemical Engineers.)

The compositions of the liquid and vapor streams exiting the flash chamber can be calculated from the simultaneous solution of Eqns (9.3-5), (9.3-6), and (9.3-14). The resulting equations are

$$x_i = \frac{z_i}{1 + \frac{V}{F}(K_i - 1)}, \quad (9.3-17)$$

$$y_i = \frac{z_i K_i}{1 + \frac{V}{F}(K_i - 1)}. \quad (9.3-18)$$

The parameters involved in the flash calculations are the temperature (T) and pressure (P) of the flash chamber, feed composition (z_i), V/F (or L/F) ratio, overhead product composition (y_i), and bottoms product composition (x_i). Among these variables, VLE calculations involve various possibilities, some of which are listed below:

- Given T, P, z_i ; calculate $V/F, x_i, y_i$.
- Given $T, V/F, z_i$; calculate P, x_i, y_i .
- Given P, z_i, y_i (or x_i); calculate $T, V/F, x_i$ (or y_i).

Case (i): Given T, P, z_i ; calculate $V/F, x_i, y_i$.

The summation of Eqns (9.3-17) and (9.3-18) over all components present in the system gives

$$\sum_{i=1}^k x_i = 1 = \sum_{i=1}^k \frac{z_i}{1 + \frac{V}{F}(K_i - 1)}, \quad (9.3-19)$$

$$\sum_{i=1}^k y_i = 1 = \sum_{i=1}^k \frac{z_i K_i}{1 + \frac{V}{F}(K_i - 1)}. \quad (9.3-20)$$

Subtraction of Eqn (9.3-19) from Eqn (9.3-20) results in⁷

$$\sum_{i=1}^k \frac{z_i(K_i - 1)}{1 + \frac{V}{F}(K_i - 1)} = 0. \quad (9.3-21)$$

The procedure to calculate the unknown quantities is given as follows:

1. Determine the vapor pressures of pure components at the given temperature.
2. Calculate K_i values [Eqns (9.3-15), (9.3-16), or DePriester chart].
3. Use Eqn (9.3-21) to determine the V/F value.
4. Calculate x_i and y_i values from Eqns (9.3-17) and (9.3-18), respectively.

⁷In the literature, Eqn (9.3-21) is also known as the *Rachford-Rice equation* (1952).

Example 9.6 A liquid mixture containing 65 mol% benzene (1) and 35% toluene (2) is flowing at 80 kmol/h. It is flashed across a valve and into a flash drum held at 363 K and 1 bar. Determine the compositions and flow rates of the overhead and bottoms products.

Solution

From Example 9.5, the vapor pressures are

$$P_1^{\text{vap}} = 1.355 \text{ bar} \quad P_2^{\text{vap}} = 0.540 \text{ bar}.$$

The K -values are

$$K_1 = \frac{1.355}{1} = 1.355,$$

$$K_2 = \frac{0.540}{1} = 0.540.$$

From Eqn (9.3-21)

$$\frac{(0.65)(1.355 - 1)}{1 + \frac{V}{F}(1.355 - 1)} + \frac{(0.35)(0.540 - 1)}{1 + \frac{V}{F}(0.540 - 1)} = 0. \quad (1)$$

The solution of Eqn (1) gives

$$\frac{V}{F} = 0.43 \Rightarrow V = 34.4 \text{ kmol/h} \quad \text{and} \quad L = 45.6 \text{ kmol/h}.$$

The use of Eqn (9.3-17) gives

$$x_1 = \frac{(0.65)}{1 + (0.43)(1.355 - 1)} = 0.564 \Rightarrow x_2 = 1 - 0.564 = 0.436.$$

From Eqn (9.3-18)

$$y_1 = \frac{(0.65)(1.355)}{1 + (0.43)(1.355 - 1)} = 0.764 \Rightarrow y_2 = 1 - 0.764 = 0.236.$$

For a binary mixture, determination of V/F from Eqn (9.3-21) is rather simple. When the number of components is greater than two, the root of the equation

$$f(V/F) = \sum_{i=1}^k \frac{z_i(K_i - 1)}{1 + \frac{V}{F}(K_i - 1)} = 0 \quad (9.3-22)$$

can be calculated by the Newton-Raphson method. It is based on the expansion of the function $f(V/F)$ by Taylor series around an estimate $(V/F)_{n-1}$. The resulting expression is given by

$$(V/F)_n = (V/F)_{n-1} - \frac{f[(V/F)_{n-1}]}{\left. \frac{df}{d(V/F)} \right|_{(V/F)_{n-1}}} \quad n > 0. \quad (9.3-23)$$

The use of Eqn (9.3-22) in Eqn (9.3-23) leads to

$$(V/F)_n = (V/F)_{n-1} + \frac{\sum_{i=1}^k \frac{z_i(K_i - 1)}{1 + (V/F)_{n-1}(K_i - 1)}}{\sum_{i=1}^k \frac{z_i(K_i - 1)^2}{[1 + (V/F)_{n-1}(K_i - 1)]^2}} \quad n > 0. \quad (9.3-24)$$

Iterations start with an initial estimate $(V/F)_o$ and the required $(V/F)_n$ is obtained when

$$|(V/F)_n - (V/F)_{n-1}| < \varepsilon, \quad (9.3-25)$$

where ε is a small positive number determined by the desired accuracy.

Example 9.7 A mixture of 25 mol% propane (1), 15% *n*-butane (2), 25% *n*-pentane (3), and 35% *n*-hexane (4) is to be separated by a flash drum operating at 310 K and 1.5 bar. Estimate the compositions of the vapor and liquid streams leaving the flash drum.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Propane	369.9	42.5	0.153
<i>n</i> -Butane	425.0	38.0	0.199
<i>n</i> -Pentane	469.8	33.6	0.251
<i>n</i> -Hexane	507.6	30.2	0.299

The vapor pressures of propane and *n*-butane cannot be determined by the use of the Antoine equation as the given temperature is outside the limits of applicability. Therefore, K -values are calculated from Eqn (9.3-16) as follows:

Component	T_r	P_r	K
Propane	0.838	0.035	8.560
<i>n</i> -Butane	0.729	0.039	2.322
<i>n</i> -Pentane	0.660	0.045	0.701
<i>n</i> -Hexane	0.611	0.050	0.235

The V/F value can be calculated from Eqn (9.3-24). Starting with $(V/F)_o = 0.1$, the iterations are given in the table below:

n	$(V/F)_n$
0	0.1
1	0.273
2	0.443
3	0.494
4	0.495

Composition of the bottoms product is calculated from Eqn (9.3-17) as

$$x_1 = 0.053 \quad x_2 = 0.091 \quad x_3 = 0.293 \quad x_4 = 0.563.$$

Composition of the overhead product is calculated from Eqn (9.3-18) as

$$y_1 = 0.451 \quad y_2 = 0.211 \quad y_3 = 0.206 \quad y_4 = 0.133.$$

Case (ii): Given T , V/F , z_i ; calculate P , x_i , y_i .

The procedure to calculate the unknown quantities is given as follows:

1. Determine the vapor pressures of pure components at the given temperature.
2. Rewrite Eqn (9.3-21) in the form

$$f(P) = \sum_{i=1}^k \frac{z_i \left(\frac{P_i^{\text{vap}}}{P} - 1 \right)}{1 + \frac{V}{F} \left(\frac{P_i^{\text{vap}}}{P} - 1 \right)} = 0 \quad (9.3-26)$$

and determine P . Application of the Newton-Raphson method yields

$$P_n = P_{n-1} - \frac{f(P_{n-1})}{\left. \frac{df}{dP} \right|_{P_{n-1}}} \quad n > 0 \quad (9.3-27)$$

or

$$P_n = P_{n-1} + \frac{\sum_{i=1}^k \frac{z_i (P_i^{\text{vap}} - P_{n-1})}{P_{n-1} + \frac{V}{F} (P_i^{\text{vap}} - P_{n-1})}}{\sum_{i=1}^k \frac{z_i P_i^{\text{vap}}}{\left[P_{n-1} + \frac{V}{F} (P_i^{\text{vap}} - P_{n-1}) \right]^2}} \quad n > 0. \quad (9.3-28)$$

In the case of a binary mixture, Eqn (9.3-26) takes the form

$$\alpha_1 P^2 + \alpha_2 P + \alpha_3 = 0, \quad (9.3-29)$$

where

$$\alpha_1 = 1 - \frac{V}{F}, \quad \alpha_2 = \frac{V}{F} (P_1^{\text{vap}} + P_2^{\text{vap}}) - (z_1 P_1^{\text{vap}} + z_2 P_2^{\text{vap}}), \quad \alpha_3 = -\frac{V}{F} P_1^{\text{vap}} P_2^{\text{vap}}. \quad (9.3-30)$$

3. Determine K_i values.

4. Calculate x_i and y_i values from Eqns (9.3-17) and (9.3-18), respectively.

Example 9.8 A liquid mixture containing 65 mol% benzene (1) and 35% toluene (2) is flowing at 80 kmol/h. It is flashed across a valve and into a flash drum held at 363 K. If the vapor stream leaving the flash drum has a flow rate of 45 kmol/h, determine the drum pressure as well as the compositions of the liquid and vapor phases.

Solution

From Example 9.5, the vapor pressures are

$$P_1^{\text{vap}} = 1.355 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 0.540 \text{ bar}.$$

The V/F value is $45/80 = 0.563$. From Eqn (9.3-30)

$$\alpha_1 = 1 - 0.563 = 0.437,$$

$$\alpha_2 = (0.563)(1.355 + 0.540) - [(0.65)(1.355) + (0.35)(0.540)] \\ = -2.9 \times 10^{-3},$$

$$\alpha_3 = -(0.563)(1.355)(0.54) = -0.412.$$

Thus, Eqn (9.3-29) becomes

$$0.437 P^2 - 2.9 \times 10^{-3} P - 0.412 = 0 \quad \Rightarrow \quad P = 0.974 \text{ bar}.$$

The K -values are

$$K_1 = \frac{1.355}{0.974} = 1.39 \quad \text{and} \quad K_2 = \frac{0.540}{0.974} = 0.55.$$

The composition of the liquid phase is determined from Eqn (9.3-17) as

$$x_1 = \frac{(0.65)}{1 + (0.563)(1.39 - 1)} = 0.533 \quad \Rightarrow \quad x_2 = 1 - 0.533 = 0.467.$$

From Eqn (9.3-18)

$$y_1 = \frac{(0.65)(1.39)}{1 + (0.563)(1.39 - 1)} = 0.741 \quad \Rightarrow \quad y_2 = 1 - 0.741 = 0.259.$$

Comment: The problem can also be stated as follows: “It is required to vaporize 56.3% of a mixture containing 65 mol% benzene (1) and 35% toluene (2) at 363 K. Estimate the pressure and the composition of the liquid and vapor phases.”

Case (iii): Given P , z_i , y_i (or x_i); calculate T , V/F , x_i (or y_i).

The procedure to calculate the unknown quantities is given as follows:

1. Assume T and determine the vapor pressures of pure components.
2. Use Eqn (9.2-10) and calculate P_{calc} .
3. Compare P_{calc} with the specified P . If $P_{\text{calc}} > P$, assumed T is too high; if $P_{\text{calc}} < P$, assumed T is too low. When the calculated pressure matches the specified pressure, mole fractions in the liquid phase are calculated from

$$x_i = \frac{y_i P}{P_i^{\text{vap}}}. \quad (9.3-31)$$

4. Calculate V/F from Eqn (9.3-7), i.e.

$$\frac{V}{F} = \frac{z_i - x_i}{y_i - x_i}. \quad (9.3-32)$$

Example 9.9 A liquid mixture containing 65 mol% benzene (1) and 35% toluene (2) is flowing at 80 kmol/h. It is flashed across a valve and into a flash chamber held at a pressure of 1 bar. It is required to get a vapor stream with a composition of 80 mol% benzene and 20 mol% toluene. Determine the temperature and flow rates of the vapor and liquid streams leaving the flash chamber.

Solution

The temperature is the dew point temperature of the vapor with given composition at 1 bar. At the dew point temperature, the pressure calculated from

$$P_{\text{calc}} = \frac{1}{\frac{y_1}{P_1^{\text{vap}}} + \frac{y_2}{P_2^{\text{vap}}}}$$

must be equal to unity.

The saturation (boiling point) temperatures of pure components at 1 bar are

$$T_1^{\text{sat}} = 52.36 + \frac{2788.51}{9.2806 - \ln 1} = 352.8 \text{ K},$$

$$T_2^{\text{sat}} = 53.67 + \frac{3096.52}{9.3935 - \ln 1} = 383.3 \text{ K}.$$

As an initial temperature estimate, let us pick the mole fraction weighted average of the pure component boiling points, i.e.

$$T = (0.65)(352.8) + (0.35)(383.3) \simeq 363 \text{ K}.$$

The values of the iterative procedure are shown in the table below:

T (K)	P_1^{vap} (bar)	P_2^{vap} (bar)	P_{calc} (bar)
363	1.355	0.540	1.041
361	1.279	0.506	0.980
361.7	1.305	0.517	1.000

The composition of the liquid phase is

$$x_1 = \frac{y_1 P}{P_1^{\text{vap}}} = \frac{(0.8)(1)}{1.305} = 0.613.$$

The value of V/F is calculated as

$$\frac{V}{F} = \frac{z_1 - x_1}{y_1 - x_1} = \frac{0.65 - 0.613}{0.80 - 0.613} = 0.2.$$

The flow rates of the overhead and bottoms products are

$$V = 0.2F = (0.2)(80) = 16 \text{ kmol/h},$$

$$L = F - V = 80 - 16 = 64 \text{ kmol/h}.$$

9.3.5.1 The Use of Graphical Techniques for Flash Calculations

For a binary system, flash calculations can also be carried out graphically using the P_{xy} , T_{xy} , or x - y diagrams.

- **The use of P_{xy} and T_{xy} diagrams**

The following example shows how to use pressure-composition and temperature-composition diagrams in flash calculations.

Example 9.10 Resolve Examples 9.6, 9.8 and 9.9 using pressure-composition and temperature-composition diagrams.

Solution

For this purpose, it is first necessary to prepare P_{xy} and T_{xy} diagrams for a binary mixture of benzene (1) and toluene (2).

Pressure-composition diagram: At a temperature of 363 K, the vapor pressures are

$$P_1^{\text{vap}} = 1.355 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 0.540 \text{ bar}.$$

The bubble point pressure curve is represented by

$$P_{\text{bubble}} = 1.355x_1 + 0.54x_2,$$

which is a straight line joining P_1^{vap} and P_2^{vap} . The dew point pressure equation is

$$P_{\text{dew}} = \frac{1}{\frac{y_1}{1.355} + \frac{y_2}{0.54}}.$$

The results are given in the table below:

y_1	P_{dew} (bar)	y_1	P_{dew} (bar)	y_1	P_{dew} (bar)
0.05	0.557	0.40	0.711	0.75	0.984
0.10	0.575	0.45	0.740	0.80	1.041
0.15	0.594	0.50	0.772	0.85	1.105
0.20	0.614	0.55	0.807	0.90	1.177
0.25	0.636	0.60	0.845	0.95	1.260
0.30	0.659	0.65	0.887		
0.35	0.684	0.70	0.933		

Temperature-composition diagram: The saturation (boiling point) temperatures of pure components at 1 bar are

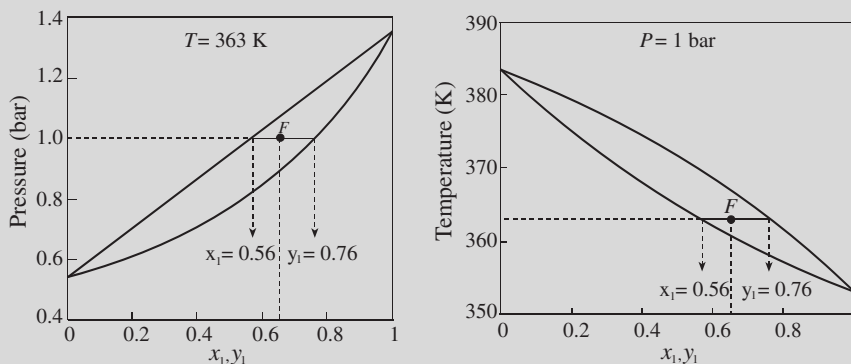
$$T_1^{\text{sat}} = 352.8 \text{ K} \quad \text{and} \quad T_2^{\text{sat}} = 383.3 \text{ K}.$$

By picking temperature values between 352.8 and 383.3 K, the calculated values of x_1 and y_1 are tabulated as follows:

T (K)	x_1	y_1	T (K)	x_1	y_1	T (K)	x_1	y_1
353	0.991	0.997	365	0.494	0.709	377	0.144	0.287
355	0.895	0.956	367	0.427	0.649	379	0.096	0.202
357	0.804	0.913	369	0.364	0.585	381	0.050	0.111
359	0.719	0.867	371	0.304	0.517	383	0.007	0.015
361	0.639	0.818	373	0.248	0.445			
363	0.564	0.765	375	0.195	0.368			

• *Example 9.6*

The problem can be solved by using either the pressure-composition or temperature-composition diagram. For this purpose, first locate the feed point F at the given temperature and pressure. Then the tie line passing through the feed point gives the compositions of the overhead and bottoms products.

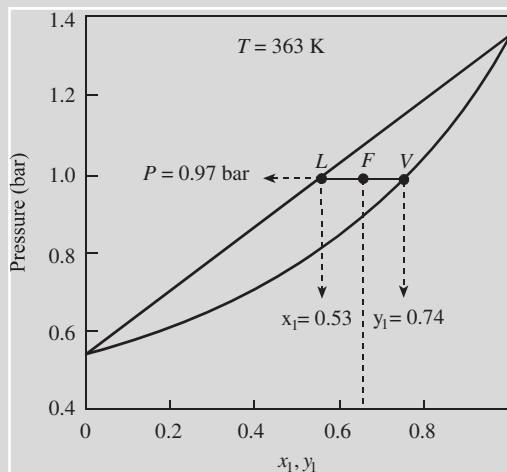


• *Example 9.8*

Since temperature is known, the problem is solved by using the pressure-composition diagram. The feed point F must lie on a vertical line passing through $z_1 = 0.65$. Within the two-phase region, the tie line passing through the feed point F should be drawn in such a way that

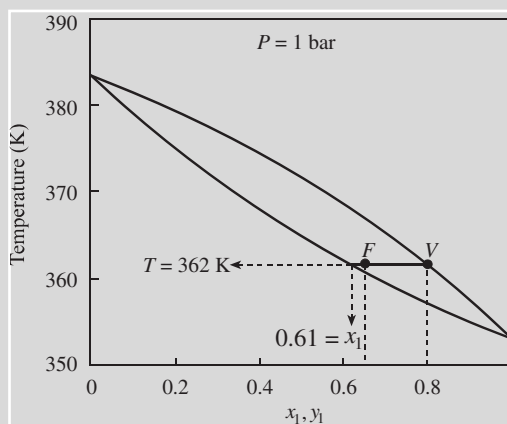
$$\frac{\overline{FV}}{\overline{LF}} = \frac{L}{V} = \frac{35}{45} = 0.78.$$

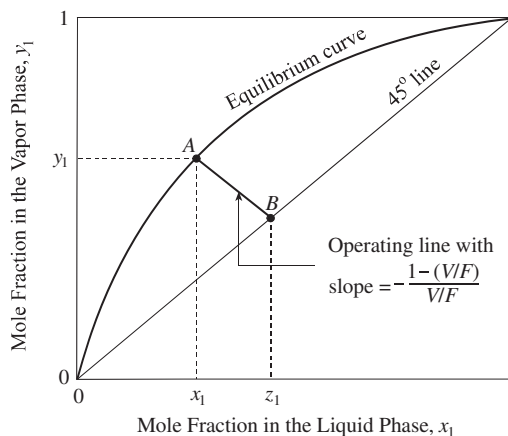
Then the pressure as well as the compositions of the streams can be easily determined as shown in the figure below:



• *Example 9.9*

Since pressure is known, the problem is solved by using the temperature-composition diagram. Using the given value of y_1 , the vapor stream V can be easily located on the dew point curve. The tie line passing through this point gives the temperature. The tie line intersects the bubble point curve at x_1 . Since z_1 is known, the feed point F can be located on the tie line. Application of the lever rule then gives the amounts of vapor and liquid streams.



Figure 9.12: The x - y diagram for a binary system.

- **The use of x - y diagrams**

Rearrangement of Eqn (9.3-7) yields

$$y_i = -\frac{1 - (V/F)}{V/F}x_i + \frac{1}{V/F}z_i, \quad (9.3-33)$$

which is known as the *operating line*. Note that the operating line is nothing more than the material balance relating compositions of the vapor and liquid streams leaving the flash chamber. Since these streams are in equilibrium with each other, point A on the equilibrium curve has the coordinates (x_1, y_1) as shown in Fig. 9.12. The operating line must pass through the point A.

When $x_i = y_i$, Eqn (9.3-33) gives

$$x_i = y_i = z_i. \quad (9.3-34)$$

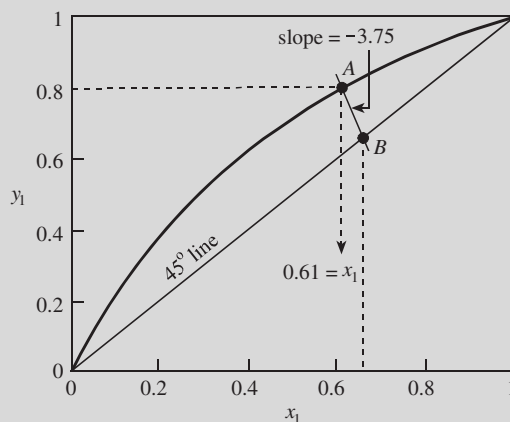
In other words, the operating line intersects the 45° line at $x_i = z_i$, i.e. point B. Note that the slope of the operating line is $-[1 - (V/F)]/(V/F)$.

Example 9.11 Resolve Example 9.9 using the x - y diagram.

Solution

The x - y diagram can be plotted by using the values given in Example 9.10. Since the vapor and liquid streams are in equilibrium with each other, y_1 and x_1 values must lie on the equilibrium curve. When $y_1 = 0.8$, point A can

be easily located on the equilibrium curve with $x_1 = 0.61$. Using the feed composition of $z_1 = 0.65$, the point B can be located on a 45° line (point B). The straight line passing through the points A and B is the operating line with slope $-[1 - (V/F)]/(V/F)$. Thus, the fraction vaporized, i.e. V/F , can be calculated from the slope as 0.21.



9.4 VLE Calculations by Numerical Techniques

When Raoult's law is applicable, VLE calculations are rather straightforward. The presence of liquid nonideality, which is usually the case, does not permit γ_i to be taken as unity in Eqn (9.2-1). Thus, when temperature and/or liquid phase composition are unknown, bubble point temperature and dew point (temperature or pressure) calculations require different trial-and-error procedures. Furthermore, if the mixture contains more than two species, which is the usual case in the chemical industry, hand calculations become very tedious. In this case, numerical solutions of the equations representing the phase equilibrium relationship are an alternative.

9.4.1 Mathematical Preliminaries

Newton's method is one of the most commonly used techniques to estimate the roots of nonlinear systems of algebraic equations. Consider two nonlinear algebraic equations in two unknowns, x and y , expressed as

$$f_1(x, y) = 0, \quad (9.4-1)$$

$$f_2(x, y) = 0. \quad (9.4-2)$$

In matrix form, Newton's iteration equation reads

$$\underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}}_{\mathbf{J}} \cdot \underbrace{\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix}}_{\mathbf{Y}} = - \underbrace{\begin{bmatrix} f_1 \\ f_2 \end{bmatrix}}_{\mathbf{F}}, \quad (9.4-3)$$

where Δ_1 and Δ_2 represent the difference between two consecutive iterations for x and y values, respectively, i.e.

$$\Delta_1 = x^{(k)} - x^{(k-1)} \quad \text{and} \quad \Delta_2 = y^{(k)} - y^{(k-1)} \quad (9.4-4)$$

with k being the iteration counter.

If the partial derivatives are difficult to evaluate analytically, numerical approximations are employed. In this respect, the forward difference approximations are

$$\frac{\partial f_1}{\partial x} = \frac{f_1(x + \Delta x, y) - f_1(x, y)}{\Delta x} \quad \text{and} \quad \frac{\partial f_1}{\partial y} = \frac{f_1(x, y + \Delta y) - f_1(x, y)}{\Delta y}, \quad (9.4-5)$$

$$\frac{\partial f_2}{\partial x} = \frac{f_2(x + \Delta x, y) - f_2(x, y)}{\Delta x} \quad \text{and} \quad \frac{\partial f_2}{\partial y} = \frac{f_2(x, y + \Delta y) - f_2(x, y)}{\Delta y}. \quad (9.4-6)$$

Furthermore, taking $\Delta x = x/100$ and $\Delta y = y/100$, the matrix $\mathbf{J}$ becomes

$$\mathbf{J} = 100 \begin{bmatrix} \frac{f_1(1.01x, y) - f_1(x, y)}{x} & \frac{f_1(x, 1.01y) - f_1(x, y)}{y} \\ \frac{f_2(1.01x, y) - f_2(x, y)}{x} & \frac{f_2(x, 1.01y) - f_2(x, y)}{y} \end{bmatrix}. \quad (9.4-7)$$

The iteration scheme is given as follows:

- Choose initial estimates $x^{(0)}$, $y^{(0)}$ and the error tolerance ϵ .
- Initialize the iteration counter $k = 0$.
- Loop:
 - (i) Increase the iteration counter by one, i.e. set $k = k + 1$.
 - (ii) Use $x^{(k-1)}$, $y^{(k-1)}$ to calculate the entries in $\mathbf{J}$ and $\mathbf{F}$.
 - (iii) Solve Eqn (9.4-3), to obtain Δ_1 and Δ_2 , i.e.

$$\mathbf{Y} = -\mathbf{J}^{-1} \cdot \mathbf{F}. \quad (9.4-8)$$

(iv) Update x and y using

$$x^{(k)} = x^{(k-1)} + \Delta_1 \quad \text{and} \quad y^{(k)} = y^{(k-1)} + \Delta_2. \quad (9.4-9)$$

(v) Choose the maximum difference, $\Delta_{\max} = \max \{|\Delta_1|, |\Delta_2|\}$.

(vi) If $\Delta_{\max} > \epsilon$ repeat the loop; if $\Delta_{\max} < \epsilon$, the set $\{x^{(k)}, y^{(k)}\}$ is the solution.

Example 9.12 Solve

$$\begin{aligned} (2x)^{2/3} + y^{2/3} &= 9^{1/3}, \\ \frac{x^2}{4} + y^2 &= 1. \end{aligned}$$

Solution

The given equations are expressed in the form

$$\begin{aligned} f_1(x, y) &= (2x)^{2/3} + y^{2/3} - 9^{1/3} = 0, \\ f_2(x, y) &= \frac{x^2}{4} + y^2 - 1 = 0. \end{aligned}$$

Since analytical differentiation of these functions is straightforward, Eqn (9.4-8) becomes

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = - \begin{bmatrix} \frac{4}{3}(2x)^{-1/3} & \frac{2}{3}y^{-1/3} \\ \frac{x}{2} & 2y \end{bmatrix}^{-1} \cdot \begin{bmatrix} (2x)^{2/3} + y^{2/3} - 9^{1/3} \\ \frac{x^2}{4} + y^2 - 1 \end{bmatrix}. \quad (1)$$

Let us choose $x^{(0)} = 1$, $y^{(0)} = 1$, and $\epsilon = 1 \times 10^{-4}$ to begin iterations for the solution.

First iteration: Eqn (1) becomes

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = - \begin{bmatrix} 1.058267 & 0.666667 \\ 0.5 & 2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0.507317 \\ 0.25 \end{bmatrix}. \quad (2)$$

The solution of Eqn (2) gives

$$\Delta_1 = -0.475531 \quad \text{and} \quad \Delta_2 = -6.117 \times 10^{-3}.$$

Therefore

$$\begin{aligned} x^{(1)} &= 1 - 0.475531 = 0.524469, \\ y^{(1)} &= 1 - 6.117 \times 10^{-3} = 0.993883. \end{aligned}$$

Since $\Delta_{\max} = 0.475531 > \epsilon$, iteration should be repeated.

Second iteration: Eqn (1) becomes

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = - \begin{bmatrix} 1.312267 & 0.668032 \\ 0.262234 & 1.987766 \end{bmatrix}^{-1} \cdot \begin{bmatrix} -0.051801 \\ 0.056570 \end{bmatrix}. \quad (3)$$

The solution of Eqn (3) gives

$$\Delta_1 = 0.057847 \quad \text{and} \quad \Delta_2 = -0.03609.$$

Therefore

$$\begin{aligned} x^{(2)} &= 0.524469 + 0.057847 = 0.582316, \\ y^{(2)} &= 0.993883 - 0.03609 = 0.957793. \end{aligned}$$

Since $\Delta_{\max} = 0.057847 > \epsilon$, iteration should be repeated. Repeating iterations until $\Delta_{\max} < \epsilon$ will yield the desired values of x and y . The results of the iterations are summarized in the following table:

k	x	y	$\Delta_{\max}$
0	1	1	—
1	0.524469	0.993883	0.475531
2	0.582316	0.957793	0.057847
3	0.584235	0.956384	1.919×10^{-3}
4	0.584237	0.956382	2.394×10^{-6}

Comment: Analytical solution of the given nonlinear system yields

$$x = \frac{8\sqrt{2/15}}{5} \quad \text{and} \quad y = \frac{7\sqrt{7/15}}{5}.$$

Example 9.13 Solve

$$\begin{aligned} &x \exp \left(16.59158 - \frac{3643.31}{y - 33.424} \right) \\ &\quad \times \exp [(1 - x)^2 (2.771 - 0.00523y)] = 40.532, \\ &(1 - x) \exp \left(14.25326 - \frac{2665.54}{y - 53.424} \right) \\ &\quad \times \exp [x^2 (2.771 - 0.00523y)] = 60.798. \end{aligned}$$

Solution

To construct nonlinear equations of the form given by Eqns (9.4-1) and (9.4-2), it is more convenient to take the logarithm of the given equations. The resulting equations are

$$f_1(x, y) = \ln x + 12.88949 - \frac{3643.31}{y - 33.424} + (1 - x)^2 (2.771 - 0.00523y) = 0, \quad (1)$$

$$f_2(x, y) = \ln(1 - x) + 10.1457 - \frac{2665.54}{y - 53.424} + x^2 (2.771 - 0.00523y) = 0. \quad (2)$$

Let us choose $x^{(0)} = 0.5$, $y^{(0)} = 350$, and $\epsilon = 1 \times 10^{-3}$ to begin iterations for the solution.

First iteration: Equation (9.4-8) becomes

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = - \underbrace{\begin{bmatrix} 1.13138 & 0.03447 \\ -1.00234 & 0.02881 \end{bmatrix}}_{\mathbf{J}}^{-1} \cdot \underbrace{\begin{bmatrix} 0.92298 \\ 0.69996 \end{bmatrix}}_{\mathbf{F}} \quad (3)$$

in which the values for the $\mathbf{J}$ matrix are calculated from Eqn (9.4-7). The solution of Eqn (3) gives

$$\Delta_1 = -0.03257 \quad \text{and} \quad \Delta_2 = -25.64771.$$

Therefore

$$x^{(1)} = 0.5 - 0.03257 = 0.46743,$$

$$y^{(1)} = 350 - 25.64771 = 324.35229.$$

Since $\Delta_{\max} = 25.64771 > \epsilon$, iteration should be repeated.

Second iteration: Equation (9.4-8) becomes

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = - \begin{bmatrix} 0.98913 & 0.04109 \\ -0.87631 & 0.03474 \end{bmatrix}^{-1} \cdot \begin{bmatrix} -0.09666 \\ -0.08170 \end{bmatrix}. \quad (4)$$

The solution of Eqn (4) gives

$$\Delta_1 = -7.3559 \times 10^{-3} \quad \text{and} \quad \Delta_2 = 2.34982.$$

Therefore

$$\begin{aligned}x^{(2)} &= 0.46743 - 7.3559 \times 10^{-3} = 0.46007, \\y^{(2)} &= 324.35229 + 2.34982 = 326.70211.\end{aligned}$$

Since $\Delta_{\max} = 2.34982 > \epsilon$, iteration should be repeated. Repeating iterations until $\Delta_{\max} < \epsilon$ will yield the desired values of x and y . The results of the iterations are summarized in the following table:

k	x	y	$\Delta_{\max}$
0	0.5	350	—
1	0.46743	324.35229	25.64771
2	0.46007	326.70211	2.34982
3	0.46019	326.69652	5.589×10^{-3}
4	0.46019	326.69657	5.271×10^{-5}

9.4.2 Governing Equations for Nonideal Liquid Mixtures

In a liquid mixture, it is most unlikely that the interactions between like molecules are identical to those between unlike molecules. Therefore, among the assumptions used in the derivation of Raoult's law, the one that liquid phase is an ideal mixture is the most questionable. A more realistic approach is to relax this assumption and modify Raoult's law as

$$y_i P = \gamma_i x_i P_i^{\text{vap}} \quad i = 1, 2, \dots, k. \quad (9.4-10)$$

The use of an activity coefficient, γ_i , on the right-hand side of Eqn (9.4-10) takes into account the nonideality in the liquid phase. Equation (9.4-10) is subject to the following constraints:

$$\sum_{i=1}^k x_i = 1 \quad \text{and} \quad \sum_{i=1}^k y_i = 1. \quad (9.4-11)$$

Since

$$P_i^{\text{vap}} = P_i^{\text{vap}}(T) \quad \text{and} \quad \gamma_i = \gamma_i(T, P, x_i) \quad (9.4-12)$$

solutions of Eqns (9.4-10) and (9.4-11) become rather complicated when temperature and/or liquid phase composition are unknown quantities.

For a binary system, combining Eqns (9.4-10) and (9.4-11) and taking the logarithm lead to

$$\ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.4-13)$$

$$\ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0. \quad (9.4-14)$$

The use of Newton's method allows one to solve Eqns (9.4-13) and (9.4-14) in a rather straightforward manner. It should be kept in mind that Newton's method requires *reasonable* initial estimates. Otherwise, convergence cannot be achieved. For this reason, an understanding of the physics of the problem is of utmost importance in the application of numerical solutions.⁸

9.5 VLE Calculations for Nonideal Liquid Mixtures

The summation of Eqn (9.4-10) over all components present in the system yields

$$\sum_{i=1}^k y_i P = \sum_{i=1}^k x_i \gamma_i P_i^{\text{vap}} \Rightarrow P \underbrace{\sum_{i=1}^k y_i}_1 = \sum_{i=1}^k x_i \gamma_i P_i^{\text{vap}}. \quad (9.5-1)$$

Therefore, the total pressure (or bubble point pressure) is given by

$$P = \sum_{i=1}^k x_i \gamma_i P_i^{\text{vap}}. \quad (9.5-2)$$

It is also possible to rearrange Eqn (9.4-10) as

$$x_i = \frac{y_i P}{\gamma_i P_i^{\text{vap}}}. \quad (9.5-3)$$

Summation of Eqn (9.5-3) over all components present in the system gives

$$\sum_{i=1}^k x_i = \sum_{i=1}^k \frac{y_i P}{\gamma_i P_i^{\text{vap}}} \Rightarrow 1 = P \sum_{i=1}^k \frac{y_i}{\gamma_i P_i^{\text{vap}}}. \quad (9.5-4)$$

Therefore, the total pressure (or dew point pressure) is given by

$$P = \frac{1}{\sum_{i=1}^k \frac{y_i}{\gamma_i P_i^{\text{vap}}}}. \quad (9.5-5)$$

It should be noted that the pressure-composition, temperature-composition, and x - y diagrams for nonideal liquid mixtures will be different from those when Raoult's law is applicable. This point will be discussed in detail in Section 9.6.

⁸As stated by Astarita (1997), students have the tendency to show a lot of respect if the solution comes out of a computer! However, numerical methods are like political candidates; they'll tell you anything you want to hear (Tao, 1989). Unless a quantity is not divided by zero, a computer (or numerical method) always yields a result. One should use engineering common sense to judge whether the result is physically possible.

Example 9.14 Prepare a pressure-composition diagram for a binary mixture of methyl glycolate (1) and ethylene glycol (2) at 343.15 K if the liquid phase nonideality is represented by the NRTL equation with $\alpha = 0.47$, $\tau_{12} = -0.060$, and $\tau_{21} = 1.378$ (Sun *et al.*, 2006).

The vapor pressures of components 1 and 2 are given as

$$\ln P_1^{\text{vap}} = 10.5394 - \frac{3982.02}{T - 45.07} \quad \text{and} \quad \ln P_2^{\text{vap}} = 12.0096 - \frac{4809.96}{T - 69.70},$$

where P is in bar and T is in K.

Solution

The pressure-composition diagram can be constructed by the following procedure:

1. Evaluate vapor pressures at 343.15 K.
2. Evaluate the parameters in the activity coefficient expressions, i.e. G_{12} and G_{21} .
3. Pick a value for liquid phase mole fraction, x_1 .
4. Evaluate the activity coefficients, γ_1 and γ_2 .
5. Calculate the bubble point pressure using Eqn (9.5-2).
6. Use Eqn (9.4-10) to calculate the mole fraction in the vapor phase, y_1 .

The vapor pressures at 343.15 K are calculated as

$$P_1^{\text{vap}} = 0.06 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 3.771 \times 10^{-3} \text{ bar}.$$

The use of Eqn (8.4-27) gives the parameters of the NRTL equation as

$$G_{12} = \exp [-(0.47)(-0.06)] = 1.029$$

$$G_{21} = \exp [-(0.47)(1.378)] = 0.523.$$

Therefore, Eqns (8.4-29) and (8.4-30) become

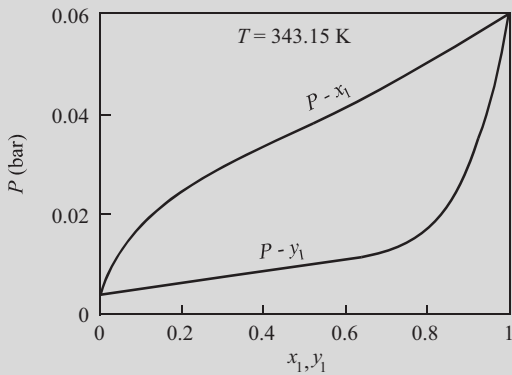
$$\ln \gamma_1 = x_2^2 \left[1.378 \left(\frac{0.523}{x_1 + 0.523x_2} \right)^2 - \frac{0.06174}{(x_2 + 1.029x_1)^2} \right],$$

$$\ln \gamma_2 = x_1^2 \left[-0.06 \left(\frac{1.029}{x_2 + 1.029x_1} \right)^2 + \frac{0.72069}{(x_1 + 0.523x_2)^2} \right].$$

The calculated values of γ_1 , γ_2 , P , and y_1 are given in the following table:

x_1	γ_1	γ_2	P (bar)	y_1	x_1	γ_1	γ_2	P (bar)	y_1
0.00	3.730	1.000	3.771×10^{-3}	0.000	0.55	1.118	1.398	0.039	0.940
0.05	2.951	1.006	0.012	0.711	0.60	1.086	1.454	0.041	0.947
0.10	2.430	1.022	0.018	0.808	0.65	1.061	1.511	0.043	0.954
0.15	2.068	1.045	0.022	0.847	0.70	1.042	1.570	0.046	0.961
0.20	1.808	1.076	0.025	0.870	0.75	1.027	1.630	0.048	0.968
0.25	1.616	1.111	0.027	0.885	0.80	1.016	1.691	0.050	0.975
0.30	1.472	1.151	0.030	0.897	0.85	1.009	1.752	0.052	0.981
0.35	1.362	1.195	0.032	0.907	0.90	1.004	1.813	0.055	0.988
0.40	1.277	1.242	0.033	0.916	0.95	1.001	1.875	0.057	0.994
0.45	1.211	1.291	0.035	0.924	1.00	1.000	1.937	0.060	1.000
0.50	1.159	1.344	0.037	0.932					

The Pxy diagram is shown below:



Example 9.15 For a binary mixture of thiophene (1) and 2,2,4-trimethylpentane (2), [Sapei *et al.* \(2007b\)](#) obtained the following data from VLE experiments at 343.15 K:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.049	1.67	1.00	0.482	1.26	1.12	0.878	1.02	1.86
0.120	1.58	1.01	0.542	1.22	1.17	0.923	1.01	2.08
0.193	1.51	1.02	0.616	1.16	1.26	0.957	1.00	2.39
0.267	1.45	1.03	0.687	1.12	1.35	0.983	1.00	2.50
0.339	1.38	1.05	0.761	1.07	1.49			
0.413	1.31	1.09	0.825	1.05	1.66			

Sapei *et al.* (2007b) also presented a pressure-composition diagram for this binary mixture at a different temperature, 353.15 K. Show how one can predict the P_{xy} diagram at 353.15 K by using the data obtained at 343.15 K; also compare your results with the experimental ones. The vapor pressures of thiophene and 2,2,4-trimethylpentane at 353.15 K are 89.34 kPa and 57.04 kPa, respectively.

Solution

Activity coefficients change as a result of the increase in temperature by 10 K. If the effect of temperature on the activity coefficients is approximated by the regular mixture theory, then

$$\frac{\ln \gamma_i|_{T_2}}{\ln \gamma_i|_{T_1}} = \frac{T_1}{T_2} \quad i = 1, 2. \quad (1)$$

The calculated values of activity coefficients at 353.15 K are given in the table below:

x_1	γ_1	γ_2	x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.049	1.646	1.000	0.482	1.252	1.116	0.878	1.019	1.828
0.120	1.560	1.010	0.542	1.213	1.165	0.923	1.010	2.037
0.193	1.492	1.019	0.616	1.155	1.252	0.957	1.000	2.332
0.267	1.435	1.029	0.687	1.116	1.339	0.983	1.000	2.436
0.339	1.367	1.049	0.761	1.068	1.473			
0.413	1.300	1.087	0.825	1.049	1.636			

The molar excess Gibbs energy is calculated from Eqn (8.3-9), i.e.

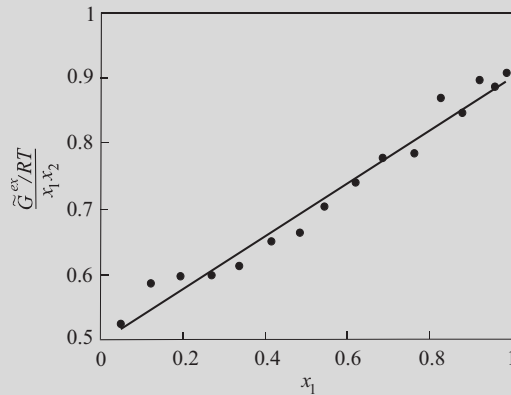
$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 \ln \gamma_1 + x_2 \ln \gamma_2$$

and the plot of $(\tilde{G}^{\text{ex}}/RT)/x_1x_2$ versus x_1 is shown below. Since the data fit a straight line with a correlation coefficient of 0.987, the system may be represented by the three-suffix Margules equation. The slope and the intercept of the best straight line passing through the data points are 0.404 and 0.499, respectively. Therefore, the parameters are

$$A = 0.701 \quad \text{and} \quad B = 0.202$$

and the activity coefficients are represented by

$$\ln \gamma_1 = x_2^2 (1.307 - 0.808x_2) \quad \text{and} \quad \ln \gamma_2 = x_1^2 (0.095 + 0.808x_1).$$



The pressure is calculated from

$$P = x_1 \gamma_1 P_1^{\text{vap}} + x_2 \gamma_2 P_2^{\text{vap}} \quad (2)$$

and the vapor phase composition is determined from

$$y_1 = \frac{x_1 \gamma_1 P_1^{\text{vap}}}{P}. \quad (3)$$

Using the liquid phase mole fractions given by [Sapei *et al.* \(2007b\)](#), the calculated values are compared with the experimental ones in the following table:

x_1	P (kPa)		y_1		x_1	P (kPa)		y_1	
	Exp.	Calc.	Exp.	Calc.		Exp.	Calc.	Exp.	Calc.
0.056	62.06	62.00	0.127	0.131	0.696	92.34	92.43	0.747	0.742
0.133	68.02	68.38	0.262	0.274	0.762	93.02	93.02	0.785	0.780
0.218	73.99	74.61	0.379	0.394	0.819	93.17	93.16	0.821	0.816
0.299	78.72	79.65	0.465	0.483	0.867	93.04	92.91	0.856	0.852
0.368	82.28	83.21	0.528	0.543	0.907	92.57	92.38	0.890	0.887
0.438	85.19	86.17	0.583	0.595	0.940	91.87	91.64	0.922	0.921
0.496	87.52	88.17	0.625	0.632	0.967	90.97	90.78	0.953	0.953
0.549	89.14	89.67	0.661	0.663	0.986	90.12	90.01	0.979	0.979
0.621	90.95	91.27	0.702	0.702					

Comment: Although the predictions are quite good in this specific example, one should be cautious in using the approximation given by Eqn (1).

9.5.1 Bubble Point Pressure Calculation

At a given temperature, it is required to calculate the pressure at which a liquid of known composition first begins to boil. The bubble point pressure calculation is straightforward and does not require numerical calculation. The calculation procedure is given below:

1. Calculate vapor pressures of pure components.
2. Calculate activity coefficients.
3. Calculate the bubble point pressure from Eqn (9.5-2).
4. Determine the composition of the vapor phase from Eqn (9.4-10), i.e.

$$y_i = \frac{x_i \gamma_i P_i^{\text{vap}}}{P}. \quad (9.5-6)$$

Example 9.16 Estimate the bubble point pressure of a liquid mixture of 15 mol% 1,3-dioxolane ($\text{C}_3\text{H}_6\text{O}_2$) and 85% 2-methyl-1-propanol [$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$] at 344 K. Also calculate the composition of the first bubble of vapor.

Data: Reyes *et al.* (2004) provided the following data for a binary mixture of 1,3-dioxolane (1) and 2-methyl-1-propanol (2):

- The activity coefficients of the mixture can be represented by the Wilson equation and the infinite dilution activity coefficients are given as

$$\gamma_1^\infty = 2.01 \quad \gamma_2^\infty = 2.35.$$

- The vapor pressures are given by

$$\begin{aligned} \log P_1^{\text{vap}} &= 6.23182 - \frac{1236.700}{T - 55.92}, \\ \log P_2^{\text{vap}} &= 6.50091 - \frac{1275.197}{T - 97.96}, \end{aligned}$$

where P is in kPa and T is in K.

Solution

The activity coefficients are given by Eqns (8.4-19) and (8.4-20) as

$$\ln \gamma_1 = - \left[\ln (x_1 + \Lambda_{12}x_2) + x_2 \left(\frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} - \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} \right) \right], \quad (1)$$

$$\ln \gamma_2 = - \left[\ln (x_2 + \Lambda_{21}x_1) + x_1 \left(\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right) \right]. \quad (2)$$

The use of Eqn (8.4-23) gives

$$\begin{aligned} \Lambda_{12} &= \frac{1}{\gamma_1^\infty} \exp \left[1 - \frac{1}{\gamma_2^\infty} \exp(1 - \Lambda_{12}) \right] \\ &= \frac{1}{2.01} \exp \left[1 - \frac{1}{2.35} \exp(1 - \Lambda_{12}) \right]. \end{aligned} \quad (3)$$

The solution of Eqn (3) yields $\Lambda_{12} = 0.807$. From Eqn (8.4-24)

$$\Lambda_{21} = 1 - \ln(\Lambda_{12}\gamma_1^\infty) = 1 - \ln[(0.807)(2.01)] = 0.516.$$

Thus, Eqns (1) and (2) take the form

$$\ln \gamma_1 = - \left[\ln (x_1 + 0.807x_2) + x_2 \left(\frac{0.516}{x_2 + 0.516x_1} - \frac{0.807}{x_1 + 0.807x_2} \right) \right] \quad (4)$$

$$\ln \gamma_2 = - \left[\ln (x_2 + 0.516x_1) + x_1 \left(\frac{0.807}{x_1 + 0.807x_2} - \frac{0.516}{x_2 + 0.516x_1} \right) \right] \quad (5)$$

At 344 K, the vapor pressures are

$$P_1^{\text{vap}} = 86.9 \text{ kPa} \quad P_2^{\text{vap}} = 20.8 \text{ kPa}.$$

At $x_1 = 0.15$, the activity coefficients are

$$\gamma_1 = 1.694 \quad \gamma_2 = 1.014.$$

The use of Eqn (9.5-2) leads to

$$\begin{aligned} P &= x_1\gamma_1 P_1^{\text{vap}} + x_2\gamma_2 P_2^{\text{vap}} \\ &= (0.15)(1.694)(86.9) + (0.85)(1.014)(20.8) = 40 \text{ kPa}. \end{aligned}$$

Composition of the first bubble of vapor can be calculated from Eqn (9.5-6), i.e.

$$\begin{aligned} y_1 &= \frac{(0.15)(1.694)(86.9)}{40} = 0.552, \\ y_2 &= \frac{(0.85)(1.014)(20.8)}{40} = 0.448. \end{aligned}$$

9.5.2 Dew Point Pressure Calculation

At a given temperature, it is required to calculate the pressure at which a vapor of known composition first begins to condense. Without the liquid composition data, activity coefficients cannot be determined *a priori*. For a binary mixture, Eqns (9.4-13) and (9.4-14) take the form

$$f_1(x_1, P) = \ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.5-7)$$

$$f_2(x_1, P) = \ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0. \quad (9.5-8)$$

Simultaneous solution of Eqns (9.5-7) and (9.5-8) using Newton's method yields pressure and liquid phase composition.

Example 9.17 Estimate the dew point pressure of a vapor mixture of 87 mol% 1,3-dioxolane (1) and 13% 2-methyl-1-propanol (2) at 328.2 K. Also calculate the composition of the first drop of liquid.

Solution

From the Antoine equations given in Example 9.16, the vapor pressures at 328.2 K are

$$P_1^{\text{vap}} = 48.9 \text{ kPa} \quad \text{and} \quad P_2^{\text{vap}} = 9.1 \text{ kPa}.$$

The activity coefficients are

$$\ln \gamma_1 = - \left[\ln (x_1 + 0.807x_2) + x_2 \left(\frac{0.516}{x_2 + 0.516x_1} - \frac{0.807}{x_1 + 0.807x_2} \right) \right], \quad (1)$$

$$\ln \gamma_2 = - \left[\ln (x_2 + 0.516x_1) + x_1 \left(\frac{0.807}{x_1 + 0.807x_2} - \frac{0.516}{x_2 + 0.516x_1} \right) \right]. \quad (2)$$

Substitution of Eqns (1) and (2) into Eqns (9.5-7) and (9.5-8), and letting $P_1^{\text{vap}} = 48.9 \text{ kPa}$, $P_2^{\text{vap}} = 9.1 \text{ kPa}$, and $y_1 = 0.87$ give

$$\begin{aligned} f_1(x_1, P) = & \ln P - 4.029 + \ln \left(\frac{0.807 + 0.193x_1}{x_1} \right) + (1 - x_1) \\ & \times \left(\frac{0.516}{1 - 0.484x_1} - \frac{0.807}{0.807 + 0.193x_1} \right) = 0, \end{aligned} \quad (3)$$

$$f_2(x_1, P) = \ln P - 4.2485 + \ln \left(\frac{1 - 0.484x_1}{1 - x_1} \right) + x_1 \times \left(\frac{0.807}{0.807 + 0.193x_1} - \frac{0.516}{1 - 0.484x_1} \right) = 0. \quad (4)$$

Equations (3) and (4) can be solved by Newton's method as explained in Section 9.4.1 in which **J** is defined by Eqn (9.4-7). The use of Eqn (9.5-5) with $\gamma_1 = \gamma_2 = 1$ provides an initial estimate for pressure as

$$P^{(0)} = \frac{1}{\frac{0.87}{(1)(48.9)} + \frac{0.13}{(1)(9.1)}} = 31.2 \text{ kPa.}$$

On the other hand, the use of Raoult's law provides an initial estimate for x_1 as

$$x_1^{(0)} = \frac{P^{(0)}y_1}{P_1^{\text{vap}}} = \frac{(31.2)(0.87)}{(48.9)} = 0.555.$$

Choosing $\epsilon = 1 \times 10^{-3}$, the results of the iterations are summarized in the following table:

k	x_1	P (kPa)	Δ_{max}
0	0.555	31.2	—
1	0.57255	37.09788	5.89788
2	0.57250	37.64553	0.54765
3	0.57250	37.64682	1.3×10^{-3}
4	0.57250	37.64681	6.5×10^{-6}

Hence, $x_1 = 0.573$ and $P = 37.647$ kPa.

9.5.3 Bubble Point Temperature Calculation

At a given pressure, it is required to calculate the temperature at which a liquid of known composition first begins to boil. Since temperature is an unknown quantity, vapor pressures of pure liquids and activity coefficients cannot be determined *a priori*. For a binary mixture, Eqns (9.4-13) and (9.4-14) take the form

$$f_1(y_1, T) = \ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.5-9)$$

$$f_2(y_1, T) = \ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0. \quad (9.5-10)$$

Simultaneous solution of Eqns (9.5-9) and (9.5-10) using Newton's method yields temperature and vapor phase composition.

Example 9.18 Estimate the bubble point temperature of a liquid mixture of 25.9 mol% methanol (1) and 74.1% 1,3-dioxolane (2) at 101.3 kPa. Also calculate the composition of the first bubble of vapor. The system is represented by the NRTL model with the following parameters (Kurihara *et al.*, 2003):

$$g_{12} - g_{22} = 2255.746 \text{ J/mol} \quad g_{21} - g_{11} = 935.446 \text{ J/mol} \quad \alpha = 0.3. \quad (1)$$

The vapor pressures are given as

$$\ln P_1^{\text{vap}} = 15.3939 - \frac{2971.37}{T - 61.940} \quad \ln P_2^{\text{vap}} = 14.1577 - \frac{2754.35}{T - 59.834}, \quad (2)$$

where P^{vap} is in kPa and T is in K.

Solution

From Eqn (8.4-28)

$$\tau_{12} = \frac{2255.746}{8.314T} = \frac{271.3190}{T} \quad \text{and} \quad \tau_{21} = \frac{935.446}{8.314T} = \frac{112.5146}{T}. \quad (3)$$

From Eqn (8.4-27)

$$G_{12} = \exp \left[-\frac{(0.3)(271.3190)}{T} \right] = \exp \left(-\frac{81.3957}{T} \right), \quad (4)$$

$$G_{21} = \exp \left[-\frac{(0.3)(112.5146)}{T} \right] = \exp \left(-\frac{33.7544}{T} \right). \quad (5)$$

Substitution of Eqns (3)–(5) into Eqns (8.4-29) and (8.4-30) leads to the following expressions for the activity coefficients:

$$\ln \gamma_1 = (1 - x_1)^2 \left\{ \frac{112.5146}{T} \left[\frac{\exp(-33.7544/T)}{x_1 + (1 - x_1) \exp(-33.7544/T)} \right]^2 + \frac{271.3190}{T} \frac{\exp(-81.3957/T)}{[1 - x_1 + x_1 \exp(-81.3957/T)]^2} \right\}, \quad (6)$$

$$\ln \gamma_2 = x_1^2 \left\{ \frac{271.3190}{T} \left[\frac{\exp(-81.3957/T)}{1 - x_1 + x_1 \exp(-81.3957/T)} \right]^2 + \frac{112.5146}{T} \frac{\exp(-33.7544/T)}{[x_1 + (1 - x_1) \exp(-33.7544/T)]^2} \right\}. \quad (7)$$

Substitution of Eqns (2), (6), and (7) into Eqns (9.5-9) and (9.5-10), and letting $P = 101.3$ kPa and $x_1 = 0.259$ give

$$f_1(y_1, T) = \ln y_1 - 9.4249 + \frac{2971.37}{T - 61.940} - \frac{61.7796}{T} \times \left[\frac{\exp(-33.7544/T)}{0.259 + 0.741 \exp(-33.7544/T)} \right]^2 - \frac{148.9761}{T} \frac{\exp(-81.3957/T)}{[0.741 + 0.259 \exp(-81.3957/T)]^2} = 0, \quad (8)$$

$$f_2(y_1, T) = \ln(1 - y_1) - 9.2399 + \frac{2754.35}{T - 59.834} - \frac{18.2003}{T} \times \left[\frac{\exp(-81.3957/T)}{0.741 + 0.259 \exp(-81.3957/T)} \right]^2 - \frac{7.5476}{T} \frac{\exp(-33.7544/T)}{[0.259 + 0.741 \exp(-33.7544/T)]^2} = 0. \quad (9)$$

Equations (8) and (9) can be solved by Newton's method as explained in Section 9.4.1 in which **J** is given by Eqn (9.4-7). At 101.3 kPa, the boiling points of pure components are

$$T_1^{\text{sat}} = 337.7\text{K} \quad \text{and} \quad T_2^{\text{sat}} = 348.6 \text{ K}.$$

The mole fraction weighted average of the pure component boiling points can be taken as an initial temperature estimate, i.e.

$$T^{(0)} = (0.259)(337.7) + (0.741)(348.6) = 345.8 \text{ K.}$$

At this temperature, $P_1^{\text{vap}} = 137.85 \text{ kPa}$ and the activity coefficient of methanol, calculated from Eqn (6), is $\gamma_1 = 1.726$. The use of Eqn (9.4-10) provides an initial estimate for y_1 as

$$y_1^{(0)} = \frac{P_1^{\text{vap}} x_1 \gamma_1}{P} = \frac{(137.85)(0.259)(1.726)}{101.3} = 0.608.$$

Choosing $\epsilon = 1 \times 10^{-3}$, the results of the iterations are summarized in the following table:

k	y_1	$T \text{ (K)}$	Δ_{max}
0	0.608	345.8	—
1	0.46246	338.85863	6.94137
2	0.45603	337.77167	1.08696
3	0.45604	337.78524	0.01357
4	0.45604	337.78507	1.6×10^{-4}

Hence, $y_1 = 0.456$ and $T = 337.8 \text{ K}$.

9.5.4 Dew Point Temperature Calculation

At a given pressure, it is required to calculate the temperature at which a vapor of known composition first begins to condense. This problem is even more complicated than the bubble point temperature since the liquid phase composition and the system temperature are unknown quantities. For a binary mixture, Eqns (9.4-13) and (9.4-14) take the form

$$f_1(x_1, T) = \ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.5-11)$$

$$f_2(x_1, T) = \ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0. \quad (9.5-12)$$

Simultaneous solution of Eqns (9.5-11) and (9.5-12) using Newton's method yields temperature and liquid phase composition.

Example 9.19 Estimate the dew point temperature of a vapor mixture of 65 mol% acetone (1) and 35% benzene (2) at a pressure of 1.013 bar. Also

calculate the composition of the first drop of liquid. The system is represented by the Wilson equation with the following energy parameters:

$$\lambda_{12} - \lambda_{11} = 1991.5 \text{ J/mol} \quad \lambda_{21} - \lambda_{22} = -569.94 \text{ J/mol.}$$

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Acetone	508	48.0	0.304
Benzene	562	48.9	0.212

From Appendix C

$$\ln P_1^{\text{vap}} = 10.0311 - \frac{2940.46}{T - 35.93} \quad \ln P_2^{\text{vap}} = 9.2806 - \frac{2788.51}{T - 52.36}. \quad (1)$$

At 1.013 bar, the boiling points of pure components are

$$T_1^{\text{sat}} = 329.4 \text{ K} \quad \text{and} \quad T_2^{\text{sat}} = 353.2 \text{ K.}$$

Choosing the mole fraction weighted average of the pure component boiling points as the initial temperature estimate gives

$$T^{(0)} = (0.65)(329.4) + (0.35)(353.2) = 337.7 \text{ K.}$$

The use of the modified Rackett equation, Eqn (5.4-9), gives the molar volumes as

$$\tilde{V}_1^L = 87.6 \text{ cm}^3/\text{mol} \quad \tilde{V}_2^L = 95.4 \text{ cm}^3/\text{mol.}$$

The Wilson parameters are calculated from Eqn (8.4-25) as

$$\begin{aligned} \Lambda_{12} &= \frac{\tilde{V}_2^L}{\tilde{V}_1^L} \exp \left(-\frac{\lambda_{12} - \lambda_{11}}{RT} \right) = \frac{95.4}{87.6} \exp \left(-\frac{1991.5}{8.314T} \right) \\ &= 1.089 \exp \left(-\frac{239.5357}{T} \right), \end{aligned} \quad (2)$$

$$\begin{aligned} \Lambda_{21} &= \frac{\tilde{V}_1^L}{\tilde{V}_2^L} \exp \left(-\frac{\lambda_{21} - \lambda_{22}}{RT} \right) = \frac{87.6}{95.4} \exp \left(\frac{569.94}{8.314T} \right) \\ &= 0.9182 \exp \left(\frac{68.5518}{T} \right). \end{aligned} \quad (3)$$

Substitution of Eqns (2) and (3) into Eqns (8.4-19) and (8.4-20) leads to the following expressions for the activity coefficients:

$$\begin{aligned} \ln \gamma_1 = & -\ln [x_1 + 1.089(1 - x_1) \exp(-239.5357/T)] - (1 - x_1) \\ & \times \left[\frac{0.9182 \exp(68.5518/T)}{1 - x_1 + 0.9182x_1 \exp(68.5518/T)} \right. \\ & \left. - \frac{1.089 \exp(-239.5357/T)}{x_1 + 1.089(1 - x_1) \exp(-239.5357/T)} \right], \end{aligned} \quad (4)$$

$$\begin{aligned} \ln \gamma_2 = & -\ln [1 - x_1 + 0.9182x_1 \exp(68.5518/T)] - x_1 \\ & \times \left[\frac{1.089 \exp(-239.5357/T)}{x_1 + 1.089(1 - x_1) \exp(-239.5357/T)} \right. \\ & \left. - \frac{0.9182 \exp(68.5518/T)}{1 - x_1 + 0.9182x_1 \exp(68.5518/T)} \right]. \end{aligned} \quad (5)$$

Substitution of Eqns (1), (4), and (5) into Eqns (9.5-11) and (9.5-12), and taking $P = 1.013$ bar and $y_1 = 0.65$ give

$$\begin{aligned} f_1(x_1, T) = & -10.4490 + \frac{2940.46}{T - 35.93} \\ & + \ln \left[\frac{x_1 + 1.089(1 - x_1) \exp(-239.5357/T)}{x_1} \right] \\ & + (1 - x_1) \left[\frac{0.9182 \exp(68.5518/T)}{1 - x_1 + 0.9182x_1 \exp(68.5518/T)} \right. \\ & \left. - \frac{1.089 \exp(-239.5357/T)}{x_1 + 1.089(1 - x_1) \exp(-239.5357/T)} \right] = 0, \end{aligned} \quad (6)$$

$$\begin{aligned} f_2(x_1, T) = & -10.3175 + \frac{2788.51}{T - 52.36} \\ & + \ln \left[\frac{1 - x_1 + 0.9182x_1 \exp(68.5518/T)}{1 - x_1} \right] \\ & + x_1 \left[\frac{1.089 \exp(-239.5357/T)}{x_1 + 1.089(1 - x_1) \exp(-239.5357/T)} \right. \\ & \left. - \frac{0.9182 \exp(68.5518/T)}{1 - x_1 + 0.9182x_1 \exp(68.5518/T)} \right] = 0. \end{aligned} \quad (7)$$

Equations (6) and (7) can be solved by Newton's method as explained in Section 9.4.1 in which $\mathbf{J}$ is given by Eqn (9.4-7). The use of Raoult's law provides an

initial estimate for x_1 as

$$x_1^{(0)} = \frac{P y_1}{P_1^{\text{vap}}}.$$

At $T^{(0)} = 337.7 \text{ K}$, $P_1^{\text{vap}} = 1.332 \text{ bar}$. Thus

$$x_1^{(0)} = \frac{(1.013)(0.65)}{1.332} = 0.494.$$

Choosing $\epsilon = 1 \times 10^{-3}$, the results of the iterations are summarized in the following table:

k	x_1	$T \text{ (K)}$	Δ_{max}
0	0.494	337.7	—
1	0.46096	336.68621	1.01379
2	0.46090	336.76265	0.07644
3	0.46090	336.76180	8.5×10^{-4}

Therefore, the dew point temperature is 336.8 K and the liquid phase composition is $x_1 = 0.461$.

9.5.5 Flash Calculation

For flash calculations, phase equilibrium relations, i.e. Eqn (9.4-10), subject to the constraints defined by Eqn (9.4-11), are also valid. Therefore, for a binary system, Eqns (9.4-13) and (9.4-14) must be solved simultaneously. Furthermore, amounts of vapor and liquid phases can be determined from one of the equations representing material balance, i.e. either Eqn (9.3-7) or Eqn (9.3-8). For example, V/F can be determined from

$$\frac{V}{F} = \frac{z_1 - x_1}{y_1 - x_1}. \quad (9.5-13)$$

As stated in Section 9.3.5, flash calculations involve several possibilities among the variables V/F (or L/F), T , P , z_i , x_i , and y_i .

Case (i): Given z_i , P , and T ; determine x_i , y_i , V/F .

Since liquid phase mole fractions are unknown quantities, activity coefficients cannot be determined *a priori*. For a binary mixture, Eqns (9.4-13) and (9.4-14) take the form

$$f_1(x_1, y_1) = \ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.5-14)$$

$$f_2(x_1, y_1) = \ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0. \quad (9.5-15)$$

Simultaneous solution of Eqns (9.5-14) and (9.5-15) by Newton's method yields compositions of the vapor and liquid streams leaving the flash chamber. Then V/F is determined from Eqn (9.5-13).

Example 9.20 A liquid mixture containing 75 mol% water (1) and 25% *n*-propanol (2) is flowing at 100 kmol/h. It is flashed across a valve into a flash chamber held at 361 K and 100 kPa. Determine the compositions and flow rates of the vapor and liquid streams leaving the flash chamber.

The binary mixture of water and *n*-propanol is represented by the NRTL equation with the following parameters:

$$g_{12} - g_{22} = 9218.80 \text{ J/mol} \quad g_{21} - g_{11} = -414.77 \text{ J/mol} \quad \alpha = 0.3.$$

Solution

From Appendix C

$$\ln P_1^{\text{vap}} = 11.6834 - \frac{3816.44}{T - 46.13} \quad \ln P_2^{\text{vap}} = 10.9237 - \frac{3166.28}{T - 80.15}.$$

At 361 K, the vapor pressures are

$$P_1^{\text{vap}} = 0.646 \text{ bar} \quad P_2^{\text{vap}} = 0.705 \text{ bar}.$$

From Eqns (8.4-27) and (8.4-28)

$$\tau_{12} = 3.072 \quad \tau_{21} = -0.138 \quad G_{12} = 0.398 \quad G_{21} = 1.042.$$

Hence, activity coefficients are expressed in the form

$$\ln \gamma_1 = x_2^2 \left[-0.138 \left(\frac{1.042}{x_1 + 1.042x_2} \right)^2 + \frac{1.223}{(x_2 + 0.398x_1)^2} \right], \quad (1)$$

$$\ln \gamma_2 = x_1^2 \left[3.072 \left(\frac{0.398}{x_2 + 0.398x_1} \right)^2 - \frac{0.144}{(x_1 + 1.042x_2)^2} \right]. \quad (2)$$

Substitution of Eqns (1) and (2) into Eqns (9.5-14) and (9.5-15), and letting $P = 1 \text{ bar}$, $P_1^{\text{vap}} = 0.646 \text{ bar}$, and $P_2^{\text{vap}} = 0.705 \text{ bar}$ give

$$f_1(x_1, y_1) = \ln y_1 + 0.437 - \ln x_1 - (1 - x_1)^2 \times \left[-0.138 \left(\frac{1.042}{1.042 - 0.042x_1} \right)^2 + \frac{1.223}{(1 - 0.602x_1)^2} \right] = 0, \quad (3)$$

$$f_2(x_1, y_1) = \ln(1 - y_1) + 0.35 - \ln(1 - x_1) - x_1^2 \times \left[3.072 \left(\frac{0.398}{1 - 0.602x_1} \right)^2 - \frac{0.144}{(1.042 - 0.042x_1)^2} \right] = 0. \quad (4)$$

Equations (3) and (4) can be solved by Newton's method as explained in Section 9.4.1 in which $\mathbf{J}$ is given by Eqn (9.4-7). Since $P_2^{\text{vap}} > P_1^{\text{vap}}$, *n*-propanol is more volatile and tends to be in the vapor phase. As a result, we expect the mole fraction of water in the liquid phase to be greater than its original value, i.e. $x_1 > 0.75$. Thus, let us choose $x_1^{(0)} = 0.9$. At this composition $\gamma_1 = 1.058$ and the initial estimate of y_1 is

$$y_1^{(0)} = \frac{P_1^{\text{vap}} x_1^{(0)} \gamma_1}{P} = \frac{(0.646)(0.9)(1.058)}{1} = 0.62.$$

Choosing $\epsilon = 1 \times 10^{-3}$, the results of the iterations are summarized in the following table:

k	x_1	y_1	Δ_{max}
0	0.9	0.62	—
1	0.95433	0.62038	0.05433
2	0.94244	0.62111	0.01190
3	0.93511	0.61987	7.3×10^{-3}
4	0.93225	0.61952	2.9×10^{-3}
5	0.93154	0.61945	7.1×10^{-4}

Hence, $x_1 = 0.932$ and $y_1 = 0.619$. The use of Eqn (9.5-13) gives

$$\frac{V}{F} = \frac{0.75 - 0.932}{0.619 - 0.932} = 0.58 \Rightarrow V = (0.58)(100) = 58 \text{ kmol/h.}$$

The flow rate of the liquid stream leaving the flash chamber is

$$L = 100 - 58 = 42 \text{ kmol/h.}$$

Case (ii): Given z_i , P , and y_i ; determine T , x_i , V/F .

Since temperature and liquid phase mole fractions are unknown quantities, vapor pressures of pure fluids and activity coefficients cannot be determined *a priori*. For a binary mixture, Eqns (9.4-13) and (9.4-14) take the form

$$f_1(x_1, T) = \ln P + \ln y_1 - \ln P_1^{\text{vap}} - \ln x_1 - \ln \gamma_1 = 0, \quad (9.5-16)$$

$$f_2(x_1, T) = \ln P + \ln(1 - y_1) - \ln P_2^{\text{vap}} - \ln(1 - x_1) - \ln \gamma_2 = 0, \quad (9.5-17)$$

which are identical with Eqns (9.5-11) and (9.5-12), i.e. dew point temperature calculation. Simultaneous solution of Eqns (9.5-16) and (9.5-17) yields temperature and composition of the liquid stream leaving the flash chamber. Then V/F is determined from (9.5-13).

In the numerical solution, the initial estimate of x_1 cannot be guessed arbitrarily. Rearrangement of Eqn (9.3-18) gives

$$K_i = \frac{y_i [1 - (V/F)]}{z_i - (V/F)y_i}. \quad (9.5-18)$$

To avoid negative values in the denominator of Eqn (9.5-18), $V/F < (z_i/y_i)_{\min}$. Once V/F is specified, then $x_1^{(0)}$ is calculated from Eqn (9.5-13), i.e.

$$x_1^{(0)} = \frac{z_1 - (V/F)y_1}{1 - (V/F)}. \quad (9.5-19)$$

Example 9.21 A liquid mixture containing 30 mol% acetone (1) and 70% water (2) is flowing at 100 kmol/h. It is flashed across a valve into a flash drum held at 1.013 bar. If the composition of the vapor product is to be $y_1 = 0.66$, calculate the temperature of the flash chamber, the liquid phase composition, and V/F .

The activity coefficients are given by

$$\ln \gamma_1 = \frac{2.05}{\left(1 + 1.367 \frac{x_1}{x_2}\right)^2} \quad \ln \gamma_2 = \frac{1.50}{\left(1 + 0.732 \frac{x_2}{x_1}\right)^2}. \quad (1)$$

Solution

From Appendix C

$$\ln P_1^{\text{vap}} = 10.0311 - \frac{2940.46}{T - 35.93} \quad \ln P_2^{\text{vap}} = 11.6834 - \frac{3816.44}{T - 46.13}. \quad (2)$$

Substitution of Eqns (1) and (2) into Eqns (9.5-16) and (9.5-17), and letting $P = 1.013$ bar and $y_1 = 0.66$ give

$$f_1(x_1, T) = -10.4337 + \frac{2940.46}{T - 35.93} - \ln x_1 - \frac{2.05}{\left(1 + 1.367 \frac{x_1}{1 - x_1}\right)^2} = 0, \quad (3)$$

$$f_2(x_1, T) = -12.7493 + \frac{3816.44}{T - 46.13} - \ln(1 - x_1) - \frac{1.50}{\left(1 + 0.732 \frac{1 - x_1}{x_1}\right)^2} = 0. \quad (4)$$

Equations (3) and (4) can be solved by Newton's method as explained in Section 9.4.1 in which $\mathbf{J}$ is given by Eqn (9.4-7). The saturation temperatures of acetone and water at 1.013 bar are 329.4 K and 373.1 K, respectively. The mole fraction weighted average of the pure component boiling points can be taken as an initial temperature estimation, i.e.

$$T^{(0)} = (0.3)(329.4) + (0.7)(373.1) \simeq 360 \text{ K.}$$

Note that

$$\frac{z_1}{y_1} = \frac{0.30}{0.66} = 0.45 \quad \text{and} \quad \frac{z_2}{y_2} = \frac{0.70}{0.34} = 2.06.$$

Therefore, the value of V/F must be less than 0.45. Taking $V/F = 0.4$, Eqn (9.5-19) gives

$$x_1^{(0)} = \frac{0.3 - (0.4)(0.66)}{1 - 0.4} = 0.06.$$

Setting $\epsilon = 1 \times 10^{-3}$, the results of the iterations are summarized in the following table:

k	x_1	T	$\Delta_{\max}$
0	0.06	360	—
1	0.06885	346.13197	13.86803
2	0.06966	346.88032	0.74835
3	0.06966	346.87356	6.8×10^{-3}
4	0.06966	346.87364	7.8×10^{-5}

Hence, $x_1 = 0.07$ and $T = 346.9 \text{ K}$. The use of Eqn (9.5-13) gives

$$\frac{V}{F} = \frac{0.3 - 0.07}{0.66 - 0.07} = 0.39.$$

9.6 Positive and Negative Deviations from Raoult's Law

An ideal mixture obeys Raoult's law and the total pressure of a binary system is given by

$$P = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}}. \quad (9.6-1)$$

When the components in a liquid mixture are chemically similar, an "ideal mixture" assumption is reasonable. For example, mixtures of alcohols or mixtures of hydrocarbons can be considered ideal. However, when the components of the mixture are

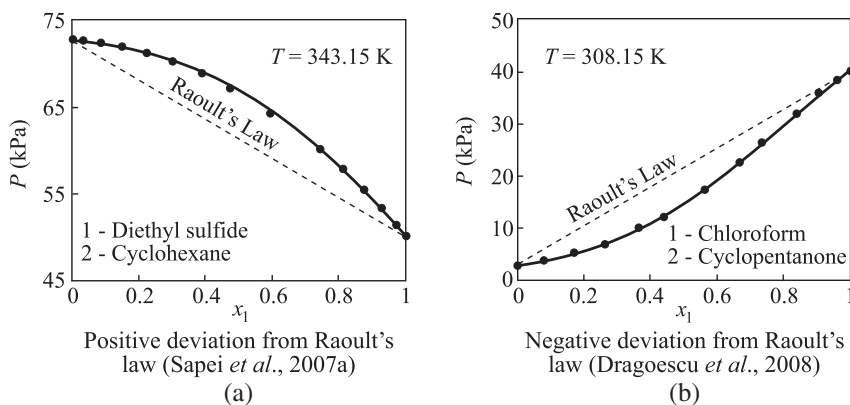


Figure 9.13: Deviations from Raoult's law.

polar or if the components have different functional groups, interactions between the components lead to nonidealities and Eqn (9.6-1) should be replaced by

$$P = \gamma_1 x_1 P_1^{\text{vap}} + \gamma_2 x_2 P_2^{\text{vap}}. \quad (9.6-2)$$

For example, a mixture of a hydrocarbon with an alcohol always forms a nonideal mixture. Water, being a highly polar substance, usually forms nonideal mixtures. However, mixtures of water with methanol and ethylene glycol may be considered almost ideal.

The total pressure of a nonideal mixture calculated from Eqn (9.6-2) may be greater or less than that predicted by Raoult's law, Eqn (9.6-1), as shown in Fig. 9.13. *Positive deviation from Raoult's law* implies that $P > \sum x_i P_i^{\text{vap}}$. This is possible only if $\gamma_i > 1$ for at least one of the components present in the mixture. On the other hand, *negative deviation from Raoult's law* implies that $P < \sum x_i P_i^{\text{vap}}$. In this case, $\gamma_i < 1$ for at least one of the components present in the mixture.

Positive deviation from Raoult's law indicates that the liquid is evaporating more easily than would be expected. This implies that interactions between unlike molecules are weaker than those between like molecules, i.e. components "dislike" each other. In this case, $\Delta V_{\text{mix}} > 0$ and $\Delta H_{\text{mix}} > 0$, i.e. heat is absorbed during mixing. Two liquids with different structures, i.e. one polar and the other nonpolar, usually exhibit this type of behavior, such as isopropyl alcohol ($\text{CH}_3\text{CHOHCH}_3$) and isopropyl ether $[(\text{CH}_3)_2\text{CHOCH}(\text{CH}_3)_2]$, water (H_2O) and benzene (C_6H_6), acetone (CH_3COCH_3) and carbon disulfide (CS_2), ethanol ($\text{C}_2\text{H}_5\text{OH}$) and benzene (C_6H_6), and phenol ($\text{C}_6\text{H}_5\text{OH}$) and water (H_2O).

Negative deviation from Raoult's law indicates that evaporation of the liquid is more difficult than would be expected. This implies that interactions between unlike molecules are stronger than those between like molecules, i.e. components "like"

each other. In this case, $\Delta V_{\text{mix}} < 0$ and $\Delta H_{\text{mix}} < 0$, i.e. heat is evolved during mixing. Liquid mixtures with similar polar structures usually exhibit this type of behavior, such as acetone (CH_3COCH_3) and chloroform (CHCl_3), hydrochloric acid (HCl) and water (H_2O), acetic acid (CH_3COOH) and dioxane ($\text{C}_4\text{H}_8\text{O}_2$), and benzaldehyde ($\text{C}_6\text{H}_5\text{CHO}$) and phenol ($\text{C}_6\text{H}_5\text{OH}$).

For certain binary mixtures when the total pressure is plotted versus the liquid composition, the resulting curve may exhibit an extremum, i.e. a minimum or a maximum. The location of this extremum can be determined from

$$\left(\frac{\partial P}{\partial x_1} \right)_T = 0. \quad (9.6-3)$$

Differentiation of Eqn (9.6-2) with respect to x_1 , keeping temperature constant, gives

$$\gamma_1 P_1^{\text{vap}} \left[1 + x_1 \left(\frac{d \ln \gamma_1}{dx_1} \right) \right] - \gamma_2 P_2^{\text{vap}} \left[1 - x_2 \left(\frac{d \ln \gamma_2}{dx_1} \right) \right] = 0. \quad (9.6-4)$$

The Gibbs-Duhem equation, Eqn (8.8-4), is

$$x_1 \left(\frac{d \ln \gamma_1}{dx_1} \right) + x_2 \left(\frac{d \ln \gamma_2}{dx_1} \right) = 0. \quad (9.6-5)$$

The use of Eqn (9.6-5) in Eqn (9.6-4) leads to

$$\gamma_1 P_1^{\text{vap}} = \gamma_2 P_2^{\text{vap}}. \quad (9.6-6)$$

The condition of equilibrium is expressed as

$$y_1 P = \gamma_1 x_1 P_1^{\text{vap}}, \quad (9.6-7)$$

$$y_2 P = \gamma_2 x_2 P_2^{\text{vap}} \quad (9.6-8)$$

from which the ratio y_1/y_2 is given by

$$\frac{y_1}{y_2} = \frac{x_1}{x_2} \frac{\gamma_1 P_1^{\text{vap}}}{\gamma_2 P_2^{\text{vap}}}. \quad (9.6-9)$$

Substitution of Eqn (9.6-6) into Eqn (9.6-9) yields

$$\frac{y_1}{y_2} = \frac{x_1}{x_2} \quad (9.6-10)$$

or

$$\frac{y_1}{1 - y_1} = \frac{x_1}{1 - x_1} \Rightarrow x_1 = y_1 \quad \text{and} \quad x_2 = y_2. \quad (9.6-11)$$

Therefore, the vapor and liquid phases have the same composition at the point where the equilibrium pressure versus mole fraction curve has either a minimum

or a maximum. Such a vapor-liquid mixture is called an *azeotrope* or an *azeotropic mixture*, and is of special interest in the distillation process. The word azeotrope comes from the Greek “zein tropos,” meaning “constant boiling.” A consequence of this constant boiling point is that one cannot entirely separate the components by distillation. Note that the bubble and dew points coincide at the azeotropic point.

Azeotropes may be classified as *minimum boiling* and *maximum boiling* as shown in Fig. 9.14. The minimum boiling azeotrope is observed if a mixture exhibits positive deviation from Raoult’s law. On the other hand, the maximum boiling azeotrope is for mixtures exhibiting negative deviation from Raoult’s law. In systems normally of interest to engineers, the minimum boiling type is encountered more often than the maximum boiling type.

Since $x_i = y_i$ at the azeotropic point, the equilibrium relation given by Eqn (9.4-10) reduces to

$$\gamma_i = \frac{P}{P_i^{\text{vap}}} \quad \text{at the azeotropic point.} \quad (9.6-12)$$

Note that, in most activity coefficient models, estimation of the parameters requires a single set of activity coefficients. Given the azeotropic composition, temperature, and pressure, one can calculate activity coefficients from Eqn (9.6-12). This information can then be used to evaluate the parameters in the activity coefficient model.

Example 9.22 A binary mixture of ethanol (1) and benzene (2) forms an azeotrope at 341.39 K and 1.013 bar. The composition at the azeotropic point is 44.8 mol% ethanol. Determine the activity coefficients as a function of composition for the ethanol-benzene mixture using the van Laar model.

Solution

From Appendix C

$$\ln P_1^{\text{vap}} = 12.2917 - \frac{3803.98}{T - 41.68} \quad \text{and} \quad \ln P_2^{\text{vap}} = 9.2806 - \frac{2788.51}{T - 52.36}.$$

At a temperature of 341.39 K, the vapor pressures are

$$P_1^{\text{vap}} = 0.670 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 0.693 \text{ bar}.$$

At the azeotropic composition, the activity coefficients can be determined from Eqn (9.6-12) as

$$\begin{aligned} \gamma_1 &= \frac{P}{P_1^{\text{vap}}} = \frac{1.013}{0.670} = 1.512 \quad \Rightarrow \quad \ln \gamma_1 = 0.4134, \\ \gamma_2 &= \frac{P}{P_2^{\text{vap}}} = \frac{1.013}{0.693} = 1.462 \quad \Rightarrow \quad \ln \gamma_2 = 0.3798. \end{aligned}$$

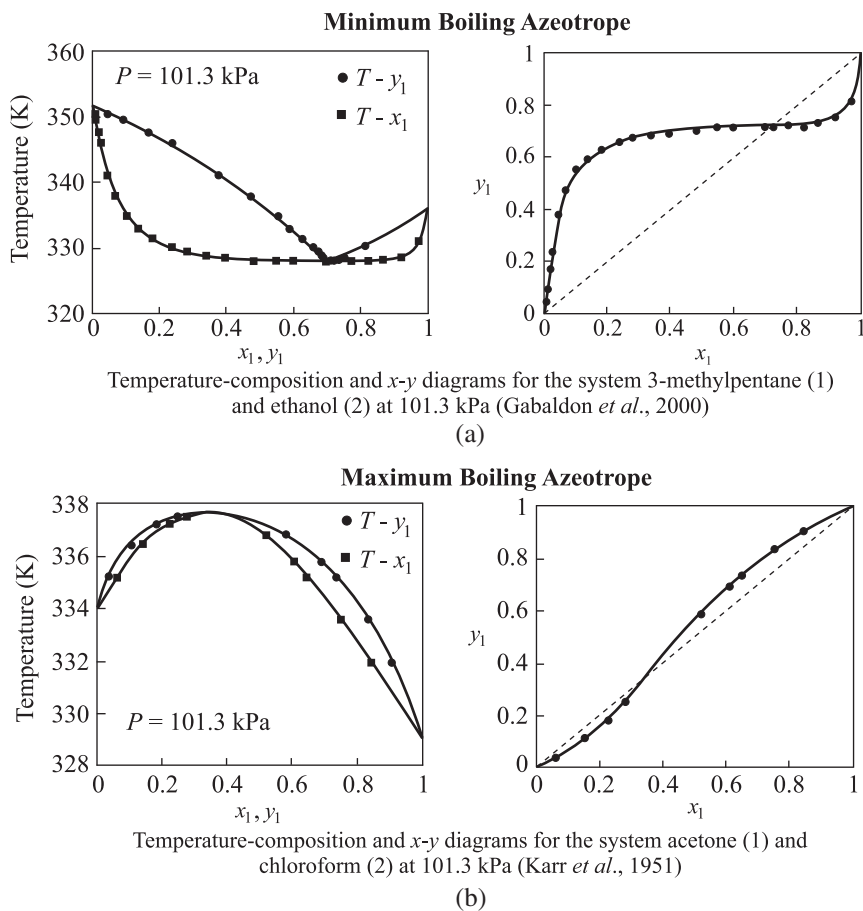


Figure 9.14: Minimum and maximum boiling azeotropes.

The parameters A and B in the van Laar model can be calculated from a single set of activity coefficient data using Eqn (8.4-16) as

$$A = \ln \gamma_1 \left(1 + \frac{x_2 \ln \gamma_2}{x_1 \ln \gamma_1} \right)^2 = (0.4134) \left[1 + \frac{(0.552)(0.3798)}{(0.448)(0.4134)} \right]^2 = 1.879,$$

$$B = \ln \gamma_2 \left(1 + \frac{x_1 \ln \gamma_1}{x_2 \ln \gamma_2} \right)^2 = (0.3798) \left[1 + \frac{(0.448)(0.4134)}{(0.552)(0.3798)} \right]^2 = 1.347.$$

Therefore, the activity coefficients are represented by

$$\ln \gamma_1 = \frac{1.879}{\left(1 + 1.395 \frac{x_1}{x_2}\right)^2} \quad \text{and} \quad \ln \gamma_2 = \frac{1.347}{\left(1 + 0.717 \frac{x_2}{x_1}\right)^2}.$$

The variation of activity coefficients as a function of composition is given in the table below:

x_1	γ_1	γ_2	x_1	γ_1	γ_2
0.1	4.090	1.025	0.5	1.388	1.579
0.2	2.809	1.094	0.6	1.217	1.853
0.3	2.087	1.207	0.7	1.109	2.199
0.4	1.656	1.367	0.8	1.044	2.634
0.448	1.512	1.462	0.9	1.010	3.176

9.7 Relative Volatility

The K -value was defined by Eqn (9.3-14) as

$$K_i = \frac{y_i}{x_i}. \quad (9.7-1)$$

When $K_i > 1$, then the component i tends to exist in the vapor phase. The use of Eqn (9.4-10) in Eqn (9.7-1) leads to

$$K_i(T, P, x_i) = \frac{\gamma_i(T, P, x_i) P_i^{\text{vap}}(T)}{P} \quad (9.7-2)$$

indicating that, besides temperature and pressure, K -values are also dependent on composition.

In distillation, the *relative volatility* of lower boiling (or more volatile) component i with respect to higher boiling (or less volatile) component j , α_{ij} , is defined by

$$\alpha_{ij} = \frac{K_i}{K_j} = \frac{y_i/x_i}{y_j/x_j}. \quad (9.7-3)$$

Substitution of Eqn (9.7-2) into Eqn (9.7-3) gives

$$\alpha_{ij}(T, P, x_i) = \frac{\gamma_i P_i^{\text{vap}}}{\gamma_j P_j^{\text{vap}}}. \quad (9.7-4)$$

Table 9.1: Relative volatilities of propane (1)–*n*-pentane (2) mixture.

<i>T</i> (K)	Vapor Pressure (bar)		α_{12}
	Propane	<i>n</i> -Pentane	
300	9.904	0.732	13.53
330	19.364	1.961	9.87
360	33.574	4.366	7.69
390	53.174	8.474	6.27

For ideal mixtures, i.e. when Raoult's law applies, Eqn (9.7-4) reduces to

$$\alpha_{ij} = \frac{P_i^{\text{vap}}}{P_j^{\text{vap}}} \quad \text{Ideal mixture.} \quad (9.7-5)$$

For a mixture of propane (1) and *n*-pentane (2), relative volatilities are calculated by using Eqn (9.7-5) at various temperatures. The results given in Table 9.1 indicate that the relative volatility decreases with increasing temperature.

On the other hand, the relative volatility decreases with increasing pressure for most systems, as will be shown in the following example. One might think of decreasing pressure to increase the relative volatility, leading to easier separation of components. However, lower pressures also increase the volumetric flow rate of vapor and, as a result, liquid droplets may be carried to the upper plate by the vapor stream, resulting in the decrease in plate efficiency. This phenomenon is known as *entrainment*.

Example 9.23 Consider a liquid mixture of 40 mol% benzene (1) and 60% toluene (2). Estimate the bubble point temperature and relative volatility as a function of pressure. The vapor pressures are given as a function of temperature in the form

$$\ln P_1^{\text{vap}}(\text{bar}) = -8.4336 \ln T - \frac{6281.04}{T} + 66.502 + 6.198 \times 10^{-6} T^2$$

$$278.68 \text{ K} < T < 562.16 \text{ K},$$

$$\ln P_2^{\text{vap}}(\text{bar}) = -8.7955 \ln T - \frac{6918.80}{T} + 69.531 + 5.755 \times 10^{-6} T^2$$

$$178.18 \text{ K} < T < 591.79 \text{ K}.$$

Solution

Since benzene and toluene are chemically similar, Raoult's law can be used in the calculations. Let us consider the case when $P = 1$ bar. Assuming $T = 367.8$ K, the vapor pressures are

$$P_1^{\text{vap}} = 1.555 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 0.630 \text{ bar}.$$

The calculated pressure is

$$P_{\text{calc}} = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} = (0.4)(1.555) + (0.6)(0.630) = 1 \text{ bar}.$$

Therefore, the assumed temperature is correct. The relative volatility is calculated from Eqn (9.7-5) as

$$\alpha_{12} = \frac{P_1^{\text{vap}}}{P_2^{\text{vap}}} = \frac{1.555}{0.630} = 2.468.$$

The relative volatilities are calculated in a similar way for other pressures and the results are given below:

P (bar)	T (K)	α_{12}	P (bar)	T (K)	α_{12}
1	367.8	2.468	6	443.2	2.019
2	393.3	2.278	7	451.3	1.986
3	410.2	2.177	8	458.6	1.959
4	423.2	2.109	9	465.3	1.936
5	433.9	2.059	10	471.4	1.916

Comment: The increase in pressure causes the relative volatility to decrease.

At the azeotropic point, mole fractions of each component in each phase are equal to each other, i.e. $x_i = y_i$, and, therefore, $\alpha_{ij} = 1$. The relative volatility α_{ij} indicates how easy or difficult it is to separate component i from component j . When the value of α_{ij} is close to unity, i.e. $0.95 < \alpha_{ij} < 1.05$, this implies that components i and j have similar boiling points and it will be difficult to separate them by distillation. When $\alpha_{ij} > 1$ or $\alpha_{ij} < 1$, then component i can be separated from component j by distillation or flashing. In distillation practice, an increase in relative volatility can be accomplished by lowering temperature and pressure. The increase in relative volatility leads to better separation at a given reflux ratio, or the same separation at a lower reflux ratio. The reduction in reflux ratio has a concomitant effect of reducing reboiler duty, i.e. energy savings.

Since the azeotropic composition limits the separation by distillation, it is necessary to change the azeotropic point. In practice, this is known as “breaking” the azeotrope and can be carried out by various means. One way of separating an azeotropic mixture is to operate two columns in sequence at different pressures, known as *pressure swing distillation*. To understand the reasoning behind this, consider a binary mixture of components 1 and 2 forming a minimum boiling azeotrope as shown in Fig. 9.15a. An increase in pressure shifts the azeotropic point to the left. Suppose the feed stream has a composition x_1^F which is less than $x_{1,az}^{\text{HP}}$. When the feed stream is sent to the low-pressure (LP) column, the bottom product is pure

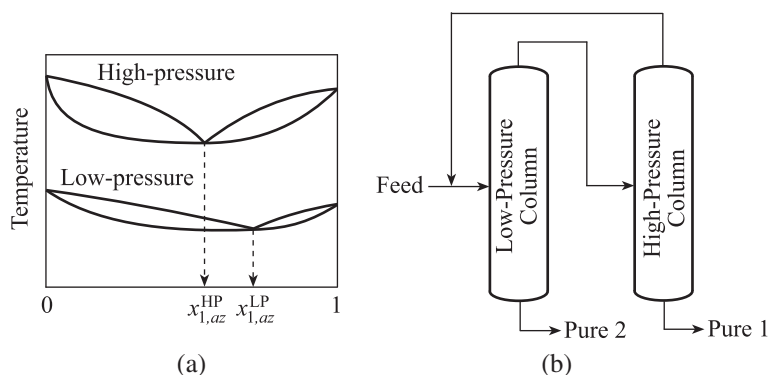


Figure 9.15: Pressure swing distillation.

2 and the overhead product has a composition around $x_{1,az}^{LP}$. This overhead product is then sent to the high-pressure (HP) column. At the bottom of the HP column, pure 1 is obtained. The top product has a composition around $x_{1,az}^{HP}$ and it is recycled back to the LP column. The schematic diagram of the overall process is shown in Fig. 9.15b.

Another way of breaking the azeotrope is to introduce a suitable solvent with a higher affinity for one component than the other. In this way, the ratio of the activity coefficients in Eqn (9.7-4) can be changed drastically, leading to easier separation of components. In the literature, this is known as *extractive distillation*. The use of volatile organic compounds as solvent makes extractive distillation less environmentally friendly than pressure swing distillation.

Example 9.24 For a binary mixture of 1 and 2, show that

$$y_1 = \frac{\alpha_{12}x_1}{1 + x_1(\alpha_{12} - 1)} \quad \text{and} \quad x_1 = \frac{y_1}{\alpha_{12} - y_1(\alpha_{12} - 1)}.$$

Also plot the x - y diagram for $\alpha_{12} = 1.5$ and $\alpha_{12} = 2.5$.

Solution

The equilibrium relationships for the components 1 and 2 are expressed as

$$Py_1 = \gamma_1 P_1^{\text{vap}} x_1, \quad (1)$$

$$Py_2 = \gamma_2 P_2^{\text{vap}} x_2. \quad (2)$$

Division of Eqn (1) by Eqn (2) leads to

$$\frac{y_1}{y_2} = \underbrace{\frac{\gamma_1 P_1^{\text{vap}}}{\gamma_2 P_2^{\text{vap}}}}_{\alpha_{12}} \frac{x_1}{x_2}. \quad (3)$$

Rearrangement of Eqn (3) gives

$$\frac{y_1}{1 - y_1} = \alpha_{12} \left(\frac{x_1}{1 - x_1} \right). \quad (4)$$

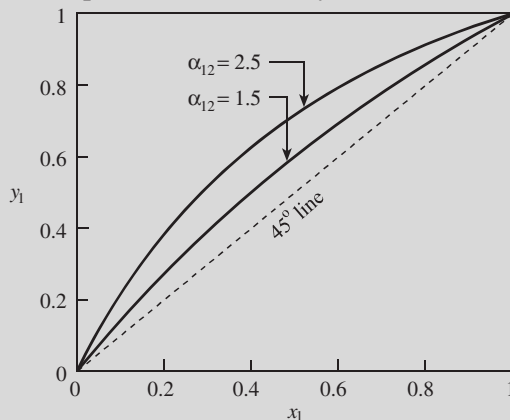
Solving Eqn (4) for y_1 gives

$$y_1 = \frac{\alpha_{12} x_1}{1 + x_1(\alpha_{12} - 1)}. \quad (5)$$

Solving Eqn (4) for x_1 gives

$$x_1 = \frac{y_1}{\alpha_{12} - y_1(\alpha_{12} - 1)}. \quad (6)$$

The plot of the x - y diagram for $\alpha_{12} = 1.5$ and $\alpha_{12} = 2.5$ is shown below. Note that the greater the difference is between the equilibrium (or x - y) curve and the 45° line, the larger the difference is in compositions of liquid and vapor phases and the easier it is to separate the mixture by distillation.



9.7.1 Criteria for Azeotrope Formation

If a binary system forms an azeotrope (either minimum or maximum boiling), the value of α_{12} on one side of the azeotrope will be greater than unity and on the other side will be less than unity. It is possible to calculate the values of α_{12} at the limiting values of $x_1 = 0$ and $x_1 = 1$ as follows:

$$(\alpha_{12})_{x_1=0} = \lim_{x_1 \rightarrow 0} \left(\frac{\gamma_1 P_1^{\text{vap}}}{\gamma_2 P_2^{\text{vap}}} \right) = \frac{\gamma_1^\infty P_1^{\text{vap}}}{P_2^{\text{vap}}} \quad (9.7-6)$$

$$(\alpha_{12})_{x_1=1} = \lim_{x_1 \rightarrow 1} \left(\frac{\gamma_1 P_1^{\text{vap}}}{\gamma_2 P_2^{\text{vap}}} \right) = \frac{P_1^{\text{vap}}}{\gamma_2^{\infty} P_2^{\text{vap}}} \quad (9.7-7)$$

Therefore, in order to check whether a given binary system has an azeotrope or not, one has to calculate $(\alpha_{12})_{x_1=0}$ and $(\alpha_{12})_{x_1=1}$. If one of the values is greater than unity and the other one is less than unity, then the system has an azeotrope.

Example 9.25 *Park et al. (2001)* studied vapor-liquid equilibria of *n*-hexane (1) and 2-methyl pyrazine (2) at 353.15 K. The activity coefficients are represented by the van Laar model with $A = 2.356$ and $B = 2.023$. At 353.15 K, the vapor pressures are given by

$$P_1^{\text{vap}} = 142.42 \text{ kPa} \quad \text{and} \quad P_2^{\text{vap}} = 16.82 \text{ kPa}.$$

Show whether or not this system exhibits an azeotrope.

Solution

One obvious way of solving this problem is to prepare the P_{xy} diagram and check for the formation of an azeotrope. This procedure, however, is tedious. A simpler procedure is to use Eqns (9.7-6) and (9.7-7) and to check the values of α_{12} at the two extreme locations, i.e. at $x_1 = 0$ and $x = 1$. From Eqn (8.4-16), the infinite dilution activity coefficients are

$$\begin{aligned} \gamma_1^{\infty} &= \exp(A) = \exp(2.356) = 10.549, \\ \gamma_2^{\infty} &= \exp(B) = \exp(2.023) = 7.561. \end{aligned}$$

From Eqn (9.7-6)

$$(\alpha_{12})_{x_1=0} = \frac{\gamma_1^{\infty} P_1^{\text{vap}}}{P_2^{\text{vap}}} = \frac{(10.549)(142.42)}{16.82} = 89.3.$$

From Eqn (9.7-7)

$$(\alpha_{12})_{x_1=1} = \frac{P_1^{\text{vap}}}{\gamma_2^{\infty} P_2^{\text{vap}}} = \frac{142.42}{(7.561)(16.82)} = 1.1.$$

Since $\alpha_{12} > 1$ at both $x_1 = 0$ and $x_1 = 1$, the system does not form an azeotrope.

Example 9.26 A binary mixture of *n*-propanol (1) and chlorobenzene (2) consists of vapor and liquid phases in equilibrium at 1 bar. The activity coefficients

are expressed as

$$\ln \gamma_1 = x_2^2(1.474 - 0.428x_2) \quad \text{and} \quad \ln \gamma_2 = x_1^2(0.832 + 0.428x_1).$$

Does this mixture form an azeotrope?

Solution

From Appendix C, the vapor pressures are given as

$$\ln P_1^{\text{vap}} = 10.9237 - \frac{3166.28}{T - 80.15} \quad \text{and} \quad \ln P_2^{\text{vap}} = 9.4474 - \frac{3295.12}{T - 55.60}.$$

Formation of an azeotrope can be checked by preparing a Txy diagram. Since this procedure is tedious, the values of α_{12} will be checked at the two extreme locations, i.e. at $x_1 = 0$ and $x_1 = 1$.

• **At $x_1 = 0$**

At this location, i.e. pure 2, $P_2^{\text{vap}} = 1$ bar. Therefore, the temperature is

$$T = 55.60 + \frac{3295.12}{9.4474} = 404.9 \text{ K}.$$

The vapor pressure of component 1 is

$$P_1^{\text{vap}} = \exp \left(10.9237 - \frac{3166.28}{404.9 - 80.15} \right) = 3.234 \text{ bar}.$$

The activity coefficient of component 1 at infinite dilution is

$$\lim_{x_1 \rightarrow 0} \ln \gamma_1 = \ln \gamma_1^\infty = 1.474 - 0.428 = 1.046 \quad \Rightarrow \quad \gamma_1^\infty = 2.846.$$

From Eqn (9.7-6)

$$(\alpha_{12})_{x_1=0} = \frac{\gamma_1^\infty P_1^{\text{vap}}}{P_2^{\text{vap}}} = \frac{(2.846)(3.234)}{1} = 9.2.$$

• **At $x_1 = 1$**

At this location, i.e. pure 1, $P_1^{\text{vap}} = 1$ bar. Therefore, the temperature is

$$T = 80.15 + \frac{3166.28}{10.9237} = 370 \text{ K}.$$

The vapor pressure of component 2 is

$$P_2^{\text{vap}} = \exp \left(9.4474 - \frac{3295.12}{370 - 55.60} \right) = 0.356 \text{ bar}.$$

The activity coefficient of component 2 at infinite dilution is

$$\lim_{x_2 \rightarrow 0} \ln \gamma_2 = \ln \gamma_2^\infty = 0.832 + 0.428 = 1.26 \Rightarrow \gamma_2^\infty = 3.525.$$

From Eqn (9.7-7)

$$(\alpha_{12})_{x_1=1} = \frac{P_1^{\text{vap}}}{\gamma_2^\infty P_2^{\text{vap}}} = \frac{1}{(3.525)(0.356)} = 0.8.$$

Since $\alpha_{12} > 1$ at $x_1 = 0$ and $\alpha_{12} < 1$ at $x_1 = 1$, the system forms a minimum boiling azeotrope.

9.8 VLE Calculations Using the Equation of State

When the vapor and liquid phases are described by the equation of state, the condition of equilibrium between the liquid and vapor phases is represented by Eqn (9.1-4), i.e.

$$y_i \hat{\phi}_i^{\text{V}}(T, P, y_i) = x_i \hat{\phi}_i^{\text{L}}(T, P, x_i). \quad (9.8-1)$$

The so-called “ ϕ - ϕ method” is generally used at high pressures. The lack of a reliable equation of state and mixing rules to represent liquid mixtures limits the use of this approach.⁹

Example 9.27 The following vapor-liquid equilibrium data are reported by [Bezanehtak et al. \(2002\)](#) for a binary system of H₂ (1) and CO₂ (2) at 278.15 K. Assuming the system is represented by the Peng-Robinson equation of state, check the consistency of the following experimental data:

P (bar)	x_1	y_1
77.22	0.0290	0.2789
153.67	0.1026	0.4796
192.53	0.1307	0.5055

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Hydrogen	33.2	13.0	−0.216
Carbon dioxide	304.2	73.8	0.239

The given data should satisfy Eqn (9.8-1), i.e.

⁹Representation of both phases by the cubic equation of state is usually possible if the mixture exhibits slight deviation from ideal behavior, which is generally true for hydrocarbon mixtures and hydrocarbons with inorganic gases.

$$y_i \hat{\phi}_i^V(T, P, y_i) = x_i \hat{\phi}_i^L(T, P, x_i) \quad (1)$$

The parameters in the Peng-Robinson equation of state as well as the fugacity coefficients for the vapor and liquid phases are given in the table below. In the calculations, $k_{12} = k_{21}$ is taken as zero.

P (bar)	Vapor Phase					Liquid Phase				
	A_{mix}	B_{mix}	Z_{mix}	$\hat{\phi}_1^V$	$\hat{\phi}_2^V$	A_{mix}	B_{mix}	Z_{mix}	$\hat{\phi}_1^L$	$\hat{\phi}_2^L$
77.22	0.378	0.080	0.676	1.401	0.570	0.583	0.088	0.165	10.984	0.412
153.67	0.488	0.145	0.757	1.364	0.419	1.030	0.170	0.322	5.703	0.246
192.53	0.574	0.179	0.775	1.406	0.367	1.232	0.211	0.396	4.801	0.213

The values given below indicate that while Eqn (1) is satisfied for carbon dioxide the values for hydrogen are slightly off.

Component	$P = 77.22$ bar		$P = 153.67$ bar		$P = 192.53$ bar	
	$y_i \hat{\phi}_i^V$	$x_i \hat{\phi}_i^L$	$y_i \hat{\phi}_i^V$	$x_i \hat{\phi}_i^L$	$y_i \hat{\phi}_i^V$	$x_i \hat{\phi}_i^L$
Hydrogen	0.391	0.319	0.654	0.585	0.711	0.628
Carbon dioxide	0.411	0.400	0.218	0.220	0.181	0.185

Once the compositions of vapor and liquid phases are known, as in Example 9.27, calculation of fugacity coefficients is a straightforward task. If one of the phase compositions is unknown, then the use of Eqn (9.8-1) requires a numerical solution. For example, consider the calculation of pressure at which a liquid of known composition and temperature first begins to boil, i.e. bubble point pressure. For a binary system, Eqn (9.8-1) takes the form

$$f_1(P, y_1) = y_1 \hat{\phi}_1^V - x_1 \hat{\phi}_1^L = 0, \quad (9.8-2)$$

$$f_2(P, y_1) = (1 - y_1) \hat{\phi}_2^V - (1 - x_1) \hat{\phi}_2^L = 0. \quad (9.8-3)$$

Simultaneous solution of Eqns (9.8-2) and (9.8-3) by Newton's method yields vapor phase composition and pressure.

Example 9.28 Estimate the bubble point pressure of a liquid mixture of 20.4 mol% ethylene (1) and 79.6% carbon dioxide (2) at 263 K. Also determine the composition of the first bubble of vapor. The mixture is represented by the Peng-Robinson equation of state with $k_{12} = 0.055$. At 263 K, the vapor pressures of ethylene and carbon dioxide are 30.68 bar and 26.66 bar, respectively.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Ethylene	282.5	50.6	0.089
Carbon dioxide	304.2	73.8	0.239

The governing equations, i.e. Eqns (9.8-2) and (9.8-3) take the form

$$f_1(P, y_1) = y_1 \hat{\phi}_1^V - 0.204 \hat{\phi}_1^L = 0, \quad (1)$$

$$f_2(P, y_1) = (1 - y_1) \hat{\phi}_2^V - 0.796 \hat{\phi}_2^L = 0. \quad (2)$$

The initial estimates of pressure and vapor phase mole fraction can be determined with the help of Raoult's law, i.e.

$$P = x_1 P_1^{\text{vap}} + x_2 P_2^{\text{vap}} = (0.204)(30.68) + (0.796)(26.66) = 27.48 \text{ bar},$$

$$y_1 = \frac{x_1 P_1^{\text{vap}}}{P} = \frac{(0.204)(30.68)}{27.48} = 0.228.$$

Let us choose ϵ as 1×10^{-3} .

First iteration: The fugacity coefficients of ethylene and carbon dioxide in the vapor and liquid phases are determined from Eqns (7.5-16) and (7.6-4), respectively, as

$$\begin{aligned} \hat{\phi}_1^V &= 0.7783776 \quad \text{and} \quad \hat{\phi}_2^V = 0.7896419, \\ \hat{\phi}_1^L &= 1.0822406 \quad \text{and} \quad \hat{\phi}_2^L = 0.7775177. \end{aligned}$$

Therefore, Eqn (9.4-8) takes the form

$$\begin{aligned} \begin{pmatrix} \Delta_1 \\ \Delta_2 \end{pmatrix} &= - \begin{pmatrix} 0.0053811 & 0.767026 \\ 0.0152007 & -0.778199 \end{pmatrix}^{-1} \cdot \begin{pmatrix} -0.0433070 \\ -0.0093005 \end{pmatrix} \\ &= \begin{pmatrix} 2.5768568 \\ 0.0383829 \end{pmatrix}. \end{aligned}$$

The pressure and the mole fraction in the vapor phase become

$$P = 27.48 + 2.5768568 = 30.0568568 \text{ bar},$$

$$y_1 = 0.228 + 0.0383829 = 0.2663829.$$

Since $\Delta_{\max} = 2.5769 > \epsilon$, iteration should be repeated. The results of the iterations are summarized in the following table:

k	P (bar)	y_1	$\Delta_{\max}$
0	27.4800000	0.2280000	–
1	30.0568568	0.2663829	2.5768568
2	30.4160778	0.2675967	0.3592210
3	30.4171707	0.2676017	0.0010929
4	30.4171540	0.2676017	0.0000167

Hence, the bubble point pressure is 30.417 bar and the composition of the first bubble of vapor is $y_1 = 0.268$. For more details on this problem, see [Bae et al. \(1982\)](#).

Problems

Problems Related to Section 9.3

- 9.1** For a binary mixture of components A and B at a specified pressure, the bubble point and the dew point temperatures are expressed as a function of composition in the form

$$T_{\text{bubble}} = 320 + 40x_A^2,$$

$$T_{\text{dew}} = 320 + 40y_A(2 - y_A),$$

where T is in K.

- Sketch a representative Txy diagram of the system and label the liquid, vapor, and two-phase regions.
- A vapor mixture of composition $y_A = 0.43$ is to be cooled at constant pressure. What is the composition of the first drop of liquid?

(Answer: (b) $x_A = 0.822$)

- 9.2** A binary vapor mixture of 60 mol% n -pentane (1) and 40% n -hexane (2) is initially at a temperature of 318 K.

- At what pressure does condensation occur if this mixture is compressed isothermally? What is the composition of the first liquid droplet?
- What is the minimum pressure required for complete condensation of this mixture?

(Answer: (a) 0.749 bar, $x_1 = 0.331$ and (b) 0.993 bar)

- 9.3** A closed tank contains 10 mol of propane (1), 16 mol of *n*-butane (2), and 8 mol of *n*-pentane (3) at 320 K and 8 bar. How many phases are present within the tank?

(Answer: Liquid)

- 9.4** A mixture of 5 mol% methane (1), 10% ethane (2), 10% propane (3), 30% *n*-butane (4), and 45% *n*-pentane (5) is to be separated by a flash drum operating at 1.5 bar. Estimate the drum temperature as well as the compositions of the liquid and vapor phases if 85% of the feed is vaporized.

(Answer: 300.2 K; $x_1 = 1.244 \times 10^{-4}$, $x_2 = 4.319 \times 10^{-3}$, $x_3 = 0.017$, $x_4 = 0.186$, $x_5 = 0.792$; $y_1 = 0.059$, $y_2 = 0.117$, $y_3 = 0.115$, $y_4 = 0.32$, $y_5 = 0.39$)

- 9.5** The calculation of bubble point and dew point temperatures is frequently encountered in chemical engineering practice. Since a trial-and-error procedure is required, calculations may require excessive computer time in the case of multicomponent mixtures. The use of a suitable numerical technique reduces the computer time.

The calculation of bubble point temperature requires the solution of

$$f(T) = \sum_{i=1}^k x_i P_i^{\text{vap}}(T) - P = 0 \quad (1)$$

using an iterative procedure. The use of the Newton-Raphson method leads to

$$T_n = T_{n-1} - \frac{f(T_{n-1})}{\left. \frac{df}{dT} \right|_{T_{n-1}}} \quad n > 0. \quad (2)$$

If the first derivative of the function is not easy to evaluate, then the following iteration scheme is suitable (Tosun, 2007):

$$T_n = T_{n-1} - \frac{0.02T_{n-1}f(T_{n-1})}{f(1.01T_{n-1}) - f(0.99T_{n-1})} \quad n > 0. \quad (3)$$

The calculation of dew point temperature requires the solution of the following equation:

$$f(T) = \frac{1}{\sum_{i=1}^k \frac{y_i}{P_i^{\text{vap}}(T)}} - P = 0. \quad (4)$$

The use of the iteration scheme given by Eqn (3) yields the dew point temperature.

Resolve Example 9.4 by using Eqn (3).

9.6 An overhead vapor stream from the distillation column consists of 25 mol% *n*-butane, 35% isobutane, 30% *n*-pentane, and 10% *n*-hexane.

- (a) If the overhead product is in equilibrium with the liquid on the top plate of the column at 1 bar, determine the temperature of the overhead vapor.
- (b) Estimate the maximum operating temperature of the condenser so as to completely condense the overhead product. The condenser pressure is 1 bar.

(Answer: (a) 301.6 K and (b) 276.5 K)

9.7 A mixture of 60 mol% *n*-butane (1), 25% isobutane (2), and 15% *n*-pentane (3) is at 1 bar and 310 K. This gas mixture is compressed in an isothermal compressor.

- (a) Determine the maximum pressure at the outlet of the compressor without generating liquid.
- (b) If the actual outlet pressure is 90% of that obtained in part (a), calculate the minimum work required to run the compressor. The gas mixture is represented by the Peng-Robinson equation of state with $k_{ij} = 0$. Use Eqn (4) of Problem 5.18 to calculate vapor pressure.

(Answer: (a) 2.734 bar and (b) 2221 J/mol)

9.8 A liquid mixture containing 80 mol% *n*-hexane (1) and 20% *n*-heptane (2) is to be separated in a flash drum held at 370 K. It is required to have equal molar flow rates of liquid and vapor streams from the flash drum. Determine the drum pressure as well as the compositions of the liquid and vapor streams.

(Answer: 1.921 bar, $x_1 = 0.734$, $y_1 = 0.866$)

9.9 A binary mixture of isopropanol (1) and *n*-pentanol (2) is to be separated by a two-stage flash operation as follows: A liquid mixture containing 90 mol% isopropanol enters the first stage with a flow rate of 60 mol/min. The flash drum operates at 313.5 K and 8 kPa. The liquid stream leaving the first stage is sent to the second stage operating at 313.5 K. The liquid and vapor streams leaving the second stage have equal flow rates. Using the pressure-composition data provided below (Mara *et al.*, 1997), determine:

- (a) The compositions of the streams leaving the first stage.
 (b) The operating pressure as well as the compositions of the streams leaving the second stage.

x_1	P (kPa)	x_1	P (kPa)	x_1	P (kPa)
0.0000	0.889	0.3521	5.454	0.7020	10.017
0.0323	1.311	0.4021	6.103	0.7517	10.598
0.0613	1.681	0.4523	6.755	0.8014	11.292
0.1322	2.596	0.5022	7.406	0.8516	11.966
0.1491	2.816	0.5533	8.067	0.9019	12.587
0.2004	3.484	0.6027	8.713	0.9412	13.134
0.2511	4.129	0.6031	8.710	0.9722	13.527
0.3010	4.780	0.6524	9.371	1.0000	13.872

(Answer: (a) $x_1 = 0.55$, $y_1 = 0.95$ and (b) $P = 4$ kPa, $x_1 = 0.25$, $y_1 = 0.85$)

9.10 A junior engineer is assigned to study the distillation column that is used for the separation of isopropanol (1) and *n*-hexane (2). On his/her first day in the job, after going over his/her calculations for the reboiler, the engineer rushes to his/her boss' office out of breath and says,

- “There’s something wrong! We must change the operating conditions of the reboiler quickly. I double-checked my calculations and there shouldn’t be any vapor produced under these conditions!”
- “What are the existing operating conditions?” the boss asks.
- The engineer replies, “the stream entering the reboiler contains 15% isopropanol, and the operating temperature and pressure are 335 K and 0.944 bar, respectively.”

Realizing that the engineer has used Raoult’s law in his/her calculations, the boss laughs and pats him/her on the shoulder. “Don’t worry” the boss says, “everything is alright. But it seems to me that you were daydreaming in Dr. Tosun’s thermodynamics classes!”

Show that the use of Raoult’s law does indeed lead to no vapor generation.

9.11 A binary mixture consisting of 50.7 mol% *n*-butane (1) and 49.3% isobutane (2) enters a flash chamber operating at 344 K and 9.6 bar. Estimate the compositions of the vapor and liquid streams leaving the flash chamber assuming ideal mixtures of vapor and liquid. The Poynting correction factor can be considered to be equal to unity. At 344 K, [Connolly \(1962\)](#) reported the following

data for vapor pressures and second virial coefficients:

$$P_1^{\text{vap}} = 8.307 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 11.143 \text{ bar},$$

$$B_{11} = -517.0 \text{ cm}^3/\text{mol} \quad \text{and} \quad B_{22} = -452.7 \text{ cm}^3/\text{mol}.$$

(Answer: $x_1 = 0.538$, $y_1 = 0.477$)

9.12 A flash chamber operating at 376 K separates a benzene (1) and toluene (2) mixture. The liquid stream exiting the flash chamber contains 20 mol% benzene and 80% toluene.

- Estimate the flash chamber pressure.
- While the flash chamber operates at the pressure as calculated in part (a), the liquid product specification is changed to 10% benzene and 90% toluene. Would you increase or decrease temperature and by how much?

Data: Use the vapor pressure data given in Example 9.23.

(Answer: (a) 1.037 bar and (b) 380 K)

9.13 A flash chamber operating at 1.5 bar separates a benzene (1) and toluene (2) mixture. The vapor stream leaving the flash chamber contains 85 mol% benzene and 15% toluene.

- Estimate the flash chamber temperature.
- While the flash chamber operates at 1.5 bar, the vapor product specification is changed to 90% benzene and 10% toluene. Would you increase or decrease temperature and by how much?

Data: Use the vapor pressure data given in Example 9.23.

(Answer: (a) 373.5 K and (b) 371.3 K)

Problems Related to Section 9.5

9.14 Prepare T_{xy} diagram for a binary mixture of trichloroethylene (1) and n -pentanol (2) at 1 bar. The liquid phase nonideality is represented by the three-suffix Margules equation with $A = 0.755$ and $B = 0.348$. The vapor pressures are given by

$$\ln P_1^{\text{vap}} = 14.2231 - \frac{3030.47}{T - 44.232},$$

$$\ln P_2^{\text{vap}} = 15.8163 - \frac{3708.62}{T - 79.441},$$

where P is in kPa and T is in K.

Compare your results with the following experimental data of Dejoz *et al.* (1999):

x_1	y_1	T (K)	x_1	y_1	T (K)	x_1	y_1	T (K)
0.015	0.094	407.85	0.317	0.795	378.15	0.704	0.939	364.25
0.038	0.217	404.45	0.365	0.830	375.25	0.765	0.947	363.15
0.080	0.387	398.85	0.419	0.859	372.75	0.822	0.955	362.25
0.125	0.512	393.85	0.477	0.884	370.55	0.874	0.964	361.35
0.170	0.610	389.35	0.532	0.903	368.75	0.923	0.974	360.55
0.218	0.688	385.05	0.586	0.918	367.05	0.964	0.986	359.85
0.269	0.746	381.35	0.641	0.928	365.65	0.988	0.995	359.55

- 9.15** A binary liquid mixture is in contact with air at a total pressure of 1 bar. The molar excess Gibbs energy of this mixture is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2,$$

where A is dependent on temperature. The following information is available at 313 K:

- $P_1^{\text{vap}} = 0.85$ bar and $P_2^{\text{vap}} = 0.48$ bar.
- $\Delta\tilde{H}_{\text{mix}} = -8500$ J/mol and $\tilde{S}^{\text{ex}} = -3$ J/mol K when $x_1 = 0.33$.

Determine the composition of air in equilibrium with the liquid mixture of $x_1 = 0.33$ at 313 K. Assume that the solubility of air in the liquid mixture is negligible.

(Answer: $y_1 = 7.7 \times 10^{-4}$, $y_2 = 0.0769$, $y_{\text{air}} = 0.92233$)

- 9.16** For a binary mixture of methyl acetate (1) and water (2) at 333 K, the following vapor-liquid equilibrium data are measured experimentally:

P (bar)	x_1	y_1
1.2	0.230	0.844

Using the van Laar model, estimate the vapor phase composition when $x_1 = 0.45$.

(Answer: $y_1 = 0.812$)

- 9.17** Consider a binary system of thiophene (1) and n -hexane (2) at 338.15 K as given in Example 8.9. Assuming ideal gas behavior, estimate the vapor phase composition in equilibrium with a liquid containing 42 mol% thiophene. The

vapor pressures of thiophene and *n*-hexanol at 338.15 K are 54.27 kPa and 90.44 kPa, respectively.

(Answer: $y_1 = 0.341$)

- 9.18** An overhead vapor stream from the distillation column is to be condensed under atmospheric conditions and, while part of it is returned to the column as reflux, the remaining part is withdrawn as product. If the vapor consists of 16 mol% water and 84% *n*-propanol, estimate the maximum operating temperature of the condenser. The binary mixture of water (1) and *n*-propanol (2) is represented by the NRTL equation with the following parameters:

$$g_{12} - g_{22} = 9218.80 \text{ J/mol} \quad g_{21} - g_{11} = -414.77 \text{ J/mol} \quad \alpha = 0.3.$$

(Answer: 365 K)

- 9.19** Estimate the dew point temperature of a vapor mixture of 54.1 mol% methanol (1) and 45.9% benzene (2) at a pressure of 1.013 bar. Also calculate the composition of the first drop of liquid. The system is represented by the Wilson equation with

$$\lambda_{12} - \lambda_{11} = 7476.31 \text{ J/mol} \quad \text{and} \quad \lambda_{21} - \lambda_{22} = 730.83 \text{ J/mol}.$$

Take $\tilde{V}_1^L = 44.4 \text{ cm}^3/\text{mol}$ and $\tilde{V}_2^L = 94.8 \text{ cm}^3/\text{mol}$.

(Answer: 332.6 K, $x_1 = 0.207$)

- 9.20** A binary system of *tert*-amyl methyl ether (1) and *n*-propanol (2) at 313.15 K is represented by the NRTL equation with the following parameters:

$$\tau_{12} = 1.0102 \quad \tau_{21} = 0.1290 \quad \alpha = 0.3.$$

The vapor pressures of *tert*-amyl methyl ether (TAME) and *n*-propanol at 313.15 K are 19.551 kPa and 7 kPa, respectively.

- Estimate the bubble point pressure of a liquid mixture of 20 mol% TAME and 80% *n*-propanol at 313.15 K.
- Estimate the dew point pressure of a vapor mixture of 70 mol% TAME and 30% *n*-propanol at 313.15 K.

(Answer: (a) 12.987 kPa and (b) 16.105 kPa)

- 9.21** A binary liquid mixture of 20 mol% methyl glycolate (1) and 80% ethylene glycol (2) is at a temperature of 353.15 K.

- As the liquid is depressurized isothermally, at what pressure will boiling begin?

(b) What is the maximum pressure for complete vaporization of this mixture?

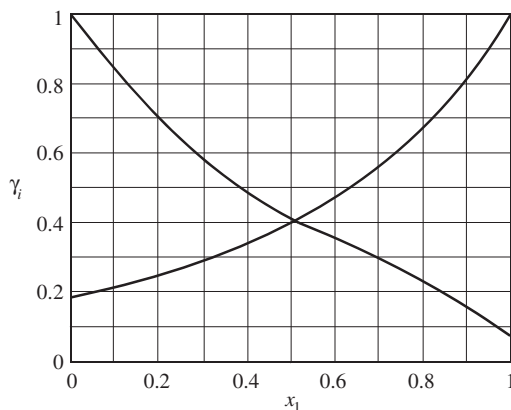
Data: The following data are provided by Sun *et al.* (2006) for methyl glycolate (1) and ethylene glycol (2) mixtures:

$$\text{At } 353.15 \text{ K } P_1^{\text{vap}} = 69.01 \text{ mmHg} \text{ and } P_2^{\text{vap}} = 5.26 \text{ mmHg},$$

$$\frac{g_{12} - g_{22}}{R} = -20.6373 \text{ K } \frac{g_{21} - g_{11}}{R} = 472.9804 \text{ K } \alpha = 0.47.$$

(Answer: (a) 29.23 mm Hg and (b) 6.54 mm Hg)

9.22 Consider a binary mixture of components 1 and 2. The following data are provided for the activity coefficients at 310 K. The vapor pressures of components 1 and 2 at 310 K are 0.95 bar and 1.67 bar, respectively. Will a liquid mixture of 20 mol% component 1 and 80% component 2 boil at a temperature of 310 K under atmospheric pressure?



(Answer: No)

9.23 Estimate the dew point temperature of a vapor mixture of 66.6 mol% methanol (1) and 33.4% ethyl methyl carbonate (2) at 101.3 kPa. Also calculate the composition of the first bubble of vapor. The system is represented by the NRTL model with the following parameters (Zhang *et al.*, 2010):

$$\frac{g_{12} - g_{22}}{R} = 487 \text{ K } \frac{g_{21} - g_{11}}{R} = -43 \text{ K } \alpha = 0.3.$$

The vapor pressures are given as

$$\ln P_1^{\text{vap}} = 16.5725 - \frac{3626.55}{T - 34.29} \quad \ln P_2^{\text{vap}} = 14.8075 - \frac{3376.6}{T - 49.461},$$

where P^{vap} is in kPa and T is in K.

(Answer: 354 K, $x_1 = 0.2$)

- 9.24** Estimate the dew point temperature of a vapor mixture of 45 mol% 1,3-dioxolane (1) and 55% water (2) at a pressure of 101.3 kPa. Also calculate the composition of the first drop of liquid. The system is represented by the Wilson equation with the following energy parameters (Kurihara *et al.*, 2003):

$$\lambda_{12} - \lambda_{11} = 1179.818 \text{ J/mol} \quad \lambda_{21} - \lambda_{22} = 7454.402 \text{ J/mol.}$$

The vapor pressures are given as

$$\ln P_1^{\text{vap}} = 14.1577 - \frac{2754.35}{T - 59.834} \quad \ln P_2^{\text{vap}} = 14.2675 - \frac{2658.44}{T - 97.619},$$

where P^{vap} is in kPa and T is in K. Take the molar volumes of liquids as

$$\tilde{V}_1^{\text{L}} = 78.4 \text{ cm}^3/\text{mol} \quad \tilde{V}_2^{\text{L}} = 18 \text{ cm}^3/\text{mol.}$$

(Answer: 357.9 K, $x_1 = 0.041$)

- 9.25** The flash point is defined as the lowest temperature at which a flammable liquid forms a combustible mixture with air. Its value is of utmost importance for the handling and storage of flammable liquids. For a binary miscible mixture of flammable liquids, Le Chatelier's rule states that

$$1 = \frac{y_1}{\text{LFL}_1} + \frac{y_2}{\text{LFL}_2}, \quad (1)$$

where y_i is the mole fraction of flammable component i in the vapor phase and LFL_i is the *lower flammability limit* of pure component i .

- (a) Using the definition of flash point, show that the lower flammability limit (LFL) of pure component i is simply the mole fraction defined by

$$\text{LFL}_i = \frac{P_i^{\text{vap}}(T_f)}{P}, \quad (2)$$

where $P_i^{\text{vap}}(T_f)$ is the vapor pressure of pure i at its flash point temperature, T_f .

- (b) Use vapor-liquid equilibrium relationships and show that Eqn (1) takes the form

$$1 = \frac{x_1 \gamma_1 P_1^{\text{vap}}(T)}{P_1^{\text{vap}}(T_{f1})} + \frac{x_2 \gamma_2 P_2^{\text{vap}}(T)}{P_2^{\text{vap}}(T_{f2})}, \quad (3)$$

where $P_i^{\text{vap}}(T)$ is the vapor pressure of pure i at the flash point temperature of the mixture, T .

- (c) Consider a binary liquid mixture consisting of 35 mol% methanol (1) and 65% methyl acetate (2). The activity coefficients of this mixture are represented by NRTL model with the following parameters:

$$\frac{g_{12} - g_{22}}{R} = 86.237 \text{ K} \quad \frac{g_{21} - g_{11}}{R} = 224.99 \text{ K} \quad \alpha = 0.271.$$

The flash point temperatures of methanol and methyl acetate are 285 K and 260 K, respectively. Estimate the flash point temperature of this mixture.

(Answer: (c) 262.4 K)

- 9.26** A binary system formed of methanol (1) and acetone (2) consists of vapor and liquid phases in equilibrium at 330 K. Assume that the vapor phase is ideal and the liquid phase is represented by the three-suffix Margules equation with the following parameters:

$$A = 0.538 \quad \text{and} \quad B = 0.028.$$

- (a) Estimate the pressure and the composition of the vapor phase when $x_1 = 0.525$.
 (b) Estimate the overall mole fraction range over which this system exists with a liquid mole fraction of $x_1 = 0.525$.

(Answer: (a) 1.008 bar, $y_1 = 0.44$ and (b) $0.44 < z_1 < 0.525$)

- 9.27** A binary system formed of methanol (1) and *n*-hexane (2) consists of vapor and liquid phases in equilibrium at 313.15 K. The system is represented by the three-suffix Margules equation with $A = 2.557$ and $B = -0.075$. The overall mole fraction of methanol is 0.75.

- (a) Estimate the pressure range over which this system exists as two phases.
 (b) For a liquid mole fraction of $x_1 = 0.9$, determine the composition and the overall mole fraction of the vapor phase.

(Answer: (a) $0.459 \text{ bar} < P < 0.701 \text{ bar}$ and (b) $y_1 = 0.534, 0.41$)

- 9.28** An equimolar liquid mixture of components 1 and 2 is represented by the three-suffix Margules equation. If 60% of this mixture is to be evaporated at 350 K, estimate the pressure as well as the compositions of the liquid and vapor phases.

Data: The following experimental data are obtained at 350 K:

At 350 K

$$P_1^{\text{vap}} = 70 \text{ kPa} \quad \text{and} \quad P_2^{\text{vap}} = 85 \text{ kPa}.$$

x_1	P (kPa)
0.3	88.33
0.7	84.35

(Answer: 87.373 kPa, $x_1 = 0.525$, $y_1 = 0.483$)

- 9.29** A liquid mixture is prepared by mixing 2 mol of component 1 with 3 mol of component 2 and the total pressure in the vapor phase is measured as 70 kPa at 353 K. The molar excess Gibbs energy of the system is represented by

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2$$

and the vapor pressures of components 1 and 2 at 353 K are 100 kPa and 65 kPa, respectively.

- Determine the composition of the vapor phase in equilibrium with this liquid mixture.
- If 50% of the liquid is evaporated at 353 K, determine the composition of the liquid and vapor phases.

(Answer: (a) $y_1 = 0.483$ and (b) $x_1 = 0.364$, $y_1 = 0.437$)

- 9.30** A liquid mixture containing 27.7 mol% diethyl sulfide (1) and 72.3% cyclohexane (2) is flashed across a valve and into a flash drum held at constant temperature and pressure. It is required to have equal flow rates of vapor and liquid streams. If the liquid stream has a composition of 30 mol% diethyl sulfide, estimate the drum temperature and pressure.

Data: Molar volumes of saturated liquids as well as the vapor pressures are given by

$$\begin{aligned}\tilde{V}_1^{\text{L}} &= 108.363 \text{ cm}^3/\text{mol} & \tilde{V}_2^{\text{L}} &= 108.860 \text{ cm}^3/\text{mol}, \\ \ln P_1^{\text{vap}} &= 7.0397 - \frac{2896.000}{T - 54.4934}, \\ \ln P_2^{\text{vap}} &= 6.848 - \frac{2767.0239}{T - 50.9565},\end{aligned}$$

where P^{vap} is in MPa and T is in K.

Diethyl sulfide–cyclohexane system is represented by the Wilson equation, and Sapei *et al.* (2007a) provided the following parameters:

$$\begin{aligned}\lambda_{12} - \lambda_{11} &= 5874.918 - 23.559T + 0.026T^2, \\ \lambda_{21} - \lambda_{22} &= 899.827 - 7.862T + 0.016T^2,\end{aligned}$$

where λ is in J/mol and T is in K.

(Answer: 356.211 K, 105.231 kPa)

- 9.31** Using the Wilson model to correlate the binary T - x data, Kirss *et al.* (2004) reported the following parameters:

Binary Mixture	Λ_{12}	Λ_{21}
Toluene (1)–Ethylbenzene (2)	0.694	1.440
Toluene (1)–Amyl acetate (2)	0.555	1.801
Ethylbenzene (1)–Amyl acetate (2)	1.084	0.923

Estimate the bubble point pressure of a ternary mixture consisting of 74 mol% toluene (1), 13% ethylbenzene (2), and 13% amyl acetate (3) at 368.45 K. Also calculate the composition of the first bubble of vapor.

Data: The vapor pressures are given by

$$\begin{aligned}\ln P_1^{\text{vap}} &= 14.0841 - \frac{3148.177}{T - 51.1715}, \\ \ln P_2^{\text{vap}} &= 13.7078 - \frac{3083.502}{T - 70.0678}, \\ \ln P_3^{\text{vap}} &= 14.0180 - \frac{3215.228}{T - 80.5320},\end{aligned}$$

where P is in kPa and T is in K.

Hint: Use Eqn (8.9-2) to determine the activity coefficients.

(Answer: $P = 52.945$ kPa, $y_1 = 0.892$, $y_2 = 0.069$)

Problems Related to Section 9.6

- 9.32** Total pressure versus composition data for a binary mixture at 358 K are found to be correlated by the following polynomial:

$$P = 80 + 50x_1 - 40x_1^2,$$

where P is the total pressure in kPa. The mixture forms an azeotrope and the liquid-phase activity coefficients can be approximated by the van Laar model. Calculate the vapor phase composition when $x_1 = 0.35$.

(Answer: $y_1 = 0.41$)

- 9.33** For a binary mixture of tetrahydrofuran (1) and *n*-hexane (2) at 313.15 K, the total vapor pressure versus liquid composition is correlated as

$$P(\text{kPa}) = 40.200x_1 + 37.220x_2 + x_1x_2[24.813 + 1.015(x_1 - x_2) + 10.920(x_1 - x_2)^2].$$

Calculate the azeotropic composition and pressure for this system. Wu and Sandler (1988) reported the azeotropic data as

$$x_1 = 0.5881 \quad \text{and} \quad P = 45.153 \text{ kPa.}$$

(Answer: $x_1 = 0.6135$, $P = 45.12 \text{ kPa}$)

- 9.34** Consider a binary mixture forming an azeotrope. The liquid phase nonideality is represented by the two-suffix Margules equation.

(a) Using $dP/dx_1 = 0$, show that the azeotropic composition is given by

$$x_1 = \frac{1}{2} \left[1 + \frac{1}{A} \ln \left(\frac{P_1^{\text{vap}}}{P_2^{\text{vap}}} \right) \right]. \quad (1)$$

(b) Show that the use of the identity

$$P_1^{\text{vap}} \gamma_1 = P_2^{\text{vap}} \gamma_2 \quad (2)$$

also leads to Eqn (1).

- (c) A binary mixture of 3-methylpentane (1) and tetrahydrofuran (2) forms an azeotrope at 333.3 K. Loras *et al.* (1999) reported that the liquid phase is represented by the two-suffix Margules equation with $A = 0.548$. The vapor pressures are given by

$$\begin{aligned} \log P_1^{\text{vap}} &= 6.24116 - \frac{1302.78}{T - 28.69} \\ \log P_2^{\text{vap}} &= 6.44102 - \frac{1384.21}{T - 27.00}, \end{aligned}$$

where P^{vap} is in kPa and T is in K. Estimate the azeotropic composition and pressure.

(Answer: (c) $x_1 = 0.589$, 101 kPa)

- 9.35** A binary mixture at a given temperature T has the following VLE data:

$$\begin{aligned} P_1^{\text{vap}} &= 1.1 \text{ bar} \quad \text{and} \quad P_2^{\text{vap}} = 1.85 \text{ bar,} \\ x_1 = y_1 &= 0.45 \quad \text{when} \quad P = 0.85 \text{ bar.} \end{aligned}$$

- (a) Sketch a representative P_{xy} diagram of the system and label the liquid, vapor, and two-phase regions.
- (b) Using the van Laar equation for the activity coefficients, calculate the vapor phase composition when $x_1 = 0.2$.
- (c) Are the like interactions stronger or weaker than the unlike interactions?

(Answer: (b) $y_1 = 0.057$)

9.36 A binary mixture of *tert*-amyl methyl ether (1) and *tert*-butyl alcohol (2) at 323 K is represented by the NRTL equation. Strothmann *et al.* (1999) reported the following data for this system:

$$\gamma_1^\infty = 1.79 \quad \gamma_2^\infty = 2.60 \quad \alpha = 0.3655$$

$$P_1^{\text{vap}} = 28.95 \text{ kPa} \quad P_2^{\text{vap}} = 23.47 \text{ kPa}.$$

Use Eqn (9.6-6) and estimate the azeotropic composition. What is the azeotropic pressure?

(Answer: $x_1 = 0.686$, $P = 32.247 \text{ kPa}$)

9.37 A liquid mixture of water (1) and dioxane (2) for which $x_1 = 0.2$ is in equilibrium with its vapor at 361 K. Determine the equilibrium pressure and vapor composition from the following information:

- At 361 K

$$P_1^{\text{vap}} = 64.72 \text{ kPa} \quad \text{and} \quad P_2^{\text{vap}} = 65.84 \text{ kPa}.$$

- The liquid mixture is represented by the van Laar model.
- The system forms an azeotrope at 361 K and atmospheric pressure for which $x_1 = y_1 = 0.53$.

(Answer: $P = 93.88 \text{ kPa}$, $y_1 = 0.41$)

9.38 A binary mixture of benzene (1) and cyclohexane (2) forms an azeotrope at 350.7 K. If the liquid mixture obeys the regular mixture theory, estimate the azeotropic composition. Hiaki *et al.* (1986) reported the azeotropic composition as $x_1 = 0.542$.

(Answer: $x_1 = 0.595$)

9.39 For a binary system of *n*-butanol (1) and octane (2), Hiaki *et al.* (1998) reported the following VLE data at 343.15 K:

- (a) Prepare the P_{xy} diagram and show that the system exhibits a minimum boiling azeotrope.

x_1	y_1	P (kPa)	x_1	y_1	P (kPa)	x_1	y_1	P (kPa)
0	0	46.82	0.3018	0.4554	73.50	0.8247	0.6560	69.34
0.0666	0.2771	62.07	0.3679	0.4655	74.53	0.8366	0.6706	68.84
0.1029	0.3346	65.94	0.4349	0.4939	75.28	0.8701	0.7027	66.97
0.1450	0.3758	68.96	0.4884	0.5095	75.67	0.9119	0.7593	64.16
0.1650	0.3869	69.90	0.5212	0.5195	75.71	0.9523	0.8361	60.04
0.1874	0.4023	70.78	0.5971	0.5441	75.48	1	1	51.89
0.2135	0.4148	71.68	0.7226	0.5943	73.49			
0.2643	0.4348	72.89	0.7722	0.6221	71.49			

(b) Note that the azeotropic composition is around $x_1 \approx 0.5$. Determine the azeotropic point by the following methods:

(i) Plot $(x_1 - y_1)$ versus x_1 around $x_1 \approx 0.5$, i.e.

$x_1 - y_1$	-0.0590	-0.0211	0.0017	0.0530
x_1	0.4349	0.4884	0.5212	0.5971

and show that the data can be expressed as

$$x_1 - y_1 = 0.690x_1 - 0.358. \quad (1)$$

When $x_1 - y_1 = 0$, the solution of Eqn (1) gives the azeotropic composition as $x_1 = 0.5188$.

(ii) Assuming $\partial P / \partial x_1 \simeq \Delta P / \Delta x_1$, plot $\partial P / \partial x_1$ versus x_1 around $x_1 \approx 0.5$, i.e.

$\partial P / \partial x_1$	9.4606	4.9826	-1.7479	-11.0228
x_1	0.4349	0.4884	0.5212	0.5971

and show that the data can be expressed as

$$\frac{\partial P}{\partial x_1} = 32.842 + 2.871x_1 - 128.385x_1^2. \quad (2)$$

When $\partial P / \partial x_1 = 0$, the solution of Eqn (2) gives the azeotropic composition as $x_1 = 0.5171$.

(iii) Consider

$$\frac{\Delta P_y}{\Delta P_x} = \frac{P - (P_1^{\text{vap}} y_1 + P_2^{\text{vap}} y_2)}{P - (P_1^{\text{vap}} x_1 + P_2^{\text{vap}} x_2)}. \quad (3)$$

Plot $\Delta P_y / \Delta P_x$ versus x_1 around $x_1 \approx 0.5$, i.e.

$\Delta P_y / \Delta P_x$	0.9886	0.9959	1.0003	1.0105
x_1	0.4349	0.4884	0.5212	0.5971

and show that the data can be expressed as

$$\frac{\Delta P_y}{\Delta P_x} = 0.135x_1 + 0.930. \quad (4)$$

When $\Delta P_y / \Delta P_x = 1$, the solution of Eqn (4) gives the azeotropic composition as $x_1 = 0.5185$.

9.40 A binary liquid mixture of ethanol (1) and ethyl acetate (2) forms an azeotrope at 333 K and at a mole fraction $x_1 = 0.4$. The molar excess Gibbs energy of this mixture is represented by

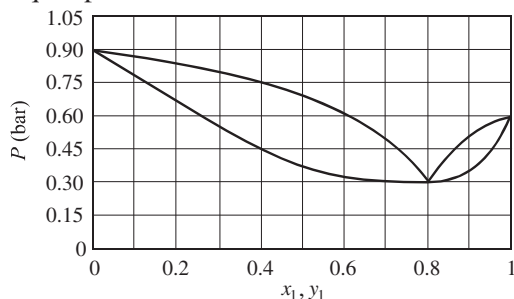
$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2.$$

- Does this mixture form a maximum boiling azeotrope or a minimum boiling azeotrope? Why?
- A liquid mixture containing 68 mol% ethanol is flashed across a valve and into a flash drum held at 333 K. It is required to get a liquid product with a composition of 75 mol% ethanol. Determine the pressure if the molar flow rates of the vapor and liquid streams leaving the flash drum are the same.

(Answer: (b) 0.6 bar)

9.41 For a binary system of components 1 and 2 the P_{xy} diagram at 300 K is shown in the figure below.

- Calculate the activity coefficients for components 1 and 2 in a liquid mixture containing 40 mol% component 1 at 300 K.
- A liquid mixture containing 40 mol% component 1 is flashed across a valve and into a flash drum held at 300 K. Estimate the drum pressure so that the liquid product contains the maximum amount of component 1.



- (c) A liquid mixture containing 40 mol% component 1 is flashed across a valve and into a flash drum held at 300 K. What should be the pressure within the drum in order to have a liquid product with $x_1 = 0.9$.
- (d) Are the like interactions stronger or weaker than the unlike interactions? Explain clearly.

(Answer: (a) $\gamma_1 = 0.3125$, $\gamma_2 = 1.25$ and (b) 0.45 bar)

- 9.42** A binary liquid mixture containing 20 mol% *n*-pentane (1) and 80% acetone (2) is flashed across a valve and into a flash drum held at 372.7 K. The following experimental data are reported by [Campbell *et al.* \(1986\)](#):

x_1	y_1	P (kPa)	x_1	y_1	P (kPa)
0.000	0.000	366.1	0.532	0.616	660.9
0.084	0.265	478.1	0.649	0.671	672.2
0.180	0.374	551.6	0.785	0.753	668.1
0.315	0.510	613.3	0.916	0.869	637.8
0.422	0.546	642.2	1.000	1.000	587.4

- (a) Plot the Pxy diagram of the system.
- (b) Determine the minimum and maximum operating pressures of the flash drum in order to have two phases.
- (c) If the drum pressure is 550 kPa, estimate the compositions of the vapor and liquid streams. What fraction of the system exists as a vapor?

(Answer: (b) 450 kPa and 557 kPa and (c) $x_1 = 0.17$, $y_1 = 0.36$, 0.16)

- 9.43** Prepare the Pxy diagram for a binary system of ethanol (1) and methyl butanoate (2) at 346.3 K, and determine the composition of the azeotropic point. Use the data given in Problem 8.21. The vapor pressures of ethanol and methyl butanoate at 346.3 K are 82.28 kPa and 37.54 kPa, respectively.

(Answer: $x_1 = 0.88$)

Problems Related to Section 9.7.

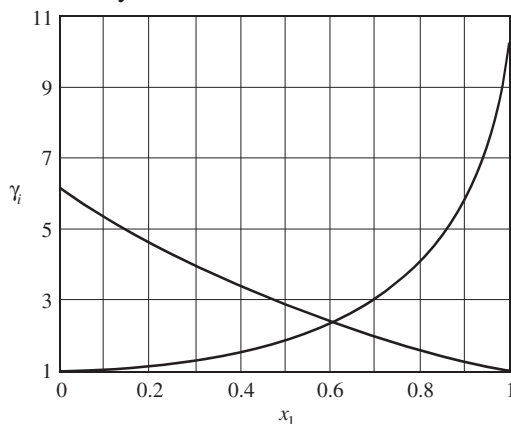
- 9.44** It is required to check whether a binary system of water (1) and 2-methyl pyrazine (2) exhibits an azeotrope at 353.15 K. [Park *et al.* \(2001\)](#) reported that the activity coefficients are represented by the van Laar model with $A = 1.20$ and $B = 2.08$. At 353.15 K, the vapor pressures are given by

$$P_1^{\text{vap}} = 47.39 \text{ kPa} \quad P_2^{\text{vap}} = 16.82 \text{ kPa}.$$

- (a) Show that the system has a minimum boiling azeotrope.
 (b) Use Eqn (9.6-6) and determine the composition at the azeotropic point.
 (c) Estimate azeotropic pressure using Eqn (9.6-12).

(Answer: (b) $x_1 = 0.827$ and (c) 51.6 kPa)

9.45 Consider a binary mixture of components 1 and 2. The following data are provided for the activity coefficients at 340 K:



The vapor pressures of components 1 and 2 at 340 K are 450 kPa and 300 kPa, respectively.

- (a) Does the binary system form an azeotrope? If it does, estimate the composition at the azeotropic point.
 (b) Are the like interactions stronger or weaker than the unlike interactions? Explain.

(Answer: (a) $x_1 = 0.7$)

9.46 Ito and Yoshida (1963) reported the following isobaric VLE data for a binary system of water (1) and formic acid (2) at 9.3 kPa:

T (K)	x_1	y_1	γ_1	γ_2	T (K)	x_1	y_1	γ_1	γ_2
311.55	0.094	0.031	0.7488	0.9895	321.65	0.598	0.624	0.9610	0.8102
316.45	0.244	0.110	0.7172	0.9190	320.55	0.734	0.816	1.0089	0.7692
317.65	0.291	0.157	0.7787	0.9102	319.45	0.826	0.900	1.0112	0.7730
320.15	0.392	0.287	0.8584	0.8709	318.85	0.907	0.953	0.9940	0.8327
321.35	0.496	0.448	0.9159	0.8448					

At 9.3 kPa, saturation temperatures of water and formic acid are 317.65 and 308.65 K, respectively.

- (a) Show that the system exhibits a maximum boiling (321.75 K) azeotrope at $x_1 = 0.564$.
 (b) Fit a fifth-order polynomial to the data points and show that

$$\ln \left(\frac{\gamma_1}{\gamma_2} \right) = 0.140 - 7.825x_1 + 44.035x_1^2 - 96.681x_1^3 + 98.126x_1^4 - 38.063x_1^5.$$

- (c) Check the consistency of the data.

9.47 Brandani (1974) proposed the following method for testing azeotrope formation of binary mixtures under isothermal conditions:

- (a) Use Eqn (9.6-12) and show that the following equation holds at the azeotropic point

$$\frac{\gamma_1}{\gamma_2} = \frac{P_2^{\text{vap}}}{P_1^{\text{vap}}}. \quad (1)$$

- (b) Show that the left-hand side of Eqn (1) takes the limiting values of γ_1^∞ and $1/\gamma_2^\infty$ for $x_1 = 0$ and $x_1 = 1$, respectively.
 (c) Conclude that for systems exhibiting positive deviations from Raoult's law

$$\gamma_1^\infty > \frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} > \frac{1}{\gamma_2^\infty}. \quad (2)$$

On the other hand, for systems exhibiting negative deviations from Raoult's law

$$\gamma_1^\infty < \frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} < \frac{1}{\gamma_2^\infty}. \quad (3)$$

- (d) The following data are obtained by Hopkins *et al.* (1994) at 303.15 K:

Components		NRTL Parameters		
1	2	τ_{12}	τ_{21}	α
Acetone	Toluene	0.1946	0.4879	0.3
Chloroform	Acetone	2.5730	-1.9615	0.3
Chloroform	Toluene	1.6416	-1.3554	0.3

Determine whether or not the above binary mixtures form an azeotrope.

(Answer: (d) No, Yes, No)

9.48 A binary system exhibits azeotropic behavior and the liquid phase is represented by the regular mixture theory.

(a) At the azeotropic point show that

$$\tilde{V}_1^L \Phi_2^2 - \tilde{V}_2^L \Phi_1^2 = \frac{RT}{(\delta_1 - \delta_2)^2} \ln \left(\frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} \right). \quad (1)$$

(b) Note that the left-hand side of Eqn (1) is a monotonic function in Φ_1 . Show that

$$\begin{aligned} \frac{RT}{(\delta_1 - \delta_2)^2} \ln \left(\frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} \right) &= \tilde{V}_1^L \quad \text{when } \Phi_1 = 0, \\ \frac{RT}{(\delta_1 - \delta_2)^2} \ln \left(\frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} \right) &= -\tilde{V}_2^L \quad \text{when } \Phi_1 = 1. \end{aligned} \quad (2)$$

Thus conclude that the criterion for azeotrope formation is given by

$$-\tilde{V}_2^L < \frac{RT}{(\delta_1 - \delta_2)^2} \ln \left(\frac{P_2^{\text{vap}}}{P_1^{\text{vap}}} \right) < \tilde{V}_1^L. \quad (3)$$

(c) Using the criterion given by Eqn (3), show that a binary mixture of benzene (1) and cyclohexane (2) forms an azeotrope at 350.7 K. Also estimate the azeotropic composition.

(Answer: (c) $x_1 = 0.595$)

Problem Related to Section 9.8

9.49 Estimate the bubble point pressure of a liquid mixture of 43.29 mol% carbon dioxide (1) and 56.71% cyclohexane (2) at 410.9 K. Also determine the composition of the first bubble of vapor. The mixture is represented by the Peng-Robinson equation of state with $k_{12} = 0.103$.

(Answer: $P = 101.769$ bar $y_1 = 0.875$)

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Chapter 10

Solubility of Gases in Liquids

In some separation processes, we are mainly interested in the transfer of a component from the gas phase to the liquid phase. For example, stringent environmental protection laws in many countries require the concentrations of pollutants in the industrial emission gases (CO_2 , H_2S , SO_2 , NO_x) to be kept below specified limits. For this purpose gas absorbers, scrubbers, and wetted wall columns are used to bring the gas stream into contact with liquid solvent so as to transfer the undesired component from the gas to the liquid. A typical gas absorber is shown in Fig. 10.1. Within the column, the contact between the gas and liquid streams is achieved either on trays, as in the case of distillation, or in a packed bed. The purpose of using a packed bed is to enhance the mass transfer between the phases by increasing the interfacial area. While the gas stream to be cleaned enters from the bottom of the column, the liquid solvent is fed from the top of the column and is distributed over the packing by a distribution plate (or a spray nozzle). Absorption in the liquid phase takes place either by physical absorption or by a chemical reaction. The effluent gas leaves the column from the top. The liquid stream leaving the absorber is first sent to a stripper to recover the dissolved solute and then recycled back to the absorption column.

In recent years, hollow fiber membrane contactors have found wide application in the biological treatment of gas streams containing volatile organic compounds (VOC's), and also in the removal of acid gas¹ from waste gas streams. These devices, in a way, resemble shell-and-tube heat exchangers. The membrane wall not only keeps liquid and gas phases separated from each other but also provides an interfacial area for the two phases. The gas stream may flow either in the tube or shell side, in cocurrent or countercurrent flow with the liquid stream. For more detail about membrane contactors, see Gabelman and Hwang (1999) and Sirkar (1997).

¹An acid gas is a gas containing significant amounts of acidic gases, such as CO_2 and H_2S .

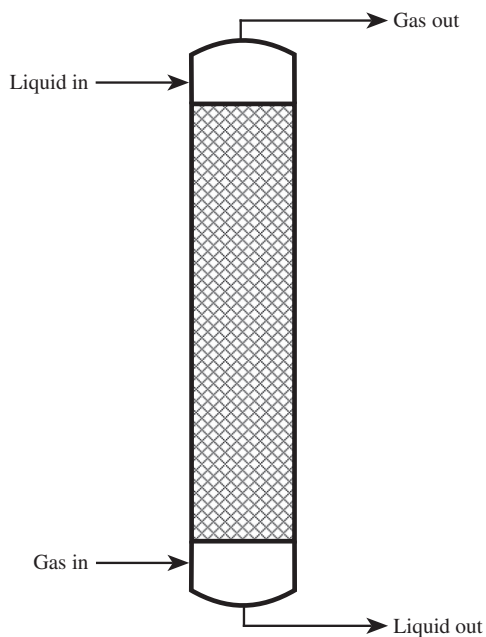


Figure 10.1: A packed gas absorber.

10.1 Henry's Law

The solubility of a gas in a liquid is determined by the equations of phase equilibrium. Consider the gaseous and liquid phases of a binary system that are in equilibrium. The condition of equilibrium states that

$$\hat{f}_i^V(T, P, y_i) = \hat{f}_i^L(T, P, x_i), \quad i = 1, 2. \quad (10.1-1)$$

Expressing the left- and right-hand sides of Eqn (10.1-1) in terms of fugacity coefficient and activity coefficient, respectively, yields

$$y_i P \hat{\phi}_i^V(T, P, y_i) = \gamma_i(T, P, x_i) x_i f_i^L(T, P). \quad (10.1-2)$$

The difficulty in using Eqn (10.1-2) comes from the fact that if the temperature of the mixture, T , is greater than the critical temperature of component i , then the pure component i exists only as a gas and it is impossible to calculate $f_i^L(T, P)$. For example, consider a rigid cylinder containing water (1) and methane (2) in equilibrium at 298 K and 1 bar. Since the critical temperature of methane is 190.6 K, methane

Table 10.1: Henry's law constants for common gases in water at 298 K.

Gas	$\mathcal{H}_i$ (atm)	Gas	$\mathcal{H}_i$ (atm)
N ₂	85.1×10^3	CO ₂	1.63×10^3
CO	58.2×10^3	H ₂ S	0.553×10^3
O ₂	42.5×10^3	SO ₂	0.461×10^2
CH ₄	39.5×10^3	NH ₃	0.988

does not exist in liquid form at the temperature of the mixture and f_2^L (298 K, 1 bar) cannot be determined.

To circumvent this problem, *Henry's law* is defined as

$$\hat{f}_i^L(T, P, x_i) = \mathcal{H}_i(T, P)x_i, \quad \text{when } x_i \rightarrow 0, \quad (10.1-3)$$

where $\mathcal{H}_i$ is called *Henry's law constant* and has the units of pressure. Henry's law "constant" is not constant but rather dependent on temperature and pressure.² It should be kept in mind that Eqn (10.1-3) is valid not only when $x_i \rightarrow 0$, but also when the species entering the liquid phase do not react or dissociate. The mole fraction of species i in the liquid phase, x_i , is generally termed as the *solubility of species i* .

Substitution of Eqn (10.1-3) into Eqn (10.1-1) gives

$$y_i P \hat{\phi}_i^V(T, P, y_i) = \mathcal{H}_i(T, P)x_i. \quad (10.1-4)$$

At low pressures, the gas phase may be considered an ideal gas mixture and Eqn (10.1-4) becomes

$$y_i P = \mathcal{H}_i x_i, \quad P \text{ is low and } x_i \ll 1 \quad (10.1-5)$$

indicating that the amount of gas dissolved in a solvent is directly proportional to the partial pressure of the gas above the solvent.

An extensive list for Henry's law constants is presented by Sander (1999). Table 10.1 gives Henry's law constants for various gases in water.

Example 10.1 Calculate the number of moles of oxygen dissolved in 1 L of water under atmospheric pressure and 298 K.

²While $\mathcal{H}_i$ is a strong function of temperature, its dependence on pressure is generally weak. Dependence of $\mathcal{H}_i$ on temperature and pressure will be discussed in detail in Section 10.2.

Solution

Air contains approximately 21 mol% O₂. Assuming that a liter of solution contains practically 1000 g of water, Eqn (10.1-5) is written as

$$(0.21)(1) = (42.5 \times 10^3) \left[\frac{n_{O_2}}{n_{O_2} + (1000/18)} \right], \quad (1)$$

where n_{O_2} is the number of moles of O₂ per liter of solution. The denominator of Eqn (1) can be approximated as

$$n_{O_2} + \frac{1000}{18} \simeq \frac{1000}{18},$$

so that the number of moles of O₂ dissolved, n_{O_2} , is given by

$$n_{O_2} = \frac{(0.21)(1000/18)}{42.5 \times 10^3} = 2.75 \times 10^{-4} \text{ mol/L.}$$

Comment: Water quality is frequently measured by dissolved oxygen amount.

Note that Henry's law constant is defined by

$$\mathcal{H}_i = \lim_{x_i \rightarrow 0} \left(\frac{\hat{f}_i^L}{x_i} \right) \quad (10.1-6)$$

indicating that $\mathcal{H}_i$ is simply a proportionality constant between the fugacity of component i and its mole fraction in the liquid phase in the limit of infinite dilution. Since both the numerator and the denominator of Eqn (10.1-6) go to zero when x_i goes to zero, the application of Hopital's rule gives

$$\mathcal{H}_i = \left(\frac{d\hat{f}_i^L}{dx_i} \right)_{x_i=0}. \quad (10.1-7)$$

Therefore, the slope of the tangent to the $\hat{f}_i^L$ versus x_i curve at $x_i = 0$ gives Henry's law constant as shown in Fig. 10.2.

In the literature, Henry's law is expressed in several different forms. Besides Eqn (10.1-5), one of the most frequently used forms is

$$c_i = \mathcal{H}_i^* P y_i. \quad (10.1-8)$$

Henry's law constant $\mathcal{H}_i^*$ has the units of (mol/L)/atm or M/atm, where M stands for molarity.³ The conversion between different units of Henry's law constant is

³The official SI unit for $\mathcal{H}_i^*$ is (mol/m³)/Pa.

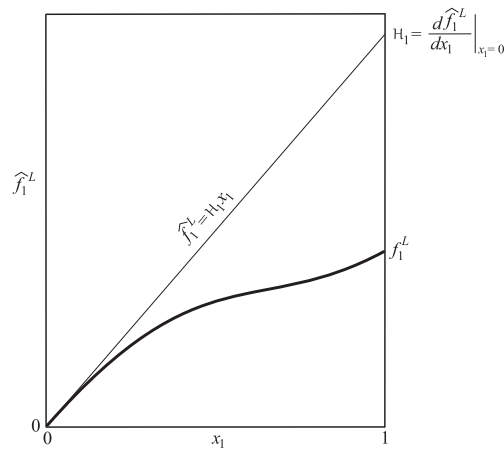


Figure 10.2: For a binary system, variation of $\hat{f}_1^L$ as a function of composition.

given as

$$\begin{aligned} \mathcal{H}_i^* \text{ (M/atm)} &= 101.325 \mathcal{H}_i^* \text{ [(mol/m}^3\text{)/Pa]}, \\ \mathcal{H}_i^* \text{ (M/atm)} &= \frac{55.3}{\mathcal{H}_i \text{ (atm)}}. \end{aligned} \quad (10.1-9)$$

Sander (1999) also provided a calculator for conversions between different units of Henry's law constant.

10.1.1 Relation Between Henry's Law and the Lewis-Randall Rule

For a binary system of components 1 and 2 at constant temperature and pressure, the use of the Gibbs-Duhem equation, Eqn (6.4-9), gives

$$x_1 \frac{d\bar{G}_1^L}{dx_1} + x_2 \frac{d\bar{G}_2^L}{dx_1} = 0 \quad (10.1-10)$$

or

$$x_1 \frac{d\bar{G}_1^L}{dx_1} - x_2 \frac{d\bar{G}_2^L}{dx_2} = 0. \quad (10.1-11)$$

The partial molar Gibbs energies are expressed by Eqn (7.3-5), i.e.

$$\bar{G}_i^L = \lambda_i(T) + RT \ln \hat{f}_i^L \Rightarrow \frac{d\bar{G}_i^L}{dx_i} = \frac{RT}{\hat{f}_i^L} \frac{d\hat{f}_i^L}{dx_i}. \quad (10.1-12)$$

Substitution of Eqn (10.1-12) into Eqn (10.1-11) and rearrangement lead to

$$\frac{\left(\frac{d\hat{f}_1^L}{dx_1}\right)}{\left(\frac{\hat{f}_1^L}{x_1}\right)} = \frac{\left(\frac{d\hat{f}_2^L}{dx_2}\right)}{\left(\frac{\hat{f}_2^L}{x_2}\right)}. \quad (10.1-13)$$

Taking the limit of Eqn (10.1-13) as $x_2 \rightarrow 0$ (or $x_1 \rightarrow 1$) gives

$$\lim_{x_1 \rightarrow 1} \frac{\left(\frac{d\hat{f}_1^L}{dx_1}\right)}{\left(\frac{\hat{f}_1^L}{x_1}\right)} = \lim_{x_2 \rightarrow 0} \frac{\left(\frac{d\hat{f}_2^L}{dx_2}\right)}{\left(\frac{\hat{f}_2^L}{x_2}\right)}. \quad (10.1-14)$$

Let us assume that component 2 obeys Henry's law, i.e. $x_1 \gg x_2$. Noting that

$$\lim_{x_1 \rightarrow 1} \left(\frac{\hat{f}_1^L}{x_1}\right) = f_1^L, \quad \lim_{x_2 \rightarrow 0} \left(\frac{d\hat{f}_2^L}{dx_2}\right) = \mathcal{H}_2, \quad \lim_{x_2 \rightarrow 0} \left(\frac{\hat{f}_2^L}{x_2}\right) = \mathcal{H}_2. \quad (10.1-15)$$

Equation (10.1-14) reduces to

$$\left(\frac{d\hat{f}_1^L}{dx_1}\right)_{x_1=1} = f_1^L, \quad (10.1-16)$$

which is nothing more than the Lewis-Randall rule, i.e. $\hat{f}_1^L = f_1^L x_1$. Hence, whenever Henry's law applies to component 2 (x_2 is very close to zero), the Lewis-Randall rule is applicable to component 1 (x_1 is close to unity). Therefore, the Lewis-Randall rule and Henry's law are not independent, but related to each other by the Gibbs-Duhem equation. In a similar way, it can be shown that when Henry's law is applicable to component 1 (x_1 is very close to zero), the Lewis-Randall rule applies to component 2 (x_2 is close to unity). Figure 10.3 shows the regions of applicability of the Lewis-Randall rule and Henry's law for components 1 and 2.

Consider an equilibrium between the vapor and liquid phases of a binary system of components 1 and 2. Let us assume that $x_1 \gg x_2$ so that Henry's law is applicable to component 2. This automatically implies that the Lewis-Randall rule applies to component 1. Therefore, the condition of equilibrium is expressed as

$$y_1 P \hat{\phi}_1^V = f_1^L x_1 = P_1^{\text{vap}} \phi_1^V(T, P_1^{\text{vap}}) \exp \left[\frac{\tilde{V}_1^L (P - P_1^{\text{vap}})}{RT} \right] x_1, \quad (10.1-17)$$

$$y_2 P \hat{\phi}_2^V = \mathcal{H}_2 x_2. \quad (10.1-18)$$

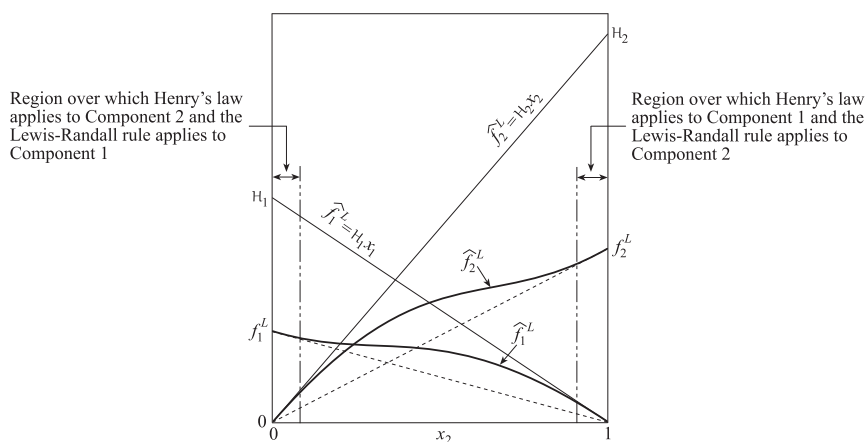


Figure 10.3: Representative variation of fugacities in a binary mixture of 1 and 2 as a function of composition.

Addition of Eqns (10.1-17) and (10.1-18) gives

$$P = \frac{P_1^{\text{vap}} \phi_1^{\text{V}}(T, P_1^{\text{vap}}) \exp[\tilde{V}_1^{\text{L}}(P - P_1^{\text{vap}})/RT]}{\hat{\phi}_1^{\text{V}}} x_1 + \frac{\mathcal{H}_2}{\hat{\phi}_2^{\text{V}}} x_2. \quad (10.1-19)$$

At low pressures, the Lewis-Randall rule leads to Raoult's law. In this case, Eqn (10.1-19) simplifies to

$$P = P_1^{\text{vap}} x_1 + \mathcal{H}_2 x_2. \quad (10.1-20)$$

Example 10.2 The following data are provided at 353 K for a binary system of components 1 and 2:

$$P_1^{\text{vap}} = 0.8 \text{ bar}, \quad P_2^{\text{vap}} = 1.5 \text{ bar}, \\ \mathcal{H}_1 = 1.53 \text{ bar}, \quad \mathcal{H}_2 = 7.38 \text{ bar}.$$

Calculate the pressure and the composition of the vapor when $x_1 = 0.99$.

Solution

Since x_1 is almost equal to unity and pressure is low, it is possible to apply Raoult's law to component 1 and Henry's law to component 2, i.e.

$$y_1 P = P_1^{\text{vap}} x_1, \quad (1)$$

$$y_2 P = \mathcal{H}_2 x_2. \quad (2)$$

Addition of Eqns (1) and (2) gives the total pressure as

$$P = P_1^{\text{vap}} x_1 + \mathcal{H}_2 x_2 = (0.8)(0.99) + (7.38)(0.01) = 0.866 \text{ bar}.$$

The composition of the vapor phase is calculated from either Eqn (1) or Eqn (2), i.e.

$$y_1 = \frac{P_1^{\text{vap}} x_1}{P} = \frac{(0.8)(0.99)}{0.866} = 0.915,$$

$$y_2 = \frac{\mathcal{H}_2 x_2}{P} = \frac{(7.38)(0.01)}{0.866} = 0.085.$$

Example 10.3 The solubility of carbon dioxide (2) in water (1) follows Henry's law and Henry's law constant at 323 K was reported as 2939 bar (Dalmolin *et al.*, 2006). The vapor is represented by the virial equation of state with the following virial coefficients:

$$B_{11} = -834.2 \text{ cm}^3/\text{mol}, \quad B_{22} = -104.2 \text{ cm}^3/\text{mol}, \quad B_{12} = -153.4 \text{ cm}^3/\text{mol}.$$

Estimate the solubility of carbon dioxide in water at 323 K and 3.65 bar. The vapor pressure of water at 323 K is 0.123 bar.

Solution

Since Henry's law applies to carbon dioxide, then the Lewis-Randall rule is applicable to water and Eqns (10.1-17) and (10.1-18) can be used for this system. Fugacity coefficients of water and carbon dioxide are calculated from Eqn (7.5-6) as

$$\begin{aligned} \hat{\phi}_1^{\text{V}} &= \exp \left\{ \frac{P}{RT} [B_{11} + y_2^2 (2B_{12} - B_{11} - B_{22})] \right\} \\ &= \exp \left(\frac{3.65}{(83.14)(323)} \{-834.2 + y_2^2 [2(-153.4) + 834.2 + 104.2]\} \right) \\ &= \exp(-0.1134 + 0.0858 y_2^2), \end{aligned} \quad (1)$$

$$\begin{aligned} \hat{\phi}_2^{\text{V}} &= \exp \left\{ \frac{P}{RT} [B_{22} + y_1^2 (2B_{12} - B_{11} - B_{22})] \right\} \\ &= \exp \left(\frac{3.65}{(83.14)(323)} \{-104.2 + y_1^2 [2(-153.4) + 834.2 + 104.2]\} \right) \\ &= \exp(-0.0142 + 0.0858 y_1^2). \end{aligned} \quad (2)$$

Fugacity coefficient of pure water under saturation condition is calculated from Eqn (5.3-5) as

$$\phi_1^{\text{V}}(T, P_1^{\text{vap}}) = \exp \left(\frac{B_{11} P_1^{\text{vap}}}{RT} \right) = \exp \left[\frac{(-834.2)(0.123)}{(83.14)(323)} \right] = 0.996. \quad (3)$$

Taking $\tilde{V}_1^L = 18 \text{ cm}^3/\text{mol}$, Eqns (10.1-17) and (10.1-18) are expressed as

$$y_1(3.65) \exp(-0.1134 + 0.0858y_2^2) = (0.123)(0.996) \\ \times \exp\left[\frac{(18)(3.65 - 0.123)}{(83.14)(323)}\right] x_1, \quad (4)$$

$$y_2(3.65) \exp(-0.0142 + 0.0858y_1^2) = (2939)x_2. \quad (5)$$

Simplification of Eqns (4) and (5) leads to

$$f_1(x_1, y_1) = y_1 \exp[-0.1134 + 0.0858(1 - y_1)^2] - 0.0336x_1 = 0, \quad (6)$$

$$f_2(x_1, y_1) = (1 - y_1) \exp(-0.0142 + 0.0858y_1^2) - 805.205(1 - x_1) = 0. \quad (7)$$

Simultaneous solution of Eqns (6) and (7) by Newton's method, as explained in Section 9.4.1, gives

$$x_1 = 0.99882 \quad \text{and} \quad y_1 = 0.03470.$$

Thus, the solubility of carbon dioxide in water at 323 K and 3.65 bar is

$$x_2 = 1 - x_1 = 1.18 \times 10^{-3}.$$

The experimental value reported by [Dalmolin *et al.* \(2006\)](#) is $x_2 = 1.28 \times 10^{-3}$.

Comment: Increased concentrations of carbon dioxide in the atmosphere lead to global warming. The purpose of “carbon dioxide sequestration” is to store CO_2 in the earth rather than releasing it into the atmosphere. For this purpose, it is proposed to dispose CO_2 on the sea floor or in sedimentary formations. Solubility of CO_2 in water has an utmost importance in the assessment of these projects.

10.1.2 Relationship Between Henry's Law Constant and Infinite Dilution Activity Coefficient

From Eqn (8.3-5)

$$\frac{\hat{f}_i^L}{x_i} = f_i^L(T, P) \gamma_i(T, P, x_i). \quad (10.1-21)$$

Substitution of Eqn (10.1-21) into Eqn (10.1-6) yields

$$\mathcal{H}_i(T, P) = \lim_{x_i \rightarrow 0} [f_i^L(T, P) \gamma_i(T, P, x_i)]. \quad (10.1-22)$$

From Eqn (5.4-8)

$$\begin{aligned}\lim_{x_i \rightarrow 0} [f_i^L(T, P)] &= \lim_{x_i \rightarrow 0} \left\{ P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\tilde{V}_i^L(P - P_i^{\text{vap}})}{RT} \right] \right\} \\ &= P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\bar{V}_i^{L\infty}(P - P_i^{\text{vap}})}{RT} \right],\end{aligned}\quad (10.1-23)$$

where $\bar{V}_i^{L\infty}$ is the partial molar volume of component i in the liquid phase at infinite dilution. On the other hand,

$$\lim_{x_i \rightarrow 0} [\gamma_i(T, P, x_i)] = \gamma_i^\infty(T, P). \quad (10.1-24)$$

Substitution of Eqns (10.1-23) and (10.1-24) into Eqn (10.1-22) leads to

$$\mathcal{H}_i = \gamma_i^\infty P_i^{\text{vap}} \phi_i^V(T, P_i^{\text{vap}}) \exp \left[\frac{\bar{V}_i^{L\infty}(P - P_i^{\text{vap}})}{RT} \right]. \quad (10.1-25)$$

At high pressures, the Poynting correction factor cannot be neglected and the estimation of $\mathcal{H}_i$ requires the value of $\bar{V}_i^{L\infty}$ to be known. In the absence of an experimental value, $\bar{V}_i^{L\infty}$ can be approximated by the molar volume of pure i , $\tilde{V}_i^L$, at its normal boiling point.

Example 10.4 The following data are provided by [Hildebrand and Scott \(1950\)](#) for the partial molar volumes of various gases in solvents at 298 K:

Solvent	$\bar{V}^{L\infty}$ (cm ³ /mol)					
	N ₂	O ₂	CO	CO ₂	SO ₂	CH ₄
Acetone	55	48	53	—	68	55
Benzene	53	46	52	—	48	52
Carbon tetrachloride	53	45	53	—	54	52
Water	40	31	36	33	—	37

The normal boiling points (NBP) of these components are given as follows:

Species	N ₂	O ₂	CO	CO ₂	SO ₂	CH ₄
NBP (K)	77.4	90.2	81.6	194.7	263.1	111.7

Calculate the molar volume of pure liquids at their normal boiling points and check the validity of approximating $\bar{V}_i^{L\infty}$ by $\tilde{V}_i^L$.

Solution

From Appendix A

Component	T_c (K)	P_c (bar)	ω
Nitrogen	126.2	34.0	0.039
Oxygen	154.6	50.4	0.025
Carbon monoxide	134.5	35.0	0.066
Carbon dioxide	304.2	73.8	0.239
Sulfur dioxide	430.3	78.8	0.256
Methane	190.6	46.1	0.011

The molar volumes of pure liquids are calculated by using the Rackett equation, Eqn (5.4-9), and are presented below:

Species	N ₂	O ₂	CO	CO ₂	SO ₂	CH ₄
$\tilde{V}^L$ (cm ³ /mol)	34.2	27.9	34.8	34.7	44.6	38.0

Note that $\tilde{V}_i^L$ values underestimate the values of $\bar{V}_i^{L\infty}$.

Example 10.5 A cylinder of propane (2) at 60 bar and 303 K contains some liquid water (1). If propane is sparingly soluble in water under these conditions, estimate Henry's law constant. The activity coefficient of propane follows the NRTL equation with $\tau_{12} = 1.156$, $\tau_{21} = 1.214$, and $\alpha = 0.3$. Propane vapor is represented by the Peng-Robinson equation of state and the vapor pressure of propane at 303 K is 10.84 bar.

Solution

From Appendix A

$$T_c = 369.9 \text{ K}, \quad P_c = 42.5 \text{ bar}, \quad \omega = 0.153.$$

Assuming $\bar{V}_2^{L\infty} \simeq \tilde{V}_2^L$, Eqn (10.1-25) takes the form

$$\mathcal{H}_2 = \gamma_2^\infty P_2^{\text{vap}} \phi_2^V(303 \text{ K}, 10.84 \text{ bar}) \exp \left[\frac{\tilde{V}_2^L(60 - 10.84)}{(8.314 \times 10^{-5})(303)} \right]. \quad (1)$$

From Eqn (8.4-32)

$$\begin{aligned} \gamma_2^\infty &= \exp [\tau_{12} + \tau_{21} \exp(-\alpha \tau_{21})] \\ &= \exp \{1.156 + 1.214 \exp[-(0.3)(1.214)]\} = 7.385. \end{aligned}$$

The use of Eqn (5.3-9) gives the fugacity coefficient of pure propane gas as

$$\phi_2^V(303 \text{ K}, 10.84 \text{ bar}) = 0.834.$$

The molar volume of liquid propane can be calculated from the modified Rackett equation, Eqn (5.4-9),

$$\tilde{V}_2^L = \frac{RT_c}{P_c} Z_{RA}^{1+(1-T_r)^{2/7}}, \quad (2)$$

where the Rackett parameter, Z_{RA} , is given by Eqn (5.4-10), i.e.

$$Z_{RA} = 0.29056 - 0.08775\omega. \quad (3)$$

The reduced temperature is

$$T_r = \frac{303}{369.9} = 0.819$$

and the Rackett parameter is calculated as

$$Z_{RA} = 0.29056 - (0.08775)(0.153) = 0.2771.$$

Therefore, Eqn (2) gives the molar volume as

$$\tilde{V}_2^L = \frac{(8.314 \times 10^{-5})(369.9)}{42.5} (0.2771)^{1+(1-0.819)^{2/7}} = 9.12 \times 10^{-5} \text{ m}^3/\text{mol}.$$

Substitution of the numerical values into Eqn (1) yields

$$\mathcal{H}_2 = (7.385)(10.84)(0.834) \exp \left[\frac{(9.12 \times 10^{-5})(60 - 10.84)}{(8.314 \times 10^{-5})(303)} \right] = 79.8 \text{ bar}.$$

At low and moderate pressures, both the fugacity coefficient of pure i and the Poynting correction factor are almost unity, and Eqn (10.1-25) simplifies to

$$\mathcal{H}_i = \gamma_i^\infty P_i^{\text{vap}}. \quad (10.1-26)$$

Example 10.6 Henry's law constant for benzene (2) in water (1) is experimentally determined at 296 K as 271 bar. Estimate the infinite dilution activity coefficient for benzene in a binary liquid mixture of benzene and water.

Solution

From Appendix C

$$\ln P_2^{\text{vap}} = 9.2806 - \frac{2788.51}{T - 52.36}.$$

At 296 K, $P_2^{\text{vap}} = 0.115$ bar. Therefore, from Eqn (10.1-26)

$$\gamma_2^\infty = \frac{\mathcal{H}_2}{P_2^{\text{vap}}} = \frac{271}{0.115} = 2357.$$

Comment: Such a high value for the infinite dilution activity coefficient indicates that benzene is practically insoluble in water. In other words, benzene and water molecules repel each other.

10.1.3 Henry's Law and Activity Coefficients

Consider a binary mixture⁴ of solvent (1) and solute (2). As long as the solute concentration is dilute, i.e. $x_2 \ll x_1$, Henry's law is applicable to the solute and the liquid phase fugacity is expressed as

$$\hat{f}_2^{\text{L}} = \mathcal{H}_2 x_2. \quad (10.1-27)$$

For concentrated mixtures, however, it is necessary to incorporate a correction factor (or activity coefficient) into Eqn (10.1-27) in the form

$$\hat{f}_2^{\text{L}} = \mathcal{H}_2 x_2 \gamma_2^*, \quad (10.1-28)$$

where the term γ_2^* is called the *asymmetric activity coefficient*. Note that γ_2^* must satisfy the following constraint:

$$\gamma_2^* \rightarrow 1, \quad x_2 \rightarrow 0. \quad (10.1-29)$$

The activity coefficient based on the pure component fugacity at the same temperature and pressure as the mixture was defined by Eqn (8.3-5) as

$$\gamma_2 = \frac{\hat{f}_2^{\text{L}}}{f_2^{\text{L}} x_2} = \frac{\text{Actual fugacity of component 2}}{\text{Fugacity of component 2 based on the Lewis-Randall rule}} \quad (10.1-30)$$

with the constraint that

$$\gamma_2 \rightarrow 1, \quad x_2 \rightarrow 1. \quad (10.1-31)$$

Rearrangement of Eqn (10.1-28) in the form

$$\gamma_2^* = \frac{\hat{f}_2^{\text{L}}}{\mathcal{H}_2 x_2} = \frac{\text{Actual fugacity of component 2}}{\text{Fugacity of component 2 based on Henry's rule}} \quad (10.1-32)$$

indicates that the standard state fugacity in the definition of the asymmetric activity coefficient is simply Henry's law constant. The standard states are shown in Fig. 10.4 for two different cases.

⁴The terms "solvent" and "solute" are used for the components present in the mixture in larger and in smaller amounts, respectively.

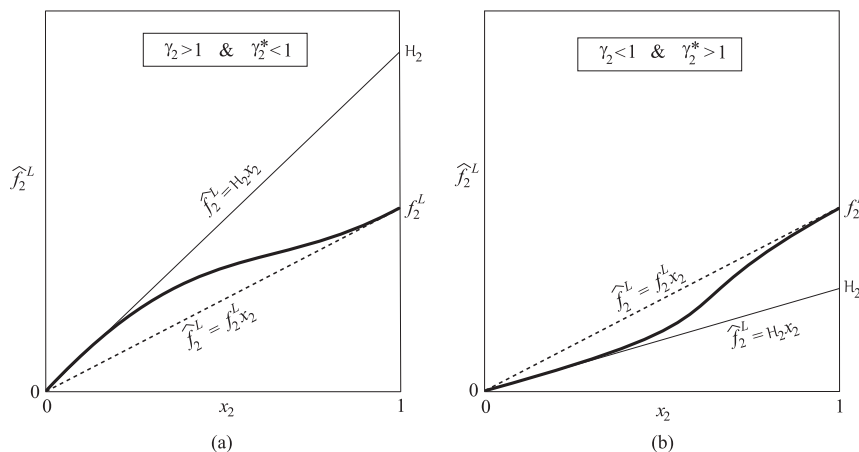


Figure 10.4: Standard states used in the definition of activity coefficients.

The relationship between γ_2^* and γ_2 can be found as follows. Comparison of Eqns (10.1-30) and (10.1-32) gives

$$\mathcal{H}_2 \gamma_2^* = f_2^L \gamma_2. \quad (10.1-33)$$

Taking the limit as $x_2 \rightarrow 0$ yields

$$\mathcal{H}_2 = f_2^L \gamma_2^\infty. \quad (10.1-34)$$

The use of Eqn (10.1-34) in Eqn (10.1-33) gives

$$\gamma_2^* = \frac{\gamma_2}{\gamma_2^\infty}. \quad (10.1-35)$$

Example 10.7 Consider a binary mixture of solvent (1) and solute (2). If the system is represented by the two-suffix Margules equation, determine the asymmetric activity coefficient, γ_2^* .

Solution

From Eqn (8.4-3)

$$\ln \gamma_2 = A x_1^2. \quad (1)$$

Taking the limit as $x_2 \rightarrow 0$ gives

$$\ln \gamma_2^\infty = A. \quad (2)$$

The use of Eqn (10.1-35) gives

$$\begin{aligned}\ln \gamma_2^* &= \ln \gamma_2 - \ln \gamma_2^\infty \\ &= Ax_1^2 - A = A(x_1^2 - 1).\end{aligned}\quad (3)$$

10.2 Factors Affecting Gas Solubility

Consider a binary mixture of solvent (1) and solute (2). The condition of equilibrium between the gas and liquid phases is expressed as

$$\hat{f}_2^V(T, P, y_2) = \hat{f}_2^L(T, P, x_2) \quad (10.2-1)$$

or

$$\hat{f}_2^V(T, P, y_2) = \mathcal{H}_2(T, P)x_2\gamma_2^*(T, P, x_2). \quad (10.2-2)$$

At a given temperature, pressure, and gas composition, the left-hand side of Eqn (10.2-2) is fixed. Therefore, there is an inverse relationship between solubility of solute, x_2 , and Henry's law constant, i.e. the higher Henry's law constant, the lower the solubility.

10.2.1 Effect of Pressure

At low pressures, the dependence of Henry's law constant on pressure may be neglected. At high pressures, however, a relationship indicating the variation of Henry's law constant with pressure is necessary. For this purpose, note that Eqn (7.7-1) gives the change in fugacity with pressure as

$$\left(\frac{\partial \ln \hat{f}_2^L}{\partial P} \right)_{T, x_j} = \frac{\bar{V}_2^L}{RT}. \quad (10.2-3)$$

Taking the limit of Eqn (10.2-3) as $x_2 \rightarrow 0$ leads to

$$\lim_{x_2 \rightarrow 0} \left(\frac{\partial \ln \hat{f}_2^L}{\partial P} \right)_{T, x_j} = \lim_{x_2 \rightarrow 0} \left(\frac{\bar{V}_2^L}{RT} \right). \quad (10.2-4)$$

Since

$$\left. \begin{aligned} \hat{f}_2^L &\rightarrow \mathcal{H}_2 x_2 \\ \bar{V}_2^L &\rightarrow \bar{V}_2^{L\infty} \end{aligned} \right\} \text{ as } x_2 \rightarrow 0, \quad (10.2-5)$$

Eqn (10.2-4) becomes

$$\left(\frac{\partial \ln \mathcal{H}_2}{\partial P} \right)_{T, x_j} = \frac{\bar{V}_2^{L\infty}}{RT}, \quad (10.2-6)$$

where $\bar{V}_2^{L\infty}$ is the partial molar volume of solute in the liquid phase at infinite dilution. Integration of Eqn (10.2-6) from the vapor pressure of solvent at the given temperature, P_1^{vap} , to any pressure P gives

$$\mathcal{H}_2 = \mathcal{H}_2(P_1^{\text{vap}}) \exp \left(\int_{P_1^{\text{vap}}}^P \frac{\bar{V}_2^{L\infty}}{RT} dP \right). \quad (10.2-7)$$

Substitution of Eqn (10.2-7) in Eqn (10.2-2) results in

$$\hat{f}_2^{\text{V}} = \mathcal{H}_2(P_1^{\text{vap}}) \exp \left(\int_{P_1^{\text{vap}}}^P \frac{\bar{V}_2^{L\infty}}{RT} dP \right) x_2 \gamma_2^*. \quad (10.2-8)$$

Krichevsky-Kasarnovsky equation: To simplify Eqn (10.2-8), Krichevsky and Kasarnovsky (1935) made the following assumptions:

- (i) $x_2 \ll 1$ so that $\gamma_2^* = 1$.
- (ii) $\bar{V}_2^{L\infty}$ is independent of pressure.

The resulting equation becomes

$$\ln \left(\frac{\hat{f}_2^{\text{V}}}{x_2} \right) = \ln \mathcal{H}_2(P_1^{\text{vap}}) + \frac{\bar{V}_2^{L\infty}}{RT} (P - P_1^{\text{vap}}), \quad (10.2-9)$$

which is known as the *Krichevsky-Kasarnovsky equation*. It predicts solubilities of sparingly soluble gases, i.e. H₂ or N₂ in water, fairly well.

When the vapor pressure of solvent is negligible, i.e. $P_1^{\text{vap}} \simeq 0$, the gas phase is almost pure solute and $\hat{f}_2^{\text{V}}$ can be substituted by f_2^{V} , fugacity of pure solute in the gas phase. Under these circumstances, Eqn (10.2-9) reduces to

$$\ln \left(\frac{f_2^{\text{V}}}{x_2} \right) = \ln \mathcal{H}_2(P_1^{\text{vap}}) + \frac{\bar{V}_2^{L\infty}}{RT} P \quad (10.2-10)$$

suggesting that when $\ln(f_2^{\text{V}}/x_2)$ is plotted as a function of P a straight line is obtained with a slope of $\bar{V}_2^{L\infty}/RT$ and an intercept of $\ln \mathcal{H}_2(P_1^{\text{vap}})$.

Example 10.8 Yuan *et al.* (2007) reported the solubilities of carbon dioxide (2) in 2-hydroxyl ethylammonium formate (1) at 313 K as a function of pressure. Using the following data, estimate Henry's law constant and partial molar volume of CO₂ at infinite dilution. Use the Soave-Redlich-Kwong equation of state to calculate fugacities.

x_2	P (bar)	x_2	P (bar)	x_2	P (bar)
0.0304	5.1	0.1099	22.2	0.2428	74.6
0.0831	16.0	0.1796	43.3	0.2437	90.9
0.1001	19.8	0.2101	58.6	0.2468	86.9

Solution

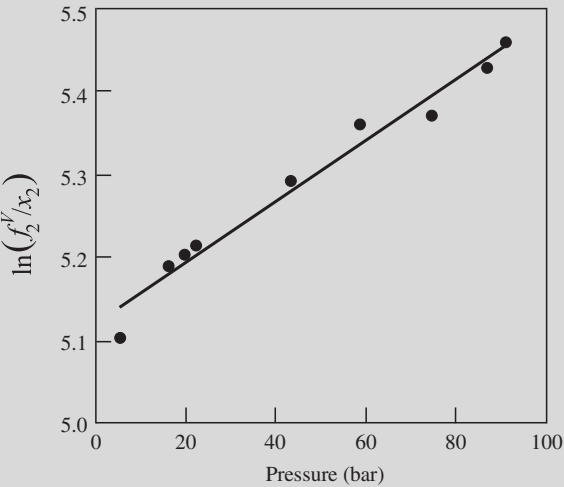
In order to plot $\ln(f_2^V/x_2)$ versus P , it is first necessary to calculate the fugacity of pure carbon dioxide using Eqn (5.3-9) at different pressures. The results are given in the following table:

x_2	P (bar)	f_2^V (bar)	x_2	P (bar)	f_2^V (bar)	x_2	P (bar)	f_2^V (bar)
0.0304	5.1	5.0	0.1099	22.2	20.2	0.2428	74.6	52.2
0.0831	16.0	14.9	0.1796	43.3	35.7	0.2437	90.9	57.2
0.1001	19.8	18.2	0.2101	58.6	44.7	0.2468	86.9	56.2

The plot of $\ln(f_2^V/x_2)$ versus P is shown below. The correlation coefficient for the best straight line passing through the data points is 0.987. The slope and the intercept of the straight line are 3.668×10^{-3} and 5.121, respectively. Thus,

$$\mathcal{H}_2 = \exp(5.121) = 167.5 \text{ bar},$$

$$\bar{V}_2^{L\infty} = (83.14)(313)(3.668 \times 10^{-3}) = 95.45 \text{ cm}^3/\text{mol}.$$



Example 10.9 *Srivatsan et al. (1998)* reported the following solubility values of methane (2) in toluene (1) at 313.2 K:

$$x_2 = 0.099 \quad \text{at 44.4 bar,}$$

$$x_2 = 0.150 \quad \text{at 70.8 bar.}$$

Estimate the solubility of methane at 55.3 bar and 313.2 K.

Solution

From the given data points, the parameters $\mathcal{H}_2(P_1^{\text{vap}})$ and $\bar{V}_2^{L\infty}$ can be calculated by using the Krichevsky-Kasarnovsky equation, Eqn (10.2-10). Using the Peng-Robinson equation of state, the fugacities of pure methane are calculated from Eqn (5.3-9) as

$$f_2^{\text{V}} = 41 \text{ bar} \quad \text{at 44.4 bar,}$$

$$f_2^{\text{V}} = 62.615 \text{ bar} \quad \text{at 70.8 bar.}$$

The use of Eqn (10.2-10) gives

$$\ln \left(\frac{41}{0.099} \right) = \ln \mathcal{H}_2(P_1^{\text{vap}}) + \frac{\bar{V}_2^{L\infty}(44.4)}{(83.14)(313.2)}, \quad (1)$$

$$\ln \left(\frac{62.615}{0.150} \right) = \ln \mathcal{H}_2(P_1^{\text{vap}}) + \frac{\bar{V}_2^{L\infty}(70.8)}{(83.14)(313.2)}. \quad (2)$$

Simultaneous solution of Eqns (1) and (2) gives

$$\ln \mathcal{H}_2(P_1^{\text{vap}}) = 6.013 \quad \text{and} \quad \bar{V}_2^{L\infty} = 7.809 \text{ cm}^3/\text{mol}.$$

Therefore, Eqn (10.2-10) takes the form

$$\ln \left(\frac{f_2^{\text{V}}}{x_2} \right) = 6.013 + \frac{7.809P}{RT}. \quad (3)$$

At $P = 55.3$ bar the fugacity of methane is 50.142 bar. Hence,

$$\ln \left(\frac{50.142}{x_2} \right) = 6.013 + \frac{(7.809)(55.3)}{(83.14)(313.2)} \Rightarrow x_2 = 0.121.$$

Comment: *Srivatsan et al. (1998)* measured the solubility of methane at 55.3 bar as 0.120.

Krichevsky-Ilinskaya equation: The assumptions used by [Krichevsky and Ilinskaya \(1945\)](#) in the simplification of Eqn (10.2-8) are as follows:

- (i) The system is represented by the two-suffix Margules equation.
- (ii) $\bar{V}_2^{L\infty}$ is independent of pressure.

For the two-suffix Margules equation, Example 10.7 indicates that $\gamma_2^* = \exp[A(x_1^2 - 1)]$. Therefore, Eqn (10.2-8) becomes

$$\ln \left(\frac{\hat{f}_2^V}{x_2} \right) = \ln \mathcal{H}_2(P_1^{\text{vap}}) + \frac{\bar{V}_2^{L\infty}}{RT} (P - P_1^{\text{vap}}) + A(x_1^2 - 1), \quad (10.2-11)$$

which is known as the *Krichevsky-Ilinskaya equation*. The use of this equation requires the evaluation of three parameters, $\mathcal{H}_2(P_1^{\text{vap}})$, $\bar{V}_2^{L\infty}$, and A .

10.2.2 Effect of Temperature

Intuitively, lower solubilities are expected as the temperature increases. From our everyday experience we notice that the amount of carbon dioxide escaping from the soda pop bottle when the bottle is opened is greater if the bottle is warm. Although solubility decreases with increasing temperature for most gases, the reverse may be true especially at very high temperatures. In general, at low temperatures Henry's law constant increases with temperature, reaches a maximum, and then decreases at higher temperatures, i.e. $T \gg T_c$. Since solubility is inversely proportional to Henry's law constant, solubility decreases with increasing temperature at low temperatures, reaches a minimum, and then increases with increasing temperature at high temperatures. As stated by [Smith and Harvey \(2007\)](#), even a variation as small as 10 K may change Henry's law constant by a factor of two, leading to serious errors in the design of process equipment.

To quantify the variation of Henry's law constant with temperature, note that Eqn (7.7-5) gives the change in fugacity with temperature as

$$\left(\frac{\partial \ln \hat{f}_2^L}{\partial T} \right)_{P, x_j} = - \frac{\bar{H}_2^L - \tilde{H}_2^{\text{IG}}}{RT^2}. \quad (10.2-12)$$

Taking the limit of Eqn (10.2-12) as $x_2 \rightarrow 0$ leads to

$$\lim_{x_2 \rightarrow 0} \left(\frac{\partial \ln \hat{f}_2^L}{\partial T} \right)_{P, x_j} = \lim_{x_2 \rightarrow 0} \left(- \frac{\bar{H}_2^L - \tilde{H}_2^{\text{IG}}}{RT^2} \right). \quad (10.2-13)$$

Since

$$\left. \begin{aligned} \hat{f}_2^L &\rightarrow \mathcal{H}_2 x_2 \\ \bar{H}_2^L &\rightarrow \bar{H}_2^{L\infty} \end{aligned} \right\} \text{ as } x_2 \rightarrow 0, \quad (10.2-14)$$

Eqn (10.2-13) becomes

$$\left(\frac{\partial \ln \mathcal{H}_2}{\partial T} \right)_{P, x_j} = - \frac{\bar{H}_2^{L\infty} - \tilde{H}_2^{\text{IG}}}{RT^2} = - \frac{\Delta \bar{H}_2^\infty}{RT^2}, \quad (10.2-15)$$

where $\bar{H}_2^{L\infty}$ is the partial molar enthalpy of solute in the liquid phase at infinite dilution and $\Delta \bar{H}_2^\infty$ is the enthalpy change associated with the transfer of one mole of a solute from an ideal gas state to an infinitely dilute solution. The term $\Delta \bar{H}_2^\infty$ can be regarded as a *heat of solution*.⁵

For small temperature differences, $\Delta \bar{H}_2^\infty$ may be considered constant. Under these circumstances, rearrangement of Eqn (10.2-15) under isobaric conditions gives

$$\frac{d \ln \mathcal{H}_2}{d(1/T)} = \frac{\Delta \bar{H}_2^\infty}{R}. \quad (10.2-16)$$

Integration of Eqn (10.2-16) gives

$$\ln \left[\frac{\mathcal{H}_2(T_2)}{\mathcal{H}_2(T_1)} \right] = \frac{\Delta \bar{H}_2^\infty}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right). \quad (10.2-17)$$

$\Delta \bar{H}_2^\infty$ can be either positive or negative, and must be measured experimentally. If it is negative, i.e. exothermic process, the solubility decreases (or $\mathcal{H}$ increases) with increasing temperature.

Besides Henry's law constants, Sander (1999) also published the values of $\Delta \bar{H}_2^\infty / R$ for various substances. In the absence of experimental information, the use of

$$\Delta \bar{H}_2^\infty \simeq -\tilde{H}_2^{\text{vap}}, \quad (10.2-18)$$

where $\tilde{H}_2^{\text{vap}}$ is the molar enthalpy of vaporization of pure solute, may provide satisfactory results.

⁵The heat effect associated with the dissolving of a solute (gas or solid) in a liquid is called a *heat of solution*. Its value is closely related to the interactions between solute and solvent. Heat of solution may be either endothermic or exothermic depending on whether more energy is used to break bonds, or more energy is released as a result of new bond formation.

Example 10.10 Indicate the conditions under which the approximation given by Eqn (10.2-18) is valid.

Solution

At low and moderate pressures Eqn (10.1-26) gives

$$\mathcal{H}_2 = \gamma_2^\infty P_2^{\text{vap}}. \quad (1)$$

Thus,

$$\frac{\mathcal{H}_2(T_2)}{\mathcal{H}_2(T_1)} = \frac{\gamma_2^\infty(T_2) P_2^{\text{vap}}(T_2)}{\gamma_2^\infty(T_1) P_2^{\text{vap}}(T_1)}. \quad (2)$$

Assuming the ratio of the infinite dilution activity coefficients to be unity, Eqn (2) can be rearranged as

$$\ln \left[\frac{\mathcal{H}_2(T_2)}{\mathcal{H}_2(T_1)} \right] = \ln \left[\frac{P_2^{\text{vap}}(T_2)}{P_2^{\text{vap}}(T_1)} \right]. \quad (3)$$

Using the Clausius-Clapeyron equation, i.e. Eqn (5.6-13), the dependence of vapor pressures on temperature is expressed as

$$\ln \left[\frac{P_2^{\text{vap}}(T_2)}{P_2^{\text{vap}}(T_1)} \right] = -\frac{\Delta \tilde{H}_2^{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right). \quad (4)$$

Substitution of Eqn (4) into Eqn (3) gives

$$\ln \left[\frac{\mathcal{H}_2(T_2)}{\mathcal{H}_2(T_1)} \right] = -\frac{\Delta \tilde{H}_2^{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right), \quad (5)$$

which is identical with Eqn (10.2-17) if $\Delta \bar{H}_2^\infty = -\Delta \tilde{H}_2^{\text{vap}}$.

10.2.3 Effect of Electrolyte Addition

Addition of a small amount of electrolyte to a solvent stabilizes the charged groups on the solute. In this way, intermolecular forces between solute and solvent becomes stronger, leading to increasing solubility of a gas in a given solvent. This phenomenon is known as *salting-in*.

At higher concentrations of electrolyte, however, solubility of a particular gas in a solvent decreases, which is known as *salting-out*. The reason for the salting-out phenomenon is the weakening of the intermolecular forces between solute and solvent as a result of the greater affinity of an electrolyte for the solvent.

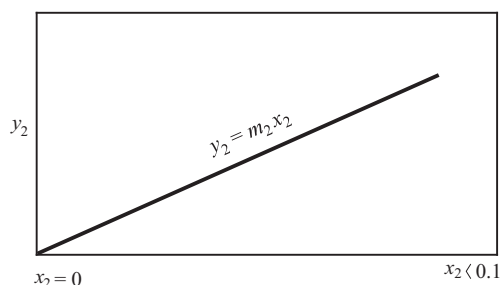


Figure 10.5: The x - y diagram for a solute.

10.2.4 Temperature and Pressure in a Gas Absorber

For a binary system of components 1 and 2, the vapor-liquid equilibrium for solute (2) gives

$$y_2 = \left(\frac{\mathcal{H}_2}{P} \right) x_2 = m_2 x_2. \quad (10.2-19)$$

The x - y diagram for a solute is given in Fig. 10.5 in which m_2 (or y_2/x_2) is the slope of the equilibrium line. During gas absorption operation, for a fixed gas composition, the purpose is to make x_2 as high as possible. This is possible by keeping m_2 as small as possible. Small values of m_2 can be achieved either by decreasing $\mathcal{H}_2$ (low temperature) or by increasing pressure. As a result, gas absorption is preferred at low temperatures and high pressures.

10.3 Applications of Henry's Law

10.3.1 Carbonated Beverages

All carbonated beverages, such as Coca Cola, Sprite, 7-Up, champagne, and sparkling wine, are bottled under pressure to increase the carbon dioxide dissolved in solution. When the bottle or can is opened, the pressure in the head space decreases. Since the liquid can no longer keep a large amount of carbon dioxide dissolved in it (in other words, the liquid is supersaturated with carbon dioxide), the solution effervesces and the carbon dioxide leaves the solution in the form of bubbles. These bubbles not only provide fizz, but also change the flavor of the drink. The CO_2 in the soda forms carbonic acid, which counteracts the sweet taste of the drink.

Example 10.11 On the average, the 330 mL soda cans have a pressure of roughly 1.2 atm when canned at 4°C, and 2.5 atm when stored at 20°C. In an unopened can, CO₂ gas dissolved in the liquid is in equilibrium with gas in the space between the can and the liquid. Once the can is opened, the pressure of CO₂ above the liquid becomes essentially zero (the partial pressure of CO₂ in the atmosphere is about 3×10^{-4} atm) and the liquid enters a metastable state, i.e. it contains CO₂ in excess of what Henry's law states. To reach a new equilibrium state corresponding to the partial pressure of CO₂ in the atmosphere, almost all the CO₂ dissolved in the liquid must escape.⁶ In other words, the soda goes “flat.” Lack of CO₂ makes “flat” soda taste much sweeter than when it is carbonated.

Calculate the concentration of CO₂ in soda (mol/L) before and after the can is opened.

Solution

Let us assume that Henry's law constant for CO₂, 1.63×10^3 atm, at 25°C is also applicable at 20°C. Choosing 1 L of solution (~1000 g of water) as a basis, the concentrations are:

Closed can:

$$2.5 = 1.63 \times 10^3 \left[\frac{n_{\text{CO}_2}}{n_{\text{CO}_2} + (1000/18)} \right] \Rightarrow n_{\text{CO}_2} = 0.085 \text{ mol/L.}$$

Open can:

$$3 \times 10^{-4} = 1.63 \times 10^3 \left[\frac{n_{\text{CO}_2}}{n_{\text{CO}_2} + (1000/18)} \right] \\ \Rightarrow n_{\text{CO}_2} = 1.02 \times 10^{-5} \text{ mol/L.}$$

Comment: Boiled water also tastes “flat” because all of the oxygen gas has been removed by heating. This is the reason why people do not prefer to prepare tea with water boiled for extended periods. Similarly, addition of boiled water to an aquarium will cause the fish to suffocate.

10.3.2 The Bends

The hydrostatic pressure in the sea increases almost linearly with depth according to the relation

$$P = P_{\text{atm}} + \rho gh, \quad (10.3-1)$$

⁶It takes a finite amount of time for the system to reach this new equilibrium state.

Table 10.2: Increase in nitrogen solubility in blood as a function of depth.

Depth (m)	P (atm)	p_{N_2} (atm)	mol N_2 /L blood
0	1	0.780	5.092×10^{-4}
2	1.198	0.935	6.102×10^{-4}
4	1.397	1.089	7.111×10^{-4}
6	1.595	1.244	8.121×10^{-4}
8	1.793	1.399	9.130×10^{-4}
10	1.991	1.553	1.014×10^{-3}
12	2.190	1.708	1.115×10^{-3}
14	2.388	1.863	1.216×10^{-3}
16	2.586	2.017	1.317×10^{-3}
18	2.784	2.172	1.418×10^{-3}
20	2.983	2.327	1.519×10^{-3}

where h is the distance measured from sea level. The density of saltwater, ρ , is about 1025 kg/m^3 . Therefore, the pressure at a depth of 40 m is⁷

$$P = 1 + (1025)(9.8)(40) \underbrace{(9.8692 \times 10^{-6})}_{\text{atm}/(\text{kg m}^{-1} \text{ s}^{-2})} = 4.97 \text{ atm.}$$

The muscles we use to expand and contract our lungs during breathing are not strong enough to overcome such a pressure.⁸ One way of overcoming this problem is to protect the diver's body from ambient pressure by placing him/her in a deep sea submersible in which pressure is maintained at 1 atm. Another way of solving this problem is to provide a pressurized breathing gas mixture to the diver. If the breathing gas supply is delivered at the same pressure as the surrounding ambient pressure, the diver's lungs do not have to work against the water pressure. Therefore, as scuba divers dive deeper, the pressure of air they breathe must match the pressure of the water around them. This increase in air pressure, however, increases the solubility of nitrogen in the blood as shown in Table 10.2. In the calculations, density of seawater is taken as 1025 kg/m^3 , blood properties are considered to be equal to those of water and the variation of Henry's law constant with temperature is considered negligible.

As a result of breathing compressed air and being subjected to high pressures caused by water depth, the amount of nitrogen dissolved in blood and other tissues increases. If the diver swims quickly to the surface, the nitrogen forms bubbles in

⁷As a rule of thumb, keep in mind that 10 m of water height creates a pressure of 1 atm.

⁸This is the reason why we cannot simply use a snorkel of the necessary length for diving.

the blood as it becomes less soluble due to a decrease in pressure. These bubbles, by blocking the capillaries in the body and stopping the blood flow, produce the symptoms of *decompression sickness* (numbness, difficulty in breathing, itching, pain in the joints or abdomen), which is also known as *the bends*. The bends is very painful and is sometimes fatal.

To avoid the effects of the bends, the diver must rise slowly and/or make intermittent stops on the way up, called decompression stops, so that the gas can come out of solution slowly.

10.3.3 Natural Disaster at Lake Nyos⁹

On August 26, 1986, officials found 1746 people and several thousand livestock lying dead mostly within 3 km of Lake Nyos, a remote lake in the highlands of northwest Cameroon. Lake Nyos rests in a volcanic crater called a maar. Maars are inverted volcanoes that form as a result of a series of powerful explosions that occur when magma ($T \approx 1400$ K) encounters cooler groundwater. The resulting steam explosions excavate a crater.

Initial reports from local authorities suggested that the tragedy was the result of a volcanic eruption. Investigators arriving at the lake several days after the incident were puzzled by the lack of evidence for hot, caustic gases that normally accompany volcanic eruptions; plants on the lake perimeter were uncharred and apparently unharmed. Later, scientists discovered that water at the bottom of the lake (maximum depth = 208 m) contained high concentrations of dissolved CO_2 gas. The amount of CO_2 dissolved in water was estimated to be $300 \times 10^6 \text{ m}^3$.

Apparently, CO_2 released from the volcano into the lake was dissolved in the lower layers of the lake by the high pressure of the water above. Application of Henry's law indicates that 1 L of water at 40 m holds 2–3 L of CO_2 , at 190 m up to 11 L of CO_2 . Scientists speculated that a landslide into the lake stirred up the waters. As the CO_2 -rich waters moved to the surface of the lake, where hydrostatic pressure was less, CO_2 solubility decreased resulting in the release of the gas, which erupted explosively from the lake surface. At Lake Nyos, the volume of the gas vented to the atmosphere was sufficient to cause a 1 m drop in lake level (the surface area of the lake is 1.49 km^2).

The research team estimated that the volume of CO_2 released was approximately equal to 1 km^3 , or about 2×10^6 tons. The exit velocity of the gas was estimated as 80 m/s. Since CO_2 (molecular weight = 44) is heavier than air (molecular weight = 29), the released gas did not simply rise into the atmosphere but formed a

⁹For more information see: <http://www.ched-ccce.org/confchem/1998/donnelly>.

“river” of CO₂ about 50 m thick that swept through the village nearby Lake Nyos, suffocating human and animal life.

To prevent the occurrence of a similar disaster, scientists have been degassing Lake Nyos since April 2001.¹⁰ For this purpose, a 205 m long polyethylene pipe (ID = 315 mm, OD = 258 mm) was placed vertically into the lake and some water was pumped up through the pipe to start the process. As the water moves up in the pipe, pressure decreases and CO₂ bubbles start to form within the liquid phase. The difference in densities of liquid and gas bubbles is large enough to keep the process going, i.e. once the movement of the bottom layers is initiated, no pumping is necessary. The resulting fountain shoots 6–10 m in the air.

10.3.4 Knuckle Cracking

Have you ever wondered where the noise comes from when you crack your knuckles? Joints are where two bones meet and they are covered by a capsule filled with synovial fluid, which is responsible for the lubrication of the joint. Gases, such as O₂, N₂, and CO₂, remain dissolved in this fluid as long as the pressure in the joint is fairly constant. When the joint is stretched, i.e. the volume of the joint is increased, the pressure in the capsule decreases, leading to the formation of gas bubbles. The formation of this bubble creates a popping sound, known as “cracking.” It takes about 25–30 min for the gas to be dissolved in the synovial fluid again. This is why a joint cannot be “re-cracked” immediately.

Problems

Problems Related to Section 10.1

10.1 A gaseous species (component 2) dissolves in liquid (component 1) to form an ideal mixture of composition $x_2 = 0.015$. If the normal boiling point of this mixture is 337 K, estimate Henry’s law constant of component 2. The vapor pressures of components 1 and 2 at 337 K are 0.98 bar and 0.35 bar, respectively.

(Answer: 3.18 bar)

10.2 Henry’s law constants of sulfur dioxide in different organic solvents at 298 K are given in the following table. Which solvent would you choose to absorb SO₂?

¹⁰See <http://mhalb.pagesperso-orange.fr/nyos/2006/index2006.htm>.

Solvent	$\mathcal{H}$ (kPa)
Tetrahydrofuran	39.2
1,2-Dichloroethane	428.0
Pyridine	8.8
Tetramethylsulfone	101.0

- 10.3** A binary system of species 1 and 2 consists of vapor and liquid phases in equilibrium at 350 K and 0.9 bar. The liquid phase is concentrated in species 2, i.e. $x_2 \gg x_1$. Estimate the compositions of the liquid and vapor phases using the following data:

$$P_1^{\text{vap}} = 0.324 \text{ bar}, \quad P_2^{\text{vap}} = 0.199 \text{ bar}, \\ \mathcal{H}_1 = 130 \text{ bar}, \quad \mathcal{H}_2 = 70 \text{ bar}.$$

(Answer: $x_1 = 0.0054$ and $y_1 = 0.78$)

- 10.4** A storage tank contains a mixture of *n*-hexane (1) and methane (2) at 310.9 K and 10.8 bar. The mole fractions of methane in the liquid and vapor phases are reported as

$$x_2 = 0.0496, \quad y_2 = 0.9613.$$

Estimate Henry's law constant of methane in *n*-hexane. The vapor phase is represented by the Peng-Robinson equation of state with $k_{12} = 0.04$.

(Answer: 205 bar)

- 10.5** When the Poynting correction factor is almost unity, Eqn (10.1-25) reduces to

$$\mathcal{H}_i = \gamma_i^\infty P_i^{\text{vap}} \phi_i^{\text{V}}(T, P_i^{\text{vap}}). \quad (1)$$

Under atmospheric pressure, Miyano (2004) reported the infinite dilution activity coefficients of *n*-butane (2) in isobutanol (1) at different temperatures as

T (K)	250	270	290
γ_2^∞	4.03	4.00	3.97

Estimate Henry's law constant for *n*-butane in isobutanol using Eqn (1). Use the virial equation of state to determine fugacity coefficients of pure *n*-butane.

(Answer: 1.545 bar, 3.524 bar, 6.993 bar)

- 10.6** From Eqn (10.1-6)

$$\mathcal{H}_i = \lim_{x_i \rightarrow 0} \left(\frac{\hat{f}_i^{\text{L}}}{x_i} \right) = \lim_{x_i \rightarrow 0} (P \hat{\phi}_i^{\text{L}}). \quad (1)$$

- (a) For a binary system of species 1 and 2, if the fugacity coefficient is calculated by the Redlich-Kwong equation of state, show that

$$\ln \mathcal{H}_1 = \ln P + \frac{B_1}{B_2}(Z_2^L - 1) - \ln(Z_2^L - B_2) - \frac{A_{22}}{B_2} \left[2(1 - k_{12}) \sqrt{\frac{A_{11}}{A_{22}}} - \frac{B_1}{B_2} \right] \ln \left(1 + \frac{B_2}{Z_2^L} \right). \quad (2)$$

- (b) Combine Eqn (2) with Eqn (3) of Problem 8.27 to obtain

$$\ln \mathcal{H}_1 = \ln \gamma_1^\infty + \ln P + Z_1^L - 1 - \ln(Z_1^L - B_1) - \frac{A_{11}}{B_1} \ln \left(1 + \frac{B_1}{Z_1^L} \right) \quad (3)$$

- (c) For a binary system of *n*-butane (1) and methanol (2) at 374 K and 1.013 bar, Miyano *et al.* (2006) reported that $\gamma_1^\infty = 9.3$. Use Eqn (3) to estimate Henry's law constant of *n*-butane in methanol.

(Answer: (c) 117.6 bar)

- 10.7** A system of water (1) and methane (2) exists as two phases at 298 K and 23.23 atm. Rigby and Prausnitz (1968) reported the water mole fraction in the vapor phase as 1.483×10^{-3} . The vapor phase is represented by the virial equation of state with the following parameters:

$$B_{11} = -1165 \text{ cm}^3/\text{mol}, \quad B_{22} = -43.4 \text{ cm}^3/\text{mol}.$$

- (a) Estimate the liquid phase composition.
 (b) Determine the second virial cross coefficient, B_{12} . The vapor pressure of water at 298 K is 0.03126 atm.

(Answer: (a) $x_2 = 5.64 \times 10^{-4}$ and (b) $-64.9 \text{ cm}^3/\text{mol}$)

- 10.8** A binary liquid mixture of 28 mol% benzene (1) and 72% *n*-heptane (2) is at a temperature of 318.15 K. Estimate the composition of the vapor phase in equilibrium with this liquid mixture if the system is represented by the three-suffix Margules equation.

Data: At 318.15 K

- Henry's law constant for benzene in *n*-heptane = 0.522 bar.
- Henry's law constant for *n*-heptane in benzene = 0.285 bar.

(Answer: $y_1 = 0.495$)

Problems Related to Section 10.2

- 10.9** The release of excessive waste heat into rivers and lakes leads to an increase in water temperature, known as “thermal pollution.” Discuss the probable harmful effects of thermal pollution on the environment.
- 10.10** Hyperbaric oxygen therapy consists of breathing 100% oxygen under increased pressure. It is especially applied to patients suffering from carbon monoxide poisoning. Why?
- 10.11** The following data are provided by O’Sullivan and Smith (1970) for the solubility of nitrogen and methane in water (1) at 324.5 K:

P (bar)	$x_2 \times 10^4$	
	Nitrogen	Methane
101.3	7.99	14.27
202.6	14.54	22.79
303.9	20.17	28.70
405.2	24.90	33.40
506.5	29.20	37.30
607.8	33.50	40.90

Use Eqn (10.2-10) and estimate $\mathcal{H}_2(P_1^{\text{vap}})$ and $\bar{V}_2^{L\infty}$ for nitrogen and methane. Use the Soave-Redlich-Kwong equation of state to calculate pure component fugacities.

(Answer: N_2 : $\mathcal{H}_2 = 1.139 \times 10^5$ bar, $\bar{V}_2^{L\infty} = 34.847$ cm³/mol; CH_4 : $\mathcal{H}_2 = 5.554 \times 10^4$ bar, $\bar{V}_2^{L\infty} = 39.37$ cm³/mol).

- 10.12** Henry’s law coefficients for propane in heavy hydrocarbons are reported by Sciamanna and Lynn (1988) as

T (°C)	$\mathcal{H}$ (kPa)	
	Dodecane	Hexadecane
25	816	775
35	985	926
45	1175	1096

Estimate the heat of solution of propane in dodecane and hexadecane.

(Answer: $-14,367$ J/mol, $-13,652$ J/mol)

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Chapter 11

Liquid-Liquid Equilibrium

When two liquids are mixed, depending on the temperature and/or composition, either they are completely miscible in each other and form a single phase, or they are partially miscible (or totally immiscible) in each other and form two separate phases. In the case of completely miscible, i.e. single phase, systems, the equations developed in Chapter 6 are applicable. In this chapter, we are interested in the case when a liquid is partially miscible (or totally immiscible) in another liquid. First, the reason for making liquids partially miscible in each other will be investigated. Then the procedure to determine the compositions in two separate liquid phases that are in equilibrium with each other will be described. Such equilibrium data are needed in various chemical and physical processes such as liquid-liquid extraction and tertiary oil recovery.

11.1 Mathematical Preliminaries

A function $f(x)$ is said to be a *concave function* in x if and only if the points on a chord connecting two points on a function lie beneath the function. On the other hand, a function $f(x)$ is said to be a *convex function* in x if and only if the points on a chord connecting two points on a function lie above the function. Concave and convex functions are shown in Fig. 11.1.

Consider a function $f(x)$ that is dependent on a single variable x . Various possibilities exist for the variation of $f(x)$ with x as shown in Fig. 11.2. Note that the functions on the left (a and c) are convex, while the functions on the right (b and d) are concave.

The first derivative of $f(x)$, $df(x)/dx$, measures the rate of change of the function $f(x)$. The second derivative of $f(x)$, $d^2f(x)/dx^2$, measures the rate of change of the first derivative, $df(x)/dx$. The signs of the first and second

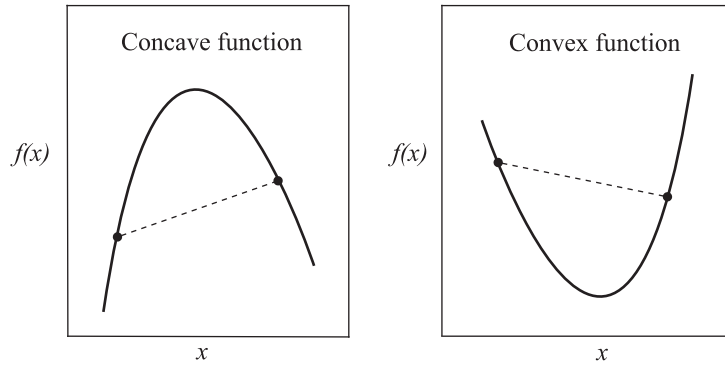


Figure 11.1: Concave and convex functions.

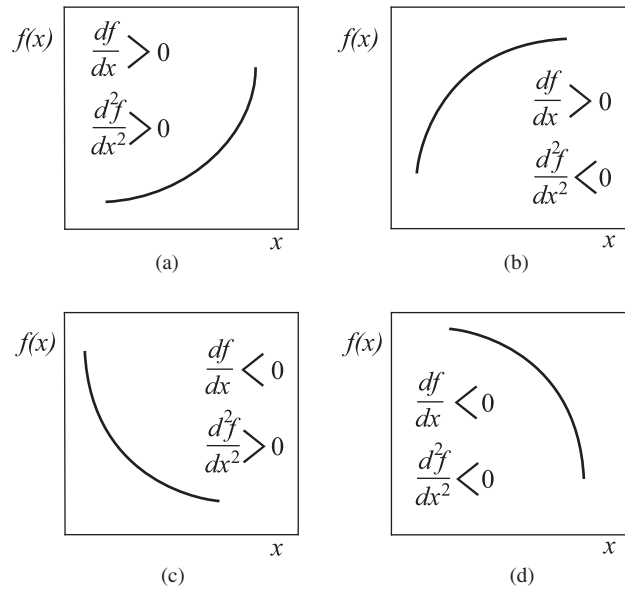


Figure 11.2: Variation of the first and second derivatives of convex and concave functions.

derivatives of $f(x)$ imply the following:

$$\frac{df(x)}{dx} \begin{cases} > 0, & \text{Value of } f(x) \text{ increases with increasing } x, \\ < 0, & \text{Value of } f(x) \text{ decreases with increasing } x, \end{cases} \quad (11.1-1)$$

$$\frac{d^2f(x)}{dx^2} \begin{cases} > 0, & \text{Slope of } f(x) \text{ versus } x \text{ tends to increase,} \\ < 0, & \text{Slope of } f(x) \text{ versus } x \text{ tends to decrease.} \end{cases} \quad (11.1-2)$$

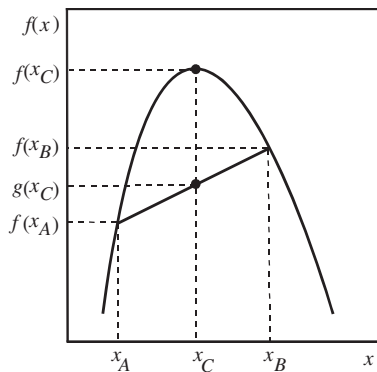


Figure 11.3: A concave function.

Note that the second derivative can be used to determine whether a function is concave or convex. The second derivatives of concave and convex functions are negative and positive, respectively.

Let us consider a concave function of a single variable, $f(x)$, and draw a chord joining any two points on the function as shown in Fig. 11.3. The chord intersects the function at x_A and x_B . The equation of the chord is

$$g(x) = \frac{f(x_B) - f(x_A)}{x_B - x_A}(x - x_A) + f(x_A). \quad (11.1-3)$$

Let x_C be any value between x_A and x_B such that

$$x_C = \lambda x_A + (1 - \lambda)x_B, \quad (11.1-4)$$

where $0 \leq \lambda \leq 1$. From Fig. 11.3 it is apparent that

$$g(x_C) < f(x_C). \quad (11.1-5)$$

When $x = x_C = \lambda x_A + (1 - \lambda)x_B$, from Eqn (11.1-3) we have

$$g(x_C) = \lambda f(x_A) + (1 - \lambda)f(x_B). \quad (11.1-6)$$

Substitution of Eqns (11.1-4) and (11.1-6) into Eqn (11.1-5) gives

$$\lambda f(x_A) + (1 - \lambda)f(x_B) < f[\lambda x_A + (1 - \lambda)x_B]. \quad (11.1-7)$$

11.2 Stability of Liquid Mixtures

For thermodynamically allowable changes at constant temperature and pressure, the criterion

$$(dG)_{T,P} \leq 0 \quad (11.2-1)$$

indicates that all irreversible changes taking place at constant temperature and pressure must decrease the total Gibbs energy of the system. When the system reaches equilibrium, no further changes can occur and Gibbs energy reaches its minimum value.

Let us consider the mixing of two liquids, 1 and 2, at constant temperature and pressure. The Gibbs energy of this binary mixture is

$$G_{\text{mix}} = n_1 \tilde{G}_1 + n_2 \tilde{G}_2 + \Delta G_{\text{mix}}. \quad (11.2-2)$$

If the Gibbs energy of the mixture is lower than the summation of the Gibbs energies of pure liquids, then liquids 1 and 2 will be miscible (totally or partially) in each other. Otherwise, liquids 1 and 2 tend to remain unmixed like olive oil and water. The condition of miscibility is mathematically expressed as

$$G_{\text{mix}} < n_1 \tilde{G}_1 + n_2 \tilde{G}_2 \quad (11.2-3)$$

or, in other words,

$$\Delta G_{\text{mix}} < 0, \quad (11.2-4)$$

which is the criterion for miscibility¹ (partial or complete).

The molar Gibbs energy change on mixing is given by

$$\Delta \tilde{G}_{\text{mix}} = \underbrace{\Delta \tilde{H}_{\text{mix}}}_X - \underbrace{T \Delta \tilde{S}_{\text{mix}}}_Y. \quad (11.2-5)$$

Since mixing increases the degree of disorder within the system, the entropy change on mixing, $\Delta \tilde{S}_{\text{mix}}$, is always positive. Thus, the term Y in Eqn (11.2-5) is always positive. The heat of mixing, $\Delta \tilde{H}_{\text{mix}}$, is related to the energetic effects. In order to make a mixture of 1 and 2 from pure components, it is necessary to break 1–1 and 2–2 bonds and form 1–2 bonds.² Depending on the magnitude of the interactions between like and unlike molecules, $\Delta \tilde{H}_{\text{mix}}$ may take positive or negative values. Therefore, it is necessary to consider the following two cases:

¹The criterion given by Eqn (11.2-4) is a necessary but not sufficient condition for *complete* miscibility of components. As will be shown later, formation of two (or more) phases may take place depending on the variation of ΔG_{mix} with composition.

²As stated in Section 6.3, while bond breaking is an endothermic process, bond formation is exothermic.

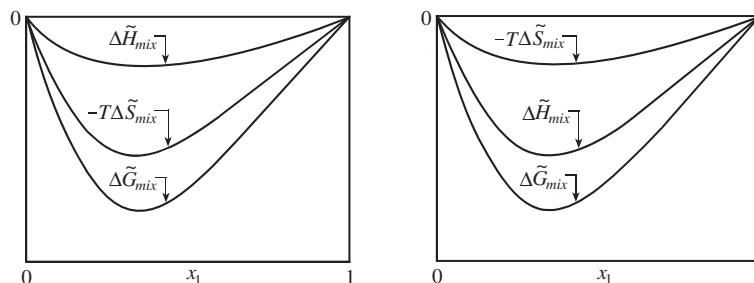


Figure 11.4: Effects of $\Delta\tilde{S}_{\text{mix}}$ and $\Delta\tilde{H}_{\text{mix}}$ on $\Delta\tilde{G}_{\text{mix}}$ when $\Delta\tilde{H}_{\text{mix}} < 0$.

Exothermic mixing ($\Delta\tilde{H}_{\text{mix}} < 0$): When $\Delta\tilde{H}_{\text{mix}} < 0$, interactions between unlike molecules are greater than those between like molecules, i.e. negative deviation from Raoult's law leading to $\gamma_i < 1$. In this case, $\Delta\tilde{G}_{\text{mix}}$ is negative for all compositions of components 1 and 2. Therefore, liquids are completely miscible in each other and form a stable single phase. This situation is illustrated in Fig. 11.4.

Endothermic mixing ($\Delta\tilde{H}_{\text{mix}} > 0$): When $\Delta\tilde{H}_{\text{mix}} > 0$, interactions between like molecules are stronger than those between unlike molecules, i.e. positive deviation from Raoult's law leading to $\gamma_i > 1$. In this case, examination of Eqn (11.2-5) indicates that the terms X and Y compete with each other and the magnitude of the temperature decides the dominant term and, hence, the sign of $\Delta\tilde{G}_{\text{mix}}$:

- At low temperatures, $\Delta\tilde{H}_{\text{mix}} > T\Delta\tilde{S}_{\text{mix}}$ and, as shown in Fig. 11.5a, this indicates that $\Delta\tilde{G}_{\text{mix}} > 0$. Therefore, liquids 1 and 2 tend to remain as two separate phases throughout the whole composition range, i.e. liquids 1 and 2 are immiscible in each other.
- When temperature is high enough so that $\Delta\tilde{G}_{\text{mix}} < 0$, i.e. $T\Delta\tilde{S}_{\text{mix}} > \Delta\tilde{H}_{\text{mix}}$, we may encounter two different situations. When $\Delta\tilde{G}_{\text{mix}}$ is a convex function of composition with only one minimum as shown in Fig. 11.5b, components are miscible in each other over the entire composition range and form a homogeneous phase.

An interesting case may arise when $\Delta\tilde{G}_{\text{mix}}$ exhibits more than one minimum as shown in Fig. 11.6. Let us tackle this case by drawing a common tangent line to the minima.³ The so-called *common tangent rule* states that when the composition of the mixture is between x_1^α and x_1^β , instead of having a homogeneous mixture, we have two separate phases, α and β , that are in equilibrium with each other. The

³Keep in mind that drawing a common tangent to the minima does not imply joining the minimum points by a straight line.

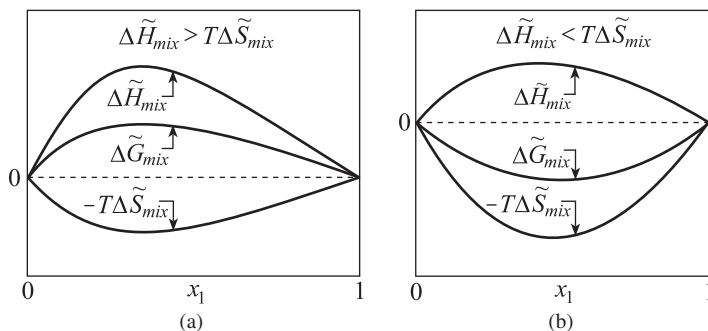


Figure 11.5: Effects of $\Delta\tilde{S}_{mix}$ and $\Delta\tilde{H}_{mix}$ on $\Delta\tilde{G}_{mix}$ when $\Delta\tilde{H}_{mix} > 0$.

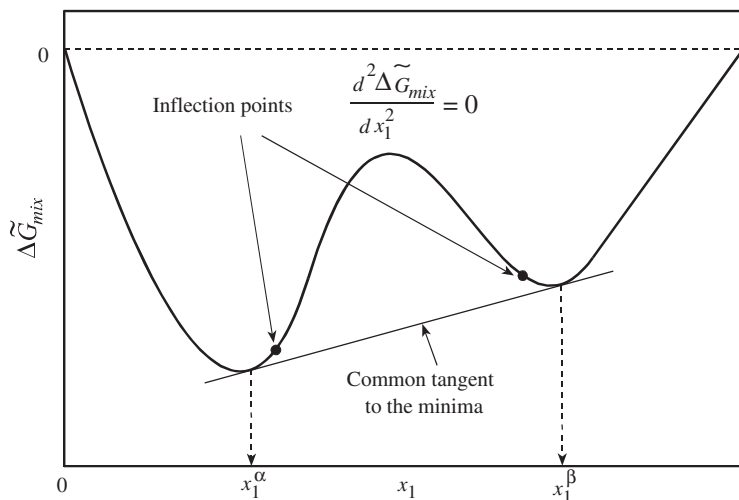


Figure 11.6: Common tangent rule.

compositions of these coexisting equilibrium phases lie at the points of co-tangency, i.e. x_1^α and x_1^β . The coexisting compositions are called *binodal points*. The reason for the phase separation can be expressed as follows.

Let x_1^* be the mole fraction of component 1 between x_1^α and x_1^β , i.e.

$$x_1^* = \lambda x_1^\alpha + (1 - \lambda)x_1^\beta, \quad (11.2-6)$$

where $0 \leq \lambda \leq 1$. The solution of Eqn (11.2-6) for λ gives

$$\lambda = \frac{x_1^\beta - x_1^*}{x_1^\beta - x_1^\alpha}. \quad (11.2-7)$$

From the lever rule (or mass balance)

$$\frac{x_1^\beta - x_1^*}{x_1^\beta - x_1^\alpha} = \frac{n_\alpha}{n_\alpha + n_\beta}. \quad (11.2-8)$$

Comparison of Eqns (11.2-7) and (11.2-8) reveals that the term λ represents the mole fraction of the α -phase.

According to Eqn (11.1-7)

$$\lambda \Delta \tilde{G}_{\text{mix}}(x_1^\alpha) + (1 - \lambda) \Delta \tilde{G}_{\text{mix}}(x_1^\beta) < \Delta \tilde{G}_{\text{mix}} \left[\lambda x_1^\alpha + (1 - \lambda) x_1^\beta \right], \quad (11.2-9)$$

which states that the combined Gibbs energies of the two separate liquid phases with compositions x_1^α and x_1^β are lower than the Gibbs energy of a homogeneous mixture with composition x_1^* . Since the system tries to minimize its Gibbs energy, over this composition range we have two separate phases of compositions x_1^α and x_1^β , and the common tangent acts as a “tie line.”

When $x_1^\alpha < x_1 < x_1^\beta$, there are two inflection points at which $d^2 \Delta \tilde{G}_{\text{mix}} / dx_1^2 = 0$. The inflection points are called *spinodal points*. Between the two inflection points, $\Delta \tilde{G}_{\text{mix}}$ is a concave function with $d^2 \Delta \tilde{G}_{\text{mix}} / dx_1^2 < 0$.

Figure 11.7 shows the variation of $\Delta \tilde{G}_{\text{mix}}$ as a function of composition at a fixed temperature, say T^* . At this temperature, it is possible to locate binodal and spinodal compositions on a temperature-composition (or solubility) diagram, as shown in Fig. 11.7. If $\Delta \tilde{G}_{\text{mix}}$ versus x_1 data are known at various temperatures, then it is possible to draw binodal and spinodal curves. The binodal and spinodal curves coincide at the *critical solution temperature* (CST) or *consolute temperature*. In this particular case, mutual solubility of components increases with increasing temperature. Thus, the resulting critical solution temperature is called the *upper critical solution temperature* (UCST). When the temperature is above the UCST, the mixture is completely miscible and forms a homogeneous phase. At temperatures less than UCST, the mixture is partially miscible and the binodal curve is the boundary between the two-phase and one-phase regions. Within the region between the spinodal and binodal curves, the mixture is metastable, and the time it takes for the phase separation is not definite. On the other hand, within the spinodal curve, the mixture is unstable and phase separation takes place immediately.

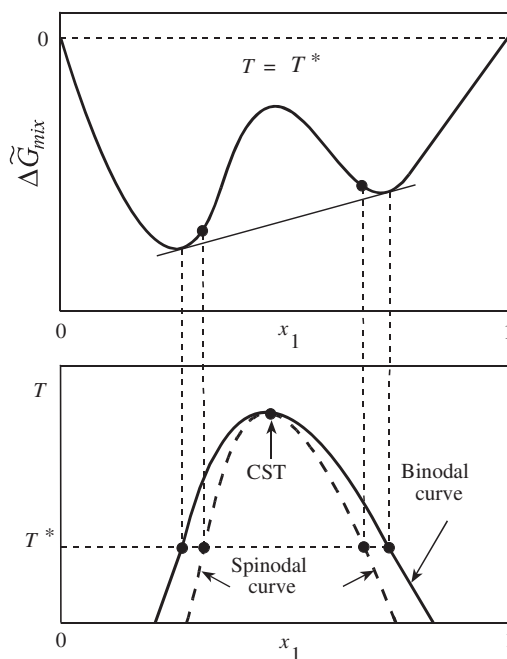


Figure 11.7: Construction of a liquid-liquid solubility diagram.

Figure 11.8 shows solubility diagrams for various binary mixtures. Although solubility diagrams of most binary mixtures exhibit a UCST (Fig. 11.8a), it is also possible to have solubility diagrams with *lower critical solution temperature* (LCST) as shown in Fig. 11.8b. In this case, the two liquids are miscible in all proportions at temperatures lower than the LCST. Binary mixtures exhibiting a closed-loop behavior have both upper and lower critical solution temperatures, as shown in Fig. 11.8c. In this case, the mixture is completely miscible at temperatures above the UCST and below the LCST.

Since the end points of the spinodal curve represent inflection points, then the condition of instability of liquid mixtures can be stated as

$$\frac{d^2 \Delta \tilde{G}_{mix}}{dx_1^2} < 0, \quad \text{Condition of instability,} \quad (11.2-10)$$

where

$$\frac{\Delta \tilde{G}_{mix}}{RT} = \frac{\Delta \tilde{G}_{mix}^{IM}}{RT} + \frac{\tilde{G}^{ex}}{RT} = \sum_{i=1}^N x_i \ln x_i + \frac{\tilde{G}^{ex}}{RT}. \quad (11.2-11)$$

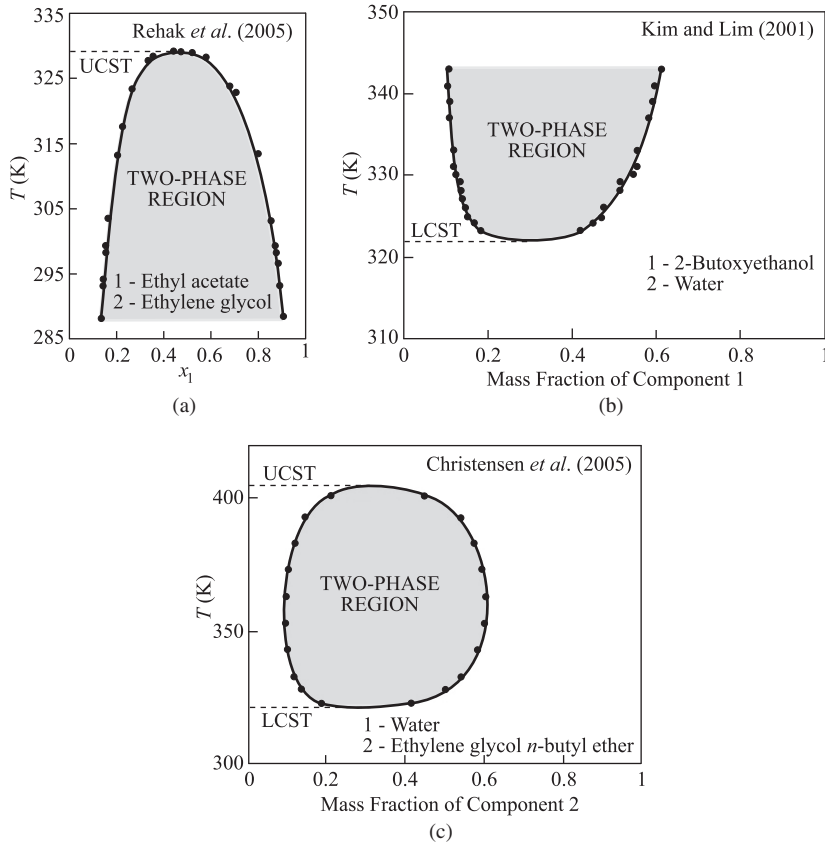


Figure 11.8: Solubility diagrams for binary mixtures.

For a binary system, substitution of Eqn (11.2-11) into Eqn (11.2-10) gives

$$\frac{1}{x_1 x_2} + \frac{d^2(\tilde{G}^{\text{ex}}/RT)}{dx_1^2} < 0. \quad (11.2-12)$$

Therefore, the concentrations at the spinodal points are obtained from the solution of the following equation:

$$\frac{1}{x_1 x_2} + \frac{d^2(\tilde{G}^{\text{ex}}/RT)}{dx_1^2} = 0. \quad (11.2-13)$$

Example 11.1 In a binary liquid mixture of 1 and 2 at constant temperature and pressure, the molar excess Gibbs energy is expressed as

$$\frac{\tilde{G}^{\text{ex}}}{RT} = x_1 x_2 [1.8 + 0.3(x_1 - x_2) - 0.85(x_1 - x_2)^2].$$

Determine the region of instability.

Solution

Substitution of $\tilde{G}^{\text{ex}}/RT$ into Eqn (11.2-13) and differentiation give

$$408x_1^4 - 852x_1^3 + 511x_1^2 - 67x_1 - 10 = 0. \quad (1)$$

The solution of Eqn (1) by MATHCAD[®] gives $x_1 = -8.54 \times 10^{-2}$, $x_1 = 0.378$, $x_1 = 0.682$, and $x_1 = 1.114$. Therefore, the limits of instability are given as

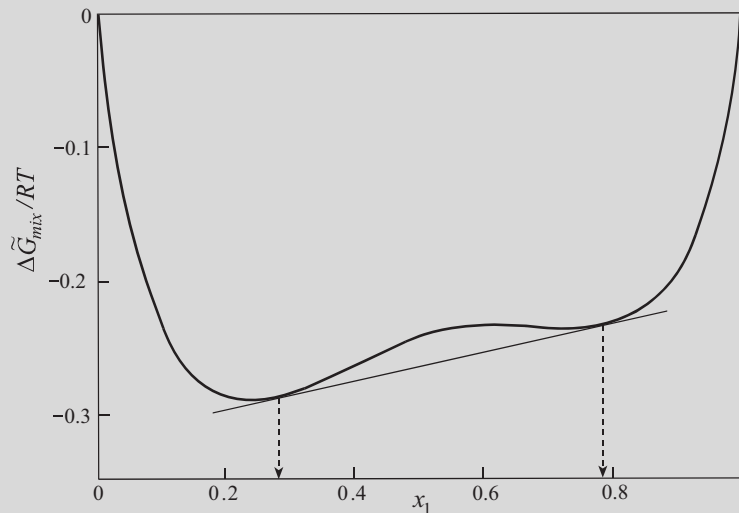
$$0.378 < x_1 < 0.682.$$

In other words, these are the mole fractions at the spinodal points.

Comment: Equation (11.2-11) gives

$$\frac{\Delta \tilde{G}_{\text{mix}}}{RT} = x_1 \ln x_1 + x_2 \ln x_2 + x_1 x_2 [1.8 + 0.3(x_1 - x_2) - 0.85(x_1 - x_2)^2].$$

The variation of $\Delta \tilde{G}_{\text{mix}}/RT$ versus x_1 is given in the figure below. The common tangent rule gives the binodal compositions as $x_1 = 0.28$ and $x_1 = 0.79$.



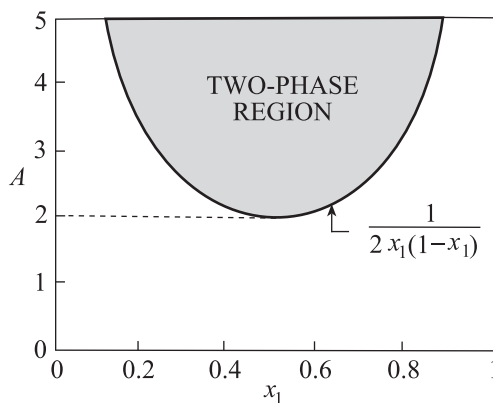


Figure 11.9: Stability diagram for the two-suffix Margules equation.

The simplest function to express molar excess Gibbs energy as a function of composition is the two-suffix Margules equation, Eqn (8.4-1),

$$\frac{\tilde{G}^{\text{ex}}}{RT} = Ax_1x_2. \quad (11.2-14)$$

Substitution of Eqn (11.2-14) into Eqn (11.2-12) and differentiation lead to

$$A > \frac{1}{2} \left(\frac{1}{x_1x_2} \right). \quad (11.2-15)$$

As shown in Fig. 11.9, the function $1/(2x_1x_2)$ becomes $+\infty$ for pure components, i.e. either $x_1 = 1$ or $x_2 = 1$, and falls to a minimum value of 2 at $x_1 = 0.5$. Therefore, a binary liquid mixture becomes unstable and exists as two separate phases when $A > 2$. From Eqn (8.4-4)

$$A = \ln \gamma_i^{\infty}, \quad i = 1, 2 \quad (11.2-16)$$

indicating that phase separation takes place when $\ln \gamma_i^{\infty} > 2$ (or $\gamma_i^{\infty} > 7.4$). As stated in Section 8.3, activity coefficients are greater than unity when components repel each other, i.e. unlike interactions are weaker than like interactions. When γ_i (or γ_i^{∞}) is very much greater than unity, it is more likely that phase separation will take place.

Substituting $A = 2$ and $x_1 = x_2 = 0.5$ into Eqn (11.2-14) gives the critical solution temperature (CST) as

$$\text{CST} = \frac{2 \tilde{G}^{\text{ex}}|_{x_1=0.5}}{R}. \quad (11.2-17)$$

The temperature calculated from Eqn (11.2-17) turns out to be the lower critical solution temperature (LCST) when A is a monotonically increasing function of T , i.e. $dA/dT > 0$. On the other hand, when A monotonically decreases with temperature, i.e. $dA/dT < 0$, the temperature calculated from Eqn (11.2-17) gives the upper critical solution temperature (UCST).

Example 11.2 A binary liquid mixture is represented by the two-suffix Margules equation. Determine the equilibrium compositions of the two liquid phases, α and β , for $A = 6.5$.

Solution

The mole fraction of component 1 can be determined from Eqn (11.2-15) as

$$x_1 = \frac{1 \pm \sqrt{1 - \frac{2}{A}}}{2}. \quad (1)$$

For $A = 6.5$, Eqn (1) gives

$$x_1^\alpha = 0.916 \quad \text{and} \quad x_1^\beta = 0.084.$$

11.3 Liquid-Liquid Equilibrium Calculations

Consider a multicomponent mixture of k species distributed in two liquid phases, α and β . The condition of equilibrium states that the fugacities of each species in α - and β -phases must be equal to each other, i.e.

$$\hat{f}_i^\alpha(T, P, x_i^\alpha) = \hat{f}_i^\beta(T, P, x_i^\beta), \quad i = 1, 2, \dots, k. \quad (11.3-1)$$

In terms of activity coefficients, Eqn (11.3-1) takes the form

$$\gamma_i^\alpha(T, P, x_i^\alpha) x_i^\alpha f_i^L(T, P) = \gamma_i^\beta(T, P, x_i^\beta) x_i^\beta f_i^L(T, P) \quad (11.3-2)$$

or

$$\gamma_i^\alpha(T, P, x_i^\alpha) x_i^\alpha = \gamma_i^\beta(T, P, x_i^\beta) x_i^\beta, \quad i = 1, 2, \dots, k. \quad (11.3-3)$$

In each phase, the mole fractions are related to each other by the following equations:

$$\sum_{i=1}^k x_i^\alpha = 1.0 \quad \text{and} \quad \sum_{i=1}^k x_i^\beta = 1.0. \quad (11.3-4)$$

Simultaneous solution of Eqns (11.3-3) and (11.3-4) gives the coexistence curve for the two-phase system.

For a binary system, Eqns (11.3-3) and (11.3-4) become

$$\ln \left(\frac{\gamma_1^\alpha}{\gamma_1^\beta} \right) = \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right), \quad (11.3-5)$$

$$\ln \left(\frac{\gamma_2^\alpha}{\gamma_2^\beta} \right) = \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right). \quad (11.3-6)$$

In practice, one encounters two types of liquid-liquid equilibrium calculations:

- When compositions are known, Eqns (11.3-5) and (11.3-6) can be solved for the parameters of an activity coefficient model.
- When parameters of an activity coefficient model are known, Eqns (11.3-5) and (11.3-6) can be solved for the compositions in α - and β -phases.

Example 11.3 Diethyl ether (1) and water (2) form two partially miscible liquid phases. At 308 K and atmospheric pressure, Villamanan *et al.* (1984) reported the following compositions of the two phases:

$$x_1^\alpha = 0.01172 \quad \text{and} \quad x_1^\beta = 0.9500.$$

If the system is represented by the three-suffix Margules equation, estimate the parameters A and B .

Solution

Substitution of Eqns (8.4-6) and (8.4-7) into Eqns (11.3-5) and (11.3-6), respectively, gives

$$A \left[(1 - x_1^\alpha)^2 - (1 - x_1^\beta)^2 \right] + B \left[(1 - x_1^\alpha)^2 (4x_1^\alpha - 1) - (1 - x_1^\beta)^2 (4x_1^\beta - 1) \right] = \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right), \quad (1)$$

$$A \left[(x_1^\alpha)^2 - (x_1^\beta)^2 \right] + B \left[(x_1^\alpha)^2 (4x_1^\alpha - 3) - (x_1^\beta)^2 (4x_1^\beta - 3) \right] = \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right). \quad (2)$$

The use of matrix algebra gives the parameters A and B as

$$\begin{pmatrix} A \\ B \end{pmatrix} = \begin{bmatrix} (1-x_1^\alpha)^2 - (1-x_1^\beta)^2 & (1-x_1^\alpha)^2(4x_1^\alpha - 1) - (1-x_1^\beta)^2(4x_1^\beta - 1) \\ (x_1^\alpha)^2 - (x_1^\beta)^2 & (x_1^\alpha)^2(4x_1^\alpha - 3) - (x_1^\beta)^2(4x_1^\beta - 3) \end{bmatrix}^{-1} \times \begin{bmatrix} \ln\left(\frac{x_1^\beta}{x_1^\alpha}\right) \\ \ln\left(\frac{1-x_1^\beta}{1-x_1^\alpha}\right) \end{bmatrix} \quad (11.3-3)$$

or

$$A = \frac{X(\Lambda - \Theta) - 6\Lambda(x_1^\alpha + x_1^\beta - 1)}{2(x_1^\alpha - x_1^\beta)^3} \quad \text{and} \quad B = \frac{(x_1^\alpha + x_1^\beta)(\Theta - \Lambda) + 2\Lambda}{2(x_1^\alpha - x_1^\beta)^3}, \quad (11.3-4)$$

where

$$X = (x_1^\alpha + x_1^\beta)(4x_1^\alpha + 4x_1^\beta - 3) - 4x_1^\alpha x_1^\beta, \quad \Theta = \ln\left(\frac{x_1^\beta}{x_1^\alpha}\right), \quad \Lambda = \ln\left(\frac{1-x_1^\beta}{1-x_1^\alpha}\right). \quad (11.3-5)$$

Substitution of the numerical values into Eqns (11.3-4) and (11.3-5) gives

$$A = 3.854 \quad \text{and} \quad B = -0.683.$$

Comment: The use of Eqn (8.4-9) gives

$$\gamma_1^\infty = 93.4 \quad \text{and} \quad \gamma_2^\infty = 23.8.$$

Since $\gamma_i^\infty \gg 1$, components repel each other and phase separation takes place.

Example 11.4 Estimate the compositions of the coexisting liquid phases in a mixture of *n*-pentane (1) and sulfolane (2) at 374.11 K. The system is represented by the NRTL model and [Ko et al. \(2007\)](#) reported the following parameters:

$$\tau_{12} = 2.329, \quad \tau_{21} = 1.061, \quad \alpha = 0.3.$$

Solution

From Eqn (8.4-27)

$$G_{12} = \exp(-\alpha\tau_{12}) = \exp[-(0.3)(2.329)] = 0.497,$$

$$G_{21} = \exp(-\alpha\tau_{21}) = \exp[-(0.3)(1.061)] = 0.727.$$

Therefore, activity coefficients defined by Eqns (8.4-29) and (8.4-30) become

$$\ln \gamma_1 = (1 - x_1)^2 \left[1.061 \left(\frac{0.727}{0.727 + 0.273x_1} \right)^2 + \frac{1.158}{(1 - 0.503x_1)^2} \right], \quad (1)$$

$$\ln \gamma_2 = x_1^2 \left[2.329 \left(\frac{0.497}{1 - 0.503x_1} \right)^2 + \frac{0.771}{(0.727 + 0.273x_1)^2} \right]. \quad (2)$$

Substitution of Eqns (1) and (2) into Eqns (11.3-5) and (11.3-6), respectively, results in two highly nonlinear equations given by

$$\begin{aligned} f_1(x_1^\alpha, x_1^\beta) = & (1 - x_1^\alpha)^2 \left[1.061 \left(\frac{0.727}{0.727 + 0.273x_1^\alpha} \right)^2 + \frac{1.158}{(1 - 0.503x_1^\alpha)^2} \right] \\ & - (1 - x_1^\beta)^2 \left[1.061 \left(\frac{0.727}{0.727 + 0.273x_1^\beta} \right)^2 + \frac{1.158}{(1 - 0.503x_1^\beta)^2} \right] \\ & - \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right) = 0, \end{aligned} \quad (3)$$

$$\begin{aligned} f_2(x_1^\alpha, x_1^\beta) = & (x_1^\alpha)^2 \left[2.329 \left(\frac{0.497}{1 - 0.503x_1^\alpha} \right)^2 + \frac{0.771}{(0.727 + 0.273x_1^\alpha)^2} \right] \\ & - (x_1^\beta)^2 \left[2.329 \left(\frac{0.497}{1 - 0.503x_1^\beta} \right)^2 + \frac{0.771}{(0.727 + 0.273x_1^\beta)^2} \right] \\ & - \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right) = 0. \end{aligned} \quad (4)$$

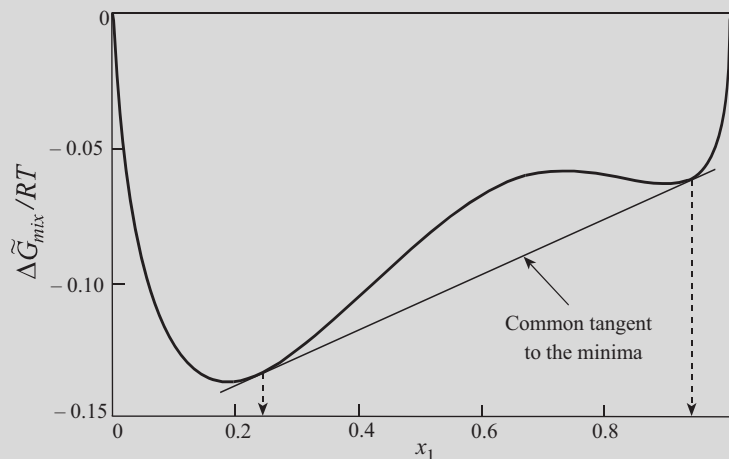
Good initial estimates for x_1^α and x_1^β are necessary for the simultaneous solution of Eqns (3) and (4) by Newton's method. These values can be determined graphically by drawing a common tangent to the minima of $\Delta \tilde{G}_{\text{mix}}/RT$ versus the x_1 curve. The molar excess Gibbs energy for the NRTL model is given by Eqn (8.4-26). Thus, the molar Gibbs energy change on mixing is given by

$$\frac{\Delta \tilde{G}_{\text{mix}}}{RT} = x_1 x_2 \left(\frac{0.771}{x_1 + 0.727x_2} + \frac{1.158}{x_2 + 0.497x_1} \right) + x_1 \ln x_1 + x_2 \ln x_2. \quad (5)$$

The values of $\Delta\tilde{G}_{\text{mix}}/RT$ as a function of x_1 are given in the table below:

x_1	$\Delta\tilde{G}_{\text{mix}}/RT$	x_1	$\Delta\tilde{G}_{\text{mix}}/RT$	x_1	$\Delta\tilde{G}_{\text{mix}}/RT$
0.00	0	0.35	-0.114	0.70	-0.059
0.05	-0.093	0.40	-0.104	0.75	-0.059
0.10	-0.123	0.45	-0.093	0.80	-0.060
0.15	-0.135	0.50	-0.083	0.85	-0.062
0.20	-0.137	0.55	-0.074	0.90	-0.063
0.25	-0.132	0.60	-0.067	0.95	-0.056
0.30	-0.124	0.65	-0.062	1.00	0

The figure shown below shows the variation in $\Delta\tilde{G}_{\text{mix}}/RT$ with composition. The equilibrium compositions in the two-phase region can be determined by drawing a line tangent to the minima with the result $x_1^\alpha = 0.23$ and $x_1^\beta = 0.92$.



Using these values as initial estimates, Newton's method yields

$$x_1^\alpha = 0.234 \quad \text{and} \quad x_1^\beta = 0.920.$$

Liquid-liquid equilibrium separations are analogous to the flash calculations mentioned in Chapter 9. Consider a binary liquid mixture with a solubility diagram as shown in Fig. 11.10a. Initially the liquid mixture is at a temperature of T_1 and the overall mole fraction of component 1 is z_1 . This liquid mixture enters a separation tank with a molar flow rate of F as shown in Fig. 11.10b. Phase separation is initiated⁴ within the tank by decreasing temperature to T_2 . The molar flow rates of the

⁴Thermodynamics is not concerned with the question of "How long will it take for the phase separation to take place?"

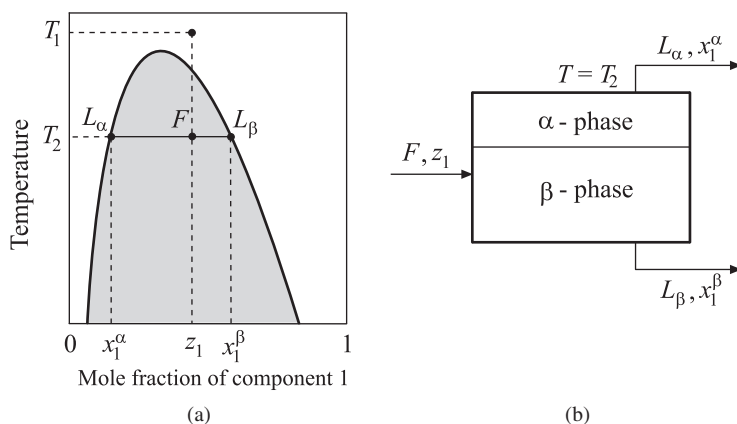


Figure 11.10: Liquid-liquid flash problem.

resulting α - and β -phases are L_α and L_β , respectively. Throughout the process, the pressure is kept constant at a value above the bubble point pressure of the mixture so that no vapor phase is present.

The overall and component material balances around the separation chamber are given by

$$F = L_\alpha + L_\beta, \quad (11.3-7)$$

$$Fz_i = L_\alpha x_i^\alpha + L_\beta x_i^\beta, \quad i = 1, 2. \quad (11.3-8)$$

For component 1, combination of Eqns (11.3-7) and (11.3-8) leads to

$$\frac{L_\alpha}{F} = \frac{x_1^\beta - z_1}{x_1^\beta - x_1^\alpha}. \quad (11.3-9)$$

Using the solubility diagram given in Fig. 11.10, Eqn (11.3-9) can also be obtained by the application of the lever rule. Note also that Eqn (11.3-9) is nothing more than Eqn (11.2-8).

From the equilibrium relations given by Eqn (11.3-5) and (11.3-6)

$$f_1(x_1^\alpha, x_1^\beta) = \ln \gamma_1^\alpha - \ln \gamma_1^\beta - \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right) = 0, \quad (11.3-10)$$

$$f_2(x_1^\alpha, x_1^\beta) = \ln \gamma_2^\alpha - \ln \gamma_2^\beta - \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right) = 0. \quad (11.3-11)$$

Compositions of the liquid phases can be determined by the simultaneous solution of Eqns (11.3-10) and (11.3-11) by Newton's method. Then L_α/F is calculated from Eqn (11.3-9).

Example 11.5 A binary mixture of 3-methylpentane (1) and *N*-formylmorpholine (2) exhibits partial miscibility at 306.4 K and atmospheric pressure. The system is represented by the NRTL model and Wang *et al.* (2012) reported the following parameters:

$$\tau_{12} = 3.060, \quad \tau_{21} = 1.714, \quad \alpha = 0.2.$$

If a feed stream consisting of 20 mol% 3-methylpentane and 80% *N*-formylmorpholine is held at 306.4 K, what will be the compositions and fractions of the two liquid phases?

Solution

From Eqn (8.4-27)

$$G_{12} = \exp(-\alpha\tau_{12}) = \exp[-(0.2)(3.060)] = 0.542,$$

$$G_{21} = \exp(-\alpha\tau_{21}) = \exp[-(0.2)(1.714)] = 0.710.$$

Therefore, activity coefficients defined by Eqns (8.4-29) and (8.4-30) become

$$\ln \gamma_1 = (1 - x_1)^2 \left[1.714 \left(\frac{0.710}{0.710 + 0.290x_1} \right)^2 + \frac{1.659}{(1 - 0.458x_1)^2} \right], \quad (1)$$

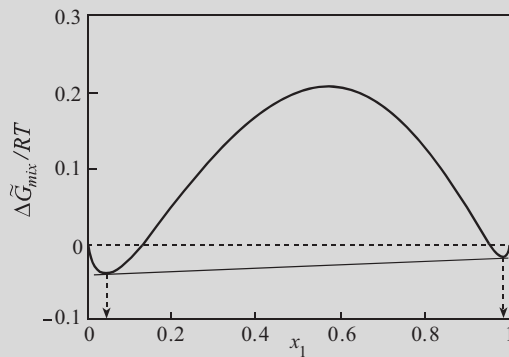
$$\ln \gamma_2 = x_1^2 \left[3.060 \left(\frac{0.542}{1 - 0.458x_1} \right)^2 + \frac{1.217}{(0.710 + 0.290x_1)^2} \right]. \quad (2)$$

Substitution of Eqns (1) and (2) into Eqns (11.3-10) and (11.3-11), respectively, results in two highly nonlinear equations given by

$$\begin{aligned} f_1(x_1^\alpha, x_1^\beta) &= (1 - x_1^\alpha)^2 \left[1.714 \left(\frac{0.710}{0.710 + 0.290x_1^\alpha} \right)^2 + \frac{1.659}{(1 - 0.458x_1^\alpha)^2} \right] \\ &\quad - (1 - x_1^\beta)^2 \left[1.714 \left(\frac{0.710}{0.710 + 0.290x_1^\beta} \right)^2 + \frac{1.659}{(1 - 0.458x_1^\beta)^2} \right] \\ &\quad - \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right) = 0, \end{aligned} \quad (3)$$

$$\begin{aligned}
 f_2(x_1^\alpha, x_1^\beta) = & (x_1^\alpha)^2 \left[3.060 \left(\frac{0.542}{1 - 0.458x_1^\alpha} \right)^2 + \frac{1.217}{(0.710 + 0.290x_1^\alpha)^2} \right] \\
 & - (x_1^\beta)^2 \left[3.060 \left(\frac{0.542}{1 - 0.458x_1^\beta} \right)^2 + \frac{1.217}{(0.710 + 0.290x_1^\beta)^2} \right] \\
 & - \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right) = 0.
 \end{aligned} \tag{11.3-4}$$

Initial estimates for x_1^α and x_1^β can be determined graphically by drawing a common tangent to the minima of $\Delta\tilde{G}_{\text{mix}}/RT$ versus the x_1 curve as shown below.



Taking $x_1^\alpha = 0.05$ and $x_1^\beta = 0.98$, simultaneous solution of Eqns (3) and (11.3-4) by Newton's method yields

$$x_1^\alpha = 0.045 \quad \text{and} \quad x_1^\beta = 0.984.$$

From Eqn (11.3-9)

$$\frac{L_\alpha}{F} = \frac{0.984 - 0.2}{0.984 - 0.045} = 0.835, \quad N\text{-formylmorpholine-rich phase.}$$

Therefore

$$\frac{L_\beta}{F} = 1 - 0.835 = 0.165, \quad 3\text{-methylpentane-rich phase.}$$

11.4 Liquid-Liquid Extraction

Contrary to the case of distillation, in which separation is based on the differences in boiling points, separation by liquid-liquid extraction is based on the differences in

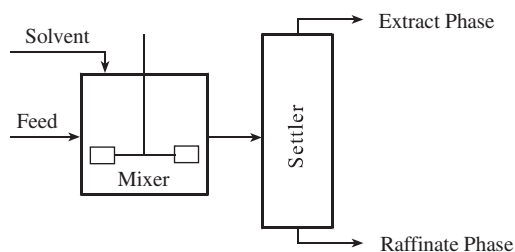


Figure 11.11: Single-stage mixer-settler.

solubilities of a solute in two immiscible (or partially miscible) liquids. It is a process in which a liquid feed stream consisting of a solute and a carrier liquid is contacted with a solvent that has an appreciable affinity for the solute but is immiscible or slightly miscible with the carrier liquid. In this way, most of the solute is transferred to the solvent. The resulting two liquid phases are then separated as a result of differences in densities. The solvent-rich phase, containing most of the solute and a small amount of the carrier liquid, is called the *extract phase* and the carrier liquid-rich phase, containing small quantities of the solute and the solvent, is called the *raffinate phase*. In cases when distillation cannot be used, i.e. close boiling point mixtures or azeotropic mixtures, liquid-liquid extraction is an alternative separation technique. It is apparent that energy costs related to extraction are considerably less than those of distillation.

Extraction can be carried out using various equipment. The simplest type, known as a single-stage mixer-settler, is shown in Fig. 11.11. It consists of two parts: a mixer for contacting the feed stream and the solvent, and a settler for mechanical separation due to density difference. The process may be either batch or continuous.

Phase equilibrium calculations involving ternary systems are most conveniently handled with an equilateral triangular diagram, shown in Fig. 11.12a.⁵ The three corners represent pure components, *A* (solute), *C* (carrier liquid), and *S* (solvent). The sides of the triangle represent binary mixtures. The point *M* represents a mixture having a composition of 30 wt.% *A*, 30% *C*, and 40% *S*.

Let us mix the solvent *S* with the binary mixture *F*, containing *A* and *C*. The composition of the resulting mixture, *M*, can be determined from the overall and component material balances:

$$F + S = M, \quad (11.4-1)$$

$$F\omega_{i,F} + S\omega_{i,S} = M\omega_{i,M}. \quad (11.4-2)$$

⁵In liquid-liquid extraction, compositions are generally expressed on a weight percentage basis.

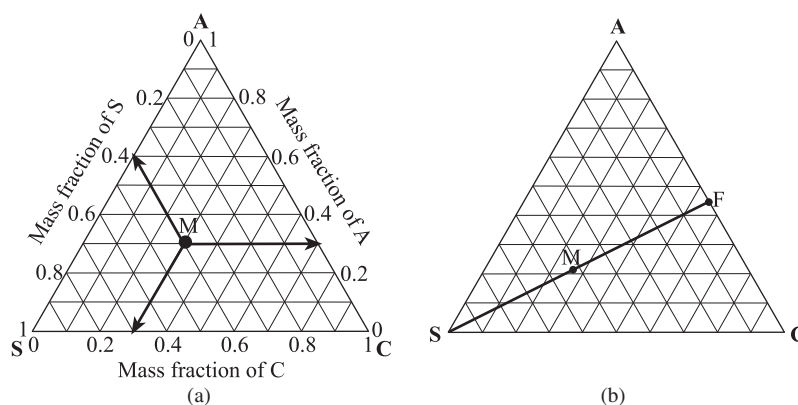


Figure 11.12: Triangular diagram.

Simultaneous solution of Eqns (11.4-1) and (11.4-2) gives

$$\frac{F}{M} = \frac{\omega_{i,M} - \omega_{i,S}}{\omega_{i,F} - \omega_{i,S}}. \quad (11.4-3)$$

It is apparent from Eqn (11.4-3) that the mixture M lies on a straight line joining the points S and F , as shown in Fig. 11.12b, and its location can be determined from the application of the lever rule, i.e.

$$\frac{F}{M} = \frac{\overline{SM}}{\overline{SF}}. \quad (11.4-4)$$

A typical liquid-liquid phase equilibrium data for a ternary mixture of A , C , and S is shown in Fig. 11.13. While S and C are partially miscible, A dissolves completely in C or S . The dome-shaped shaded region indicates the two-phase region, the boundary of which is determined from the equilibrium data, i.e. binodal curve. Within the two-phase region, the tie lines join the equilibrium compositions of the separate phases. The lengths of the tie lines decrease towards the apex of the dome-shaped region. At the *plait point*, the tie line shrinks to a point and, as a result, the two phases become identical. Outside the dome-shaped region all components are miscible in each other, i.e. one-phase region.

The calculation procedure for a single-stage extraction is shown in Fig. 11.14. When the feed stream F , containing a mixture of a solute and a carrier liquid, is mixed with the solvent, S , the resulting mixture, M , lies on a straight line joining the two streams. The location of M can be determined by the application of the lever rule. The tie line passing through the point M gives the compositions of the extract and raffinate phases, which are represented by the points E and R , respectively. The

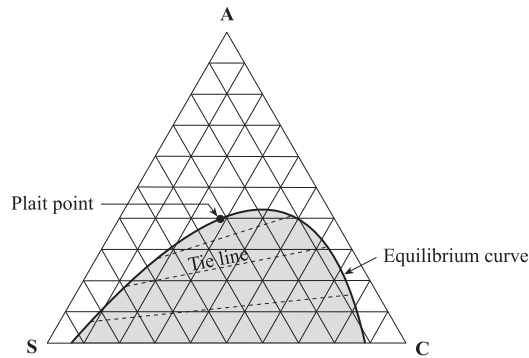


Figure 11.13: Liquid-liquid phase diagram.

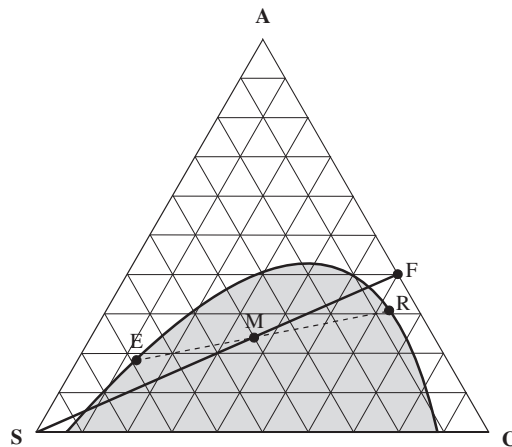


Figure 11.14: Graphical calculation procedure for a single-stage extraction.

amounts of the extract and raffinate phases can then be determined from either the material balance or the lever rule.

Example 11.6 A liquid feed stream of 1500 kg/h consisting of 35 wt.% acetonitrile (1) and 65% water (2) is to be extracted using isobutyl methyl ketone (3) as the solvent. If the flow rate of the solvent is 500 kg/h, determine the flow rates and compositions of the exit streams. The overall process takes place at a constant temperature of 303 K.

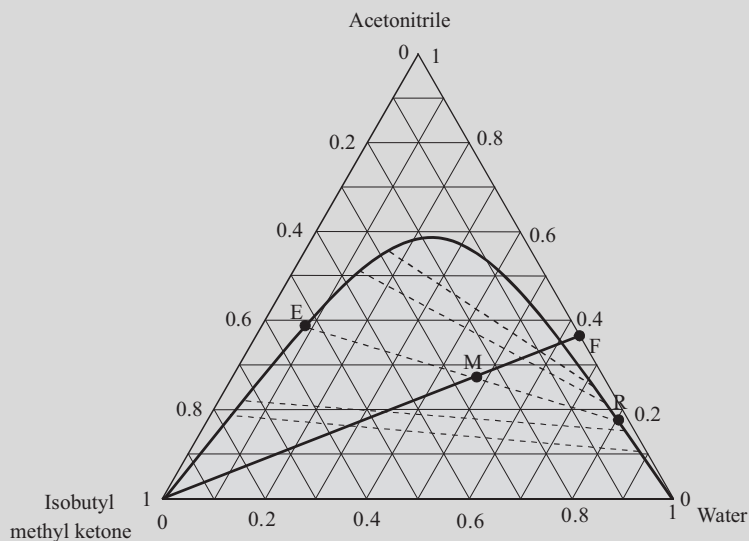
Data: For the acetonitrile-water-isobutyl methyl ketone system, [Rao *et al.* \(1979\)](#) reported the following liquid-liquid equilibrium data at 303 K (compositions are on weight percent basis):

Isobutyl methyl ketone	Acetonitrile	Water	Isobutyl methyl ketone	Acetonitrile	Water
0.4	0.00	99.60	17.50	58.50	24.00
0.8	8.5	90.70	25.88	57.00	17.12
1.65	16.94	81.41	35.58	51.84	12.58
2.61	24.62	72.77	45.25	44.59	10.16
3.84	32.72	63.44	56.01	37.02	6.97
5.65	41.33	53.02	66.50	28.21	5.29
9.50	49.50	42.00	77.40	18.76	3.78
12.66	56.25	31.19	87.97	9.76	2.27
14.59	57.02	28.39	99.25	0.00	0.75

Water layer			Organic layer		
Isobutyl methyl ketone	Acetonitrile	Water	Isobutyl methyl ketone	Acetonitrile	Water
1.00	8.50	90.50	79.60	17.50	2.90
1.10	11.00	87.90	79.00	18.00	3.00
1.70	15.20	83.10	72.50	23.50	4.00
2.00	17.70	80.30	54.50	38.00	7.50
2.20	20.50	77.30	42.00	47.50	10.50
2.40	22.60	75.00	35.00	52.00	13.00
2.70	25.10	72.20	25.20	57.30	17.50

Solution

Using the given data, the phase diagram of the given system is plotted in the figure below.



Once the feed and the solvent streams are mixed, the resulting mixture composition is

$$\begin{aligned}\omega_{\text{acetonitrile}} &= \frac{(1500)(0.35)}{2000} = 0.2625, \\ \omega_{\text{water}} &= \frac{(1500)(0.65)}{2000} = 0.4875, \\ \omega_{\text{isobutyl methyl ketone}} &= \frac{500}{2000} = 0.25.\end{aligned}$$

Thus, the point M can be located on the straight line joining the solvent and the feed streams. The tie line passing through M gives the compositions of the extract and raffinate phases as follows:

Component	Extract phase (wt.%)	Raffinate phase (wt.%)
Acetonitrile	0.38	0.17
Water	0.08	0.80
Isobutyl methyl ketone	0.54	0.03

The material balance for acetonitrile is

$$0.38E + 0.17R = (1500)(0.35). \quad (1)$$

The material balance for water is

$$0.08E + 0.8R = (1500)(0.65). \quad (2)$$

Simultaneous solution of Eqns (1) and (2) yields

$$E = 875 \text{ kg/h} \quad \text{and} \quad R = 1131 \text{ kg/h}.$$

11.5 Applications of Liquid-Liquid Equilibrium

11.5.1 Octanol-Water Partition Coefficient

Octanol [$\text{CH}_3(\text{CH}_2)_7\text{OH}$] and water are partially immiscible and the distribution of an organic compound i between these two phases is known as the *octanol-water partition coefficient*, K_i^{ow} , i.e.

$$K_i^{\text{ow}} = \frac{c_i^{\text{o}}}{c_i^{\text{w}}}. \quad (11.5-1)$$

Since K_i^{ow} values may range from 10^{-4} to 10^8 (encompassing 12 orders of magnitude), it is usually reported as $\log K_i^{\text{ow}}$. Octanol-water partition coefficients of various substances are given in Table 11.1.

Table 11.1: Octanol-water partition coefficients of various substances.

Substance	Chemical formula	$\log K^{\text{ow}}$
Methanol	CH_3OH	-0.77
Chloroform	CHCl_3	1.97
Benzene	C_6H_6	2.13
1,1,2,2-Tetrachloroethane	$\text{C}_2\text{H}_2\text{Cl}_4$	2.39
1,1,1-Trichloroethane	CH_3CCl_3	2.49
Naphthalene	C_{10}H_8	3.29
Hexachlorobenzene	C_6Cl_6	6.18

^a Compiled from US National Library of Medicine, *Hazardous Substances Data Bank* (HSDB), URL: <http://toxnet.nlm.nih.gov>. See also *A Databank of Evaluated Octanol-Water Partition Coefficients* (LOGKOW), URL: <http://logkow.cisti.nrc.ca/logkow/index.jsp>.

Cells are mainly made of lipids and they are generally modeled as a lipid bilayer model, with a long hydrophobic (water disliking) chain and a polar hydrophilic (water liking) end. The reason for choosing *n*-octanol is the fact that it exhibits both a hydrophobic and a hydrophilic character, and its carbon/oxygen ratio is similar to that of lipids. In other words, *n*-octanol mimics the structure and properties of cells and organisms.

Since octanol-water partition coefficient quantifies how a substance distributes itself between lipid and water, it is extensively used to describe lipophilic (lipid liking) and hydrophilic properties of a particular substance. In that respect, it is one of the key physical/chemical properties, such as vapor pressure and solubility in water, used to assess the impact of agricultural and industrial chemicals on the environment. For example, polychlorinated biphenyls (PCBs) have low solubilities in water and high octanol-water partition coefficients. If PCBs are accidentally released into the lake, then they are most probably found in higher concentrations in the sediment layer.

11.5.2 Steam Distillation

If two immiscible liquids are placed in a tank, the one with the lower density floats on the top layer and solely contributes to the pressure in the vapor phase. The liquid with the higher density has no contribution to the pressure in the vapor phase since it is located in the lower layer. Therefore, when we talk about the *vapor-liquid-liquid equilibrium* (VLLE) calculations, we are implicitly assuming that the liquid phases are continuously agitated in such a way that there will be droplets of both liquids on the surface, which is in contact with the vapor phase.

A pure liquid starts to boil when its vapor pressure equals the surrounding pressure. For example, using the values given in Appendix C, the vapor pressures

of toluene and water are expressed as

$$\ln P_{\text{toluene}}^{\text{vap}} = 9.3935 - \frac{3096.52}{T - 53.67},$$

$$\ln P_{\text{water}}^{\text{vap}} = 11.6834 - \frac{3816.44}{T - 46.13},$$

where P^{vap} is in bar and T is in K. The normal boiling point temperatures of pure toluene and pure water are

$$T_{\text{toluene}}^{\text{sat}} = 53.67 + \frac{3096.52}{9.3935 - \ln 1.01325} = 383.78 \text{ K},$$

$$T_{\text{water}}^{\text{sat}} = 46.13 + \frac{3816.44}{11.6834 - \ln 1.01325} = 373.15 \text{ K}.$$

In the case of immiscible liquids, each component contributes to the vapor pressure of the mixture. The total pressure is simply the sum of the vapor pressures of each component, i.e.

$$P = \sum_{i=1}^k P_i^{\text{vap}}, \quad \text{Immiscible mixture.} \quad (11.5-2)$$

Since toluene and water are essentially immiscible as liquids, Eqn (11.5-2) is expressed as

$$\exp\left(9.3935 - \frac{3096.52}{T - 53.67}\right) + \exp\left(11.6834 - \frac{3816.44}{T - 46.13}\right) = 1.013.$$

The solution of the above equation gives $T = 357.48 \text{ K}$. Thus, under atmospheric pressure, a toluene-water mixture boils at 357.48 K, lower than the boiling points of pure toluene (383.78 K) and water (373.15 K).

High boiling point liquids at atmospheric pressure, i.e. essential oils,⁶ waxes, and complex fats, may decompose at high temperatures and cannot be purified by distillation. Since oils are usually insoluble in water, then the mixture of oil and steam boils at a temperature well below the boiling point of pure oil. Therefore, to decrease the boiling point, steam is directly injected into the distillation column. When the resulting vapor is condensed, the two immiscible liquid phases are separated easily. Such a process is called *steam distillation*.

⁶Essential oils are the concentrated extracts of plants and herbs.

Problems

Problems Related to Section 11.2

11.1 Show that an ideal mixture always forms a homogeneous phase over the entire composition range.

11.2 Start with Eqn (8.4-18) and show that

$$\frac{d^2(\tilde{G}^{\text{ex}}/RT)}{dx_1^2} = \frac{\Lambda_{12}^2}{x_1(x_1 + \Lambda_{12}x_2)^2} + \frac{\Lambda_{21}^2}{x_2(x_2 + \Lambda_{21}x_1)^2} \quad (1)$$

for the Wilson model. Since Eqn (1) is always positive (Why?), conclude that the Wilson model cannot be used to predict phase separation.

11.3 For a binary liquid mixture of water (1) and diacetyl (2), [Baudot and Marin \(1996\)](#) reported the following activity coefficients at infinite dilution:

$$\gamma_1^\infty = 3.3, \quad \gamma_2^\infty = 13, \quad 303 < T < 323.$$

Estimate the spinodal compositions if the system is represented by the three-suffix Margules model.

(Answer: $x_1 = 0.531$ and $x_1 = 0.799$)

Problems Related to Section 11.3

11.4 When compositions, i.e. x_1^α and x_1^β , are known, the parameters of an activity coefficient model can be estimated by solving Eqns (11.3-5) and (11.3-6).

(a) If the binary mixture is represented by the two-suffix Margules model, show that

$$A = \frac{\ln(x_1^\beta/x_1^\alpha)}{(1 - x_1^\alpha)^2 - (1 - x_1^\beta)^2} = \frac{\ln \left[(1 - x_1^\beta)/(1 - x_1^\alpha) \right]}{(x_1^\alpha - x_1^\beta)(x_1^\alpha + x_1^\beta)}. \quad (1)$$

(b) If the binary mixture is represented by the van Laar model, show that

$$A = -\frac{(X - Y)(X\Theta + \Lambda)^2(Y\Theta + \Lambda)^2}{[\Theta(X + Y) + 2\Lambda][Y\Lambda + X(2Y\Theta + \Lambda)]^2}, \quad (2)$$

$$B = \frac{(X - Y)(X\Theta + \Lambda)^2(Y\Theta + \Lambda)^2}{[\Theta(X + Y) + 2\Lambda]^2[Y\Lambda + X(2Y\Theta + \Lambda)]}, \quad (3)$$

where

$$X = \frac{x_1^\alpha}{1 - x_1^\alpha}, \quad Y = \frac{x_1^\beta}{1 - x_1^\beta}, \quad \Theta = \ln \left(\frac{x_1^\beta}{x_1^\alpha} \right), \quad \Lambda = \ln \left(\frac{1 - x_1^\beta}{1 - x_1^\alpha} \right). \quad (4)$$

- (c) [Abedinzadegan and Meisen \(1996\)](#) studied liquid-liquid equilibrium of diethanolamine (1) and octadecane (2) mixtures and reported the following solubility values at 492 K and atmospheric pressure:

$$x_1^\alpha = 0.0959 \quad \text{and} \quad x_1^\beta = 0.9924.$$

If the system is represented by the van Laar model, estimate the parameters A and B .

(Answer: (c) $A = 2.607$, $B = 4.933$)

- 11.5** A mixture of acetonitrile (1) and n -hexadecane (2) forms two partially miscible liquid phases. Liquid-liquid equilibrium data on such systems are needed for the design of the oxidative desulfurization process. [Furuya *et al.* \(2007\)](#) obtained the following NRTL parameters at 333 K:

$$\tau_{12} = 4.852, \quad \tau_{21} = 0.2818, \quad \alpha = 0.2.$$

- (a) If the composition of acetonitrile in the acetonitrile-rich phase is 0.994, estimate its composition in the n -hexadecane-rich phase.
 (b) Are the like interactions stronger or weaker than the unlike interactions?

(Answer: (a) 0.168)

- 11.6** A binary liquid mixture is represented by the three-suffix Margules equation with the following parameters:

$$A = 2.5 \quad \text{and} \quad B = 0.5.$$

Show that the compositions of the coexisting liquid phases are given by

$$x_1^\alpha = 0.208 \quad \text{and} \quad x_1^\beta = 0.928.$$

- 11.7** Infinite dilution activity coefficients of various liquids in benzene at 293 K are given as follows:

Species	Acetone	Acetonitrile	Carbon tetrachloride	n -Hexane
γ^∞	1.77	3.21	1.13	2.21

Arrange these species in order of decreasing solubility in benzene.

- 11.8** A binary liquid mixture of components 1 and 2 is represented by the NRTL model. At 300 K, the parameters are

$$\tau_{12} = 0.8, \quad \tau_{21} = 3.0, \quad \alpha = 0.3.$$

The molar excess enthalpy of the mixture at 300 K is given as

$$\tilde{H}^{\text{ex}} \text{ (J/mol)} = x_1 x_2 [1400 - 380(x_1 - x_2)].$$

When 2 mol of component 1 at 300 K are mixed with 6 mol of component 2 at 300 K, estimate the heat that must be removed or added so as to keep the mixture temperature constant at 300 K.

(Answer: 923.1 J)

- 11.9** Since the solubility of water in *n*-hexadecane is 0.0059 mol fraction and that of *n*-hexadecane in water is 0.0072 mol fraction at 298.15 K, the *n*-hexadecane-water system is regarded as a system containing the two pure solvents. Consider a solute (1) that is partitioned between *n*-hexadecane (α -phase) and water (β -phase). The following data are provided:

Component	ρ (kg/m ³)	Molecular weight
<i>n</i> -Hexadecane	770.20	226.44
Water	997.05	18.02

The condition of equilibrium states that

$$x_1^\alpha \gamma_1^\alpha = x_1^\beta \gamma_1^\beta. \quad (1)$$

- (a) The hexadecane-water partition coefficient, $K_1^{\alpha\beta}$, is defined as the molar concentration of the solute in the *n*-hexadecane phase to the molar concentration of the solute in the water phase, i.e.

$$K_1^{\alpha\beta} = \frac{c_1^\alpha}{c_1^\beta}. \quad (2)$$

Combine Eqns (1) and (2) to obtain

$$\gamma_1^\beta = 16.3 K_1^{\alpha\beta} \gamma_1^\alpha. \quad (3)$$

If the solute is infinitely dilute in this system, note that Eqn (3) takes the form

$$\gamma_1^{\beta\infty} = 16.3 K_1^{\alpha\beta} \gamma_1^{\alpha\infty}. \quad (4)$$

If the infinite-dilution activity coefficient of solute in *n*-hexadecane and the hexadecane-water partition coefficient are known, then Eqn (4) can be used to estimate the infinite-dilution activity coefficient of solute in water.

- (b) The infinite-dilution activity coefficient of toluene in *n*-hexadecane is given as a function of temperature as follows (Schult *et al.*, 2001):

Temperature (K)	323.2	343.2	353.2	373.2
γ^∞	0.941	0.870	0.846	0.808

The hexadecane-water partition coefficient of toluene is (Abraham *et al.*, 1990)

$$K = 575.4.$$

Express the infinite-dilution activity coefficient in the form

$$\ln \gamma^\infty = A + \frac{B}{T}$$

and estimate the infinite-dilution activity coefficient of toluene in water at 298 K.

(Answer: (b) 9667)

11.10 Consider two liquid phases, α and β , in which the α -phase is almost pure 1 and the β -phase is almost pure 2.

- (a) Using the condition of equilibrium show that

$$\gamma_1^{\beta\infty} = \frac{1}{x_1^\beta} \quad \text{and} \quad \gamma_2^{\alpha\infty} = \frac{1}{x_2^\alpha}. \quad (1)$$

Therefore, determination of solubility enables one to estimate the infinite-dilution activity coefficient. This method, known as *inverse solubility*, is suitable for organics that are sparingly soluble in water, i.e. $x_i < 10^{-3}$.

- (b) The solubility of benzene (2) in water (1) at 298.15 K is reported by May *et al.* (1983) as

$$x_2 = 0.4129 \times 10^{-3}.$$

Estimate the infinite-dilution activity coefficient of benzene in water.

(Answer: (b) 2422)

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Chapter 12

Solid-Liquid Equilibrium

Crystallization is the formation of solid particles from within a homogeneous phase and it is one of the unit operations used to separate substances in pure solid form from a liquid mixture. Therefore, reliable data on the solubility of solids in liquid solvents are needed when designing crystallizers. This chapter first provides equations to estimate solid solubility in solvents. Then how boiling and freezing points of liquids are affected by the presence of dissolved solids will be discussed. Finally, the osmosis phenomenon will be explained.

12.1 Equilibrium Between a Pure Solid and a Liquid Mixture

Let substance i be present as a pure solid and also as a component in a liquid solution at constant temperature and pressure. The condition of equilibrium states that

$$f_i^S(T, P) = \hat{f}_i^L(T, P, x_i) = f_i^L(T, P)\gamma_i(T, P, x_i)x_i. \quad (12.1-1)$$

Rearrangement of Eqn (12.1-1) gives

$$\ln(\gamma_i x_i) = \ln \left[\frac{f_i^S(T, P)}{f_i^L(T, P)} \right]. \quad (12.1-2)$$

The fugacities of pure solid and liquid are related to the molar Gibbs energies by Eqn (5.2-1), i.e.

$$\ln \left[\frac{f_i^S(T, P)}{P} \right] = \frac{\tilde{G}_i^S(T, P) - \tilde{G}_i^{\text{IG}}(T, P)}{RT}, \quad (12.1-3)$$

$$\ln \left[\frac{f_i^L(T, P)}{P} \right] = \frac{\tilde{G}_i^L(T, P) - \tilde{G}_i^{\text{IG}}(T, P)}{RT}. \quad (12.1-4)$$

Thus,

$$\begin{aligned} \ln \left[\frac{f_i^S(T, P)}{f_i^L(T, P)} \right] &= \frac{\tilde{G}_i^S(T, P) - \tilde{G}_i^L(T, P)}{RT} = -\frac{\Delta \tilde{G}_i^{\text{fus}}(T, P)}{RT} \\ &= -\frac{\Delta \tilde{H}_i^{\text{fus}}(T, P) - T \Delta \tilde{S}_i^{\text{fus}}(T, P)}{RT}. \end{aligned} \quad (12.1-5)$$

The use of Eqn (12.1-5) in Eqn (12.1-2) leads to

$$\ln (\gamma_i x_i) = -\frac{\Delta \tilde{H}_i^{\text{fus}}(T, P) - T \Delta \tilde{S}_i^{\text{fus}}(T, P)}{RT}. \quad (12.1-6)$$

Since the solution temperature, T , is usually much lower than the triple point temperature, T_t , of pure solid, a pure substance does not exist in the liquid form at the solution temperature. Thus, extrapolation of the liquid-vapor pressure curve from the triple point temperature to the solid region is necessary until the temperature reaches the solution temperature, as shown in Fig. 12.1.

Systematic extrapolation can be carried out by devising a following path as shown in Fig. 12.2:

Step 1: Solid is heated at constant pressure from solution temperature, T , to triple point temperature, T_t .

Step 2: Solid is melted to form a liquid ($\Delta \tilde{H}_i^{\text{fus}}$ and $\Delta \tilde{S}_i^{\text{fus}}$ at T_t are known).

Step 3: Liquid is subcooled without solidification from T_t to T .

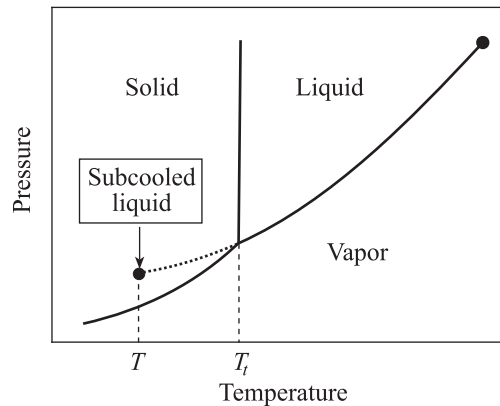


Figure 12.1: Extrapolation of vapor-liquid pressure curve to the solution temperature, T .

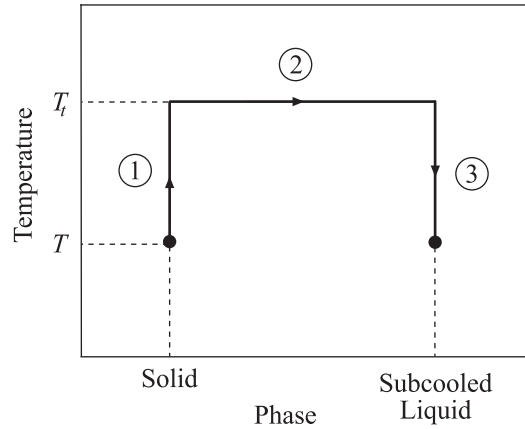


Figure 12.2: Calculation path for $\Delta\tilde{H}_i^{\text{fus}}(T)$ and $\Delta\tilde{S}_i^{\text{fus}}(T)$.

For steps 1 and 3, the changes in enthalpy and entropy can be calculated from Eqns (3.3-1) and (3.4-2), respectively. Since pressure remains constant, the changes in enthalpy and entropy for the overall process are expressed as

$$\begin{aligned}\Delta\tilde{H}_i^{\text{fus}}(T) &= \int_T^{T_t} \tilde{C}_{P_i}^S dT + \Delta\tilde{H}_i^{\text{fus}}(T_t) + \int_{T_t}^T \tilde{C}_{P_i}^L dT \\ &= \Delta\tilde{H}_i^{\text{fus}}(T_t) + \int_{T_t}^T \Delta\tilde{C}_{P_i}^{\text{LS}} dT,\end{aligned}\quad (12.1-7)$$

$$\begin{aligned}\Delta\tilde{S}_i^{\text{fus}}(T) &= \int_T^{T_t} \frac{\tilde{C}_{P_i}^S}{T} dT + \Delta\tilde{S}_i^{\text{fus}}(T_t) + \int_{T_t}^T \frac{\tilde{C}_{P_i}^L}{T} dT \\ &= \Delta\tilde{S}_i^{\text{fus}}(T_t) + \int_{T_t}^T \frac{\Delta\tilde{C}_{P_i}^{\text{LS}}}{T} dT,\end{aligned}\quad (12.1-8)$$

where

$$\Delta\tilde{C}_{P_i}^{\text{LS}} = \tilde{C}_{P_i}^L - \tilde{C}_{P_i}^S. \quad (12.1-9)$$

At the triple point, solid and liquid phases of pure component i are in equilibrium with each other, i.e.

$$\Delta\tilde{G}_i^{\text{fus}}(T_t) = \Delta\tilde{H}_i^{\text{fus}}(T_t) - T_t \Delta\tilde{S}_i^{\text{fus}}(T_t) = 0 \quad \Rightarrow \quad \Delta\tilde{S}_i^{\text{fus}}(T_t) = \frac{\Delta\tilde{H}_i^{\text{fus}}(T_t)}{T_t}. \quad (12.1-10)$$

The use of Eqn (12.1-10) in Eqn (12.1-8) leads to

$$\Delta \tilde{S}_i^{\text{fus}}(T) = \frac{\Delta \tilde{H}_i^{\text{fus}}(T_t)}{T_t} + \int_{T_t}^T \frac{\Delta \tilde{C}_{P_i}^{\text{LS}}}{T} dT. \quad (12.1-11)$$

Substitution of Eqns (12.1-7) and (12.1-11) into Eqn (12.1-6) results in

$$\ln(\gamma_i x_i) = \frac{\Delta \tilde{H}_i^{\text{fus}}(T_t)}{RT_t} \left(1 - \frac{T_t}{T}\right) - \frac{1}{RT} \int_{T_t}^T \Delta \tilde{C}_{P_i}^{\text{LS}} dT + \frac{1}{R} \int_{T_t}^T \frac{\Delta \tilde{C}_{P_i}^{\text{LS}}}{T} dT. \quad (12.1-12)$$

For most substances, the melting (or fusion) curve has a very steep slope. As a result, the triple point temperature is very close to the normal melting point, T_m . Moreover, it is plausible to assume that the enthalpy of fusion is essentially the same at these two temperatures. Under these circumstances, Eqn (12.1-12) becomes

$$\ln(\gamma_i x_i) = \frac{\Delta \tilde{H}_i^{\text{fus}}(T_m)}{RT_m} \left(1 - \frac{T_m}{T}\right) - \frac{1}{RT} \int_{T_m}^T \Delta \tilde{C}_{P_i}^{\text{LS}} dT + \frac{1}{R} \int_{T_m}^T \frac{\Delta \tilde{C}_{P_i}^{\text{LS}}}{T} dT. \quad (12.1-13)$$

Equation (12.1-13) can be used twofold: When it is applied to a solid, it gives the solubility of solid in a solvent. On the other hand, if it is applied to a solvent, it gives its freezing point depression.

12.1.1 Solubility of Solid in a Solvent

Let i be the solid (2) dissolved in a solvent (1). Then Eqn (12.1-13) gives the *mole fraction solubility*, x_2 , of solute as

$$\ln(\gamma_2 x_2) = \frac{\Delta \tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right) - \frac{1}{RT} \int_{T_{m_2}}^T \Delta \tilde{C}_{P_2}^{\text{LS}} dT + \frac{1}{R} \int_{T_{m_2}}^T \frac{\Delta \tilde{C}_{P_2}^{\text{LS}}}{T} dT. \quad (12.1-14)$$

The term γ_2 takes into account the intermolecular forces between the solid and the solvent. As a result, the solubility of solid is dependent on the type of solvent. Activity coefficient is estimated from either experimental data or a liquid solution model, i.e. regular mixture theory given in Section 8.5.

In the literature, solubility is also expressed as “g of solid per 100 g of solvent,” S , which is related to the mole fraction based solubility, x_2 , in the form

$$S = 100 \left(\frac{x_2}{1 - x_2} \right) \left(\frac{M_2}{M_1} \right), \quad (12.1-15)$$

where M_1 and M_2 are the molecular weights of solvent and solid, respectively.

When $\Delta\tilde{C}_{P_2}^{\text{LS}}$ is constant, Eqn (12.1-14) simplifies to

$$\ln(\gamma_2 x_2) = \frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right) - \frac{\Delta\tilde{C}_{P_2}^{\text{LS}}}{R} \left[1 - \frac{T_{m_2}}{T} + \ln\left(\frac{T_{m_2}}{T}\right)\right]. \quad (12.1-16)$$

When $\Delta\tilde{C}_{P_2}^{\text{LS}}$ is close to zero,¹ the second term on the right-hand side of Eqn (12.1-16) vanishes, with the result

$$\ln(\gamma_2 x_2) = \frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right). \quad (12.1-17)$$

On the other hand, if T and T_{m_2} are close to each other, expansion of the term $\ln(T_{m_2}/T)$ in Taylor series yields

$$\ln\left(\frac{T_{m_2}}{T}\right) = \left(\frac{T_{m_2}}{T} - 1\right) - \frac{1}{2}\left(\frac{T_{m_2}}{T} - 1\right)^2 + \frac{1}{3}\left(\frac{T_{m_2}}{T} - 1\right)^3 - \dots \quad (12.1-18)$$

Negligible when T and T_m do not differ greatly

The use of Eqn (12.1-18) in Eqn (12.1-16) also leads to Eqn (12.1-17). In the literature, Eqn (12.1-17) is known as the *Schröder equation*² and it is applicable to compounds that do not ionize or dissociate in solution, and when the solid phase is pure.

When the solubility is very low, i.e. x_2 is of the order of 10^{-3} or less, it is plausible to replace γ_2 by γ_2^∞ and Eqn (12.1-17) becomes

$$\ln(\gamma_2^\infty x_2) = \frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right), \quad x_2 \leq 10^{-3}. \quad (12.1-19)$$

The term *ideal solubility*, x_2^{ideal} , is used when the liquid mixture is ideal, i.e. $\gamma_2 = 1$. In that case Eqn (12.1-17) reduces to

$$\ln x_2^{\text{ideal}} = \frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right) \quad (12.1-20)$$

indicating that the ideal solubility is independent of the type of solvent. It follows from Eqn (12.1-20) that the solubility of a solid increases with an increase in temperature.

¹For compounds with high melting temperatures, i.e. pharmaceuticals and organic pollutants, instead of taking $\Delta\tilde{C}_{P_2}^{\text{LS}}$ as zero, the approximation $\Delta\tilde{C}_{P_2}^{\text{LS}} = \Delta\tilde{S}_2^{\text{fus}}(T_m)$ gives better results (Pappa *et al.*, 2005).

²The Schröder equation can also be derived by combining Eqn (5) of Example 5.13 with Eqn (12.1-2).

It can be shown from Eqns (12.1-17) and (12.1-20) that

$$x_2 = \frac{x_2^{\text{ideal}}}{\gamma_2} \quad (12.1-21)$$

indicating the higher the activity coefficient of solid in a solvent, the lower its solubility.

Example 12.1 Calculate the ideal solubility of naphthalene (2) in a solvent (1) at 313.15 K. For naphthalene, $T_m = 353.35$ K and $\Delta\tilde{H}^{\text{fus}} = 18,803$ J/mol.

Solution

The use of Eqn (12.1-20) gives

$$\ln x_2^{\text{ideal}} = \frac{18,803}{(8.314)(353.35)} \left(1 - \frac{353.35}{313.15} \right) \Rightarrow x_2^{\text{ideal}} = 0.44.$$

Comment: While the ideal solubility is independent of the type of solvent, solubilities of naphthalene (2) in various solvents at 313.15 K are reported by Gmehling *et al.* (1978) as follows:

Solvent	x_2	Solvent	x_2
Methanol	0.044	<i>n</i> -Hexane	0.222
Ethanol	0.073	Cyclohexanol	0.225
<i>n</i> -Propanol	0.094	Acetic acid	0.117
Isopropanol	0.076	Acetone	0.378
<i>n</i> -Butanol	0.116	Chloroform	0.473

Example 12.2 At 313.15 K, solubility of naphthalene (2) in *n*-hexane (1) is 0.222. Estimate the solubility of naphthalene in *n*-hexane at 325 K if its activity coefficient is represented by the van Laar model. The molar volumes of liquid naphthalene and *n*-hexane are given as 125.03 and 131.61 cm³/mol, respectively.

Solution

A single set of activity coefficient data must be known to determine the parameters in the van Laar model. From Example 12.1, $x_2^{\text{ideal}} = 0.44$. The use of Eqn (12.1-21) gives the activity coefficient of naphthalene at $x_2 = 0.222$ as

$$\gamma_2 = \frac{x_2^{\text{ideal}}}{x_2} = \frac{0.44}{0.222} = 1.982.$$

For the van Laar model, Eqn (8.4-17) gives

$$\frac{B}{A} = \frac{\tilde{V}_2^L}{\tilde{V}_1^L} = \frac{125.03}{131.61} = 0.95. \quad (1)$$

The use of Eqn (8.4-14) gives the parameter B as

$$B = \left[1 + \frac{B}{A} \left(\frac{x_2}{1 - x_2} \right) \right]^2 \ln \gamma_2 = \left[1 + 0.95 \left(\frac{0.222}{1 - 0.222} \right) \right]^2 \ln(1.982) = 1.105.$$

Therefore, the activity coefficient of naphthalene is expressed as

$$\ln \gamma_2 = \frac{1.105}{\left[1 + 0.95 \left(\frac{x_2}{1 - x_2} \right) \right]^2}. \quad (2)$$

Substitution of Eqn (2) into Eqn (12.1-17) leads to

$$\ln x_2 + \frac{1.105}{\left[1 + 0.95 \left(\frac{x_2}{1 - x_2} \right) \right]^2} = \frac{18,803}{(8.314)(353.35)} \left(1 - \frac{353.35}{T} \right). \quad (3)$$

When $T = 325$ K, the solution of Eqn (3) by MATHCAD[®] gives $x_2 = 0.356$.

Comment: IUPAC-NIST Solubility Database³ gives the solubility of naphthalene in *n*-hexane as a function of temperature as

$$\ln x_2 = -\frac{3849.6}{T} + 10.789. \quad (4)$$

When $T = 325$ K, Eqn (4) gives $x_2 = 0.348$.

Example 12.3 Estimate the solubility of anthracene (2) in chlorobenzene (1) at 298 K. The liquid mixture obeys the regular mixture theory and the following data are provided:

Chlorobenzene: $\tilde{V}_1^L = 102 \text{ cm}^3/\text{mol}$, $\delta_1 = 19.4 \text{ (J/cm}^3)^{1/2}$.

Anthracene: $\tilde{V}_2^L = 150 \text{ cm}^3/\text{mol}$, $T_{m2} = 490 \text{ K}$, $\Delta \tilde{H}_2^{\text{fus}} = 29,372 \text{ J/mol}$

$$\ln P_2^{\text{sub}} = 19.9724 - \frac{11,467}{T},$$

where P^{sub} is in bar and T is in K.

³<http://srdata.nist.gov/solubility>.

Solution

To calculate the solubility of anthracene in chlorobenzene with the help of Eqn (12.1-17), the activity coefficient of anthracene must be expressed as a function of composition by using Eqns (8.5-2) and (8.5-3). For this purpose, it is first necessary to determine the solubility parameter of anthracene by using Eqn (8.5-5), i.e.

$$\delta_2 = \sqrt{\frac{\Delta \tilde{H}_2^{\text{vap}} - RT}{\tilde{V}_2^{\text{L}}}}. \quad (1)$$

The heat of vaporization can be indirectly calculated as

$$\Delta \tilde{H}_2^{\text{vap}} = \Delta \tilde{H}_2^{\text{sub}} - \Delta \tilde{H}_2^{\text{fus}}. \quad (2)$$

The heat of sublimation of anthracene can be calculated from Eqn (5.6-30), i.e.

$$\frac{d \ln P_2^{\text{sub}}}{dT} = \frac{\Delta \tilde{H}_2^{\text{sub}}}{RT^2} = \frac{11,467}{T^2}$$

or

$$\Delta \tilde{H}_2^{\text{sub}} = (11,467)(8.314) = 95,337 \text{ J/mol}.$$

Therefore, the heat of vaporization and the solubility parameter are calculated as

$$\Delta \tilde{H}_2^{\text{vap}} = 95,337 - 29,372 = 65,965 \text{ J/mol}$$

and

$$\delta_2 = \sqrt{\frac{65,965 - (8.314)(298)}{150}} = 20.6 \text{ (J/cm}^3\text{)}^{1/2}.$$

Equation (12.1-17) is expressed as

$$\frac{\tilde{V}_2^{\text{L}}(\delta_1 - \delta_2)^2}{RT} \left(\frac{x_1 \tilde{V}_1^{\text{L}}}{x_1 \tilde{V}_1^{\text{L}} + x_2 \tilde{V}_2^{\text{L}}} \right)^2 + \ln x_2 = \frac{\Delta \tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T} \right) \quad (3)$$

or

$$\begin{aligned} & \frac{(150)(19.4 - 20.6)^2}{(8.314)(298)} \left[\frac{102(1 - x_2)}{102(1 - x_2) + 150x_2} \right]^2 + \ln x_2 \\ &= \frac{29,372}{(8.314)(490)} \left(1 - \frac{490}{298} \right). \end{aligned} \quad (4)$$

The solution of Eqn (4) by MATHCAD[®] gives $x_2 = 8.824 \times 10^{-3}$.

As stated in Section 8.3.2, activity coefficients are generally estimated from vapor-liquid equilibrium (VLE) measurements. If the boiling points of the components are very high, then it is difficult to obtain VLE data. In this case, it is more convenient to obtain activity coefficients from solid-liquid equilibrium (SLE) measurements.⁴

12.1.2 Freezing Point Depression

In this case, Eqn (12.1-13) is applied to a solvent (1). Assuming the second and third terms on the right-hand side of Eqn (12.1-13) to be negligible, the result is

$$\ln(\gamma_1 x_1) = \frac{\Delta \tilde{H}_1^{\text{fus}}}{RT_{m_1}} \left(1 - \frac{T_{m_1}}{T} \right). \quad (12.1-22)$$

When the freezing point depression is small, i.e. $T \simeq T_{m_1}$, then $TT_{m_1} \simeq T_{m_1}^2$. Consequently, Eqn (12.1-22) takes the form

$$T_{m_1} - T = -\frac{RT_{m_1}^2}{\Delta \tilde{H}_1^{\text{fus}}} \ln(\gamma_1 x_1). \quad (12.1-23)$$

If solute (2) is dilute enough that the liquid can be treated as an ideal mixture, i.e. $\gamma_1 \simeq 1$, then

$$\ln(\gamma_1 x_1) = \ln x_1 = \ln(1 - x_2) = -x_2. \quad (12.1-24)$$

Substitution of Eqn (12.1-24) into Eqn (12.1-23) gives

$$T_{m_1} - T = \frac{RT_{m_1}^2}{\Delta \tilde{H}_1^{\text{fus}}} x_2 \quad (12.1-25)$$

indicating that the freezing point depression depends on the mole fraction and not on the chemical structure of solute.⁵

The mole fraction of solute, x_2 , is related to the molality of solute,⁶ m_2 , as

$$x_2 = \frac{m_2}{m_2 + \frac{1000}{M_1}} \simeq m_2 \left(\frac{M_1}{1000} \right). \quad (12.1-26)$$

⁴See Problem 12.4.

⁵The freezing point depression is known as a colligative property. *Colligative* means depending on the number of particles and not on the nature of particles. Other colligative properties will be discussed in Section 12.2.

⁶*Molality* of a solute is defined as the number of moles of solute per kg of solvent.

Table 12.1: Freezing point depression constants for various solvents.

Solvent	Normal Freezing Point (°C)	K_f [°C/(mol/kg)]
Acetic acid	16.6	3.90
Benzene	5.5	5.12
CCl ₄	−22.8	5.22
Chloroform	−63.5	4.68
Ethanol	−114.6	1.99
Water	0	1.86

The use of Eqn (12.1-26) in Eqn (12.1-25) leads to

$$T_{m1} - T = \left(\frac{RT_{m1}^2}{1000\Delta\hat{H}_1^{\text{fus}}} \right) m_2. \quad (12.1-27)$$

In the literature, Eqn (12.1-27) is usually expressed as

$$T_{m1} - T = iK_fm_2, \quad (12.1-28)$$

where i is called the *van't Hoff factor*⁷ and K_f is the *freezing point depression constant* of a solvent defined by

$$K_f = \frac{RT_{m1}^2}{1000\Delta\hat{H}_1^{\text{fus}}}. \quad (12.1-29)$$

The van't Hoff factor, i , is associated with the degree of dissociation of the solute in the solvent and it represents the moles of particles in solution per mole of solute dissolved. For example,

$$i = \begin{cases} 1 & \text{Nonelectrolytes,} \\ 2 & \text{Electrolytes that ionize into two ions, i.e. NaCl,} \\ 3 & \text{Electrolytes that ionize into three ions, i.e. MgCl}_2. \end{cases} \quad (12.1-30)$$

Freezing point depression constants for some solvents are given in Table 12.1.

Note that either Eqn (12.1-25) or Eqn (12.1-28) can be used not only to estimate the freezing point depression but also to estimate the heat of fusion of a solvent, $\Delta\hat{H}_1^{\text{fus}}$, or the molecular weight of a solute.

⁷Jacobus Henricus van't Hoff was a Dutch physical and organic chemist who lectured at various institutions (Veterinary College in Utrecht, University of Amsterdam, University of Berlin). He was the recipient in 1901 of the first Nobel prize for chemistry.

Example 12.4 Calculate the freezing point of a solution prepared by dissolving 171 g of $\text{Al}_2(\text{SO}_4)_3$ in 750 g of water.

Solution

The molality of $\text{Al}_2(\text{SO}_4)_3$ (molecular weight = 342 g/mol) is

$$m_2 = \frac{171/342}{0.75} = 0.667 \text{ mol/kg.}$$

If $\text{Al}_2(\text{SO}_4)_3$ completely dissociates, i.e.



then $i = 5$. The use of Eqn (12.1-28) gives

$$0 - T = (5)(1.86)(0.667) = 6.203^\circ\text{C.}$$

Thus, the freezing point temperature is $T = -6.203^\circ\text{C}$.

12.1.3 Solid-Liquid Equilibrium Phase Diagrams

Solid-liquid equilibrium data are obtained experimentally by cooling a liquid mixture of known composition and recording the temperature continuously as a function of time.⁸ A break point in this curve indicates the formation of a solid phase. The temperature at this point is the solid-liquid equilibrium temperature, and the composition of the liquid mixture gives the solubility of solute in the solvent. SLE data are generally presented in the form of temperature versus solubility phase diagrams under isobaric conditions. Binary SLE phase diagrams can be classified as (i) systems exhibiting eutectic behavior and (ii) systems forming solid solutions.

12.1.3.1 Systems Exhibiting Eutectic Behavior

Among the phase diagrams exhibiting eutectic behavior, the simplest one is shown in Fig. 12.3a. The temperatures at points A and C represent the freezing (or melting) temperatures of pure 1 and pure 2, respectively. Only homogeneous liquid phase exists over the curve AEC. The curve AE represents the *liquidus curve* on which pure solid 1 is in equilibrium with the liquid mixture. Similarly, the curve CE is the liquidus curve on which pure solid 2 is in equilibrium with the liquid mixture. In

⁸Besides this traditional technique of cooling curve and visual measurement, the differential scanning calorimetry (DSC) method is also used (Kousksou *et al.*, 2007).

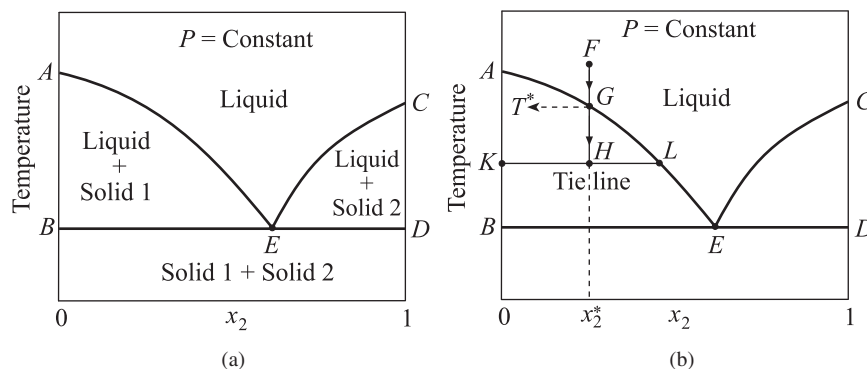


Figure 12.3: Solid-liquid phase diagram for a binary mixture.

other words, while the curve AE can be regarded as the solubility curve of solid 1 in solvent 2, the curve CE represents the solubility curve of solid 2 in solvent 1. Therefore, liquidus curves are determined from the solubility equation, Eqn (12.1-17). The curves AE and CE intersect at the point E, known as the *eutectic point*. At the eutectic point, the liquid phase is in equilibrium with the two solid phases. Since their compositions are the same, no separation is possible at the eutectic point. The corresponding eutectic temperature represents the minimum freezing (or melting) temperature of the mixture. Note that the eutectic point is analogous to the azeotropic point in vapor-liquid equilibrium. The curve BED represents the *solidus curve* below which nonhomogeneous two-phase solid consisting of pure 1 and pure 2 exists.

Consider a mixture at point F as shown in Fig. 12.3b. As this liquid mixture is cooled at constant pressure, pure solid 1 forms when the temperature drops to T^* (point G) and the relationship between T^* and x_2^* is given by Eqn (12.1-17), i.e.

$$\ln [\gamma_1(1 - x_2^*)] = \frac{\Delta \tilde{H}_1^{\text{fus}}}{RT_{m1}} \left(1 - \frac{T_{m1}}{T^*} \right). \quad (12.1-31)$$

As the temperature is further decreased, pure solid 1 continues to form and the liquid mixture becomes richer in component 2. For example, at point H, the amount of solid 1 and liquid mixture can be calculated by the application of the lever rule in the form

$$\frac{\text{Amount of solid 1}}{\text{Amount of liquid mixture}} = \frac{\overline{HL}}{\overline{HK}}. \quad (12.1-32)$$

Example 12.5 Prepare a solid-liquid phase diagram for naphthalene (1) and *p*-dichlorobenzene (2) if the mixture is represented by the regular mixture theory. The following data are provided for this system:

Component	T_m (K)	$\Delta\tilde{H}_i^{\text{fus}}$ (J/mol)	$\tilde{V}^L$ (cm ³ /mol)	δ (J/cm ³) ^{1/2}
Naphthalene	353.35	18,803	125.03	20.3
<i>p</i> -dichlorobenzene	326.24	18,082	101.8	19.8

Solution

The prediction of the liquidus curve requires the determination of the solubility of solid i in solvent j . Assuming $\Delta\tilde{C}_P^{\text{LS}} \simeq 0$, Eqn (12.1-17) gives the solubility of solid i in solvent j in the form

$$\ln \gamma_i + \ln x_i = \frac{\Delta\tilde{H}_i^{\text{fus}}}{RT_{m_i}} \left(1 - \frac{T_{m_i}}{T} \right) \quad (1)$$

in which the activity coefficient of component i is given by Eqns (8.5-2) and (8.5-3) as

$$\ln \gamma_i = \frac{\tilde{V}_i^L (\delta_j - \delta_i)^2}{RT} \left(\frac{x_j \tilde{V}_j^L}{x_j \tilde{V}_j^L + x_i \tilde{V}_i^L} \right)^2. \quad (2)$$

Thus, the solubility of solid naphthalene in *p*-dichlorobenzene ($i = 1, j = 2$) is

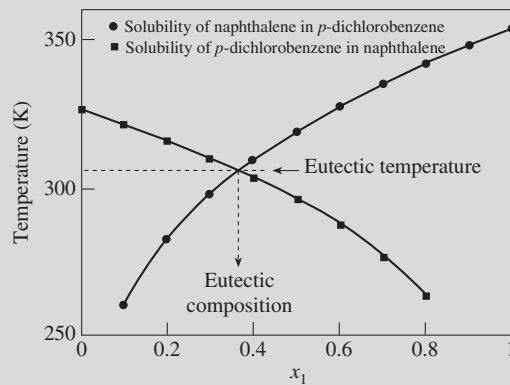
$$\begin{aligned} & \frac{(125.03)(19.8 - 20.3)^2}{8.314T} \left(\frac{101.8x_2}{101.8x_2 + 125.03x_1} \right)^2 + \ln x_1 \\ &= \frac{18,803}{(8.314)(353.35)} \left(1 - \frac{353.35}{T} \right). \end{aligned} \quad (3)$$

On the other hand, the solubility of solid *p*-dichlorobenzene in naphthalene ($i = 2, j = 1$) is

$$\begin{aligned} & \frac{(101.8)(20.3 - 19.8)^2}{8.314T} \left(\frac{125.03x_1}{101.8x_2 + 125.03x_1} \right)^2 + \ln x_2 \\ &= \frac{18,082}{(8.314)(326.24)} \left(1 - \frac{326.24}{T} \right). \end{aligned} \quad (4)$$

The easiest way of preparing a solid-liquid equilibrium diagram is to give values to x_1 (or x_2) and to calculate the corresponding temperatures from Eqns (3) and (4). The results presented below are plotted in the following figure.

x_1	T (K)	
	Solid naphthalene in p -dichlorobenzene	Solid p -dichlorobenzene in naphthalene
0	—	326.24
0.1	260.2	321.2
0.2	282.6	315.7
0.3	297.6	309.7
0.4	309.3	303.1
0.5	318.9	295.6
0.6	327.3	287.0
0.7	334.7	276.5
0.8	341.5	263.1
0.9	347.6	242.8
1.0	353.35	—



From the point of intersection of the liquidus curves, composition and temperature at the eutectic point are $x_1 = 0.36$ and 306 K, respectively. The experimental values are $x_1 = 0.39$ and 302.85 K (Wei and Jin, 2009).

Comment: A similar problem can be rephrased as follows: “A liquid mixture consisting of 20 mol% naphthalene (1) and 80% p -dichlorobenzene is cooled at constant pressure. Estimate the temperature at which the solid phase first appears and identify the solid.”

For this purpose it is necessary to calculate the freezing (or melting) temperatures of naphthalene and p -dichlorobenzene at the given composition. For naphthalene from Eqn (3)

$$\begin{aligned}
 & \frac{(125.03)(19.8 - 20.3)^2}{8.314T} \left[\frac{(101.8)(0.8)}{(101.8)(0.8) + (125.03)(0.2)} \right]^2 + \ln(0.2) \\
 &= \frac{18,803}{(8.314)(353.35)} \left(1 - \frac{353.35}{T} \right) \Rightarrow T = 282.6 \text{ K.} \quad (5)
 \end{aligned}$$

For *p*-dichlorobenzene from Eqn (4)

$$\begin{aligned} & \frac{(101.8)(20.3 - 19.8)^2}{8.314T} \left[\frac{(125.03)(0.2)}{(101.8)(0.8) + (125.03)(0.2)} \right]^2 + \ln(0.8) \\ &= \frac{18,082}{(8.314)(326.24)} \left(1 - \frac{326.24}{T} \right) \Rightarrow T = 315.7 \text{ K.} \end{aligned} \quad (6)$$

Therefore, pure *p*-dichlorobenzene solidifies first at 315.7 K. Further cooling will result in an increase in the concentration of naphthalene in the solution.

12.1.3.2 Systems Forming Solid Solution

The components of a solid solution completely dissolve in one another in a manner similar to liquids forming homogeneous solution. As a result, the system does not have a eutectic point and the solid-liquid phase diagram shown in Fig. 12.4 resembles the temperature-composition diagram for a vapor-liquid mixture, i.e. Fig. 9.3. The two-phase region in the middle is bounded by the liquidus and solidus curves. Contrary to the case exhibiting eutectic behavior, components 1 and 2 are completely miscible in the solid phase.

12.2 Colligative Properties

Colligative means *depending on the number of particles and not on the nature of particles*. Colligative properties can only be applied to solutions and, as the name

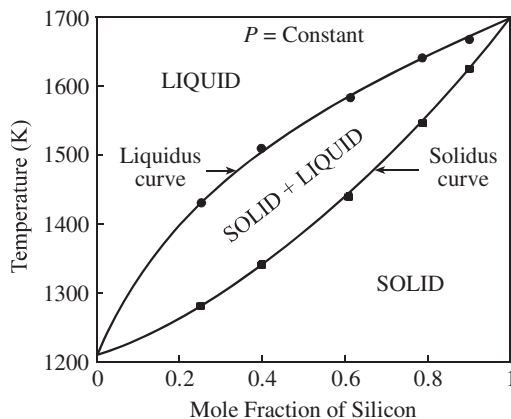


Figure 12.4: Phase diagram of germanium and silicon (Thurmond, 1953).

implies, they are independent of the identity of the solute. In Section 12.1.2, we showed that the freezing point depression is a colligative property. Now, we'll consider two other major colligative properties, namely boiling point elevation and osmotic pressure.

12.2.1 Boiling Point Elevation

The boiling point of a pure liquid is the temperature at which the vapor pressure of the liquid is equal to the surrounding pressure. Consider a pure solvent (1) at a given temperature T . At the surface of a liquid there is competition between the kinetic energy of the molecules and the intermolecular forces. If the kinetic energy of the molecules is high enough to overcome the intermolecular forces, then the molecules will escape into the gas phase. In the meantime, gas molecules may return to the liquid phase when they collide with the surface. When the solvent is in equilibrium with the gas phase above it, then the partial pressure of the solvent in the gas phase is P_1^{vap} . In this case the number of molecules leaving the surface is equal to those returning to the surface.

When a nonvolatile solute (2) is added to the solvent, some of the solvent molecules on the surface of the liquid will be replaced by solute molecules. Since there are fewer solvent molecules on the surface to escape, then the vapor pressure goes down. Since x_2 is small, x_1 is close to unity and we can use Raoult's law for the solvent, i.e.

$$Py_1 = x_1 P_1^{\text{vap}} = (1 - x_2) P_1^{\text{vap}} \quad (12.2-1)$$

indicating a decrease in the partial pressure of solvent.

The decrease in vapor pressure implies an increase in the boiling temperature. To quantify the elevation of boiling point, let us consider the condition of equilibrium between vapor and liquid phases. Since the vapor phase contains only solvent, then we can write

$$f_1^{\text{V}}(T, P) = \hat{f}_1^{\text{L}}(T, P, x_1) = \gamma_1 x_1 f_1^{\text{L}}(T, P) \quad (12.2-2)$$

or

$$\ln(\gamma_1 x_1) = \ln \left[\frac{f_1^{\text{V}}(T, P)}{f_1^{\text{L}}(T, P)} \right]. \quad (12.2-3)$$

Fugacities of pure solvent in the vapor and liquid phases are related to molar Gibbs energies by Eqn (5.2-1), i.e.

$$\ln \left[\frac{f_1^{\text{V}}(T, P)}{P} \right] = \frac{\tilde{G}_1^{\text{V}}(T, P) - \tilde{G}_1^{\text{IG}}(T, P)}{RT}, \quad (12.2-4)$$

$$\ln \left[\frac{f_1^L(T, P)}{P} \right] = \frac{\tilde{G}_1^L(T, P) - \tilde{G}_1^{\text{IG}}(T, P)}{RT}. \quad (12.2-5)$$

Therefore,

$$\begin{aligned} \ln \left[\frac{f_1^V(T, P)}{f_1^L(T, P)} \right] &= \frac{\tilde{G}_1^V(T, P) - \tilde{G}_1^L(T, P)}{RT} = \frac{\Delta \tilde{G}_1^{\text{vap}}(T, P)}{RT} \\ &= \frac{\Delta \tilde{H}_1^{\text{vap}}(T, P) - T \Delta \tilde{S}_1^{\text{vap}}(T, P)}{RT}. \end{aligned} \quad (12.2-6)$$

The use of Eqn (12.2-6) in Eqn (12.2-3) leads to

$$\ln(\gamma_1 x_1) = \frac{\Delta \tilde{H}_1^{\text{vap}}(T, P) - T \Delta \tilde{S}_1^{\text{vap}}(T, P)}{RT}. \quad (12.2-7)$$

The enthalpy and entropy of vaporization at the normal boiling point T_b are generally known. To find the enthalpy and entropy of vaporization at any given temperature, it is necessary to devise an alternative path as shown in Fig. 12.5.

The change in enthalpy is expressed as

$$\Delta \tilde{H}_1^{\text{vap}}(T) = \int_T^{T_b} \tilde{C}_{P_1}^L dT + \Delta \tilde{H}_1^{\text{vap}}(T_b) + \int_{T_b}^T \tilde{C}_{P_1}^V dT \quad (12.2-8)$$

or

$$\Delta \tilde{H}_1^{\text{vap}}(T) = \Delta \tilde{H}_1^{\text{vap}}(T_b) + \int_{T_b}^T \Delta \tilde{C}_{P_1}^{\text{VL}} dT, \quad (12.2-9)$$

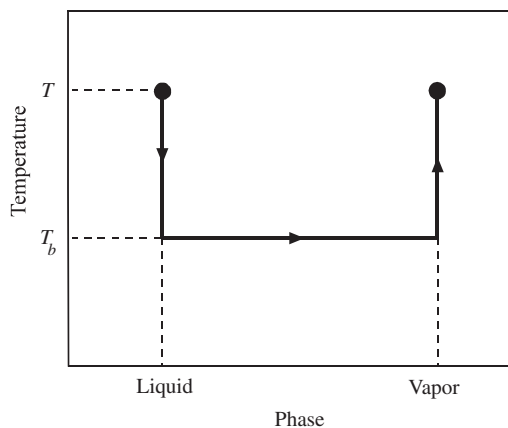


Figure 12.5: Calculation path for $\Delta \tilde{H}_1^{\text{vap}}(T)$ and $\Delta \tilde{S}_1^{\text{vap}}(T)$.

where

$$\Delta \tilde{C}_{P_1}^{\text{VL}} = \tilde{C}_{P_1}^{\text{V}} - \tilde{C}_{P_1}^{\text{L}}. \quad (12.2-10)$$

On the other hand, the entropy change is

$$\Delta \tilde{S}_1^{\text{vap}}(T) = \int_T^{T_b} \frac{\tilde{C}_{P_1}^{\text{L}}}{T} dT + \Delta \tilde{S}_1^{\text{vap}}(T_b) + \int_{T_b}^T \frac{\tilde{C}_{P_1}^{\text{V}}}{T} dT. \quad (12.2-11)$$

At a temperature of T_b , vapor and liquid phases of pure solvent are in equilibrium with each other, i.e.

$$\Delta \tilde{G}_1^{\text{vap}}(T_b) = \Delta \tilde{H}_1^{\text{vap}}(T_b) - T_b \Delta \tilde{S}_1^{\text{vap}}(T_b) = 0 \Rightarrow \Delta \tilde{S}_1^{\text{vap}}(T_b) = \frac{\Delta \tilde{H}_1^{\text{vap}}(T_b)}{T_b}. \quad (12.2-12)$$

The use of Eqn (12.2-12) in Eqn (12.2-11) leads to

$$\Delta \tilde{S}_1^{\text{vap}}(T) = \frac{\Delta \tilde{H}_1^{\text{vap}}(T_b)}{T_b} + \int_{T_b}^T \frac{\Delta \tilde{C}_{P_1}^{\text{VL}}}{T} dT. \quad (12.2-13)$$

Substitution of Eqns (12.2-9) and (12.2-13) into Eqn (12.2-7) gives

$$\ln(\gamma_1 x_1) = \frac{\Delta \tilde{H}_1^{\text{vap}}(T_b)}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right) + \frac{1}{RT} \int_{T_b}^T \Delta \tilde{C}_{P_1}^{\text{VL}} dT - \frac{1}{R} \int_{T_b}^T \frac{\Delta \tilde{C}_{P_1}^{\text{VL}}}{T} dT. \quad (12.2-14)$$

If heat capacities are independent of temperature, Eqn (12.2-14) simplifies to

$$\ln(\gamma_1 x_1) = \frac{\Delta \tilde{H}_1^{\text{vap}}}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right) + \frac{\Delta \tilde{C}_{P_1}^{\text{VL}}}{R} \left[1 - \frac{T_b}{T} + \ln \left(\frac{T_b}{T} \right) \right]. \quad (12.2-15)$$

The term $\ln(T_b/T)$ is expanded in Taylor series as

$$\ln \left(\frac{T_b}{T} \right) = \left(\frac{T_b}{T} - 1 \right) - \frac{1}{2} \left(\frac{T_b}{T} - 1 \right)^2 + \frac{1}{3} \left(\frac{T_b}{T} - 1 \right)^3 - \dots \quad (12.2-16)$$

Negligible when T and T_b do not differ greatly

If solute (2) is dilute, the solvent (1) can be treated as an ideal solution, i.e. $\gamma_1 = 1$, and the boiling point elevation is small, i.e. $T \simeq T_b$. Consequently, Eqn (12.2-15) reduces to

$$\ln x_1 = \ln(1 - x_2) = \frac{\Delta \tilde{H}_1^{\text{vap}}}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right). \quad (12.2-17)$$

Table 12.2: Boiling point depression constants for various solvents.

Solvent	Normal Boiling Point (°C)	K_b [°C/(mol/kg)]
Acetic acid	118.1	3.07
Benzene	80.1	2.63
CCl ₄	76.8	5.22
Chloroform	61.3	3.63
Ethanol	78.4	1.23
Water	100	0.513

Further simplification of Eqn (12.2-17) with $\ln(1 - x_2) \simeq -x_2$ and $TT_b \simeq T_b^2$ leads to

$$T - T_b = \frac{RT_b^2}{\Delta \tilde{H}_1^{\text{vap}}} x_2 \quad (12.2-18)$$

indicating that the boiling point elevation is independent of the chemical nature of a solute. Equation (12.2-18) can be used not only for calculating the boiling point elevation, $T - T_b$, but also for determining the heat of vaporization (or latent heat) of a solvent, $\Delta \tilde{H}_1^{\text{vap}}$, or the molecular weight of a solute.

The mole fraction of solute, x_2 , is related to the molality of solute, m_2 , by Eqn (12.1-26). Substitution of Eqn (12.1-26) into Eqn (12.2-18) gives

$$T - T_b = \left(\frac{RT_b^2}{1000 \Delta \hat{H}_1^{\text{vap}}} \right) m_2. \quad (12.2-19)$$

In an analogous fashion to Eqn (12.1-28), Eqn (12.2-19) is usually expressed as

$$T - T_b = iK_b m_2, \quad (12.2-20)$$

where i is the *van't Hoff factor* taking into account the degree of dissociation of the solute in the solvent, and K_b is the *boiling point elevation constant* of a solvent defined by

$$K_b = \frac{RT_b^2}{1000 \Delta \hat{H}_1^{\text{vap}}}. \quad (12.2-21)$$

Boiling point elevation constants for some solvents are given in Table 12.2.

Example 12.6 Calculate the boiling point of a solution prepared by dissolving 68.4 g of sucrose (molecular weight = 342 g/mol) in 1000 g of water.

Solution

The molality of sucrose is

$$m_2 = \frac{68.4/342}{1} = 0.2 \text{ mol/kg.}$$

Sucrose is a nonelectrolyte, i.e. $i = 1$, and $K_b = 0.513$ for water. The use of Eqn (12.2-20) gives

$$T - 100 = (1)(0.513)(0.2) = 0.103^\circ\text{C}.$$

Therefore, the boiling point is $T = 100.103^\circ\text{C}$.

Example 12.7 Calculate the boiling point of seawater if the concentration of NaCl is 3.5% by weight.

Solution

The molality of salt (molecular weight = 58.5 g/mol) is

$$m_2 = \frac{3.5/58.5}{(100 - 3.5)/1000} = 0.62 \text{ mol/kg.}$$

If we assume that the NaCl completely dissociates into sodium and chloride,⁹ i.e.



then $i = 2$. The use of Eqn (12.2-20) gives

$$T - 100 = (2)(0.513)(0.62) = 0.636^\circ\text{C}.$$

Therefore, seawater boils at 100.636°C .

12.2.2 Osmosis

Consider two compartments separated by a semipermeable membrane as shown in Fig. 12.6. While the compartment on the left is filled with pure solvent (1), the other compartment contains a dilute amount of solute (2) in solvent (1). The membrane allows passage of solvent but prevents flow of the solute.

Since the fugacity of pure solvent in the left compartment is greater than the fugacity of solvent in the solution, solvent molecules start passing from the left to the right compartment until the fugacities (or partial molar Gibbs energies) become

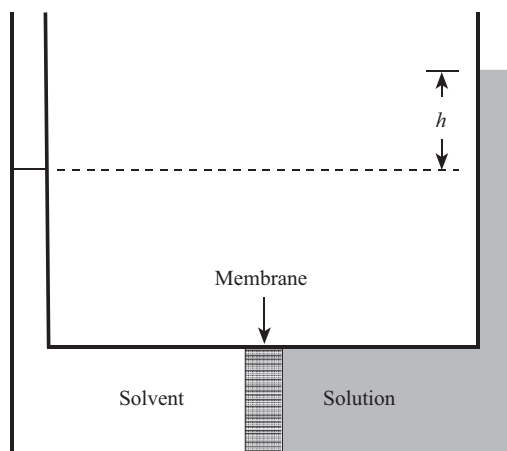


Figure 12.6: Osmotic pressure at equilibrium.

equal to each other. The transport of a pure solvent into a concentrated solution through a semipermeable membrane is called *osmosis*.¹⁰ In general, solvent moves from a solution with lower solute concentration into a solution with higher solute concentration.

When the movement of solvent stops, i.e. the system reaches equilibrium, the pressure of the solution, P^{soln} , is greater than the pressure of the pure solvent, P . This extra pressure, i.e. $P^{\text{soln}} - P$, is called the *osmotic pressure*, Π , i.e.

$$\Pi = P^{\text{soln}} - P = \rho_{\text{soln}}gh. \quad (12.2-22)$$

The flow direction of the solvent can only be reversed by applying a pressure in excess of osmotic pressure to the compartment on the right-hand side. This process is called *reverse osmosis*.

At the membrane, the condition of equilibrium indicates that the fugacity of pure solvent is equal to the fugacity of solvent in the solution, i.e.

$$f_1^{\text{L}}(T, P) = \hat{f}_1^{\text{soln}}(T, P^{\text{soln}}, x_1) = \gamma_1 x_1 f_1^{\text{L}}(T, P^{\text{soln}}) \quad (12.2-23)$$

or

$$-\ln(\gamma_1 x_1) = \ln \left[\frac{f_1^{\text{L}}(T, P^{\text{soln}})}{f_1^{\text{L}}(T, P)} \right]. \quad (12.2-24)$$

¹⁰The word “osmosis” comes from the Greek word for “push.”

The use of Eqn (5.4-11) gives

$$\frac{f_1^L(T, P^{\text{soln}})}{f_1^L(T, P)} = \exp \left[\frac{\tilde{V}_1(P^{\text{soln}} - P)}{RT} \right]. \quad (12.2-25)$$

Substitution of Eqn (12.2-25) into Eqn (12.2-24) yields

$$-\ln(\gamma_1 x_1) = \frac{\tilde{V}_1(P^{\text{soln}} - P)}{RT}. \quad (12.2-26)$$

If solute is dilute enough, the mixture forms an ideal solution, i.e. $\gamma_1 \simeq 1$, and the left-hand side of Eqn (12.2-26) is simplified as

$$\ln(\gamma_1 x_1) \simeq \ln x_1 = \ln(1 - x_2) \simeq -x_2. \quad (12.2-27)$$

Therefore, Eqn (12.2-26) becomes

$$\Pi = \frac{RTx_2}{\tilde{V}_1} \quad (12.2-28)$$

indicating that the osmotic pressure of a solution depends on the number (or mole fraction) of solute in the solution and not on the chemical structure of the solute. In the literature, Eqn (12.2-28) is known as the *van't Hoff equation for osmotic pressure*.

The mole fraction of solute is expressed as

$$x_2 = \frac{n_2}{n_1 + n_2} \simeq \frac{n_2}{n_1}. \quad (12.2-29)$$

On the other hand, molar volume of solvent is given by

$$\tilde{V}_1 = \frac{V_1}{n_1}. \quad (12.2-30)$$

Substitution of Eqns (12.2-29) and (12.2-30) into Eqn (12.2-28) gives

$$\Pi = \left(\frac{n_2}{V_1} \right) RT, \quad (12.2-31)$$

which is analogous to the ideal gas equation of state, i.e. $P = (n/V)RT$. Note that the term n_2/V_1 represents the molarity of solute.

Example 12.8 Calculate the osmotic pressure of 1 M solution of any nonelectrolyte, i.e. glucose, dextrose, or sucrose, in water at 298 K.

Solution

Taking $R = 8.314 \times 10^{-2}$ bar L/(mol K), the use of Eqn (12.2-31) leads to

$$\Pi = (1)(8.314 \times 10^{-2})(298) = 24.8 \text{ bar.}$$

Comment: How fast this osmotic pressure is generated is not a subject in thermodynamics.

For ionized solutes, Eqn (12.2-31) is modified as

$$\Pi = i \left(\frac{n_2}{V_1} \right) RT, \quad (12.2-32)$$

where i is the van't Hoff factor defined by Eqn (12.1-30).

Example 12.9 List the osmotic pressures of the following solutions in decreasing order: 0.1 M NaNO₃, 0.1 M KNO₃, 0.1 M NaCl, 0.1 M glucose, 0.1 M K₂SO₄.

Solution

Since molarities are equal, osmotic pressure will be dependent on the van't Hoff factor, i . The van't Hoff factors of the species are:

Species	NaNO ₃	KNO ₃	NaCl	Glucose	K ₂ SO ₄
i	2	2	2	1	3

Therefore,

$$\Pi_{\text{glucose}} < \Pi_{\text{NaNO}_3} = \Pi_{\text{KNO}_3} = \Pi_{\text{NaCl}} < \Pi_{\text{K}_2\text{SO}_4}.$$

Example 12.10 It is required to obtain fresh water from seawater by reverse osmosis. Seawater contains about 3.5 wt.% NaCl.

- What is the minimum pressure required to carry out this process at 298 K?
- Estimate the minimum work required to obtain 10 L of fresh water.

Solution

(a) Taking the density of water as 1 g/cm³ and the molecular weight of NaCl as 58.5 g/mol, the use of Eqn (12.2-32) gives

$$\Pi = (2) \underbrace{\left[\frac{(3.5/58.5)}{(96.5/1) \times 10^{-3}} \right]}_{\frac{\text{mol}}{\text{L}}} \underbrace{(8.314 \times 10^{-2})}_{\frac{\text{bar L}}{\text{mol K}}} \underbrace{(298)}_{\text{K}} = 30.7 \text{ bar.}$$

The process of reverse osmosis requires an applied pressure just in excess of the osmotic pressure. The minimum applied pressure is then 30.7 bar. In practice, desalination of seawater is carried out at 50–60 bar.

(b) The minimum work required is

$$W = -\Pi \Delta V = -(30.7)(10 \times 10^{-3}) = -0.307 \text{ bar m}^3 = -30.7 \text{ kJ}.$$

The minus sign indicates that the work must be supplied by external means.

Comment: In the case of a batch process, the solution concentration increases as the water is removed. Hence, the applied pressure must be continuously increased.

Note that Eqn (12.2-31) can also be expressed in the form

$$\Pi = \left(\frac{m_2/M_2}{V_1} \right) RT. \quad (12.2-33)$$

If c_2 is the mass of solute per unit volume of solvent, i.e. m_2/V_1 , then Eqn (12.2-33) reduces to

$$\Pi = \frac{RTc_2}{M_2}. \quad (12.2-34)$$

It should be kept in mind that Eqn (12.2-34) is only valid for very dilute solutions, i.e.

$$\lim_{c_2 \rightarrow 0} \frac{\Pi}{c_2} = \frac{RT}{M_2}. \quad (12.2-35)$$

For finite concentrations, expansion of the right-hand side of Eqn (12.2-34) by power series in terms of c_2 gives

$$\frac{\Pi}{c_2} = RT \left(\frac{1}{M_2} + B^*c_2 + C^*c_2^2 + \cdots \right), \quad (12.2-36)$$

where B^* and C^* are the second and third virial coefficients, respectively. Once the osmotic pressure data are obtained as a function of solute concentration, a plot of reduced osmotic pressure, $(\Pi/RT)/c_2$, versus c_2 gives a straight line with slope B^* and an intercept $1/M_2$. A negative value of B^* is indicative of a good solvent. A positive value of B^* indicates the solute is insoluble in the solvent.

Measurement of osmotic pressure is often used to determine the molecular weights of macromolecules. The instrument for carrying out these measurements

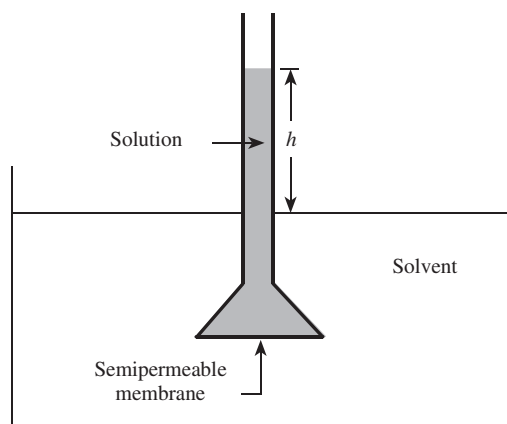


Figure 12.7: A simple osmometer.

is called an *osmometer*. In a simple osmometer, shown in Fig. 12.7, the pressure generated across a semipermeable membrane with solvent on one side and the solution in question on the other is measured in terms of the height of a solution. The molecular weight of the solute is then calculated from Eqn (12.2-34).

Example 12.11 To determine the molecular weight of insulin, 0.58 g of insulin is first dissolved in 100 mL of water. It is then placed in an osmometer operating at 291 K. If the height of the dilute solution rises 28 mm above pure water, determine the molecular weight of insulin.

Solution

The osmotic pressure is

$$\Pi = \rho gh = (1000)(9.8)(28 \times 10^{-3}) = 274.4 \text{ Pa.}$$

The mass concentration of the insulin is

$$c_2 = \frac{0.58}{100} = 5.8 \times 10^{-3} \text{ g/cm}^3 = 5.8 \times 10^3 \text{ g/m}^3.$$

From Eqn (12.2-34)

$$M_2 = \frac{RTc_2}{\Pi} = \frac{(8.314)(291)(5.8 \times 10^3)}{274.4} = 51,138 \text{ g/mol.}$$

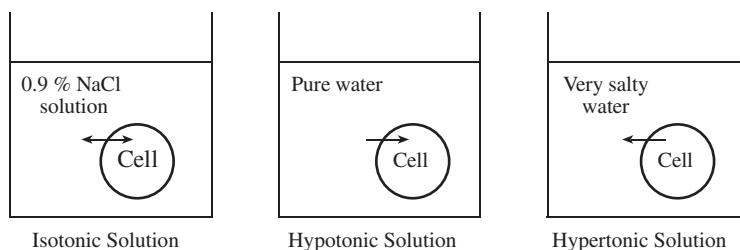


Figure 12.8: Direction of water movement as a result of osmosis.

12.2.2.1 Applications of Osmosis

Biology: Cells are composed of three parts: membrane, cytoplasm, and nucleus. The watery or aqueous component of the cytoplasm is the cytosol, which includes ions and soluble macromolecules, for example enzymes. The insoluble constituents of the cytoplasm include the organelles and the cytoskeleton, which gives to the cytoplasm a gel-like structure and consistency.

Solutions that contain the same concentration of water and solutes as the cell cytoplasm are called *isotonic* solutions. Cells placed in an isotonic solution will neither shrink nor swell since there is no net gain or loss of water. Cell cytoplasm, body fluids (including blood, sweat, and tears), and 0.9 wt.% NaCl solution contain approximately the same solute concentration. Therefore, they are all isotonic solutions and blood freezes at -0.52°C lower than water.¹¹

Solutions that contain a lower concentration of solute than the cytoplasm are said to be *hypotonic* solutions. When a cell is placed in a hypotonic solution, the water diffuses into the cell as a result of osmosis, causing the cell to swell and possibly explode. This phenomenon is called *lysis*. On the other hand, solutions that contain a higher concentration of solute than the cell cytoplasm are called *hypertonic* solutions. When a cell is placed in a hypertonic solution, the water diffuses out of the cell as a result of osmosis, causing the cell to dehydrate and collapse. For example, people drinking ocean water could die as a result of dehydration.

The three cases described above are schematically shown in Fig. 12.8. Note that the body loses water and minerals during physical activity. To make up for these losses, an isotonic beverage, i.e. Gatorade[®], must be drunk.

Example 12.12 If a red blood cell is placed in a large volume of 0.25 M NaCl solution, what happens to the volume of the cell?

¹¹Taking $i = 1.8$, $T = -(1.8)(1.86) \left(\frac{0.9/58.5}{0.0991} \right) = -0.52^\circ\text{C}$.

Solution

The concentration of NaCl in the cell is approximately 0.9 wt.%. The corresponding molarity is

$$\frac{0.9/58.5}{[(100 - 0.9)/1] \times 10^{-3}} = 0.155 \text{ M.}$$

Since the cell is exposed to a hypertonic solution, water moves out of the cell with a concomitant decrease in cell volume. The cell shrinks until the inside solute concentration reaches 0.25 M. Since the total mass of solutes within the cell does not change, then

$$\frac{V_f}{V_i} = \frac{0.155}{0.25} = 0.62,$$

where V_f and V_i represent the cell volume at the final and initial states.

When making pickles, the use of hypertonic salt solution is preferred since it causes the cucumbers to lose water as the lack of water prevents bacterial activities. This is the reason why plain old cucumbers become infested with mold if they remain in the refrigerator over a long period of time.

Osmotic power plant: The world's first osmotic power plant has been in operation in Norway since 2009. In the prototype built by Statkraft, seawater and fresh water, filtered to remove silt, are fed into pipes leading to membrane modules, made up of spiral coils to maximize surface area. In the membrane module, 80–90% of fresh water is transferred by osmosis across the membrane into the seawater. As a result, the volumetric flow rate and pressure on the seawater side increase. The seawater leaving the membrane module is then sent to a hydropower turbine to generate electricity. The details of the plant can be found on Statkraft's web page.¹² Interested readers may also refer to Skilhagen *et al.* (2008) and Gerstandt *et al.* (2008).

Osmotic pump: The most general form of an osmotic pump is shown in Fig. 12.9. A collapsible partition separates the drug and osmotic driving agent chambers. When the osmotic pump is placed in an aqueous medium, water moves across the semipermeable membrane into the chamber containing osmotic driving agent, leading to an increase in pressure. As a result of the pressure exerted on the collapsible partition, the contents of the drug compartment are displaced through the delivery orifice. Osmotic pumps manufactured by ALZET®¹³ are extensively used for research in mice, rats, and other laboratory animals.

¹²<http://www.statkraft.com>.

¹³<http://www.alzet.com>.

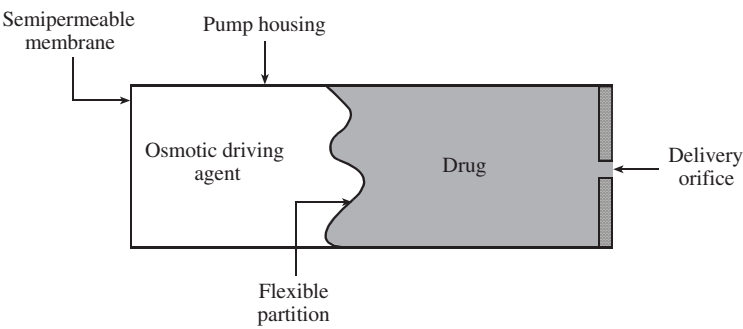


Figure 12.9: A typical osmotic pump (Theeuwes and Yum, 1976).

Problems

Problems Related to Section 12.1

12.1 Using the following experimental data for the solubility of drug (2) in a solvent (1) as a function of temperature, estimate the heat of fusion of the drug. The molecular weights of the solvent and drug are 72.1 and 284.5 g/mol, respectively.

<i>S</i> (g/100 g solvent)	<i>T</i> (°C)	<i>S</i> (g/100 g solvent)	<i>T</i> (°C)	<i>S</i> (g/100 g solvent)	<i>T</i> (°C)
0.725	8.5	2.274	18.5	5.190	26.1
1.547	15.0	2.770	20.7	7.441	29.5

(Answer: 77.7 kJ/mol)

12.2 For a binary system of stearic acid (2) and acetone (1), Goff *et al.* (2005) estimated the following three-suffix Margules parameters from VLE measurements:

$$A = 0.045, \quad B = 0.585$$

Estimate the solubility of stearic acid in acetone at 323 K. The heat of fusion and the melting point temperature of stearic acid are 61,300 J/mol and 341 K, respectively.

(Answer: $x_2 = 0.315$)

12.3 Verduzco *et al.* (2007) studied solid-liquid equilibria of dibenzothiophene (2) in cyclohexane (1). The properties of dibenzothiophene are

$$\Delta \tilde{C}_P^{\text{LS}} = 31.4 \text{ J/mol K}, \quad T_{\text{m}} = 372 \text{ K}, \quad \Delta \tilde{H}^{\text{fus}} = 21,000 \text{ J/mol}.$$

The activity coefficients of this binary liquid mixture are represented by the Wilson model with

$$\lambda_{12} - \lambda_{11} = 1969 \text{ J/mol}, \quad \lambda_{21} - \lambda_{22} = 3709 \text{ J/mol}.$$

The molar volumes of liquid cyclohexane and dibenzothiophene are 119.8 and 163.6 cm³/mol, respectively. Calculate the solubility of dibenzothiophene in cyclohexane at 328.8 K. The experimental value is reported as 0.1291.

(Answer: 0.1302)

- 12.4** When the constituents of a mixture have high boiling points, determination of activity coefficients from vapor-liquid equilibrium experiments is difficult. In this case, the solid-liquid equilibrium data in the form of “solubility versus temperature” provide the means to estimate activity coefficients as a function of composition. The following experimental data are provided by [Verduzco et al. \(2007\)](#) for the solubility of dibenzothiophene (2) in *n*-hexadecane (1) at atmospheric conditions:

x_2	T (K)	x_2	T (K)	x_2	T (K)
0.0463	291.0	0.1008	315.0	0.3737	350.9
0.0475	292.2	0.1201	319.9	0.5021	356.9
0.0487	293.0	0.1783	331.5	0.5094	357.0
0.0504	293.7	0.2000	334.7	0.6047	360.1
0.0726	304.4	0.2565	341.5	0.6954	362.7

The properties of dibenzothiophene are reported as

$$\Delta \tilde{C}_P^{\text{LS}} = 31.4 \text{ J/mol K}, \quad T_m = 372 \text{ K}, \quad \Delta \tilde{H}^{\text{fus}} = 21,000 \text{ J/mol}.$$

Determine the activity coefficients of dibenzothiophene as a function of composition.

- 12.5** When $\Delta \tilde{C}_{P_2}^{\text{LS}} \neq 0$, calculation of ideal solubility from Eqn (12.1-20) is valid as long as T and T_m do not differ greatly. Otherwise, the second and third terms on the right-hand side of Eqn (12.1-14) should be taken into consideration.

(a) If $\Delta \tilde{C}_{P_2}^{\text{LS}}$ varies linearly with temperature, i.e.

$$\Delta \tilde{C}_{P_2}^{\text{LS}} = \alpha + \beta T \quad (1)$$

show that the ideal solubility is given by

$$\ln x_2^{\text{ideal}} = \frac{\Delta \tilde{H}_2^{\text{fus}}}{RT_{m_2}} \left(1 - \frac{T_{m_2}}{T}\right) + \frac{(\beta T - \alpha)}{R} \left(1 - \frac{T_{m_2}}{T}\right) - \frac{\beta T}{2R} \left[1 - \left(\frac{T_{m_2}}{T}\right)^2\right] - \frac{\alpha}{R} \ln \left(\frac{T_{m_2}}{T}\right). \quad (2)$$

- (b) Use Eqn (2) and calculate the ideal solubility of paracetamol at 298.2 K. The following data are provided by [Hojjati and Rohani \(2006\)](#) for paracetamol:

$$\Delta \tilde{H}^{\text{fus}} = 28.1 \text{ kJ/mol}, \quad T_m = 442.2 \text{ K},$$

$$\Delta \tilde{C}_p^{\text{LS}} \text{ (J/mol K)} = 302.54 - 0.4589T.$$

Compare the calculated value with the one obtained from Eqn (12.1-20).

(Answer: (b) 0.093 and 0.025)

- 12.6** [Nti-Gyabaah et al. \(2008\)](#) reported the solubility of lovastatin (2) in *n*-butanol (1) as

<i>T</i> (K)	$x_2 \times 10^3$	<i>T</i> (K)	$x_2 \times 10^3$
285.70	4.6046	304.10	7.6196
288.70	4.9877	305.20	7.8606
290.65	5.2554	306.70	8.2028
296.20	6.1126	308.70	8.6850
301.20	7.0227	310.10	9.0410

The heat of fusion and the melting temperature of lovastatin are 43,136 J/mol and 445.5 K, respectively. It is required to express the infinite dilution activity coefficients as a function of temperature in the form

$$\ln \gamma_2^\infty = A + \frac{B}{T}. \quad (1)$$

- (a) If $\Delta \tilde{C}_{P_2}^{\text{LS}} = 255 \text{ J/mol K}$, use Eqn (12.1-16) with $\gamma_2 = \gamma_2^\infty$ (Why?) and show that

$$A = -3.898 \quad \text{and} \quad B = 1798.$$

- (b) Assume $\Delta \tilde{C}_{P_2}^{\text{LS}} = 0$ and use Eqn (12.1-19) to obtain

$$A = 8.456 \quad \text{and} \quad B = -2737.$$

- (c) If $\Delta\tilde{C}_{P_2}^{\text{LS}}$ is approximated to be equal to the entropy of fusion at the melting temperature, show that Eqn (12.1-16) simplifies to

$$\ln(\gamma_2 x_2) = -\frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} \ln\left(\frac{T_{m_2}}{T}\right). \quad (2)$$

Use Eqn (2) with $\gamma_2 = \gamma_2^\infty$ and show that

$$A = 3.765 \quad \text{and} \quad B = -1015.$$

- 12.7** For low solubility values, i.e. $x_2 \ll 1$, it is plausible to replace γ_2 by γ_2^∞ in Eqn (12.1-16). The temperature dependence of the infinite dilution activity coefficient can be expressed as

$$\ln \gamma_2^\infty = A + \frac{B}{T}. \quad (1)$$

- (a) Show that the substitution of Eqn (1) into Eqn (12.1-16) and rearrangement lead to

$$\ln x_2 = \alpha_1 + \frac{\alpha_2}{T} + \alpha_3 \ln T \quad (2)$$

where

$$\alpha_1 = \frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m_2}} - \frac{\Delta\tilde{C}_{P_2}^{\text{LS}}}{R} (1 + \ln T_{m_2}) - A, \quad (3)$$

$$\alpha_2 = \frac{\Delta\tilde{C}_{P_2}^{\text{LS}} T_{m_2}}{R} - \frac{\Delta\tilde{H}_2^{\text{fus}}}{R} - B, \quad (4)$$

$$\alpha_3 = \frac{\Delta\tilde{C}_{P_2}^{\text{LS}}}{R}. \quad (5)$$

Expressing solubility in the form of Eqn (2) is known as the *Apelblat model*.

- (b) Wang and Dang (2009) reported the following data for the solubility of erythromycin (2) in acetone (1) as a function of temperature:

$x_2 \times 10^4$	83.50	130.5	195.2	339.9	444.2	654.0
T (K)	298	303	308	313	318	323

Fit the data to Eqn (2) by using the method of least squares as explained in Example 6.15 and show that

$$\alpha_1 = -98.633, \quad \alpha_2 = 4358, \quad \alpha_3 = 14.747.$$

- 12.8** Irganox 1010 ($\text{C}_{73}\text{H}_{108}\text{O}_{12}$) is a widely used phenolic antioxidant in plastic, rubber, synthetic fiber, and elastomer. Its solubility in various solvents is of interest for the proper design of a crystallizer. [Wei *et al.* \(2009\)](#) reported that the binary system of *n*-hexane (1) and Irganox 1010 (2) follows the Wilson model with the following parameters:

$$\lambda_{12} - \lambda_{11} = 3289.60 \text{ J/mol} \quad \text{and} \quad \lambda_{21} - \lambda_{22} = 4482.60 \text{ J/mol}.$$

Estimate the solubility of Irganox 1010 in *n*-hexane at 328.15 K. The characteristic data for pure components are given by

Component	$\Delta \tilde{H}^{\text{fus}}$ (kJ/mol)	T_{m} (K)	$\tilde{V}^{\text{L}}$ (cm^3/mol)
Irganox 1010	65.95	385.75	1125.7
<i>n</i> -Hexane			134.9

The experimental value reported by [Wei *et al.* \(2009\)](#) is $x_2 = 3.703 \times 10^{-3}$.
(**Answer:** $x_2 = 3.625 \times 10^{-3}$)

- 12.9** [Nti-Gyabaah *et al.* \(2009\)](#) reported the solubility of simvastatin (2), a lipid-lowering agent, in methyl acetate (1) at 285.3 K as $x_2 = 18.86 \times 10^{-3}$. Using the following data, estimate the solubility of simvastatin in butyl acetate at the same temperature:

Data: Methyl acetate (1)-simvastatin (2) and butyl acetate (3)-simvastatin (2) mixtures are represented by the NRTL model with the following parameters:

$$g_{12} - g_{22} = 5516 \text{ J/mol}, \quad g_{21} - g_{11} = -369 \text{ J/mol},$$

$$g_{32} - g_{22} = 6718 \text{ J/mol}, \quad g_{23} - g_{33} = -1466 \text{ J/mol},$$

$$\alpha = 0.4.$$

(**Answer:** $x_2 = 24.48 \times 10^{-3}$)

- 12.10** Solubility of acenaphthene (2) in chloroform (1) at 305 K is reported by [Kotula and Marciniak \(2001\)](#) as $x_2 = 0.2322$. Estimate the solubility of acenaphthene in 1,1-dichloroethane (3) at the same temperature. Chloroform-acenaphthene and 1,1-dichloroethane-acenaphthene systems are represented by the regular mixture theory with the following parameters:

Component	$\tilde{V}^L$ (cm ³ /mol)	δ (J/cm ³) ^{1/2}
Chloroform	88.27	16.616
Acenaphthene	149.8	18.93
1,1-Dichloroethane	93.28	15.984

(Answer: $x_2 = 0.2066$)

12.11 Solubility of sulfur (2) in diethylene glycol dimethyl ether (1) at 359 K is reported by Sciamanna and Lynn (1988) as $x_2 = 0.0113$. It is required to estimate the solubility of sulfur in diethylene glycol (3) at the same temperature. The following data are given at 359 K:

- Average activity coefficient of sulfur in diethylene glycol dimethyl ether = 39.
- Average activity coefficient of sulfur in diethylene glycol = 452.

Estimate the solubility of sulfur in diethylene glycol at 359 K.

(Answer: $x_2 = 9.75 \times 10^{-4}$)

12.12 Prediction of solid-liquid equilibrium is of importance in the cryogenics industry. It is required to estimate the solubility of solid *n*-hexane (2) in liquid methane (1) at 150 K.

- (a) Estimate the solubility by using the regular mixture theory with the following data:

Component	$\Delta\tilde{H}^{\text{fus}}$ (kJ/mol)	T_m (K)	$\tilde{V}^L$ (cm ³ /mol)	δ (J/cm ³) ^{1/2}
Methane			52	11.6
<i>n</i> -Hexane	13.08	177.8	134.9	14.9

- (b) The activity coefficient of species *i* in a mixture is defined by Eqn (8.3-5) as

$$\gamma_i = \frac{\hat{\phi}_i^L(T, P, x_i)}{\phi_i^L(T, P)} \Rightarrow \ln \gamma_i = \ln \hat{\phi}_i^L(T, P, x_i) - \ln \phi_i^L(T, P). \quad (1)$$

If the liquid phase is represented by the Redlich-Kwong equation of state, show that the substitution of Eqns (5.4-12) and (7.6-2) into Eqn (1)

leads to

$$\begin{aligned} \ln \gamma_i = & \frac{B_i}{B_{\text{mix}}} (Z_{\text{mix}}^L - 1) - (Z_i^L - 1) + \ln \left(\frac{Z_i^L - B_i}{Z_{\text{mix}}^L - B_{\text{mix}}} \right) + \frac{A_{ii}}{B_i} \\ & \times \ln \left(1 + \frac{B_i}{Z_i^L} \right) - \frac{A_{\text{mix}}}{B_{\text{mix}}} \left(\frac{2 \sum_{j=1}^k x_j A_{ij}}{A_{\text{mix}}} - \frac{B_i}{B_{\text{mix}}} \right) \\ & \times \ln \left(1 + \frac{B_{\text{mix}}}{Z_{\text{mix}}^L} \right). \end{aligned} \quad (2)$$

Estimate the solubility of solid *n*-hexane in liquid methane using the Redlich-Kwong equation of state. De Mateo and Kurata (1975) reported that $k_{12} = 0.0567$. In the calculation of compressibility factors, use the vapor pressure of pure methane, which is 10.1 bar at 150 K, as the pressure.

(Answer: (a) $x_2 = 9.3 \times 10^{-2}$ and (b) $x_2 = 1.25 \times 10^{-2}$)

12.13 The “proof” of an alcoholic beverage is twice the volume percent of its alcoholic (ethyl alcohol) content. As a chemical engineer, you want to determine the temperature at which a bottle of vodka (65 proof) will first start to freeze.

- Use Eqn (12.1-28) to estimate the freezing point temperature.
- Take $\gamma_1 = 1$ and use Eqn (12.1-22) to estimate the freezing point temperature.
- Which equation yields more accurate result and why?
- Estimate the temperature at which a bottle will completely freeze.

Data:

Component	$\Delta \tilde{H}^{\text{fus}}$ (kJ/mol)	ρ (g/cm ³)	T_m (K)
Water	6.02	1	273.15
Ethyl alcohol	5.02	0.79	158.8

(Answer: (a) 257.8 K, (b) 259.5 K and (d) 153.3 K)

12.14 When 8 g of nonelectrolyte X is dissolved in 92 g of benzene, the observed freezing point is 2.8 °C. Estimate the molecular weight of X.

(Answer: 164.9 g/mol)

- 12.15** A liquid mixture contains 70 mol% benzoic acid (1) and 30% 1-naphthalenol (2). If this mixture is cooled at constant pressure, estimate the temperature at which the solid phase first appears and identify the solid. The mixture is represented by the Wilson equation with the following parameters (Huang and Chen, 2000):

$$\Lambda_{12} = 1.5484 \quad \text{and} \quad \Lambda_{21} = 0.6326.$$

For pure components

Component	$\Delta \tilde{H}^{\text{fus}}$ (J/mol)	T_m (K)
Benzoic acid	16,230	395.4
1-Naphthalenol	23,220	369.0

(Answer: 368.1 K, Benzoic acid)

- 12.16** A liquid mixture consisting of 80 mol% 1-naphthalenol (1) and 20% 2-naphthalenol (2) is slowly cooled at atmospheric pressure. At what temperature does a solid phase appear? Also identify the solid.

Data:

Component	$\Delta \tilde{H}^{\text{fus}}$ (J/mol)	T_m (K)
1-Naphthalenol	23,220	369.0
2-Naphthalenol	18,790	393.6

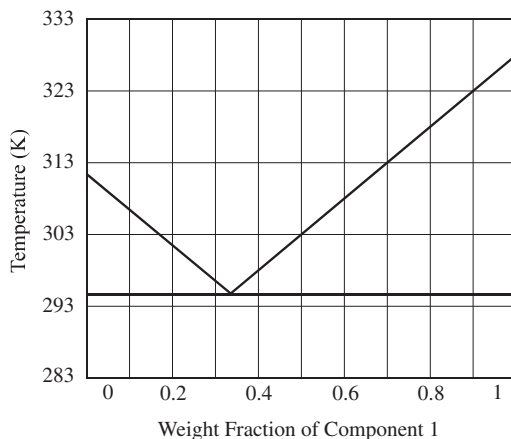
The activity coefficients are expressed in the form (Rastogi and Varma, 1958)

$$RT \ln \gamma_1 = x_2^2 [283 - 81(x_1 - x_2)(x_2 - 5x_1)],$$

$$RT \ln \gamma_2 = x_1^2 [283 - 81(x_1 - x_2)(5x_2 - x_1)].$$

(Answer: 358.7 K, 1-naphthalenol)

- 12.17** 180 kg of a liquid mixture consisting of 70 wt.% component 1 and 30% component 2 is fed to a crystallizer. Within the crystallizer, the temperature of the feed solution is slowly reduced from 323 K to 303 K and one-third of the liquid mixture is removed from the crystallizer. The contents of the crystallizer are then heated up to 313 K. Using the phase diagram given below, estimate the amount of component 1 crystals at the end of the overall process.



(Answer: 24 kg)

12.18 Consider a binary mixture of components 1 and 2 forming an ideal mixture when liquid, but are totally immiscible as solids.

(a) Show that the composition at the eutectic point, $(x_1)_e$, is given by

$$\frac{(x_1)_e^{1/\Delta\tilde{H}_1^{\text{fus}}}}{[1 - (x_1)_e]^{1/\Delta\tilde{H}_2^{\text{fus}}}} = \exp \left[\frac{1}{R} \left(\frac{1}{T_{m1}} - \frac{1}{T_{m2}} \right) \right]. \quad (1)$$

(b) Estimate the eutectic composition and temperature for a binary mixture of naphthalene (1) and biphenyl (2). The system may be considered to form an ideal mixture in the liquid phase as a result of nonpolar nature of both components and of their similar molar volumes.

Data:

Component	$\Delta\tilde{H}^{\text{fus}}$ (J/mol)	T_m (K)
Naphthalene	18,803	353.35
Biphenyl	19,900	343.3

(Answer: (b) $x_1 = 0.462$ and $T = 315.3$ K)

12.19 Consider an equilibrium between the liquid and solid phases of a multicomponent system. The condition of equilibrium is expressed as

$$\hat{f}_i^{\text{L}} = \hat{f}_i^{\text{S}} \Rightarrow f_i^{\text{L}} x_i^{\text{L}} \gamma_i^{\text{L}} = f_i^{\text{S}} x_i^{\text{S}} \gamma_i^{\text{S}}. \quad (1)$$

(a) Show that

$$\frac{x_i^S}{x_i^L} = \frac{\gamma_i^L}{\gamma_i^S} \exp \left[-\frac{\Delta \tilde{H}_i^{\text{fus}}}{RT_{m_i}} \left(1 - \frac{T_{m_i}}{T} \right) + \frac{1}{RT} \int_{T_{m_i}}^T \Delta \tilde{C}_{P_i}^{\text{LS}} dT - \frac{1}{R} \int_{T_{m_i}}^T \frac{\Delta \tilde{C}_{P_i}^{\text{LS}}}{T} dT \right]. \quad (2)$$

(b) For a multicomponent mixture, the activity coefficients for the liquid and solid phases can be expressed in terms of the solubility parameters based on regular mixture theory as

$$\ln \gamma_i^L = \frac{\tilde{V}_i^L (\langle \delta \rangle^L - \delta_i^L)^2}{RT} \quad \text{and} \quad \ln \gamma_i^S = \frac{\tilde{V}_i^S (\langle \delta \rangle^S - \delta_i^S)^2}{RT}, \quad (3)$$

where average solubility parameters, $\langle \delta \rangle$, are defined as a function of volume fractions, Φ , as

$$\langle \delta \rangle^L = \sum_i \Phi_i^L \delta_i^L \quad \text{and} \quad \langle \delta \rangle^S = \sum_i \Phi_i^S \delta_i^S, \quad (4)$$

$$\Phi_i^L = \frac{x_i^L \tilde{V}_i^L}{\sum_i x_i^L \tilde{V}_i^L} \quad \text{and} \quad \Phi_i^S = \frac{x_i^S \tilde{V}_i^S}{\sum_i x_i^S \tilde{V}_i^S}. \quad (5)$$

(c) Assume that the heat capacities and molar volumes of liquid and solid phases are, more or less, equal to each other, i.e.

$$\Delta \tilde{C}_{P_i}^{\text{LS}} = 0 \quad \text{and} \quad \tilde{V}_i^L = \tilde{V}_i^S = \tilde{V}_i. \quad (6)$$

Under these circumstances, show that Eqn (2) takes the form

$$\frac{x_i^S}{x_i^L} = \exp \left[-\frac{\Delta \tilde{H}_i^{\text{fus}}}{RT_{m_i}} \left(1 - \frac{T_{m_i}}{T} \right) + \frac{\tilde{V}_i}{RT} [(\langle \delta \rangle^L - \delta_i^L)^2 - (\langle \delta \rangle^S - \delta_i^S)^2] \right]. \quad (7)$$

(d) Consider solid-liquid equilibrium of a binary mixture of tetradecane (1) and hexadecane (2) at 283.8 K. The following empirical expressions are suggested by Pedersen *et al.* (1991) for the solubility parameters of paraffinic and naphthenic waxes:

$$\delta_i^L = 7.41 + 0.5914 \ln \left(\frac{n}{7} \right) \quad \text{and} \quad \delta_i^S = 8.50 + 5.763 \ln \left(\frac{n}{7} \right), \quad (8)$$

where δ_i is in $(\text{cal}/\text{cm}^3)^{1/2}$, and n denotes the carbon number. For $i = 1$ and $i = 2$, solve Eqn (7) simultaneously by using Newton's method and calculate the compositions of the liquid and solid phases.

Data:

Component	Formula	$\Delta\tilde{H}^{\text{fus}}$ (J/mol)	T_m (K)	$\tilde{V}$ (cm^3/mol)
Tetradecane	$\text{C}_{14}\text{H}_{30}$	45,070	278.45	259.9
Hexadecane	$\text{C}_{16}\text{H}_{34}$	47,545	291.05	292.8

(e) For an ideal mixture containing two species, show that

$$x_1^L = \frac{1 - \exp(\beta)}{\exp(\alpha) - \exp(\beta)} \quad \text{and} \quad x_1^S = \left[\frac{1 - \exp(\beta)}{\exp(\alpha) - \exp(\beta)} \right] \exp(\alpha), \quad (9)$$

where

$$\alpha = -\frac{\Delta\tilde{H}_1^{\text{fus}}}{RT_{m1}} \left(1 - \frac{T_{m1}}{T} \right) \quad \text{and} \quad \beta = -\frac{\Delta\tilde{H}_2^{\text{fus}}}{RT_{m2}} \left(1 - \frac{T_{m2}}{T} \right). \quad (10)$$

(Answer: (d) $x_1^L = 0.326$, $x_1^S = 0.187$)

Problems Related to Section 12.2

12.20 When 38 g of polymer X is dissolved in 80 g of benzene, the normal boiling point of benzene increases by 1.014 K. Calculate the molecular weight of X if the only available data are the variation of vapor pressure of benzene as a function of temperature in the form

$$\ln P^{\text{vap}} = 9.2806 - \frac{2788.51}{T - 52.36},$$

where P is in bar and T is in K.

(Answer: 1151 g/mol)

12.21 When 1.3 g of an unknown protein is dissolved in 125 cm^3 of water at 298 K, the osmotic pressure is measured as 0.005153 bar. Estimate the molecular weight of the protein.

(Answer: 50,000 g/mol)

12.22 When you soak your hands in water for a long period of time, you notice that they go wrinkly. Explain.

12.23 Explain what happens to your eyesight when you swim

- (a) in a pool,
- (b) in the ocean. Why?

(Answer: (a) short-sightedness and (b) long-sightedness)

12.24 Explain the following:

- (a) What is the purpose of using excessive sugar in making jams?
- (b) What is the cause of diarrhea?

12.25 Determine the molarity of a nonelectrolyte having an osmotic pressure of 70 bar at 293 K.

(Answer: 2.87 mol/L)

12.26 Moon *et al.* (2000) reported the following osmotic pressure data for dilute concentrations of lysozyme (2) in ammonium sulfate solutions at 298.15 K for an ionic strength of 1 M at pH 8:

c_2 (g/L)	Π (mmHg)	c_2 (g/L)	Π (mmHg)
4.06	4.40	14.26	14.55
6.13	6.50	16.31	16.30
8.18	8.65	18.38	18.35
10.42	10.75	20.44	20.15
12.26	12.55		

Determine the molecular weight of lysozyme.

(Answer: 16,910 g/mol)

12.27 Consider a long hollow pipe with a reverse osmosis membrane attached to its bottom end.

- (a) How deep should this pipe be lowered into the ocean so as to initiate reverse osmosis? Assume uniform temperature and salinity of the ocean and use the following data:
 - Osmotic pressure of seawater = 26 bar.
 - Atmospheric pressure = 1 bar.
 - Density of seawater = 1025 kg/m³.
 - Density of pure water = 1000 kg/m³.
 - Acceleration of gravity = 9.8 m/s².
- (b) If the pipe is lowered to a depth of 500 m, estimate the level of fresh water within the pipe.
- (c) Calculate the length of the pipe required to bring fresh water just to the ocean surface. Is this a feasible project?

For more details on this problem see [Levenspiel and de Nevers \(1974\)](#).

(**Answer:** (a) 259 m, (b) 253 m below the sea level, and (c) 10,612 m)

12.28 The primary function of red blood cells (erythrocytes) is to carry oxygen to the tissues and to remove waste. The cytoplasm of red blood cells is congested with protein molecules, called hemoglobin, occupying a quarter of the cell volume. Red blood cells get their color from hemoglobin, which is bright red. Consider a spherical red blood cell with a diameter of 8.4×10^{-6} m. The average diameter of a hemoglobin molecule is about 6×10^{-9} m.

(a) If a red blood cell is placed in a large volume of pure water at 298 K, estimate the osmotic pressure experienced by the cell membrane.

Hint: 1 mol of hemoglobin contains 6.02×10^{23} molecules, i.e. Avogadro's number.

(b) The surface tension, σ , of a spherical droplet of radius R is given by

$$\sigma = \frac{R\Delta P}{2},$$

where ΔP represents the pressure differential across the droplet surface. If the cell membrane can withstand a maximum tension of 2×10^{-4} N/m, determine whether lysis of the red blood cell takes place.

(**Answer:** (a) 12.1 kPa)

12.29 Experimental determination of the freezing point depression is much easier than that of the osmotic pressure. For this reason, it is generally preferred to estimate the osmotic pressure of a solution from its freezing point depression. When the solvent is water at a temperature of 273.15 K, combine Eqns (12.1-27) and (12.2-31) to show that

$$\Pi = 12.2(T_{m1} - T),$$

where Π is in bar and $(T_{m1} - T)$ is in K (or °C).

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Chapter 13

Chemical Reaction Equilibrium

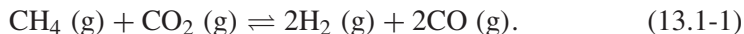
A chemical reactor is the heart of a chemical plant in which reactants are converted into products. Design of a reactor should provide answers to questions like: Which products will be obtained from the reaction(s) of given reactants? What are the best conditions, i.e. temperature, pressure, addition of inerts, etc., to convert more of the reactants into products? What quantities of products will be formed from the given amounts of reactants?

This chapter starts with the fundamentals such as stoichiometry, molar extent of reaction, and the condition of chemical reaction equilibrium. Calculation of the equilibrium constant is then introduced. Estimation of the equilibrium extent for reactions taking place in a single phase, as well as factors affecting it, will be discussed.

13.1 Stoichiometry of a Chemical Reaction

Balancing of a chemical equation is based on the conservation of mass for a closed thermodynamic system. If a chemical reaction takes place in a closed container, the mass does not change even if there is an exchange of energy with the surroundings.

Consider carbon dioxide reforming¹ of methane described by the reaction



If $A_1 = \text{CH}_4$, $A_2 = \text{CO}_2$, $A_3 = \text{H}_2$, and $A_4 = \text{CO}$, Eqn (13.1-1) is expressed as

$$A_1 + A_2 = 2A_3 + 2A_4. \quad (13.1-2)$$

¹The term “reforming” is used for the production of hydrogen from hydrocarbons.

It is convenient to write all the chemical species on one side of the equation and give a positive sign to the species that are regarded as the products of the reaction. Thus,

$$2A_3 + 2A_4 - A_1 - A_2 = 0. \quad (13.1-3)$$

In general, any chemical reaction is expressed as

$$\sum_{i=1}^k \alpha_i A_i = 0, \quad (13.1-4)$$

where α_i is the stoichiometric coefficient of the i th chemical species (positive if species is a product, negative if species is a reactant), k is the total number of species in the reaction, and A_i is the chemical symbol for the i th chemical species, representing the molecular weight of species. Equation (13.1-4) is simply a mathematical statement representing the *conservation of mass*.² In matrix notation, Eqn (13.1-4) is expressed in the form

$$\begin{bmatrix} A_1 & A_2 & A_3 & \cdots & A_k \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_k \end{bmatrix} = [0]. \quad (13.1-5)$$

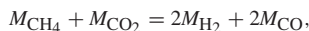
Each chemical species, A_i , is the sum of the chemical elements, E_j , such that

$$A_i = \sum_{j=1}^t \beta_{ji} E_j, \quad (13.1-6)$$

where β_{ji} represents the number of chemical elements E_j in the chemical species A_i , and t is the total number of chemical elements. Substitution of Eqn (13.1-6) into Eqn (13.1-4) gives

$$\sum_{i=1}^k \alpha_i \left(\sum_{j=1}^t \beta_{ji} E_j \right) = \sum_{j=1}^t \left(\sum_{i=1}^k \beta_{ji} \alpha_i \right) E_j = 0. \quad (13.1-7)$$

²For the reaction given by Eqn (13.1-1), Eqn (13.1-4) reads



where M represents the molecular weight.

Since all the E_j are linearly independent,³ then

$$\sum_{i=1}^k \beta_{ji} \alpha_i = 0, \quad j = 1, 2, \dots, t \quad (13.1-8)$$

In matrix notation, Eqn (13.1-8) is expressed in the form

$$\underbrace{\begin{bmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \cdots & \beta_{1k} \\ \beta_{21} & \beta_{22} & \beta_{23} & \cdots & \beta_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{t1} & \beta_{t2} & \beta_{t3} & \cdots & \beta_{tk} \end{bmatrix}}_{\boldsymbol{\beta}} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_t \end{bmatrix} = [0], \quad (13.1-9)$$

where the *element-by-species matrix*, $[\boldsymbol{\beta}]$, is formed as

$$[\boldsymbol{\beta}] = \begin{array}{c} \text{Species} \longrightarrow \\ \begin{matrix} E_1 \\ E_2 \\ \vdots \\ E_t \end{matrix} \end{array} \begin{array}{ccccc} A_1 & A_2 & A_3 & & A_k \\ \left[\begin{array}{ccccc} \beta_{11} & \beta_{12} & \beta_{13} & \cdots & \beta_{1k} \\ \beta_{21} & \beta_{22} & \beta_{23} & \cdots & \beta_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{t1} & \beta_{t2} & \beta_{t3} & \cdots & \beta_{tk} \end{array} \right] \end{array}. \quad (13.1-10)$$

Using the property of a “reduced row echelon form” of a matrix as explained in Appendix F, i.e. Eqns (F.6-10) and (F.6-11), Eqn (13.1-9) can be rewritten in the form

$$\text{rref}[\boldsymbol{\beta}] \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_t \end{bmatrix} = [0], \quad (13.1-11)$$

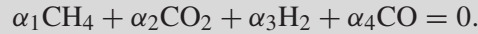
where $\text{rref}[\boldsymbol{\beta}]$ represents the reduced row echelon form of the element-by-species matrix. Equation (13.1-11) is used to balance chemical reactions.

³The expression

$$\sum_{i=1}^n \alpha_i x_i = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n,$$

where $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a set of scalars, is called a *linear combination* of the elements of the set $S = \{x_1, x_2, \dots, x_n\}$. The elements of the set S are said to be *linearly dependent* if there exists a set of scalars $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ with elements α_i not all equal to zero, such that the linear combination $\sum_{i=1}^n \alpha_i x_i = 0$ holds. If $\sum_{i=1}^n \alpha_i x_i = 0$ holds for all $\alpha_i = 0$, then the set S is *linearly independent*. See also Appendix F.

Example 13.1 Consider carbon dioxide reforming of methane, i.e.



Balance this equation by applying Eqn (13.1-11).

Solution

The element-by-species matrix is written as

$$[\beta] = \begin{array}{c} \text{Species} \rightarrow \\ \begin{array}{c} \text{C} \\ \text{H} \\ \text{O} \end{array} \end{array} \begin{array}{cc} \text{CH}_4 & \text{CO}_2 & \text{H}_2 & \text{CO} \\ \left[\begin{array}{cccc} 1 & 1 & 0 & 1 \\ 4 & 0 & 2 & 0 \\ 0 & 2 & 0 & 1 \end{array} \right] \end{array} \quad (1)$$

Reduced row echelon form of $[\beta]$ is

$$\text{rref}[\beta] = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 \end{bmatrix}. \quad (2)$$

Therefore, Eqn (13.1-11) becomes

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (3)$$

Multiplication of the matrices yields

$$\alpha_1 + \frac{1}{2}\alpha_4 = 0, \quad (4)$$

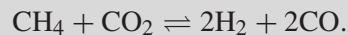
$$\alpha_2 + \frac{1}{2}\alpha_4 = 0, \quad (5)$$

$$\alpha_3 - \alpha_4 = 0. \quad (6)$$

Since there are four unknowns and three equations, expressing α_1 , α_2 , and α_3 in terms of α_4 , a free variable, gives

$$\alpha_1 = -\frac{1}{2}\alpha_4, \quad \alpha_2 = -\frac{1}{2}\alpha_4, \quad \alpha_3 = \alpha_4. \quad (7)$$

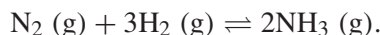
If we take $\alpha_4 = 2$, then $\alpha_1 = \alpha_2 = -1$, and $\alpha_3 = 2$. Hence, the reaction becomes



Comment: Stoichiometric coefficients have units. For example, in the above equation the stoichiometric coefficient of H_2 indicates that there are 2 mol of H_2 per mole of CH_4 .

13.2 The Law of Combining Proportions

Stoichiometric coefficients have the units of moles of i per mole of basis species, where basis species is arbitrarily chosen. Consider, for example, a reaction between nitrogen and hydrogen to form ammonia, i.e.



If N_2 is chosen as the basis species, then the stoichiometric coefficients 3 and 2 represent the “moles of H_2 per mole of N_2 ” and “moles of NH_3 per mole of N_2 ,” respectively. Suppose that the reactor is initially charged with $(n_{N_2})_o$ moles of N_2 , $(n_{H_2})_o$ moles of H_2 , and $(n_{NH_3})_o$ moles of NH_3 . At any given instant during the reaction, let the number of moles of N_2 , H_2 , and NH_3 in the reacting mixture be n_{N_2} , n_{H_2} , and n_{NH_3} , respectively. Therefore, the number of moles of N_2 and H_2 reacted as well as the number of moles of NH_3 formed can be expressed as

$$\begin{aligned} \text{Moles of } N_2 \text{ reacted} &= (n_{N_2})_o - n_{N_2}, \\ \text{Moles of } H_2 \text{ reacted} &= (n_{H_2})_o - n_{H_2}, \\ \text{Moles of } NH_3 \text{ formed} &= n_{NH_3} - (n_{NH_3})_o. \end{aligned} \quad (13.2-1)$$

The moles of N_2 reacted can also be expressed by using either the moles of H_2 reacted or the moles of NH_3 formed as

$$\text{Moles of } N_2 \text{ reacted} = (n_{N_2})_o - n_{N_2} = \frac{(n_{H_2})_o - n_{H_2}}{3} = \frac{n_{NH_3} - (n_{NH_3})_o}{2}. \quad (13.2-2)$$

Rearrangement of Eqn (13.2-2) gives

$$\text{Moles of } N_2 \text{ reacted} = \frac{n_{N_2} - (n_{N_2})_o}{-1} = \frac{n_{H_2} - (n_{H_2})_o}{-3} = \frac{n_{NH_3} - (n_{NH_3})_o}{2}. \quad (13.2-3)$$

Equation (13.2-3) can be generalized as

$$\text{moles of basis species} = \frac{\text{moles of } i \text{ reacted}}{(\text{moles of } i / \text{mole of basis species})}, \quad (13.2-4)$$

which is known as the *law of combining proportions*. The mathematical expression for the law of combining proportions is given by

$$\frac{n_i - n_{i_o}}{\alpha_i} = \epsilon, \quad (13.2-5)$$

where ϵ is called the *molar extent* of the reaction.⁴ Rearrangement of Eqn (13.2-5) gives

$$n_i = n_{i_o} + \alpha_i \epsilon. \quad (13.2-6)$$

Once ϵ has been determined, the number of moles of any chemical species participating in the reaction can be determined by using Eqn (13.2-6).

The molar extent of the reaction should not be confused with the *fractional conversion*, which can only take values between 0 and 1. The molar extent of the reaction is an extensive property measured in moles and its value can be greater than unity.

It is also important to note that the fractional conversion may be different for each reacting species, i.e.

$$X_i = \frac{n_{i_o} - n_i}{n_{i_o}}. \quad (13.2-7)$$

On the other hand, molar extent is unique for a given reaction. Comparison of Eqns (13.2-6) and (13.2-7) indicates that

$$\epsilon = \frac{n_{i_o}}{(-\alpha_i)} X_i. \quad (13.2-8)$$

The total number of moles, n_T , of a reacting mixture at any instant can be calculated by the summation of Eqn (13.2-6) over all species, i.e.

$$n_T = n_{T_o} + \bar{\alpha} \epsilon, \quad (13.2-9)$$

where n_{T_o} is the initial total number of moles and $\bar{\alpha}$ is defined by

$$\bar{\alpha} = \sum_i \alpha_i. \quad (13.2-10)$$

Example 13.2 A reactor is initially charged with 2 mol of C_2H_4 , 5 mol of H_2O , and 1 mol of H_2 . After setting up the temperature and pressure, the following reaction takes place:



⁴The term ϵ has been given various names in the literature, such as degree of advancement, reaction of coordinate, degree of reaction, and progress variable.

- (a) Determine expressions for the number of moles, n_i , and the mole fractions, y_i , of each species as a function of the molar extent of the reaction.
- (b) Calculate the number of moles and mole fractions of each species when $\epsilon = 1.8$.

Solution

- (a) The use of Eqn (13.2-6) expresses number of moles of each species as a function of the molar extent of the reaction in the form

$$\begin{array}{rcl}
 n_{\text{C}_2\text{H}_4} & = & 2 - \epsilon \\
 n_{\text{H}_2\text{O}} & = & 5 - 2\epsilon \\
 n_{\text{CH}_4} & = & \epsilon \\
 n_{\text{CO}_2} & = & \epsilon \\
 n_{\text{H}_2} & = & 1 + 2\epsilon \\
 \hline
 n_{\text{T}} & = & 8 + \epsilon.
 \end{array}$$

The mole fractions can be calculated from $y_i = n_i/n_{\text{T}}$ as

$$\begin{array}{ll}
 y_{\text{C}_2\text{H}_4} = \frac{2 - \epsilon}{8 + \epsilon}, & y_{\text{H}_2\text{O}} = \frac{5 - 2\epsilon}{8 + \epsilon}, \\
 y_{\text{CH}_4} = y_{\text{CO}_2} = \frac{\epsilon}{8 + \epsilon}, & y_{\text{H}_2} = \frac{1 + 2\epsilon}{8 + \epsilon}.
 \end{array}$$

- (b) When $\epsilon = 1.8$, number of moles of each species becomes

$$\begin{array}{ll}
 n_{\text{C}_2\text{H}_4} = 2 - 1.8 = 0.2 \text{ mol}, & n_{\text{H}_2\text{O}} = 5 - 2(1.8) = 1.4 \text{ mol}, \\
 n_{\text{CH}_4} = n_{\text{CO}_2} = 1.8 \text{ mol}, & n_{\text{H}_2} = 1 + 2(1.8) = 4.6 \text{ mol}.
 \end{array}$$

The mole fractions are

$$y_{\text{C}_2\text{H}_4} = 0.020, \quad y_{\text{H}_2\text{O}} = 0.143, \quad y_{\text{CH}_4} = y_{\text{CO}_2} = 0.184, \quad y_{\text{H}_2} = 0.469.$$

Comment: The value of ϵ that makes $n_i = 0$ for a reactant gives the greatest possible value of ϵ . For the given reaction

$$\begin{array}{ll}
 n_{\text{C}_2\text{H}_4} = 0 & \Rightarrow \quad \epsilon = 2, \\
 n_{\text{H}_2\text{O}} = 0 & \Rightarrow \quad \epsilon = 2.5.
 \end{array}$$

Therefore, ethylene is the limiting reactant and ϵ cannot exceed 2.

The molar concentration of the i th species, c_i , is defined by

$$c_i = \frac{n_i}{V}. \quad (13.2-11)$$

Dividing Eqn (13.2-6) by the volume V gives

$$\frac{n_i}{V} = \frac{n_{i_o}}{V} + \alpha_i \left(\frac{\epsilon}{V} \right) \quad (13.2-12)$$

or

$$c_i = c_{i_o} + \alpha_i \xi, \quad (13.2-13)$$

where c_{i_o} is the initial molar concentration of the i th species and ξ is the *intensive extent of the reaction* in moles per unit volume. Note that ξ is related to fractional conversion by

$$\xi = \frac{c_{i_o}}{(-\alpha_i)} X_i. \quad (13.2-14)$$

The total molar concentration, c , of a reacting mixture at any instant can be calculated by the summation of Eqn (13.2-13) over all species. The result is

$$c = c_o + \bar{\alpha} \xi, \quad (13.2-15)$$

where c_o is the initial total molar concentration.

When more than one reaction takes place in a reactor, Eqn (13.2-6) takes the form

$$n_{ij} = n_{ij_o} + \alpha_{ij} \epsilon_j, \quad (13.2-16)$$

where

n_{ij} = number of moles of the i th species in the j th reaction,

n_{ij_o} = initial number of moles of the i th species in the j th reaction,

α_{ij} = stoichiometric coefficient of the i th species in the j th reaction,

ϵ_j = extent of the j th reaction.

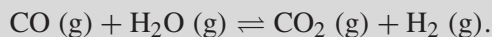
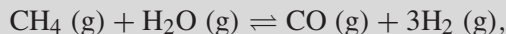
Summation of Eqn (13.2-16) over all reactions taking place in a reactor gives

$$\sum_j n_{ij} = \sum_j n_{ij_o} + \sum_j \alpha_{ij} \epsilon_j \quad (13.2-17)$$

or

$$n_i = n_{i_o} + \sum_j \alpha_{ij} \epsilon_j. \quad (13.2-18)$$

Example 13.3 The following two reactions occur simultaneously in a batch reactor:



A mixture of 20 mol% CH_4 , 70% H_2O , and 10% inerts is fed into a reactor and the reactions proceed until 6% CO and 43% H_2 are formed. Determine the percentage of each species in a reacting mixture.

Solution

Basis: One mole of a feed mixture.

Let ϵ_1 and ϵ_2 be the extents of the first and second reactions, respectively. Then the number of moles of each species can be expressed as

$$n_{\text{CH}_4} = 0.2 - \epsilon_1$$

$$n_{\text{H}_2\text{O}} = 0.7 - \epsilon_1 - \epsilon_2$$

$$n_{\text{CO}} = \epsilon_1 - \epsilon_2$$

$$n_{\text{H}_2} = 3\epsilon_1 + \epsilon_2$$

$$n_{\text{CO}_2} = \epsilon_2$$

$$n_{\text{inert}} = 0.1$$

$$n_{\text{T}} = 1 + 2\epsilon_1.$$

The mole fractions of CO and H_2 are given in the problem statement. These values are used to determine the extent of the reactions as

$$y_{\text{CO}} = \frac{\epsilon_1 - \epsilon_2}{1 + 2\epsilon_1} = 0.06 \quad \Rightarrow \quad 0.88\epsilon_1 - \epsilon_2 = 0.06,$$

$$y_{\text{H}_2} = \frac{3\epsilon_1 + \epsilon_2}{1 + 2\epsilon_1} = 0.43 \quad \Rightarrow \quad 2.14\epsilon_1 + \epsilon_2 = 0.43.$$

Simultaneous solution of these two equations gives

$$\epsilon_1 = 0.162 \quad \text{and} \quad \epsilon_2 = 0.083.$$

Therefore, the mole fractions of CH_4 , H_2O , CO_2 , and the inerts are

$$y_{\text{CH}_4} = \frac{0.2 - \epsilon_1}{1 + 2\epsilon_1} = \frac{0.2 - 0.162}{1 + (2)(0.162)} = 0.029,$$

$$y_{\text{H}_2\text{O}} = \frac{0.7 - \epsilon_1 - \epsilon_2}{1 + 2\epsilon_1} = \frac{0.7 - 0.162 - 0.083}{1 + (2)(0.162)} = 0.344,$$

$$y_{\text{CO}_2} = \frac{\epsilon_2}{1 + 2\epsilon_1} = \frac{0.083}{1 + (2)(0.162)} = 0.063,$$

$$y_{\text{inert}} = \frac{0.1}{1 + 2\epsilon_1} = \frac{0.1}{1 + (2)(0.162)} = 0.076.$$

Comment: Although inert species do not take part in the reactions, they must be considered in determining the total number of moles.

13.3 Equilibrium for a Single Reaction

Consider a single chemical reaction taking place in a single phase. The change in the total Gibbs energy of the system is given by Eqn (7.1-13), i.e.

$$dG = V dP - S dT + \sum_{i=1}^k \bar{G}_i dn_i. \quad (13.3-1)$$

The changes in the number of moles occur as a result of a chemical reaction and from Eqn (13.2-6)

$$dn_i = \alpha_i d\epsilon. \quad (13.3-2)$$

Substitution of Eqn (13.3-2) into Eqn (13.3-1) gives

$$dG = V dP - S dT + \sum_{i=1}^k \alpha_i \bar{G}_i d\epsilon. \quad (13.3-3)$$

Differentiation of Eqn (13.3-3) with respect to the molar extent of reaction, ϵ , by keeping temperature and pressure constant gives

$$\left(\frac{\partial G}{\partial \epsilon} \right)_{T,P} = \sum_{i=1}^k \alpha_i \bar{G}_i, \quad (13.3-4)$$

which indicates that the rate of change of the total Gibbs energy of the system with the molar extent of reaction is equal to $\sum \alpha_i \bar{G}_i$. Since $(dG)_{T,P} = 0$ at equilibrium as shown in Fig. 13.1, then it follows that

$$\sum_{i=1}^k \alpha_i \bar{G}_i = 0 \quad \text{Criterion for chemical reaction equilibrium.} \quad (13.3-5)$$

In other words, when a chemical reaction reaches equilibrium, total Gibbs energy of a reacting mixture reaches its minimum value.

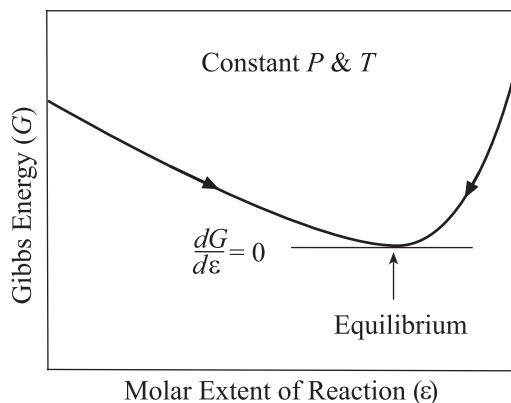


Figure 13.1: Variation of Gibbs energy with molar extent of reaction.

13.3.1 The Equilibrium Constant

The solution of Eqn (13.3-5) requires an expression for the partial molar Gibbs energy. From Eqn (7.3-5)

$$\bar{G}_i(T, P, x_i) = \lambda_i(T) + RT \ln \hat{f}_i(T, P, x_i). \quad (13.3-6)$$

The molar Gibbs energy of a pure component i at the temperature of the system, T , and at some reference pressure, P^o , is

$$\tilde{G}_i^o(T, P^o) = \lambda_i(T) + RT \ln f_i^o(T, P^o). \quad (13.3-7)$$

The superscript “ o ” indicates the *standard state* conditions, i.e. T, P^o , and the phase. Subtraction of Eqn (13.3-7) from Eqn (13.3-6) gives

$$\bar{G}_i(T, P, x_i) = \tilde{G}_i^o(T, P^o) + RT \ln \hat{a}_i(T, P, x_i) \quad (13.3-8)$$

in which the activity, $\hat{a}_i$, is defined by

$$\hat{a}_i(T, P, x_i) = \frac{\hat{f}_i(T, P, x_i)}{f_i^o(T, P^o)}. \quad (13.3-9)$$

The use of Eqn (13.3-8) in Eqn (13.3-5) leads to

$$\begin{aligned} \sum_{i=1}^k \alpha_i \bar{G}_i &= 0 = \sum_{i=1}^k \alpha_i \tilde{G}_i^o + RT \sum_{i=1}^k \alpha_i \ln \hat{a}_i \\ &= \sum_{i=1}^k \alpha_i \tilde{G}_i^o + RT \sum_{i=1}^k \ln \hat{a}_i^{\alpha_i} = \sum_{i=1}^k \alpha_i \tilde{G}_i^o + RT \ln \prod_{i=1}^k \hat{a}_i^{\alpha_i}. \end{aligned} \quad (13.3-10)$$

The *equilibrium constant*, K_a , and the *standard Gibbs energy change of reaction*, ΔG_{rxn}^o , are defined by

$$K_a = \prod_{i=1}^k \hat{a}_i^{\alpha_i}, \quad (13.3-11)$$

$$\Delta G_{\text{rxn}}^o(T) = \sum_{i=1}^k \alpha_i \tilde{G}_i^o(T, P^o), \quad (13.3-12)$$

so that Eqn (13.3-10) takes the form

$$K_a(T) = \exp \left[-\frac{\Delta G_{\text{rxn}}^o(T)}{RT} \right]. \quad (13.3-13)$$

It is important to keep in mind that the equilibrium constant, K_a , is independent of pressure, and varies only with temperature. The presence of a catalyst does not affect the equilibrium constant. A catalyst influences reaction rate.

13.4 Evaluation of the Equilibrium Constant

Calculation of the equilibrium constant, K_a , from Eqn (13.3-13) requires the standard Gibbs energy change of reaction, ΔG_{rxn}^o , to be known at the temperature of reaction, T . Standard Gibbs energy change of reaction can be calculated from the relation

$$\Delta G_{\text{rxn}}^o = \Delta H_{\text{rxn}}^o - T \Delta S_{\text{rxn}}^o, \quad (13.4-1)$$

where ΔH_{rxn}^o , standard enthalpy change of reaction (or standard heat of reaction), and ΔS_{rxn}^o , standard entropy change of reaction, ΔS_{rxn}^o , are defined by

$$\Delta H_{\text{rxn}}^o = \sum_{i=1}^k \alpha_i \tilde{H}_i^o, \quad (13.4-2)$$

$$\Delta S_{\text{rxn}}^o = \sum_{i=1}^k \alpha_i \tilde{S}_i^o. \quad (13.4-3)$$

Standard heat of reaction is the difference between the enthalpies of pure products and reactants in their standard states. Since it is not practical to list ΔH_{rxn}^o and ΔS_{rxn}^o for every reaction, calculations of ΔH_{rxn}^o and ΔG_{rxn}^o are based on the formation reactions of compounds involved in the reaction.

The difference between the enthalpy of 1 mol of a pure compound i and the total enthalpy of the elements of which it is composed at the temperature of the system, T , and the standard state pressure of 1 bar is called the *standard enthalpy of formation* of the compound, $(\Delta \tilde{H}_f^o)_i$. The enthalpies of formation of elements

in their standard state are defined to be zero at any temperature. As a result, the standard enthalpy of formation of a compound is just the standard heat of reaction in which 1 mol of it is formed from elements. Thus,

$$\tilde{H}_i^o = (\Delta \tilde{H}_f^o)_i. \quad (13.4-4)$$

Substitution of Eqn (13.4-4) into Eqn (13.4-2) leads to

$$\Delta H_{\text{rxn}}^o = \sum_{i=1}^k \alpha_i (\Delta \tilde{H}_f^o)_i. \quad (13.4-5)$$

The difference between the Gibbs energy of 1 mol of a pure compound and the total Gibbs energy of the elements of which it is composed at the temperature of the system, T , and the standard state pressure of 1 bar is called the *standard Gibbs energy of formation* of the compound, $(\Delta \tilde{G}_f^o)_i$. Gibbs energies of formation of elements in their standard state are defined to be zero at any temperature. Thus,

$$\tilde{G}_i^o = (\Delta \tilde{G}_f^o)_i. \quad (13.4-6)$$

Substitution of Eqn (13.4-6) into Eqn (13.3-12) leads to

$$\Delta G_{\text{rxn}}^o = \sum_{i=1}^k \alpha_i (\Delta \tilde{G}_f^o)_i. \quad (13.4-7)$$

In the literature, standard enthalpy and Gibbs energy of formation data are generally tabulated at 298 K and 1 bar. The values of $\Delta \tilde{H}_f^o$ and $\Delta \tilde{G}_f^o$ for a number of common substances are given in Appendix E.

Example 13.4 Calculate ΔH_{rxn}^o and ΔG_{rxn}^o for the following reactions at 298 K:



Solution

From Appendix E, enthalpies and Gibbs energies of formation of the species taking place in the reactions are as follows:

Species	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
CH ₄ (g)	-74.90	-50.87
H ₂ O (g)	-242.00	-228.80
H ₂ O (l)	-285.83	-237.14
CO (g)	-110.60	-137.40
CH ₃ COOH (l)	-484.09	-389.23
C ₂ H ₅ OH (l)	-276.98	-173.99
CH ₃ COOC ₂ H ₅ (l)	-479.86	-332.93
CO ₂ (g)	-393.80	-394.60

• *Reaction 1:*

$$\Delta H_{\text{rxn},298}^o = -110.60 + 74.90 + 242.00 = 206.30 \text{ kJ},$$

$$\Delta G_{\text{rxn},298}^o = -137.40 + 50.87 + 228.80 = 142.27 \text{ kJ}.$$

• *Reaction 2:*

$$\Delta H_{\text{rxn},298}^o = -479.86 - 285.83 + 484.09 + 276.98 = -4.62 \text{ kJ},$$

$$\Delta G_{\text{rxn},298}^o = -332.93 - 237.14 + 389.23 + 173.99 = -6.85 \text{ kJ}.$$

• *Reaction 3:*

$$\Delta H_{\text{rxn},298}^o = -393.80 + (2)(110.6) = -172.6 \text{ kJ},$$

$$\Delta G_{\text{rxn},298}^o = -394.60 + (2)(137.40) = -119.80 \text{ kJ}.$$

Once $\Delta G_{\text{rxn},298}^o$ is known, K_a at 298 K can be easily calculated from Eqn (13.3-13). To determine K_a at any other temperature, it is first necessary to differentiate Eqn (13.3-13) with respect to temperature to get

$$\begin{aligned} \frac{d \ln K_a}{dT} &= -\frac{1}{R} \frac{d}{dT} \left(\frac{\Delta G_{\text{rxn}}^o}{T} \right) = -\frac{1}{R} \left(\frac{1}{T} \underbrace{\frac{d\Delta G_{\text{rxn}}^o}{dT}}_{-\Delta S_{\text{rxn}}^o} - \frac{\Delta G_{\text{rxn}}^o}{T^2} \right) \\ &= \frac{\overbrace{T \Delta S_{\text{rxn}}^o}^{\Delta H_{\text{rxn}}^o} + \Delta G_{\text{rxn}}^o}{RT^2} \end{aligned} \quad (13.4-8)$$

or

$$\frac{d \ln K_a}{dT} = \frac{\Delta H_{\text{rxn}}^o}{RT^2}, \quad (13.4-9)$$

which is known as the *van't Hoff equation*. The van't Hoff equation is valid for both homogeneous and heterogeneous reactions.⁵

If heat is evolved in the reaction, the reaction is called *exothermic*. If heat is absorbed, the reaction is called *endothermic*. Therefore,

$$\Delta H_{\text{rxn}}^o \begin{cases} > 0 \text{ for an endothermic reaction,} \\ < 0 \text{ for an exothermic reaction.} \end{cases} \quad (13.4-10)$$

When the reaction is exothermic, K_a decreases with increasing temperature. On the other hand, if the reaction is endothermic, K_a increases with increasing temperature. Integration of Eqn (13.4-9) is dependent on whether ΔH_{rxn}^o remains constant or not.

13.4.1 ΔH_{rxn}^o is Independent of Temperature

For small temperature ranges ΔH_{rxn}^o can be considered independent of temperature. In this case, integration of Eqn (13.4-9) from 298 K to any T gives

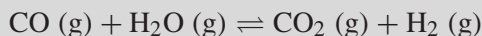
$$\ln \left(\frac{K_a}{K_{a,298}} \right) = -\frac{\Delta H_{\text{rxn}}^o}{R} \left(\frac{1}{T} - \frac{1}{298} \right). \quad (13.4-11)$$

Rearrangement of Eqn (13.4-11) gives

$$\ln K_a = -\frac{\Delta H_{\text{rxn}}^o}{R} \frac{1}{T} + \frac{\Delta S_{\text{rxn}}^o}{R}. \quad (13.4-12)$$

Equations (13.4-11) and (13.4-12) both indicate that a plot of $\ln K_a$ versus the reciprocal of the absolute temperature, $1/T$, is a straight line with a slope $-\Delta H_{\text{rxn}}^o/R$ as shown in Fig. 13.2.

Example 13.5 Calculate the equilibrium constant for the water-gas shift reaction, i.e.



at 350 K by assuming ΔH_{rxn}^o to be independent of temperature.

⁵A *homogeneous* reaction occurs throughout the given phase. A *heterogeneous* reaction occurs at an interface between at least two different phases and is usually a catalytic reaction.

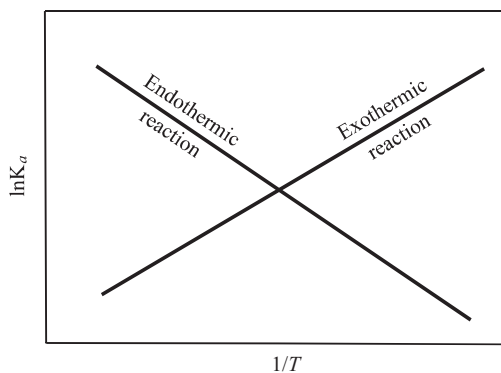


Figure 13.2: $\ln K_a$ versus $1/T$ when ΔH_{rxn}^o is independent of temperature.

Solution

From Appendix E

Species	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
CO	-110.60	-137.40
H ₂ O	-242.00	-228.80
CO ₂	-393.80	-394.60

Therefore, ΔH_{rxn}^o and ΔG_{rxn}^o at 298 K are

$$\Delta H_{\text{rxn},298}^o = -393.80 + 110.60 + 242.00 = -41.20 \text{ kJ} = -41,200 \text{ J},$$

$$\Delta G_{\text{rxn},298}^o = -394.60 + 137.40 + 228.80 = -28.40 \text{ kJ} = -28,400 \text{ J}.$$

The equilibrium constant at 298 K is

$$K_{a,298} = \exp \left[\frac{28,400}{(8.314)(298)} \right] = 95,114.$$

The use of Eqn (13.4-11) gives the equilibrium constant at 350 K as

$$\ln \left(\frac{K_a}{95,114} \right) = \frac{41,200}{8.314} \left(\frac{1}{350} - \frac{1}{298} \right) \Rightarrow K_a = 8040.$$

13.4.2 ΔH_{rxn}^o is Dependent on Temperature

The change in ΔH_{rxn}^o with temperature is expressed as

$$\frac{d\Delta H_{\text{rxn}}^o}{dT} = \frac{d}{dT} \sum_{i=1}^k \alpha_i \tilde{H}_i^o = \sum_{i=1}^k \alpha_i \frac{d\tilde{H}_i^o}{dT} = \sum_{i=1}^k \alpha_i \tilde{C}_{P_i}^o = \Delta C_P^o. \quad (13.4-13)$$

In Appendix B, heat capacities are expressed as a function of temperature in the form

$$\tilde{C}_{P_i}^o = a_i + b_i T + c_i T^2 + d_i T^3 + e_i T^4. \quad (13.4-14)$$

Substitution of Eqn (13.4-14) into Eqn (13.4-13) gives

$$\Delta C_p^o = \Delta a + \Delta b T + \Delta c T^2 + \Delta d T^3 + \Delta e T^4, \quad (13.4-15)$$

where

$$\begin{aligned} \Delta a &= \sum_{i=1}^k \alpha_i a_i, & \Delta b &= \sum_{i=1}^k \alpha_i b_i, & \Delta c &= \sum_{i=1}^k \alpha_i c_i, \\ \Delta d &= \sum_{i=1}^k \alpha_i d_i, & \Delta e &= \sum_{i=1}^k \alpha_i e_i. \end{aligned} \quad (13.4-16)$$

Substitution of Eqn (13.4-15) into Eqn (13.4-13) and integration from 298 K to any temperature T give

$$\begin{aligned} \Delta H_{\text{rxn}}^o &= \Delta H_{\text{rxn},298}^o + \Delta a (T - 298) + \frac{\Delta b}{2} (T^2 - 298^2) + \frac{\Delta c}{3} (T^3 - 298^3) \\ &\quad + \frac{\Delta d}{4} (T^4 - 298^4) + \frac{\Delta e}{5} (T^5 - 298^5). \end{aligned} \quad (13.4-17)$$

The use of Eqn (13.4-17) in the van't Hoff equation, Eqn (13.4-9), leads to

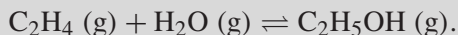
$$R \ln K_a = \Delta a \ln T + \frac{\Delta b}{2} T + \frac{\Delta c}{6} T^2 + \frac{\Delta d}{12} T^3 + \frac{\Delta e}{20} T^4 + \frac{\Lambda}{T} - \Omega, \quad (13.4-18)$$

where

$$\Lambda = 298 \Delta a + \frac{298^2 \Delta b}{2} + \frac{298^3 \Delta c}{3} + \frac{298^4 \Delta d}{4} + \frac{298^5 \Delta e}{5} - \Delta H_{\text{rxn},298}^o, \quad (13.4-19)$$

$$\begin{aligned} \Omega &= (1 + \ln 298) \Delta a + 298 \Delta b + \frac{298^2 \Delta c}{2} + \frac{298^3 \Delta d}{3} + \frac{298^4 \Delta e}{4} \\ &\quad - \frac{\Delta H_{\text{rxn},298}^o - \Delta G_{\text{rxn},298}^o}{298}. \end{aligned} \quad (13.4-20)$$

Example 13.6 The vapor phase hydration of ethylene to ethanol is represented by the reaction



Calculate the equilibrium constant at 500 K.

Solution

From Appendix E

Species	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
C ₂ H ₄	52.34	68.16
H ₂ O	-242.00	-228.80
C ₂ H ₅ OH	-235.00	-168.40

Therefore, ΔH_{rxn}^o and ΔG_{rxn}^o at 298 K are

$$\Delta H_{\text{rxn},298}^o = -235.00 - 52.34 + 242.00 = -45.34 \text{ kJ} = -45,340 \text{ J},$$

$$\Delta G_{\text{rxn},298}^o = -168.40 - 68.16 + 228.80 = -7.76 \text{ kJ} = -7760 \text{ J}.$$

From Appendix B, the coefficients of the equation for the molar heat capacity are

Species	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
C ₂ H ₄	17.562	0.692	0.936	-1.293	4.294
H ₂ O	33.763	-0.006	0.224	-0.100	0.110
C ₂ H ₅ OH	19.959	1.428	0.776	-1.513	5.366

Therefore,

$$\Delta a = 19.959 - 17.562 - 33.763 = -31.366,$$

$$\Delta b = (1.428 - 0.692 + 0.006) \times 10^{-1} = 7.42 \times 10^{-2},$$

$$\Delta c = (0.776 - 0.936 - 0.224) \times 10^{-4} = -3.84 \times 10^{-5},$$

$$\Delta d = (-1.513 + 1.293 + 0.100) \times 10^{-7} = -1.2 \times 10^{-8},$$

$$\Delta e = (5.366 - 4.294 - 0.110) \times 10^{-11} = 9.62 \times 10^{-12}.$$

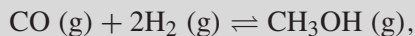
Substitution of the numerical values into Eqn (13.4-18) gives

$$\begin{aligned} \ln K_a = & -3.773 \ln T + 4.462 \times 10^{-3} T - 7.698 \times 10^{-7} T^2 - 1.203 \times 10^{-10} T^3 \\ & + 5.785 \times 10^{-14} T^4 + \frac{4682}{T} + 7.654. \end{aligned}$$

When $T = 500 \text{ K}$

$$\ln K_a = -4.4 \Rightarrow K_a = 0.012.$$

Example 13.7 For the methanol synthesis reaction



the equilibrium constant is given by

$$\ln K_a = 11.988 + \frac{9143.6}{T} - 7.492 \ln T + 4.076 \times 10^{-3}T - 7.161 \times 10^{-8}T^2,$$

where T is in K. Estimate the standard heat of reaction at 473 K and 573 K.

Solution

Once K_a is expressed as a function of temperature, ΔH_{rxn}^o can be calculated from the van't Hoff equation, Eqn (13.4-9):

$$\frac{d \ln K_a}{dT} = -\frac{9143.6}{T^2} - \frac{7.492}{T} + 4.076 \times 10^{-3} - 14.322 \times 10^{-8}T = \frac{\Delta H_{\text{rxn}}^o}{RT^2}. \quad (1)$$

Solving Eqn (1) for ΔH_{rxn}^o yields

$$\Delta H_{\text{rxn}}^o = R(-9143.6 - 7.492T + 4.076 \times 10^{-3}T^2 - 14.322 \times 10^{-8}T^3).$$

Hence

$$T = 473 \text{ K}, \quad \Delta H_{\text{rxn}}^o = -98.03 \text{ kJ},$$

$$T = 573 \text{ K}, \quad \Delta H_{\text{rxn}}^o = -100.81 \text{ kJ}.$$

Comment: Since $\Delta H_{\text{rxn}}^o < 0$, the reaction is exothermic.

13.4.3 An Alternative Way of Evaluating K_a

Once standard Gibbs energy change of reaction, ΔG_{rxn}^o , is known at the temperature of reaction, then the equilibrium constant can be easily calculated from Eqn (13.3-13). For this purpose, it is first necessary to obtain a relationship between $(\Delta \tilde{G}_f^o)_i$ and $(\Delta \tilde{G}_{f,298}^o)_i$. Note that

$$(\Delta \tilde{H}_f^o)_i = (\Delta \tilde{H}_{f,298}^o)_i + \int_{298}^T \Delta \tilde{C}_{P_i}^o dT, \quad (13.4-21)$$

where

$$\Delta \tilde{C}_{P_i}^o = \tilde{C}_{P_i}^o - \sum_e \alpha_{ei} \tilde{C}_{P_e}^o \quad (13.4-22)$$

in which α_{ei} is the number of atoms of an element e in species i , and $\tilde{C}_{p_e}^o$ is the molar heat capacity of element e . For example, for ethanol ($\text{C}_2\text{H}_5\text{OH}$), $\alpha_{eC} = 2$, $\alpha_{e\text{H}_2} = 3$, and $\alpha_{e\text{O}_2} = 0.5$. The use of Eqn (13.4-21) in the Gibbs-Helmholtz equation, Eqn (5.7-8), gives the standard Gibbs energy of formation at the temperature of reaction as

$$\frac{(\Delta\tilde{G}_f^o)_i}{T} = \frac{(\Delta\tilde{G}_{f,298}^o)_i}{298} - \int_{298}^T \left[(\Delta\tilde{H}_{f,298}^o)_i + \int_{298}^T \Delta\tilde{C}_{p_i}^o dT \right] \frac{dT}{T^2}. \quad (13.4-23)$$

The use of Eqn (13.4-23) requires heat capacities of elements to be known. Heat capacities of elements in the gaseous form can be obtained from Appendix B. The molar heat capacity of carbon (graphite) is given as (Kubaschewski and Alcock, 1979)

$$\tilde{C}_p^o = 17.152 + 4.273 \times 10^{-3}T - 8.7879 \times 10^5 T^{-2} \quad \text{For carbon,} \quad (13.4-24)$$

where $\tilde{C}_p^o$ is in J/mol K, and T is in K.

Example 13.8 Calculate the equilibrium constant of the reaction given in Example 13.6 by calculating ΔG_{rxn}^o at 500 K.

Solution

For ethylene (C_2H_4), the use of Eqn (13.4-22) gives

$$\begin{aligned} \Delta\tilde{C}_{p_{\text{C}_2\text{H}_4}}^o &= [17.562 - (2)(17.152) - (2)(27.004)] \\ &\quad + [0.692 - (2)(4.273 \times 10^{-2}) - (2)(0.119)] \times 10^{-1}T \\ &\quad + [0.936 + (2)(0.241)] \times 10^{-4}T^2 + [-1.293 - (2)(0.215)] \\ &\quad \times 10^{-7}T^3 + [4.294 + (2)(0.615)] \times 10^{-11}T^4 \\ &\quad + (2)(8.7879 \times 10^5)T^{-2}. \end{aligned} \quad (1)$$

Substitution of Eqn (1) into Eqn (13.4-23) and carrying out the integrations give standard Gibbs energy of formation at 500 K as

$$(\Delta\tilde{G}_f^o)_{\text{C}_2\text{H}_4} = 80,702 \text{ J/mol.}$$

Similarly,

$$(\Delta\tilde{G}_f^o)_{\text{H}_2\text{O}} = -219,399 \text{ J/mol}, \quad (\Delta\tilde{G}_f^o)_{\text{C}_2\text{H}_5\text{OH}} = -120,408 \text{ J/mol.}$$

Standard Gibbs energy change of reaction at 500 K is calculated from Eqn (13.4-7) as

$$\Delta G_{\text{rxn},500}^o = -120,408 - 80,702 + 219,399 = 18,289 \text{ J.}$$

The use of Eqn (13.3-13) gives the equilibrium constant as

$$K_a = \exp \left[-\frac{18,289}{(8.314)(500)} \right] = 0.012.$$

13.5 Gas Phase Reactions

For gas phase reactions, the standard state is defined as pure components at the temperature of the system and standard state pressure, P^o . The standard state pressure is chosen in such a way that a pure component behaves as an ideal gas.⁶ Under these conditions, fugacity is equal to pressure, i.e. $f_i^o(T, P^o) = P^o$. Thus, the equilibrium constant is expressed as

$$K_a = \prod_{i=1}^k \left[\frac{\hat{f}_i(T, P, y_i)}{f_i^o(T, P^o)} \right]^{\alpha_i} = \prod_{i=1}^k \left[\frac{\hat{f}_i(T, P, y_i)}{P^o} \right]^{\alpha_i}. \quad (13.5-1)$$

Substitution of

$$\hat{f}_i = \hat{\phi}_i y_i P \quad (13.5-2)$$

into Eqn (13.5-1) gives

$$K_a = \underbrace{\prod_{i=1}^k \hat{\phi}_i^{\alpha_i}}_{K_\phi} \underbrace{\prod_{i=1}^k y_i^{\alpha_i}}_{K_y} \prod_{i=1}^k \left(\frac{P}{P^o} \right)^{\alpha_i}. \quad (13.5-3)$$

The third term on the right-hand side of Eqn (13.5-3) is expressed as

$$\prod_{i=1}^k \left(\frac{P}{P^o} \right)^{\alpha_i} = \left(\frac{P}{P^o} \right)^{\sum \alpha_i} = \left(\frac{P}{P^o} \right)^{\bar{\alpha}}, \quad (13.5-4)$$

so that Eqn (13.5-3) becomes

$$K_a = K_\phi K_y \left(\frac{P}{P^o} \right)^{\bar{\alpha}}. \quad (13.5-5)$$

The units of P and P^o must be consistent so that the ratio P/P^o is a dimensionless quantity.

⁶The standard state pressure is usually chosen as 1 bar.

For many gas phase systems, the Lewis-Randall rule is applicable. In that case the fugacity coefficient becomes

$$\hat{\phi}_i(T, P, y_i) = \frac{\hat{f}_i(T, P, y_i)}{y_i P} = \frac{y_i f_i(T, P)}{y_i P} = \frac{f_i(T, P)}{P} = \phi_i(T, P), \quad (13.5-6)$$

where f_i is the fugacity of pure i at the temperature and pressure of the reacting system. The use of Eqn (13.5-6) in Eqn (13.5-5) leads to

$$K_a = K_{f/P} K_y \left(\frac{P}{P^o} \right)^{\bar{\alpha}} \quad \text{Ideal mixture of nonideal gases,} \quad (13.5-7)$$

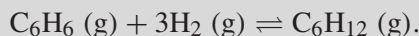
where

$$K_{f/P} = \prod_{i=1}^k \left(\frac{f_i}{P} \right)^{\alpha_i} = \prod_{i=1}^k \phi_i^{\alpha_i}. \quad (13.5-8)$$

If pure gases behave ideally, then $K_{f/P} = 1$ and Eqn (13.5-7) reduces to

$$K_a = K_y \left(\frac{P}{P^o} \right)^{\bar{\alpha}} \quad \text{Ideal gas mixture.} \quad (13.5-9)$$

Example 13.9 Cyclohexane is mainly produced from the hydrogenation of benzene according to the reaction



The reactor temperature is 550 K and hydrogen to benzene feed mole ratio is 4.5:1. Estimate the gas composition under equilibrium conditions if the reactor pressure is

- (a) 1 bar,
- (b) 15 bar.

Solution

(a) Since the pressure is low, we can assume ideal gas behavior. Choosing $P^o = 1$ bar, Eqn (13.5-9) reduces to

$$K_a = K_y P^{\bar{\alpha}}. \quad (1)$$

Note that

$$\bar{\alpha} = \sum_{i=1}^3 \alpha_i = 1 - 1 - 3 = -3 \quad \text{and} \quad P = 1 \text{ bar,}$$

so that Eqn (1) reduces to

$$K_a = K_y = \frac{n_{\text{C}_6\text{H}_{12}}/n_{\text{T}}}{(n_{\text{C}_6\text{H}_6}/n_{\text{T}})(n_{\text{H}_2}/n_{\text{T}})^3} = \frac{n_{\text{C}_6\text{H}_{12}}n_{\text{T}}^3}{n_{\text{C}_6\text{H}_6}n_{\text{H}_2}^3}. \quad (2)$$

From Appendix E, enthalpies and Gibbs energies of formation of the species taking place in the reaction are as follows:

Species	$\Delta \tilde{H}_{\text{f}}^o$ (kJ/mol)	$\Delta \tilde{G}_{\text{f}}^o$ (kJ/mol)
C ₆ H ₆	82.98	129.70
C ₆ H ₁₂	−123.20	31.78

Therefore, ΔH_{rxn}^o and ΔG_{rxn}^o at 298 K are

$$\begin{aligned} \Delta H_{\text{rxn},298}^o &= -123.20 - 82.98 = -206.18 \text{ kJ} = -206,180 \text{ J}, \\ \Delta G_{\text{rxn},298}^o &= 31.78 - 129.70 = -97.92 \text{ kJ} = -97,920 \text{ J}. \end{aligned}$$

From Appendix B, the coefficients of the equation for the molar heat capacity are

Species	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
C ₆ H ₆	−60.711	6.267	−5.795	2.799	−5.493
H ₂	27.004	0.119	−0.241	0.215	−0.615
C ₆ H ₁₂	−63.733	6.444	−2.633	−0.378	3.796

Therefore,

$$\begin{aligned} \Delta a &= -63.733 + 60.711 - (3 \times 27.004) = -84.034, \\ \Delta b &= [6.444 - 6.267 - (3 \times 0.119)] \times 10^{-1} = -0.018, \\ \Delta c &= [-2.633 + 5.795 + (3 \times 0.241)] \times 10^{-4} = 3.885 \times 10^{-4}, \\ \Delta d &= [-0.378 - 2.799 - (3 \times 0.215)] \times 10^{-7} = -3.822 \times 10^{-7}, \\ \Delta e &= [3.796 + 5.493 + (3 \times 0.615)] \times 10^{-11} = 1.113 \times 10^{-10}. \end{aligned}$$

Substitution of the numerical values into Eqn (13.4-18) gives

$$\begin{aligned} \ln K_a &= -10.108 \ln T - 1.083 \times 10^{-3}T + 7.788 \times 10^{-6}T^2 \\ &\quad - 3.831 \times 10^{-9}T^3 + 6.696 \times 10^{-13}T^4 \\ &\quad + \frac{22,020}{T} + 22.945. \end{aligned} \quad (3)$$

When $T = 550 \text{ K}$

$$K_a = 1.47.$$

Choosing 1 mol of benzene as a basis, the number of moles of each species present in the reactor can be expressed as a function of the molar extent of the reaction from Eqn (13.2-6) as

$$\begin{aligned} n_{\text{C}_6\text{H}_6} &= 1 - \epsilon \\ n_{\text{H}_2} &= 4.5 - 3\epsilon \\ n_{\text{C}_6\text{H}_{12}} &= \epsilon \\ \hline n_{\text{T}} &= 5.5 - 3\epsilon. \end{aligned}$$

Therefore, Eqn (2) can be expressed as

$$1.47 = \left(\frac{\epsilon}{1 - \epsilon} \right) \left(\frac{5.5 - 3\epsilon}{4.5 - 3\epsilon} \right)^3 \Rightarrow \epsilon = 0.4.$$

The mole fractions of each species are

$$\begin{aligned} y_{\text{C}_6\text{H}_6} &= \frac{1 - 0.4}{5.5 - (3)(0.4)} = 0.140, \\ y_{\text{H}_2} &= \frac{4.5 - (3)(0.4)}{5.5 - (3)(0.4)} = 0.767, \\ y_{\text{C}_6\text{H}_{12}} &= \frac{0.4}{5.5 - (3)(0.4)} = 0.093. \end{aligned}$$

(b) Since the pressure is moderately high, it is plausible to assume that nonideal gases form an ideal mixture, i.e. the Lewis-Randall rule applies. Taking $P^o = 1$ bar, Eqn (13.5-7) reduces to

$$K_a = K_{f/P} K_y P^{-3} \Rightarrow K_y = \frac{K_a P^3}{K_{f/P}}. \quad (4)$$

Fugacity coefficients of pure components, calculated from the Peng-Robinson equation of state, Eqn (5.3-9), are given in the following table:

Species	T_c (K)	P_c (bar)	ω_i	ϕ_i
C_6H_6	562.0	48.9	0.212	0.879
H_2	33.2	13.0	-0.216	1.004
C_6H_{12}	554.0	40.7	0.212	0.862

Therefore, $K_{f/P}$ is

$$K_{f/P} = \frac{\phi_{\text{C}_6\text{H}_{12}}}{\phi_{\text{C}_6\text{H}_6} \phi_{\text{H}_2}^3} = \frac{0.862}{(0.879)(1.004)^3} = 0.969.$$

Substitution of numerical values into Eqn (4) gives

$$K_y = \frac{(1.47)(15)^3}{0.969} = 5120.$$

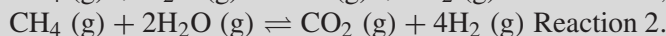
The molar extent of the reaction is calculated from

$$K_y = 5120 = \left(\frac{\epsilon}{1 - \epsilon} \right) \left(\frac{5.5 - 3\epsilon}{4.5 - 3\epsilon} \right)^3 \Rightarrow \epsilon = 0.999,$$

which indicates that percent conversion of benzene to cyclohexane is almost 100% and the mole fractions are

$$y_{\text{C}_6\text{H}_6} = 0.0004, \quad y_{\text{C}_6\text{H}_{12}} = 0.3991, \quad y_{\text{H}_2} = 0.6005.$$

Example 13.10 A mixture of methane and steam in the mole ratio 1:5 enters a reactor operating at 800 K and 2 bar. Estimate the composition of the product stream at equilibrium if the following reactions take place within the reactor:



Solution

From Appendix E

Species	$\Delta \tilde{H}_f^\circ$ (kJ/mol)	$\Delta \tilde{G}_f^\circ$ (kJ/mol)
CH ₄	−74.90	−50.87
H ₂ O	−242.00	−228.80
CO	−110.60	−137.40
CO ₂	−393.80	−394.60

From Appendix B

Species	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
CH ₄	36.155	−0.511	2.215	−1.824	4.899
H ₂ O	33.763	−0.006	0.224	−0.100	0.110
CO	29.651	−0.007	0.183	−0.094	0.108
H ₂	27.004	0.119	−0.241	0.215	−0.615
CO ₂	29.268	−0.224	2.653	−4.153	20.057

Following the procedure outlined in Example 13.6, the equilibrium constants at 800 K are calculated as

$$(K_a)_1 = 0.031 \quad \text{and} \quad (K_a)_2 = 0.096.$$

Let ϵ_1 and ϵ_2 be the extents of the first and second reactions, respectively. The number of moles of each species present in the reactor can be expressed as a function of the molar extents of the reactions in the form

$$\begin{array}{rcl} n_{\text{CH}_4} & = & 1 - \epsilon_1 - \epsilon_2 \\ n_{\text{H}_2\text{O}} & = & 5 - \epsilon_1 - 2\epsilon_2 \\ n_{\text{CO}} & = & \epsilon_1 \\ n_{\text{CO}_2} & = & \epsilon_2 \\ n_{\text{H}_2} & = & 3\epsilon_1 + 4\epsilon_2 \\ \hline n_{\text{T}} & = & 6 + 2\epsilon_1 + 2\epsilon_2. \end{array}$$

Assuming an ideal gas mixture and taking $P^o = 1$ bar, Eqn (13.5-9) gives

$$(K_a)_1 = (K_y)_1 P^2 \quad (1)$$

and

$$(K_a)_2 = (K_y)_2 P^2. \quad (2)$$

K_y values are given by

$$\begin{aligned} (K_y)_1 &= \frac{(n_{\text{CO}}/n_{\text{T}}) (n_{\text{H}_2}/n_{\text{T}})^3}{(n_{\text{CH}_4}/n_{\text{T}}) (n_{\text{H}_2\text{O}}/n_{\text{T}})} = \frac{n_{\text{CO}} n_{\text{H}_2}^3}{n_{\text{CH}_4} n_{\text{H}_2\text{O}} n_{\text{T}}^2} \\ &= \frac{\epsilon_1 (3\epsilon_1 + 4\epsilon_2)^3}{(1 - \epsilon_1 - \epsilon_2) (5 - \epsilon_1 - 2\epsilon_2) (6 + 2\epsilon_1 + 2\epsilon_2)^2}, \end{aligned} \quad (3)$$

$$\begin{aligned} (K_y)_2 &= \frac{(n_{\text{CO}_2}/n_{\text{T}}) (n_{\text{H}_2}/n_{\text{T}})^4}{(n_{\text{CH}_4}/n_{\text{T}}) (n_{\text{H}_2\text{O}}/n_{\text{T}})^2} = \frac{n_{\text{CO}_2} n_{\text{H}_2}^4}{n_{\text{CH}_4} n_{\text{H}_2\text{O}}^2 n_{\text{T}}^2} \\ &= \frac{\epsilon_2 (3\epsilon_1 + 4\epsilon_2)^4}{(1 - \epsilon_1 - \epsilon_2) (5 - \epsilon_1 - 2\epsilon_2)^2 (6 + 2\epsilon_1 + 2\epsilon_2)^2}. \end{aligned} \quad (4)$$

Substitution of Eqns (3) and (4) into Eqns (1) and (2), respectively, yields

$$0.031 = \left[\frac{\epsilon_1 (3\epsilon_1 + 4\epsilon_2)^3}{(1 - \epsilon_1 - \epsilon_2) (5 - \epsilon_1 - 2\epsilon_2) (6 + 2\epsilon_1 + 2\epsilon_2)^2} \right] (2)^2, \quad (5)$$

$$0.096 = \left[\frac{\epsilon_2 (3\epsilon_1 + 4\epsilon_2)^4}{(1 - \epsilon_1 - \epsilon_2) (5 - \epsilon_1 - 2\epsilon_2)^2 (6 + 2\epsilon_1 + 2\epsilon_2)^2} \right] (2)^2. \quad (6)$$

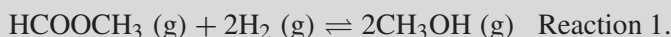
Simultaneous solution of Eqns (5) and (6) by MATHCAD® gives

$$\epsilon_1 = 0.078 \quad \text{and} \quad \epsilon_2 = 0.463.$$

Therefore, the mole fractions of the gas mixture under equilibrium conditions are given by

$$y_{\text{CH}_4} = 0.065, \quad y_{\text{H}_2\text{O}} = 0.564, \quad y_{\text{CO}} = 0.011, \quad y_{\text{CO}_2} = 0.065, \quad y_{\text{H}_2} = 0.295.$$

Example 13.11 You are given the task of producing methanol from hydrogenolysis of methyl formate (HCOOCH_3) over copper-containing catalyst, i.e.



Your boss is suspicious about the possibility of the following decarbonylation reaction



taking place in the reactor. If the hydrogen to methyl formate feed mole ratio is 4:1, and the reactor temperature and pressure are 500 K and 40 bar, respectively, estimate the gas composition under equilibrium conditions.

Data: For methyl formate

$$T_c = 487.2 \text{ K}, \quad P_c = 60 \text{ bar}, \quad \omega = 0.257.$$

The gases can be represented by the Soave-Redlich-Kwong equation of state ($k_{ij} = 0$). Kim *et al.* (1990) reported the equilibrium constants as

$$\ln K_{a_1} = \frac{5772.5}{T} - 4.81 \ln T + 1.72 \times 10^{-3}T - 6.79 \times 10^{-7}T^2 + 19.1,$$

$$\ln K_{a_2} = -\frac{5130}{T} + 3.16 \ln T - 4.023 \times 10^{-3}T + 1.55 \times 10^{-6}T^2 - 14.15,$$

where T is in K.

Solution

Since equilibrium constants are given as a function of temperature, ΔH_{rxn}^o can be calculated from Eqn (13.4-9) by differentiating given equilibrium constant expressions with respect to temperature as shown in Example 13.7. The results are

$$\begin{aligned}
\Delta H_{\text{rxn}_1}^o &= (8.314) [-5772.5 - (4.81)(500) \\
&\quad + (1.72 \times 10^{-3})(500)^2 - (13.58 \times 10^{-7})(500)^3] \\
&= -65,824 \text{ J}, \\
\Delta H_{\text{rxn}_2}^o &= (8.314) [5130 + (3.16)(500) - (4.023 \times 10^{-3})(500)^2 \\
&\quad + (3.1 \times 10^{-6})(500)^3] \\
&= 50,647 \text{ J}
\end{aligned}$$

indicating that while the first reaction is highly exothermic the second is highly endothermic. Calculation of the equilibrium constants gives

$$K_{a_1} = 4.233 \quad \text{and} \quad K_{a_2} = 0.012.$$

Since $K_{a_1} \gg K_{a_2}$, it seems that at the reaction temperature of 500 K the first reaction is dominant and the effect of the second reaction on the equilibrium gas composition can be safely neglected.⁷

Taking $P^o = 1$ bar, Eqn (13.5-5) becomes

$$K_a = K_\phi K_y P^{-1} = 4.233. \quad (1)$$

Since the reaction pressure is 40 bar, the assumption of ideal mixture behavior is doubtful. In this case, direct calculation of K_ϕ is impossible since determination of fugacity coefficients requires *a priori* knowledge of the equilibrium gas composition. For this reason, iterative calculations will be carried out indirectly as follows.

Choosing 1 mol of methyl formate as a basis, the number of moles of each species present in the reactor can be expressed as a function of the molar extent of the reaction in the form

$$\begin{array}{rcl}
n_{\text{HCOOCH}_3} &= & 1 - \epsilon \\
n_{\text{H}_2} &= & 4 - 2\epsilon \\
n_{\text{CH}_3\text{OH}} &= & 2\epsilon \\
\hline
n_{\text{T}} &= & 5 - \epsilon.
\end{array}$$

Therefore, mole fractions of the species are expressed as a function of the molar extent of reaction as

$$y_{\text{HCOOCH}_3} = \frac{1 - \epsilon}{5 - \epsilon}, y_{\text{H}_2} = \frac{4 - 2\epsilon}{5 - \epsilon}, y_{\text{CH}_3\text{OH}} = \frac{2\epsilon}{5 - \epsilon}. \quad (2)$$

⁷At higher temperatures, however, this assumption is certainly not valid. For example, $K_{a_1} = 0.283$ and $K_{a_2} = 0.145$ at 600 K, indicating that the two reactions are competing with each other. At higher temperatures, due to its endothermic behavior, the second reaction is expected to be the dominant one. Indeed, the equilibrium constants at 800 K are $K_{a_1} = 7.477 \times 10^{-3}$ and $K_{a_2} = 4.706$.

The expression for K_y becomes

$$K_y = \frac{y_{\text{CH}_3\text{OH}}^2}{y_{\text{HCOOCH}_3} y_{\text{H}_2}^2} = \frac{4\epsilon^2(5 - \epsilon)}{(1 - \epsilon)(4 - 2\epsilon)^2}. \quad (13.5-3)$$

The iterative procedure is given as follows:

1. Assume ϵ .
2. Calculate gas phase composition from Eqn (2).
3. Calculate K_y from Eqn (13.5-3).
4. Once fugacity coefficients are determined from Eqn (7.5-15), calculate K_ϕ from

$$K_\phi = \frac{\hat{\phi}_{\text{CH}_3\text{OH}}^2}{\hat{\phi}_{\text{HCOOCH}_3} \hat{\phi}_{\text{H}_2}^2}.$$

5. Check whether the following equation is satisfied:

$$f(\epsilon) = K_a - K_\phi K_y P^{-1} = 0.$$

6. If not, repeat the calculations with

$$\epsilon_{\text{new}} = \epsilon - \frac{0.02\epsilon f(\epsilon)}{f(1.01\epsilon) - f(0.99\epsilon)}.$$

According to the stoichiometry of the reaction, 1 mol of HCOOCH_3 reacts with 2 mol of H_2 . Since H_2 is in excess, HCOOCH_3 is the limiting reactant. If all HCOOCH_3 were depleted, the maximum value of ϵ would be 1. Thus, ϵ may take values less than unity. The results are shown in the table below:

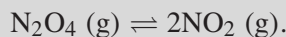
ϵ	K_y	Z_{mix}	$\hat{\phi}_{\text{HCOOCH}_3}$	$\hat{\phi}_{\text{H}_2}$	$\hat{\phi}_{\text{CH}_3\text{OH}}$	$K_a - K_y K_\phi P^{-1}$
0.980	185.5	0.965	0.867	1.065	0.871	0.650
0.983	220.8	0.965	0.867	1.065	0.871	-0.028

Thus, the equilibrium gas phase composition is

$$y_{\text{HCOOCH}_3} = \frac{1 - 0.983}{5 - 0.983} = 0.004, \quad y_{\text{H}_2} = \frac{4 - 2(0.983)}{5 - 0.983} = 0.506,$$

$$y_{\text{CH}_3\text{OH}} = \frac{2(0.983)}{5 - 0.983} = 0.489.$$

Example 13.12 A 100 L constant-volume reactor is evacuated and then filled with 1 mol of dinitrogen tetroxide (N_2O_4) at 298 K. It decomposes according to the reaction



Estimate the equilibrium gas composition if the temperature is kept constant at 298 K.

Solution

From Appendix E

Species	$\Delta \tilde{G}_f^o$ (kJ/mol)
N_2O_4	97.79
NO_2	51.26

Therefore, ΔG_{rxn}^o at 298 K is

$$\Delta G_{\text{rxn},298}^o = (2)(51.26) - 97.79 = 4.73 \text{ kJ} = 4730 \text{ J}.$$

The equilibrium constant at 298 K is

$$K_{a,298} = \exp\left(-\frac{\Delta G_{\text{rxn},298}^o}{RT}\right) = \exp\left[\frac{-4730}{(8.314)(298)}\right] = 0.148.$$

The initial pressure, P_o , within the reactor is

$$P_o = \frac{n_{T_o}RT}{V} = \frac{(1)(8.314 \times 10^{-2})(298)}{100} = 0.248 \text{ bar}.$$

Assuming ideal gas behavior and choosing $P^o = 1 \text{ bar}$, Eqn (13.5-9) reduces to

$$K_a = K_y P = \frac{(n_{\text{NO}_2}/n_T)^2}{n_{\text{N}_2\text{O}_4}/n_T} P = \frac{n_{\text{NO}_2}^2}{n_{\text{N}_2\text{O}_4} n_T} P. \quad (1)$$

The number of moles of each species present in the reactor can be expressed as a function of the molar extent of the reaction as

$$\begin{array}{rcl} n_{\text{N}_2\text{O}_4} & = & 1 - \epsilon \\ n_{\text{NO}_2} & = & 2\epsilon \\ \hline n_T & = & 1 + \epsilon. \end{array}$$

In this case, the pressure within the reactor is not constant and changes as a function of the molar extent of reaction in the form

$$P = \left(\frac{n_T}{n_{T_o}} \right) P_o = 0.248 (1 + \epsilon) \text{ bar.} \quad (2)$$

Thus, Eqn (1) takes the form

$$0.148 = \left[\frac{4\epsilon^2}{(1 - \epsilon)(1 + \epsilon)} \right] 0.248 (1 + \epsilon) \Rightarrow \frac{\epsilon^2}{1 - \epsilon} = 0.149.$$

The solution of the above quadratic equation gives $\epsilon = 0.319$. The mole fractions are

$$y_{\text{N}_2\text{O}_4} = \frac{1 - 0.319}{1 + 0.319} = 0.516 \quad \text{and} \quad y_{\text{NO}_2} = \frac{(2)(0.319)}{1 + 0.319} = 0.484.$$

13.5.1 Variables Affecting the Extent of Reaction

For simplicity let us consider an ideal gas mixture. Taking the standard state pressure as 1 bar, Eqn (13.5-9) reduces to

$$K_y = K_a P^{-\bar{\alpha}}. \quad (13.5-10)$$

Note that while K_a depends only on temperature K_y is a function of both temperature and pressure. Before investigating the effect of various factors, i.e. temperature, pressure, and the amount of inerts, on the extent of reaction, it is first necessary to determine how K_y varies with the extent of reaction.

The mole fraction of each species is represented by

$$y_i = \frac{n_i}{n_T} = \frac{n_{i_o} + \alpha_i \epsilon}{n_{T_o} + \bar{\alpha} \epsilon} = \frac{y_{i_o} + \alpha_i \epsilon^*}{1 + \bar{\alpha} \epsilon^*}, \quad (13.5-11)$$

where

$$\epsilon^* = \frac{\epsilon}{n_{T_o}}. \quad (13.5-12)$$

By definition

$$K_y = \prod_{i=1}^k y_i^{\alpha_i} \Rightarrow \ln K_y = \sum_{i=1}^k \alpha_i \ln y_i. \quad (13.5-13)$$

Differentiation of Eqn (13.5-13) with respect to ϵ^* yields

$$\frac{d \ln K_y}{d \epsilon^*} = \sum_{i=1}^k \alpha_i \frac{d \ln y_i}{d \epsilon^*} = \sum_{i=1}^k \frac{\alpha_i}{y_i} \frac{dy_i}{d \epsilon^*}. \quad (13.5-14)$$

From Eqn (13.5-11)

$$\frac{dy_i}{d \epsilon^*} = \frac{\alpha_i (1 + \bar{\alpha} \epsilon^*) - \bar{\alpha} (y_{i_o} + \alpha_i \epsilon^*)}{(1 + \bar{\alpha} \epsilon^*)^2} = \frac{\alpha_i}{1 + \bar{\alpha} \epsilon^*} - \frac{\bar{\alpha} y_i}{1 + \bar{\alpha} \epsilon^*}. \quad (13.5-15)$$

Substitution of Eqn (13.5-15) into Eqn (13.5-14) results in

$$\frac{d \ln K_y}{d\epsilon^*} = \frac{1}{1 + \bar{\alpha}\epsilon^*} \left[\sum_{i=1}^k \frac{\alpha_i^2}{y_i} - \left(\sum_{i=1}^k \alpha_i \right)^2 \right]. \quad (13.5-16)$$

According to Cauchy's inequality

$$\left(\sum_{i=1}^k a_i^2 \right) \left(\sum_{i=1}^k b_i^2 \right) \geq \left(\sum_{i=1}^k a_i b_i \right)^2. \quad (13.5-17)$$

If we let

$$a_i = \frac{\alpha_i}{\sqrt{y_i}} \quad \text{and} \quad b_i = \sqrt{y_i}, \quad (13.5-18)$$

then Eqn (13.5-17) gives

$$\left(\sum_{i=1}^k \frac{\alpha_i^2}{y_i} \right) \underbrace{\left(\sum_{i=1}^k y_i \right)}_1 \geq \left(\sum_{i=1}^k \alpha_i \right)^2 \quad (13.5-19)$$

indicating that the terms in brackets in Eqn (13.5-16) are always greater than zero. Also note that

$$1 + \bar{\alpha}\epsilon^* = \frac{n_T}{n_{T_o}} > 0. \quad (13.5-20)$$

Therefore, one can conclude that

$$\frac{d \ln K_y}{d\epsilon^*} > 0 \Rightarrow \frac{dK_y}{d\epsilon^*} > 0 \Rightarrow \frac{dK_y}{d\epsilon} > 0, \quad (13.5-21)$$

which implies that the molar extent of reaction increases with increasing K_y or vice versa.

Substitution of Eqn (13.3-13) into Eqn (13.5-10) results in

$$K_y = \underbrace{\exp \left(-\frac{\Delta G_{\text{rxn}}^o}{RT} \right)}_{K_a} P^{-\bar{\alpha}} \quad (13.5-22)$$

indicating that

Large negative values of $\Delta G_{\text{rxn}}^o \rightarrow \text{Large } K_a \rightarrow \text{Large } K_y \rightarrow \text{Large } \epsilon$,

Large positive values of $\Delta G_{\text{rxn}}^o \rightarrow \text{Small } K_a \rightarrow \text{Small } K_y \rightarrow \text{Small } \epsilon$.

Effect of temperature at constant pressure: Differentiation of the molar extent of the reaction with respect to temperature, while keeping pressure constant, is given by the chain rule as

$$\left(\frac{\partial \epsilon}{\partial T}\right)_P = \frac{d\epsilon}{dK_y} \left(\frac{\partial K_y}{\partial T}\right)_P = \frac{d\epsilon}{dK_y} \left(\frac{dK_a}{dT} P^{-\bar{\alpha}}\right). \quad (13.5-23)$$

The van't Hoff equation, Eqn (13.4-9), gives

$$\frac{d \ln K_a}{dT} = \frac{\Delta H_{\text{rxn}}^o}{RT^2} \Rightarrow \frac{dK_a}{dT} = K_a \left(\frac{\Delta H_{\text{rxn}}^o}{RT^2}\right). \quad (13.5-24)$$

Substitution of Eqn (13.5-24) into Eqn (13.5-23) yields

$$\left(\frac{\partial \epsilon}{\partial T}\right)_P = \left(\frac{d\epsilon}{dK_y} \frac{K_a}{RT^2} P^{-\bar{\alpha}}\right) \Delta H_{\text{rxn}}^o. \quad (13.5-25)$$

In Eqn (13.5-25), the terms in parentheses are always greater than zero. Therefore, the value of $(\partial \epsilon / \partial T)_P$ is dependent on the sign of ΔH_{rxn}^o , i.e.

$$\begin{aligned} \text{Exothermic rxn} &\Rightarrow \Delta H_{\text{rxn}}^o < 0 \Rightarrow (\partial \epsilon / \partial T)_P < 0 \\ &\Rightarrow \epsilon \text{ decreases with increasing } T, \end{aligned}$$

$$\begin{aligned} \text{Endothermic rxn} &\Rightarrow \Delta H_{\text{rxn}}^o > 0 \Rightarrow (\partial \epsilon / \partial T)_P > 0 \\ &\Rightarrow \epsilon \text{ increases with increasing } T. \end{aligned}$$

For a highly exothermic reaction, while equilibrium conversion decreases at higher temperatures, the reaction rate increases. Thus, the optimum reactor temperature should be determined as a compromise between reaction kinetics and thermodynamics.⁸

Effect of pressure at constant temperature: Differentiation of the molar extent of the reaction with respect to pressure at constant temperature is given by the chain rule as

$$\left(\frac{\partial \epsilon}{\partial P}\right)_T = \frac{d\epsilon}{dK_y} \left(\frac{\partial K_y}{\partial P}\right)_T = \left(\frac{d\epsilon}{dK_y} \frac{K_y}{P}\right) (-\bar{\alpha}). \quad (13.5-26)$$

Therefore, if the summation of the stoichiometric coefficients, $\bar{\alpha}$, is negative, increase in pressure at constant temperature results in an increase in the equilibrium extent of the reaction. On the other hand, if $\bar{\alpha}$ is positive, then an increase in pressure at constant temperature causes a decrease in the equilibrium extent of the reaction.

⁸Thermodynamics, reaction kinetics, and economics must be considered together in the design and operation of chemical reactors.

It should be kept in mind that the value of K_a does not change with pressure. However, the position of equilibrium may change with a change in pressure.

Effect of inerts addition at constant temperature and pressure: Since K_y is fixed at constant temperature and pressure, then

$$K_y = \text{constant} = \prod_{i=1}^k y_i^{\alpha_i} = \prod_{i=1}^k \left(\frac{n_i}{n_T} \right)^{\alpha_i} = \frac{\prod_{i=1}^k n_i^{\alpha_i}}{n_T^{\bar{\alpha}}}. \quad (13.5-27)$$

Since the total number of moles, n_T , increases with the addition of inerts, the variation of ϵ will be dependent on the summation of the stoichiometric coefficients, $\bar{\alpha}$.

When $\bar{\alpha} > 0$, addition of inerts increases the denominator of Eqn (13.5-27). Thus, the numerator, $\prod_{i=1}^k n_i^{\alpha_i}$, should increase so as to satisfy the equality, leading to an increase in ϵ .

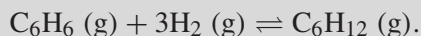
When $\bar{\alpha} < 0$, rearrangement of Eqn (13.5-27) gives

$$K_y = \text{constant} = n_T^{-\bar{\alpha}} \prod_{i=1}^k n_i^{\alpha_i}. \quad (13.5-28)$$

In this case, addition of inerts increases $n_T^{-\bar{\alpha}}$. Thus, $\prod_{i=1}^k n_i^{\alpha_i}$ should decrease to satisfy the equality, leading to a decrease in ϵ .

When $\bar{\alpha} = 0$, addition of inerts has no influence on the molar extent of the reaction.

Example 13.13 Consider the production of cyclohexane from the hydrogenation of benzene as given in Example 13.9 according to the reaction



Estimate the equilibrium extent of the reaction under the different sets of conditions given below:

- The hydrogen to benzene feed mole ratio is 4.5:1, and the reaction takes place at 600 K and 1 bar.
- The hydrogen to benzene feed mole ratio is 4.5:1, and the reaction takes place at 600 K and 2 bar.
- The hydrogen to benzene feed mole ratio is 3:1, and the reaction takes place at 600 K and 1 bar.
- The hydrogen to benzene feed mole ratio is 6:1, and the reaction takes place at 600 K and 1 bar.

Solution

In Example 13.9, the equilibrium constant is expressed as a function of temperature in the form

$$\ln K_a = -10.108 \ln T - 1.083 \times 10^{-3} T + 7.788 \times 10^{-6} T^2 - 3.831 \times 10^{-9} T^3 + 6.696 \times 10^{-13} T^4 + \frac{22,020}{T} + 22.945. \quad (1)$$

When $T = 600$ K

$$K_a = 0.027.$$

Assuming ideal gas behavior and taking $P^o = 1$ bar, Eqn (13.5-9) becomes

$$K_a = \frac{K_y}{P^3}, \quad (2)$$

where

$$K_y = \frac{n_{\text{C}_6\text{H}_{12}}}{n_{\text{C}_6\text{H}_6}} \left(\frac{n_{\text{T}}}{n_{\text{H}_2}} \right)^3. \quad (3)$$

(a) In Example 13.9, the equilibrium constant at 550 K was calculated as 1.47. Since the increase in temperature from 550 K to 600 K decreases K_a from 1.47 to 0.027, it is obvious that the equilibrium extent of the reaction will decrease. The same conclusion can also be reached by the application of *Le Chatelier's principle*, which states that “if a stress (or disturbance) is applied to a system at equilibrium, the system will react so as to relieve the stress.” In this specific case, the reaction is exothermic ($\Delta H_{\text{rxn},298}^o = -206,180$ J) and, as a result of the reaction, the heat produced can be regarded as a “product.” If the reaction temperature is increased, the reaction will shift to the left so as to relieve the disturbance with a concomitant decrease in the yield of cyclohexane.

To quantify this conclusion, let us choose 1 mol of benzene as a basis. Thus, the number of moles of each species present in the reactor can be expressed as a function of the molar extent of the reaction in the form

$$\begin{array}{rcl} n_{\text{C}_6\text{H}_6} & = & 1 - \epsilon \\ n_{\text{H}_2} & = & 4.5 - 3\epsilon \\ n_{\text{C}_6\text{H}_{12}} & = & \epsilon \\ \hline n_{\text{T}} & = & 5.5 - 3\epsilon. \end{array}$$

Thus, Eqn (2) takes the form

$$0.027 = \left(\frac{\epsilon}{1 - \epsilon} \right) \left(\frac{5.5 - 3\epsilon}{4.5 - 3\epsilon} \right)^3 \Rightarrow \epsilon = 0.014.$$

(b) In this case, the pressure is different from unity and Eqn (2) becomes

$$0.027 = \left(\frac{\epsilon}{1 - \epsilon} \right) \left(\frac{5.5 - 3\epsilon}{4.5 - 3\epsilon} \right)^3 \frac{1}{(2)^3} \Rightarrow \epsilon = 0.102$$

indicating an increase in the molar extent of the reaction with an increase in pressure. The same conclusion can also be reached by the application of Le Chatelier's principle. Since there are 4 mol of gaseous reactants on the left-hand side yielding 1 mol of gaseous cyclohexane on the right-hand side, an increase in pressure will shift the reaction to the right.

(c) In this case, benzene and hydrogen are introduced to the reactor in stoichiometric proportions. The number of moles of each species present in the reactor can be expressed as

$$\begin{array}{r} n_{\text{C}_6\text{H}_6} = 1 - \epsilon \\ n_{\text{H}_2} = 3 - 3\epsilon \\ n_{\text{C}_6\text{H}_{12}} = \epsilon \\ \hline n_{\text{T}} = 4 - 3\epsilon \end{array}$$

and Eqn (2) becomes

$$0.027 = \frac{\epsilon (4 - 3\epsilon)^3}{27(1 - \epsilon)^4} \Rightarrow \epsilon = 0.011.$$

In part (a), i.e. when the hydrogen to benzene mole ratio was 4.5:1, the molar extent of the reaction was calculated as 0.014. Thus, increasing the number of moles of hydrogen increases the molar extent of the reaction. This phenomenon is also explained by Le Chatelier's principle. When a reactant is added to a system at equilibrium, part of it will be consumed by the reaction as it establishes a new equilibrium state. This offsets some of the stress of the increase in reactant.

(d) The number of moles of each species present in the reactor can be expressed as

$$\begin{array}{r} n_{\text{C}_6\text{H}_6} = 1 - \epsilon \\ n_{\text{H}_2} = 6 - 3\epsilon \\ n_{\text{C}_6\text{H}_{12}} = \epsilon \\ \hline n_{\text{T}} = 7 - 3\epsilon \end{array}$$

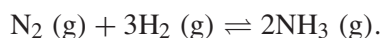
and Eqn (2) becomes

$$0.027 = \left(\frac{\epsilon}{1 - \epsilon} \right) \left(\frac{7 - 3\epsilon}{6 - 3\epsilon} \right)^3 \Rightarrow \epsilon = 0.017.$$

Therefore, as hydrogen is introduced in excess of the stoichiometric amount, the extent of the reaction slightly increases.

13.5.2 Exceptions to Le Chatelier's Principle

Consider the reaction between nitrogen and hydrogen to form ammonia, i.e.



Once equilibrium is established at the given temperature and pressure, what happens if more nitrogen is added to the system?

According to Le Chatelier's principle, one can immediately conclude that the reaction shifts to the right, producing more NH_3 , so as to relieve the disturbance. This conclusion, however, is not generally true as will be shown in the following analysis.

Assuming an ideal gas mixture and taking the standard state pressure as 1 bar, Eqn (13.5-9) becomes

$$K_a = \frac{y_{\text{NH}_3}^2}{y_{\text{N}_2} y_{\text{H}_2}^3} P^{-2}. \quad (13.5-29)$$

Since temperature and pressure are kept constant, the use of $y_i = n_i/n_T$ and rearrangement reduce Eqn (13.5-29) to

$$J = K_a P^2 = \text{constant} = \frac{n_{\text{NH}_3}^2 n_T^2}{n_{\text{N}_2} n_{\text{H}_2}^3}. \quad (13.5-30)$$

The variation of J with respect to the moles of nitrogen is given by

$$\frac{dJ}{dn_{\text{N}_2}} = \frac{n_{\text{NH}_3}^2}{n_{\text{H}_2}^3} \frac{d}{dn_{\text{N}_2}} \left[\frac{(n_{\text{N}_2} + n_{\text{H}_2} + n_{\text{NH}_3})^2}{n_{\text{N}_2}} \right] = \underbrace{\left(\frac{n_{\text{NH}_3}^2 n_T}{n_{\text{H}_2}^3 n_{\text{N}_2}^2} \right)}_{\text{Always positive}} (2n_{\text{N}_2} - n_T). \quad (13.5-31)$$

Thus, the sign of dJ/dn_{N_2} is dependent on the sign of $(2n_{\text{N}_2} - n_T)$.

When

$$2n_{\text{N}_2} - n_T > 0 \quad \Rightarrow \quad y_{\text{N}_2} = \frac{n_{\text{N}_2}}{n_T} > 0.5$$

addition of nitrogen leads to an increase in J . To re-establish the constancy of J , the reaction must shift to the left, producing more nitrogen. This is in contradiction with Le Chatelier's principle.

When

$$2n_{\text{N}_2} - n_T < 0 \quad \Rightarrow \quad y_{\text{N}_2} < 0.5$$

addition of nitrogen leads to a decrease in J . To re-establish the constancy of J , the reaction must proceed to the right, producing more ammonia as predicted by Le Chatelier's principle.

For a more detailed discussion on the subject, see [de Heer \(1957\)](#), [Katz \(1961\)](#), and [Corti and Franses \(2003\)](#).

13.6 Liquid (or Solid) Phase Reactions

The equilibrium constant, K_a , is defined by Eqn (13.3-11), i.e.

$$K_a = \prod_{i=1}^k \left[\frac{\hat{f}_i(T, P, x_i)}{f_i^o(T, P^o)} \right]^{\alpha_i} \quad (13.6-1)$$

Substitution of

$$\hat{f}_i(T, P, x_i) = \gamma_i(T, P, x_i) x_i f_i(T, P) \quad (13.6-2)$$

into Eqn (13.6-1) gives

$$K_a = \underbrace{\prod_{i=1}^k \gamma_i^{\alpha_i}}_{K_\gamma} \underbrace{\prod_{i=1}^k x_i^{\alpha_i}}_{K_x} \prod_{i=1}^k \left[\frac{f_i(T, P)}{f_i^o(T, P^o)} \right]^{\alpha_i} \quad (13.6-3)$$

Note that while f_i is the fugacity of pure liquid (or solid) at the temperature and pressure of the system f_i^o is the fugacity of pure liquid (or solid) at the temperature of the system but at standard state pressure, P^o . These two quantities are related by Eqn (5.4-11), i.e.

$$\frac{f_i(T, P)}{f_i^o(T, P^o)} = \exp \left[\frac{\tilde{V}_i(P - P^o)}{RT} \right] \quad (13.6-4)$$

Substitution of Eqn (13.6-4) into Eqn (13.6-3) gives

$$K_a = K_\gamma K_x \prod_{i=1}^k \left\{ \exp \left[\frac{\tilde{V}_i(P - P^o)}{RT} \right] \right\}^{\alpha_i} = K_\gamma K_x \exp \left[\frac{(P - P^o)}{RT} \sum_{i=1}^k \alpha_i \tilde{V}_i \right] \quad (13.6-5)$$

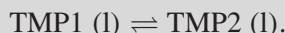
Except for high pressures, the exponential term in Eqn (13.6-5) will approach unity and may be neglected. Hence,

$$K_a = K_\gamma K_x \quad \text{Pressure correction neglected.} \quad (13.6-6)$$

For an ideal mixture, $K_y = 1$ and Eqn (13.6-6) reduces to

$$K_a = K_x \quad \text{Ideal mixture.} \quad (13.6-7)$$

Example 13.14 Methyl *tert*-butyl ether (MTBE) has been used in gasoline as an octane enhancer since the late 1970s. It is typically manufactured in petroleum refineries by reacting isobutene with methanol over a catalyst. Since it can easily mix with water, leading to groundwater pollution, there have been restrictions on the use of MTBE over the last decade. As a result, refineries have been seeking new uses for isobutene, one of which is to use it as a raw material in the production of isooctene. For this purpose, isobutene is first dimerized, 2,4,4-trimethyl-1-pentene (TMP1) and 2,4,4-trimethyl-2-pentene (TMP2) being the main dimeric products. The isomerization reaction between TMP1 and TMP2 is given by



Karinen *et al.* (2001) studied this reaction and represented the equilibrium constant as

$$\ln K_a = -\frac{421.67}{T} - 0.056.$$

Determine the equilibrium composition of the mixture if isomerization takes place at 333 K and atmospheric pressure.

Solution

The equilibrium constant K_a is

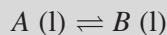
$$K_a = \exp\left(-\frac{421.67}{333} - 0.056\right) = 0.267. \quad (1)$$

Assuming that the reacting species form an ideal mixture, from Eqn (13.6-7)

$$K_a = 0.267 = K_x = \frac{x_{\text{TMP2}}}{x_{\text{TMP1}}} = \frac{x_{\text{TMP2}}}{1 - x_{\text{TMP2}}}. \quad (2)$$

The solution of Eqn (2) gives

$$x_{\text{TMP2}} = 0.211 \quad \text{and} \quad x_{\text{TMP1}} = 0.789.$$

Example 13.15 The isomerization reaction

takes place at 298 K. The molar excess Gibbs energy of the liquid mixture is given by

$$\frac{\tilde{G}^{\text{ex}}}{RT} = 0.8x_Ax_B.$$

Determine the equilibrium composition of the mixture if $\Delta G_{\text{rxn},298}^o = -2500$ J.

Solution

When pressure correction is neglected, the expression for K_a is given by Eqn (13.6-6), i.e.

$$K_a = K_\gamma K_x. \quad (1)$$

The equilibrium constant K_a is

$$K_a = \exp\left(-\frac{\Delta G_{\text{rxn}}^o}{RT}\right) = \exp\left[\frac{2500}{(8.314)(298)}\right] = 2.743. \quad (2)$$

From the given expression for the excess Gibbs energy (two-suffix Margules equation), the activity coefficients are

$$\ln \gamma_A = 0.8x_B^2 \quad \text{and} \quad \ln \gamma_B = 0.8x_A^2. \quad (3)$$

Therefore, K_γ is

$$K_\gamma = \frac{\gamma_B}{\gamma_A} = \exp[0.8(x_A^2 - x_B^2)] = \exp[0.8(2x_A - 1)]. \quad (4)$$

The expression for K_x is

$$K_x = \frac{x_B}{x_A} = \frac{1 - x_A}{x_A}. \quad (5)$$

The use of Eqns (2), (4), and (5) in Eqn (1) leads to

$$2.743 = \left(\frac{1 - x_A}{x_A}\right) \exp[0.8(2x_A - 1)]. \quad (6)$$

The solution of this nonlinear equation gives $x_A = 0.179$.

Comment: If we were to take $K_\gamma = 1$, the result would be $x_A = 0.267$.

The presence of two species in Examples 13.14 and 13.15 certainly simplifies the calculations. Liquid phase reactions usually involve more than two species and the resulting mixture cannot be considered ideal in most cases. As a result, calculation of activity coefficients and their dependence on the mole fraction of species complicate the calculations. In this case, the following procedure should be used:

1. Calculate K_a from Eqn (13.3-13).
2. Express mole fractions of each species, x_i , in terms of the extent of the reaction, ϵ .
3. Assume ϵ .
4. Calculate x_i and K_x .
5. Calculate activity coefficients of species involved in the reaction.⁹
6. Calculate K_γ .
7. Check whether the following equation is satisfied:

$$f(\epsilon) = K_a - K_\gamma K_x = 0.$$

8. If not, go to step 4 and repeat the calculations with

$$\epsilon_{\text{new}} = \epsilon - \frac{0.02\epsilon f(\epsilon)}{f(1.01\epsilon) - f(0.99\epsilon)}.$$

13.7 An Alternative Way of Calculating Equilibrium Composition

In Section 13.5, equilibrium compositions of the gas phase reactions are determined by first expressing the number of moles of each species in terms of the molar extent of reaction and then substituting these into the equilibrium constant expression(s).

An alternative way of calculating equilibrium composition is to first express the equilibrium constant in terms of the mole fractions of each species. The constraint on the mole fractions, i.e.

$$\sum_{i=1}^k y_i = 1 \quad (13.7-1)$$

⁹Usually UNIFAC is used to estimate γ_i values in a liquid mixture.

provides one additional equation. The conservation of elements is expressed in the form

$$E_j = \sum_{i=1}^k \beta_{ji} n_i, \quad j = 1, 2, \dots, t, \quad (13.7-2)$$

where t represents the number of elements. Dividing Eqn (13.7-2) by the total number of moles, n_T , gives

$$\frac{E_j}{n_T} = \sum_{i=1}^k \beta_{ji} y_i. \quad (13.7-3)$$

In matrix notation, Eqn (13.7-3) is expressed as

$$\begin{bmatrix} E_1/n_T \\ E_2/n_T \\ \vdots \\ E_t/n_T \end{bmatrix} = \underbrace{\begin{bmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \cdots & \beta_{1k} \\ \beta_{21} & \beta_{22} & \beta_{23} & \cdots & \beta_{2k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{t1} & \beta_{t2} & \beta_{t3} & \cdots & \beta_{tk} \end{bmatrix}}_{\beta} \cdot \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}, \quad (13.7-4)$$

where $[\beta]$ is the element-by-species matrix defined by Eqn (13.1-10). Simultaneous solution of Eqns (13.7-1) and (13.7-4) with the equilibrium constant expression gives the composition under equilibrium conditions. This approach is generally preferred for heterogeneous reactions.

Example 13.16 Resolve part (a) of Example 13.9 using an alternative approach.

Solution

The equilibrium constant expression reduces to

$$K_a = K_y = \frac{y_{\text{C}_6\text{H}_{12}}}{y_{\text{C}_6\text{H}_6} y_{\text{H}_2}^3} = 1.47. \quad (1)$$

To determine four unknowns, i.e. $y_{\text{C}_6\text{H}_{12}}$, $y_{\text{C}_6\text{H}_6}$, y_{H_2} , and n_T , three more equations are needed besides Eqn (1). One of these equations is Eqn (13.7-1), i.e.

$$y_{\text{C}_6\text{H}_6} + y_{\text{H}_2} + y_{\text{C}_6\text{H}_{12}} = 1. \quad (2)$$

The remaining two equations come from the conservation of elements. The element-by-species matrix is written as

$$[\beta] = \begin{array}{c} \text{Species} \rightarrow \\ \begin{array}{c} \text{C} \\ \text{H} \end{array} \end{array} \begin{array}{ccc} \text{C}_6\text{H}_6 & \text{H}_2 & \text{C}_6\text{H}_{12} \\ \left[\begin{array}{ccc} 6 & 0 & 6 \\ 6 & 2 & 12 \end{array} \right] \end{array} \quad (3)$$

Hydrogen to benzene feed ratio is 4.5:1. Choosing 1 mol of C_6H_6 as a basis, 6 atoms of carbon and 15 atoms of hydrogen enter the reactor. Hence, Eqn (13.7-4) takes the form

$$\begin{bmatrix} 6/n_T \\ 15/n_T \end{bmatrix} = \begin{bmatrix} 6 & 0 & 6 \\ 6 & 2 & 12 \end{bmatrix} \cdot \begin{bmatrix} y_{\text{C}_6\text{H}_6} \\ y_{\text{H}_2} \\ y_{\text{C}_6\text{H}_{12}} \end{bmatrix} \quad (4)$$

or

$$6y_{\text{C}_6\text{H}_6} + 6y_{\text{C}_6\text{H}_{12}} = \frac{6}{n_T}, \quad (5)$$

$$6y_{\text{C}_6\text{H}_6} + 2y_{\text{H}_2} + 12y_{\text{C}_6\text{H}_{12}} = \frac{15}{n_T}. \quad (6)$$

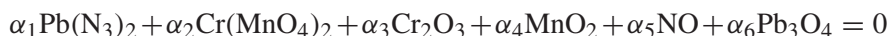
Simultaneous solution of Eqns (1), (2), (5), and (6) by MATHCAD[®] yields

$$y_{\text{C}_6\text{H}_6} = 0.140, \quad y_{\text{H}_2} = 0.768, \quad y_{\text{C}_6\text{H}_{12}} = 0.093, \quad n_T = 4.302 \text{ mol.}$$

Problems

Problem Related to Section 13.1

13.1 Balance the reaction

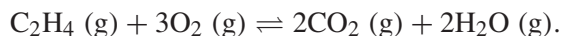


using Eqn (13.1-11).

(Answer: $\alpha_1 = -15$, $\alpha_2 = -44$, $\alpha_3 = 22$, $\alpha_4 = 88$, $\alpha_5 = 90$, $\alpha_6 = 5$)

Problems Related to Section 13.2

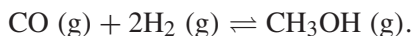
13.2 A system containing 3 mol of C_2H_4 and 10 mol of O_2 undergoes the following reaction:



Calculate the mole fractions of each species when $\epsilon = 1.5$.

(**Answer:** $y_{\text{C}_2\text{H}_4} = 0.115$, $y_{\text{O}_2} = 0.423$, $y_{\text{CO}_2} = y_{\text{H}_2\text{O}} = 0.231$)

- 13.3** A mixture of 25 mol% CO and 75% H₂ is fed to a reactor in which the following reaction takes place:



The reaction proceeds until the mole fraction of H₂ in the gas mixture drops to 0.6. Estimate the mole fractions of CO and CH₃OH.

(**Answer:** $y_{\text{CO}} = 0.1$, $y_{\text{CH}_3\text{OH}} = 0.3$)

Problem Related to Section 13.3

- 13.4** Rearrange Eqn (13.3-13) in the form

$$\ln K_a = -\frac{\Delta H_{\text{rxn}}^o}{RT} + \frac{\Delta S_{\text{rxn}}^o}{R}. \quad (1)$$

In his Priestley Medal Address, [Rossini \(1971\)](#) stated that the equilibrium constant, which is a measure of the reaction's spontaneity, increases with either an increase in ΔS_{rxn}^o or a decrease in ΔH_{rxn}^o . An increase in ΔS_{rxn}^o leads to an increase in randomness and a decrease in ΔH_{rxn}^o implies bond formation leading to a more stable system. Thus, by analogy, Rossini related the terms $\Delta S_{\text{rxn}}^o/R$ and $-\Delta H_{\text{rxn}}^o/RT$ to personal freedom and personal security, respectively. He then concluded that one cannot have a maximum of freedom and a maximum of security at the same time. In other words, an increase in security is accompanied by a decrease in freedom. Do you agree with Rossini's statement? For further discussion on the subject, see [Leonard \(2006\)](#), [Wojcik \(2006\)](#), and [Silverstein \(2006\)](#).

Problems Related to Section 13.4

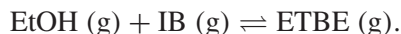
- 13.5** Assume ΔH_{rxn}^o to be independent of temperature and show that the equilibrium constant of the reaction



is given by

$$\ln K_a = -\frac{26,606}{T} - 21.63.$$

13.6 Ethyl *tert*-butyl ether (ETBE), a good octane booster, is obtained by the gas phase reaction of ethanol (EtOH) and isobutylene (IB) according to the reaction



- (a) *Iborra et al. (1989)* reported enthalpy and Gibbs energy of formation data in the gas phase as follows:

Species	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
EtOH	−234.81	−168.28
IB	−16.90	58.07
ETBE	−311.83	−124.60

Heat capacities are expressed in the form

$$\tilde{C}_p^o \text{ (J/mol K)} = a + bT + cT^2 + dT^3, \quad T \text{ in K}$$

with the following parameters:

Species	a	$b \times 10^2$	$c \times 10^5$	$d \times 10^9$
EtOH	9.008	21.393	−8.385	1.372
IB	16.041	28.024	−10.908	9.092
ETBE	−0.649	63.388	−34.974	242.253

Show that the equilibrium constant of the reaction is given by

$$\ln K_a = -2.173 + \frac{6944.8}{T} - 3.091 \ln T + 8.402 \times 10^{-3}T - 3.143 \\ \times 10^{-6}T^2 + 2.323 \times 10^{-9}T^3.$$

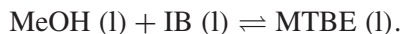
- (b) *Iborra et al. (1989)* also reported the equilibrium composition of the gas mixture as

T (K)	P (bar)	y_{ETBE}	y_{EtOH}	y_{IB}	y_{N_2}
373.4	1.069	0.007	0.063	0.150	0.723
363.5	1.097	0.009	0.077	0.085	0.828
353.4	1.059	0.012	0.049	0.101	0.839
343.9	1.085	0.016	0.052	0.066	0.866
333.9	1.074	0.025	0.038	0.082	0.855
338.9	1.079	0.026	0.048	0.094	0.833
328.9	1.071	0.026	0.030	0.070	0.874
323.7	1.080	0.037	0.025	0.093	0.845
319.8	1.078	0.032	0.021	0.072	0.876

Using these experimental data show that the equilibrium constant is represented by

$$\ln K_a = \frac{7424}{T} - 20.2.$$

- 13.7** Liquid phase synthesis of methyl *tert*-butyl ether (MTBE) from methanol (MeOH) and isobutene (IB) is given by



Zhang and Datta (1995) reported enthalpy and Gibbs energy of formation data in the liquid phase as follows:

Species	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
MeOH	-238.91	-166.64
IB	-37.70	60.672
MTBE	-315.13	-119.87

Liquid phase heat capacities are expressed in the form

$$\tilde{C}_p^o \text{ (J/mol K)} = a + bT + cT^2 + dT^3, \quad T \text{ in K}$$

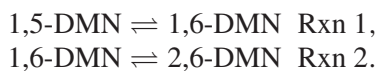
with the following parameters:

Species	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$
MeOH	7.696	1.617	2.058	2.847
IB	35.44	8.020	-31.24	50.45
MTBE	53.41	7.335	-16.25	21.52

Show that the equilibrium constant of the reaction is given by

$$\ln K_a = -13.496 + \frac{4391}{T} + 1.2357 \ln T - 0.014T + 2.592 \times 10^{-5}T^2 - 3.185 \times 10^{-8}T^3.$$

- 13.8** 2,6-Dimethylnaphthalene (2,6-DMN) is an intermediate for producing high-performance thermoplastic polyethylene naphthalate. It is obtained by the liquid phase isomerization of 1,5-DMN according to the following reactions:



The equilibrium constants for these reactions are reported by Kraikul *et al.* (2005) as

$$\ln K_{a,1} = 3.872 - \frac{1263}{T},$$

$$\ln K_{a,2} = 0.6746 - \frac{319.5}{T}.$$

For 2,6-DMN

$$\Delta \tilde{H}_f^\circ = 15,196 \text{ J/mol}, \quad \Delta \tilde{G}_f^\circ = 184,924 \text{ J/mol}.$$

Estimate $\Delta \tilde{H}_f^\circ$ and $\Delta \tilde{G}_f^\circ$ for 1,5-DMN.

(Answer: 2039 J/mol and 183,031 J/mol)

- 13.9** The standard Gibbs energy of formation values for PbO(s) are reported by Ganesan *et al.* (2003) as a function of temperature as follows:

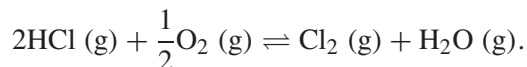
$\Delta \tilde{G}_f^\circ$ (kJ/mol)	−149.2	−129.3	−109.4
T (K)	700	900	1100

Calculate $\Delta \tilde{H}_f^\circ$ for PbO(s) at 800 K.

(Answer: −218.8 kJ/mol)

Problems Related to Section 13.5

- 13.10** Oxidation of HCl by air (oxygen) in the presence of a catalyst (CuCl_2) to produce Cl_2 is known as the *Deacon process* and is represented by



The feed consisting of 80 mol% HCl, 19.5% O_2 , and 0.5% N_2 enters the reactor operating at 700 K and 1 bar. Estimate the composition of the equilibrium mixture.

(Answer: HCl: 25.51%, O_2 : 5.79%, $\text{Cl}_2 = \text{H}_2\text{O}$: 34.06%, N_2 : 0.58%)

- 13.11** Estimate the temperature at which the fractional conversion of propane in the following reaction:



is 90%. The pressure is 1 bar. Assume ideal gases and ΔH_{rxn}^o to be independent of temperature.

(Answer: 655.7 K)

- 13.12** One of the current commercially important processes to produce acetone ($\text{C}_3\text{H}_6\text{O}$) is the catalytic dehydrogenation of isopropanol ($\text{C}_3\text{H}_8\text{O}$) in the vapor phase at 1 bar according to the reaction



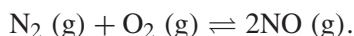
- (a) Using the procedure outlined in Section 13.4.3, estimate $\Delta \tilde{G}_f^o$ for isopropanol and acetone at 400, 450, 500, 550, 600, and 650 K.
- (b) Calculate K_a at these temperatures.
- (c) Plot $\ln K_a$ versus $1/T$ and show that

$$\ln K_a = -\frac{6823.7}{T} + 14.559.$$

- (d) Neglecting other reactions, estimate the minimum temperature for 95% conversion of isopropanol to acetone.

(Answer: (d) 553.3 K)

- 13.13** Nitric oxide, NO, is a common air pollutant produced by automobile engines and power plants according to the reaction



A constant-volume reactor of 100 L is initially charged with 4 mol of N_2 and 1 mol of O_2 and then temperature is increased to 800 K. Estimate the equilibrium extent of the reaction if the pressure is kept constant at 1 bar.

(Answer: $\epsilon = 5.567 \times 10^{-6}$)

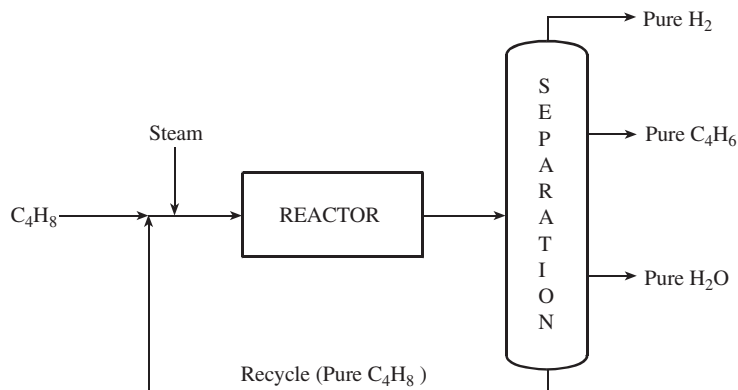
- 13.14** Butadiene (C_4H_6) is produced by the thermal cracking of 1-butene (C_4H_8) according to the reaction



The equilibrium constant is given by

$$\ln K_a = -\frac{13,998}{T} + 14.343.$$

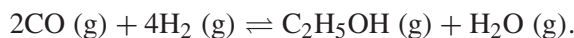
Before entering the reactor, the 1-butene is diluted with steam (5 mol of steam per mole of 1-butene) to suppress side reactions. A simplified flowsheet of the process is given below.



- If the reactor operates at 900 K and 1 bar, determine the composition of the stream exiting the reactor.
- When you report to work one morning, you notice that the recycle ratio of C_4H_8 has decreased. What conditions or situations might have caused such a response and why?

(Answer: (a) $y_{\text{C}_4\text{H}_8} = 0.04$, $y_{\text{C}_4\text{H}_6} = y_{\text{H}_2} = 0.109$, $y_{\text{H}_2\text{O}} = 0.743$)

13.15 Ethanol synthesis from syngas is represented by the reaction



An equimolar mixture of CO and H_2 enters the reactor operating at 573 K. Estimate the gas composition under equilibrium conditions if the reactor pressure is

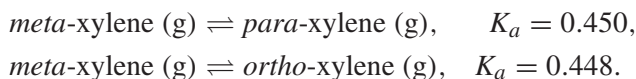
- 1 bar,
- 15 bar (assume ideal mixture behavior and use the Peng-Robinson equation of state).

(Answer: (a) $y_{\text{CO}} = 0.5$, $y_{\text{H}_2} = 0.418$, $y_{\text{C}_2\text{H}_5\text{OH}} = y_{\text{H}_2\text{O}} = 0.041$ and (b) $y_{\text{CO}} = 0.5$, $y_{\text{H}_2} = 0.053$, $y_{\text{C}_2\text{H}_5\text{OH}} = y_{\text{H}_2\text{O}} = 0.224$)

13.16 The following xylene isomerization scheme has been proposed by [Hanson and Engel \(1967\)](#) over silica-alumina catalyst:



They also reported the equilibrium constants at 644 K as

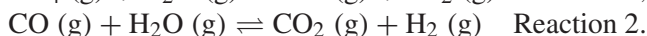


If 1 mol of *ortho*-xylene is fed to a reactor operating at 1 bar and 644 K, determine the equilibrium gas phase composition.

(**Answer:** *ortho* = 0.236, *meta* = 0.527, and *para* = 0.237)

Problem Related to Section 13.7

13.17 The following reactions take place during the steam reforming of methane at 1000 K and 1 bar:



The equilibrium constants are given by

$$K_{a1} = \exp \left(-\frac{23,703}{T} + 3.66 \ln T + 1.57 \right),$$

$$K_{a2} = \exp \left(\frac{5161.7}{T} + 0.98 \ln T - 11.56 \right).$$

The $\text{H}_2\text{O}/\text{CH}_4$ molar ratio of the feed introduced into the reactor is 5.

(a) Use the procedure explained in Section 13.5 and show that the equilibrium gas phase composition is given as

$$y_{\text{CO}} = 0.052, \quad y_{\text{CO}_2} = 0.073, \quad y_{\text{H}_2} = 0.446,$$

$$y_{\text{CH}_4} = 4.644 \times 10^{-4}, \quad y_{\text{H}_2\text{O}} = 0.428.$$

(b) Estimate the equilibrium gas composition by using the procedure explained in Section 13.7.

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Chapter 14

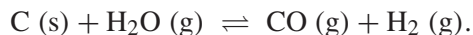
Heterogeneous Reactions and Multireaction Equilibria

In general, the reactions taking place in a reactor are numerous. Moreover, most of them are heterogeneous. This chapter starts with how to express equilibrium constants for heterogeneous reactions. Determination of independent reactions is then explained. Feasibility of a reaction and carbon deposition are also discussed. The chapter ends with the calculation of chemical equilibrium by minimization of Gibbs energy.

14.1 Heterogeneous Reactions

While reactants and products are in the same phase for homogeneous reactions, they are in different phases for heterogeneous reactions. As in the case of homogeneous reactions, the choices of standard states for heterogeneous reactions are completely arbitrary, except for the temperature, which must always be the temperature of the system. Therefore, it is possible to choose different standard states for different compounds.

Equations developed in Section 13.4 are applicable to all (homogeneous and heterogeneous) reactions. To calculate the equilibrium constant of a heterogeneous reaction, one must start with Eqn (13.3-11) and simplify this expression depending on the reaction conditions. Consider, for example, conversion of hydrocarbons to carbon monoxide and hydrogen (synthesis gas), which is represented by the water-carbon reaction as



From Eqn (13.3-11), the equilibrium constant is

$$K_a = \frac{\hat{a}_{\text{CO}}\hat{a}_{\text{H}_2}}{\hat{a}_{\text{C}}\hat{a}_{\text{H}_2\text{O}}} = \frac{\left(\frac{\hat{f}_{\text{CO}}}{f_{\text{CO}}^o}\right)\left(\frac{\hat{f}_{\text{H}_2}}{f_{\text{H}_2}^o}\right)}{\left(\frac{\hat{f}_{\text{C}}}{f_{\text{C}}^o}\right)\left(\frac{\hat{f}_{\text{H}_2\text{O}}}{f_{\text{H}_2\text{O}}^o}\right)}. \quad (14.1-1)$$

If the solubility of gases (H_2O , CO , H_2) in the solid carbon is negligible, then carbon is present as a pure solid phase. As a result, the activity of carbon is expressed as

$$\hat{a}_{\text{C}} = \frac{\hat{f}_{\text{C}}}{f_{\text{C}}^o} = \frac{f_{\text{C}}(T, P)}{f_{\text{C}}^o(T, P^o)} = a_{\text{C}} \quad (14.1-2)$$

in which f_{C} represents the fugacity of carbon at system temperature and pressure, and f_{C}^o is the fugacity of carbon at system temperature and standard state pressure. These two quantities are related by Eqn (5.5-9), i.e.

$$\frac{f_{\text{C}}(T, P)}{f_{\text{C}}^o(T, P^o)} = \exp\left[\frac{\tilde{V}_{\text{C}}(P - P^o)}{RT}\right]. \quad (14.1-3)$$

For solids, standard state pressure is taken as either 1 bar or sublimation pressure. Unless the pressure is very high, the exponential term in Eqn (14.1-3) is almost equal to unity. Hence, Eqn (14.1-1) simplifies to

$$K_a = \frac{\left(\frac{\hat{f}_{\text{CO}}}{f_{\text{CO}}^o}\right)\left(\frac{\hat{f}_{\text{H}_2}}{f_{\text{H}_2}^o}\right)}{\frac{\hat{f}_{\text{H}_2\text{O}}}{f_{\text{H}_2\text{O}}^o}}. \quad (14.1-4)$$

Standard states for gases are chosen at the T of the system and P^o (usually 1 bar) so as to make pure component fugacities equal to unity, i.e.

$$f_{\text{CO}}^o = f_{\text{H}_2}^o = f_{\text{H}_2\text{O}}^o = 1. \quad (14.1-5)$$

Thus, Eqn (14.1-4) becomes

$$K_a = \frac{\hat{f}_{\text{CO}}\hat{f}_{\text{H}_2}}{\hat{f}_{\text{H}_2\text{O}}}. \quad (14.1-6)$$

At low pressures, it is plausible to assume that the gas mixture behaves ideally, i.e. component fugacities are equal to partial pressures. In this case, Eqn (14.1-6) simplifies to

$$K_a = \frac{p_{\text{CO}}p_{\text{H}_2}}{p_{\text{H}_2\text{O}}} = \left(\frac{y_{\text{CO}}y_{\text{H}_2}}{y_{\text{H}_2\text{O}}}\right)P. \quad (14.1-7)$$

The mole fractions appearing in the K_a expression refer to the *mole fraction in the gas phase* and not the total mole fraction.

Note that solids do not appear in the heterogeneous equilibrium constant expression. As a result, if solid is a limiting reactant, then the reaction proceeds until the solid is completely consumed. In the case of homogeneous reactions, however, complete consumption of a limiting reactant implies $K_a = \infty$. Therefore, reactions in a homogeneous phase can never be 100% complete.

The equilibrium constant, K_a , can be calculated by following the procedure outlined in Section 13.4. This requires heat capacities of solids to be known as a function of temperature. Heat capacities of solids are generally expressed as

$$\tilde{C}_{P_i}^o = a_i + b_i T + c_i T^2 + d_i T^3 + \frac{f_i}{T^2} \quad (14.1-8)$$

which is known as the *Shomate equation*. The coefficients of Eqn (14.1-8) can be found from either NIST Chemistry WebBook¹ or Kubaschewski and Alcock (1979). For gases, heat capacities are expressed by Eqn (13.4-14), i.e.

$$\tilde{C}_{P_i}^o = a_i + b_i T + c_i T^2 + d_i T^3 + e_i T^4 \quad (14.1-9)$$

the coefficients for which are given in Appendix B. These values, which are valid up to 800 K, cannot be used for reactions taking place at temperatures higher than 800 K. At higher temperatures, heat capacities of gases are also expressed in the form of the Shomate equation.

The use of the generalized heat capacity expression in the form

$$\tilde{C}_{P_i}^o = a_i + b_i T + c_i T^2 + d_i T^3 + e_i T^4 + \frac{f_i}{T^2} \quad (14.1-10)$$

in Eqn (13.4-13) yields the following expression for the equilibrium constant:

$$R \ln K_a = \Delta a \ln T + \frac{\Delta b}{2} T + \frac{\Delta c}{6} T^2 + \frac{\Delta d}{12} T^3 + \frac{\Delta e}{20} T^4 + \frac{\Delta f}{2T^2} + \frac{\Lambda}{T} - \Omega, \quad (14.1-11)$$

where

$$\begin{aligned} \Lambda = & 298\Delta a + \frac{298^2\Delta b}{2} + \frac{298^3\Delta c}{3} + \frac{298^4\Delta d}{4} + \frac{298^5\Delta e}{5} \\ & - \frac{\Delta f}{298} - \Delta H_{\text{rxn},298}^o, \end{aligned} \quad (14.1-12)$$

$$\begin{aligned} \Omega = & (1 + \ln 298)\Delta a + 298\Delta b + \frac{298^2\Delta c}{2} + \frac{298^3\Delta d}{3} + \frac{298^4\Delta e}{4} \\ & - \frac{1}{2} \frac{\Delta f}{298^2} - \frac{\Delta H_{\text{rxn},298}^o - \Delta G_{\text{rxn},298}^o}{298}. \end{aligned} \quad (14.1-13)$$

¹<http://webbook.nist.gov/chemistry>.

In various references (NIST-JANAF Thermochemical Tables, 1998; Barin, 1989; Kubaschewski and Alcock, 1979) standard Gibbs energy of formation, $\Delta \tilde{G}_f^o$, values are given as a function of temperature. Table 14.1 lists $\Delta \tilde{G}_f^o$ values for a limited number of compounds. Once ΔG_{rxn}^o is determined from

$$\Delta G_{\text{rxn}}^o = \sum_{i=1}^k \alpha_i (\Delta \tilde{G}_f^o)_i \quad (14.1-14)$$

the equilibrium constant can be calculated from Eqn (13.3-13), i.e.

$$K_a(T) = \exp \left[-\frac{\Delta G_{\text{rxn}}^o(T)}{RT} \right]. \quad (14.1-15)$$

Example 14.1 Express the equilibrium constants of the following reactions as a function of temperature:

- (a) $\text{CO (g)} + \text{H}_2 \text{ (g)} \rightleftharpoons \text{C (s)} + \text{H}_2\text{O (g)}.$
- (b) $2\text{CO (g)} \rightleftharpoons \text{C (s)} + \text{CO}_2 \text{ (g)}.$
- (c) $\text{C (s)} + 2\text{H}_2\text{O (g)} \rightleftharpoons \text{CO}_2 \text{ (g)} + 2\text{H}_2 \text{ (g)}.$
- (d) $\text{C (s)} + 2\text{H}_2\text{O (g)} \rightleftharpoons \text{CH}_4 \text{ (g)} + \text{O}_2 \text{ (g)}.$

Solution

- (a) This reaction is known as a “heterogeneous water-gas reaction.” Using the values given in Table 14.1, Eqn (14.1-14) yields

$$\begin{aligned} \Delta G_{\text{rxn}}^o &= -239,690 + 8.15T \ln T - 9.25T - (-111,786 - 87.71T) \\ &= -127,904 + 8.15T \ln T + 78.46T. \end{aligned}$$

The equilibrium constant is determined from Eqn (14.1-15) as

$$\ln K_a = \frac{15,384}{T} - 0.98 \ln T - 9.44.$$

- (b) This reaction is known as the “Boudouard reaction.” Using the values given in Table 14.1, Eqn (14.1-14) yields

$$\begin{aligned} \Delta G_{\text{rxn}}^o &= -394,390 - 0.84T - (2)(-111,786 - 87.71T) \\ &= -170,818 + 174.58T. \end{aligned}$$

The equilibrium constant is determined from Eqn (14.1-15) as

$$\ln K_a = \frac{20,546}{T} - 21.$$

(c) Using the values given in Table 14.1, Eqn (14.1-14) yields

$$\begin{aligned}\Delta G_{\text{rxn}}^o &= -394,390 - 0.84T - (2)(-239,690 + 8.15T \ln T - 9.25T) \\ &= 84,990 - 16.3T \ln T + 17.66T.\end{aligned}$$

The equilibrium constant is determined from Eqn (14.1-15) as

$$\ln K_a = -\frac{10,223}{T} + 1.96 \ln T - 2.12.$$

(d) Using the values given in Table 14.1, Eqn (14.1-14) yields

$$\begin{aligned}\Delta G_{\text{rxn}}^o &= -69,165 + 22.27T \ln T - 65.40T \\ &\quad - (2)(-239,690 + 8.15T \ln T - 9.25T) \\ &= 410,215 + 5.97T \ln T - 46.9T.\end{aligned}$$

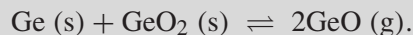
The equilibrium constant is determined from Eqn (14.1-15) as

$$\ln K_a = -\frac{49,340}{T} - 0.72 \ln T + 5.64.$$

Table 14.1: Standard Gibbs energy change of formation for selected compounds (Kubaschewski and Alcock, 1979).

Substance	$\Delta \tilde{G}_f^o$ (J/mol)	Temp. Range (K)
CH ₄ (g)	$-69,165 + 22.27T \ln T - 65.40T$	298–1200
CO ₂ (g)	$-394,390 - 0.84T$	298–2000
CO (g)	$-111,786 - 87.71T$	298–2500
H ₂ O (g)	$-239,690 + 8.15T \ln T - 9.25T$	298–2500
Cr ₂ O ₃ (s)	$-1,160,519 + 222.9T$	298–2500
NiO (s)	$-234,499 + 85.28T$	298–1725
FeO (s)	$-265,062 + 1.88T$	298–1642
SO ₂ (g)	$-362,654 + 72.47T$	298–2000
SO ₃ (g)	$-457,274 + 161.9T$	318–1800

Example 14.2 Sublimation of GeO from a mixture of germanium and germanium dioxide is represented by the following reaction:



The equilibrium pressures of GeO are recorded as a function of temperature as follows:

T (K)	650	700	750	800	850
P_{GeO} (bar)	5.867×10^{-6}	1.188×10^{-4}	1.610×10^{-3}	1.575×10^{-2}	1.177×10^{-1}

Determine the heat of reaction.

Solution

The standard states are chosen as pure components (either solid or gas) at the temperature of reaction and 1 bar. If Ge and GeO₂ exist as two separate solid phases and if GeO behaves as an ideal gas, then

$$\hat{a}_{\text{Ge}} = a_{\text{Ge}} = 1 \quad \hat{a}_{\text{GeO}_2} = a_{\text{GeO}_2} = 1 \quad a_{\text{GeO}} = \frac{P_{\text{GeO}}}{1 \text{ bar}}.$$

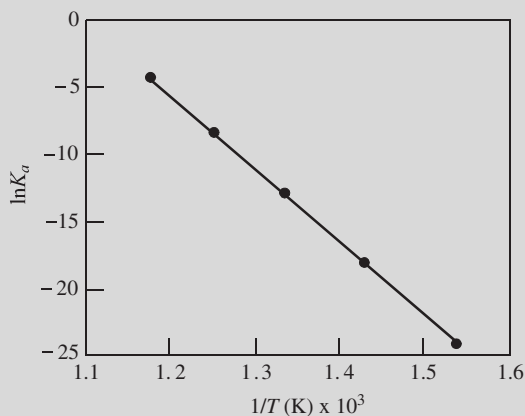
Hence, the equilibrium constant expression simplifies to

$$K_a = P_{\text{GeO}}^2. \quad (1)$$

If $\Delta H_{\text{rxn}}^\circ$ remains essentially constant over the temperature range, the van't Hoff equation, Eqn (13.4-9), reduces to

$$d \ln K_a = -\frac{\Delta H_{\text{rxn}}^\circ}{R} d\left(\frac{1}{T}\right) \quad (2)$$

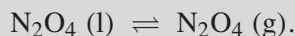
indicating that $\ln K_a$ versus $1/T$ yields a straight line with a slope $-\Delta H_{\text{rxn}}^\circ/R$. Using the given data, K_a values are calculated by using Eqn (1) and the plot of $\ln K_a$ versus $1/T$ is shown below.



The slope of the straight line is -5.473×10^4 . Therefore, ΔH_{rxn}^o is

$$\Delta H_{\text{rxn}}^o = (5.473 \times 10^4)(8.314) = 455,025 \text{ J} = 455 \text{ kJ}.$$

Example 14.3 Estimate the vapor pressure of nitrogen tetroxide (N_2O_4) at 298 K using the following reaction:



Solution

The standard states are chosen as pure components (either liquid or gas) at 298 K and 1 bar. Therefore,

$$a_{\text{N}_2\text{O}_4 (\text{l})} = 1 \quad \text{and} \quad a_{\text{N}_2\text{O}_4 (\text{g})} = \frac{P_{\text{N}_2\text{O}_4}^{\text{vap}} (\text{bar})}{1 \text{ bar}}$$

and the equilibrium constant expression becomes

$$K_a = P_{\text{N}_2\text{O}_4}^{\text{vap}}. \quad (1)$$

Using the $\Delta \tilde{G}_f^o$ values given in Appendix E, ΔG_{rxn}^o is

$$\Delta G_{\text{rxn}}^o = 97.79 - 97.51 = 0.28 \text{ kJ} = 280 \text{ J}.$$

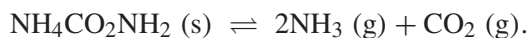
From Eqn (14.1-15)

$$K_a = \exp \left[-\frac{280}{(8.314)(298)} \right] = 0.893 \quad (2)$$

substitution of Eqn (2) into Eqn (1) yields $P_{\text{N}_2\text{O}_4}^{\text{vap}} = 0.893 \text{ bar}$.

14.1.1 Exceptions to Le Chatelier's Principle-Revisited

Consider the decomposition of ammonium carbamate into an evacuated space according to the reaction



Once equilibrium is established at the given temperature and pressure, what happens if more ammonia or carbon dioxide is added to the system? According to Le Chatelier's principle, one can immediately conclude that the reaction shifts to the left so as to relieve the disturbance. This conclusion, however, is not generally true as will be shown in the following analysis.

Assuming negligible solubility of gases in the solid and ideal gas mixture, the equilibrium constant takes the form

$$K_a = p_{\text{NH}_3}^2 p_{\text{CO}_2} = y_{\text{NH}_3}^2 y_{\text{CO}_2} P^3. \quad (14.1-16)$$

Since temperature and pressure are kept constant, rearrangement of Eqn (14.1-16) gives

$$J = K_a P^{-3} = \text{constant} = y_{\text{NH}_3}^2 y_{\text{CO}_2}. \quad (14.1-17)$$

The summation of the mole fractions in the gas phase is equal to unity, i.e.

$$y_{\text{NH}_3} + y_{\text{CO}_2} = 1. \quad (14.1-18)$$

Hence, Eqn (14.1-17) can be expressed as either

$$J = \text{constant} = y_{\text{NH}_3}^2 (1 - y_{\text{NH}_3}). \quad (14.1-19)$$

or

$$J = \text{constant} = (1 - y_{\text{CO}_2})^2 y_{\text{CO}_2}. \quad (14.1-20)$$

The variation of J with respect to the mole fraction of ammonia is

$$\frac{dJ}{dy_{\text{NH}_3}} = y_{\text{NH}_3} (2 - 3y_{\text{NH}_3}). \quad (14.1-21)$$

When

$$2 - 3y_{\text{NH}_3} < 0 \quad \Rightarrow \quad y_{\text{NH}_3} > \frac{2}{3} \quad (14.1-22)$$

addition of ammonia leads to a decrease in J . To re-establish the constancy of J , the reaction must shift to the right, forming more NH_3 . This is in contradiction with Le Chatelier's principle.

The variation of J with respect to the mole fraction of carbon dioxide is

$$\frac{dJ}{dy_{\text{CO}_2}} = \underbrace{(1 - y_{\text{CO}_2})}_{\text{Always positive}} (1 - 3y_{\text{CO}_2}). \quad (14.1-23)$$

When

$$1 - 3y_{\text{CO}_2} < 0 \quad \Rightarrow \quad y_{\text{CO}_2} > \frac{1}{3} \quad (14.1-24)$$

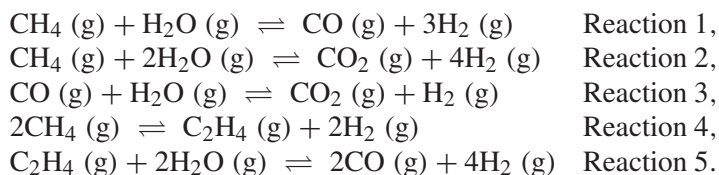
addition of carbon dioxide decreases J . To re-establish the constancy of J , the reaction should proceed to the right, and more CO_2 would be formed. This is in contradiction with Le Chatelier's principle.

For a more thorough discussion on the subject see [Katz \(1961\)](#).

14.2 Multiple Chemical Reactions

The first step in determining the composition of an equilibrium system involving several reactions is to decide on the number of independent reactions to be considered. The term *independent chemical reactions* gives the smallest collection of reactions that includes all possible chemical reactions among the species present. As a result, no reaction in the set is a linear combination of others.

For example, consider the following five reactions:



To determine the independent reactions, it is first necessary to form the *stoichiometric matrix*, $[\alpha]$, in which rows and columns indicate reactions and species, respectively:

$$[\alpha] = \begin{array}{c} \text{Species} \rightarrow \\ \begin{array}{c} \text{Rxn1} \\ \text{Rxn2} \\ \text{Rxn3} \\ \text{Rxn4} \\ \text{Rxn5} \end{array} \end{array} \begin{bmatrix} \text{CH}_4 & \text{H}_2\text{O} & \text{CO} & \text{H}_2 & \text{CO}_2 & \text{C}_2\text{H}_4 \\ -1 & -1 & 1 & 3 & 0 & 0 \\ -1 & -2 & 0 & 4 & 1 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \\ -2 & 0 & 0 & 2 & 0 & 1 \\ 0 & -2 & 2 & 4 & 0 & -1 \end{bmatrix} \quad (14.2-1)$$

The order of chemical species in forming the stoichiometric matrix is immaterial. Note that whether or not the i th reaction is a linear combination of the other reactions is the same as whether or not the i th row of the stoichiometric matrix is a linear combination of the other rows. In other words, the linear independence of the reactions in a given set of reactions is equivalent to the linear independence of the rows in the corresponding stoichiometric matrix.

As shown in Appendix F, the rank of a matrix shows the maximum number of linearly independent rows. Thus, the rank of $[\alpha]$ gives the number of independent reactions. The rank of the stoichiometric matrix $[\alpha]$ defined by Eqn (14.2-1) is 3. Therefore, three of the given five reactions constitute an independent set.

In Appendix F, it is also shown that the number of nonzero rows in the reduced row echelon form of a matrix is equivalent to its rank. The reduced row echelon

form of the stoichiometric matrix is

$$\text{rref}[\alpha] = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -0.5 \\ 0 & 1 & 0 & -1.5 & -0.5 & 0.25 \\ 0 & 0 & 1 & 0.5 & -0.5 & -0.25 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (14.2-2)$$

indicating that the number of independent reactions is 3.

The next question to answer is which three reactions are linearly independent. The linearly independent reactions must be chosen such that the rank of the resulting stoichiometric matrix is also equal to 3.

If reactions 1, 2, and 3 are chosen as an independent set, the stoichiometric matrix is

$$\begin{bmatrix} -1 & -1 & 1 & 3 & 0 & 0 \\ -1 & -2 & 0 & 4 & 1 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{bmatrix}$$

and its rank is 2. Since

$$\text{Rxn 3} = \text{Rxn 2} - \text{Rxn 1}$$

reactions 1, 2, and 3 cannot be considered as an independent set.

If one chooses reactions 1, 2, and 4 as an independent set, the stoichiometric matrix is

$$\begin{bmatrix} -1 & -1 & 1 & 3 & 0 & 0 \\ -1 & -2 & 0 & 4 & 1 & 0 \\ -2 & 0 & 0 & 2 & 0 & 1 \end{bmatrix}$$

and its rank is 3. These three independent reactions, however, are not unique. Other sets of three independent reactions can also be formed. For example, it is also possible to consider reactions 2, 4, and 5 as an independent set. In this case, the stoichiometric matrix is

$$\begin{bmatrix} -1 & -2 & 0 & 4 & 1 & 0 \\ -2 & 0 & 0 & 2 & 0 & 1 \\ 0 & -2 & 2 & 4 & 0 & -1 \end{bmatrix}$$

and its rank is 3. The equilibrium composition, however, does not depend on the choice of independent set of reactions.

14.3 Determination of Independent Reactions From The Given Species

If the possible equations taking place within the reactor are known, independent reactions can be determined by calculating the rank (or reduced row echelon form)

of the stoichiometric matrix as shown in Section 14.2. It is often necessary, however, to obtain a proper set of independent reactions representing the system from a given list of chemical species. For example, what are the representative reactions for the system consisting of the following species: CO, H₂, H₂O, CH₄, CH₃OH? To answer such a question, one has to consider the conservation of mass together with Eqn (13.1-11) representing the conservation of elements.

With the help of Eqn (13.1-5), the conservation of mass is expressed as

$$\begin{bmatrix} \text{CO} & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 & \text{CH}_3\text{OH} \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = [0], \quad (14.3-1)$$

where α_1 , α_2 , α_3 , α_4 , and α_5 are the stoichiometric coefficients of the species CO, H₂, H₂O, CH₄, and CH₃OH, respectively.

The element-by-species matrix, $[\beta]$, is written as

$$\begin{array}{lcl} \text{Species} \rightarrow & \text{H}_2 & \text{CO} & \text{CH}_4 & \text{H}_2\text{O} & \text{CH}_3\text{OH} \\ [\beta] = \begin{array}{l} \text{C} \\ \text{H} \\ \text{O} \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 2 & 0 & 4 & 2 & 4 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} & \end{array} \quad (14.3-2)$$

The reduced row echelon form of $[\beta]$ is

$$\text{rref}[\beta] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix}, \quad (14.3-3)$$

so that Eqn (13.1-11) takes the form

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad (14.3-4)$$

which is equivalent to the set of following equations:

$$\begin{aligned} \alpha_1 + \alpha_4 + \alpha_5 &= 0, \\ \alpha_2 + 3\alpha_4 + 2\alpha_5 &= 0, \\ \alpha_3 - \alpha_4 &= 0. \end{aligned} \quad (14.3-5)$$

Since there are three variables and five unknowns, the above equations can be solved for any three variables in terms of the remaining two variables, which are named the *free variables*. Choosing α_4 and α_5 as the free variables, α_1 , α_2 , and α_3 are expressed as

$$\begin{aligned}\alpha_1 &= -\alpha_4 - \alpha_5, \\ \alpha_2 &= -3\alpha_4 - 2\alpha_5, \\ \alpha_3 &= \alpha_4.\end{aligned}\tag{14.3-6}$$

The column vector representing stoichiometric coefficients can then be expressed as

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ 1 \\ 1 \\ 0 \end{bmatrix} \alpha_4 + \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \alpha_5.\tag{14.3-7}$$

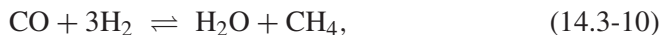
Substituting Eqn (14.3-7) into Eqn (14.3-1), noting that α_4 and α_5 are independent of each other, gives

$$\begin{bmatrix} \text{CO} & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 & \text{CH}_3\text{OH} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -3 \\ 1 \\ 1 \\ 0 \end{bmatrix} = [0]\tag{14.3-8}$$

and

$$\begin{bmatrix} \text{CO} & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 & \text{CH}_3\text{OH} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} = [0].\tag{14.3-9}$$

Equations (14.3-8) and (14.3-9) result in the following independent reactions, respectively:



In this specific example, stoichiometric coefficients of CH_4 and CH_3OH , i.e. α_4 and α_5 , are chosen as free variables. A question that could be raised after this example is, does one have the freedom of choosing any two of the five species as free variables? The choice of free variables is *not* completely arbitrary. Free variables

should be chosen from among the products of the reactions and should contain all the elements. For a more through discussion on the subject see [Whitwell and Dartt \(1973\)](#).

Example 14.4 One of the ways of reforming methane to synthesis gas (syngas), a mixture of CO and H₂, is CO₂ reforming, known as “dry reforming.” This primary product is accompanied by a side reaction producing water vapor. Determine the independent set of reactions for this system.

Solution

Conservation of mass is expressed as

$$\begin{bmatrix} \text{CH}_4 & \text{CO}_2 & \text{H}_2 & \text{CO} & \text{H}_2\text{O} \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = [0]. \quad (1)$$

The element-by-species matrix, $[\beta]$, is written as

$$\begin{array}{rcl} \text{Species} \rightarrow & \text{CH}_4 & \text{CO}_2 & \text{H}_2 & \text{CO} & \text{H}_2\text{O} \\ [\beta] = & \begin{array}{c} \text{C} \\ \text{H} \\ \text{O} \end{array} & \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 4 & 0 & 2 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{bmatrix} & \end{array} \quad (2)$$

The reduced row echelon form of $[\beta]$ is

$$\text{ref}[\beta] = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -1 & 2 \end{bmatrix}. \quad (3)$$

Thus, Eqn (13.1-11) takes the form

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (4)$$

Multiplication of the matrices results in the following set of equations:

$$\begin{aligned}\alpha_1 + \frac{1}{2}\alpha_4 - \frac{1}{2}\alpha_5 &= 0, \\ \alpha_2 + \frac{1}{2}\alpha_4 + \frac{1}{2}\alpha_5 &= 0, \\ \alpha_3 - \alpha_4 + 2\alpha_5 &= 0.\end{aligned}\tag{5}$$

There are five species and three equations. Considering α_4 and α_5 as free variables, α_1 , α_2 , and α_3 are expressed as

$$\begin{aligned}\alpha_1 &= -\frac{1}{2}\alpha_4 + \frac{1}{2}\alpha_5, \\ \alpha_2 &= -\frac{1}{2}\alpha_4 - \frac{1}{2}\alpha_5, \\ \alpha_3 &= \alpha_4 - 2\alpha_5.\end{aligned}\tag{6}$$

The column vector representing stoichiometric coefficients can then be expressed as

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \\ 1 \\ 0 \end{bmatrix} \alpha_4 + \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -2 \\ 0 \\ 1 \end{bmatrix} \alpha_5.\tag{7}$$

Substituting Eqn (7) into Eqn (1), noting that α_4 and α_5 are independent of each other, gives

$$\begin{bmatrix} \text{CH}_4 & \text{CO}_2 & \text{H}_2 & \text{CO} & \text{H}_2\text{O} \end{bmatrix} \cdot \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \\ 1 \\ 0 \end{bmatrix} = [0]\tag{8}$$

and

$$\begin{bmatrix} \text{CH}_4 & \text{CO}_2 & \text{H}_2 & \text{CO} & \text{H}_2\text{O} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -2 \\ 0 \\ 1 \end{bmatrix} = [0].\tag{9}$$

Equations (8) and (9) result in the following independent reactions, respectively:



Comment: Addition of Eqns (10) and (11) results in



The reactions given by Eqns (10) and (12), or Eqns (11) and (12) also constitute an independent set.

Example 14.5 Coal gasification takes place at 1 bar and 1000 K. The species expected at equilibrium are CO_2 , CO , H_2O , H_2 , CH_4 , and O_2 . If 3 mol of steam is used per mole of O_2 , determine the gas composition under equilibrium conditions.

Solution

Conservation of mass is expressed as

$$\begin{bmatrix} \text{CO}_2 & \text{CO} & \text{H}_2\text{O} & \text{H}_2 & \text{O}_2 & \text{C} & \text{CH}_4 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \\ \alpha_7 \end{bmatrix} = [0]. \quad (1)$$

The element-by-species matrix, $[\beta]$, is written as

$$[\beta] = \begin{array}{c} \text{Species} \rightarrow \\ \begin{array}{c} \text{C} \\ \text{H} \\ \text{O} \end{array} \end{array} \begin{array}{c} \text{CO}_2 \quad \text{CO} \quad \text{H}_2\text{O} \quad \text{H}_2 \quad \text{O}_2 \quad \text{C} \quad \text{CH}_4 \\ \left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 & 0 & 0 & 4 \\ 2 & 1 & 1 & 0 & 2 & 0 & 0 \end{array} \right] \end{array} \quad (2)$$

The reduced row echelon form of $[\beta]$ is

$$\text{rref}[\beta] = \begin{bmatrix} 1 & 0 & 0 & -1 & 2 & -1 & -3 \\ 0 & 1 & 0 & 1 & -2 & 2 & 4 \\ 0 & 0 & 1 & 1 & 0 & 0 & 2 \end{bmatrix}, \quad (3)$$

so that Eqn (13.1-11) takes the form

$$\begin{bmatrix} 1 & 0 & 0 & -1 & 2 & -1 & -3 \\ 0 & 1 & 0 & 1 & -2 & 2 & 4 \\ 0 & 0 & 1 & 1 & 0 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \\ \alpha_7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (4)$$

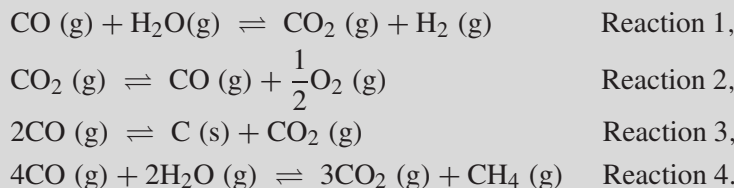
Choosing α_4 , α_5 , α_6 , and α_7 as free variables, α_1 , α_2 , and α_3 are expressed as

$$\begin{aligned} \alpha_1 &= \alpha_4 - 2\alpha_5 + \alpha_6 + 3\alpha_7, \\ \alpha_2 &= -\alpha_4 + 2\alpha_5 - 2\alpha_6 - 4\alpha_7, \\ \alpha_3 &= -\alpha_4 - 2\alpha_7. \end{aligned} \quad (5)$$

The column vector representing stoichiometric coefficients can then be expressed as

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \\ \alpha_7 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \alpha_4 + \begin{bmatrix} -2 \\ 2 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \alpha_5 + \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \alpha_6 + \begin{bmatrix} 3 \\ -4 \\ -2 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \alpha_7. \quad (6)$$

Substituting Eqn (6) into Eqn (1), noting that α_4 , α_5 , α_6 , and α_7 are independent of each other, leads to the following set of independent reactions:



The equilibrium constants of Reactions 1, 2, and 4 can be determined from $\Delta \tilde{G}_f^o$ values given in Table 14.1. The results are

$$K_{a1} = \exp\left(\frac{5161.7}{1000} + 0.98 \ln 1000 - 11.56\right) = 1.45,$$

$$K_{a2} = \exp\left(-\frac{33,991}{1000} + 10.45\right) = 5.97 \times 10^{-11},$$

$$K_{a4} = \exp\left(\frac{39,188}{1000} - 0.72 \ln 1000 - 36.25\right) = 0.13.$$

Reaction 3 is the Boudouard reaction. From Example 14.1

$$K_{a3} = \exp\left(\frac{20,546}{1000} - 21\right) = 0.64.$$

Since K_{a2} is very small, oxygen concentration under equilibrium conditions is practically zero.² As a result, the second reaction will be ignored. The equilibrium constant expressions for the remaining reactions are expressed as

$$K_{a1} = 1.45 = \frac{y_{\text{CO}_2} y_{\text{H}_2}}{y_{\text{CO}} y_{\text{H}_2\text{O}}}, \quad (7)$$

$$K_{a3} = 0.64 = \frac{y_{\text{CO}_2}}{y_{\text{CO}}^2}, \quad (8)$$

$$K_{a4} = 0.13 = \frac{y_{\text{CO}_2}^3 y_{\text{CH}_4}}{y_{\text{CO}}^4 y_{\text{H}_2\text{O}}^2}, \quad (9)$$

The summation of the mole fractions is equal to unity, i.e.

$$y_{\text{CO}_2} + y_{\text{CO}} + y_{\text{H}_2} + y_{\text{H}_2\text{O}} + y_{\text{CH}_4} = 1. \quad (10)$$

Choosing 1 mol of O_2 as a basis, 5 atoms of oxygen and 6 atoms of hydrogen enter the reactor. Oxygen balance gives

$$2y_{\text{CO}_2} + y_{\text{CO}} + y_{\text{H}_2\text{O}} = \frac{5}{n_T}. \quad (11)$$

On the other hand, hydrogen balance gives

$$2y_{\text{H}_2} + 2y_{\text{H}_2\text{O}} + 4y_{\text{CH}_4} = \frac{6}{n_T}. \quad (12)$$

Simultaneous solution of Eqns (7)-(12) by MATHCAD[®] leads to

$$\begin{aligned} y_{\text{CO}_2} &= 0.123 & y_{\text{CO}} &= 0.438 & y_{\text{H}_2} &= 0.358 \\ y_{\text{H}_2\text{O}} &= 0.069 & y_{\text{CH}_4} &= 0.012 & n_T &= 6.643 \text{ mol.} \end{aligned}$$

²This is in accordance with the known kinetic fact that oxygen virtually disappears in the fuel bed within a few solid particle diameters.

14.3.1 The Gibbs Phase Rule for Reactive Components

As given by Eqn (1.1-4), the Gibbs phase rule specifies the number of degrees of freedom, $\mathcal{F}$, as

$$\mathcal{F} = \mathcal{C} + 2 - \mathcal{P}, \quad (14.3-12)$$

where $\mathcal{P}$ is the number of phases³ and $\mathcal{C}$ is the number of components. For reacting systems, the term $\mathcal{C}$ is expressed as

$$\mathcal{C} = N - \mathcal{R} - \mathcal{S} \quad (14.3-13)$$

in which N is the total number of species, $\mathcal{R}$ is the number of independent reactions, and $\mathcal{S}$ is the number of stoichiometric constraints on the concentrations of species.⁴ If some species are introduced into the system only by a chemical reaction that produces them, and if two or more of these exist in the same phase, then their concentrations will be related by stoichiometric ratios and $\mathcal{S} \neq 0$.

Example 14.6 Determine the number of degrees of freedom of a gaseous system consisting of ethylene, methane, ethane, and propane.

Solution

To determine $\mathcal{F}$, it is first necessary to find out the number of independent reactions. Conservation of mass is expressed as

$$\begin{bmatrix} \text{C}_2\text{H}_4 & \text{CH}_4 & \text{C}_2\text{H}_6 & \text{C}_3\text{H}_8 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = [0]. \quad (1)$$

The element-by-species matrix, $[\beta]$, is written as

$$\begin{array}{c} \text{Species} \rightarrow \\ [\beta] = \begin{array}{c} \text{C} \\ \text{H} \end{array} \begin{array}{cccc} \text{C}_2\text{H}_4 & \text{CH}_4 & \text{C}_2\text{H}_6 & \text{C}_3\text{H}_8 \end{array} \\ \left[\begin{array}{cccc} 2 & 1 & 2 & 3 \\ 4 & 4 & 6 & 8 \end{array} \right] \end{array} \quad (2)$$

The reduced row echelon form of $[\beta]$ is

$$\text{rref}[\beta] = \begin{bmatrix} 1 & 0 & \frac{1}{2} & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}. \quad (3)$$

³While there can be at most one gas phase, there can be more than one solid or liquid phase.

⁴In a given phase, there is *no* constraint that summation of the mole fractions of components must add up to unity.

Thus, Eqn (13.1-11) becomes

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (4)$$

Multiplication of the matrices results in the following set of equations:

$$\begin{aligned} \alpha_1 + \frac{1}{2}\alpha_3 + \alpha_4 &= 0, \\ \alpha_2 + \alpha_3 + \alpha_4 &= 0. \end{aligned} \quad (5)$$

There are four species and two equations. Considering α_3 and α_4 as free variables, α_1 and α_2 are expressed as

$$\begin{aligned} \alpha_1 &= -\frac{1}{2}\alpha_3 - \alpha_4, \\ \alpha_2 &= -\alpha_3 - \alpha_4. \end{aligned} \quad (6)$$

The column vector representing stoichiometric coefficients can then be expressed as

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -1 \\ 1 \\ 0 \end{bmatrix} \alpha_3 + \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \alpha_4. \quad (7)$$

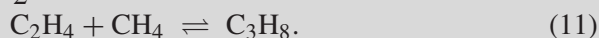
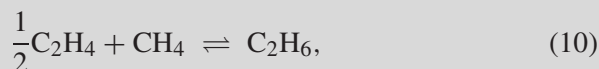
Substituting Eqn (7) into Eqn (1), noting that α_4 and α_5 are independent of each other, gives

$$\begin{bmatrix} \text{C}_2\text{H}_4 & \text{CH}_4 & \text{C}_2\text{H}_6 & \text{C}_3\text{H}_8 \end{bmatrix} \cdot \begin{bmatrix} -\frac{1}{2} \\ -1 \\ 1 \\ 0 \end{bmatrix} = [0] \quad (8)$$

and

$$\begin{bmatrix} \text{C}_2\text{H}_4 & \text{CH}_4 & \text{C}_2\text{H}_6 & \text{C}_3\text{H}_8 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = [0]. \quad (9)$$

Equations (8) and (9) result in the following independent reactions, respectively:



Noting that $N = 4$, $\mathcal{R} = 2$, and $\mathcal{S} = 0$, Eqn (14.3-13) gives $\mathcal{C} = 2$. Since $\mathcal{P} = 1$, Eqn (14.3-12) indicates that $\mathcal{F} = 3$. Therefore, the independent intensive variables are temperature, pressure, and $(n_{\text{C}_2\text{H}_4})_o/(n_{\text{CH}_4})_o$, where subscripts o indicate initial values.

Example 14.7 Ammonium hydrogen sulfide is put into an evacuated reactor and the reactor is sealed. Upon heating, it decomposes according to the reaction



Determine the number of degrees of freedom for this system.

Solution

The number of chemical species, N , is 3 and $\mathcal{R} = 1$. NH_3 and H_2S form as a result of the dissociation reaction and they exist in the gas phase in equimolar proportions, i.e. $y_{\text{NH}_3} = y_{\text{H}_2\text{S}} = 0.5$. Since this is a stoichiometric constraint, $\mathcal{S} = 1$. From Eqn (14.3-13)

$$\mathcal{C} = N - \mathcal{R} - \mathcal{S} = 3 - 1 - 1 = 1.$$

The system consists of two phases (solid and gas). Thus, application of Eqn (14.3-12) gives the number of degrees of freedom as

$$\mathcal{F} = \mathcal{C} + 2 - \mathcal{P} = 1 + 2 - 2 = 1.$$

It is necessary to fix either temperature or pressure.

Comment: Prior to the reaction if certain amounts of NH_3 and/or H_2S are added besides solid NH_4HS to the reactor and then the system is heated, there is no special constraint and $\mathcal{F} = 2$.

In the literature, determination of the term $\mathcal{C}$ by Eqn (14.3-13) is known as *Jouguet's method*. In an equilibrium mixture, estimation of N and $\mathcal{R}$ may not be as straightforward as in Examples 14.6 and 14.7. For example, presence of various species in the equilibrium mixture as a result of chemical reactions, ionization, and association makes the determination of N very difficult. On the other hand, an increase in N causes a tremendous increase in the number of chemical reactions. As a result, it becomes extremely difficult to take all the possible reactions into consideration and to determine the independent ones. For a more thorough discussion on the subject see Björnbohm (1975, 1981), Zhao *et al.* (1992), Alberty (1993), Weltin (1994), and Alper (1999).

14.4 The Enthalpy and Gibbs Energy Change of Reaction

14.4.1 The Enthalpy Change for a Reaction

The term ΔH_{rxn}^o represents the standard-state enthalpy change of reaction, i.e. enthalpy change between products and reactants in their respective standard states. It is *not* the enthalpy change when the reactants and products are present in an equilibrium mixture. For an equilibrium mixture, the enthalpy change for the reaction, ΔH_{rxn} , is given by

$$\Delta H_{\text{rxn}} = \sum_{i=1}^k \alpha_i \bar{H}_i, \quad (14.4-1)$$

where $\bar{H}_i$ is the partial molar enthalpy of component i at the temperature and pressure of the system and in the equilibrium mixture.

Note that $\Delta H_{\text{rxn}} = \Delta H_{\text{rxn}}^o$ only if enthalpy is independent of pressure, i.e. $(\partial H / \partial P)_T = 0$, and enthalpy change on mixing, ΔH_{mix} , is equal to zero. For ideal gas mixtures, there are no heat effects of mixing or pressure effects on enthalpy so that in this special case

$$\Delta H_{\text{rxn}} = \Delta H_{\text{rxn}}^o \quad \text{Ideal gas mixture.} \quad (14.4-2)$$

14.4.2 The Feasibility of a Chemical Reaction

In a closed system at constant temperature and pressure, an infinitesimal change can only take place spontaneously if the Gibbs energy of the system decreases. If a single chemical reaction is taking place in a closed system at constant temperature and pressure, the following criteria for the feasibility of the reaction hold:

$$(\Delta G)_{T,P} \begin{cases} < 0 & \text{The reaction proceeds as written,} \\ = 0 & \text{The reaction is in equilibrium,} \\ > 0 & \text{The reaction proceeds in the reverse direction.} \end{cases} \quad (14.4-3)$$

It should be kept in mind, however, that the value of ΔG_{rxn} may be changed by changing the pressure of the system. Thus, the use of ΔG_{rxn}^o may be a better measurement since it eliminates the effect of pressure. In this case, however, the difficulty arises when the system reaches equilibrium. At equilibrium

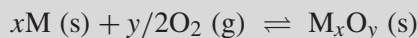
$$\Delta G_{\text{rxn}}^o = -RT \ln K_a \quad (14.4-4)$$

and ΔG_{rxn}^o is not equal to zero (unless $K_a = 1$). [Dodge \(1944\)](#) proposed the following values for the rough estimate of the feasibility of a given reaction:

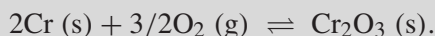
- When $\Delta G_{\text{rxn}}^o < 0$, the reaction is proceeding as written.

- When $0 < \Delta G_{\text{rxn}}^o < 42 \text{ kJ}$, the reaction is doubtful to proceed as written; however, it needs further study.
- When $\Delta G_{\text{rxn}}^o > 42 \text{ kJ}$, the reaction will not proceed as written.⁵

Example 14.8 In general, oxidation of a metal M, given by the reaction,



is accompanied by large negative values of ΔG_{rxn}^o . As a result, the above reaction will proceed as written. This is the reason why oxides are much more stable than the elemental metals. For example, consider oxidation of chromium according to the reaction



From Table 14.1

$$\Delta G_{\text{rxn}}^o = -1,160,519 + 222.9T,$$

where ΔG_{rxn}^o is in J and T is in K. Noting that

$$\Delta G_{\text{rxn}}^o = -1094 \text{ kJ at } T = 298 \text{ K},$$

$$\Delta G_{\text{rxn}}^o = -937.6 \text{ kJ at } T = 1000 \text{ K},$$

one can easily conclude that the reaction shifts to the right, leading to oxidation of chromium.

Standard Gibbs energy change of reaction, ΔG_{rxn}^o , is expressed as

$$\Delta G_{\text{rxn}}^o = \Delta H_{\text{rxn}}^o - T\Delta S_{\text{rxn}}^o. \quad (14.4-5)$$

While ΔG_{rxn}^o is dependent on the sign of ΔH_{rxn}^o at low temperatures, the sign of ΔS_{rxn}^o becomes dominant at high temperatures. For endothermic reactions, i.e. $\Delta H_{\text{rxn}}^o > 0$, a reaction is possible only if $T\Delta S_{\text{rxn}}^o$ is positive⁶ and larger than ΔH_{rxn}^o .

The partial molar Gibbs energy of species i in a mixture is given by Eqn (13.3-8), i.e.

$$\bar{G}_i = \tilde{G}_i^o + RT \ln \hat{a}_i. \quad (14.4-6)$$

Multiplication of Eqn (14.4-6) by α_i and summation over all species present in the reacting mixture give

$$\sum_{i=1}^k \alpha_i \bar{G}_i = \sum_{i=1}^k \alpha_i \tilde{G}_i^o + RT \sum_{i=1}^k \alpha_i \ln \hat{a}_i = \sum_{i=1}^k \alpha_i \tilde{G}_i^o + RT \ln \prod_{i=1}^k \hat{a}_i^{\alpha_i}. \quad (14.4-7)$$

⁵Mathematically speaking, no reaction is impossible. For an impossible reaction, either the value of K_a should be equal to zero or ΔG_{rxn}^o should be infinite.

⁶For example, the change from an ordered solid phase to the random gaseous phase is always accompanied by an increase in entropy.

Equation (14.4-7) can be expressed in the form

$$\Delta G_{\text{rxn}} = \Delta G_{\text{rxn}}^o + RT \ln Q_a, \quad (14.4-8)$$

where Q_a is defined by

$$Q_a = \prod_{i=1}^k \hat{a}_i^{\alpha_i}. \quad (14.4-9)$$

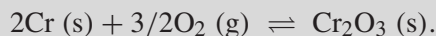
The term Q_a is called the *apparent equilibrium constant* or *reaction quotient*. It is the value of the equilibrium expression using *nonequilibrium* concentrations. The use of Eqn (14.4-4) in Eqn (14.4-8) leads to

$$\Delta G_{\text{rxn}} = RT \ln \left(\frac{Q_a}{K_a} \right). \quad (14.4-10)$$

Therefore, one can conclude that

$$\begin{aligned} Q_a > K_a &\Rightarrow \Delta G_{\text{rxn}} > 0 \Rightarrow \text{Reaction proceeds in the reverse direction} \\ &\quad \text{until equilibrium is established,} \\ Q_a < K_a &\Rightarrow \Delta G_{\text{rxn}} < 0 \Rightarrow \text{Reaction proceeds as written until} \\ &\quad \text{equilibrium is established.} \end{aligned}$$

Example 14.9 Chromium metal is placed in a chamber kept at 1500 K. Estimate the maximum permissible air pressure to prevent oxidation of chromium according to the reaction



Solution

Oxidation can be prevented or, in other words, the reaction proceeds in the reverse direction if

$$Q_a > K_a. \quad (1)$$

From the data given in Example 14.8

$$K_a = \exp \left[\frac{1,160,519}{(8.314)(1500)} - \frac{222.9}{8.314} \right] = 5.9 \times 10^{28}. \quad (2)$$

The standard states are chosen as pure components (either solid or gas) at 1500 K and 1 bar. If Cr and Cr_2O_3 exist as two separate solid phases and if O_2 behaves as an ideal gas, then

$$\hat{a}_{\text{Cr}} = a_{\text{Cr}} = 1 \quad \hat{a}_{\text{Cr}_2\text{O}_3} = a_{\text{Cr}_2\text{O}_3} = 1 \quad a_{\text{O}_2} = \frac{p_{\text{O}_2}}{1 \text{ bar}}.$$

Hence, Q_a expression simplifies to

$$Q_a = \frac{1}{p_{O_2}^{3/2}}. \quad (3)$$

Substitution of Eqns (2) and (3) into Eqn (1) gives

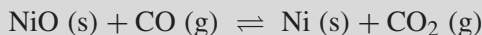
$$\frac{1}{p_{O_2}^{3/2}} > 5.9 \times 10^{28} \Rightarrow p_{O_2} < 6.6 \times 10^{-20} \text{ bar}.$$

Since air contains approximately 21 mol% O_2 , the maximum air pressure is

$$P = \frac{p_{O_2}}{y_{O_2}} = \frac{6.6 \times 10^{-20}}{0.21} = 3.1 \times 10^{-19} \text{ bar}.$$

A pressure higher than this will allow spontaneous oxidation of chromium.

Example 14.10 Heating nickel oxide with carbon monoxide reduces it to metallic nickel according to the reaction



Will a gaseous mixture of 20 mol % CO_2 , 3% CO , and 77% N_2 oxidize nickel at 800 K?

Solution

Using the values given in Table 14.1, Eqn (14.1-14) yields

$$\begin{aligned} \Delta G_{\text{rxn}}^o &= -394,390 - 0.84T + 234,499 - 85.28T + 111,786 + 87.71T \\ &= -48,105 + 1.59T. \end{aligned}$$

The use of Eqn (14.1-15) gives the equilibrium constant as

$$K_a = \exp\left(\frac{5786}{T} - 0.19\right).$$

Thus, the equilibrium constant at 800 K is

$$K_a = 1144.$$

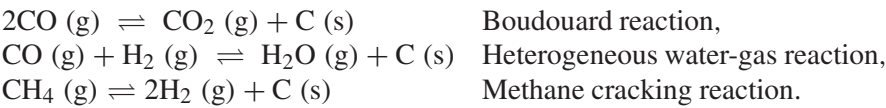
The apparent equilibrium constant is

$$Q_a = \frac{y_{CO_2}}{y_{CO}} = \frac{0.2}{0.03} = 6.67.$$

Since $Q_a < K_a$, the reaction proceeds as written. In other words, nickel cannot be oxidized at the given temperature.

14.5 Carbon Formation

Carbon may form during decomposition of hydrocarbons. Carbon deposition on metallic surfaces, i.e. walls and/or tube surfaces, adversely affects heat transfer efficiency by lowering heat transfer coefficient. Moreover, carbon deposition on the catalyst surface blocks the active sites, leading to catalyst poisoning. In the literature, the following three reactions are usually considered for carbon formation:



The equilibrium constants of these reactions are given in Table 14.2.

Since the Boudouard and heterogeneous water-gas reactions are exothermic, they are the major pathways for carbon deposition when $T < 850$ K. On the other hand, being an endothermic reaction, the methane cracking reaction is solely responsible for carbon deposition when $T > 1000$ K. Equilibrium constants of these reactions at different temperatures are given in Table 14.3 and are also plotted in Fig. 14.1.

Table 14.2: Equilibrium constants of major carbon deposition reactions.

Reaction	K_a
Boudouard	$\exp\left(\frac{20,546}{T} - 21\right)$
Heterogeneous water-gas	$\exp\left(\frac{15,384}{T} - 0.98 \ln T - 9.44\right)$
Methane cracking	$\exp\left(-\frac{8319.1}{T} + 2.68 \ln T - 7.87\right)$

Table 14.3: K_a values of major carbon deposition reactions as a function of temperature.

T (K)	K_a		
	Boudouard rxn	Heterogeneous Water-Gas Rxn	Methane Cracking Rxn
700	4236	453.5	0.111
750	598.6	97.94	0.295
800	108.0	25.51	0.702
850	23.85	7.76	1.52
900	6.23	2.68	3.06
950	1.87	1.04	5.75
1000	0.635	0.438	10.21
1050	0.239	0.201	17.30
1100	0.098	0.099	28.09

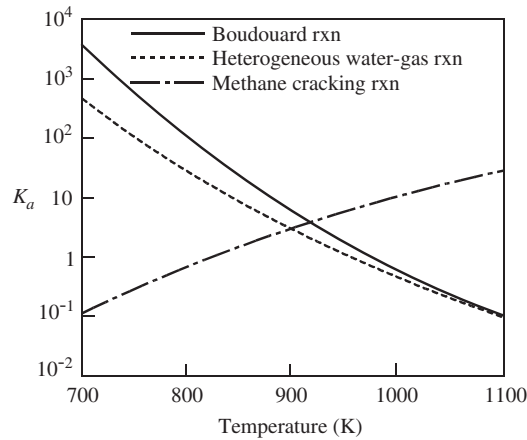


Figure 14.1: Variation of the equilibrium constants of major carbon deposition reactions as a function of temperature.

Assuming negligible solubility of gases in carbon, i.e. C is present as a pure solid, and neglecting the effect of pressure on the fugacity of pure carbon result in $a_C = 1$. Choosing standard state pressure for gases so as to make pure component fugacities equal to unity and assuming an ideal gas mixture result in the following apparent equilibrium constant (or reaction quotient) expressions for the carbon deposition reactions:

- *Boudouard reaction*

$$Q_a = \frac{p_{\text{CO}_2}}{p_{\text{CO}}^2} = \left(\frac{y_{\text{CO}_2}}{y_{\text{CO}}^2} \right) \frac{1}{P}. \quad (14.5-1)$$

- *Heterogeneous water-gas reaction*

$$Q_a = \frac{p_{\text{H}_2\text{O}}}{p_{\text{CO}} p_{\text{H}_2}} = \left(\frac{y_{\text{H}_2\text{O}}}{y_{\text{CO}} y_{\text{H}_2}} \right) \frac{1}{P}. \quad (14.5-2)$$

- *Methane cracking reaction*

$$Q_a = \frac{p_{\text{H}_2}^2}{p_{\text{CH}_4}} = \left(\frac{y_{\text{H}_2}^2}{y_{\text{CH}_4}} \right) P. \quad (14.5-3)$$

Carbon formation takes place when $Q_a < K_a$. Carbon formation can be avoided by adjusting conditions so as to make Q_a greater than K_a .

Example 14.11 You are asked to design a heat exchanger to increase the temperature of a gas mixture at 2 bar with a composition of 75 mol% CH₄, 8% H₂, and 17% N₂. What is the maximum outlet temperature of the gas mixture so that no carbon will deposit on the tube surface?

Solution

Carbon may deposit as a result of the methane cracking reaction, i.e.



The equilibrium constant is given by

$$K_a = \exp \left(-\frac{8319.1}{T} + 2.68 \ln T - 7.87 \right). \quad (1)$$

Carbon deposition can be prohibited if

$$Q_a > K_a, \quad (2)$$

where Q_a is given by Eqn (14.5-3). Thus, Eqn (2) takes the form

$$\left(\frac{y_{\text{H}_2}^2}{y_{\text{CH}_4}} \right) P > \exp \left(-\frac{8319.1}{T} + 2.68 \ln T - 7.87 \right). \quad (3)$$

Substitution of the numerical values into Eqn (3) gives

$$\frac{(0.08)^2(2)}{0.75} > \exp \left(-\frac{8319.1}{T} + 2.68 \ln T - 7.87 \right). \quad (4)$$

Solution of Eqn (4) by MATHCAD[®] yields

$$T < 619 \text{ K}.$$

Example 14.12 Is there any danger of carbon formation if the reactions given in Example 13.10 reach equilibrium?

Solution

In Example 13.10, the equilibrium gas phase composition was calculated as $y_{\text{CH}_4} = 0.065$ $y_{\text{H}_2\text{O}} = 0.564$ $y_{\text{CO}} = 0.011$ $y_{\text{CO}_2} = 0.065$ $y_{\text{H}_2} = 0.295$.

Boudouard reaction

From Table 14.3 $K_a = 108$ at 800 K. The use of Eqn (14.5-1) gives Q_a as

$$Q_a = \left[\frac{0.065}{(0.011)^2} \right] \frac{1}{2} = 268.6.$$

Since $Q_a > K_a$, carbon deposition does not take place.

Heterogeneous water-gas reaction

From Table 14.3 $K_a = 25.51$ at 800 K. The use of Eqn (14.5-2) gives Q_a as

$$Q_a = \left[\frac{0.564}{(0.011)(0.295)} \right] \frac{1}{2} = 86.9.$$

Since $Q_a > K_a$, carbon deposition does not take place.

Methane cracking reaction

From Table 14.3 $K_a = 0.702$ at 800 K. The use of Eqn (14.5-3) gives Q_a as

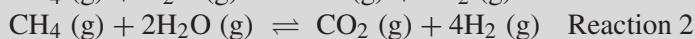
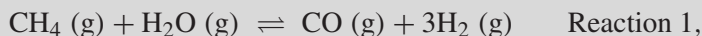
$$Q_a = \left[\frac{(0.295)^2}{0.065} \right] (2) = 2.68.$$

Since $Q_a > K_a$, carbon deposition does not take place.

Example 14.13 Resolve Example 13.10 if the methane to steam mole ratio is 1:1.

Solution

The reactions taking place in the reactor are



and the number of moles of each species present in the reactor can be expressed as a function of the molar extents of the reactions in the form

$$n_{\text{CH}_4} = 1 - \epsilon_1 - \epsilon_2$$

$$n_{\text{H}_2\text{O}} = 1 - \epsilon_1 - 2\epsilon_2$$

$$n_{\text{CO}} = \epsilon_1$$

$$n_{\text{CO}_2} = \epsilon_2$$

$$n_{\text{H}_2} = 3\epsilon_1 + 4\epsilon_2$$

$$n_T = 2 + 2\epsilon_1 + 2\epsilon_2$$

Thus, Eqns (5) and (6) of Example 13.10 take the form

$$0.031 = \left[\frac{\epsilon_1 (3\epsilon_1 + 4\epsilon_2)^3}{(1 - \epsilon_1 - \epsilon_2) (1 - \epsilon_1 - 2\epsilon_2) (2 + 2\epsilon_1 + 2\epsilon_2)^2} \right] (2)^2, \quad (1)$$

$$0.096 = \left[\frac{\epsilon_2 (3\epsilon_1 + 4\epsilon_2)^4}{(1 - \epsilon_1 - \epsilon_2) (1 - \epsilon_1 - 2\epsilon_2)^2 (2 + 2\epsilon_1 + 2\epsilon_2)^2} \right] (2)^2. \quad (2)$$

Simultaneous solution of Eqns (1) and (2) by MATHCAD® gives

$$\epsilon_1 = 0.055 \quad \text{and} \quad \epsilon_2 = 0.147.$$

Hence, the composition of the equilibrium gas mixture is calculated as

$$y_{\text{CH}_4} = 0.332 \quad y_{\text{H}_2\text{O}} = 0.271 \quad y_{\text{CO}} = 0.023 \quad y_{\text{CO}_2} = 0.061 \quad y_{\text{H}_2} = 0.313.$$

Now it is necessary to check whether carbon deposition reactions take place.

Boudouard reaction

$K_a = 108$ at 800 K. The use of Eqn (14.5-1) gives Q_a as

$$Q_a = \left[\frac{0.061}{(0.023)^2} \right] \frac{1}{2} = 57.7.$$

Since $Q_a < K_a$, the Boudouard reaction leads to carbon deposition.

Heterogeneous water-gas reaction

$K_a = 25.51$ at 800 K. The use of Eqn (14.5-2) gives Q_a as

$$Q_a = \left[\frac{0.271}{(0.023)(0.313)} \right] \frac{1}{2} = 18.8.$$

Since $Q_a < K_a$, the heterogeneous water-gas reaction leads to carbon deposition.

Methane cracking reaction

$K_a = 0.702$ at 800 K. The use of Eqn (14.5-3) gives Q_a as

$$Q_a = \left[\frac{(0.313)^2}{0.332} \right] (2) = 0.59.$$

Since $Q_a < K_a$, methane cracking reaction leads to carbon deposition.

Therefore, it is necessary to recalculate the equilibrium gas phase composition by taking the following carbon deposition reactions into account:



The stoichiometric matrix is

$$[\alpha] = \begin{array}{c} \text{Species} \rightarrow \\ \text{Rxn1} \\ \text{Rxn2} \\ \text{Rxn3} \\ \text{Rxn4} \\ \text{Rxn5} \end{array} \begin{bmatrix} \text{CH}_4 & \text{H}_2\text{O} & \text{CO} & \text{H}_2 & \text{CO}_2 & \text{C} \\ -1 & -1 & 1 & 3 & 0 & 0 \\ -1 & -2 & 0 & 4 & 1 & 0 \\ 0 & 0 & -2 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 & 0 & 1 \\ -1 & 0 & 0 & 2 & 0 & 1 \end{bmatrix}$$

and its rank is 3. Since two of the carbon deposition reactions is redundant, calculations will be carried out by considering reaction 1, 2, and 3.

The equilibrium constants of these reactions are expressed as

$$K_{a_1} = 0.031 = \frac{y_{\text{CO}} y_{\text{H}_2}^3}{y_{\text{CH}_4} y_{\text{H}_2\text{O}}} (2)^2, \quad (1)$$

$$K_{a_2} = 0.096 = \frac{y_{\text{CO}_2} y_{\text{H}_2}^4}{y_{\text{CH}_4} y_{\text{H}_2\text{O}}^2} (2)^2, \quad (2)$$

$$K_{a_3} = 108 = \frac{y_{\text{CO}_2}}{y_{\text{CO}}^2} \frac{1}{2}. \quad (3)$$

The summation of mole fractions is equal to unity, i.e.

$$y_{\text{CH}_4} + y_{\text{H}_2\text{O}} + y_{\text{CO}} + y_{\text{H}_2} + y_{\text{CO}_2} = 1. \quad (4)$$

Choosing 1 mol of H_2O as a basis, 6 atoms of hydrogen and 1 atom of oxygen enter the reactor. Hydrogen balance gives

$$4y_{\text{CH}_4} + 2y_{\text{H}_2\text{O}} + 2y_{\text{H}_2} = \frac{6}{n_T}. \quad (5)$$

Oxygen balance gives

$$y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2} = \frac{1}{n_T}. \quad (6)$$

Simultaneous solution of Eqns (1)–(6) by MATHCAD[®] gives

$$\begin{aligned} y_{\text{CH}_4} &= 0.257 & y_{\text{H}_2\text{O}} &= 0.324 & y_{\text{CO}} &= 0.012 \\ y_{\text{H}_2} &= 0.373 & y_{\text{CO}_2} &= 0.034 & n_T &= 2.477 \text{ mol.} \end{aligned}$$

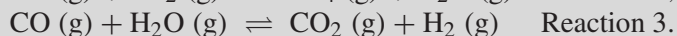
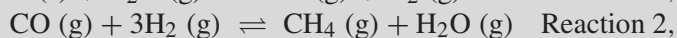
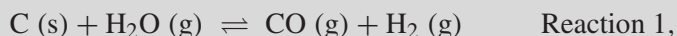
Comment: Carbon deposition can be prevented by increasing steam to carbon ratio of the feed stream.

14.6 The Inverse Problem

If the reactions taking place within the reactor are known and if the operating conditions are specified, then the composition of the gas mixture under equilibrium conditions and the yield of the desired product can be calculated in a rather straightforward manner as will be shown in Example 14.14. Under these circumstances the solution is unique.

The inverse problem deals with the determination of the operating conditions once the reactions taking place in the reactor are given and the yield of the desired product is specified. As will be shown in Example 14.15, this problem does not have a unique solution and various sets of operating conditions may lead to the desired yield.

Example 14.14 Consider the following reactions during coal methanation, i.e. conversion of coal to methane:



Calculate the gas composition under equilibrium conditions and the moles of methane produced per mole of carbon reacted for the following operating conditions:

- The hydrogen to steam mole ratio is 17.5:1, and the reactor temperature and pressure are 1000 K and 5 bar, respectively.
- The hydrogen to steam mole ratio is 3.5:1, and the reactor temperature and pressure are 800 K and 2.5 bar, respectively.

Solution

Using the values given in Table 14.1, the equilibrium constants are expressed as

$$K_{a_1} = \exp \left(-\frac{15,384}{T} + 0.98 \ln T + 9.44 \right),$$

$$K_{a_2} = \exp \left(\frac{23,703}{T} - 3.66 \ln T - 1.57 \right),$$

$$K_{a_3} = \exp \left(\frac{5161.7}{T} + 0.98 \ln T - 11.56 \right).$$

- At 1000 K, the equilibrium constants are

$$K_{a_1} = 2.283 \quad K_{a_2} = 0.043 \quad K_{a_3} = 1.450.$$

Assuming negligible solubility of gases in carbon and neglecting the effect of pressure on the fugacity of pure carbon result in $a_{\text{C}} = 1$. Choosing standard state pressure for gases so as to make pure component fugacities equal to unity and assuming an ideal gas mixture result in the following equilibrium constant expressions for the given reactions:

$$K_{a1} = 2.283 = \frac{y_{\text{CO}}y_{\text{H}_2}}{y_{\text{H}_2\text{O}}} (5), \quad (1)$$

$$K_{a2} = 0.043 = \frac{y_{\text{CH}_4}y_{\text{H}_2\text{O}}}{y_{\text{CO}}y_{\text{H}_2}^3} \frac{1}{5^2}, \quad (2)$$

$$K_{a3} = 1.450 = \frac{y_{\text{CO}_2}y_{\text{H}_2}}{y_{\text{CO}}y_{\text{H}_2\text{O}}}. \quad (3)$$

The summation of mole fractions is equal to unity, i.e.

$$y_{\text{CO}} + y_{\text{H}_2} + y_{\text{H}_2\text{O}} + y_{\text{CH}_4} + y_{\text{CO}_2} = 1. \quad (4)$$

Choosing 1 mol of H_2O as a basis, 37 atoms of hydrogen and 1 atom of oxygen enter the reactor. Hydrogen balance gives

$$4y_{\text{CH}_4} + 2y_{\text{H}_2\text{O}} + 2y_{\text{H}_2} = \frac{37}{n_T}. \quad (5)$$

Oxygen balance gives

$$y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2} = \frac{1}{n_T}. \quad (6)$$

Simultaneous solution of Eqns (1)–(6) by MATHCAD[®] yields the equilibrium gas phase composition as

$$\begin{aligned} y_{\text{CH}_4} &= 0.239 & y_{\text{H}_2} &= 0.698 & y_{\text{CO}} &= 0.024 \\ y_{\text{H}_2\text{O}} &= 0.037 & y_{\text{CO}_2} &= 1.895 \times 10^{-3} \end{aligned}$$

and the total number of moles of the gas phase as

$$n_T = 15.258 \text{ mol.}$$

The number of moles of carbon reacted, n_{C} , is calculated from the carbon balance as

$$\begin{aligned} n_{\text{C}} &= n_T(y_{\text{CH}_4} + y_{\text{CO}} + y_{\text{CO}_2}) \\ &= (15.258)(0.239 + 0.024 + 1.895 \times 10^{-3}) = 4.042 \text{ mol.} \end{aligned}$$

The number of moles of methane produced is given by

$$n_{\text{CH}_4} = (15.258)(0.239) = 3.647 \text{ mol.}$$

Hence, the number of moles of methane produced per mole of carbon reacted is

$$\frac{3.647}{4.042} = 0.902.$$

(b) At 800 K, the equilibrium constants are

$$K_{a_1} = 0.039 \quad K_{a_2} = 36.346 \quad K_{a_3} = 4.234.$$

Following the procedure presented in part (a), the governing equations are

$$0.039 = \frac{y_{\text{CO}} y_{\text{H}_2}}{y_{\text{H}_2\text{O}}} (2.5), \quad (7)$$

$$36.346 = \frac{y_{\text{CH}_4} y_{\text{H}_2\text{O}}}{y_{\text{CO}} y_{\text{H}_2}^3} \frac{1}{2.5^2}, \quad (8)$$

$$4.234 = \frac{y_{\text{CO}_2} y_{\text{H}_2}}{y_{\text{CO}} y_{\text{H}_2\text{O}}}, \quad (9)$$

$$y_{\text{CO}} + y_{\text{H}_2} + y_{\text{H}_2\text{O}} + y_{\text{CH}_4} + y_{\text{CO}_2} = 1, \quad (10)$$

$$4y_{\text{CH}_4} + 2y_{\text{H}_2\text{O}} + 2y_{\text{H}_2} = \frac{9}{n_T}, \quad (11)$$

$$y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2} = \frac{1}{n_T}. \quad (12)$$

Simultaneous solution of Eqns (7)–(12) by MATHCAD[®] yields the equilibrium gas phase composition as

$$y_{\text{CH}_4} = 0.396 \quad y_{\text{H}_2} = 0.334 \quad y_{\text{CO}} = 0.011 \quad y_{\text{H}_2\text{O}} = 0.229 \quad y_{\text{CO}_2} = 0.031$$

and the total number of moles of the gas phase as

$$n_T = 3.323 \text{ mol.}$$

The number of moles of carbon reacted is

$$n_C = (3.323)(0.396 + 0.011 + 0.031) = 1.455 \text{ mol.}$$

The number of moles of methane produced is

$$n_{\text{CH}_4} = (3.323)(0.396) = 1.316 \text{ mol.}$$

Hence, the number of moles of methane produced per mole of carbon reacted is

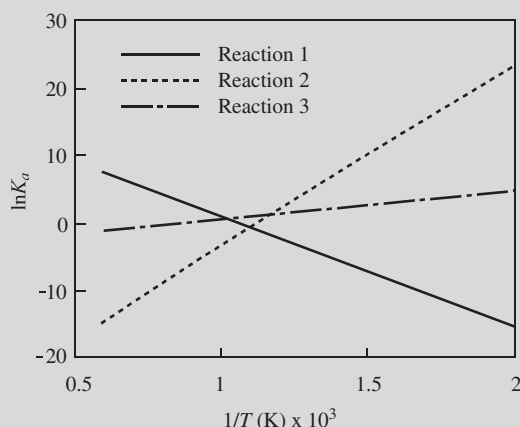
$$\frac{1.316}{1.455} = 0.904.$$

Comment: For the two different sets of operating conditions, approximately 90% of the coal is converted to methane.

Example 14.15 Specify the operating conditions during coal methanation to convert 90% of the coal to methane, i.e. 0.9 mol of methane produced per mole of carbon reacted.⁷

Solution

In this case, temperature, pressure, and the hydrogen to steam mole ratio are unknown quantities. To get an idea about the temperature range, it is first necessary to plot $\ln K_a$ versus $1/T$ as shown in the figure below. Formation of methane is dependent on reactions 1 and 2. While reaction 1 is endothermic, reaction 2 is exothermic. It is therefore necessary to choose a temperature that is a compromise between these two reactions. A temperature range of 600–1200 K seems reasonable in this respect.



The reactor pressure, P , is the sum of the partial pressures of species in the gas phase, i.e.

$$P = p_{\text{CH}_4} + p_{\text{H}_2} + p_{\text{CO}} + p_{\text{CO}_2} + p_{\text{H}_2\text{O}}. \quad (1)$$

Since P is an unknown, it is easier to express equilibrium constants of reactions in terms of partial pressures. Thus

$$K_{a1} = \frac{p_{\text{CO}}p_{\text{H}_2}}{p_{\text{H}_2\text{O}}}, \quad (2)$$

$$K_{a2} = \frac{p_{\text{CH}_4}p_{\text{H}_2\text{O}}}{p_{\text{CO}}p_{\text{H}_2}^3}, \quad (3)$$

$$K_{a3} = \frac{p_{\text{CO}_2}p_{\text{H}_2}}{p_{\text{CO}}p_{\text{H}_2\text{O}}}. \quad (4)$$

⁷This problem is suggested by Helfferich (1989).

The requirement that 0.9 mol of methane should be produced per mole of carbon reacted is expressed as

$$\frac{y_{\text{CH}_4}}{y_{\text{CH}_4} + y_{\text{CO}} + y_{\text{CO}_2}} = 0.9. \quad (5)$$

Simplification of Eqn (5) leads to

$$\frac{y_{\text{CH}_4}}{y_{\text{CO}} + y_{\text{CO}_2}} = 9 \implies \frac{p_{\text{CH}_4}}{p_{\text{CO}} + p_{\text{CO}_2}} = 9. \quad (6)$$

With the help of Eqns (2), (3), (4), and (6), one can express the partial pressures of each component in terms of p_{CH_4} , partial pressure of the desired component. From Eqns (2) and (3)

$$p_{\text{H}_2} = \sqrt{\frac{p_{\text{CH}_4}}{K_{a1}K_{a2}}}. \quad (7)$$

From Eqns (2) and (4)

$$p_{\text{CO}} = \sqrt{\left(\frac{K_{a1}}{K_{a3}}\right) p_{\text{CO}_2}}. \quad (8)$$

Elimination of p_{CO_2} between Eqns (6) and (8) results in

$$p_{\text{CO}}^2 + \left(\frac{K_{a1}}{K_{a3}}\right) p_{\text{CO}} - \frac{1}{9} \left(\frac{K_{a1}}{K_{a3}}\right) p_{\text{CH}_4} = 0. \quad (9)$$

Once p_{CO} is determined from Eqn (9), then the partial pressure of carbon dioxide is easily determined from Eqn (6), i.e.

$$p_{\text{CO}_2} = \frac{p_{\text{CH}_4}}{9} - p_{\text{CO}}. \quad (10)$$

After determining p_{CO} and p_{H_2} , the partial pressure of water is calculated from Eqn (2) as

$$p_{\text{H}_2\text{O}} = \frac{p_{\text{CO}}p_{\text{H}_2}}{K_{a1}}. \quad (11)$$

The procedure to calculate the pressure and the mole fractions is as follows:

1. Specify temperature.
2. Choose p_{CH_4} .
3. Calculate p_{H_2} , p_{CO} , p_{CO_2} , and $p_{\text{H}_2\text{O}}$ from Eqns (7), (9), (10), and (11), respectively.
4. Calculate P from Eqn (1).
5. Use $y_i = p_i/P$ to calculate mole fractions in the gas phase.

Material balances should be utilized to determine the moles of hydrogen entering the reactor per mole of steam. Since the only source of oxygen is water, the oxygen balance gives

$$\text{Moles of H}_2\text{O in} = n_T(y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2}). \quad (12)$$

On the other hand, the hydrogen balance is

$$(\text{Moles of H}_2 + \text{Moles of H}_2\text{O}) \text{ in} = n_T(y_{\text{H}_2\text{O}} + y_{\text{H}_2} + 2y_{\text{CH}_4}). \quad (13)$$

From Eqns (12) and (13)

$$\frac{\text{Moles of H}_2 \text{ in}}{\text{Moles of H}_2\text{O in}} = \frac{y_{\text{H}_2\text{O}} + y_{\text{H}_2} + 2y_{\text{CH}_4}}{y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2}} - 1 = \frac{p_{\text{H}_2\text{O}} + p_{\text{H}_2} + 2p_{\text{CH}_4}}{p_{\text{H}_2\text{O}} + p_{\text{CO}} + 2p_{\text{CO}_2}} - 1. \quad (14)$$

Let us specify reactor temperature as 1000 K. The equilibrium constants are

$$K_{a1} = 2.283 \quad K_{a2} = 0.043 \quad K_{a3} = 1.450.$$

Choosing $p_{\text{CH}_4} = 1.2$ bar, from Eqn (7)

$$p_{\text{H}_2} = \sqrt{\frac{1.2}{(2.283)(0.043)}} = 3.496 \text{ bar.}$$

From Eqn (9)

$$p_{\text{CO}}^2 + 1.574p_{\text{CO}} - 0.210 = 0 \implies p_{\text{CO}} = 0.124 \text{ bar.}$$

From Eqn (10)

$$p_{\text{CO}_2} = \frac{1.2}{9} - 0.124 = 9.333 \times 10^{-3} \text{ bar.}$$

From Eqn (11)

$$p_{\text{H}_2\text{O}} = \frac{(0.124)(3.496)}{2.283} = 0.190 \text{ bar.}$$

Thus, the reactor pressure is calculated from Eqn (1) as

$$P = 1.2 + 3.496 + 0.124 + 9.333 \times 10^{-3} + 0.190 = 5.019 \text{ bar.}$$

The use of $y_i = p_i/P$ gives the mole fractions as

$$y_{\text{CH}_4} = 0.239 \quad y_{\text{H}_2} = 0.697 \quad y_{\text{CO}} = 0.025 \quad y_{\text{H}_2\text{O}} = 0.038 \quad y_{\text{CO}_2} = 1.860 \times 10^{-3}.$$

From Eqn (14)

$$\frac{\text{Moles of H}_2 \text{ in}}{\text{Moles of H}_2\text{O in}} = \frac{0.190 + 3.496 + (2)(1.2)}{0.190 + 0.124 + (2)(9.333 \times 10^{-3})} - 1 = 17.3.$$

If one chooses $p_{\text{CH}_4} = 3$ bar, the results are

$$P = 9.546 \text{ bar,}$$

$$y_{\text{CH}_4} = 0.314 \quad y_{\text{H}_2} = 0.579 \quad y_{\text{CO}} = 0.030 \quad y_{\text{H}_2\text{O}} = 0.072 \quad y_{\text{CO}_2} = 5.238 \times 10^{-3},$$

$$\frac{\text{Moles of H}_2 \text{ in}}{\text{Moles of H}_2\text{O in}} = 10.371.$$

For a more thorough discussion on the problem see [Helfferich \(1989\)](#).

Comment: If the reactor temperature is less than the critical temperature of water (647 K), there is always a danger of water condensation. Water condenses when $p_{\text{H}_2\text{O}} > P_{\text{H}_2\text{O}}^{\text{vap}}$ at the specified temperature, leading to corrosion. In this case, calculations must be repeated with another value of T and/or p_{CH_4} to make $p_{\text{H}_2\text{O}} < P_{\text{H}_2\text{O}}^{\text{vap}}$.

14.7 Chemical Equilibrium Calculations by the Minimization of Gibbs Energy

In Example 13.9, the extent of the reaction for the hydrogenation of benzene to produce cyclohexane at 550 K and 1 bar was calculated as 0.4. Let us show that the Gibbs energy of the system is indeed a minimum at a reaction extent of 0.4.

The total Gibbs energy of the system is expressed as

$$G_{\text{mix}} = n_{\text{C}_6\text{H}_6} \bar{G}_{\text{C}_6\text{H}_6} + n_{\text{H}_2} \bar{G}_{\text{H}_2} + n_{\text{C}_6\text{H}_{12}} \bar{G}_{\text{C}_6\text{H}_{12}}. \quad (14.7-1)$$

The partial molar Gibbs energy of species i in a mixture is given by Eq. (13.3-8) as

$$\bar{G}_i = \tilde{G}_i^o + RT \ln \hat{a}_i. \quad (14.7-2)$$

where

$$\tilde{G}_i^o = (\Delta \tilde{G}_f^o)_i \quad \text{and} \quad \hat{a}_i = \frac{\hat{\phi}_i y_i P}{P^o}. \quad (14.7-3)$$

Choosing $P^o = 1$ bar and assuming an ideal gas mixture, Eqn (14.7-2) reduces to

$$\bar{G}_i = (\Delta \tilde{G}_f^o)_i + RT \ln(P y_i). \quad (14.7-4)$$

Substitution of Eqn (14.7-4) into Eqn (14.7-1) and expressing number of moles and mole fractions of each species in terms of the extent of reaction result in

$$G_{\text{mix}} = (1 - \epsilon) \left[(\Delta \tilde{G}_f^o)_{\text{C}_6\text{H}_6} + RT \ln \left(\frac{1 - \epsilon}{5.5 - 3\epsilon} \right) \right] + (4.5 - 3\epsilon) \left[(\Delta \tilde{G}_f^o)_{\text{H}_2} + RT \ln \left(\frac{4.5 - 3\epsilon}{5.5 - 3\epsilon} \right) \right] + \epsilon \left[(\Delta \tilde{G}_f^o)_{\text{C}_6\text{H}_{12}} + RT \ln \left(\frac{\epsilon}{5.5 - 3\epsilon} \right) \right]. \quad (14.7-5)$$

In Eqn (14.7-5), $(\Delta \tilde{G}_f^o)_i$ values should be calculated at the reaction temperature using Eqn (13.4-23).

From Eqn (13.4-22)

$$\begin{aligned} \Delta \tilde{C}_{P_{\text{C}_6\text{H}_6}}^o = & -60.711 + 6.267 \times 10^{-1}T - 5.795 \times 10^{-4}T^2 + 2.799 \times 10^{-7}T^3 \\ & - 5.493 \times 10^{-11}T^4 - [6(17.152 + 0.043 \times 10^{-1}T \\ & - 8.788 \times 10^5T^{-2}) + 3(27.004 + 0.119 \times 10^{-1}T - 0.241 \times 10^{-4}T^2 \\ & + 0.215 \times 10^{-7}T^3 - 0.615 \times 10^{-11}T^4)]. \end{aligned} \quad (14.7-6)$$

Simplification gives

$$\begin{aligned} \Delta \tilde{C}_{P_{\text{C}_6\text{H}_6}}^o = & -244.64 + 5.652 \times 10^{-1}T - 5.072 \times 10^{-4}T^2 \\ & + 2.154 \times 10^{-7}T^3 - 3.648 \times 10^{-11}T^4 + 52.728 \times 10^5T^{-2}. \end{aligned} \quad (14.7-7)$$

Substitution of Eqn (14.7-7) into Eqn (13.4-23) and carrying out the integrations lead to

$$(\Delta \tilde{G}_f^o)_{\text{C}_6\text{H}_6} = 173,835 \text{ J/mol.}$$

Similarly

$$(\Delta \tilde{G}_f^o)_{\text{H}_2} = 0 \text{ J/mol} \quad \text{and} \quad (\Delta \tilde{G}_f^o)_{\text{C}_6\text{H}_{12}} = 172,073 \text{ J/mol.}$$

The use of these values in Eqn (14.7-5) enables one to calculate the Gibbs energy of the system as a function of the extent of reaction. The results are given in Table 14.4 and are also plotted in Fig. 14.2. The Gibbs energy of the mixture reaches its minimum value when $\epsilon = 0.4$.

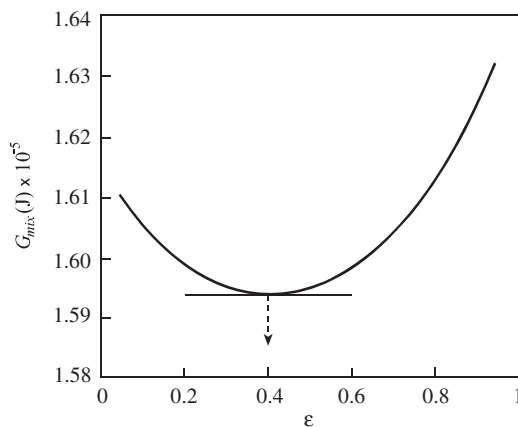
14.7.1 Mathematical Preliminaries

To find the maximum (or minimum) value of $f(x_1, x_2, \dots, x_n)$ if the variables are connected by a constraint

$$g(x_1, x_2, \dots, x_n) = 0 \quad (14.7-8)$$

Table 14.4: Variation of G_{mix} as a function of ϵ .

ϵ	G_{mix} (J)	ϵ	G_{mix} (J)	ϵ	G_{mix} (J)
0.05	161,054	0.40	159,389	0.75	160,769
0.10	160,532	0.45	159,417	0.80	161,224
0.15	160,146	0.50	159,499	0.85	161,776
0.20	159,857	0.55	159,635	0.90	162,418
0.25	159,645	0.60	159,828	0.95	163,222
0.30	159,500	0.65	160,078		
0.35	159,416	0.70	160,390		

Figure 14.2: Variation of G_{mix} as a function of ϵ .

it is first necessary to define a function F in the form

$$F = f(x_1, x_2, \dots, x_n) + \lambda g(x_1, x_2, \dots, x_n), \quad (14.7-9)$$

where λ is called the *Lagrange (or Lagrangian) multiplier*. By solving the following $(n + 1)$ equations

$$\left. \begin{aligned} \frac{\partial F}{\partial x_i} &= \frac{\partial f}{\partial x_i} + \lambda \frac{\partial g}{\partial x_i} = 0 \quad i = 1, 2, \dots, n, \\ g &= 0, \end{aligned} \right\} \quad (14.7-10)$$

one can get the n values of the x_i for which f is a maximum (or minimum).

Example 14.16 How does one construct a conical container (no top) of a given volume V with the minimum surface area A ?

Solution

Consider a conical container with a bottom radius r and a height h . It is required to find the minimum of

$$A = \pi r \sqrt{r^2 + h^2} \quad (1)$$

with the following constraint

$$\underbrace{\frac{1}{3}\pi r^2 h - V}_{g(r,h)} = 0. \quad (2)$$

The function F is defined by

$$F = \underbrace{\pi r \sqrt{r^2 + h^2}}_A + \lambda \underbrace{\left(\frac{1}{3}\pi r^2 h - V \right)}_g \quad (3)$$

Application of Eqn (14.7-10) gives

$$\begin{aligned} \frac{\partial A}{\partial r} + \lambda \frac{\partial g}{\partial r} = 0 &\Rightarrow \pi \sqrt{r^2 + h^2} + \frac{\pi r^2}{\sqrt{r^2 + h^2}} + \lambda \frac{2}{3}\pi r h = 0, \\ \frac{\partial A}{\partial h} + \lambda \frac{\partial g}{\partial h} = 0 &\Rightarrow \frac{\pi r h}{\sqrt{r^2 + h^2}} + \lambda \frac{1}{3}\pi r^2 = 0, \\ g = 0 &\Rightarrow \frac{1}{3}\pi r^2 h - V = 0. \end{aligned} \quad (4)$$

Simultaneous solution of these three equations leads to

$$r = \sqrt[3]{\frac{3V}{\sqrt{2}\pi}} \quad \text{and} \quad h = \sqrt{2} \sqrt[3]{\frac{3V}{\sqrt{2}\pi}}. \quad (5)$$

14.7.2 Minimization of Gibbs Energy

At a specified temperature and pressure, the total Gibbs energy of a reacting mixture consisting of k species is given by

$$G_{\text{mix}}(n_1, n_2, \dots, n_k) = \sum_{i=1}^k n_i \bar{G}_i, \quad (14.7-11)$$

where the partial molar Gibbs energy of species i in a mixture is given by Eqn (13.3-8) as

$$\bar{G}_i = \tilde{G}_i^o + RT \ln \hat{a}_i. \quad (14.7-12)$$

The conservation of elements is expressed in the form

$$E_j = \sum_{i=1}^k \beta_{ji} n_i \quad j = 1, 2, \dots, t, \quad (14.7-13)$$

where t represents the number of elements. To find the set $\{n_i\}$ that minimizes the total Gibbs energy subject to the constraint given by Eqn (14.7-13), it is first necessary to define the function F as

$$F = \sum_{i=1}^k n_i \bar{G}_i + \sum_{j=1}^t \lambda_j \left(\sum_{i=1}^k \beta_{ji} n_i - E_j \right). \quad (14.7-14)$$

Setting the partial derivatives of Eqn (14.7-14) equal to zero, i.e.

$$\frac{\partial F}{\partial n_i} = 0 \quad i = 1, 2, \dots, k \quad (14.7-15)$$

yields

$$\bar{G}_i + \sum_{j=1}^t \beta_{ji} \lambda_j = 0 \quad (14.7-16)$$

or

$$\tilde{G}_i^o + RT \ln \hat{a}_i + \sum_{j=1}^t \beta_{ji} \lambda_j = 0. \quad (14.7-17)$$

Note that

$$\tilde{G}_i^o = (\Delta \tilde{G}_f^o)_i \quad \text{and} \quad \hat{a}_i = \frac{\hat{\phi}_i y_i P}{P^o}. \quad (14.7-18)$$

Therefore, the simultaneous solution of the following $(k+t+1)$ equations minimizes the total Gibbs energy of the reacting mixture:

$$\begin{aligned} \ln \left[\hat{\phi}_i y_i \left(\frac{P}{P^o} \right) \right] + \sum_{j=1}^t \beta_{ji} \lambda_j^* &= -\frac{(\Delta \tilde{G}_f^o)_i}{RT} \quad i = 1, 2, \dots, k \\ \sum_{i=1}^k \beta_{ji} y_i &= \frac{E_j}{n_T} \quad j = 1, 2, \dots, t \\ \sum_{i=1}^k y_i &= 1, \end{aligned} \quad (14.7-19)$$

where

$$\lambda_j^* = \frac{\lambda_j}{RT} \quad (14.7-20)$$

The values of $(\Delta \tilde{G}_f^o)_i$ should be calculated at the reaction temperature using Eqn (13.4-23).

Example 14.17 Steam reforming of methane at 800 K and 1 bar produces a gas phase consisting of CH_4 , H_2O , CO , CO_2 , and H_2 . Estimate the equilibrium composition if the water to methane feed mole ratio is 4:1.

Solution

Since pressure is atmospheric and temperature is high, it is plausible to assume that the gas mixture behaves ideally. Thus, Eqn (14.7-19) reduces to

$$\ln y_i + \sum_{j=1}^3 \beta_{ji} \lambda_j^* = -\frac{(\Delta \tilde{G}_f^o)_i}{RT} \quad i = 1, 2, \dots, 5, \quad (1)$$

$$\sum_{i=1}^5 \beta_{ji} y_i = \frac{E_j}{n_T} \quad j = 1, 2, 3, \quad (2)$$

$$\sum_{i=1}^5 y_i = 1. \quad (3)$$

From Appendix E

Species	$\Delta \tilde{H}_f^o, 298$ (kJ/mol)	$\Delta \tilde{G}_f^o, 298$ (kJ/mol)
CH_4	-74.90	-50.87
H_2O	-242.00	-228.80
CO	-110.60	-137.40
CO_2	-393.80	-394.60

From Appendix B

Species	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
CH_4	36.155	-0.511	2.215	-1.824	4.899
H_2O	33.763	-0.006	0.224	-0.100	0.110
CO	29.651	-0.007	0.183	-0.094	0.108
CO_2	29.268	-0.224	2.653	-4.153	20.057
H_2	27.004	0.119	-0.241	0.215	-0.615
O_2	29.705	-0.099	0.399	-0.339	0.918

From Eqn (13.4-24)

$$\tilde{C}_p^o = 17.152 + 4.273 \times 10^{-3}T - 8.7879 \times 10^5 T^{-2} \quad \text{for carbon.}$$

It is first necessary to calculate the values of $(\Delta \tilde{G}_f^o)_i$ at 800 K. For example, for CO, the use of Eqn (13.4-22) gives

$$\begin{aligned}\Delta \tilde{C}_{P_{CO}}^o &= [29.651 - 17.152 - (0.5)(29.705)] \\ &\quad + [-0.007 - 0.043 + (0.5)(0.099)] \times 10^{-1}T \\ &\quad + [0.183 - (0.5)(0.399)] \times 10^{-4}T^2 \\ &\quad + [-0.094 + (0.5)(0.339)] \times 10^{-7}T^3 \\ &\quad + [0.108 - (0.5)(0.918)] \times 10^{-11}T^4 + 8.788 \times 10^5 T^{-2}\end{aligned}$$

or

$$\begin{aligned}\Delta \tilde{C}_{P_{CO}}^o &= -2.3535 - 5 \times 10^{-5}T - 1.65 \times 10^{-6}T^2 + 7.55 \times 10^{-9}T^3 \\ &\quad - 3.51 \times 10^{-12}T^4 + 8.788 \times 10^5 T^{-2}.\end{aligned}\quad (4)$$

Substitution of Eqn (4) into Eqn (13.4-23) and carrying out the integrations give

$$(\Delta \tilde{G}_f^o)_{CO} = -183,500 \text{ J/mol}.$$

Similarly

$$\begin{aligned}(\Delta \tilde{G}_f^o)_{CH_4} &= -2093 \text{ J/mol} \quad (\Delta \tilde{G}_f^o)_{H_2O} = -204,500 \text{ J/mol} \\ (\Delta \tilde{G}_f^o)_{CO_2} &= -395,400 \text{ J/mol}.\end{aligned}$$

Therefore, Eqn (1) can be expressed for each species as

$$\ln y_{CH_4} + \lambda_C^* + 4\lambda_H^* = \frac{2093}{(8.314)(800)}, \quad (5)$$

$$\ln y_{H_2O} + 2\lambda_H^* + \lambda_O^* = \frac{204,500}{(8.314)(800)}, \quad (6)$$

$$\ln y_{CO} + \lambda_C^* + \lambda_O^* = \frac{183,500}{(8.314)(800)}, \quad (7)$$

$$\ln y_{CO_2} + \lambda_C^* + 2\lambda_O^* = \frac{395,400}{(8.314)(800)}, \quad (8)$$

$$\ln y_{H_2} + 2\lambda_H^* = 0. \quad (9)$$

On the other hand, the element-by-species matrix is written as

$$\begin{array}{c} \text{Species} \rightarrow \\ \begin{array}{ccccc} CH_4 & H_2O & CO & CO_2 & H_2 \end{array} \\ \begin{array}{c} C \\ H \\ O \end{array} \left[\begin{array}{ccccc} 1 & 0 & 1 & 1 & 0 \\ 4 & 2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 2 & 0 \end{array} \right] \end{array} \quad (10)$$

Choosing 1 mol of methane as a basis, 1 atom of carbon, 12 atoms of hydrogen, and 4 atoms of oxygen enter the reactor and Eqn (2) can be written as

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 4 & 2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} y_{\text{CH}_4} \\ y_{\text{H}_2\text{O}} \\ y_{\text{CO}} \\ y_{\text{CO}_2} \\ y_{\text{H}_2} \end{bmatrix} = \begin{bmatrix} 1/n_T \\ 12/n_T \\ 4/n_T \end{bmatrix}. \quad (11)$$

Carrying out the multiplication gives

$$\text{Carbon balance: } y_{\text{CH}_4} + y_{\text{CO}} + y_{\text{CO}_2} = \frac{1}{n_T}. \quad (12)$$

$$\text{Hydrogen balance: } 4y_{\text{CH}_4} + 2y_{\text{H}_2\text{O}} + 2y_{\text{H}_2} = \frac{12}{n_T}. \quad (13)$$

$$\text{Oxygen balance: } y_{\text{H}_2\text{O}} + y_{\text{CO}} + 2y_{\text{CO}_2} = \frac{4}{n_T}. \quad (14)$$

Eqn (3) is also written by

$$y_{\text{CH}_4} + y_{\text{H}_2\text{O}} + y_{\text{CO}} + y_{\text{CO}_2} + y_{\text{H}_2} = 1. \quad (15)$$

Simultaneous solution of nine equations, i.e. Eqns (5)–(9) and (12)–(15), by Newton's method leads to

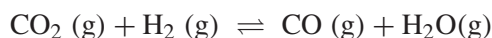
$$\begin{aligned} y_{\text{CH}_4} &= 0.065 & y_{\text{H}_2\text{O}} &= 0.473 & y_{\text{CO}} &= 0.020 & y_{\text{CO}_2} &= 0.077 & y_{\text{H}_2} &= 0.366, \\ \lambda_{\text{C}}^* &= 1.035 & \lambda_{\text{H}}^* &= 0.503 & \lambda_{\text{O}}^* &= 30.49 & n_T &= 6.193 \text{ mol.} \end{aligned}$$

Comment: See Problem 14.31 for the details of the solution.

Problems

Problems related to Section 14.1

14.1 Using the $\Delta G_{\text{f}}^{\circ}$ values given in Table 14.1, show that the equilibrium constant of the reaction



is expressed as a function of temperature as

$$\ln K_a = -\frac{5161.7}{T} - 0.98 \ln T + 11.56.$$

- 14.2** Limestone is put into an evacuated reactor and the reactor is sealed. Upon heating, it decomposes according to the reaction



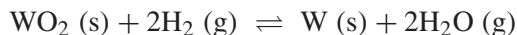
Kubaschewski and Alcock (1979) reported that

$$\Delta G_{\text{rxn}}^{\circ} = 168,516 - 143.02T \quad 449 < T < 1150,$$

where $\Delta G_{\text{rxn}}^{\circ}$ is in J and T is in K. Estimate the pressure of CO_2 within the reactor at 1000 K.

(Answer: 0.047 bar)

- 14.3** At 900 K and 1 bar, the mole fraction of hydrogen at equilibrium in the reaction



is 0.55. Calculate the equilibrium constant of this reaction at 900 K.

(Answer: 0.669)

- 14.4** Ammonium chloride decomposes into an evacuated space at a given temperature according to the reaction



Calculate K_a if the measured total pressure under equilibrium conditions is 2.6 bar.

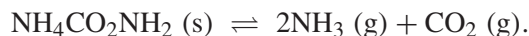
(Answer: 1.69)

- 14.5** Consider the methane cracking reaction, i.e.



Show that if β is the equilibrium mole fraction of CH_4 , then $\beta/(1 - \beta)^2$ is proportional to the pressure.

- 14.6** One of the methods for determining $\Delta H_{\text{rxn}}^{\circ}$, $\Delta S_{\text{rxn}}^{\circ}$, and $\Delta G_{\text{rxn}}^{\circ}$ is to measure the dissociation pressures of pure solid compounds at various temperatures. Consider the decomposition of ammonium carbamate into an evacuated space according to the reaction



(a) Show that the equilibrium constant is given by

$$K_a = \left(\frac{4}{27} \right) P^3,$$

where P is the total equilibrium pressure.

(b) *Joncich et al. (1967)* reported the following data:

T (°C)	25	30	35	40	45	50
P (bar)	0.1173	0.1707	0.2380	0.3293	0.4533	0.6293

Plot $\ln K_a$ versus $1/T$ and show that

$$\ln K_a = -\frac{19,236}{T} + 56.232.$$

(c) Determine $\Delta H_{\text{rxn}}^\circ$ and $\Delta S_{\text{rxn}}^\circ$ for the decomposition of ammonium carbamate.

(Answer: (c) 159,928 J and 467.5 J)

14.7 Ammonium hydrogen sulfide decomposes by the following reaction:

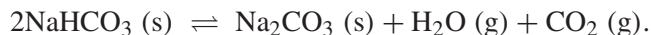


At 298 K, the total pressure over solid NH_4HS is reported as 0.668 bar.

- (a) Calculate K_a for the decomposition reaction at 298 K.
 (b) Pure NH_4HS is to be decomposed in a reactor operating at 1 bar using a gas feed consisting of 70 mol % NH_3 and 30% N_2 . If the operation is isothermal at 298 K, how many moles of feed gas must be supplied per mole of NH_4HS decomposed?

(Answer: (a) 0.112 and (b) 3.61)

14.8 Thermal decomposition of sodium bicarbonate is given by the following reaction:



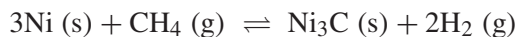
The equilibrium constant for this reaction is given by

$$\ln K_a = 39.81 - \frac{15,381}{T}$$

where T is in K. An evacuated rigid tank having a volume of 5 m^3 and containing 250 kg of sodium bicarbonate is maintained at 400 K. Calculate the amount of NaHCO_3 remaining in the tank when the system reaches equilibrium.

(Answer: 200.3 kg)

14.9 Browning and Emmett (1952) studied the reaction



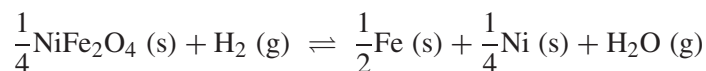
between 500 and 600 K at subatmospheric pressure and reported the following data:

T (K)	P (bar)	y_{H_2}	T (K)	P (bar)	y_{H_2}	T (K)	P (bar)	y_{H_2}
499.55	0.0128	0.0050	520.65	0.0194	0.0078	538.95	0.0230	0.0152
500.75	0.0155	0.0049	513.35	0.0278	0.0076	538.95	0.0499	0.0101
509.65	0.0241	0.0053	502.15	0.0515	0.0023	544.65	0.0460	0.0118
499.95	0.0208	0.0042	520.65	0.0413	0.0066	551.05	0.0557	0.0148
514.05	0.0165	0.0052	526.35	0.0368	0.0070	546.05	0.0520	0.0132
513.45	0.0424	0.0036	528.45	0.0297	0.0090	550.55	0.0464	0.0146
513.45	0.0259	0.0048	527.85	0.0263	0.0096	556.65	0.0493	0.0171
501.75	0.0227	0.0036	529.05	0.0322	0.0088	558.15	0.0406	0.0206
507.05	0.0199	0.0040	522.95	0.0363	0.0085			

Determine the heat of reaction.

(Answer: 165.2 kJ)

14.10 The reduction of nickel ferrite (NiFe_2O_4) to iron and nickel by hydrogen is given by the reaction

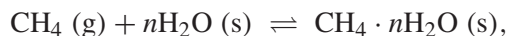
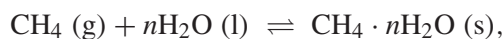


in which $\Delta G_{\text{rxn}}^\circ = -10.836$ kJ at 1000 K (Svoboda *et al.*, 2008). Calculate the mole fraction of hydrogen under equilibrium conditions.

(Answer: 0.214)

14.11 Clathrate hydrates (or commonly known as gas hydrates) are crystalline solid compounds consisting of water and at least one other compound, usually small molecules like methane. Each m^3 of gas hydrate can hold approximately 180 m^3 of natural gas at standard temperature and pressure (STP). In that respect, gas hydrates have the potential of storing various gases in large quantities. In the oil and gas industry, however, formation of gas hydrates causes clogging of pipelines.

Depending on temperature, formation of methane gas hydrates is represented by the following reactions:



where the hydration number, n , varies between 5.77 and 7.18.

Makogon and Sloan (1994) reported the methane hydrate equilibrium data as follows:

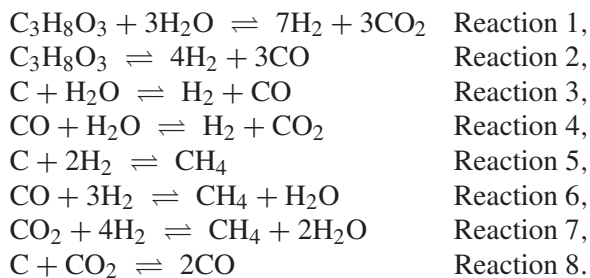
T (K)	190.15	198.15	208.15	218.15	262.40
P (bar)	0.8471	1.349	2.278	3.666	18.47

Considering water and the hydrate as two separate pure solid phases, estimate the equilibrium partial pressure of methane for hydrate formation at 243.15 K.

(Answer: 9.825 bar)

Problems related to Section 14.2

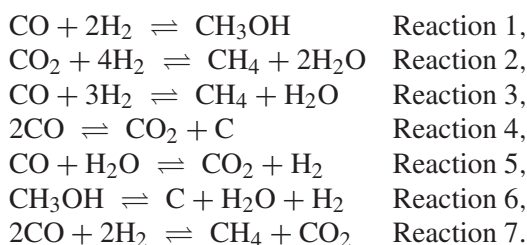
14.12 The following reactions are suggested for the steam reforming of glycerol ($C_3H_8O_3$):



Determine the number of independent reactions representing this system and identify one set of independent reactions.

(Answer: Reactions 1, 2, 3, and 5)

14.13 The probable reactions taking place during the methanol synthesis are suggested as follows:

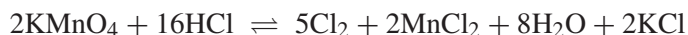


Determine the number of independent reactions representing this system and identify one set of independent reactions.

(Answer: Reactions 1, 2, 3, and 4)

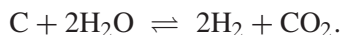
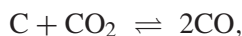
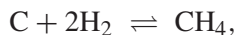
Problems related to Section 14.3

- 14.14** For a mixture consisting of KMnO_4 , HCl , Cl_2 , MnCl_2 , H_2O , KCl , and NaCl , show that the reaction

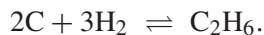
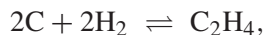
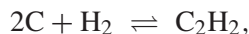


represents this system. Note that NaCl does not take part in the reaction.

- 14.15** Coal and steam react at high temperature to form hydrogen, methane, carbon monoxide, and carbon dioxide. Show that one set of probable reactions is given as follows:

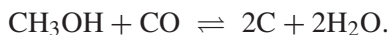
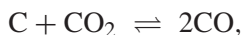
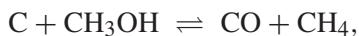


- 14.16** For a system composed of C , H_2 , C_2H_2 , C_2H_4 , and C_2H_6 , show that the following reactions constitute an independent set:

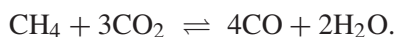
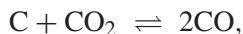
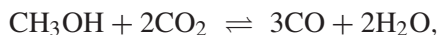


- 14.17** Consider a system composed of CH_3OH , C , CO , CH_4 , CO_2 , and H_2O .

(a) Show that one set of probable reactions is given as follows:

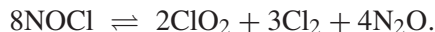
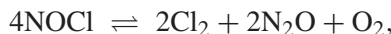
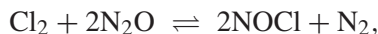


(b) Show that another possible set is given by

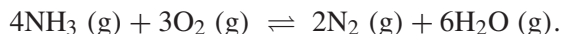
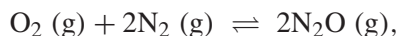
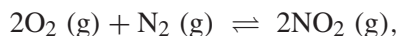
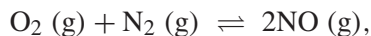


- 14.18** Explosion of ammonium perchlorate (NH_4ClO_4) yields the following species present in the equilibrium mixture: Cl_2 , N_2O , NOCl , HCl , H_2O , N_2 ,

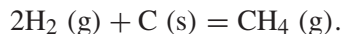
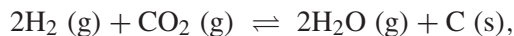
O₂, and ClO₂. Show that the following chemical reactions represent this system:



- 14.19** Combustion of ammonia over a platinum catalyst yields a gas mixture consisting of NH₃, O₂, N₂, NO, NO₂, N₂O, and H₂O. Show that one set of probable reactions is given as follows:



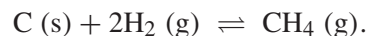
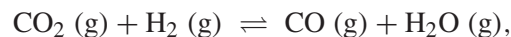
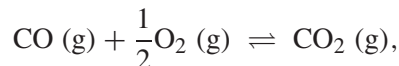
- 14.20** Conversion of carbon dioxide produced by the crew to oxygen is of utmost importance for long-duration manned space missions. For this purpose NASA developed the Bosch process in which carbon dioxide is catalytically reduced with hydrogen to form carbon and water. The water is subsequently electrolyzed to yield hydrogen for reuse in the Bosch reactor and oxygen for the crew. The following reactions take place within the Bosch reactor:



Estimate the equilibrium gas composition when $T = 900 \text{ K}$, $P = 1 \text{ bar}$, and O/H atomic ratio is 0.5.

(**Answer:** $y_{\text{H}_2} = 0.412$, $y_{\text{H}_2\text{O}} = 0.187$, $y_{\text{CO}} = 0.169$, $y_{\text{CO}_2} = 0.177$, $y_{\text{CH}_4} = 0.056$)

- 14.21** In Example 14.5, the following four reactions can also be considered as an independent set:



Estimate the equilibrium composition of the gas phase using these reactions.

- 14.22** Determine the number of degrees of freedom of a gaseous system consisting of C_2H_4 , C_6H_6 , C_7H_8 , C_8H_{10} , C_9H_{12} , $\text{C}_{10}\text{H}_{14}$, and $\text{C}_{11}\text{H}_{16}$.

(Answer: 3)

- 14.23** Determine the number of degrees of freedom of a system consisting of Si (s), H_2 , HCl , SiCl_4 , SiCl_2 , SiHCl_3 , SiH_2Cl_2 , SiH_3Cl , and SiH_4 at 1500 K.

(Answer: 3)

- 14.24** Limestone is put into an evacuated reactor and the reactor is sealed. Upon heating, it decomposes according to the reaction



Determine the number of degrees of freedom for this system.

(Answer: 1)

- 14.25** Consider a system that contains the gases ethylene (C_2H_4), methane (CH_4), ethane (C_2H_6), and propane (C_3H_8).

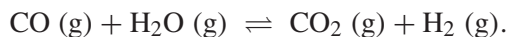
(a) Show that the following set of reactions represents this system:



(b) Conclude that $\mathcal{F} = 3$. Indicate the intensive variables that must be specified for this system.

Problems related to Section 14.4

- 14.26** A gaseous mixture consisting of CO , CO_2 , H_2 , and H_2O is heated to 1500 K at a constant pressure of 1 bar. The following reaction is expected to take place under these conditions:



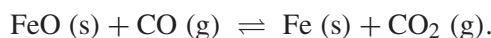
- (a) Calculate the equilibrium constant of this reaction with the help of Table 14.1.
- (b) If the initial composition is 25 mol% CO , 25% CO_2 , 35% H_2 , and 15% H_2O , estimate the final equilibrium composition of the gas mixture.
- (c) If the initial composition is 35 mol% CO , 15% CO_2 , 10% H_2 , and 40% H_2O , estimate the final equilibrium composition of the gas mixture.

(**Answer:** (a) 0.386 (b) $y_{\text{CO}_2} = 0.144$, $y_{\text{H}_2} = 0.244$, $y_{\text{CO}} = 0.356$, $y_{\text{H}_2\text{O}} = 0.256$ and c) $y_{\text{CO}_2} = 0.217$, $y_{\text{H}_2} = 0.167$, $y_{\text{CO}} = 0.283$, $y_{\text{H}_2\text{O}} = 0.333$)

- 14.27** An evacuated rigid tank having a volume of 3 m^3 and containing 3 kg of solid carbon is maintained at 900 K. Estimate the equilibrium pressure at 900 K after introducing 1 kmol of H_2 into the tank.

(**Answer:** 18.7 bar)

- 14.28** Calculate the maximum allowable partial pressure of CO_2 for the reduction of FeO to metallic iron at 1100 K and 1 bar according to the reaction



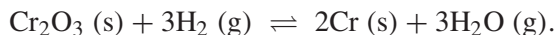
The equilibrium constant for this reaction is given by

$$\ln K_a = \frac{2108.6}{T} - 2.581,$$

where T is in K.

(**Answer:** 0.34 bar)

- 14.29** The following equilibrium data have been determined for the reaction:



$T \text{ (K)}$	1100	1200	1300	1400	1500
K_a	7.326×10^{-14}	3.033×10^{-12}	7.080×10^{-11}	1.054×10^{-9}	1.094×10^{-8}

Would an atmosphere of 0.01 mol% H_2O , 12% H_2 , and 87.99% N_2 oxidize chromium at 1485 K?

(**Answer:** No)

Problem Related to Section 14.5

- 14.30** A gaseous mixture consists of 15 mol% CO , 10% CO_2 , and 75% N_2 .

- (a) Is there any danger of carbon deposition if the temperature and pressure of the gaseous mixture are 1000 K and 1 bar, respectively?
 (b) What will be the effect of increasing pressure?

(**Answer:** (a) No)

Problems Related to Section 14.7

14.31 Letting $y_{\text{CH}_4} = X_1$, $y_{\text{H}_2\text{O}} = X_2$, $y_{\text{CO}} = X_3$, $y_{\text{CO}_2} = X_4$, $y_{\text{H}_2} = X_5$, $\lambda_{\text{C}}^* = X_6$, $\lambda_{\text{H}}^* = X_7$, $\lambda_{\text{O}}^* = X_8$, and $n_T = X_9$ and taking 8.314 as 8314/1000, Eqns (5)–(9) and (12)–(15) of Example 14.17 become

$$\ln X_1 + X_6 + 4X_7 = \frac{10,465}{33,256}, \quad (1)$$

$$\ln X_2 + 2X_7 + X_8 = \frac{255,625}{8314}, \quad (2)$$

$$\ln X_3 + X_6 + X_8 = \frac{229,375}{8314}, \quad (3)$$

$$\ln X_4 + X_6 + 2X_8 = \frac{247,125}{4157}, \quad (4)$$

$$\ln X_5 + 2X_7 = 0, \quad (5)$$

$$X_1 + X_2 + X_4 = \frac{1}{X_9}, \quad (6)$$

$$4X_1 + 2X_2 + 2X_5 = \frac{12}{X_9}, \quad (7)$$

$$X_2 + X_3 + 2X_4 = \frac{4}{X_9}, \quad (8)$$

$$X_1 + X_2 + X_3 + X_4 + X_5 = 1. \quad (9)$$

The numerical solution requires initial estimates for the unknown quantities. While the values of mole fractions lie between 0 and 1, it is not easy to estimate the initial values for $\lambda_{\text{C}}^*(X_6)$, $\lambda_{\text{H}}^*(X_7)$, and $\lambda_{\text{O}}^*(X_8)$. Therefore, it is recommended to express X_6 , X_7 , and X_8 in terms of other unknown quantities.

Note that Eqns (1)–(3) can be expressed in matrix notation as

$$\begin{bmatrix} 1 & 4 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_6 \\ X_7 \\ X_8 \end{bmatrix} = \begin{bmatrix} -\ln X_1 + \frac{10,465}{33,256} \\ -\ln X_2 + \frac{255,625}{8314} \\ -\ln X_3 + \frac{229,375}{8314} \end{bmatrix}. \quad (10)$$

The solution gives

$$X_6 = \frac{1}{3} \left(-\ln X_1 + 2 \ln X_2 - 2 \ln X_3 - \frac{199,535}{33,256} \right), \quad (11)$$

$$X_7 = \frac{1}{6} \left(-\ln X_1 - \ln X_2 + \ln X_3 + \frac{115,465}{33,256} \right), \quad (12)$$

$$X_8 = \frac{1}{3} \left(\ln X_1 - 2 \ln X_2 - \ln X_3 + \frac{2,952,035}{33,256} \right). \quad (13)$$

Substitution of Eqns (11) and (13) into Eqn (4) gives

$$X_4 = \exp \left(\frac{-33,256 \ln X_1 + 66,512 \ln X_2 + 133,024 \ln X_3 + 226,465}{99,768} \right). \quad (14)$$

Substitution of Eqn (12) into Eqn (5) gives

$$X_5 = \exp \left(\frac{33,256 \ln X_1 + 33,256 \ln X_2 - 33,256 \ln X_3 - 115,465}{99,768} \right). \quad (15)$$

Knowing X_4 , X_5 , X_6 , X_7 , and X_8 in terms of X_1 , X_2 , and X_3 , it is possible to establish the following four nonlinear equations in four unknowns (X_1 , X_2 , X_3 , X_9) by using Eqns (6)–(9):

$$f_1(X_1, X_2, X_3, X_9) = X_1 + X_2 - \frac{1}{X_9} + \exp \left(\frac{-33,256 \ln X_1 + 66,512 \ln X_2 + 133,024 \ln X_3 + 226,465}{99,768} \right) = 0,$$

$$f_2(X_1, X_2, X_3, X_9) = 4X_1 + 2X_2 - \frac{12}{X_9} + 2 \exp \left(\frac{33,256 \ln X_1 + 33,256 \ln X_2 - 33,256 \ln X_3 - 115,465}{99,768} \right) = 0,$$

$$f_3(X_1, X_2, X_3, X_9) = X_2 + X_3 - \frac{4}{X_9} + 2 \exp \left(\frac{-33,256 \ln X_1 + 66,512 \ln X_2 + 133,024 \ln X_3 + 226,465}{99,768} \right) = 0,$$

$$f_4(X_1, X_2, X_3, X_9) = -1 + X_1 + X_2 + X_3 + \exp \left(\frac{-33,256 \ln X_1 + 66,512 \ln X_2 + 133,024 \ln X_3 + 226,465}{99,768} \right) + \exp \left(\frac{33,256 \ln X_1 + 33,256 \ln X_2 - 33,256 \ln X_3 - 115,465}{99,768} \right) = 0.$$

Starting with the initial estimates

$$X_1^{(0)} = 0.02 \quad X_2^{(0)} = 0.4 \quad X_3^{(0)} = 0.02 \quad X_9^{(0)} = 5$$

the solution is

$$X_1 = 0.065132 \quad X_2 = 0.472732 \quad X_3 = 0.019534 \quad X_9 = 6.19324.$$

Then we calculate $X_4 = 0.076800$, $X_5 = 0.365802$, $X_6 = 1.034686$, $X_7 = 0.502832$, and $X_8 = 30.489894$.

14.32 Hydrogen production is becoming increasingly important as a source of fuel for fuel cells. In that respect, steam reforming of ethanol offers an environmentally clean and renewable source for hydrogen production. Steam reforming of ethanol at 1000 K and 1 bar yields the following species: $\text{C}_2\text{H}_5\text{OH}$, H_2O , CO_2 , H_2 , CO , CH_4 , and CH_3CHO . If 3 mol of steam is introduced per mole of ethanol, estimate the composition of the gas mixture using minimization of Gibbs energy. The following data are provided at 1000 K:

Species	$\text{C}_2\text{H}_5\text{OH}$	H_2O	CO_2	CO	CH_4	CH_3CHO
$\Delta \tilde{G}_f^\circ$ (kJ/mol)	8.558	-192.713	-395.810	-200.261	19.572	-41.190

(Answer: $y_{\text{C}_2\text{H}_5\text{OH}} = 7.977 \times 10^{-14}$, $y_{\text{H}_2\text{O}} = 0.182$, $y_{\text{CO}_2} = 0.077$, $y_{\text{H}_2} = 0.564$, $y_{\text{CO}} = 0.17$, $y_{\text{CH}_4} = 6.403 \times 10^{-3}$, $y_{\text{CH}_3\text{CHO}} = 5.61 \times 10^{-11}$)

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Appendix A

Critical Constants and Acentric Factors

Compiled From

1. NIST (National Institute of Standards and Technology) Chemistry WebBook
www.nist.gov/chemistry
2. Korea Thermophysical Properties Data Bank (KDB)
www.thermo.org/research/kdb

Name	Formula	Molecular			
		Weight	T_c (K)	P_c (bar)	ω
Elements					
Bromine	Br ₂	159.81	588.0	103.4	0.108
Chlorine	Cl ₂	70.91	417.0	79.9	0.090
Hydrogen	H ₂	2.02	33.2	13.0	−0.216
Iodine	I ₂	253.81	819.0	116.5	0.229
Neon	Ne	20.18	44.4	27.6	−0.029
Nitrogen	N ₂	28.01	126.2	34.0	0.039
Oxygen	O ₂	32.00	154.6	50.4	0.025
Hydrocarbons					
Acetylene	C ₂ H ₂	26.04	308.3	61.4	0.190
Benzene	C ₆ H ₆	78.11	562.0	48.9	0.212
<i>n</i> -Butane	C ₄ H ₁₀	58.12	425.0	38.0	0.199
Cumene	C ₉ H ₁₂	120.19	631.0	32.1	0.326
Cyclohexane	C ₆ H ₁₂	84.16	554.0	40.7	0.212
Cyclopentane	C ₅ H ₁₀	70.13	511.7	45.1	0.196
<i>n</i> -Decane	C ₁₀ H ₂₂	142.28	617.8	21.1	0.489
Ethane	C ₂ H ₆	30.07	305.3	49.0	0.099
Ethylbenzene	C ₈ H ₁₀	106.17	617.0	36.4	0.302
Ethylene	C ₂ H ₄	28.05	282.5	50.6	0.089
<i>n</i> -Heptane	C ₇ H ₁₆	100.20	540.0	27.4	0.349
1-Heptene	C ₇ H ₁₄	98.19	537.3	29.2	0.358
<i>n</i> -Hexane	C ₆ H ₁₄	86.18	507.6	30.2	0.299
1-Hexene	C ₆ H ₁₄	84.16	504.0	32.1	0.285
Isobutane	C ₄ H ₁₀	58.12	407.7	36.5	0.183
Isopentane	C ₅ H ₁₂	72.15	461.0	33.8	0.227
Methane	CH ₄	16.04	190.6	46.1	0.011
Naphthalene	C ₁₀ H ₈	128.17	748.0	41.0	0.302
<i>n</i> -Octane	C ₈ H ₁₈	114.23	568.9	24.9	0.398
1-Octene	C ₈ H ₁₆	112.21	567.0	26.8	0.386
<i>n</i> -Pentane	C ₅ H ₁₂	72.15	469.8	33.6	0.251
Propane	C ₃ H ₈	44.10	369.9	42.5	0.153
Propylene	C ₃ H ₆	42.08	365.2	46.0	0.144
Styrene	C ₈ H ₈	104.15	647.0	39.9	0.257
Toluene	C ₇ H ₈	92.14	593.0	41.0	0.263
<i>m</i> -Xylene	C ₈ H ₁₀	106.17	618.0	35.4	0.325
<i>o</i> -Xylene	C ₈ H ₁₀	106.17	631.0	37.0	0.310
<i>p</i> -Xylene	C ₈ H ₁₀	106.17	617.0	35.0	0.320

Name	Molecular				
	Formula	Weight	T_c (K)	P_c (bar)	ω
Inorganic compounds					
Ammonia	NH ₃	17.03	405.4	113.0	0.250
Carbon dioxide	CO ₂	44.10	304.2	73.8	0.239
Carbon disulfide	CS ₂	76.14	552.0	79.0	0.109
Carbon monoxide	CO	28.01	134.5	35.0	0.066
Carbon tetrachloride	CCl ₄	153.82	556.4	44.9	0.193
Chloroform	CHCl ₃	119.38	537.0	53.3	0.218
Hydrogen chloride	HCl	36.46	324.7	82.6	0.133
Hydrogen cyanide	HCN	27.03	456.7	53.9	0.388
Hydrogen sulfide	H ₂ S	34.08	373.3	89.7	0.081
Nitric oxide	NO	30.01	180.0	64.8	0.588
Nitrous oxide	N ₂ O	44.01	309.6	72.4	0.165
Sulfur dioxide	SO ₂	64.06	430.3	78.8	0.256
Sulfur trioxide	SO ₃	80.06	491.0	82.0	0.481
Water	H ₂ O	18.02	647.0	220.6	0.344
Organic compounds					
Acetaldehyde	C ₂ H ₄ O	44.05	466.0	55.7	0.303
Acetic acid	C ₂ H ₄ O ₂	60.05	593.0	57.8	0.447
Acetone	C ₃ H ₆ O	58.08	508.0	48.0	0.304
Acetonitrile	C ₂ H ₃ N	41.05	545.0	48.7	0.327
Aniline	C ₆ H ₇ N	93.13	698.8	53.1	0.384
<i>n</i> -Butanol	C ₄ H ₁₀ O	74.12	562.0	45.0	0.593
Chlorobenzene	C ₆ H ₅ Cl	112.56	632.4	45.2	0.249
Diethyl ether	C ₄ H ₁₀ O	74.12	467.0	36.0	0.281
Dimethyl ether	C ₂ H ₆ O	46.07	401.0	54.0	0.200
Ethanol	C ₂ H ₆ O	46.07	514.0	63.0	0.644
Ethyl acetate	C ₄ H ₈ O ₂	88.11	530.0	38.8	0.362
Ethylene oxide	C ₂ H ₄ O	44.05	468.9	72.3	0.202
Formaldehyde	CH ₂ O	30.03	408.0	65.9	0.253
Furfural	C ₅ H ₄ O ₂	96.08	670.0	55.1	0.383
<i>n</i> -Heptanol	C ₇ H ₁₆ O	116.20	632.6	30.6	0.560
Isobutanol	C ₄ H ₁₀ O	74.12	548.0	45.0	0.592
Isopropanol	C ₃ H ₈ O	60.10	509.0	49.0	0.665
Methanol	CH ₄ O	32.04	513.0	81.0	0.556
Methyl acetate	C ₃ H ₆ O ₂	74.08	510.0	47.5	0.326
Methyl ethyl ketone	C ₄ H ₈ O	72.11	535.0	42.0	0.320
Phenol	C ₆ H ₆ O	94.11	694.3	59.3	0.438
<i>n</i> -Propanol	C ₃ H ₈ O	60.10	536.9	52.0	0.623

Appendix B

Heat Capacity of Ideal Gases

Compiled From

Korea Thermophysical Properties Data Bank (KDB)

www.thermo.com/thermo/kdb

The heat capacity is expressed as a function of temperature in the form

$$\tilde{C}_p^* = a + bT + cT^2 + dT^3 + eT^4,$$

where $\tilde{C}_p^*$ is in J/molK and T is in K. The coefficients are valid for $200\text{ K} \leq T \leq 800\text{ K}$.

Name	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
Elements					
Bromine	33.860	0.113	-0.119	453.400	0
Chlorine	26.468	0.340	-0.373	0.159	-0.172
Hydrogen	27.004	0.119	-0.241	0.215	-0.615
Iodine	35.590	0.065	-0.070	0.028	0
Nitrogen	29.802	-0.070	0.174	-0.085	0.093
Oxygen	29.705	-0.099	0.399	-0.339	0.918
Hydrocarbons					
Acetylene	8.709	1.764	-2.419	1.689	-4.437
Benzene	-60.711	6.267	-5.795	2.799	-5.493
<i>n</i> -Butane	24.258	2.335	1.279	-2.437	8.552
Cumene	-42.658	8.017	-5.444	1.643	-1.346
Cyclohexane	-63.733	6.444	-2.633	-0.378	3.796
Cyclopentane	-61.877	5.809	-3.631	1.020	-0.770
<i>n</i> -Decane	121.188	-0.759	23.586	-32.069	134.425
Ethane	33.313	-0.111	3.566	-3.762	11.983
Ethylbenzene	39.308	0.751	11.746	-16.549	67.943
Ethylene	17.562	0.692	0.936	-1.293	4.294
<i>n</i> -Heptane	37.663	3.941	2.587	-4.744	16.635
1-Heptene	28.940	4.001	1.935	-4.420	16.620
<i>n</i> -Hexane	32.365	3.447	2.026	-3.806	13.306
Isobutane	24.108	2.091	2.166	-3.372	11.619
Isopentane	4.817	4.159	-0.809	-1.008	4.688
Methane	36.155	-0.511	2.215	-1.824	4.899
Naphthalene	-47.760	7.213	-3.873	-0.012	4.422
<i>n</i> -Octane	37.209	4.810	2.419	-5.167	18.845
1-Octene	33.168	4.544	2.477	-5.425	20.411
<i>n</i> -Pentane	33.780	2.485	2.535	-3.838	12.977
Propane	29.595	0.838	3.256	-3.958	13.129
Propylene	17.905	1.478	68.773	-1.384	4.845
Styrene	-51.641	7.580	-6.943	3.422	-7.005
Toluene	-20.488	4.800	-1.636	-0.887	5.445
<i>m</i> -Xylene	-33.289	6.354	-3.678	0.687	0.750
<i>o</i> -Xylene	-12.618	5.660	-2.796	0.203	1.721
<i>p</i> -Xylene	-39.704	6.734	-4.317	1.133	-0.380

Name	a	$b \times 10^1$	$c \times 10^4$	$d \times 10^7$	$e \times 10^{11}$
Inorganic compounds					
Ammonia	34.236	-0.221	1.213	-1.088	3.203
Carbon dioxide	29.268	-0.224	2.653	-4.153	20.057
Carbon disulfide	27.440	0.813	-0.767	0.267	0
Carbon monoxide	29.651	-0.007	0.183	-0.094	0.108
Carbon tetrachloride	40.720	2.049	-2.270	0.884	0
Chloroform	24.000	1.893	-1.841	0.666	0
Hydrogen chloride	29.179	-0.002	-0.026	0.093	-0.408
Hydrogen iodide	31.160	-0.143	0.297	-0.135	0
Hydrogen sulfide	33.876	-0.119	0.538	-0.377	0.860
Nitric oxide	34.152	-0.310	0.708	-0.541	1.418
Nitrous oxide	51.060	-0.512	1.154	-0.887	2.347
Sulfur dioxide	31.666	0.142	0.703	-0.931	3.244
Sulfur trioxide	19.210	1.374	-1.176	0.370	0
Water	33.763	-0.006	0.224	-0.100	0.110
Organic compounds					
Acetaldehyde	7.716	1.823	-1.007	0.238	0
Acetic acid	4.840	2.549	-1.753	0.495	0
Acetone	6.301	2.606	-1.253	0.204	0
Acetonitrile	20.480	1.196	-0.449	0.032	0
Aniline	-40.520	6.385	-5.133	1.633	0
<i>n</i> -Butanol	11.707	3.460	-0.251	-1.515	6.499
Chlorobenzene	-33.890	5.631	-4.522	1.426	0
Diethyl ether	39.252	2.467	1.055	-2.084	6.953
Dimethyl ether	17.020	1.791	-0.523	-0.019	0
Ethanol	19.959	1.428	0.776	-1.513	5.366
Ethyl acetate	7.235	4.072	-2.092	0.286	0
Ethylene oxide	-7.519	2.222	-1.256	0.259	0
Formaldehyde	23.480	0.316	0.299	-0.230	0
<i>n</i> -Heptanol	49.070	6.778	-3.447	0.605	0
Isobutanol	-7.708	4.689	-2.884	0.723	0
Isopropanol	24.208	2.088	1.035	-2.322	8.683
Methanol	34.375	-0.134	2.156	-2.090	6.212
Methyl acetate	16.550	2.245	-0.434	0.291	0
Methyl ethyl ketone	41.846	1.881	1.212	-2.057	6.926
Phenol	-27.718	5.545	-4.151	1.277	-0.685
<i>n</i> -Propanol	18.884	2.216	0.618	-1.733	6.408

Appendix C

Antoine Constants

Compiled From

R.C. Reid, J.M. Prausnitz and T.K. Sherwood, *The Properties of Gases and Liquids*, 3rd Ed., McGraw-Hill, New York, 1977.

The vapor pressure is expressed as a function of temperature in the form

$$\ln P^{\text{vap}} = A - \frac{B}{T + C},$$

where P^{vap} is in bar and T is in K.

Name	A	B	C	T _{max}	T _{min}
Elements					
Argon	8.6128	700.51	−5.84	94	81
Bromine	9.2239	2582.32	−51.16	354	259
Chlorine	9.3408	1978.32	−27.01	264	172
Hydrogen	7.0131	164.90	3.19	25	14
Nitrogen	8.3340	588.72	−6.60	90	54
Oxygen	8.7873	734.55	−6.45	100	63
Hydrocarbons					
Acetylene	9.7279	1637.14	−19.77	202	194
Benzene	9.2806	2788.51	−52.36	377	280
<i>n</i> -Butane	9.0580	2154.90	−34.42	290	195
Cumene	9.3520	3363.60	−63.37	454	311
Cyclohexane	9.1325	2766.63	−50.50	380	280
Cyclopentane	9.2372	2588.48	−41.79	345	230
<i>n</i> -Decane	9.3912	3456.80	−78.67	476	330
Ethane	9.0435	1511.42	−17.16	199	130
Ethylbenzene	9.3993	3279.47	−59.95	450	300
Ethylene	8.9166	1347.01	−18.15	182	120
<i>n</i> -Heptane	9.2535	2911.32	−56.51	400	270
1-Heptene	9.2692	2895.51	−53.97	400	265
<i>n</i> -Hexane	9.2164	2697.55	−48.78	370	245
Isobutane	8.9179	2032.73	−33.15	280	187
Isopentane	9.0136	2348.67	−40.05	322	216
Methane	8.6041	597.84	−7.16	120	93
Naphthalene	9.5224	3992.01	−71.29	525	360
<i>n</i> -Octane	9.3224	3120.29	−63.63	425	292
1-Octene	9.3428	3116.52	−60.39	420	288
<i>n</i> -Pentane	9.2131	2477.07	−39.94	330	220
Propane	9.1058	1872.46	−25.16	249	164
Propylene	9.0825	1807.53	−26.15	240	160
Styrene	9.3991	3328.57	−63.72	460	305
Toluene	9.3935	3096.52	−53.67	410	280
<i>m</i> -Xylene	9.5188	3366.99	−58.04	440	300
<i>o</i> -Xylene	9.4954	3395.57	−59.46	445	305
<i>p</i> -Xylene	9.4761	3346.65	−57.84	440	300

Name	A	B	C	T _{max}	T _{min}
Inorganic compounds					
Ammonia	10.3729	2132.50	−32.98	261	179
Carbon dioxide	15.9696	3103.39	−0.16	204	154
Carbon disulfide	9.3642	2690.85	−31.62	342	228
Carbon monoxide	7.7484	530.22	−13.15	108	63
Carbon tetrachloride	9.2540	2808.19	−45.99	374	253
Chloroform	8.3530	2696.79	−46.16	370	260
Hydrogen chloride	9.8838	1714.25	−14.45	200	137
Hydrogen sulfide	9.4838	1768.69	−26.06	230	190
Nitric oxide	13.5112	1572.52	−4.88	140	95
Nitrous oxide	9.5069	1506.49	−25.99	200	144
Sulfur dioxide	10.1478	2302.35	−35.97	280	195
Sulfur trioxide	14.2201	3995.70	−36.66	332	290
Water	11.6834	3816.44	−46.13	441	284
Organic compounds					
Acetaldehyde	9.6729	2465.15	−37.15	320	210
Acetic acid	10.1878	3405.57	−56.34	430	290
Acetone	10.0311	2940.46	−35.93	350	241
Acetonitrile	9.6672	2945.47	−49.15	390	260
Aniline	10.0546	3857.52	−73.15	500	340
<i>n</i> -Butanol	10.5958	3137.02	−94.43	404	288
Chlorobenzene	9.4474	3295.12	−55.60	420	320
Diethyl ether	9.4626	2511.29	−41.95	340	225
Dimethyl ether	10.2265	2361.44	−17.10	265	179
Ethanol	12.2917	3803.98	−41.68	369	270
Ethyl acetate	9.5314	2790.50	−57.15	385	260
Ethylene oxide	10.1198	2567.61	−29.01	310	200
Formaldehyde	9.8573	2204.13	−30.15	271	185
<i>n</i> -Heptanol	8.6866	2626.42	−146.6	449	333
Isobutanol	10.2510	2874.73	−100.3	388	293
Isopropanol	12.0727	3640.20	−53.54	374	273
Methanol	11.9673	3626.55	−34.29	364	257
Methyl acetate	9.5093	2601.92	−56.15	360	245
Methyl ethyl ketone	9.9784	3150.42	−36.65	376	257
Phenol	9.8077	3490.89	−98.59	481	345
<i>n</i> -Propanol	10.9237	3166.28	−80.15	400	285

Appendix D

Corresponding States Using the Lee-Kesler Equation of State*

At a given temperature and pressure, first determine the reduced temperature and pressure

$$T_r = \frac{T}{T_c} \quad \text{and} \quad P_r = \frac{P}{P_c}. \quad (\text{D-1})$$

Compressibility Factor

Let $\tilde{V}_r^{(0)}$ be the ideal reduced volume of a simple fluid, i.e. $\omega = 0$. The value of $\tilde{V}_r^{(0)}$ is estimated from the solution of the following equation

$$\begin{aligned} \frac{P_r \tilde{V}_r^{(0)}}{T_r} &= 1 + \frac{B}{\tilde{V}_r^{(0)}} + \frac{C}{(\tilde{V}_r^{(0)})^2} + \frac{D}{(\tilde{V}_r^{(0)})^5} + \frac{c_4}{T_r^3 (\tilde{V}_r^{(0)})^2} \\ &\times \left[\beta + \frac{\gamma}{(\tilde{V}_r^{(0)})^2} \right] \exp \left[-\frac{\gamma}{(\tilde{V}_r^{(0)})^2} \right], \end{aligned} \quad (\text{D-2})$$

where

$$B = b_1 - \frac{b_2}{T_r} - \frac{b_3}{T_r^2} - \frac{b_4}{T_r^3}, \quad (\text{D-3})$$

$$C = c_1 - \frac{c_2}{T_r} + \frac{c_3}{T_r^3}, \quad (\text{D-4})$$

$$D = d_1 + \frac{d_2}{T_r}, \quad (\text{D-5})$$

*Lee, B.I. and M.G. Kesler, 1975, *AIChE Journal* **21**, 510–525.

Table D.1: Constants in Eqn (D-2).

Constants	Simple Fluid	Reference Fluid
b_1	0.1181193	0.2026579
b_2	0.265728	0.331511
b_3	0.154790	0.027655
b_4	0.030323	0.203488
c_1	0.0236744	0.0313385
c_2	0.0186984	0.0503618
c_3	0	0.016901
c_4	0.042724	0.041577
$d_1 \times 10^4$	0.155488	0.48736
$d_2 \times 10^4$	0.623689	0.0740336
β	0.65392	1.226
γ	0.060167	0.03754

and the constants are given in Table D.1 for a simple fluid. Once $\tilde{V}_r^{(0)}$ is determined, the value of $Z^{(0)}$ is calculated as

$$Z^{(0)} = \frac{P_r \tilde{V}_r^{(0)}}{T_r}. \quad (\text{D-6})$$

For a reference fluid, the term $\tilde{V}_r^{(0)}$ in Eqn (D-2) is replaced by $\tilde{V}_r^{(R)}$ together with the reference fluid constants in Table D.1. Once the value of $\tilde{V}_r^{(R)}$ is determined from the solution of Eqn (D-2), the value of $Z^{(R)}$ is calculated from

$$Z^{(R)} = \frac{P_r \tilde{V}_r^{(R)}}{T_r}. \quad (\text{D-7})$$

Then, the value of $Z^{(1)}$ is given by

$$Z^{(1)} = \frac{Z^{(R)} - Z^{(0)}}{0.3978}. \quad (\text{D-8})$$

Enthalpy Departure Function

For a simple fluid

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(0)}}{RT_c} = T_r \left[Z^{(0)} - 1 - \frac{1}{T_r \tilde{V}_r^{(0)}} \left(b_2 + \frac{2b_3}{T_r} + \frac{3b_4}{T_r^2} \right) - \frac{1}{2T_r (\tilde{V}_r^{(0)})^2} \left(c_2 - \frac{3c_3}{T_r^2} \right) + \frac{d_2}{5T_r (\tilde{V}_r^{(0)})^5} + 3E_o \right], \quad (\text{D-9})$$

where

$$E_o = \frac{c_4}{2T_r^3 \gamma} \left\{ \beta + 1 - \left[\beta + 1 + \frac{\gamma}{(\tilde{V}_r^{(0)})^2} \right] \exp \left[-\frac{\gamma}{(\tilde{V}_r^{(0)})^2} \right] \right\}. \quad (\text{D-10})$$

For a reference fluid

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(R)}}{RT_c} = T_r \left[Z^{(R)} - 1 - \frac{1}{T_r \tilde{V}_r^{(R)}} \left(b_2 + \frac{2b_3}{T_r} + \frac{3b_4}{T_r^2} \right) - \frac{1}{2T_r (\tilde{V}_r^{(R)})^2} \left(c_2 - \frac{3c_3}{T_r^2} \right) + \frac{d_2}{5T_r (\tilde{V}_r^{(R)})^5} + 3E_R \right], \quad (\text{D-11})$$

where

$$E_R = \frac{c_4}{2T_r^3 \gamma} \left\{ \beta + 1 - \left[\beta + 1 + \frac{\gamma}{(\tilde{V}_r^{(R)})^2} \right] \exp \left[-\frac{\gamma}{(\tilde{V}_r^{(R)})^2} \right] \right\}. \quad (\text{D-12})$$

Then, the value of $(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(1)} / RT_c$ is given by

$$\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(1)}}{RT_c} = \frac{1}{0.3978} \left[\frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(R)}}{RT_c} - \frac{(\tilde{H} - \tilde{H}^{\text{IG}})_{T,P}^{(0)}}{RT_c} \right]. \quad (\text{D-13})$$

Entropy Departure Function

For a simple fluid

$$\begin{aligned} \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(0)}}{R} = \ln Z^{(0)} - \frac{1}{\tilde{V}_r^{(0)}} \left(b_1 + \frac{b_3}{T_r^2} + \frac{2b_4}{T_r^3} \right) \\ - \frac{1}{2(\tilde{V}_r^{(0)})^2} \left(c_1 - \frac{2c_3}{T_r^3} \right) - \frac{d_1}{5(\tilde{V}_r^{(0)})^5} + 2E_o. \end{aligned} \quad (\text{D-14})$$

For a reference fluid

$$\begin{aligned} \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(R)}}{R} = \ln Z^{(R)} - \frac{1}{\tilde{V}_r^{(R)}} \left(b_1 + \frac{b_3}{T_r^2} + \frac{2b_4}{T_r^3} \right) \\ - \frac{1}{2(\tilde{V}_r^{(R)})^2} \left(c_1 - \frac{2c_3}{T_r^3} \right) - \frac{d_1}{5(\tilde{V}_r^{(R)})^5} + 2E_R. \end{aligned} \quad (\text{D-15})$$

Then, the value of $(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(1)} / R$ is given by

$$\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(1)}}{R} = \frac{1}{0.3978} \left[\frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(R)}}{R} - \frac{(\tilde{S} - \tilde{S}^{\text{IG}})_{T,P}^{(0)}}{R} \right]. \quad (\text{D-16})$$

Appendix E

Enthalpy and Gibbs Energy of Formation at 298 K and 1 bar

Compiled From

1. NIST (National Institute of Standards and Technology) Chemistry WebBook
www.nist.gov/chemistry
2. Korea Thermophysical Properties Data Bank (KDB)
www.thermo.com/research/kdb

Name	Formula	Phase	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
Hydrocarbons				
Acetylene	C ₂ H ₂	G	226.90	209.30
Benzene	C ₆ H ₆	G	82.98	129.70
Benzene	C ₆ H ₆	L	48.95	124.53
<i>n</i> -Butane	C ₄ H ₁₀	G	−126.2	−16.10
Cumene	C ₉ H ₁₂	G	3.94	137.10
Cyclohexane	C ₆ H ₁₂	G	−123.20	31.78
Cyclohexane	C ₆ H ₁₂	L	−157.70	26.89
<i>n</i> -Decane	C ₁₀ H ₂₂	G	−249.80	33.24
<i>n</i> -Decane	C ₁₀ H ₂₂	L	−301.00	17.25
Ethane	C ₂ H ₆	G	−84.74	−32.95
Ethylbenzene	C ₈ H ₁₀	G	29.81	130.7
Ethylbenzene	C ₈ H ₁₀	L	−12.47	119.95
Ethylene	C ₂ H ₄	G	52.34	68.16
<i>n</i> -Heptane	C ₇ H ₁₆	G	−187.90	8.00
<i>n</i> -Heptane	C ₇ H ₁₆	L	−224.39	1.33
<i>n</i> -Hexane	C ₆ H ₁₄	G	−167.30	−0.17
<i>n</i> -Hexane	C ₆ H ₁₄	L	−198.78	−4.04
1-Hexene	C ₆ H ₁₂	G	−41.70	87.50
Isobutane	C ₄ H ₁₀	G	−134.60	−20.90
Isobutane	C ₄ H ₁₀	L	−158.55	−21.98
Isopentane	C ₅ H ₁₂	G	−154.60	−14.82
Methane	CH ₄	G	−74.90	−50.87
Naphthalene	C ₁₀ H ₈	G	151.1	223.7
Naphthalene	C ₁₀ H ₈	S	78.13	201.18
<i>n</i> -Octane	C ₈ H ₁₈	G	−208.6	16.40
<i>n</i> -Octane	C ₈ H ₁₈	L	−249.95	6.71
1-Octene	C ₈ H ₁₆	G	−82.98	104.3
1-Octene	C ₈ H ₁₆	L	−123.59	94.16
<i>n</i> -Pentane	C ₅ H ₁₂	G	−146.5	−8.37
<i>n</i> -Pentane	C ₅ H ₁₂	L	−173.33	−9.46
Propane	C ₃ H ₈	G	−103.9	−23.49
Propylene	C ₃ H ₆	G	20.43	62.76
Styrene	C ₈ H ₈	G	147.5	213.9
Toluene	C ₇ H ₈	G	50.03	122.1
Toluene	C ₇ H ₈	L	12.01	113.96
<i>m</i> -Xylene	C ₈ H ₁₀	G	17.25	118.9
<i>m</i> -Xylene	C ₈ H ₁₀	L	−25.41	107.73
<i>o</i> -Xylene	C ₈ H ₁₀	G	19.00	122.20
<i>o</i> -Xylene	C ₈ H ₁₀	L	−24.39	110.76
<i>p</i> -Xylene	C ₈ H ₁₀	G	17.96	121.20
<i>p</i> -Xylene	C ₈ H ₁₀	L	−24.45	110.03

Name	Formula	Phase	$\Delta \tilde{H}_f^o$ (kJ/mol)	$\Delta \tilde{G}_f^o$ (kJ/mol)
Inorganic compounds				
Ammonia	NH ₃	G	−45.72	−16.16
Calcium carbonate	CaCO ₃	S	−1206.92	−1128.81
Calcium chloride	CaCl ₂	G	−471.54	−479.15
Calcium chloride	CaCl ₂	S	−795.80	−748.11
Calcium hydroxide	Ca(OH) ₂	S	−986.09	−898.47
Calcium oxide	CaO	S	−634.92	−603.51
Calcium sulfate	CaSO ₄	S	−1434.11	−1321.68
Carbon dioxide	CO ₂	G	−393.80	−394.60
Carbon disulfide	CS ₂	G	117.10	66.95
Carbon monoxide	CO	G	−110.60	−137.40
Carbon tetrachloride	CCl ₄	L	−128.40	−65.20
Chloroform	CHCl ₃	G	−101.30	−68.58
Chloroform	CHCl ₃	L	−134.30	−71.89
Hydrogen chloride	HCl	G	−92.36	−95.30
Hydrogen cyanide	HCN	G	130.60	120.20
Hydrogen iodide	HI	G	26.38	1.59
Hydrogen sulfide	H ₂ S	G	−20.18	−33.08
Nitric oxide	NO	G	90.43	86.75
Nitrogen dioxide	NO ₂	G	33.1	51.26
Nitrous oxide	N ₂ O	G	81.60	103.70
Dinitrogen tetroxide	N ₂ O ₄	G	9.08	97.79
Dinitrogen tetroxide	N ₂ O ₄	L	−19.56	97.51
Sodium carbonate	Na ₂ CO ₃	S	−1130.77	−1048.01
Sodium chloride	NaCl	S	−411.12	−384.02
Sodium hydroxide	NaOH	G	−197.76	−200.19
Sodium hydroxide	NaOH	S	−425.93	−379.74
Sulfur dioxide	SO ₂	G	−297.10	−300.40
Sulfur trioxide	SO ₃	G	−396.00	−371.30
Sulfuric acid	H ₂ SO ₄	G	−735.13	−653.37
Sulfuric acid	H ₂ SO ₄	L	−813.99	−689.89
Tungsten	W	G	851.03	808.90
Tungsten dioxide	WO ₂	G	76.57	62.36
Tungsten dioxide	WO ₂	S	−589.69	−533.86
Water	H ₂ O	G	−242.00	−228.80
Water	H ₂ O	L	−285.83	−237.14
Zinc oxide	ZnO	S	−350.46	−320.48
Zinc sulfate	ZnSO ₄	S	−980.14	−871.45
Zinc sulfide	ZnS	S	−205.98	−201.29

Name	Formula	Phase	$\Delta \tilde{H}_f^\circ$ (kJ/mol)	$\Delta \tilde{G}_f^\circ$ (kJ/mol)
Organic compounds				
Acetaldehyde	C ₂ H ₄ O	G	−164.40	−133.40
Acetic acid	C ₂ H ₄ O ₂	G	−435.10	−376.90
Acetic acid	C ₂ H ₄ O ₂	L	−484.09	−389.23
Acetone	C ₃ H ₆ O	G	−217.70	−153.20
Acetone	C ₃ H ₆ O	L	−248.11	−155.26
Acetonitrile	C ₂ H ₃ N	G	87.92	105.7
Acetonitrile	C ₂ H ₃ N	L	53.17	98.93
Aniline	C ₆ H ₇ N	G	86.92	166.80
Aniline	C ₆ H ₇ N	L	31.11	149.18
<i>n</i> -Butanol	C ₄ H ₁₀ O	G	−274.90	−150.90
<i>n</i> -Butanol	C ₄ H ₁₀ O	L	−326.03	−161.19
Ethanol	C ₂ H ₆ O	G	−235.00	−168.40
Ethanol	C ₂ H ₆ O	L	−276.98	−173.99
Ethyl acetate	C ₄ H ₈ O ₂	G	−443.20	−327.60
Ethyl acetate	C ₄ H ₈ O ₂	L	−479.86	−332.93
Ethylene oxide	C ₂ H ₄ O	G	−52.67	−13.10
Ethylene oxide	C ₂ H ₄ O	L	−77.46	−11.43
Formaldehyde	CH ₂ O	G	−116.00	−110.00
<i>n</i> -Heptanol	C ₇ H ₁₆ O	G	−332.00	−121.00
Isobutanol	C ₄ H ₁₀ O	G	−283.40	−167.40
Isopropanol	C ₃ H ₈ O	G	−272.60	−173.50
Methanol	CH ₄ O	G	−201.30	−162.60
Methanol	CH ₄ O	L	−238.57	−166.15
Methyl ethyl ketone	C ₄ H ₈ O	G	−238.50	−146.20
Phenol	C ₆ H ₆ O	G	−96.42	−32.90
Phenol	C ₆ H ₆ O	S	−165.02	−50.22
<i>n</i> -Propanol	C ₃ H ₈ O	G	−256.60	−161.90
<i>n</i> -Propanol	C ₃ H ₈ O	L	−304.76	−170.78

Appendix F

Matrices

F.1 Matrix Definition

A rectangular array of elements or functions is called a *matrix*. If the array has m rows and n columns, it is called an $m \times n$ matrix and expressed in the form

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}. \quad (\text{F.1-1})$$

The numbers or functions a_{ij} are called the *elements of a matrix*. Equation (F.1-1) is also expressed as

$$\mathbf{A} = (a_{ij}) \quad (\text{F.1-2})$$

in which the subscripts i and j represent the row and the column of the matrix, respectively.

A matrix having only one row is called a *row matrix* (or *row vector*) while a matrix having only one column is called a *column matrix* (or *column vector*). For example,

$$\mathbf{b} = [b_1 \quad b_2 \quad b_3 \quad \cdots \quad b_n] \quad \text{Row matrix (row vector),} \quad (\text{F.1-3})$$

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_m \end{bmatrix} \quad \text{Column matrix (column vector).} \quad (\text{F.1-4})$$

For convenience in writing, the elements of a one-column matrix are frequently arranged horizontally, i.e.

$$\mathbf{b} = \{ b_1 \ b_2 \ b_3 \ \cdots \ b_n \}. \quad (\text{F.1-5})$$

The use of braces then being necessary to indicate the transposition.

F.2 Types of Matrices

Square matrix: When the number of rows and the number of columns are the same, i.e. $m = n$, the matrix is called a *square matrix* or a matrix of order n . For example,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \quad (\text{F.2-1})$$

In a square matrix, the diagonal containing elements of equal indices ($a_{11}, a_{22}, \dots, a_{nn}$) is called the principal diagonal of this matrix.

Diagonal matrix: A square matrix in which all the elements except those on the major diagonal line are zero, is called a *diagonal matrix*. For example,

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}. \quad (\text{F.2-2})$$

A diagonal matrix in which the diagonal elements are all unity is called the *unit matrix* or the *identity matrix*, $\mathbf{I}$, i.e.

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (\text{F.2-3})$$

Triangular Matrix: A matrix having all elements on one side of the major diagonal line are zero is called a *triangular matrix*. If zero elements are below the major diagonal line, then the matrix is called an upper tridiagonal matrix, $\mathbf{U}$. If zero elements are above the major diagonal line, then the matrix is called a *lower tridiagonal* matrix, $\mathbf{L}$, i.e.

$$\mathbf{U} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} \quad \mathbf{L} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \quad (\text{F.2-4})$$

Sparse Matrix: A matrix in which most elements are zero is called a *sparse matrix*.

Null Matrix: A matrix whose elements are all zero is called a *null matrix* and is denoted by 0.

F.3 Matrix Algebra

F.3.1 Addition and Subtraction

If $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ have the same order, the sum of $\mathbf{A}$ and $\mathbf{B}$ is defined as

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij}). \quad (\text{F.3-1})$$

If $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are the matrices of the same order, addition is commutative and associative, i.e.

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad (\text{F.3-2})$$

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}. \quad (\text{F.3-3})$$

If $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ have the same order, the difference of $\mathbf{A}$ and $\mathbf{B}$ is defined as

$$\mathbf{A} - \mathbf{B} = (a_{ij} - b_{ij}). \quad (\text{F.3-4})$$

Example F.1 If

$$\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 2 & -4 \\ 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

determine $\mathbf{A} + \mathbf{B}$ and $\mathbf{A} - \mathbf{B}$.

Solution

The use of Eqn (F.3-1) gives

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 2+2 & -1-4 \\ 1+3 & 0+0 \\ 3+0 & 5+1 \end{bmatrix} = \begin{bmatrix} 4 & -5 \\ 4 & 0 \\ 3 & 6 \end{bmatrix}.$$

Application of Eqn (F.3-4) yields

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} 2-2 & -1+4 \\ 1-3 & 0-0 \\ 3-0 & 5-1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ -2 & 0 \\ 3 & 4 \end{bmatrix}.$$

F.3.2 Multiplication

Scalar multiplication: If $\mathbf{A} = (a_{ij})$ and λ is any number, the product of $\mathbf{A}$ by λ is defined as

$$\lambda \mathbf{A} = \mathbf{A} \lambda = (\lambda a_{ij}). \quad (\text{F.3-5})$$

Example F.2 If

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 4 & 2 \end{bmatrix},$$

determine $2\mathbf{A} - \mathbf{B}$.

Solution

$$2\mathbf{A} - \mathbf{B} = 2 \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} (2)(2) - 1 & (2)(1) + 1 \\ (2)(3) - 4 & (2)(5) - 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 8 \end{bmatrix}.$$

Matrix multiplication: The product of two matrices $\mathbf{A}$ and $\mathbf{B}$, $\mathbf{A} \cdot \mathbf{B}$, is defined only if the number of columns in $\mathbf{A}$ is equal to the number of rows in $\mathbf{B}$. In this case, the two matrices are said to be *conformable* in the order stated. For example, if $\mathbf{A}$ is of order (4×2) and $\mathbf{B}$ is of order (2×3) , then the product $\mathbf{A} \cdot \mathbf{B}$ is

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} & a_{31}b_{13} + a_{32}b_{23} \\ a_{41}b_{11} + a_{42}b_{21} & a_{41}b_{12} + a_{42}b_{22} & a_{41}b_{13} + a_{42}b_{23} \end{bmatrix}. \end{aligned} \quad (\text{F.3-6})$$

In general, if a matrix of order $(m \times r)$ is multiplied by a matrix of order $(r \times n)$, the product is a matrix of order $(m \times n)$. Symbolically, this may be expressed as

$$(m \times r) \cdot (r \times n) = (m \times n). \quad (\text{F.3-7})$$

Example F.3 If

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \\ -1 & 5 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

determine $\mathbf{A} \cdot \mathbf{B}$.

Solution

$$\mathbf{A} \cdot \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (1)(1) + (-1)(2) \\ (2)(1) + (0)(2) \\ (-1)(1) + (5)(2) \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 9 \end{bmatrix}.$$

A matrix $\mathbf{A}$ can be multiplied by itself if and only if it is a square matrix. The product $\mathbf{A} \cdot \mathbf{A}$ can be expressed as $\mathbf{A}^2$. If the relevant products are defined, multiplication of matrices is associative, i.e.

$$\mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C}) = (\mathbf{A} \cdot \mathbf{B}) \cdot \mathbf{C}, \quad (\text{F.3-8})$$

and distributive, i.e.

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}, \quad (\text{F.3-9})$$

$$(\mathbf{B} + \mathbf{C}) \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{A} + \mathbf{C} \cdot \mathbf{A}, \quad (\text{F.3-10})$$

but, in general, not commutative, i.e. $\mathbf{A} \cdot \mathbf{B} \neq \mathbf{B} \cdot \mathbf{A}$. For example, if $\mathbf{A}$ is of order $(m \times n)$ and $\mathbf{B}$ is of order $(n \times m)$, then the product $\mathbf{A} \cdot \mathbf{B} = \mathbf{C}$ is a matrix of order $(m \times m)$. On the other hand, the product $\mathbf{B} \cdot \mathbf{A} = \mathbf{D}$ is a matrix of order $(n \times n)$. Even if the matrices $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order, in general, $\mathbf{C}$ will be different from $\mathbf{D}$.

For any unit (or identity) matrix $\mathbf{I}$

$$\mathbf{I} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{I} = \mathbf{A}. \quad (\text{F.3-11})$$

F.3.3 Some Properties of Matrices

- Any row (or column) in a matrix can be modified by multiplying or dividing by a nonzero scalar without affecting the matrix.
- Any row (or column) can be added or subtracted from another row (or column) without affecting the matrix.
- Any two rows (or columns) can be interchanged without affecting the matrix.

F.4 Determinants

For each square matrix $\mathbf{A}$, it is possible to associate a scalar quantity called the determinant of $\mathbf{A}$, $|\mathbf{A}|$. If the matrix $\mathbf{A}$ in Eqn (F.1-1) is a square matrix, then the

determinant of $\mathbf{A}$ is given by

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}. \quad (\text{F.4-1})$$

If the row and column containing an element a_{ij} in a square matrix $\mathbf{A}$ are deleted, the determinant of the remaining square array is called the *minor* of a_{ij} and denoted by M_{ij} . The *cofactor* of a_{ij} , denoted by A_{ij} , is then defined by the relation

$$A_{ij} = (-1)^{i+j} M_{ij}. \quad (\text{F.4-2})$$

Thus, if the sum of the row and column indices of an element is even, the cofactor and the minor of that element are identical; otherwise they differ in sign.

The determinant of a square matrix $\mathbf{A}$ can be calculated by the following formula:

$$|\mathbf{A}| = \sum_{k=1}^n a_{ik} A_{ik} = \sum_{k=1}^n a_{kj} A_{kj}, \quad (\text{F.4-3})$$

where i and j may stand for any row and column, respectively. Therefore, the determinants of (2×2) and (3×3) matrices are

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}, \quad (\text{F.4-4})$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}. \quad (\text{F.4-5})$$

Example F.4 Determine $|\mathbf{A}|$ if

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 3 & 2 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

Solution

Expanding on the first row, i.e. $i = 1$, gives

$$|\mathbf{A}| = 1 \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} - 0 \begin{vmatrix} 3 & 1 \\ -1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} = -1 + 5 = 4.$$

F.4.1 Some Properties of Determinants

- If all elements in a row or column are zero, the determinant is zero, i.e.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 0 & 0 & 0 \end{vmatrix} = 0 \quad \begin{vmatrix} 0 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & b_3 & c_3 \end{vmatrix} = 0. \quad (\text{F.4-6})$$

- The value of a determinant is not altered when the rows are changed to columns and the columns to rows, i.e. when the rows and columns are interchanged.
- The interchange of any two columns or any two rows of a determinant changes the sign of the determinant.
- If two columns or two rows of a determinant are identical, the determinant is equal to zero.
- If each element in any column or row of a determinant is expressed as the sum of two quantities, the determinant can be expressed as the sum of two determinants of the same order, i.e.

$$\begin{vmatrix} a_1 + d_1 & b_1 & c_1 \\ a_2 + d_2 & b_2 & c_2 \\ a_3 + d_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}. \quad (\text{F.4-7})$$

- Adding the same multiple of each element of one row to the corresponding element of another row does not change the value of the determinant. The same holds true for the columns.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} (a_1 + nb_1) & b_1 & c_1 \\ (a_2 + nb_2) & b_2 & c_2 \\ (a_3 + nb_3) & b_3 & c_3 \end{vmatrix}. \quad (\text{F.4-8})$$

- If all the elements in any column or row are multiplied by any factor, the determinant is multiplied by that factor, i.e.

$$\begin{vmatrix} \lambda a_1 & b_1 & c_1 \\ \lambda a_2 & b_2 & c_2 \\ \lambda a_3 & b_3 & c_3 \end{vmatrix} = \lambda \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \quad (\text{F.4-9})$$

$$\begin{vmatrix} (1/\lambda)a_1 & b_1 & c_1 \\ (1/\lambda)a_2 & b_2 & c_2 \\ (1/\lambda)a_3 & b_3 & c_3 \end{vmatrix} = \frac{1}{\lambda} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}. \quad (\text{F.4-10})$$

F.5 Special Matrices

Transpose of a matrix: The matrix which is obtained from $\mathbf{A}$ by interchanging rows and columns is called the *transpose* of $\mathbf{A}$ and denoted by $\mathbf{A}^T$. For example,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ -1 & 1 \end{bmatrix} \Rightarrow \mathbf{A}^T = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 1 \end{bmatrix}.$$

A matrix and its transpose are always conformable for multiplication, i.e.

$$\mathbf{A} \cdot \mathbf{A}^T = \mathbf{A}^T \cdot \mathbf{A} \quad (\text{F.5-1})$$

and if

$$\mathbf{A} \cdot \mathbf{A}^T = \mathbf{A}^T \cdot \mathbf{A} = \mathbf{I}, \quad (\text{F.5-2})$$

then the matrix $\mathbf{A}$ is said to be *orthogonal*.

The transpose of the product $\mathbf{A} \cdot \mathbf{B}$ is the product of the transposes in the form

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T. \quad (\text{F.5-3})$$

Symmetric and skew-symmetric matrices: A square matrix $\mathbf{A}$ is said to be *symmetric* if

$$\mathbf{A} = \mathbf{A}^T \quad \text{or} \quad a_{ij} = a_{ji}. \quad (\text{F.5-4})$$

A square matrix $\mathbf{A}$ is said to be *skew-symmetric* (or *antisymmetric*) if

$$\mathbf{A} = -\mathbf{A}^T \quad \text{or} \quad a_{ij} = -a_{ji}. \quad (\text{F.5-5})$$

Equation (F.5-5) implies that the diagonal elements of a skew-symmetric matrix are all zero.

Singular matrix: A square matrix $\mathbf{A}$ for which the determinant $|\mathbf{A}|$ of its elements is zero, is termed a singular matrix. If $|\mathbf{A}| \neq 0$, then $\mathbf{A}$ is *nonsingular*.

Inverse of a matrix: If the determinant $|\mathbf{A}|$ of a square matrix $\mathbf{A}$ does not vanish, i.e. nonsingular matrix, it then possesses an *inverse* (or, *reciprocal*) matrix $\mathbf{A}^{-1}$ such that

$$\mathbf{A} \cdot \mathbf{A}^{-1} = \mathbf{A}^{-1} \cdot \mathbf{A} = \mathbf{I}. \quad (\text{F.5-6})$$

The inverse of a matrix $\mathbf{A}$ is defined by

$$\mathbf{A}^{-1} = \frac{\text{Adj } \mathbf{A}}{|\mathbf{A}|}, \quad (\text{F.5-7})$$

where $\text{Adj } \mathbf{A}$ is called the *adjoint* of $\mathbf{A}$. It is obtained from a square matrix $\mathbf{A}$ by replacing each element by its cofactor and then interchanging rows and columns.

Example F.5 Find the inverse of the matrix $\mathbf{A}$ given in Example F.4.

Solution

The minor of $\mathbf{A}$ is given by

$$M_{ij} = \begin{bmatrix} \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} & \begin{vmatrix} 3 & 1 \\ -1 & 0 \end{vmatrix} & \begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 5 \\ -1 & 1 & 1 \\ -2 & -2 & 2 \end{bmatrix}.$$

The cofactor matrix is

$$A_{ij} = \begin{bmatrix} -1 & -1 & 5 \\ 1 & 1 & -1 \\ -2 & 2 & 2 \end{bmatrix}.$$

The transpose of the cofactor matrix gives the adjoint of $\mathbf{A}$ as

$$\text{Adj } \mathbf{A} = \begin{bmatrix} -1 & 1 & -2 \\ -1 & 1 & 2 \\ 5 & -1 & 2 \end{bmatrix}.$$

Since $|\mathbf{A}| = 4$, the use of (F.5-7) gives the inverse of $\mathbf{A}$ in the form

$$\mathbf{A}^{-1} = \frac{\text{Adj } \mathbf{A}}{|\mathbf{A}|} = \begin{bmatrix} -0.25 & 0.25 & -0.5 \\ -0.25 & 0.25 & 0.5 \\ 1.25 & -0.25 & 0.5 \end{bmatrix}.$$

Consider the following matrix equation

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{C}. \quad (\text{F.5-8})$$

Premultiplication of the both sides of Eqn (F.5-8) by $\mathbf{A}^{-1}$ gives

$$\mathbf{A}^{-1} \cdot (\mathbf{A} \cdot \mathbf{B}) = \mathbf{A}^{-1} \cdot \mathbf{C},$$

$$(\mathbf{A}^{-1} \cdot \mathbf{A}) \cdot \mathbf{B} = \mathbf{A}^{-1} \cdot \mathbf{C},$$

$$\mathbf{I} \cdot \mathbf{B} = \mathbf{A}^{-1} \cdot \mathbf{C},$$

$$\mathbf{B} = \mathbf{A}^{-1} \cdot \mathbf{C}. \quad (\text{F.5-9})$$

This procedure is especially useful in solving systems of linear equations.

Example F.6 Solve the following system of algebraic equations

$$\begin{aligned} x + y + z + w &= 7, \\ 2x + y - 4z + w &= -1, \\ 3x - y + 2z + 2w &= 5, \\ 2x - 3y - z - 3w &= -34. \end{aligned}$$

Solution

In matrix notation, the given equations are expressed as

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -3 & -1 & -3 \end{bmatrix}}_{\mathbf{A}} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 5 \\ -34 \end{bmatrix}, \quad (1)$$

where $\mathbf{A}$ is the coefficient matrix. Premultiplication of the both sides of Eqn (1) by $\mathbf{A}^{-1}$ gives

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -3 & -1 & -3 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 7 \\ -1 \\ 5 \\ -34 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 1 \\ 7 \end{bmatrix}.$$

Therefore, $x = -3$, $y = 2$, $z = 1$, and $w = 7$.

The inverse of the product of two nonsingular square matrices $\mathbf{A}$ and $\mathbf{B}$ is the product of the inverses in the form

$$(\mathbf{A} \cdot \mathbf{B})^{-1} = \mathbf{B}^{-1} \cdot \mathbf{A}^{-1}. \quad (\text{F.5-10})$$

The inverse of a transpose of a matrix is equal to the transpose of its inverse, i.e.

$$(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T. \quad (\text{F.5-11})$$

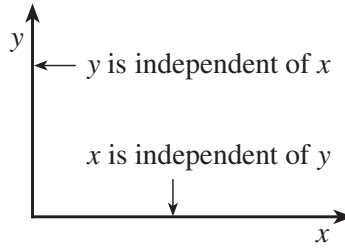


Figure F.1: Cartesian coordinate system.

F.6 Linear Dependence

Consider the x - y coordinate system as shown in Fig. F.1. The linear combination of x and y is expressed as $\lambda_1 x + \lambda_2 y$, where λ_1 and λ_2 are nonzero scalars. Let the linear combination of x and y be equal to zero, i.e.

$$\lambda_1 x + \lambda_2 y = 0. \quad (\text{F.6-1})$$

Note that $y = 0$ on the x -axis, and Eqn (F.6-1) gives

$$\lambda_1 x = 0. \quad (\text{F.6-2})$$

Since Eqn (F.6-2) must hold for any x , then $\lambda_1 = 0$. On the other hand, $x = 0$ on the y -axis and Eqn (F.6-1) reduces to

$$\lambda_2 y = 0. \quad (\text{F.6-3})$$

Since Eqn (F.6-3) must hold for any y , then $\lambda_2 = 0$. Therefore, when x and y are *independent*, then the only set of multipliers λ_1 and λ_2 which will make $\lambda_1 x + \lambda_2 y = 0$ are $\lambda_1 = \lambda_2 = 0$.

On the other hand, let us consider the case in which y is related to x through the following relationship

$$y = mx. \quad (\text{F.6-4})$$

When the linear combination of x and y is equal to zero, Eqn (F.6-1) is still valid. Substitution of Eqn (F.6-4) into Eqn (F.6-1) gives

$$(\lambda_1 + m\lambda_2)x = 0 \Rightarrow \lambda_1 + m\lambda_2 = 0 \Rightarrow \lambda_1 = -m\lambda_2. \quad (\text{F.6-5})$$

Therefore, when x and y are *dependent*, the equation $\lambda_1 x + \lambda_2 y = 0$ is satisfied when $\lambda_1 = -m\lambda_2$.

F.6.1 Rank of a Matrix

Let $\mathbf{A}$ be a matrix of order $(m \times n)$ in which m is less than n . The determinants formed from the m rows and any m of the n columns are called *minors* of the matrix $\mathbf{A}$.

The *rank of a matrix*, denoted by $\text{rank}(\mathbf{A})$, is equal to the order of the highest order nonvanishing minor. For a square matrix of order $(n \times n)$, the first minor will be the determinant of the matrix and of order n . The second minor will be of order $(n - 1)$, and so onto the minor of order 1.

Example F.7 Determine the rank of the matrix

$$\begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ 5 & -2 & 4 \end{bmatrix}.$$

Solution

The first minor is

$$\begin{vmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ 5 & -2 & 4 \end{vmatrix} = 2 \begin{vmatrix} 1 & 3 \\ -2 & 4 \end{vmatrix} + 3 \begin{vmatrix} 3 & 3 \\ 5 & 4 \end{vmatrix} + 1 \begin{vmatrix} 3 & 1 \\ 5 & -2 \end{vmatrix} \\ = 2(4 + 6) + 3(12 - 15) + 1(-6 - 5) = 0,$$

indicating that the rank is not equal to 3. For one of the determinants of order 2, i.e.

$$\begin{vmatrix} 1 & 3 \\ -2 & 4 \end{vmatrix} = 10 \neq 0.$$

Therefore, the rank of the given matrix is 2.

Comment: Note that the first minor is zero since the third row is the summation of the first and second rows. As a result, the rank of a matrix is defined as the number of linearly independent rows (or columns) in the matrix.

Example F.8 Determine the rank of the coefficient matrix $\mathbf{A}$ in Example F.6.

Solution

The first minor is

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -3 & -1 & -3 \end{vmatrix} = 81 \neq 0.$$

Therefore, the rank of the coefficient matrix is 4.

Example F.9 Solve the following system of algebraic equations

$$\begin{aligned}x + y + z + w &= 7, \\2x + y - 4z + w &= -1, \\3x - y + 2z + 2w &= 5, \\2x - y + 7z + 2w &= 13.\end{aligned}$$

Solution

In matrix notation, the given equations are expressed as

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -1 & 7 & 2 \end{bmatrix}}_{\mathbf{A}} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 5 \\ 13 \end{bmatrix}, \quad (1)$$

where $\mathbf{A}$ is the coefficient matrix. Premultiplication of the both sides of Eqn (1) by $\mathbf{A}^{-1}$ gives

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -1 & 7 & 2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 7 \\ -1 \\ 5 \\ 13 \end{bmatrix}. \quad (2)$$

The solution, however, does not exist because $\mathbf{A}^{-1}$ cannot be calculated. The reason for this is the fact that the determinant of the coefficient matrix is zero. Since the fourth equation can be obtained as

$$\text{Fourth eq.} = \text{First eq.} + \text{Third eq.} - \text{Second eq.},$$

then only three of the given four equations are independent and, the number of unknowns exceeds the number of independent equations.

Comment: Note that the rank of the coefficient matrix is three. In other words, the rank of a matrix indicates the number of independent rows.

F.6.2 Reduced Row Echelon Form of a Matrix

A matrix is said to be in *reduced row echelon form* if it satisfies the following criteria:

1. In each row that does not consist of all zero elements, the first nonzero element in this row is a 1, called “leading 1.”
2. In any two successive rows with nonzero elements, the leading 1 of the lower row occurs farther to the right than the leading 1 of the higher row.
3. All rows consisting entirely of 0 are at the bottom of the matrix.
4. In each column that contains a leading 1 of some row, all other elements of this column are zeros.

For example, while the following matrices **A** and **B** are in reduced row echelon form

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

the matrix **C**

$$\mathbf{C} = \begin{bmatrix} 1 & -1 & -3 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 1 & \frac{1}{2} & -5 \end{bmatrix}$$

is not in reduced row echelon form since the leading 1 in the third row is not in the right of the leading 1 in the second row. Besides, all the other entries above the leading 1 in the second column are not zero.

Every matrix of order $(m \times n)$ can be transformed to a unique matrix in reduced row echelon form by elementary row operations.

Example F.10 Obtain the reduced row echelon form of the following matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 3 & 7 & 0 \\ 1 & 3 & -2 & 17 \end{bmatrix}.$$

Solution

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 3 & 7 & 0 \\ 1 & 3 & -2 & 17 \end{bmatrix}$$

$\downarrow$ Row 2 = Row 2 - 2(Row 1)
 $\downarrow$ Row 3 = Row 3 - Row 1

$$\begin{aligned}
 & \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & 5 & -6 \\ 0 & 2 & -3 & 14 \end{bmatrix} \\
 & \quad \downarrow \begin{array}{l} \text{Row 1} = \text{Row 1} - \text{Row 2} \\ \text{Row 3} = \text{Row 3} - 2(\text{Row 2}) \end{array} \\
 & \begin{bmatrix} 1 & 0 & -4 & 9 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & -13 & 26 \end{bmatrix} \\
 & \quad \downarrow \text{Row 3} = \text{Row 3} / (-13) \\
 & \begin{bmatrix} 1 & 0 & -4 & 9 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 1 & -2 \end{bmatrix} \\
 & \quad \downarrow \begin{array}{l} \text{Row 1} = \text{Row 1} + 4(\text{Row 3}) \\ \text{Row 2} = \text{Row 2} - 5(\text{Row 3}) \end{array} \\
 & \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} = \text{rref}[\mathbf{A}].
 \end{aligned}$$

Comment: The reduced row echelon form of a matrix can be easily obtained by using MATHCAD®.

Consider a system of linear equations given by

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\
 a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\
 &\vdots \\
 a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m.
 \end{aligned}$$

In matrix notation, these equations are expressed as

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_{\mathbf{A}} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \quad (\text{F.6-6})$$

where $\mathbf{A}$ represents the coefficient matrix. The *augmented matrix* is a coefficient matrix having an extra column containing the constant terms and this extra column

is separated by a vertical line, i.e.

$$[A | b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]. \quad (\text{F.6-7})$$

Once the augmented matrix $[A | b]$ can be transformed to the matrix in reduced row echelon form $[C | d]$ by elementary row operations, i.e.

$$\text{rref} \{ [A | b] \} = [C | d] = \left[\begin{array}{cccc|c} c_{11} & c_{12} & \cdots & c_{1n} & d_1 \\ c_{21} & c_{22} & \cdots & c_{2n} & d_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} & d_m \end{array} \right], \quad (\text{F.6-8})$$

then, the solutions for the linear system corresponding to $[C | d]$ is exactly the same as the one corresponding to $[A | b]$. In other words,

$$\left[\begin{array}{cccc} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{array} \right] \cdot \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right] = \left[\begin{array}{c} d_1 \\ d_2 \\ \vdots \\ d_m \end{array} \right]. \quad (\text{F.6-9})$$

Example F.11 Solve the system of algebraic equations given in Example F.6.

Solution

In matrix notation, the given equations are expressed as

$$\underbrace{\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 2 & 1 & -4 & 1 \\ 3 & -1 & 2 & 2 \\ 2 & -3 & -1 & -3 \end{array} \right]}_A \left[\begin{array}{c} x \\ y \\ z \\ w \end{array} \right] = \left[\begin{array}{c} 7 \\ -1 \\ 5 \\ -34 \end{array} \right]. \quad (1)$$

The augmented matrix, i.e. $[A | b]$, is given by

$$[A | b] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 7 \\ 2 & 1 & -4 & 1 & -1 \\ 3 & -1 & 2 & 2 & 5 \\ 2 & -3 & -1 & -3 & -34 \end{array} \right]. \quad (2)$$

The reduced row echelon form (rref) of the *augmented matrix*, i.e. $[C | d]$, is given by

$$[C | d] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 7 \end{array} \right]. \quad (3)$$

Since the linear system corresponding to $[C | d]$ is exactly the same as the one corresponding to $[A | b]$, then

$$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 1 \\ 7 \end{bmatrix}. \quad (4)$$

Multiplication yields

$$x = -3 \quad y = 2 \quad z = 1 \quad w = 7.$$

The following two properties related to the reduced row echelon form of a matrix find widespread application in chemical reaction equilibrium:

- The rank of a matrix **A** is equal to the number of nonzero rows in the reduced row echelon form of the matrix **A**.

Example F.12 Determine the rank of the following matrix:

$$\mathbf{A} = \left[\begin{array}{ccccc} 6 & 2 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 2 & 0 & 1 & 1 & 1 \\ 4 & 1 & 2 & 1 & 1 \end{array} \right].$$

Solution

The rank of the matrix **A** is three. On the other, the reduced row echelon form of the matrix **A** is given by

$$\text{rref}[\mathbf{A}] = \left[\begin{array}{ccccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

The number of nonzero rows in the reduced row echelon form is also three.

- If multiplication of two matrices is equal to zero, i.e.

$$\mathbf{A} \cdot \mathbf{B} = 0, \quad (\text{F.6-10})$$

then

$$\text{rref}[\mathbf{A}] \cdot \mathbf{B} = 0. \quad (\text{F.6-11})$$

Example F.13 Consider the system of algebraic equations expressed as

$$\begin{aligned} 4x_1 + x_2 + x_3 &= 0, \\ 2x_1 - x_2 + 3x_3 + x_4 &= 0, \\ -x_1 + 2x_2 + x_3 - 3x_4 &= 0. \end{aligned}$$

In matrix notation, the given equations are written as

$$\underbrace{\begin{bmatrix} 4 & 1 & 1 & 0 \\ 2 & -1 & 3 & 1 \\ -1 & 2 & 1 & -3 \end{bmatrix}}_{\mathbf{A}} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (1)$$

The reduced row echelon form of the matrix $\mathbf{A}$ is

$$\text{rref}[\mathbf{A}] = \begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{7}{2} \\ 0 & 0 & 1 & \frac{5}{2} \end{bmatrix}. \quad (2)$$

Therefore, Eqn (F.6-11) is written as

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{7}{2} \\ 0 & 0 & 1 & \frac{5}{2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (3)$$

or, the given set of equations are expressed as

$$\begin{aligned} x_1 - \frac{3}{2}x_4 &= 0, \\ x_2 + \frac{7}{2}x_4 &= 0, \\ x_3 + \frac{5}{2}x_4 &= 0. \end{aligned}$$

Appendix G

Databanks, Simulation Programs, Books, Websites

Finding thermochemical data and selecting proper thermodynamic model are major challenges, among others, in the design of a chemical plant. Some of the commercial and/or free-to-use literature on the subject are provided below.

Databanks

1. CODATA (The Committee on Data for Science and Technology)
 - <http://www.codata.org/resources/databases/key1.html>.
 - It provides internationally agreed values for the thermodynamic properties of key chemical substances.
2. DETHERM
 - <http://i-systems.dechema.de/detherm>.
 - It provides thermophysical properties of pure substances and mixtures for design and optimization of chemical plants and apparatus.
3. DIPPR (Design Institute for Physical Properties)
 - <http://dippr.byu.edu/student.asp>.
 - Student version of the database provides students to access 100 compounds.
4. Dortmund Data Bank Software Package (DDBSP)
 - <http://www.ddbst.de>.
 - It provides thermodynamic and transport properties of pure components and mixtures for use in process design and development as well as special programs for data correlation, estimation, and process synthesis.

5. FactSage
 - <http://factsage.com>.
 - It is an integrated database computing system in chemical thermodynamics.
6. Kaye and Laby Online
 - <http://www.kayelaby.npl.co.uk>.
 - It provides tables of physical and chemical constants.
7. KDB (Korea Thermophysical Properties Data Bank)
 - <http://www.therc.org/research/kdb>.
 - It provides information about thermophysical properties and calculation methods for hydrocarbons, light gases, polymers, and electrolyte solutions commonly encountered in chemical engineering practices.
8. NIST (National Institute of Standards and Technology) Chemistry WebBook
 - <http://webbook.nist.gov/chemistry>.
 - It provides chemical and physical property data for various chemical species.
9. NIST/TRC Web Thermo Tables (WTT)
 - <http://wtt-pro.nist.gov/cgi-bin/openindex.cgi>.
 - It provides thermodynamic property data primarily for pure organic compounds.

Simulation Programs

1. Aspen HYSYS
 - <http://www.aspentech.com/hysys>.
 - Process simulation of chemical plants and oil refineries. It includes tools for estimation of physical properties and liquid-vapor phase equilibria, heat and material balances, and simulation of many types of chemical engineering equipment.
2. CALPHAD (CALculation of PHase Diagrams)
 - <http://www.calphad.org>.
 - Simulation software for performing thermodynamic and phase diagram calculations.

3. CHEMCAD
 - <http://www.chemstations.net>.
 - Chemical engineering process simulation.
4. CompuTherm LLC
 - <http://www.computherm.com>.
 - It provides phase diagram and thermodynamic calculation software tools for industrial, academic, and educational use.
5. EQS4WIN (Chemical EQUilibrium Software 4WINdows)
 - <http://www.mathtrek.com>.
 - Calculates equilibrium compositions for reacting systems.
6. MTDATA
 - <http://www.npl.co.uk/advanced-materials/measurement-techniques/modelling/mtdata>.
 - Software tool for the calculation of phase equilibria and thermodynamic properties.
7. Physical Property Data Service (PPDS)
 - <http://www.tuvnel.com/contents/PPDS.aspx>.
 - It provides key information on more than 15,000 chemicals, constant and variable physical properties, thermodynamic model sets, and flash calculations.
8. PRODE
 - <http://www.prode.com>.
 - Solves phase equilibrium separations, prints phase envelopes and phase diagrams, calculates temperature or pressure at dew point, bubble point or specified liquid fraction.
9. ProSim
 - <http://www.prosim.net>.
 - Process simulation and optimization for design and operation of industrial processes.
10. Thermo-Calc
 - <http://www.thermocalc.com>.
 - Simulation software for performing thermodynamic calculations.

11. WinSim

- <http://www.winsim.com>.
- Process simulation for chemical and hydrocarbon processes including refining, refrigeration, petrochemical, gas processing, gas treating, pipelines, fuel cells, ammonia, methanol, sulfur, and hydrogen facilities.

Note: It is extremely important to keep in mind that the values obtained from simulation packages do not guarantee correct results. One should understand the thermodynamic assumptions on which the simulation program is based on. For more details on the subject the reader may refer to:

- Agarwal, R., Y.K. Li, O. Santollani, M.A. Satyro and A. Vieler, 2001, Uncovering the realities of simulation—Part 1, *Chemical Engineering Progress* **97** (5), 42–52.
- Agarwal, R., Y.K. Li, O. Santollani, M.A. Satyro and A. Vieler, 2001, Uncovering the realities of simulation—Part 2, *Chemical Engineering Progress* **97** (6), 64–72.

Books

1. Assael, M.J., J.P.M. Trusler, T.F. Tsolakis, *Thermophysical Properties of Fluids: An Introduction to Their Prediction*, Imperial College Press, 1996.
2. Barin, I., *Thermophysical Data of Pure Substances*, VCH, Weinheim, 1989.
3. DECHEMA—Chemistry Data Series

- | | |
|-----------|--|
| Vol. I | Vapor-Liquid Equilibrium Data Collection |
| Vol. II | Critical Data of Pure Substances |
| Vol. III | Heats of Mixing Data Collection |
| Vol. IV | Recommended Data of Selected Compounds and Binary Mixtures |
| Vol. V | Liquid-Liquid Equilibrium Data Collection |
| Vol. VI | Vapor-Liquid Equilibria for Mixtures of Low Boiling Substances |
| Vol. VIII | Solid-Liquid Equilibrium Data Collection |
| Vol. IX | Activity Coefficients at Infinite Dilution |
| Vol. X | Thermal Conductivity and Viscosity Data of Fluid Mixtures |
| Vol. XII | Electrolyte Data Collection |
| Vol. XIV | Polymer Solution Data Collection |
| Vol. XV | Solubility and Related Properties of Large Complex Chemicals |

4. Millat, J., J.H. Dymond, C.A. Nieto de Castro, *Transport Properties of Fluids*, Cambridge University Press, 1996.

5. *NIST-JANAF Thermochemical Tables*, 4th Ed., Chase, M.W. Jr. (Ed.), Springer, 1998.
6. Poling, B.E., J.M. Prausnitz, J.P. O'Connell, *The Properties of Gases and Liquids*, 5th Ed., McGraw-Hill, New York, 2001.
7. TRC Thermodynamic Tables—Hydrocarbons, Ed. Frenkel, M., NIST, Boulder, CO, Standard Reference Data Program, Publication Series NSRDS-NIST-75 (1942–2007), Gaithersburg, MD.
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9. Vargaftik, N.B., Y.K. Vinogradov, V.S. Yargin, *Handbook of Physical Properties of Liquids and Gases*, Third Augmented and Revised Ed., Begell House, New York, 1996.
10. Yaws, C.L., *Thermodynamic and Physical Property Data*, Gulf Pub. Company, Houston, Texas, 1992.

Websites

1. Chemical Reaction Stoichiometry (CRS)
<http://www.chemical-stoichiometry.net>.
2. Gibbs Models
<http://www.iastate.edu/~jolls>.
3. Macatea Productions—Thermo Workshop
<http://www.macatea.com/wshop>.
4. Nano, Quantum and Statistical Mechanics and Thermodynamics Educational Sites
http://tiger.uic.edu/~mansoori/Thermodynamics.Educational.Sites_html.
5. Thermographics
<http://www.owl.net.rice.edu/~wgchap/Phase/v3dcmnt.htm>.

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